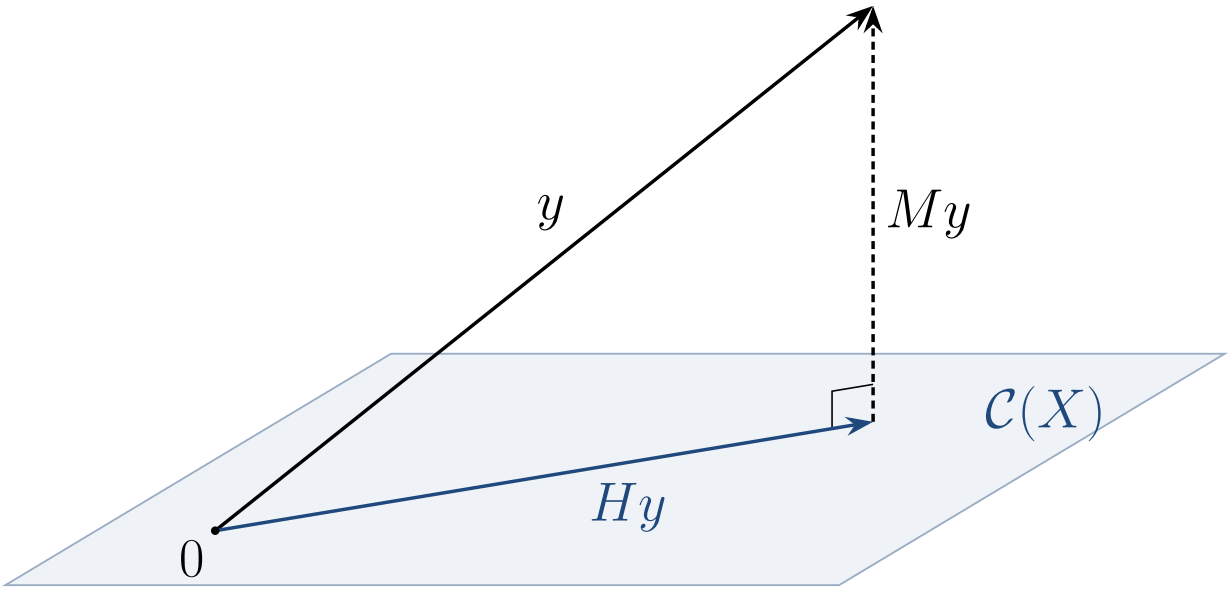
Linear Algebra

Lecture Notes



Chapter 1. Matrices

Matrices and products

Definition 1.1. Let m, n be positive integers. An $m \times n$ **matrix** $A = (a_{ij})$ is a rectangular array of real numbers, its **entries**; a_{ij} stands in row i and column j ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$). The set of all $m \times n$ matrices is $\mathbb{R}^{m\times n}$.

- $A = B$ means that A and B have the same size and $a_{ij} = b_{ij}$ for all i, j.
- $\mathbb{R}^n = \mathbb{R}^{n\times 1}$ is the set of **column vectors** $x = (x_i)$, and $\mathbb{R}^{1\times n}$ the set of **row vectors**. A 1×1 matrix is identified with its entry.
- $\mathrm{col}_j(A) \in \mathbb{R}^m$ is the jth column and $\mathrm{row}_i(A) = (a_{i1}, \ldots, a_{in}) \in \mathbb{R}^{1\times n}$ the ith row of A. We write $a_j = \mathrm{col}_j(A)$, so $A = [a_1 \ \cdots \ a_n]$.
- Sums and scalar multiples are taken entrywise: $(A + B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$ ($c \in \mathbb{R}$); $-A = (-1)A$ and $A - B = A + (-B)$. They obey the laws of numbers, such as $A + B = B + A$ and $c(A + B) = cA + cB$, because these hold entry by entry.
- The **zero matrix** 0 has all entries 0. The **identity matrix** is $I_n = (\delta_{ij}) \in \mathbb{R}^{n\times n}$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$; its columns $e_1, \ldots, e_n$ are the **standard unit vectors** of $\mathbb{R}^n$.

In statistics, measurements of k variables on n units are stored in an $n \times k$ **data matrix** (x_{ij}): row i holds the measurements on unit i, column j holds the n values of variable j, and x_{ij} is the value of variable j on unit i.

Definition 1.2. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. The **product** $AB \in \mathbb{R}^{m\times p}$ is defined by

$$(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \qquad (1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant p)$$

It is defined only when A has as many columns as B has rows.

Theorem 1.3. Whenever the sizes allow the operations,

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad A(cB) = c(AB) = (cA)B,$$

$$I_mA = A = AI_n, \qquad 0A = 0, \qquad A0 = 0 \qquad (A \in \mathbb{R}^{m\times n})$$

Proof. Let $A \in \mathbb{R}^{m\times n}$. In each rule both sides have the same size, so by Definition 1.1 it suffices to show that they have the same (i, j) entry. Every entry below is computed from Definition 1.2 and from the entrywise definition of sums and scalar multiples.

- $A(BC) = (AB)C$ for $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times q}$. Since $(BC)_{rj} = \sum_{s=1}^{p} b_{rs}c_{sj}$,

$$[A(BC)]_{ij} = \sum_{r=1}^{n} a_{ir}(BC)_{rj} = \sum_{r=1}^{n} a_{ir}\Big(\sum_{s=1}^{p} b_{rs}c_{sj}\Big) = \sum_{r=1}^{n}\sum_{s=1}^{p} a_{ir}b_{rs}c_{sj}$$

  Since $(AB)_{is} = \sum_{r=1}^{n} a_{ir}b_{rs}$,

$$[(AB)C]_{ij} = \sum_{s=1}^{p}(AB)_{is}c_{sj} = \sum_{s=1}^{p}\Big(\sum_{r=1}^{n} a_{ir}b_{rs}\Big)c_{sj} = \sum_{s=1}^{p}\sum_{r=1}^{n} a_{ir}b_{rs}c_{sj}$$

  Both double sums add the same np numbers $a_{ir}b_{rs}c_{sj}$ ($1 \leqslant r \leqslant n$, $1 \leqslant s \leqslant p$), only in a different order, so they are equal.

- $A(B+C) = AB + AC$ for $B, C \in \mathbb{R}^{n\times p}$:

$$[A(B+C)]_{ij} = \sum_{r=1}^{n} a_{ir}(b_{rj} + c_{rj}) = \sum_{r=1}^{n} a_{ir}b_{rj} + \sum_{r=1}^{n} a_{ir}c_{rj} = (AB)_{ij} + (AC)_{ij} = (AB + AC)_{ij}$$

- $(A+B)C = AC + BC$ for $B \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times p}$:

$$[(A+B)C]_{ij} = \sum_{r=1}^{n} (a_{ir} + b_{ir})c_{rj} = \sum_{r=1}^{n} a_{ir}c_{rj} + \sum_{r=1}^{n} b_{ir}c_{rj} = (AC + BC)_{ij}$$

- $A(cB) = c(AB) = (cA)B$ for $B \in \mathbb{R}^{n\times p}$, $c \in \mathbb{R}$:

$$[A(cB)]_{ij} = \sum_{r=1}^{n} a_{ir}(cb_{rj}) = c\sum_{r=1}^{n} a_{ir}b_{rj} = [c(AB)]_{ij},$$
$$[(cA)B]_{ij} = \sum_{r=1}^{n} (ca_{ir})b_{rj} = c\sum_{r=1}^{n} a_{ir}b_{rj} = [c(AB)]_{ij}$$

- $I_m A = A = AI_n$. In $(I_m A)_{ij} = \sum_{r=1}^{m} \delta_{ir}a_{rj}$ every term with $r \neq i$ is 0, because $\delta_{ir} = 0$, and the term with $r = i$ is $1 \cdot a_{ij}$; so $(I_m A)_{ij} = a_{ij}$. In the same way $(AI_n)_{ij} = \sum_{r=1}^{n} a_{ir}\delta_{rj} = a_{ij}$.
- $0A = 0$ and $A0 = 0$: every term of $(0A)_{ij} = \sum_r 0 \cdot a_{rj}$ and of $(A0)_{ij} = \sum_r a_{ir} \cdot 0$ is 0. □

By the first rule, a product of several matrices needs no parentheses. For $A \in \mathbb{R}^{n\times n}$ we put $A^0 = I_n$ and $A^{k+1} = A^k A$ $(k \geqslant 0)$.

Theorem 1.4. Let $A \in \mathbb{R}^{m\times n}$ have columns $a_1, \ldots, a_n$ and $B \in \mathbb{R}^{n\times p}$ columns $b_1, \ldots, b_p$.

(a) $AB = [Ab_1 \ \cdots \ Ab_p]$, that is, $\mathrm{col}_j(AB) = Ab_j$.

(b) For $x \in \mathbb{R}^n$, $Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n$. In particular $Ae_j = a_j$.

(c) $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B = a_{i1}\,\mathrm{row}_1(B) + \cdots + a_{in}\,\mathrm{row}_n(B)$.

(d) $(AB)_{ij} = \mathrm{row}_i(A)\, b_j$, and $AB = a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$, a sum of n products of a column and a row (the **outer product expansion**).

Proof. Write $B = (b_{rj})$. The column a_r of A has entries $a_{1r}, \ldots, a_{mr}$, the column b_j of B has entries $b_{1j}, \ldots, b_{nj}$, and $\mathrm{row}_r(B) = (b_{r1}, \ldots, b_{rp})$. By Definition 1.2,

$$(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Each part compares entries with this sum (Definition 1.1).

- (a) Fix j. The product Ab_j of $A \in \mathbb{R}^{m\times n}$ and $b_j \in \mathbb{R}^{n\times 1}$ lies in $\mathbb{R}^m$, and by Definition 1.2 its ith entry is

$$(Ab_j)_i = \sum_{r=1}^{n} a_{ir}(b_j)_r = \sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij} \qquad (1 \leqslant i \leqslant m)$$

  So the vector Ab_j and $\mathrm{col}_j(AB)$, whose entries are $(AB)_{1j}, \ldots, (AB)_{mj}$, have the same entries:

$$\mathrm{col}_j(AB) = \begin{pmatrix} a_{11}b_{1j} + \cdots + a_{1n}b_{nj} \\ \vdots \\ a_{m1}b_{1j} + \cdots + a_{mn}b_{nj} \end{pmatrix} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{1j} \\ \vdots \\ b_{nj} \end{pmatrix} = Ab_j$$

- (b) By Definition 1.2 (with $p = 1$), the ith entry of Ax is $a_{i1}x_1 + \cdots + a_{in}x_n$. By the entrywise definition of scalar multiples and sums (Definition 1.1), the ith entry of $x_1a_1 + \cdots + x_na_n$ is $x_1a_{i1} + \cdots + x_na_{in}$, the same number. Hence

$$Ax = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = x_1a_1 + \cdots + x_na_n$$

  For $x = e_j$ we have $x_j = 1$ and $x_r = 0$ for $r \neq j$, so $Ae_j = 1 \cdot a_j = a_j$.

- (c) Fix i. The product of $\mathrm{row}_i(A) = (a_{i1}, \ldots, a_{in}) \in \mathbb{R}^{1\times n}$ and $B \in \mathbb{R}^{n\times p}$ lies in $\mathbb{R}^{1\times p}$, and by Definition 1.2 its jth entry is $\sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij}$, the jth entry of $\mathrm{row}_i(AB)$. So $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B$. The jth entry of the row vector $a_{i1}\,\mathrm{row}_1(B) + \cdots + a_{in}\,\mathrm{row}_n(B)$ is $a_{i1}b_{1j} + \cdots + a_{in}b_{nj}$, the same sum. Hence

$$\mathrm{row}_i(AB) = \Big(\sum_{r=1}^{n} a_{ir}b_{r1},\ \ldots,\ \sum_{r=1}^{n} a_{ir}b_{rp} \Big) = \sum_{r=1}^{n} a_{ir}\,(b_{r1}, \ldots, b_{rp}) = \sum_{r=1}^{n} a_{ir}\,\mathrm{row}_r(B)$$

- (d) The product of $\mathrm{row}_i(A) \in \mathbb{R}^{1\times n}$ and $b_j \in \mathbb{R}^{n\times 1}$ is the 1×1 matrix with entry $\sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij}$. Next fix r. The product of $a_r \in \mathbb{R}^{m\times 1}$ and $\mathrm{row}_r(B) \in \mathbb{R}^{1\times p}$ is the $m \times p$ matrix whose (i, j) entry is the one-term sum $a_{ir}b_{rj}$ (Definition 1.2 with inner size 1):

$$a_r\,\mathrm{row}_r(B) = \begin{pmatrix} a_{1r} \\ \vdots \\ a_{mr} \end{pmatrix} (b_{r1},\ \ldots,\ b_{rp}) = \begin{pmatrix} a_{1r}b_{r1} & \cdots & a_{1r}b_{rp} \\ \vdots & & \vdots \\ a_{mr}b_{r1} & \cdots & a_{mr}b_{rp} \end{pmatrix}$$

  Adding these n matrices entry by entry, the (i, j) entry of $a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$ is $a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = (AB)_{ij}$. □

Example 1.5. Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 1 \\ 2 & 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 4}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 2 & -1 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

By Definition 1.2, each entry of $AB \in \mathbb{R}^{3\times 2}$ is a sum of four products:

$$\begin{aligned} (AB)_{11} &= 1\cdot 1 + 2\cdot 0 + 0\cdot 1 + 1\cdot 2 = 3, & (AB)_{12} &= 1\cdot 0 + 2\cdot 1 + 0\cdot 1 + 1\cdot(-1) = 1, \\ (AB)_{21} &= 0\cdot 1 + 1\cdot 0 + 3\cdot 1 + 1\cdot 2 = 5, & (AB)_{22} &= 0\cdot 0 + 1\cdot 1 + 3\cdot 1 + 1\cdot(-1) = 3, \\ (AB)_{31} &= 2\cdot 1 + 0\cdot 0 + 1\cdot 1 + 1\cdot 2 = 5, & (AB)_{32} &= 2\cdot 0 + 0\cdot 1 + 1\cdot 1 + 1\cdot(-1) = 0 \end{aligned}$$

so

$$AB = \begin{pmatrix} 3 & 1 \\ 5 & 3 \\ 5 & 0 \end{pmatrix}$$

The three views of Theorem 1.4 produce the same matrix.

- Columns, by (a) and (b): column j of AB is the combination of the four columns of A with coefficients from column j of B,

$$Ab_1 = 1\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 0\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 5 \end{pmatrix},$$

$$Ab_2 = 0\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} - 1\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

- Rows, by (c): row i of AB is the combination of the four rows of B with coefficients from row i of A,

$$\begin{aligned}\mathrm{row}_1(AB) &= 1\,(1,0) + 2\,(0,1) + 0\,(1,1) + 1\,(2,-1) = (3,\ 1),\\ \mathrm{row}_2(AB) &= 0\,(1,0) + 1\,(0,1) + 3\,(1,1) + 1\,(2,-1) = (5,\ 3),\\ \mathrm{row}_3(AB) &= 2\,(1,0) + 0\,(0,1) + 1\,(1,1) + 1\,(2,-1) = (5,\ 0)\end{aligned}$$

- Outer products, by (d): AB is the sum of four 3×2 matrices $a_r\,\mathrm{row}_r(B)$,

$$\begin{aligned}AB &= \begin{pmatrix}1\\0\\2\end{pmatrix}(1,\ 0) + \begin{pmatrix}2\\1\\0\end{pmatrix}(0,\ 1) + \begin{pmatrix}0\\3\\1\end{pmatrix}(1,\ 1) + \begin{pmatrix}1\\1\\1\end{pmatrix}(2,\ -1)\\ &= \begin{pmatrix}1&0\\0&0\\2&0\end{pmatrix} + \begin{pmatrix}0&2\\0&1\\0&0\end{pmatrix} + \begin{pmatrix}0&0\\3&3\\1&1\end{pmatrix} + \begin{pmatrix}2&-1\\2&-1\\2&-1\end{pmatrix} = \begin{pmatrix}3&1\\5&3\\5&0\end{pmatrix}\end{aligned}$$

Example 1.6. (a) $\begin{pmatrix}1&0&0\\-3&1&0\\2&0&1\end{pmatrix}\begin{pmatrix}1&-1&2&1\\3&4&8&5\\-2&3&-1&0\end{pmatrix} = \begin{pmatrix}1&-1&2&1\\0&7&2&2\\0&1&3&2\end{pmatrix}$. By Theorem 1.4(c), its second row is $-3\,(1,-1,2,1) + 1\,(3,4,8,5) = (0,7,2,2)$, and its third row is $2\,(1,-1,2,1) + 1\,(-2,3,-1,0) = (0,1,3,2)$.

(b) $\begin{pmatrix}2&1&0\\5&4&1\\1&3&2\\0&1&3\end{pmatrix}\begin{pmatrix}2\\-1\\1\end{pmatrix} = 2\begin{pmatrix}2\\5\\1\\0\end{pmatrix} - 1\begin{pmatrix}1\\4\\3\\1\end{pmatrix} + 1\begin{pmatrix}0\\1\\2\\3\end{pmatrix} = \begin{pmatrix}3\\7\\1\\2\end{pmatrix}$, by Theorem 1.4(b).

(c) For $N = \begin{pmatrix}0&1&0\\0&0&0\\0&0&0\end{pmatrix}$ and $M = \begin{pmatrix}1&-5&2\\2&3&4\\9&-1&3\end{pmatrix}$,

$$NM = \begin{pmatrix}2&3&4\\0&0&0\\0&0&0\end{pmatrix}, \qquad MN = \begin{pmatrix}0&1&0\\0&2&0\\0&9&0\end{pmatrix}$$

Row 1 of NM is row 2 of M, and column 2 of MN is column 1 of M. So $NM \neq MN$, although both are 3×3.

Remark 1.7. Matrix products differ from products of numbers. In general $AB \neq BA$ (Example 1.6(c)); square matrices with $AB = BA$ are said to **commute**. A product can be 0 although no factor is 0: for

$$X = \begin{pmatrix}1&1&3\\2&2&6\\-1&-1&-3\end{pmatrix}, \qquad X^2 = 0$$

To see this, put $w = \mathrm{row}_1(X) = (1,1,3)$ and $u = \mathrm{col}_1(X)$, the column with entries $1, 2, -1$. Rows $1,2,3$ of X are $1, 2, -1$ times w, that is, $\mathrm{row}_i(X) = u_i w$; columns $1,2,3$ of X are $1,1,3$ times u, that is, $\mathrm{col}_j(X) = w_j u$. By Theorem 1.4(d) and the rule $A(cB) = c(AB) = (cA)B$ of Theorem 1.3,

$$(X^2)_{ij} = \mathrm{row}_i(X)\,\mathrm{col}_j(X) = (u_i w)(w_j u) = u_i w_j\,(wu), \qquad wu = 1\cdot 1 + 1\cdot 2 + 3\cdot(-1) = 0,$$

so every entry of X^2 is 0. Hence $AB = 0$ does not force $A = 0$ or $B = 0$ (take $A = B = X$), and $AB = AC$ does not force $B = C$ (take $A = B = X$ and $C = 0$).

Transpose and trace

Definition 1.8. The **transpose** of $A \in \mathbb{R}^{m\times n}$ is $A' \in \mathbb{R}^{n\times m}$ with $(A')_{ij} = a_{ji}$: the rows of A' are the transposed columns of A. We write column vectors as $x = (x_1, \dots, x_n)'$. A square A is **symmetric** if $A' = A$.

Theorem 1.9. For matrices of sizes for which the operations are defined and $c \in \mathbb{R}$,

$$(A')' = A, \quad (A+B)' = A' + B', \quad (cA)' = cA', \quad (AB)' = B'A', \quad (ABC)' = C'B'A'$$

For $x, y \in \mathbb{R}^n$, $x'y = y'x = \sum_{i=1}^n x_i y_i$; in particular $x'x = \sum_i x_i^2$. For every $B \in \mathbb{R}^{m\times n}$, $B'B$ and BB' are symmetric.

Proof. In each identity both sides have the same size; we compare their entries, using $(A')_{ij} = a_{ji}$ (Definition 1.8).

- For $A, B \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}$, Definitions 1.8 and 1.1 give
$$((A')')_{ij} = (A')_{ji} = a_{ij}, \qquad ((cA)')_{ij} = (cA)_{ji} = ca_{ji} = (cA')_{ij},$$
$$((A+B)')_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A' + B')_{ij}$$
- Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. Then $(AB)'$ and $B'A'$ are both $p \times m$. For $1 \leqslant k \leqslant p$ and $1 \leqslant i \leqslant m$, Definition 1.2 gives
$$(B'A')_{ki} = \sum_{j=1}^n (B')_{kj}(A')_{ji} = \sum_{j=1}^n b_{jk}a_{ij} = \sum_{j=1}^n a_{ij}b_{jk} = (AB)_{ik} = ((AB)')_{ki}$$
- For a third matrix C with ABC defined, the previous bullet applied first to the pair AB, C and then to the pair A, B gives
$$(ABC)' = ((AB)C)' = C'(AB)' = C'B'A'$$
- For $x, y \in \mathbb{R}^n = \mathbb{R}^{n\times 1}$, $x' \in \mathbb{R}^{1\times n}$ has entries $(x')_{1i} = x_i$, so $x'y$ is 1×1 with entry $\sum_{i=1}^n x_i y_i$. In the same way $y'x = \sum_{i=1}^n y_i x_i$, the same number since $y_i x_i = x_i y_i$. With $y = x$, $x'x = \sum_i x_i^2$.
- For $B \in \mathbb{R}^{m\times n}$, the product rule and $(B')' = B$ give $(B'B)' = B'(B')' = B'B$ and $(BB')' = (B')'B' = BB'$. □

Definition 1.10. The **trace** of $A \in \mathbb{R}^{n\times n}$ is $\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$.

Theorem 1.11. (a) $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$, $\operatorname{tr}(cA) = c\operatorname{tr}(A)$ and $\operatorname{tr}(A') = \operatorname{tr}(A)$ for $A, B \in \mathbb{R}^{n\times n}$.

(b) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$.

(c) $\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times m}$.

(d) $\operatorname{tr}(A'A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$ for $A \in \mathbb{R}^{m\times n}$; hence $\operatorname{tr}(A'A) = 0$ only if $A = 0$.

Proof. • (a) The ith diagonal entries of $A + B$, cA and A' are $a_{ii} + b_{ii}$, ca_{ii} and $(A')_{ii} = a_{ii}$ (Definitions 1.1 and 1.8). Hence, by Definition 1.10,

$$\operatorname{tr}(A+B) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii}, \qquad \operatorname{tr}(cA) = \sum_{i=1}^n ca_{ii} = c\sum_{i=1}^n a_{ii}, \qquad \operatorname{tr}(A') = \sum_{i=1}^n a_{ii}$$

- (b) Both $AB \in \mathbb{R}^{m\times m}$ and $BA \in \mathbb{R}^{n\times n}$ are square. By Definitions 1.10 and 1.2,
$$\operatorname{tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ji}, \qquad \operatorname{tr}(BA) = \sum_{j=1}^n (BA)_{jj} = \sum_{j=1}^n \sum_{i=1}^m b_{ji}a_{ij}$$
Both double sums add the same mn numbers $a_{ij}b_{ji}$ ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$), so they are equal.

- (c) Since $AB \in \mathbb{R}^{m\times p}$ and $C \in \mathbb{R}^{p\times m}$, (b) applied to the pair AB, C gives $\operatorname{tr}((AB)C) = \operatorname{tr}(C(AB))$, that is, $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$. Since $A \in \mathbb{R}^{m\times n}$ and $BC \in \mathbb{R}^{n\times m}$, (b) applied to the pair A, BC gives $\operatorname{tr}(A(BC)) = \operatorname{tr}((BC)A)$, that is, $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$. Parentheses are dropped by Theorem 1.3.
- (d) The matrix $A'A$ is $n \times n$, and its jth diagonal entry is $(A'A)_{jj} = \sum_{i=1}^{m}(A')_{ji}a_{ij} = \sum_{i=1}^{m} a_{ij}^2$. Summing over $j = 1, \dots, n$ gives $\operatorname{tr}(A'A) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2$. Every term a_{ij}^2 is $\geqslant 0$. If some $a_{kl} \neq 0$, then $\operatorname{tr}(A'A) \geqslant a_{kl}^2 > 0$. So $\operatorname{tr}(A'A) = 0$ forces every $a_{ij} = 0$, that is, $A = 0$. □

Special matrices; means and sums of squares

Diagonal and triangular matrices. $D = \operatorname{diag}(d_1, \dots, d_n)$ is the $n \times n$ matrix with diagonal entries $d_1, \dots, d_n$ and all other entries 0. Left multiplication by a diagonal matrix scales the rows, right multiplication scales the columns:

$$\operatorname{row}_i(DA) = d_i \operatorname{row}_i(A), \qquad \operatorname{col}_j(AD) = d_j \operatorname{col}_j(A)$$

Indeed, the (i, r) entry of D is d_i if $r = i$ and 0 if $r \neq i$, so in each sum of Definition 1.2 only one term survives:

$$(DA)_{ij} = \sum_{r=1}^{n} D_{ir}a_{rj} = d_i a_{ij}, \qquad (AD)_{ij} = \sum_{r=1}^{n} a_{ir}D_{rj} = a_{ij}d_j$$

In particular, for $F = \operatorname{diag}(f_1, \dots, f_n)$, $(DF)_{ij} = d_iF_{ij}$ is d_if_i for $i = j$ and 0 for $i \neq j$, that is, $\operatorname{diag}(d_1, \dots, d_n)\operatorname{diag}(f_1, \dots, f_n) = \operatorname{diag}(d_1f_1, \dots, d_nf_n)$. A square A is **upper triangular** if $a_{ij} = 0$ for $i > j$ and **lower triangular** if $a_{ij} = 0$ for $i < j$. A product of two upper (lower) triangular matrices is upper (lower) triangular, with diagonal entries $a_{ii}b_{ii}$. For upper triangular $A, B \in \mathbb{R}^{n\times n}$, the term $a_{ik}b_{kj}$ is 0 if $k < i$ (then $a_{ik} = 0$) and if $k > j$ (then $b_{kj} = 0$); so only the terms with $i \leqslant k \leqslant j$ remain:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=i}^{j} a_{ik}b_{kj}$$

For $i > j$ no k satisfies $i \leqslant k \leqslant j$, so $(AB)_{ij} = 0$; for $i = j$ only $k = i$ remains, so $(AB)_{ii} = a_{ii}b_{ii}$. For lower triangular A, B the term $a_{ik}b_{kj}$ is 0 if $k > i$ or $k < j$, so only the terms with $j \leqslant k \leqslant i$ remain, and $(AB)_{ij} = 0$ for $i < j$, $(AB)_{ii} = a_{ii}b_{ii}$.

Definition 1.12. The **summing vector** $\mathbf{1}_n \in \mathbb{R}^n$ has all entries 1. Put

$$J_n = \mathbf{1}_n\mathbf{1}_n' \in \mathbb{R}^{n\times n}, \qquad \bar{J}_n = \tfrac{1}{n}J_n, \qquad C_n = I_n - \bar{J}_n$$

Every entry of J_n is 1, and C_n is the **centering matrix**. Subscripts are dropped when the size is clear. A square matrix A with $A^2 = A$ is **idempotent**.

Proposition 1.13. $\mathbf{1}'\mathbf{1} = n$, $J_n^2 = nJ_n$, and $\bar{J}_n$ and $C = C_n$ are symmetric and idempotent:

$$\bar{J}_n' = \bar{J}_n = \bar{J}_n^2, \qquad C = C' = C^2, \qquad C\mathbf{1} = 0$$

Proof. Write $\mathbf{1} = \mathbf{1}_n$, $J = J_n$, $\bar{J} = \bar{J}_n$, $I = I_n$.

- By Theorem 1.9, $\mathbf{1}'\mathbf{1} = \sum_{i=1}^{n} 1 \cdot 1 = n$.
- By Theorem 1.3 (associativity, and $A(cB) = c(AB)$ for the number $\mathbf{1}'\mathbf{1} = n$),

$$J^2 = (\mathbf{1}\mathbf{1}')(\mathbf{1}\mathbf{1}') = \mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{1}\, n\, \mathbf{1}' = n\,\mathbf{1}\mathbf{1}' = nJ, \qquad \bar{J}^2 = \tfrac{1}{n^2}J^2 = \tfrac{1}{n^2}\,nJ = \tfrac{1}{n}J = \bar{J}$$

- By Theorem 1.9, $J' = (\mathbf{1}\mathbf{1}')' = (\mathbf{1}')'\mathbf{1}' = \mathbf{1}\mathbf{1}' = J$, so $\bar{J}' = \frac{1}{n}J' = \bar{J}$. Also $I' = I$, because $\delta_{ji} = \delta_{ij}$. Hence $C' = I' - \bar{J}' = I - \bar{J} = C$.

- By Theorem 1.3 and $\bar{J}^2 = \bar{J}$,

$$C^2 = (I - \bar{J})(I - \bar{J}) = I \cdot I - I\bar{J} - \bar{J}I + \bar{J}^2 = I - 2\bar{J} + \bar{J} = I - \bar{J} = C$$

- By Theorem 1.3 and $\mathbf{1}'\mathbf{1} = n$, $C\mathbf{1} = \mathbf{1} - \frac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1} - \frac{1}{n}\, n\, \mathbf{1} = 0$. □

Example 1.14. For data $x = (x_1, \ldots, x_n)'$, $\mathbf{1}'x = x_1 + \cdots + x_n$ (Theorem 1.9), so the mean is $\bar{x} = \frac{1}{n}\mathbf{1}'x$, and $\mathbf{1}'x = n\bar{x}$. By Theorem 1.3,

$$Cx = (I - \tfrac{1}{n}\mathbf{1}\mathbf{1}')x = x - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'x) = x - \bar{x}\mathbf{1} = (x_1 - \bar{x},\ \ldots,\ x_n - \bar{x})',$$

the vector of **deviations** from the mean. Moreover $x'\mathbf{1} = \mathbf{1}'x = n\bar{x}$ (Theorem 1.9), so

$$x'Cx = x'x - \tfrac{1}{n}(x'\mathbf{1})(\mathbf{1}'x) = \sum_{i=1}^{n} x_i^2 - \tfrac{1}{n}(n\bar{x})(n\bar{x}) = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

Expanding each square and using $\sum_i x_i = n\bar{x}$,

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i^2 - 2\bar{x}x_i + \bar{x}^2) = \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + n\bar{x}^2 = \sum_{i=1}^{n} x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

Hence the sum of squares about the mean is

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = x'x - n\bar{x}^2 = x'Cx$$

Example 1.15. Four units ($n = 4$) are measured on two variables ($k = 2$); x_{ij} is the value of variable j on unit i:

$$X_0 = (x_{ij}) = \begin{pmatrix} 2 & 3 \\ 0 & 3 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 2}, \qquad \mathbf{1}'X_0 = (4,\ 8), \qquad \bar{x} = \tfrac{1}{4}X_0'\mathbf{1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The vector $\bar{x} = (\bar{x}_1, \bar{x}_2)'$ of column means $\bar{x}_j = \frac{1}{n}\sum_i x_{ij}$ is the **mean vector**. By Theorem 1.4(a), $\mathrm{col}_j(CX_0) = C\,\mathrm{col}_j(X_0)$, which by Example 1.14 is the vector of deviations of variable j from its mean $\bar{x}_j$. In matrix form, by Theorem 1.3 and $\mathbf{1}'X_0 = (X_0'\mathbf{1})' = n\bar{x}'$ (Theorem 1.9),

$$CX_0 = X_0 - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'X_0) = X_0 - \mathbf{1}\bar{x}'$$

By Proposition 1.13, $C = C^2 = C'C$, so $X_0'CX_0 = X_0'C'CX_0 = (CX_0)'(CX_0)$ (Theorem 1.9). The centered data matrix and the matrix of **corrected sums of squares and products** are therefore

$$CX_0 = \begin{pmatrix} 2-1 & 3-2 \\ 0-1 & 3-2 \\ 2-1 & 1-2 \\ 0-1 & 1-2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix}, \qquad X_0'CX_0 = (CX_0)'(CX_0) = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

Here the (j, l) entry of $(CX_0)'(CX_0)$ is $\mathrm{col}_j(CX_0)'\,\mathrm{col}_l(CX_0)$ (Theorem 1.4(d)): $1 + 1 + 1 + 1 = 4$ for $(1, 1)$ and $(2, 2)$, and $1 - 1 - 1 + 1 = 0$ for $(1, 2)$ and $(2, 1)$. The same matrix comes from the uncorrected sums: in general, by Theorem 1.3 and $X_0'\mathbf{1} = n\bar{x}$,

$$X_0'CX_0 = X_0'(I - \tfrac{1}{n}\mathbf{1}\mathbf{1}')X_0 = X_0'X_0 - \tfrac{1}{n}(X_0'\mathbf{1})(\mathbf{1}'X_0) = X_0'X_0 - \tfrac{1}{n}(n\bar{x})(n\bar{x}') = X_0'X_0 - n\bar{x}\bar{x}',$$

and here

$$X_0'X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 8 & 16 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

Exercises

Exercise 1.1. Let $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 1 & 0 & 1 \end{pmatrix}$, and let B be the matrix obtained from A by interchanging its first two rows. Compute AB', BA', $A'B$ and $B'A$, and compare their traces.

Solution. Write $r_1 = (1,2,1,0)$, $r_2 = (0,1,2,1)$, $r_3 = (2,1,0,1)$ for the rows of A. The rows of B are r_2, r_1, r_3, so $B = \left(\begin{smallmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{smallmatrix}\right)$. Since $\mathrm{col}_j(B') = \mathrm{row}_j(B)'$, the (i,j) entry of AB' is $\mathrm{row}_i(A)\,\mathrm{row}_j(B)'$ (Theorem 1.4(d)). The six numbers $r_i r_k'$ needed are

$$\begin{aligned} r_1 r_1' &= 1+4+1+0 = 6, \quad & r_2 r_2' &= 0+1+4+1 = 6, \quad & r_3 r_3' &= 4+1+0+1 = 6, \\ r_1 r_2' &= 0+2+2+0 = 4, \quad & r_1 r_3' &= 2+2+0+0 = 4, \quad & r_2 r_3' &= 0+1+0+1 = 2, \end{aligned}$$

and $r_k r_i' = r_i r_k'$ (Theorem 1.9). Hence

$$AB' = \begin{pmatrix} r_1 r_2' & r_1 r_1' & r_1 r_3' \\ r_2 r_2' & r_2 r_1' & r_2 r_3' \\ r_3 r_2' & r_3 r_1' & r_3 r_3' \end{pmatrix} = \begin{pmatrix} 4 & 6 & 4 \\ 6 & 4 & 2 \\ 2 & 4 & 6 \end{pmatrix}, \qquad BA' = (AB')' = \begin{pmatrix} 4 & 6 & 2 \\ 6 & 4 & 4 \\ 4 & 2 & 6 \end{pmatrix}$$

by Theorem 1.9. Since $\mathrm{col}_k(A') = \mathrm{row}_k(A)' = r_k'$, the outer product expansion (Theorem 1.4(d)) gives

$$A'B = \mathrm{col}_1(A')\,\mathrm{row}_1(B) + \mathrm{col}_2(A')\,\mathrm{row}_2(B) + \mathrm{col}_3(A')\,\mathrm{row}_3(B) = r_1' r_2 + r_2' r_1 + r_3' r_3$$

This sum is symmetric, since $(r_1' r_2)' = r_2' r_1$ and $(r_3' r_3)' = r_3' r_3$ (Theorem 1.9). Its (i,j) entry is $\mathrm{col}_i(A')\,\mathrm{col}_j(B)$; for instance $(A'B)_{23} = 2\cdot 2 + 1\cdot 1 + 1\cdot 0 = 5$. Computing all sixteen entries in this way,

$$A'B = \begin{pmatrix} 4 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 \\ 2 & 5 & 4 & 1 \\ 3 & 3 & 1 & 1 \end{pmatrix} = (A'B)' = B'A$$

So $A'B$ is symmetric while AB' is not. The reason: $B = SA$ with the symmetric matrix $S = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, since $\mathrm{row}_1(SA) = \mathrm{row}_2(A)$, $\mathrm{row}_2(SA) = \mathrm{row}_1(A)$ and $\mathrm{row}_3(SA) = \mathrm{row}_3(A)$ (Theorem 1.4(c)). Then $A'B = A'SA$ and $(A'SA)' = A'S'A = A'SA$, whereas $AB' = AA'S$ need not be symmetric.

The four products have the same trace 14 ($4+4+6$ and $4+5+4+1$); this is no coincidence: $\mathrm{tr}(AB') = \mathrm{tr}(B'A)$ and $\mathrm{tr}(BA') = \mathrm{tr}(A'B)$ by Theorem 1.11(b), and $\mathrm{tr}(BA') = \mathrm{tr}\big((AB')'\big) = \mathrm{tr}(AB')$ by Theorems 1.9 and 1.11(a).

Exercise 1.2. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(a) Compute AB and BA.

(b) Write AB as the sum of the four products $\mathrm{col}_j(A)\,\mathrm{row}_j(B)$ of Theorem 1.4(d).

(c) Compute $B'A'$, $\mathrm{tr}(AB)$, $\mathrm{tr}(BA)$ and $\mathrm{tr}(A'A)$, and show that $(BA)^2 = BA$.

Solution. (a) By Theorem 1.4(c), $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B$ is a combination of the rows $(1,-1,1)$, $(0,1,-1)$, $(0,0,1)$, $(0,0,0)$ of B:

$$\begin{aligned} \mathrm{row}_1(AB) &= (1,-1,1) + (0,1,-1) = (1,\ 0,\ 0), \\ \mathrm{row}_2(AB) &= (0,1,-1) + (0,0,1) = (0,\ 1,\ 0), \\ \mathrm{row}_3(AB) &= (0,0,1) + (0,0,0) = (0,\ 0,\ 1) \end{aligned}$$

Likewise each row of $BA \in \mathbb{R}^{4\times 4}$ is a combination of the rows of A:

$$\begin{aligned}
\mathrm{row}_1(BA) &= (1,1,0,0) - (0,1,1,0) + (0,0,1,1) = (1,\ 0,\ 0,\ 1),\\
\mathrm{row}_2(BA) &= (0,1,1,0) - (0,0,1,1) = (0,\ 1,\ 0,\ -1),\\
\mathrm{row}_3(BA) &= (0,0,1,1), \qquad \mathrm{row}_4(BA) = (0,0,0,0)
\end{aligned}$$

Hence

$$AB = I_3, \qquad BA = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq I_4$$

(b) $\mathrm{col}_4(A)\,\mathrm{row}_4(B) = 0$, and

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}(1,\ -1,\ 1) + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}(0,\ 1,\ -1) + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}(0,\ 0,\ 1) = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$

(c) $B'A' = (AB)' = I_3' = I_3$ (Theorem 1.9). Next $\mathrm{tr}(AB) = 3$ and $\mathrm{tr}(BA) = 1+1+1+0 = 3$, equal by Theorem 1.11(b); and by Theorem 1.11(d), $\mathrm{tr}(A'A)$ is the sum of the squares of the entries of A, which is 6. Finally, by Theorem 1.3, $(BA)^2 = B(AB)A = BI_3A = BA$.

Exercise 1.3. For $i, j \in \{1,2,3\}$ let $E_{ij} = e_i e_j' \in \mathbb{R}^{3\times 3}$.

(a) Show that E_{ij} has entry 1 in position (i,j) and 0 elsewhere, and that $E_{ij}E_{kl} = \delta_{jk}E_{il}$.

(b) Compute $\mathrm{tr}(E_{12}E_{23}E_{31})$ and $\mathrm{tr}(E_{12}E_{31}E_{23})$.

Solution. (a) The (r,s) entry of e_ie_j' is $(e_i)_r(e_j)_s$ (Definition 1.2), which is 1 if $r=i$ and $s=j$, and 0 otherwise. By Theorem 1.9, $e_j'e_k = \sum_{r=1}^3 (e_j)_r(e_k)_r$; the only possibly nonzero term is the one with $r=j$, which is $(e_k)_j = \delta_{jk}$. By Theorem 1.3 (associativity, then $A(cB) = c(AB)$ for the number $c = e_j'e_k$),

$$E_{ij}E_{kl} = (e_ie_j')(e_ke_l') = e_i(e_j'e_k)e_l' = \delta_{jk}\, e_ie_l' = \delta_{jk}E_{il}$$

(b) By (a), $E_{12}E_{23}E_{31} = E_{13}E_{31} = E_{11}$, with trace 1, whereas $E_{12}E_{31} = \delta_{23}E_{11} = 0$, so $E_{12}E_{31}E_{23} = 0$, with trace 0. Thus $\mathrm{tr}(ABC)$ is unchanged when the factors are moved cyclically (Theorem 1.11(c)), but it can change when two factors are interchanged: here $A = E_{12}$, $B = E_{23}$, $C = E_{31}$ and $\mathrm{tr}(ABC) = 1 \neq 0 = \mathrm{tr}(ACB)$.

Exercise 1.4. Let $u = (1,2,3,4)'$, $v = (1,1,1,-1)'$, $w = (1,1,-1,0)'$, and put $A = uv'$ and $B = uw'$.

(a) Write out A and B, and compute $v'u$ and $w'u$.

(b) Show that $A^2 = 2A$, $A^3 = 4A$, $B^2 = 0$, $AB = 2B$ and $BA = 0$.

(c) Compute $\mathrm{tr}(A)$ and $\mathrm{tr}(B)$.

Solution. (a) Row i of uv' is u_iv' (Theorem 1.4(c)), so

$$A = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 2 & 2 & 2 & -2 \\ 3 & 3 & 3 & -3 \\ 4 & 4 & 4 & -4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 2 & 2 & -2 & 0 \\ 3 & 3 & -3 & 0 \\ 4 & 4 & -4 & 0 \end{pmatrix},$$

and $v'u = 1+2+3-4 = 2$, $w'u = 1+2-3+0 = 0$.

(b) By Theorem 1.3, a product of two such matrices collapses to a number times a column times a row: for instance $A^2 = (uv')(uv') = u(v'u)v' = (v'u)\,uv'$, because $v'u$ is a number. In this way

$$A^2 = u(v'u)v' = 2A, \qquad A^3 = A^2A = 2A^2 = 4A, \qquad B^2 = u(w'u)w' = 0,$$
$$AB = u(v'u)w' = 2B, \qquad BA = u(w'u)v' = 0$$

So A and B do not commute, and $B \neq 0$ has $B^2 = 0$.

(c) $\operatorname{tr}(A) = 1 + 2 + 3 - 4 = 2$ and $\operatorname{tr}(B) = 1 + 2 - 3 + 0 = 0$. In general $\operatorname{tr}(uv') = \operatorname{tr}(v'u) = v'u$ by Theorem 1.11(b).

Exercise 1.5. Let

$$N = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Compute N^2, N^3 and N^4.

(b) Show that the matrices $B \in \mathbb{R}^{4\times 4}$ with $NB = BN$ are exactly

$$B = \alpha I_4 + \beta N + \gamma N^2 + \delta N^3 = \begin{pmatrix} \alpha & \beta & \gamma & \delta \\ 0 & \alpha & \beta & \gamma \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \alpha \end{pmatrix} \qquad (\alpha, \beta, \gamma, \delta \in \mathbb{R})$$

Solution. (a) By Theorem 1.4(a),(b), $Ne_1 = 0$ and $Ne_j = e_{j-1}$ for $j = 2, 3, 4$. Hence $N^2e_j = e_{j-2}$ for $j = 3, 4$ and $N^2e_j = 0$ for $j = 1, 2$; $N^3e_4 = e_1$ and $N^3e_j = 0$ for $j \leqslant 3$; and $N^4e_j = 0$ for all j. Since $\operatorname{col}_j(N^k) = N^ke_j$,

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

This gives the matrix form of $\alpha I_4 + \beta N + \gamma N^2 + \delta N^3$ in (b).

(b) Let $B = (b_{ij})$. Since $\operatorname{row}_i(N) = e'_{i+1}$ for $i \leqslant 3$ and $\operatorname{row}_4(N) = 0$, Theorem 1.4(c) gives $\operatorname{row}_i(NB) = \operatorname{row}_{i+1}(B)$ for $i \leqslant 3$ and $\operatorname{row}_4(NB) = 0$. By (a), $\operatorname{col}_j(BN) = BNe_j = Be_{j-1} = \operatorname{col}_{j-1}(B)$ for $j \geqslant 2$ and $\operatorname{col}_1(BN) = 0$. So

$$(NB)_{ij} = \begin{cases} b_{i+1,j} & i \leqslant 3 \\ 0 & i = 4 \end{cases}, \qquad (BN)_{ij} = \begin{cases} b_{i,j-1} & j \geqslant 2 \\ 0 & j = 1 \end{cases}$$

Suppose $NB = BN$. Comparing the entries:

- row 4, $j \geqslant 2$: $b_{4,j-1} = 0$, so $b_{41} = b_{42} = b_{43} = 0$;
- column 1, $i \leqslant 3$: $b_{i+1,1} = 0$, so $b_{21} = b_{31} = b_{41} = 0$;
- $i \leqslant 3$, $j \geqslant 2$: $b_{i+1,j} = b_{i,j-1}$, so every entry equals the entry diagonally above and to its left.

The last condition gives $b_{32} = b_{21} = 0$; so all entries below the diagonal vanish. It also gives $b_{11} = b_{22} = b_{33} = b_{44}$, $b_{12} = b_{23} = b_{34}$ and $b_{13} = b_{24}$. With $\alpha = b_{11}$, $\beta = b_{12}$, $\gamma = b_{13}$, $\delta = b_{14}$, B has the stated form. Conversely, by Theorem 1.3,

$$N(\alpha I + \beta N + \gamma N^2 + \delta N^3) = \alpha N + \beta N^2 + \gamma N^3 + \delta N^4 = (\alpha I + \beta N + \gamma N^2 + \delta N^3)N$$

for all $\alpha, \beta, \gamma, \delta$.

Exercise 1.6. Let

$$S = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

so that $Se_1 = e_2$, $Se_2 = e_3$, $Se_3 = e_4$, $Se_4 = e_1$.

(a) Compute S^2, S^3 and S^4, and show that $S' = S^3$ and $S'S = I_4$.

(b) Let $C = I_4 + 2S + 3S^2 + 4S^3$. Write out C and C', show that $CC' = C'C$, and compute CC'.

Solution. (a) Applying S repeatedly moves each unit vector one step along the cycle $e_1 \to e_2 \to e_3 \to e_4 \to e_1$. So $S^2 e_j = e_{j+2}$ and $S^3 e_j = e_{j+3}$, indices taken cyclically ($e_5 = e_1$, $e_6 = e_2$, ...), and $S^4 e_j = e_j$. With $\operatorname{col}_j(S^k) = S^k e_j$ (Theorem 1.4(b)),

$$S^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad S^3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad S^4 = I_4$$

The entries 1 of S stand in positions $(2,1)$, $(3,2)$, $(4,3)$, $(1,4)$, so by $(S')_{ij} = s_{ji}$ the entries 1 of S' stand in positions $(1,2)$, $(2,3)$, $(3,4)$, $(4,1)$, all other entries being 0. These are exactly the positions of the entries 1 of S^3, so $S' = S^3$, and then $S'S = S^3 S = S^4 = I_4$.

(b) Adding the entries of I, $2S$, $3S^2$, $4S^3$,

$$C = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}, \qquad C' = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

By Theorem 1.9 and (a), $(S^2)' = (SS)' = S'S' = (S^3)^2 = S^6 = S^4 S^2 = S^2$ and $(S^3)' = (S')^3 = S^9 = (S^4)^2 S = S$, so $C' = I + 4S + 3S^2 + 2S^3$. Both C and C' are combinations of powers of S, and $S^a S^b = S^{a+b} = S^b S^a$; so $CC' = C'C$ by Theorem 1.3. Expanding and using $S^4 = I$, the coefficient of S^k in $CC' = (I + 2S + 3S^2 + 4S^3)(I + 4S + 3S^2 + 2S^3)$ collects the pairs of exponents with sum k or $k + 4$:

$$I:\ 1\cdot 1 + 2\cdot 2 + 3\cdot 3 + 4\cdot 4 = 30, \qquad S:\ 1\cdot 4 + 2\cdot 1 + 3\cdot 2 + 4\cdot 3 = 24,$$
$$S^2:\ 1\cdot 3 + 2\cdot 4 + 3\cdot 1 + 4\cdot 2 = 22, \qquad S^3:\ 1\cdot 2 + 2\cdot 3 + 3\cdot 4 + 4\cdot 1 = 24$$

Hence

$$CC' = 30I + 24S + 22S^2 + 24S^3 = \begin{pmatrix} 30 & 24 & 22 & 24 \\ 24 & 30 & 24 & 22 \\ 22 & 24 & 30 & 24 \\ 24 & 22 & 24 & 30 \end{pmatrix}$$

Its $(1,1)$ entry is $\operatorname{row}_1(C)\operatorname{row}_1(C)' = 1 + 16 + 9 + 4 = 30$.

Exercise 1.7. Let

$$N = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Compute N^2, N^3 and N^4.

(b) Let $G = I_4 - N + N^2 - N^3$. Show that $(I_4 + N)G = I_4 = G(I_4 + N)$, and write out G.

Solution. (a) By Theorem 1.4(c), $\operatorname{row}_i(N^{k+1}) = \operatorname{row}_i(N^k)N$ is the combination of the rows of N with coefficients $\operatorname{row}_i(N^k)$. The rows of N are $(0,1,1,1)$, $(0,0,2,1)$, $(0,0,0,3)$, 0. So

$$\operatorname{row}_1(N^2) = 1\,(0,0,2,1) + 1\,(0,0,0,3) = (0,0,2,4), \qquad \operatorname{row}_2(N^2) = 2\,(0,0,0,3) = (0,0,0,6),$$
$$\operatorname{row}_1(N^3) = 2\,(0,0,0,3) = (0,0,0,6), \qquad \operatorname{row}_2(N^3) = 6\,\operatorname{row}_4(N) = 0,$$

while $\operatorname{row}_3(N^2) = 3\,\operatorname{row}_4(N) = 0$, $\operatorname{row}_4(N^2) = \operatorname{row}_4(N)N = 0$, and rows $3, 4$ of N^3 are $\operatorname{row}_3(N^2)N = 0$ and $\operatorname{row}_4(N^2)N = 0$. Hence

$$N^2 = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 6\,e_1 e_4', \qquad N^4 = 6\,e_1 e_4' N = 6\,e_1 \operatorname{row}_4(N) = 0$$

(b) By Theorem 1.3 the terms cancel in pairs:

$$(I+N)(I-N+N^2-N^3) = I - N + N^2 - N^3 + N - N^2 + N^3 - N^4 = I - N^4 = I,$$

and the product in the other order expands to the same sum. Adding the entries,

$$G = \begin{pmatrix} 1 & -1 & 1 & -3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

For instance $g_{14} = -1 + 4 - 6 = -3$ and $g_{24} = -1 + 6 = 5$. As a check, $\operatorname{row}_1(I+N)\operatorname{col}_4(G) = 1\cdot(-3) + 1\cdot 5 + 1\cdot(-3) + 1\cdot 1 = 0$.

Exercise 1.8. Let $L \in \mathbb{R}^{4\times 4}$ be the lower triangular matrix with all entries on and below the diagonal equal to 1, and $D = I_4 - Z$, where Z has entries $z_{21} = z_{32} = z_{43} = 1$ and all others 0:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

(a) For $x \in \mathbb{R}^4$, compute Lx and Dx. Deduce that $DL = I_4 = LD$.

(b) Show that $LL' = (\min(i,j))$ and $L'L = (5 - \max(i,j))$, and write out both.

(c) Compute $D'D$ and show that $(LL')(D'D) = I_4$.

Solution. (a) $\operatorname{row}_i(L)x = x_1 + \cdots + x_i$, so $Lx = (x_1,\ x_1+x_2,\ x_1+x_2+x_3,\ x_1+x_2+x_3+x_4)'$ is the vector of partial sums, and $Dx = x - Zx = (x_1,\ x_2 - x_1,\ x_3 - x_2,\ x_4 - x_3)'$ is the vector of differences. The differences of the partial sums of x are the entries of x, and the partial sums of the differences of x are again the entries of x: $D(Lx) = x = L(Dx)$ for every x. With $x = e_j$, $\operatorname{col}_j(DL) = DLe_j = e_j$ and $\operatorname{col}_j(LD) = e_j$ (Theorem 1.4(a),(b)); so $DL = I_4 = LD$.

(b) $(LL')_{ij} = \operatorname{row}_i(L)\operatorname{row}_j(L)'$ counts the indices k with $k \leqslant i$ and $k \leqslant j$, which is $\min(i,j)$ (Theorems 1.4(d) and 1.9). $(L'L)_{ij} = \operatorname{col}_i(L)'\operatorname{col}_j(L)$ counts the indices $k \leqslant 4$ with $k \geqslant i$ and $k \geqslant j$, which is $5 - \max(i,j)$. So

$$LL' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \qquad L'L = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 3 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix},$$

both with trace 10.

(c) The columns of D are $d_1 = e_1 - e_2$, $d_2 = e_2 - e_3$, $d_3 = e_3 - e_4$, $d_4 = e_4$, and $(D'D)_{ij} = d_i' d_j$ (Theorems 1.4(d) and 1.9). Here $d_1'd_1 = d_2'd_2 = d_3'd_3 = 1 + 1 = 2$, $d_4'd_4 = 1$, $d_1'd_2 = d_2'd_3 = d_3'd_4 = -1$, and $d_i'd_j = 0$ for $|i - j| \geqslant 2$, since d_i and d_j then have no nonzero entry in the same position:

$$D'D = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

By Theorems 1.3 and 1.9 and (a), $(LL')(D'D) = L(L'D')D = L(DL)'D = LI_4D = LD = I_4$.

Exercise 1.9. Let

$$H = [h_1\ h_2\ h_3\ h_4] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

(a) Show that $H' = H$ and $H^2 = 4I_4$.

(b) Show that $HDH = d_1h_1h_1' + d_2h_2h_2' + d_3h_3h_3' + d_4h_4h_4'$ for every $D = \operatorname{diag}(d_1, d_2, d_3, d_4)$.

(c) Let $E = \operatorname{diag}(1, 1, 0, 0)$ and $P = \frac{1}{4}HEH$. Write out P, and show that $P^2 = P$, $\operatorname{tr}(P) = 2$ and $\frac{1}{4}H(I_4 - E)H = I_4 - P$.

Solution. (a) Row i of H equals h_i' for each i, so $H' = H$. Then $(H^2)_{ij} = \operatorname{row}_i(H)\operatorname{col}_j(H) = h_i'h_j$ (Theorem 1.4(d)). Each $h_i'h_i = 1 + 1 + 1 + 1 = 4$, and the six products with $i < j$ vanish:

$$h_1'h_2 = 1 - 1 + 1 - 1 = 0, \quad h_1'h_3 = 1 + 1 - 1 - 1 = 0, \quad h_1'h_4 = 1 - 1 - 1 + 1 = 0,$$
$$h_2'h_3 = 1 - 1 - 1 + 1 = 0, \quad h_2'h_4 = 1 + 1 - 1 - 1 = 0, \quad h_3'h_4 = 1 - 1 + 1 - 1 = 0$$

So $H^2 = 4I_4$.

(b) $\operatorname{col}_j(HD) = d_jh_j$ (right multiplication by D scales the columns) and $\operatorname{row}_j(H) = h_j'$ by (a). The outer product expansion (Theorem 1.4(d)) of $(HD)H$ is $\sum_j \operatorname{col}_j(HD)\operatorname{row}_j(H) = \sum_j d_jh_jh_j'$.

(c) By (b) with $d_1 = d_2 = 1$, $d_3 = d_4 = 0$, $P = \frac{1}{4}(h_1h_1' + h_2h_2')$. Here $h_1h_1' = J_4$. The ith entry of h_2 is $(-1)^{i+1}$, so the (i, j) entry of h_2h_2' is $(-1)^{i+1}(-1)^{j+1} = (-1)^{i+j}$: 1 where $i + j$ is even and -1 where $i + j$ is odd. Adding, $h_1h_1' + h_2h_2'$ has entry 2 where $i + j$ is even and 0 where $i + j$ is odd; so

$$P = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

By Theorem 1.3, $H^2 = 4I$ from (a), and $E^2 = E$,

$$P^2 = \tfrac{1}{16}(HEH)(HEH) = \tfrac{1}{16}HE(H^2)EH = \tfrac{1}{16}HE(4I)EH = \tfrac{1}{4}HE^2H = \tfrac{1}{4}HEH = P$$

Next $\operatorname{tr}(P) = \frac{1}{2}(1+1+1+1) = 2$. Finally, by Theorem 1.3 and (a), $\frac{1}{4}H(I - E)H = \frac{1}{4}H^2 - \frac{1}{4}HEH = \frac{1}{4}\,4I - P = I - P$.

Exercise 1.10. With $h_1, \dots, h_4$ as in Exercise 1.9, let

$$A = [h_2\ h_3\ h_4] = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 3}$$

(a) Compute $A'A$ and AA'.

(b) Show that $AA' = 4C_4$, $\mathbf{1}_4'A = 0$ and $C_4A = A$.

(c) Compute $\mathrm{tr}(A'A)$ and $\mathrm{tr}(AA')$.

Solution. (a) $(A'A)_{jl} = h_{j+1}'h_{l+1}$, so $A'A = 4I_3$ by the products listed in Exercise 1.9(a). By Theorem 1.4(d), $AA' = h_2h_2' + h_3h_3' + h_4h_4'$, and by Exercise 1.9(a),(b) with $D = I_4$, $h_1h_1' + h_2h_2' + h_3h_3' + h_4h_4' = H^2 = 4I_4$. Since $h_1h_1' = J_4$,

$$AA' = 4I_4 - J_4 = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}$$

(b) $4I_4 - J_4 = 4(I_4 - \frac{1}{4}J_4) = 4C_4$ (Definition 1.12). Next $\mathbf{1}_4'A = (h_1'h_2,\ h_1'h_3,\ h_1'h_4) = 0$, so $C_4A = A - \frac{1}{4}\mathbf{1}_4(\mathbf{1}_4'A) = A$: each column of A already has mean 0.

(c) $\mathrm{tr}(A'A) = 12$ and $\mathrm{tr}(AA') = 4 \cdot 3 = 12$, equal by Theorem 1.11(b); both are the sum of the squares of the twelve entries ± 1 of A (Theorem 1.11(d)).

Exercise 1.11. Four units are measured on two variables ($n = 4$, $k = 2$), with data matrix

$$X_0 = \begin{pmatrix} 1 & 0 \\ 3 & 2 \\ 2 & 0 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

Find

(a) the mean vector $\bar{x}$ and the centered data matrix C_4X_0;

(b) the matrix $X_0'C_4X_0$ of corrected sums of squares and products, both as $(C_4X_0)'(C_4X_0)$ and as $X_0'X_0 - 4\bar{x}\bar{x}'$.

Solution. (a) $\mathbf{1}'X_0 = (1+3+2+2,\ 0+2+0+2) = (8,\ 4)$, so $\bar{x} = \frac{1}{4}X_0'\mathbf{1} = (2,\ 1)'$. As in Example 1.15, $C_4X_0 = X_0 - \mathbf{1}\bar{x}'$:

$$C_4X_0 = \begin{pmatrix} 1-2 & 0-1 \\ 3-2 & 2-1 \\ 2-2 & 0-1 \\ 2-2 & 2-1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}$$

(b) The (j,l) entry of $(C_4X_0)'(C_4X_0)$ is $\mathrm{col}_j(C_4X_0)'\,\mathrm{col}_l(C_4X_0)$ (Theorems 1.4(d) and 1.9): $1+1+0+0 = 2$ for $(1,1)$, $1+1+0+0 = 2$ for $(1,2)$ and $(2,1)$, and $1+1+1+1 = 4$ for $(2,2)$. On the other hand

$$X_0'X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - 4\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - \begin{pmatrix} 16 & 8 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$$

(for instance $18 = 1+9+4+4$ and $10 = 0+6+0+4$). So $X_0'C_4X_0 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 4 \end{smallmatrix}\right)$: the sums of squares about the means are $\sum_i (x_{i1} - \bar{x}_1)^2 = 2$ and $\sum_i (x_{i2} - \bar{x}_2)^2 = 4$, and the corrected sum of products is $\sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) = 2$.

Exercise 1.12. Five units are measured on three variables ($n = 5$, $k = 3$), with data matrix

$$X_0 = \begin{pmatrix} 2 & 2 & 5 \\ 3 & 2 & 2 \\ 1 & 4 & 2 \\ 3 & 6 & 8 \\ 1 & 6 & 8 \end{pmatrix} \in \mathbb{R}^{5\times 3}$$

(a) Find the mean vector $\bar{x}$ and the centered data matrix C_5X_0.

(b) Find $S = X_0'C_5X_0$, both as $(C_5X_0)'(C_5X_0)$ and as $X_0'X_0 - 5\bar{x}\bar{x}'$.

(c) Let $D = \mathrm{diag}(\frac{1}{2}, \frac{1}{4}, \frac{1}{6})$. Compute DSD, and explain why its diagonal entries equal 1.

Solution. (a) $\mathbf{1}'X_0 = (10,\ 20,\ 25)$, so $\bar{x} = \frac{1}{5}X_0'\mathbf{1} = (2,\ 4,\ 5)'$. As in Example 1.15, $C_5X_0 = X_0 - \mathbf{1}\bar{x}'$:

$$C_5X_0 = \begin{pmatrix} 0 & -2 & 0 \\ 1 & -2 & -3 \\ -1 & 0 & -3 \\ 1 & 2 & 3 \\ -1 & 2 & 3 \end{pmatrix}$$

(b) Write c_1, c_2, c_3 for the columns of C_5X_0. The (j, l) entry of $(C_5X_0)'(C_5X_0)$ is $c_j'c_l$ (Theorems 1.4(d) and 1.9):

$$c_1'c_1 = 0+1+1+1+1 = 4, \quad c_2'c_2 = 4+4+0+4+4 = 16, \quad c_3'c_3 = 0+9+9+9+9 = 36,$$
$$c_1'c_2 = 0-2+0+2-2 = -2, \quad c_1'c_3 = 0-3+3+3-3 = 0, \quad c_2'c_3 = 0+6+0+6+6 = 18$$

On the other hand

$$X_0'X_0 - 5\bar{x}\bar{x}' = \begin{pmatrix} 24 & 38 & 50 \\ 38 & 96 & 118 \\ 50 & 118 & 161 \end{pmatrix} - \begin{pmatrix} 20 & 40 & 50 \\ 40 & 80 & 100 \\ 50 & 100 & 125 \end{pmatrix} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 16 & 18 \\ 0 & 18 & 36 \end{pmatrix} = S$$

(for instance $38 = 4+6+4+18+6$ and $161 = 25+4+4+64+64$).

(c) Left multiplication by D scales the rows and right multiplication scales the columns, so $(DSD)_{ij} = d_is_{ij}d_j$:

$$DSD = \begin{pmatrix} 1 & -\frac{1}{4} & 0 \\ -\frac{1}{4} & 1 & \frac{3}{4} \\ 0 & \frac{3}{4} & 1 \end{pmatrix},$$

for instance $(DSD)_{23} = \frac{1}{4}\cdot 18 \cdot \frac{1}{6} = \frac{3}{4}$. The diagonal entries are $d_i^2 s_{ii} = \frac{4}{4}, \frac{16}{16}, \frac{36}{36}$, all equal to 1, because $d_i = 1/\sqrt{s_{ii}}$ for $i = 1, 2, 3$.

Problems

Problem 1.1. Give examples of real 2×2 matrices such that: (a) $A^2 = -I$; (b) $B^2 = 0$ ($B \neq 0$); (c) $CD = -DC$ ($CD \neq 0$); (d) $EF = 0$ (no entries of E or F are zero).

Solution. (a) No real number a has $a^2 = -1$, but $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ gives $A^2 = \left(\begin{smallmatrix} 0\cdot 0+1\cdot(-1) & 0\cdot 1+1\cdot 0 \\ (-1)\cdot 0+0\cdot(-1) & (-1)\cdot 1+0\cdot 0 \end{smallmatrix}\right) = -I$.

(b) $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \neq 0$ satisfies $B^2 = \left(\begin{smallmatrix} 0\cdot 0+1\cdot 0 & 0\cdot 1+1\cdot 0 \\ 0\cdot 0+0\cdot 0 & 0\cdot 1+0\cdot 0 \end{smallmatrix}\right) = 0$.

(c) For $C = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$,

$$CD = \begin{pmatrix} 1-1 & -1-1 \\ 1+1 & -1+1 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad DC = \begin{pmatrix} 1-1 & 1+1 \\ -1-1 & -1+1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = -CD \neq 0$$

(d) For $E = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $F = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$, every entry of EF is $1\cdot 1 + 1\cdot(-1) = 0$, so $EF = 0$.

Problem 1.2. Let $A, B \in \mathbb{R}^{n\times n}$. Show that $(A+B)(A-B) = A^2 - B^2$ if and only if A and B commute.

Solution. By Theorem 1.3, $(A+B)(A-B) = A(A-B) + B(A-B) = A^2 - AB + BA - B^2$. Thus $(A+B)(A-B) = A^2 - B^2$ if and only if $-AB + BA = 0$, that is, if and only if $AB = BA$.

Problem 1.3. Let $A, B \in \mathbb{R}^{n\times n}$.

(a) Show that $\operatorname{tr}(AB - BA) = 0$.

(b) Deduce that there are no $A, B \in \mathbb{R}^{n\times n}$ with $AB - BA = I_n$.

Solution. (a) By Theorem 1.11(a),(b), $\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$.

(b) $\operatorname{tr}(I_n) = n \neq 0$, so by (a) I_n is not of the form $AB - BA$. In Exercise 1.4, for instance, $AB - BA = 2B$ and $\operatorname{tr}(2B) = 0$.

Problem 1.4. (a) Show that the product AB of two symmetric matrices $A, B \in \mathbb{R}^{n\times n}$ is symmetric if and only if A and B commute.

(b) Give an example of two symmetric matrices of the same size whose product is not symmetric.

Solution. (a) Since A and B are symmetric, $(AB)' = B'A' = BA$ (Theorem 1.9). So $AB = (AB)'$ holds if and only if $AB = BA$.

(b) Take $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 0 \end{smallmatrix}\right) \neq \left(\begin{smallmatrix} 0 & 2 \\ 1 & 0 \end{smallmatrix}\right) = BA$, so by (a) AB is not symmetric.

Problem 1.5. Show that for any $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{p\times m}$, $\operatorname{tr}(ABC) = \operatorname{tr}(B'A'C') = \operatorname{tr}(A'C'B')$.

Solution. By Theorem 1.11(c), then (a), then Theorem 1.9, and finally (c) again,

$$\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}\Big((CAB)'\Big) = \operatorname{tr}(B'A'C') = \operatorname{tr}(A'C'B')$$

In the last step the factors $B' \in \mathbb{R}^{p\times n}$, $A' \in \mathbb{R}^{n\times m}$, $C' \in \mathbb{R}^{m\times p}$ are moved cyclically.

Problem 1.6. Let $A \in \mathbb{R}^{n\times n}$ and let $D = \operatorname{diag}(a_{11}, \ldots, a_{nn})$. Show that the sum of all entries of A is

$$\mathbf{1}_n' A \mathbf{1}_n = \mathbf{1}_n' D \mathbf{1}_n + \operatorname{tr}\Big((\mathbf{1}_n \mathbf{1}_n' - I_n)A\Big)$$

Solution. By Theorem 1.11(a),(b),

$$\operatorname{tr}\Big((\mathbf{1}\mathbf{1}' - I_n)A\Big) = \operatorname{tr}(\mathbf{1}\mathbf{1}'A) - \operatorname{tr}(A) = \operatorname{tr}(\mathbf{1}'A\mathbf{1}) - \operatorname{tr}(A) = \mathbf{1}'A\mathbf{1} - \mathbf{1}'D\mathbf{1}$$

Here $\operatorname{tr}(\mathbf{1}\mathbf{1}'A) = \operatorname{tr}(\mathbf{1}'A\mathbf{1})$ moves the $n \times 1$ factor $\mathbf{1}$ past the $1 \times n$ factor $\mathbf{1}'A$; the trace of the 1×1 matrix $\mathbf{1}'A\mathbf{1}$ is that number; and $\mathbf{1}'D\mathbf{1} = \sum_i a_{ii} = \operatorname{tr}(A)$, because $D\mathbf{1} = (a_{11}, \ldots, a_{nn})'$. Rearranging gives the identity. Also $\mathbf{1}'A\mathbf{1} = \sum_i \sum_j a_{ij}$: by Theorem 1.4(b), $A\mathbf{1} = a_1 + \cdots + a_n$ has ith entry $\sum_j a_{ij}$, and by Theorem 1.9, $\mathbf{1}'(A\mathbf{1})$ is the sum of these n entries.

Problem 1.7. Let $A \in \mathbb{R}^{m\times n}$.

(a) Show that $A'A = 0$ implies $A = 0$.

(b) Show that $A'AB = A'AC$ implies $AB = AC$ for $B, C \in \mathbb{R}^{n\times p}$.

(c) Show that $BA'A = CA'A$ implies $BA' = CA'$ for $B, C \in \mathbb{R}^{p\times n}$.

(d) Let $m = n$ and A be symmetric. Show that $A^k = 0$ for some $k \geqslant 1$ implies $A = 0$. (Compare the non-symmetric $X \neq 0$ with $X^2 = 0$ of Remark 1.7.)

Solution. (a) If $A'A = 0$, then $\operatorname{tr}(A'A) = 0$, so $A = 0$ by Theorem 1.11(d).

(b) Put $F = A(B - C) \in \mathbb{R}^{m\times p}$. By Theorems 1.3 and 1.9, $F'F = (B - C)'A'A(B - C) = (B - C)'(A'AB - A'AC) = 0$. By (a) applied to F, $F = 0$, that is, $AB = AC$.

(c) Transposing $BA'A = CA'A$ gives $A'AB' = A'AC'$ (Theorem 1.9, with $(A'A)' = A'A$). By (b), $AB' = AC'$, and transposing back, $BA' = CA'$.

(d) Every power of A is symmetric: $(A^1)' = A' = A$, and if $(A^r)' = A^r$, then $(A^{r+1})' = (A^rA)' = A'(A^r)' = AA^r = A^{r+1}$ by Theorem 1.9, since $A^rA = AA^r$ (both equal the product of $r + 1$ factors A, Theorem 1.3).

- If G is symmetric and $G^2 = 0$, then $G'G = G^2 = 0$, so $G = 0$ by (a).
- Choose $s \geqslant 0$ with $2^s \geqslant k$. Then $A^{2^s} = A^{2^s-k}A^k = 0$.
- For $j = s, s - 1, \ldots, 1$: if $A^{2^j} = 0$, then $G = A^{2^{j-1}}$ is symmetric with $G^2 = A^{2^j} = 0$, so $A^{2^{j-1}} = 0$ by the first bullet.

Descending from $j = s$ to $j = 1$ gives $A = A^{2^0} = 0$.

Problem 1.8. Let $N \in \mathbb{R}^{n\times n}$ be **strictly upper triangular**: $n_{ij} = 0$ for $i \geqslant j$.

(a) Show that, for every $k \geqslant 1$, $(N^k)_{ij} = 0$ whenever $j < i + k$.

(b) Deduce that $N^n = 0$ and $(I_n - N)(I_n + N + \cdots + N^{n-1}) = I_n = (I_n + N + \cdots + N^{n-1})(I_n - N)$.

Solution. (a) Induction on k. For $k = 1$, $j < i + 1$ means $i \geqslant j$, and $n_{ij} = 0$ by hypothesis. Assume the claim for k. Then $(N^{k+1})_{ij} = \sum_{r=1}^{n}(N^k)_{ir}n_{rj}$. A term can be nonzero only if $r \geqslant i + k$ (induction hypothesis) and $j \geqslant r + 1$ (hypothesis on N), which forces $j \geqslant i + k + 1$. So if $j < i + (k + 1)$, every term is 0.

(b) For $k = n$ and all $i, j \in \{1, \ldots, n\}$, $j \leqslant n < i + n$; so $N^n = 0$ by (a). By Theorem 1.3,

$$(I - N)\sum_{r=0}^{n-1} N^r = \sum_{r=0}^{n-1} N^r - \sum_{r=1}^{n} N^r = I - N^n = I,$$

and the product in the other order is the same sum. Replacing N by the strictly upper triangular $-N$ gives $(I + N)(I - N + N^2 - \cdots + (-N)^{n-1}) = I$, which is Exercise 1.7(b) for $n = 4$.

Problem 1.9. (a) Show that the only symmetric idempotent 2×2 matrices are I_2, 0, and uu' with $u \in \mathbb{R}^2$, $u'u = 1$.

(b) Give an example of an $n \times n$ idempotent matrix ($n \geqslant 2$) that is not symmetric.

Solution. (a) The symmetric matrix $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$ is idempotent if and only if $\left(\begin{smallmatrix} a^2+b^2 & b(a+d) \\ b(a+d) & b^2+d^2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, that is, $a^2 + b^2 = a$, $b(a + d) = b$, $d^2 + b^2 = d$.

- If $a + d \neq 1$, the second equation gives $b = 0$; then $a^2 = a$ and $d^2 = d$, so $a, d \in \{0, 1\}$, and $a + d \neq 1$ leaves $a = d = 0$ or $a = d = 1$: $A = 0$ or $A = I_2$.

- If $a+d=1$, then $d=1-a$ and the first equation gives $b^2=a(1-a)$, so $0\leqslant a\leqslant 1$ and $b=\pm\sqrt{a(1-a)}$; the third equation then holds, since $(1-a)^2+a(1-a)=1-a$. Hence

$$A=\begin{pmatrix} a & \pm\sqrt{a(1-a)} \\ \pm\sqrt{a(1-a)} & 1-a \end{pmatrix}=\begin{pmatrix} \sqrt{a} \\ \pm\sqrt{1-a} \end{pmatrix}\begin{pmatrix} \sqrt{a} & \pm\sqrt{1-a} \end{pmatrix}=uu',$$

  with the same sign throughout, and $u'u=a+(1-a)=1$.

Conversely, I_2 and 0 are symmetric idempotent, and if $u'u=1$, then uu' is symmetric and $(uu')(uu')=u(u'u)u'=uu'$.

(b) Let $a,b\in\mathbb{R}^n$ with $b'a=1$, and $A=ab'$. Then $A^2=a(b'a)b'=ab'=A$. This A is symmetric only when b is a multiple of a: if $ab'=ba'$, multiplying on the right by a gives $a(b'a)=b(a'a)$, that is, $a=(a'a)b$; here $a\neq 0$ because $b'a=1$, so $a'a>0$ and $b=a/(a'a)$. Conversely, if $b=ca$, then $ab'=c\,aa'$ is symmetric. For a concrete example take $a=e_1$ and $b=e_1+e_2$: $b'a=1$, and $A=e_1e_1'+e_1e_2'$ has $(1,1)$ and $(1,2)$ entries 1 and all other entries 0; for $n=2$, $A=\left(\begin{smallmatrix}1&1\\0&0\end{smallmatrix}\right)$, with $A^2=A\neq A'$.

Problem 1.10. Let $P,Q\in\mathbb{R}^{n\times n}$ be idempotent.

(a) Show that I_n-P is idempotent and $P(I_n-P)=(I_n-P)P=0$.

(b) Show that $P+Q$ is idempotent if and only if $PQ=QP=0$.

(c) Show that if $PQ=QP$, then PQ is idempotent.

Solution. (a) By Theorem 1.3 and $P^2=P$, $(I-P)^2=I-2P+P^2=I-P$, and $P(I-P)=P-P^2=0=(I-P)P$.

(b) $(P+Q)^2=P^2+PQ+QP+Q^2=P+Q+PQ+QP$, so $P+Q$ is idempotent if and only if $PQ+QP=0$.

- If $PQ=QP=0$, then $PQ+QP=0$.
- Conversely, let $PQ+QP=0$. Multiplying on the left by P gives $PQ+PQP=0$; multiplying on the right by P gives $PQP+QP=0$. Subtracting, $PQ=QP$. Then $2PQ=PQ+QP=0$, so $PQ=QP=0$.

(c) $(PQ)^2=P(QP)Q=P(PQ)Q=P^2Q^2=PQ$.

Problem 1.11. (a) Let $X_0\in\mathbb{R}^{n\times k}$ be a data matrix with mean vector $\bar{x}$ and $C=C_n$. Show that for every $b\in\mathbb{R}^k$

$$(X_0-\mathbf{1}b')'(X_0-\mathbf{1}b')=X_0'CX_0+n(\bar{x}-b)(\bar{x}-b)'$$

(b) Deduce that for $x\in\mathbb{R}^n$ and $a\in\mathbb{R}$, $\sum_{i=1}^n(x_i-a)^2=\sum_{i=1}^n(x_i-\bar{x})^2+n(\bar{x}-a)^2$; hence $\sum_i(x_i-a)^2$ is smallest exactly when $a=\bar{x}$.

Solution. (a) Put $c=\bar{x}-b$. Since $\bar{x}'=\frac{1}{n}\mathbf{1}'X_0$, $CX_0=X_0-\frac{1}{n}\mathbf{1}(\mathbf{1}'X_0)=X_0-\mathbf{1}\bar{x}'$, so $X_0-\mathbf{1}b'=CX_0+\mathbf{1}c'$. By Theorems 1.3 and 1.9,

$$(CX_0+\mathbf{1}c')'(CX_0+\mathbf{1}c')=X_0'C'CX_0+X_0'C'\mathbf{1}c'+c\,\mathbf{1}'CX_0+c\,(\mathbf{1}'\mathbf{1})c'$$

By Proposition 1.13, $C'C=C^2=C$, $C'\mathbf{1}=C\mathbf{1}=0$, $\mathbf{1}'C=(C'\mathbf{1})'=0$ and $\mathbf{1}'\mathbf{1}=n$. The right side is therefore $X_0'CX_0+n\,cc'$.

(b) Take $k=1$, $X_0=x$, $b=a$. The left side of (a) is $(x-a\mathbf{1})'(x-a\mathbf{1})=\sum_i(x_i-a)^2$ (Theorem 1.9), and $x'Cx=\sum_i(x_i-\bar{x})^2$ by Example 1.14. Since $n(\bar{x}-a)^2\geqslant 0$, with equality only for $a=\bar{x}$, the sum $\sum_i(x_i-a)^2$ is smallest exactly at $a=\bar{x}$.

Chapter 2. Gauss elimination, RREF, and inverses

Linear systems and augmented matrices

Definition 2.1. Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. The system of m linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

is the matrix equation $Ax = b$, since $(Ax)_i = \sum_j a_{ij}x_j$. A is the **coefficient matrix** and $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$, the matrix A with the column b appended, is the **augmented matrix**. A **solution** is an $x \in \mathbb{R}^n$ with $Ax = b$. The system is **consistent** if it has a solution and **inconsistent** otherwise. It is **homogeneous** if $b = 0$; then $x = 0$ is a solution, the **trivial solution**.

Elementary row operations

Definition 2.2. The **elementary row operations** on a matrix with m rows are, for row indices $r \neq s$:

(1) multiply row r by a number $c \neq 0$, written $R_r \leftarrow cR_r$;

(2) add c times row s to row r ($c \in \mathbb{R}$), written $R_r \leftarrow R_r + cR_s$;

(3) interchange rows r and s, written $R_r \leftrightarrow R_s$.

When several operations label one arrow, they are performed one after another, in the order written (upper label first). B is **row-equivalent** to A if B is obtained from A by finitely many elementary row operations.

Theorem 2.3. (a) Each elementary row operation is undone by an operation of the same type: $R_r \leftarrow cR_r$ by $R_r \leftarrow c^{-1}R_r$, $R_r \leftarrow R_r + cR_s$ by $R_r \leftarrow R_r - cR_s$, and $R_r \leftrightarrow R_s$ by itself.

(b) If row operations take $[A \mid b]$ to $[B \mid z]$, then the systems $Ax = b$ and $Bx = z$ have exactly the same solutions.

Proof. (a) is checked row by row. (b) It suffices to consider one operation. Each equation of the new system is a multiple of an equation of the old one, or a sum of an old equation and a multiple of another, so every solution of $Ax = b$ solves $Bx = z$. By (a), the old system is obtained back from the new one by a row operation, so every solution of $Bx = z$ solves $Ax = b$. □

So we solve a system by simplifying its augmented matrix with row operations; the next definition says how far to simplify.

Echelon form and RREF

Definition 2.4. A matrix R is in **reduced row echelon form** (**RREF**) if

(a) the first nonzero entry of each nonzero row is 1, its **pivot**;

(b) each column that contains a pivot has all its other entries 0;

(c) every zero row lies below every nonzero row;

(d) if rows $1, \ldots, r$ are the nonzero rows and the pivot of row i lies in column k_i, then $k_1 < k_2 < \cdots < k_r$.

The columns $k_1, \dots, k_r$ are the **pivot columns**; column k_i of R is the standard unit vector e_i. A matrix that satisfies (a), (c), (d), but not necessarily (b), is in **echelon form**; then every entry below a pivot is 0.

For example, I_n and 0 are in RREF, and so is

$$R = \begin{pmatrix} 0 & 1 & -3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad (r = 2,\ k_1 = 2,\ k_2 = 4)$$

The matrix $\left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{smallmatrix}\right)$ is in echelon form but not in RREF (column 3 violates (b)), and $\left(\begin{smallmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ violates (a) and (d).

Definition 2.5 (Gauss–Jordan elimination)**.** Let M be any matrix (for instance an augmented matrix). Start with $i = 1$.

Step 1. Find the first column that has a nonzero entry in rows $i, i+1, \dots$; if there is none, go to Step 5. This is the next pivot column.

Step 2. If the entry of row i in this column is 0, interchange row i with a lower row that has a nonzero entry there.

Step 3. Multiply row i by the reciprocal of this entry, so that the pivot is 1.

Step 4. Add suitable multiples of row i to the rows below it, so that every entry below the pivot is 0. Increase i by 1 and return to Step 1.

Step 5. (*Backward elimination.*) For each pivot, from the last one to the first, add suitable multiples of its row to the rows above it, so that every entry above the pivot is 0.

Steps 1–4 (**forward elimination**) produce an echelon form; Step 5 produces a matrix in RREF. The scaling in Step 3 may also be postponed, to the end of forward elimination or even to the very end (the pivots become 1 only then); this often avoids fractions. Every matrix is thus row-equivalent to a matrix in RREF.

Example 2.6. Solve the homogeneous system

$$x_1 + x_2 + x_3 - 2x_4 = 0, \qquad x_1 + x_2 + 2x_3 - 3x_4 = 0, \qquad 2x_1 + x_2 + 5x_3 - 4x_4 = 0$$

Solution. Row-reduce the coefficient matrix A (the column $b = 0$ never changes, so it is not written).

$$\begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & -3 \\ 2 & 1 & 5 & -4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 3 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + 3R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

After the first step the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3, and Step 3 multiplies the new row 2 by -1. The first matrix of the second line is an echelon form (end of forward elimination); the last matrix, R, is in RREF, with pivot columns $1, 2, 3$. The system $Rx = 0$ reads $x_1 + 2x_4 = 0$, $x_2 - 3x_4 = 0$, $x_3 - x_4 = 0$, so $x_4 = t$ is free and all solutions are

$$x = t\,(-2,\ 3,\ 1,\ 1)' \qquad (t \in \mathbb{R})$$

Check: $-2 + 3 + 1 - 2 = 0$, $-2 + 3 + 2 - 3 = 0$, $-4 + 3 + 5 - 4 = 0$.

Reading off the solutions

Proposition 2.7. Let row operations take $[A \mid b]$ ($A \in \mathbb{R}^{m\times n}$) to $[R \mid z]$ with R in RREF, nonzero rows $1,\dots,r$ and pivot columns $k_1 < \cdots < k_r$. Call $x_{k_1},\dots,x_{k_r}$ the **pivot unknowns** and the other $n-r$ unknowns the **free unknowns**.

(a) $Ax = b$ is consistent if and only if $z_i = 0$ for every $i > r$, that is, no row of $[R \mid z]$ reads $(0 \ \cdots \ 0 \mid z_i)$ with $z_i \neq 0$.

(b) If it is consistent, all solutions are obtained by choosing arbitrary values for the free unknowns and computing each pivot unknown from its row: $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij}x_j$ $(1 \leqslant i \leqslant r)$.

Proof. By Theorem 2.3, $Ax = b$ and $Rx = z$ have the same solutions. (a) Row $i > r$ of $Rx = z$ reads $0 = z_i$. So there is no solution if one of these z_i is not 0; if all of them are 0, these rows hold for every x, and the argument for (b) gives a solution. (b) For $i \leqslant r$, column k_l of R is e_l, so the pivot unknown x_{k_i} occurs only in equation i, with coefficient 1; each choice of the free unknowns therefore gives exactly one solution. □

Proposition 2.8 (Uniqueness of the RREF)**.** Let R and S be matrices in RREF that are both row-equivalent to $A \in \mathbb{R}^{m\times n}$. Then $R = S$. So A is row-equivalent to exactly one matrix in RREF, called **the RREF of** A.

Proof. • By Theorem 2.3(a), the operations that take A to R can be undone, so A is obtained from R by row operations, and then S from A. By Theorem 2.3(b) (with $b = 0$), $Rx = 0$ and $Sx = 0$ have the same solutions.

- Let R have nonzero rows $1,\dots,r$ and pivot columns $k_1 < \cdots < k_r$. For $1 \leqslant j \leqslant n$: j is not a pivot column of R if and only if some solution x of $Rx = 0$ has $x_j = 1$ and $x_l = 0$ for all $l > j$. Indeed, if x_j is free, Proposition 2.7(b) with $x_j = 1$ and the other free unknowns 0 gives a solution with $x_{k_i} = -R_{ij}$; here $R_{ij} = 0$ whenever $k_i > j$, since row i is 0 before column k_i (Definition 2.4(a)), so $x_l = 0$ for every $l > j$. If $j = k_i$ is a pivot column, equation i of $Rx = 0$ reads $x_{k_i} + \sum_{l>k_i} R_{il}x_l = 0$, so a solution with $x_l = 0$ for all $l > j$ has $x_j = 0 \neq 1$.
- This describes the pivot columns by the solutions of $Rx = 0$, which are those of $Sx = 0$. So R and S have the same pivot columns $k_1 < \cdots < k_r$, hence the same number r of nonzero rows, the same zero rows $r+1,\dots,m$, and the same columns $e_1,\dots,e_r$ in the pivot positions.
- Let j be a non-pivot column and $1 \leqslant i \leqslant r$. The solution x of $Rx = 0$ with $x_j = 1$ and the other free unknowns 0 has $x_{k_i} = -R_{ij}$ (Proposition 2.7(b)). It also solves $Sx = 0$, and S has the same free unknowns, so the same proposition applied to S gives $x_{k_i} = -S_{ij}$. Hence $R_{ij} = S_{ij}$. All entries of R and S agree, and $R = S$. □

Example 2.9. For the RREF R displayed after Definition 2.4, the system $Rx = 0$ is $x_2 - 3x_3 + 2x_5 = 0$, $x_4 + 2x_5 = 0$; the pivot unknowns are x_2, x_4 and the free unknowns x_1, x_3, x_5. With $x_1 = a$, $x_3 = b$, $x_5 = c$, all solutions are

$$x = \begin{pmatrix} a \\ 3b-2c \\ b \\ -2c \\ c \end{pmatrix} = a\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c\begin{pmatrix} 0 \\ -2 \\ 0 \\ -2 \\ 1 \end{pmatrix} \qquad (a,b,c \in \mathbb{R})$$

Example 2.10. For which $y = (y_1, y_2, y_3, y_4)'$ is $Ax = y$ consistent, where

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 1 \end{pmatrix}?$$

Solve it for $y = (4, 7, 4, 7)'$.

Solution. Row-reduce the augmented matrix, keeping $y_1, \dots, y_4$ as letters:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 2 & 3 & 1 & 1 & y_2 \\ 1 & 1 & 1 & 1 & y_3 \\ 2 & 2 & 2 & 1 & y_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & -1 & 1 & -1 & y_2 - 2y_1 \\ 0 & -1 & 1 & 0 & y_3 - y_1 \\ 0 & -2 & 2 & -1 & y_4 - 2y_1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 + R_2,\ R_4 \leftarrow R_4 + 2R_2]{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & 1 & -1 & 1 & 2y_1 - y_2 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 1 & 2y_1 - 2y_2 + y_4 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & 1 & -1 & 1 & 2y_1 - y_2 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right)$$

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 0 & y_2 - y_3 \\ 0 & 1 & -1 & 0 & y_1 - y_3 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2y_1 + y_2 + y_3 \\ 0 & 1 & -1 & 0 & y_1 - y_3 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right)$$

The left part of the last matrix is the RREF of A, with $r = 3$ and pivot columns $1, 2, 4$, and $z_4 = y_1 - y_2 - y_3 + y_4$. By Proposition 2.7, the system is consistent if and only if $y_1 - y_2 - y_3 + y_4 = 0$; then $x_3 = c$ is free and

$$x_1 = -2y_1 + y_2 + y_3 - 2c, \qquad x_2 = y_1 - y_3 + c, \qquad x_3 = c, \qquad x_4 = y_1 - y_2 + y_3 \qquad (c \in \mathbb{R})$$

For $y = (4, 7, 4, 7)'$, $4 - 7 - 4 + 7 = 0$, and the solutions are $x = (3, 0, 0, 1)' + c\,(-2, 1, 1, 0)'$: the solution $(3, 0, 0, 1)'$ plus the multiples of $(-2, 1, 1, 0)'$, which solve $Ax = 0$. Check: $A(3, 0, 0, 1)' = (4, 7, 4, 7)'$, and $A(-2, 1, 1, 0)' = 0$ says $\mathrm{col}_3(A) = 2\,\mathrm{col}_1(A) - \mathrm{col}_2(A)$. For $y = (1, 0, 0, 0)'$, $y_1 - y_2 - y_3 + y_4 = 1 \neq 0$, so $Ax = y$ has no solution.

Homogeneous systems

Theorem 2.11. (a) If $A \in \mathbb{R}^{m \times n}$ with $m < n$ (fewer equations than unknowns), then $Ax = 0$ has a nontrivial solution.

(b) For $A \in \mathbb{R}^{n \times n}$, $Ax = 0$ has only the trivial solution if and only if the RREF of A is I_n.

Proof. Let R be the RREF of A, with r nonzero rows. (a) $r \leqslant m < n$, so there is a free unknown; giving it the value 1 and the other free unknowns the value 0 gives a solution $x \neq 0$ (Proposition 2.7(b)). (b) If $R = I$, then $Rx = x$, so $x = 0$ is the only solution (Theorem 2.3(b)). If $Ax = 0$ has only the trivial solution, there is no free unknown, so $r = n$; then $1 \leqslant k_1 < \cdots < k_n \leqslant n$ forces $k_i = i$, so column i of R is e_i and $R = I$. □

Inverses

Definition 2.12. $A \in \mathbb{R}^{n \times n}$ is **invertible** if $AB = BA = I_n$ for some $B \in \mathbb{R}^{n \times n}$. Such a B is unique: if $BA = I$ and $AC = I$, then $B = B(AC) = (BA)C = C$. It is called the **inverse** of A, written A^{-1}. For example, $\mathrm{diag}(d_1, \dots, d_n)$ with all $d_i \neq 0$ has inverse $\mathrm{diag}(d_1^{-1}, \dots, d_n^{-1})$, since the products in both orders are $\mathrm{diag}(1, \dots, 1) = I$.

Proposition 2.13. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2\times 2}$. Then A is invertible if and only if $ad - bc \neq 0$, and in that case

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Proof. Put $\Delta = ad - bc$ and $B = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$. Multiplying out,

$$AB = \begin{pmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{pmatrix} = \Delta I_2, \qquad BA = \begin{pmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{pmatrix} = \Delta I_2$$

- If $\Delta \neq 0$, then $A(\Delta^{-1}B) = \Delta^{-1}AB = I_2$ and $(\Delta^{-1}B)A = \Delta^{-1}BA = I_2$. So A is invertible with $A^{-1} = \Delta^{-1}B$ (Definition 2.12).
- If $\Delta = 0$, then $AB = 0$. Choose $x \neq 0$ with $Ax = 0$: if $B \neq 0$, let x be a nonzero column of B, so that Ax is the corresponding column of $AB = 0$ (Theorem 1.4(a)); if $B = 0$, then $a = b = c = d = 0$, so $A = 0$ and $x = e_1$ works. If A were invertible, then $x = A^{-1}(Ax) = A^{-1}0 = 0$ ✘. So A is not invertible. □

Example 2.14. For $A = \left(\begin{smallmatrix} 3 & 5 \\ 1 & 2 \end{smallmatrix}\right)$, $ad - bc = 6 - 5 = 1$, so Proposition 2.13 gives $A^{-1} = \left(\begin{smallmatrix} 2 & -5 \\ -1 & 3 \end{smallmatrix}\right)$. For $C = \left(\begin{smallmatrix} 2 & 6 \\ 1 & 3 \end{smallmatrix}\right)$, $ad - bc = 6 - 6 = 0$, so C is not invertible; the first column $x = (3, -1)'$ of $\left(\begin{smallmatrix} 3 & -6 \\ -1 & 2 \end{smallmatrix}\right)$ satisfies $Cx = (6 - 6,\ 3 - 3)' = 0$.

Theorem 2.15. Let $A, B \in \mathbb{R}^{n\times n}$ be invertible.

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; more generally $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$ for invertible $A_1, \ldots, A_k$.

(c) A' is invertible and $(A')^{-1} = (A^{-1})'$.

Proof. (a) $AA^{-1} = A^{-1}A = I$ says that A is the inverse of A^{-1}. (b) $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and similarly $(B^{-1}A^{-1})(AB) = I$; for k factors repeat this. (c) $A'(A^{-1})' = (A^{-1}A)' = I$ and $(A^{-1})'A' = (AA^{-1})' = I$ by Theorem 1.9. □

Elementary matrices

Definition 2.16. An **elementary matrix** is a matrix $E = \omega(I_m)$ obtained from I_m by one elementary row operation ω. For $r \neq s$ we write:

- $D_r(c)$ ($c \neq 0$): I_m with its (r, r) entry replaced by c (from $R_r \leftarrow cR_r$);
- $T_{rs}(c)$: I_m with its (r, s) entry replaced by c (from $R_r \leftarrow R_r + cR_s$);
- P_{rs}: I_m with rows r and s interchanged (from $R_r \leftrightarrow R_s$).

For example, for $m = 3$:

$$P_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad T_{31}(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix}, \qquad D_3(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{pmatrix}$$

Theorem 2.17. Let ω be an elementary row operation on matrices with m rows and $E = \omega(I_m)$.

(a) $\omega(A) = EA$ for every $A \in \mathbb{R}^{m\times n}$: a row operation is a left multiplication.

(b) E is invertible, and E^{-1} is the elementary matrix of the inverse operation: $D_r(c)^{-1} = D_r(c^{-1})$, $T_{rs}(c)^{-1} = T_{rs}(-c)$, $P_{rs}^{-1} = P_{rs}$.

(c) If row operations $\omega_1, \dots, \omega_k$ (in this order) take A to B, then $B = PA$ with $P = E_k \cdots E_2 E_1$, $E_j = \omega_j(I_m)$; and P is invertible.

Proof. (a) By Theorem 1.4(c), $\mathrm{row}_i(EA) = \mathrm{row}_i(E)A$. An unchanged row of E is e_i', and $e_i'A = \mathrm{row}_i(A)$. For type (2), $\mathrm{row}_r(E) = e_r' + ce_s'$, so $\mathrm{row}_r(EA) = \mathrm{row}_r(A) + c\,\mathrm{row}_s(A)$; types (1) and (3) are the same computation. (b) Let $\tilde{\omega}$ be the inverse operation (Theorem 2.3(a)) and $\tilde{E} = \tilde{\omega}(I)$. By (a), $E\tilde{E} = \omega(\tilde{\omega}(I)) = I$ and $\tilde{E}E = \tilde{\omega}(\omega(I)) = I$. (c) By (a), $B = E_k(\cdots(E_1A)) = (E_k \cdots E_1)A$, and a product of invertible matrices is invertible (Theorem 2.15(b)). □

Deciding invertibility and computing inverses

Theorem 2.18. For $A \in \mathbb{R}^{n\times n}$ the following are equivalent:

(i) A is invertible;

(ii) the RREF of A is I_n;

(iii) $Ax = 0$ has only the trivial solution;

(iv) $Ax = b$ has a solution for every $b \in \mathbb{R}^n$;

(v) A is a product of elementary matrices.

Moreover, if $BA = I$ or $AB = I$ for some $B \in \mathbb{R}^{n\times n}$, then A is invertible and $B = A^{-1}$.

Proof. Let $R = PA$ be the RREF of A, with $P = E_k \cdots E_1$ (Theorem 2.17(c)).

- If $R = I$, then $A = P^{-1} = E_1^{-1} \cdots E_k^{-1}$ is a product of elementary matrices, hence invertible (Theorem 2.15(b)); $Ax = 0$ has only the trivial solution (Theorem 2.11(b)); and $x = A^{-1}b$ solves $Ax = b$.
- If $R \neq I$, then R has a zero row, hence a zero last row (by (c) of Definition 2.4); otherwise $r = n$ and $R = I$ as in the proof of Theorem 2.11(b). Then $Ax = 0$ has a solution $x \neq 0$ (Theorem 2.11(b)), so A has no inverse (else $x = A^{-1}Ax = 0$ ✘), and A is not a product of elementary matrices (such a product is invertible, Theorems 2.17(b) and 2.15(b)). For $b = P^{-1}e_n$, the system $Ax = b$ has the same solutions as $Rx = Pb = e_n$, whose last equation reads $0 = 1$; so (iv) fails.

Thus each of (i)–(v) holds exactly when $R = I$. If $BA = I$, then $Ax = 0$ gives $x = BAx = 0$, so A is invertible by (iii), and $B = BAA^{-1} = A^{-1}$. If $AB = I$, then B is invertible with $B^{-1} = A$ by the case just shown, so $A = B^{-1}$ is invertible with $A^{-1} = B$. □

Computing A^{-1}. Row-reduce the $n \times 2n$ matrix $[A \mid I_n]$ by Gauss–Jordan elimination.

- If the left half becomes I_n, the right half is A^{-1}: $[A \mid I] \to [I \mid B]$ means $E[A \mid I] = [EA \mid E] = [I \mid B]$ for the product E of the elementary matrices used (Theorem 2.17(c); E multiplies each column separately, Theorem 1.4(a)), so $EA = I$ and $B = E = A^{-1}$ by Theorem 2.18.
- If a zero row appears in the left half, A is not invertible. The left half is then EA, with E the (invertible) product of the elementary matrices used so far (Theorem 2.17(c)). For every $C \in \mathbb{R}^{n\times n}$, EAC has the same zero row (Theorem 1.4(c)), so $EAC \neq I_n$ and EA is not invertible; if A were invertible, so would be EA (Theorem 2.15(b)).

Example 2.19. Find the inverse of

$$A = \begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -2 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -1 & 1 & 0 \end{pmatrix}$$

and write A and A^{-1} as products of elementary matrices.

Solution. Forward elimination of $[A \mid I_4]$:

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ -1 & 1 & -2 & 1 & 0 & 1 & 0 & 0 \\ 2 & -1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

After the first step the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3, and Step 3 multiplies the new row 3 by -1. Backward elimination, clearing columns 4, 3, 2 in turn:

$$\xrightarrow[R_2 \leftarrow R_2 - R_4]{R_1 \leftarrow R_1 + R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

The left half has become I_4, so

$$A^{-1} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Check: $\mathrm{row}_3(A)A^{-1} = (2,-1,2,-1)A^{-1} = (0+1-2+1,\ 2+0-2+0,\ 2-1+0+0,\ 0+1+0-1) = (0,0,1,0)$, and similarly for the other rows. The nine operations have the elementary matrices

$$E_1 = T_{21}(1),\ E_2 = T_{31}(-2),\ E_3 = T_{41}(-1),\ E_4 = P_{23},\ E_5 = D_3(-1),$$
$$E_6 = T_{14}(1),\ E_7 = T_{24}(-1),\ E_8 = T_{13}(-1),\ E_9 = T_{12}(1),$$

and by Theorem 2.17(c), $A^{-1} = E_9E_8\cdots E_1$. By Theorems 2.15(b) and 2.17(b),

$$A = E_1^{-1}E_2^{-1}\cdots E_9^{-1} = T_{21}(-1)\,T_{31}(2)\,T_{41}(1)\,P_{23}\,D_3(-1)\,T_{14}(-1)\,T_{24}(1)\,T_{13}(1)\,T_{12}(-1)$$

Example 2.20. Find the inverse of

$$P = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{pmatrix}$$

(each entry outside row 1 and column 1 is the sum of the entry above it and the entry to its left; P is the 4×4 **Pascal matrix**).

Solution. Every pivot is 1, so no scaling is needed. Forward elimination:

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 6 & 10 & 0 & 0 & 1 & 0 \\ 1 & 4 & 10 & 20 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & 9 & -1 & 0 & 1 & 0 \\ 0 & 3 & 9 & 19 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 10 & 2 & -3 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - 3R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

Backward elimination, clearing columns 4, 3, 2 in turn:

$$\xrightarrow[R_1\leftarrow R_1-R_4]{R_3\leftarrow R_3-3R_4,\ R_2\leftarrow R_2-3R_4} \left(\begin{array}{cccc|cccc} 1&1&1&0&2&-3&3&-1\\ 0&1&2&0&2&-8&9&-3\\ 0&0&1&0&4&-11&10&-3\\ 0&0&0&1&-1&3&-3&1 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_3]{R_2\leftarrow R_2-2R_3} \left(\begin{array}{cccc|cccc} 1&1&0&0&-2&8&-7&2\\ 0&1&0&0&-6&14&-11&3\\ 0&0&1&0&4&-11&10&-3\\ 0&0&0&1&-1&3&-3&1 \end{array}\right) \xrightarrow{R_1\leftarrow R_1-R_2} \left(\begin{array}{cccc|cccc} 1&0&0&0&4&-6&4&-1\\ 0&1&0&0&-6&14&-11&3\\ 0&0&1&0&4&-11&10&-3\\ 0&0&0&1&-1&3&-3&1 \end{array}\right)$$

The left half has become I_4, so

$$P^{-1}=\begin{pmatrix} 4&-6&4&-1\\ -6&14&-11&3\\ 4&-11&10&-3\\ -1&3&-3&1 \end{pmatrix}$$

Check: $\mathrm{row}_2(P)P^{-1}=(4-12+12-4,\ -6+28-33+12,\ 4-22+30-12,\ -1+6-9+4)=(0,1,0,0)$, and similarly for the other rows. Every operation added an integer multiple of one row to another, so the right half stayed an integer matrix throughout; this is why P^{-1} has integer entries.

Example 2.21. The matrix A of Example 2.10 is not invertible: its RREF is $\left(\begin{smallmatrix} 1&0&2&0\\ 0&1&-1&0\\ 0&0&0&1\\ 0&0&0&0 \end{smallmatrix}\right)\neq I_4$ (the left part of the last matrix there). Accordingly $Ax=0$ has the nontrivial solution $x=(-2,1,1,0)'$, and $Ax=y$ has no solution for $y=(1,0,0,0)'$ (Theorem 2.18, (iii) and (iv)).

Solving $Ax=b$

In practice $Ax=b$ is solved by forward elimination of $[A\mid b]$ alone: an echelon form already shows whether there is a solution, and how many.

Theorem 2.22. Let $A\in\mathbb{R}^{m\times n}$ and $b\in\mathbb{R}^m$, and let row operations take $[A\mid b]$ to a matrix $[E\mid d]$ in echelon form ($E\in\mathbb{R}^{m\times n}$, $d\in\mathbb{R}^m$). Then $Ax=b$ is consistent if and only if $[E\mid d]$ has no row $(0\ \cdots\ 0\mid c)$ with $c\neq 0$, that is, if and only if the last column of $[E\mid d]$ is not a pivot column.

Proof.

- By Theorem 2.3, $Ax=b$ and $Ex=d$ have the same solutions. In an echelon form, a row $(0\ \cdots\ 0\mid c)$ with $c\neq 0$ is exactly a row whose pivot lies in the last column.
- If $[E\mid d]$ has such a row, the corresponding equation reads $0=c$, and no x satisfies it. The same holds if such a row appears at any stage of the elimination, so one may stop there.
- Suppose there is no such row. Then every zero row of E is a zero row of $[E\mid d]$, and its equation reads $0=0$. Let rows $1,\dots,r$ be the nonzero rows of E, with pivots in columns $k_1<\cdots<k_r$; call $x_{k_1},\dots,x_{k_r}$ the pivot unknowns and the others free, as in Proposition 2.7.
- Give the free unknowns any values. Row i of $E=(e_{ij})$ is 0 before column k_i and 1 there, so for $i=r,r-1,\dots,1$ equation i reads $x_{k_i}+\sum_{j>k_i}e_{ij}x_j=d_i$. Each x_j with $j>k_i$ is free or the pivot unknown of a lower row, already found; so the equation determines x_{k_i} (**back substitution**). The resulting x solves $Ex=d$, hence $Ax=b$.
- Every solution arises in this way from the values of its free unknowns, since the equations force the pivot unknowns. □

Theorem 2.23. Let $A\in\mathbb{R}^{m\times n}$ and $b\in\mathbb{R}^m$.

(a) If x_p is one solution of $Ax = b$ (a **particular solution**), then the solutions of $Ax = b$ are exactly the vectors $x_p + z$ with $Az = 0$.

(b) $Ax = b$ has no solution, exactly one solution, or infinitely many solutions. If it is consistent, it has exactly one solution if and only if there are no free unknowns, that is, every column of E in Theorem 2.22 is a pivot column.

(c) If A is square, $Ax = b$ has exactly one solution if and only if A is invertible; this solution is $x = A^{-1}b$.

Proof. (a) If $Az = 0$, then $A(x_p + z) = Ax_p + Az = b$. Conversely, if $Ax = b$, then $z = x - x_p$ satisfies $Az = b - b = 0$, and $x = x_p + z$.

(b) Suppose $Ax = b$ is consistent, and take $[E \mid d]$ as in Theorem 2.22.

- By the proof of that theorem, each choice of values of the free unknowns gives exactly one solution, every solution arises so, and different choices give different solutions (they differ in a free unknown, an entry of x).
- Without free unknowns there is therefore exactly one solution; with a free unknown, each of its values gives a different solution, so there are infinitely many.

(c)
- If A is invertible, $x = A^{-1}b$ is a solution, since $A(A^{-1}b) = b$; and every solution satisfies $x = A^{-1}(Ax) = A^{-1}b$.
- If $Ax = b$ has exactly one solution x_p, then $Az = 0$ only for $z = 0$, since otherwise $x_p + z$ would be a second solution by (a). So A is invertible by Theorem 2.18 ((iii) $\Rightarrow$ (i)). □

So, to solve $Ax = b$: reduce $[A \mid b]$ by forward elimination to an echelon form; if a row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ appears, there is no solution; otherwise choose the free unknowns arbitrarily and find the pivot unknowns by back substitution (or continue to the RREF and use Proposition 2.7).

Example 2.24. Solve the system $x_1 - x_2 + x_3 - x_4 = 1$, $-x_1 + x_2 - 2x_3 + x_4 = 0$, $2x_1 - x_2 + 2x_3 - x_4 = 3$, $x_1 - x_2 + x_3 = 3$.

Solution. The coefficient matrix is the matrix A of Example 2.19, and $b = (1, 0, 3, 3)'$. Forward elimination of $[A \mid b]$ uses the operations of the forward elimination there:

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -2 & 1 & 0 \\ 2 & -1 & 2 & -1 & 3 \\ 1 & -1 & 1 & 0 & 3 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

- The last matrix is an echelon form with no row $(0\ 0\ 0\ 0 \mid c)$, $c \neq 0$, so the system is consistent (Theorem 2.22).
- Every column of the left part is a pivot column, so there are no free unknowns and the solution is unique (Theorem 2.23(b)).
- Back substitution: $x_4 = 2$; $x_3 = -1$; $x_2 = 1 - x_4 = -1$; $x_1 = 1 + x_2 - x_3 + x_4 = 1 - 1 + 1 + 2 = 3$. So $x = (3, -1, -1, 2)'$.
- Check, with A^{-1} from Example 2.19 (Theorem 2.23(c)): $A^{-1}b = (0 + 0 + 3 + 0,\ -1 + 0 + 3 - 3,\ -1 + 0 + 0 + 0,\ -1 + 0 + 0 + 3)' = (3, -1, -1, 2)'$.

Example 2.25. Solve the system $x_1+x_2+x_3-2x_4=2$, $x_1+x_2+2x_3-3x_4=3$, $2x_1+x_2+5x_3-4x_4=7$.

Solution. The coefficient matrix is the matrix A of Example 2.6, and $b=(2,3,7)'$. Forward elimination of $[A \mid b]$ uses the same operations as there:

$$\left(\begin{array}{cccc|c} 1&1&1&-2&2\\ 1&1&2&-3&3\\ 2&1&5&-4&7 \end{array}\right) \xrightarrow[R_3\leftarrow R_3-2R_1]{R_2\leftarrow R_2-R_1} \left(\begin{array}{cccc|c} 1&1&1&-2&2\\ 0&0&1&-1&1\\ 0&-1&3&0&3 \end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3} \left(\begin{array}{cccc|c} 1&1&1&-2&2\\ 0&-1&3&0&3\\ 0&0&1&-1&1 \end{array}\right) \xrightarrow{R_2\leftarrow -R_2} \left(\begin{array}{cccc|c} 1&1&1&-2&2\\ 0&1&-3&0&-3\\ 0&0&1&-1&1 \end{array}\right)$$

- The last matrix is an echelon form with no row $(0\ 0\ 0\ 0 \mid c)$, $c\neq 0$: the system is consistent (Theorem 2.22).
- The pivot columns are $1,2,3$, so x_4 is free and there are infinitely many solutions (Theorem 2.23(b)).
- Back substitution with $x_4=t$: $x_3=1+t$; $x_2=-3+3x_3=3t$; $x_1=2-x_2-x_3+2x_4=2-3t-(1+t)+2t=1-2t$. So

$$x=\begin{pmatrix}1-2t\\3t\\1+t\\t\end{pmatrix}=\begin{pmatrix}1\\0\\1\\0\end{pmatrix}+t\begin{pmatrix}-2\\3\\1\\1\end{pmatrix} \qquad (t\in\mathbb{R})$$

- This is the particular solution $x_p=(1,0,1,0)'$ plus the solutions $t(-2,3,1,1)'$ of $Ax=0$ found in Example 2.6, as Theorem 2.23(a) says.

Example 2.26. Show that the system $x_1-2x_2+x_3+2x_4=1$, $x_1-x_2-x_3+x_4=2$, $x_1+x_2-5x_3-x_4=3$ has no solution.

Solution. Forward elimination of the augmented matrix $[A \mid b]$, $b=(1,2,3)'$:

$$\left(\begin{array}{cccc|c} 1&-2&1&2&1\\ 1&-1&-1&1&2\\ 1&1&-5&-1&3 \end{array}\right) \xrightarrow[R_3\leftarrow R_3-R_1]{R_2\leftarrow R_2-R_1} \left(\begin{array}{cccc|c} 1&-2&1&2&1\\ 0&1&-2&-1&1\\ 0&3&-6&-3&2 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-3R_2} \left(\begin{array}{cccc|c} 1&-2&1&2&1\\ 0&1&-2&-1&1\\ 0&0&0&0&-1 \end{array}\right) \xrightarrow{R_3\leftarrow -R_3} \left(\begin{array}{cccc|c} 1&-2&1&2&1\\ 0&1&-2&-1&1\\ 0&0&0&0&1 \end{array}\right)$$

The last matrix is an echelon form whose last column is a pivot column: row 3 reads $0=1$. So the system has no solution (Theorem 2.22). Indeed, the coefficient matrix A has $\operatorname{row}_3(A)=3\operatorname{row}_2(A)-2\operatorname{row}_1(A)$, while $3b_2-2b_1=4\neq 3=b_3$: the third equation contradicts the first two.

Exercises

Exercise 2.1. Reduce the matrix $A = \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -2 & 0 & 4 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix}$ to echelon form by elementary row operations, and record the elementary matrices used.

Solution. Follow Definition 2.5, postponing the scaling (Step 3) to the end. Column 1 is the first pivot column; since $a_{11} = 0$, interchange rows 1 and 2; then clear column 1 below the pivot, and then column 3, the next pivot column:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Finally divide row 1 by 2:

$$H = \begin{pmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = D_1(\tfrac{1}{2})\, T_{32}(-1)\, T_{31}(-1)\, P_{12}\, A$$

(Theorem 2.17(c)). H is in echelon form with pivot columns 1 and 3. One more operation, $R_1 \leftarrow R_1 + R_2$, clears the entry above the second pivot and gives the RREF $\left(\begin{smallmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{smallmatrix}\right)$.

Exercise 2.2. Find the solutions of the equations

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + 2x_5 &= 0 \\ 2x_1 + 3x_2 + x_3 + 3x_4 + 5x_5 &= 0 \\ x_1 + 2x_3 + x_4 + 2x_5 &= 0 \end{aligned}$$

Solution. Row-reduce the coefficient matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 & 5 \\ 1 & 0 & 2 & 1 & 2 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & -1 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

The pivot columns are $1, 2, 4$, so x_3 and x_5 are free unknowns (Proposition 2.7). With $x_3 = \lambda$ and $x_5 = \mu$, the rows of the RREF give $x_1 = -2\lambda - \mu$, $x_2 = \lambda$, $x_4 = -\mu$. All solutions are

$$x = \lambda(-2, 1, 1, 0, 0)' + \mu(-1, 0, 0, -1, 1)' \qquad (\lambda, \mu \in \mathbb{R})$$

Check: $(-2, 1, 1, 0, 0)'$ gives $-2 + 1 + 1 = 0$, $-4 + 3 + 1 = 0$, $-2 + 2 = 0$, and $(-1, 0, 0, -1, 1)'$ gives $-1 - 1 + 2 = 0$, $-2 - 3 + 5 = 0$, $-1 - 1 + 2 = 0$. Three equations in five unknowns always have nontrivial solutions (Theorem 2.11(a)); here there are two free unknowns.

Exercise 2.3. Solve the equations

$$\begin{pmatrix} 1 & -3 & -2 & -1 \\ -1 & 2 & 1 & 1 \\ -1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -2 \\ 1 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 2 & 3 & 1 \\ 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 3 \\ 1 \end{pmatrix}$$

by Gaussian elimination.

Solution. First system. Forward elimination:

$$\left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ -1 & 2 & 1 & 1 & -1 \\ -1 & 2 & 2 & 1 & -2 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

Backward elimination:

$$\xrightarrow{R_1 \leftarrow R_1 + R_4} \left(\begin{array}{cccc|c} 1 & -3 & -2 & 0 & 2 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + 3R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

By Proposition 2.7, the unique solution is $(x_1, x_2, x_3, x_4) = (3, 1, -1, 1)$. Check: $3 - 3 + 2 - 1 = 1$, $-3 + 2 - 1 + 1 = -1$, $-3 + 2 - 2 + 1 = -2$, $1 - 1 + 1 = 1$.

Second system. Forward elimination; after the first step the entry of row 2 in column 2 is 0, so rows 2 and 3 are interchanged (Step 2):

$$\left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 4 \\ 2 & 2 & 3 & 1 & 6 \\ 1 & 2 & 2 & 2 & 3 \\ 0 & 1 & 1 & 3 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 4 \\ 0 & 0 & -1 & -1 & -2 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 3 & 1 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & -1 & -1 & -2 \\ 0 & 1 & 1 & 3 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow -R_3]{R_4 \leftarrow R_4 - R_2} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right)$$

Every column of the left part is a pivot column and the last column is not, so there is exactly one solution (Theorems 2.22 and 2.23(b)). Back substitution: $x_4 = 0$; $x_3 = 2 - x_4 = 2$; $x_2 = -1 - x_4 = -1$; $x_1 = 4 - x_2 - 2x_3 - x_4 = 4 + 1 - 4 - 0 = 1$. The unique solution is $(x_1, x_2, x_3, x_4) = (1, -1, 2, 0)$. Check: $1 - 1 + 4 + 0 = 4$, $2 - 2 + 6 + 0 = 6$, $1 - 2 + 4 + 0 = 3$, $0 - 1 + 2 + 0 = 1$.

Exercise 2.4. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix}$. Find an elementary matrix E with $B = EA$ when: (a) B is obtained from A by interchanging the first and fourth rows; (b) B is obtained from A by multiplying the second row by 7; (c) B is obtained from A by subtracting twice the third row from the fourth row.

Solution. Perform each operation on I_4 (Theorem 2.17(a)):

$$P_{14} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad D_2(7) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_{43}(-2) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{pmatrix}$$

and

$$P_{14}A = \begin{pmatrix} 10 & 11 & 12 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{pmatrix}, \quad D_2(7)A = \begin{pmatrix} 1 & 2 & 3 \\ 28 & 35 & 42 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix}, \quad T_{43}(-2)A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ -4 & -5 & -6 \end{pmatrix}$$

Exercise 2.5. Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$.

(a) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then multiplying row 2 by 2. Does the order of the two operations matter?

(b) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then adding 3 times row 1 to row 2. Does the order matter?

(c) Do elementary matrices commute?

Solution. By Theorem 2.17, the operation done first is the factor next to A.

(a) $E = D_2(2)P_{12} = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, while $P_{12}D_2(2) = \left(\begin{smallmatrix} 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, and

$$D_2(2)P_{12}A = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 & 4 & 6 \\ 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} = P_{12}D_2(2)A$$

The order matters.

(b) $E = T_{21}(3)P_{12} = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, while $P_{12}T_{21}(3) = \left(\begin{smallmatrix} 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, and

$$T_{21}(3)P_{12}A = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 5 & 9 & 13 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 3 & 7 & 11 & 15 \\ 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} = P_{12}T_{21}(3)A$$

The order matters.

(c) No: (a) and (b) give pairs of elementary matrices that do not commute.

Exercise 2.6. Use row reduction of $[A \mid I_4]$ to find the inverse of $A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 3 & 4 & 3 & 4 \\ 1 & 1 & 1 & 2 \end{pmatrix}$.

Solution. The entry in position $(1,1)$ is 0, so Step 2 interchanges rows 1 and 2:

$$\left(\begin{array}{cccc|cccc} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 3 & 4 & 3 & 4 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 3 & 4 & 3 & 4 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 3R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow -R_3]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_2 \leftarrow R_2 - R_4]{R_1 \leftarrow R_1 - R_4} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & -1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 & 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right)$$

Hence

$$A^{-1} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -2 & 1 & -1 \\ 1 & 3 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Check: $\text{row}_3(A)A^{-1} = (3,4,3,4)A^{-1} = (-3+3,\ 3-8+9-4,\ 4-3,\ -4+4) = (0,0,1,0)$, and similarly for the other rows.

Exercise 2.7. Do $A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 5 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$ have inverses? Find the inverse when it exists.

Solution. (A) Row-reduce A; after the first step the entry of row 2 in column 2 is 0, so rows 2 and 3 are interchanged:

$$A \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 2 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

A zero row appears, so A has no inverse (the computation of A^{-1} after Theorem 2.18). (Indeed row 4 of A is the sum of rows 2 and 3.)

(B) Row-reduce $[B \mid I_4]$. Each step subtracts one row from the next one, and every pivot is 1:

$$\left(\begin{array}{cccc|cccc} 1&1&0&0&1&0&0&0\\ 1&2&1&0&0&1&0&0\\ 0&1&2&1&0&0&1&0\\ 0&0&1&2&0&0&0&1 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_3]{R_2\leftarrow R_2-R_1,\ R_3\leftarrow R_3-R_2} \left(\begin{array}{cccc|cccc} 1&1&0&0&1&0&0&0\\ 0&1&1&0&-1&1&0&0\\ 0&0&1&1&1&-1&1&0\\ 0&0&0&1&-1&1&-1&1 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_2]{R_3\leftarrow R_3-R_4,\ R_2\leftarrow R_2-R_3} \left(\begin{array}{cccc|cccc} 1&0&0&0&4&-3&2&-1\\ 0&1&0&0&-3&3&-2&1\\ 0&0&1&0&2&-2&2&-1\\ 0&0&0&1&-1&1&-1&1 \end{array}\right)$$

The left block is I_4, so B is invertible and

$$B^{-1} = \begin{pmatrix} 4&-3&2&-1\\ -3&3&-2&1\\ 2&-2&2&-1\\ -1&1&-1&1 \end{pmatrix}$$

(Theorem 2.18). Check: the second row of BB^{-1} is $(4{-}6{+}2,\ -3{+}6{-}2,\ 2{-}4{+}2,\ -1{+}2{-}1) = (0,1,0,0)$, and similarly for the other rows.

Exercise 2.8. Use Proposition 2.13 to decide which of $\left(\begin{smallmatrix}5&3\\3&2\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}4&6\\6&9\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}7&-3\\-5&2\end{smallmatrix}\right)$ are invertible, and find the inverses that exist. Then find the inverse of $\begin{pmatrix}0&1&1&1\\1&1&1&1\\0&1&1&2\\0&0&1&1\end{pmatrix}$.

Solution. (2×2) By Proposition 2.13:

- $5\cdot 2 - 3\cdot 3 = 1$, so $\left(\begin{smallmatrix}5&3\\3&2\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}2&-3\\-3&5\end{smallmatrix}\right)$.
- $4\cdot 9 - 6\cdot 6 = 0$, so $\left(\begin{smallmatrix}4&6\\6&9\end{smallmatrix}\right)$ is not invertible.
- $7\cdot 2 - (-3)(-5) = -1$, so $\left(\begin{smallmatrix}7&-3\\-5&2\end{smallmatrix}\right)^{-1} = -\left(\begin{smallmatrix}2&3\\5&7\end{smallmatrix}\right) = \left(\begin{smallmatrix}-2&-3\\-5&-7\end{smallmatrix}\right)$.

Check: $\left(\begin{smallmatrix}7&-3\\-5&2\end{smallmatrix}\right)\left(\begin{smallmatrix}-2&-3\\-5&-7\end{smallmatrix}\right) = \left(\begin{smallmatrix}-14+15&-21+21\\10-10&15-14\end{smallmatrix}\right) = I_2$.

(4×4) Call the second matrix K and row-reduce $[K \mid I_4]$. The entry in position $(1,1)$ is 0, so Step 2 interchanges rows 1 and 2; after clearing column 2, the entry in position $(3,3)$ is 0, so rows 3 and 4 are interchanged:

$$\left(\begin{array}{cccc|cccc} 0&1&1&1&1&0&0&0\\ 1&1&1&1&0&1&0&0\\ 0&1&1&2&0&0&1&0\\ 0&0&1&1&0&0&0&1 \end{array}\right) \xrightarrow{R_1\leftrightarrow R_2} \left(\begin{array}{cccc|cccc} 1&1&1&1&0&1&0&0\\ 0&1&1&1&1&0&0&0\\ 0&1&1&2&0&0&1&0\\ 0&0&1&1&0&0&0&1 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-R_2} \left(\begin{array}{cccc|cccc} 1&1&1&1&0&1&0&0\\ 0&1&1&1&1&0&0&0\\ 0&0&0&1&-1&0&1&0\\ 0&0&1&1&0&0&0&1 \end{array}\right) \xrightarrow{R_3\leftrightarrow R_4} \left(\begin{array}{cccc|cccc} 1&1&1&1&0&1&0&0\\ 0&1&1&1&1&0&0&0\\ 0&0&1&1&0&0&0&1\\ 0&0&0&1&-1&0&1&0 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_4]{R_3\leftarrow R_3-R_4,\ R_2\leftarrow R_2-R_4} \left(\begin{array}{cccc|cccc} 1&1&1&0&1&1&-1&0\\ 0&1&1&0&2&0&-1&0\\ 0&0&1&0&1&0&-1&1\\ 0&0&0&1&-1&0&1&0 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_3]{R_2\leftarrow R_2-R_3} \left(\begin{array}{cccc|cccc} 1&1&0&0&0&1&0&-1\\ 0&1&0&0&1&0&0&-1\\ 0&0&1&0&1&0&-1&1\\ 0&0&0&1&-1&0&1&0 \end{array}\right) \xrightarrow{R_1\leftarrow R_1-R_2} \left(\begin{array}{cccc|cccc} 1&0&0&0&-1&1&0&0\\ 0&1&0&0&1&0&0&-1\\ 0&0&1&0&1&0&-1&1\\ 0&0&0&1&-1&0&1&0 \end{array}\right)$$

So

$$K^{-1} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix}$$

Check: $\mathrm{row}_3(K)K^{-1} = (0,1,1,2)K^{-1} = (1+1-2,\ 0,\ -1+2,\ -1+1) = (0,0,1,0)$, and similarly for the other rows.

Exercise 2.9. Let A be the coefficient matrix of Example 2.26.

(a) For which $b = (b_1, b_2, b_3)'$ is $Ax = b$ consistent?

(b) Find all solutions of $Ax = b$ for $b = (1, 2, 4)'$.

Solution. (a) Row-reduce $[A \mid b]$, keeping b_1, b_2, b_3 as letters:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 1 & -1 & -1 & 1 & b_2 \\ 1 & 1 & -5 & -1 & b_3 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 3 & -6 & -3 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 2b_2 - b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array}\right)$$

The left part of the last matrix is the RREF of A, with $r = 2$. By Proposition 2.7(a), $Ax = b$ is consistent if and only if $2b_1 - 3b_2 + b_3 = 0$. (In Example 2.26, $2 - 6 + 3 = -1 \neq 0$.)

(b) For $b = (1, 2, 4)'$, $2 - 6 + 4 = 0$, so the system is consistent, and the last matrix becomes

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 3 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The free unknowns are $x_3 = s$, $x_4 = t$, and by Proposition 2.7(b) $x_1 = 3 + 3s$, $x_2 = 1 + 2s + t$. So

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 3 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \qquad (s, t \in \mathbb{R}),$$

infinitely many solutions (Theorem 2.23(b)): the particular solution $(3, 1, 0, 0)'$ plus the solutions of $Ax = 0$ (Theorem 2.23(a)). Check: $A(3, 1, 0, 0)' = (3 - 2,\ 3 - 1,\ 3 + 1)' = (1, 2, 4)'$.

Exercise 2.10. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 5 & 3 & 3 \\ 1 & 2 & 2 & 3 & 3 & 1 \\ 3 & 6 & 4 & 7 & 3 & 5 \end{pmatrix} \in \mathbb{R}^{4\times 6}$$

(a) Show that $Ax = b$, $b = (b_1, b_2, b_3, b_4)'$, is consistent if and only if $2b_2 = b_3 + b_4$.

(b) Find all solutions of $Ax = b$ for $b = (2, 4, 3, 5)'$.

Solution. (a) Forward elimination of $[A \mid b]$, keeping $b_1, \ldots, b_4$ as letters:

$$\left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 2 & 4 & 3 & 5 & 3 & 3 & b_2 \\ 1 & 2 & 2 & 3 & 3 & 1 & b_3 \\ 3 & 6 & 4 & 7 & 3 & 5 & b_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 1 & 1 & 2 & 0 & b_3 - b_1 \\ 0 & 0 & 1 & 1 & 0 & 2 & b_4 - 3b_1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 1 & -1 & b_1 - b_2 + b_3 \\ 0 & 0 & 0 & 0 & -1 & 1 & b_4 - b_1 - b_2 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 1 & -1 & b_1 - b_2 + b_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2b_2 + b_3 + b_4 \end{array}\right)$$

The last matrix is in echelon form, with pivots in columns $1, 3, 5$ of A. By Theorem 2.22, $Ax = b$ is consistent if and only if $-2b_2 + b_3 + b_4 = 0$, that is, $2b_2 = b_3 + b_4$.

(b) For $b = (2, 4, 3, 5)'$, $2 \cdot 4 = 3 + 5$, and the last matrix has rows $(1, 2, 1, 2, 1, 1 \mid 2)$, $(0, 0, 1, 1, 1, 1 \mid 0)$, $(0, 0, 0, 0, 1, -1 \mid 1)$ and a zero row. Backward elimination:

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccccc|c} 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The free unknowns are $x_2 = \lambda$, $x_4 = \mu$, $x_6 = \nu$, and Proposition 2.7(b) gives $x_1 = 2 - 2\lambda - \mu$, $x_3 = -1 - \mu - 2\nu$, $x_5 = 1 + \nu$:

$$x = \begin{pmatrix} 2 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \nu \begin{pmatrix} 0 \\ 0 \\ -2 \\ 0 \\ 1 \\ 1 \end{pmatrix} \qquad (\lambda, \mu, \nu \in \mathbb{R}),$$

the particular solution $(2, 0, -1, 0, 1, 0)'$ plus the solutions of $Ax = 0$ (Theorem 2.23(a)). Check: $A(2, 0, -1, 0, 1, 0)' = 2\,\mathrm{col}_1(A) - \mathrm{col}_3(A) + \mathrm{col}_5(A) = (2-1+1,\ 4-3+3,\ 2-2+3,\ 6-4+3)' = (2, 4, 3, 5)'$.

Exercise 2.11. For which values of k and h does the system

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 1 \\ x_1 + 2x_2 + 2x_3 + 2x_4 &= 2 \\ x_1 + 2x_2 + 3x_3 + 3x_4 &= 4 \\ x_1 + 2x_2 + 3x_3 + kx_4 &= h \end{aligned}$$

have no solution, exactly one solution, or infinitely many solutions? Give the solutions when they exist.

Solution. Forward elimination; each step subtracts a row from all rows below it:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 & 4 \\ 1 & 2 & 3 & k & h \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 1 & 2 & k-1 & h-1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & k-2 & h-2 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & k-3 & h-4 \end{array}\right)$$

- $k \neq 3$: $R_4 \leftarrow \frac{1}{k-3} R_4$ gives an echelon form with pivots in columns $1, 2, 3, 4$ and none in the last column. So there is exactly one solution (Theorems 2.22 and 2.23(b)). Back substitution with $s = \frac{h-4}{k-3}$: $x_4 = s$, $x_3 = 2 - s$, $x_2 = 1 - x_3 - x_4 = -1$, $x_1 = 1 - x_2 - x_3 - x_4 = 0$, that is, $x = (0, -1, 2, 0)' + s\,(0, 0, -1, 1)'$.
- $k = 3$, $h \neq 4$: row 4 reads $0 = h - 4 \neq 0$, so there is no solution.
- $k = 3$, $h = 4$: row 4 is 0, and the first three rows form an echelon form with pivots in columns $1, 2, 3$; $x_4 = t$ is free, so there are infinitely many solutions: $x = (0, -1, 2, 0)' + t\,(0, 0, -1, 1)'$ $(t \in \mathbb{R})$.

Check: $(0, -1, 2, 0)'$ satisfies the first three equations ($0-1+2+0 = 1$, $0-2+4+0 = 2$, $0-2+6+0 = 4$), and $(0, 0, -1, 1)'$ satisfies their homogeneous versions ($-1 + 1 = 0$, $-2 + 2 = 0$, $-3 + 3 = 0$).

Problems

Problem 2.1. Show that the product of two elementary matrices is, in general, not elementary.

Solution. For example,

$$P_{13} D_2(7) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 7 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

This is not elementary: a matrix $D_r(c)$ or $T_{rs}(c)$ differs from I_3 in one entry only, while this product differs from I_3 in five entries; and every P_{rs} has all entries 0 or 1, while this product has the entry 7.

Problem 2.2. Let $A \in \mathbb{R}^{n \times n}$ be invertible. Show that:

(a) $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}$ $(\lambda \neq 0)$;

(b) if A and B commute and B is invertible, then A and B^{-1} commute.

Solution. (a) We check both products (Definition 2.12): $(\lambda A)(\frac{1}{\lambda} A^{-1}) = AA^{-1} = I$ and $(\frac{1}{\lambda} A^{-1})(\lambda A) = A^{-1}A = I$. (b) Multiply $AB = BA$ on the left and on the right by B^{-1}: $B^{-1}ABB^{-1} = B^{-1}BAB^{-1}$, that is, $B^{-1}A = AB^{-1}$.

Problem 2.3. Let $A \in \mathbb{R}^{n\times n}$.

(a) If $A^2 = 0$, show that $I - A$ is invertible.

(b) If $A^3 = 0$, show that $I - A$ is invertible.

(c) In general, if $A^k = 0$ for some positive integer k, show that $I - A$ is invertible.

(d) Suppose that $A^2 + 2A + I = 0$. Show that A is invertible.

(e) Suppose that $A^3 - A + I = 0$. Show that A is invertible.

Solution. (a)–(c) For every positive integer k, by Theorem 1.3,

$$(I - A)(I + A + A^2 + \cdots + A^{k-1}) = (I + A + \cdots + A^{k-1}) - (A + A^2 + \cdots + A^k) = I - A^k,$$

and in the same way $(I + A + \cdots + A^{k-1})(I - A) = I - A^k$. If $A^k = 0$, both products equal I, so $I - A$ is invertible with inverse $I + A + \cdots + A^{k-1}$. Parts (a) and (b) are the cases $k = 2$ and $k = 3$: $(I - A)^{-1} = I + A$ and $I + A + A^2$, respectively.

(d) $A(-A - 2I) = (-A - 2I)A = -A^2 - 2A = I$, since $A^2 + 2A + I = 0$. So $A^{-1} = -A - 2I$.

(e) $A(-A^2 + I) = (-A^2 + I)A = -A^3 + A = I$, since $A^3 - A + I = 0$. So $A^{-1} = I - A^2$.

Problem 2.4. Let $A, B, C \in \mathbb{R}^{m\times n}$.

(a) Show that B is row-equivalent to A if and only if $B = PA$ for some invertible $P \in \mathbb{R}^{m\times m}$.

(b) Show that A is row-equivalent to A; that if B is row-equivalent to A, then A is row-equivalent to B; and that if B is row-equivalent to A and C to B, then C is row-equivalent to A.

(c) Show that if B is row-equivalent to A, then $Ax = 0$ and $Bx = 0$ have the same solutions.

(d) Show that any two invertible matrices in $\mathbb{R}^{n\times n}$ are row-equivalent.

Solution.

(a) If row operations $\omega_1, \ldots, \omega_k$ take A to B, then $B = PA$ with $P = E_k \cdots E_1$ invertible (Theorem 2.17(c)). Conversely, let P be invertible. By Theorem 2.18 ((i) $\Rightarrow$ (v)), $P = F_1F_2 \cdots F_l$ with elementary matrices $F_j = \omega_j(I_m)$. By Theorem 2.17(a), $B = F_1(F_2(\cdots(F_lA)))$ is obtained from A by the operations $\omega_l, \ldots, \omega_2, \omega_1$, in this order.

(b) By (a): $A = I_mA$; if $B = PA$ with P invertible, then $A = P^{-1}B$ with P^{-1} invertible (Theorem 2.15(a)); if $B = PA$ and $C = QB$ with P, Q invertible, then $C = (QP)A$ with QP invertible (Theorem 2.15(b)).

(c) Write $B = PA$ as in (a). If $Ax = 0$, then $Bx = P(Ax) = 0$; if $Bx = 0$, then $Ax = P^{-1}(Bx) = 0$.

(d) If $A, B \in \mathbb{R}^{n\times n}$ are invertible, then $B = (BA^{-1})A$, and BA^{-1} is invertible (Theorem 2.15(a),(b)); apply (a).

Problem 2.5. Let $A \in \mathbb{R}^{m\times n}$.

(a) If $m < n$, show that for every $b \in \mathbb{R}^m$ the system $Ax = b$ has no solution or infinitely many solutions.

(b) If $m > n$, show that $Ax = b$ has no solution for some $b \in \mathbb{R}^m$.

(c) If $m > n$, show that there is no $C \in \mathbb{R}^{n\times m}$ with $AC = I_m$.

Solution.

(a) By Theorem 2.11(a) there is $z \neq 0$ with $Az = 0$. If x_p is a solution of $Ax = b$, then so is $x_p + tz$ for every $t \in \mathbb{R}$ (Theorem 2.23(a)), and these vectors are different for different t, since $z \neq 0$.

(b) Let $R = PA$ be the RREF of A, with P invertible (Theorem 2.17(c)). The pivots of the nonzero rows of R lie in different columns, so R has at most $n < m$ nonzero rows, and row m of R is zero. Put $b = P^{-1}e_m$. The operations that take A to R take $[A \mid b]$ to $P[A \mid b] = [PA \mid Pb] = [R \mid e_m]$ (Theorems 2.17(a) and 1.4(a)), whose last row is $(0 \ \cdots \ 0 \mid 1)$. By Proposition 2.7(a), $Ax = b$ has no solution.

(c) If $AC = I_m$, then for every $b \in \mathbb{R}^m$ the vector $x = Cb$ satisfies $Ax = (AC)b = b$, against (b).

Problem 2.6. (a) Let $A, B \in \mathbb{R}^{n\times n}$. Show that if AB is invertible, then A and B are invertible.

(b) Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$. Show that $AB = I_2$, that $BA \neq I_3$, and that no $C \in \mathbb{R}^{2\times 3}$ satisfies $BC = I_3$.

Solution.

(a) Let $C = (AB)^{-1}$. Then $A(BC) = (AB)C = I_n$, so A is invertible by Theorem 2.18 (last statement); and $(CA)B = C(AB) = I_n$, so B is invertible by the same statement.

(b) By Definition 1.2,

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2, \qquad BA = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \neq I_3$$

Since $B \in \mathbb{R}^{3\times 2}$ has more rows than columns, no C with $BC = I_3$ exists (Problem 2.5(c)). So (a) fails for matrices that are not square.

Problem 2.7 (Sherman–Morrison formula)**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible, $u, v \in \mathbb{R}^n$, and $\gamma = 1 + v'A^{-1}u$ (a number).

(a) If $\gamma \neq 0$, show that $A + uv'$ is invertible and

$$(A + uv')^{-1} = A^{-1} - \frac{1}{\gamma} A^{-1}uv'A^{-1}$$

(b) If $\gamma = 0$, show that $A + uv'$ is not invertible.

(c) Show that $(I_n + J_n)^{-1} = I_n - \frac{1}{n+1}J_n$, and that $C_n = I_n - \bar{J}_n$ is not invertible.

(d) Let $u = \mathbf{1}_4$ and $v = (1, -1, -1, -1)'$. Show that $S = I_4 + uv' = \begin{pmatrix} 2 & -1 & -1 & -1 \\ 1 & 0 & -1 & -1 \\ 1 & -1 & 0 & -1 \\ 1 & -1 & -1 & 0 \end{pmatrix}$ is its own inverse.

Solution.

(a) Since $v'A^{-1}u = \gamma - 1$ is a number, $u(v'A^{-1}u)v'A^{-1} = (\gamma - 1)\, uv'A^{-1}$. Hence, by Theorem 1.3,

$$\begin{aligned}(A + uv')\Big(A^{-1} - \tfrac{1}{\gamma}A^{-1}uv'A^{-1}\Big) &= I + uv'A^{-1} - \tfrac{1}{\gamma}\, uv'A^{-1} - \tfrac{\gamma-1}{\gamma}\, uv'A^{-1} \\ &= I + \Big(1 - \tfrac{1}{\gamma} - \tfrac{\gamma-1}{\gamma}\Big)uv'A^{-1} = I\end{aligned}$$

By Theorem 2.18 (last statement), $A + uv'$ is invertible with this inverse.

(b) If $\gamma = 0$, then $u \neq 0$ (otherwise $\gamma = 1$), so $x = A^{-1}u \neq 0$ (as $Ax = u$), and

$$(A + uv')x = u + u\,(v'A^{-1}u) = \gamma u = 0$$

So $Ax = 0$ has a nontrivial solution for the matrix $A + uv'$, which is therefore not invertible (Theorem 2.18, (i) $\Leftrightarrow$ (iii)).

(c) Take $A = I_n$ and $u = v = \mathbf{1}_n$: $uv' = J_n$ and $\gamma = 1 + \mathbf{1}'\mathbf{1} = n + 1$ (Proposition 1.13), so (a) gives $(I_n + J_n)^{-1} = I_n - \frac{1}{n+1}J_n$. Take $A = I_n$, $u = -\frac{1}{n}\mathbf{1}_n$, $v = \mathbf{1}_n$: $uv' = -\bar{J}_n$ and $\gamma = 1 - \frac{1}{n}\mathbf{1}'\mathbf{1} = 0$, so C_n is not invertible by (b) (and indeed $C_n\mathbf{1} = 0$, Proposition 1.13).

(d) Row i of uv' is $u_i v' = v'$ (Theorem 1.4(c)), which gives the displayed S. Here $v'u = 1-1-1-1 = -2$, so $\gamma = -1$, and (a) with $A = I_4$ gives $S^{-1} = I_4 - \frac{1}{-1}uv' = I_4 + uv' = S$.

Problem 2.8. Let $U = (u_{ij}) \in \mathbb{R}^{n\times n}$ be upper triangular.

(a) If $u_{ii} \neq 0$ for every i, show that U is invertible and that U^{-1} is upper triangular with diagonal entries $u_{11}^{-1}, \ldots, u_{nn}^{-1}$.

(b) If $u_{jj} = 0$ for some j, show that U is not invertible.

(c) If U has integer entries and $u_{ii} \in \{1, -1\}$ for every i, show that U^{-1} has integer entries.

(d) Show that (a)–(c) also hold for lower triangular matrices.

Solution.

(a) Reduce U to I_n by row operations.

- First $R_i \leftarrow u_{ii}^{-1}R_i$ for $i = 1, \ldots, n$. The result U_1 is upper triangular with all diagonal entries 1.
- Then clear the columns $s = n, n-1, \ldots, 2$ in turn (Step 5 of Definition 2.5): for each $r < s$, $R_r \leftarrow R_r - cR_s$, where c is the current (r, s) entry. When column s is treated, row s is e_s' (its entries left of column s are 0 by triangularity, those right of it were cleared before), so the operation changes only the (r, s) entry, to 0. At the end the matrix is I_n.
- The elementary matrices used are $D_i(u_{ii}^{-1})$, diagonal, and $T_{rs}(-c)$ with $r < s$, upper triangular with diagonal entries 1. Their product E satisfies $EU = I_n$ (Theorem 2.17(c)), so U is invertible and $U^{-1} = E$ (Theorem 2.18, last statement).
- A product of upper triangular matrices is upper triangular, and its diagonal entries are the products of the diagonal entries of the factors (Chapter 1, diagonal and triangular matrices). Only $D_i(u_{ii}^{-1})$ has a diagonal entry $\neq 1$ in position i, so $(U^{-1})_{ii} = u_{ii}^{-1}$.

(b) Suppose $u_{jj} = 0$. If $j = 1$, the first column of U is 0, so $Ue_1 = 0$. If $j \geqslant 2$, look for $x = (x_1, \ldots, x_j, 0, \ldots, 0)'$. For $i \geqslant j$ and $k \leqslant j$, $u_{ik} = 0$ (either $k < i$, or $k = i = j$), so rows $j, \ldots, n$ of Ux are 0 for every such x. Rows $1, \ldots, j-1$ of $Ux = 0$ are $j - 1$ homogeneous equations in the j unknowns $x_1, \ldots, x_j$, which have a nontrivial solution (Theorem 2.11(a)). In both cases $Ux = 0$ for some $x \neq 0$, and U is not invertible (Theorem 2.18, (i) $\Leftrightarrow$ (iii)).

(c) In (a), $u_{ii}^{-1} = u_{ii} \in \{1, -1\}$, so the first operations keep the entries integers; each clearing operation adds an integer multiple $-c$ of one row to another (c is an entry of the current matrix, an integer). So every elementary matrix used has integer entries, and so has their product $E = U^{-1}$.

(d) If L is lower triangular, L' is upper triangular with the same diagonal. If all $l_{ii} \neq 0$, L' is invertible by (a), so $L = (L')'$ is invertible with $L^{-1} = ((L')^{-1})'$ (Theorem 2.15(c)), the transpose of an upper triangular matrix with diagonal entries l_{ii}^{-1}, and with integer entries in the situation of (c). If some $l_{jj} = 0$, L' is not invertible by (b), so neither is L (otherwise L' would be, by Theorem 2.15(c)).

Problem 2.9 (LU factorization)**.** Let $A \in \mathbb{R}^{n\times n}$. Suppose that forward elimination of A uses neither interchanges nor scalings: for $s = 1, 2, \dots, n-1$ in turn, it applies $R_r \leftarrow R_r - m_{rs}R_s$ for $r = s+1, \dots, n$ (numbers m_{rs}, possibly 0), and it ends with an upper triangular matrix U. Let L be the lower triangular matrix with diagonal entries 1 and entries $l_{rs} = m_{rs}$ for $r > s$, that is, $L = I_n + \sum_{r>s} m_{rs} e_r e_s'$.

(a) Show that $A = LU$, and that $E[A \mid I_n] = [U \mid L^{-1}]$, where E is the product of the elementary matrices used.

(b) Find L and U for the matrix B of Exercise 2.7.

Solution.

(a) List the operations in the order used as $R_{r_j} \leftarrow R_{r_j} - m_j R_{s_j}$ $(j = 1, \dots, k)$, so that $s_1 \leqslant s_2 \leqslant \cdots \leqslant s_k$ and $r_j > s_j$. Their elementary matrices are $E_j = T_{r_j s_j}(-m_j)$, and $E_k \cdots E_1 A = U$ (Theorem 2.17(c)). By Theorems 2.17(b) and 2.15(b),

$$A = E_1^{-1}E_2^{-1}\cdots E_k^{-1}U, \qquad E_j^{-1} = T_{r_j s_j}(m_j) = I + m_j e_{r_j} e_{s_j}'$$

We show by induction on j that $E_1^{-1}\cdots E_j^{-1} = I + \sum_{i\leqslant j} m_i e_{r_i} e_{s_i}'$. For $j = 1$ this is the formula for E_1^{-1}. If it holds for $j-1$, then

$$\Big(I + \sum_{i<j} m_i e_{r_i} e_{s_i}'\Big)\Big(I + m_j e_{r_j} e_{s_j}'\Big) = I + \sum_{i\leqslant j} m_i e_{r_i} e_{s_i}' + \sum_{i<j} m_i m_j\, e_{r_i}(e_{s_i}' e_{r_j}) e_{s_j}'$$

and $e_{s_i}' e_{r_j} = 0$, because $r_j > s_j \geqslant s_i$. For $j = k$ the sum contains each pair $r > s$ exactly once, so $E_1^{-1}\cdots E_k^{-1} = L$ and $A = LU$. Moreover $E = E_k \cdots E_1 = (E_1^{-1}\cdots E_k^{-1})^{-1} = L^{-1}$ (Theorem 2.15(a),(b)), and $E[A \mid I] = [EA \mid E] = [U \mid L^{-1}]$ (Theorem 1.4(a)).

(b) In Exercise 2.7 the forward elimination of B is $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_2$, $R_4 \leftarrow R_4 - R_3$, so $m_{21} = m_{32} = m_{43} = 1$ and all other $m_{rs} = 0$ (the operations with multiplier 0 change nothing). Hence

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

so $B = LU = U'U$. Check: $\mathrm{row}_3(L)U = \mathrm{row}_2(U) + \mathrm{row}_3(U) = (0, 1, 2, 1) = \mathrm{row}_3(B)$, and similarly for the other rows.

Problem 2.10. Let P be the Pascal matrix of Example 2.20, and $L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{pmatrix}$.

(a) Use Problem 2.9 and the forward elimination in Example 2.20 to show that $P = LL'$, and write down L^{-1}.

(b) Deduce that $P^{-1} = (L^{-1})'L^{-1}$, and use this to recompute the $(2,2)$ entry of P^{-1}.

(c) Check $P = LL'$ directly.

Solution.

(a) The forward elimination there is of the form in Problem 2.9, with $m_{21} = m_{31} = m_{41} = 1$, $m_{32} = 2$, $m_{42} = 3$, $m_{43} = 3$: these are the entries of L below the diagonal. It ends with the left half

$U = \left(\begin{smallmatrix}1&1&1&1\\0&1&2&3\\0&0&1&3\\0&0&0&1\end{smallmatrix}\right) = L'$. So $P = LU = LL'$, and by Problem 2.9(a) the right half after forward elimination is

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{pmatrix}$$

(b) By Theorem 2.15(b),(c), $P^{-1} = (LL')^{-1} = (L')^{-1}L^{-1} = (L^{-1})'L^{-1}$. Its $(2,2)$ entry is $\operatorname{col}_2(L^{-1})'\operatorname{col}_2(L^{-1}) = 0^2 + 1^2 + (-2)^2 + 3^2 = 14$, as found in Example 2.20.

(c) $(LL')_{ij} = \operatorname{row}_i(L)\operatorname{row}_j(L)'$ (Theorem 1.4(d)). With $\ell_1 = (1,0,0,0)$, $\ell_2 = (1,1,0,0)$, $\ell_3 = (1,2,1,0)$, $\ell_4 = (1,3,3,1)$: $\ell_1\ell_j' = 1$ for all j; $\ell_2\ell_2' = 2$, $\ell_2\ell_3' = 3$, $\ell_2\ell_4' = 4$; $\ell_3\ell_3' = 1+4+1 = 6$, $\ell_3\ell_4' = 1+6+3 = 10$; $\ell_4\ell_4' = 1+9+9+1 = 20$. These are the entries of the symmetric matrix P.

Chapter 3. Vector spaces, subspaces, span, and bases

Vector spaces

Definition 3.1. A (real) **vector space** is a set V, whose elements are called **vectors**, with an **addition** that gives $u+v \in V$ for $u, v \in V$ and a **scalar multiplication** that gives $cv \in V$ for $c \in \mathbb{R}$ and $v \in V$, such that for all $u, v, w \in V$ and $c, d \in \mathbb{R}$:

(V1) $u+v = v+u$ and (V2) $u+(v+w) = (u+v)+w$;

(V3) there is a **zero vector** $0 \in V$ with $v+0 = v$ for every $v \in V$;

(V4) for each $v \in V$ there is $-v \in V$ with $v+(-v) = 0$;

(V5) $1v = v$ and (V6) $(cd)v = c(dv)$;

(V7) $c(u+v) = cu+cv$ and (V8) $(c+d)v = cv+dv$.

A vector $c_1v_1 + \cdots + c_kv_k$ ($c_i \in \mathbb{R}$) is a **linear combination** of $v_1, \ldots, v_k$.

From these rules, $0v = 0$ and $(-1)v = -v$: indeed $0v = (0+0)v = 0v+0v$, and adding $-(0v)$ to both sides gives $0 = 0v$; then $v+(-1)v = (1+(-1))v = 0v = 0$, and $-v$ is the only vector w with $v+w = 0$, since such a w equals $w+(v+(-v)) = (w+v)+(-v) = 0+(-v) = -v$. Likewise $c0 = 0$, and $cv = 0$ only if $c = 0$ or $v = 0$ (if $c \neq 0$, then $v = c^{-1}(cv) = c^{-1}0 = 0$).

Example 3.2. $\mathbb{R}^n$, $\mathbb{R}^{1\times n}$ and $\mathbb{R}^{m\times n}$, with the entrywise sum and scalar multiple of Definition 1.1, are vector spaces: (V1)–(V8) hold entry by entry, the zero vector is the zero matrix, and $-A = (-1)A$. In this course every vector space is one of these or a subspace of one of them.

Subspaces

Definition 3.3. A **subspace** of a vector space V is a subset $W \subseteq V$ that is itself a vector space with the addition and scalar multiplication of V.

Theorem 3.4. For a subset W of a vector space V the following are equivalent:

(a) W is a subspace of V;

(b) $0 \in W$, and $u+v \in W$ and $cu \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$ (W is **closed under addition and scalar multiplication**);

(c) $0 \in W$, and $cu+v \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$.

Proof. (a) $\Rightarrow$ (b): the operations of the vector space W are those of V and give vectors of W; W has a zero vector, so it contains some w, and then $0 = 0w \in W$. (b) $\Rightarrow$ (c): $cu \in W$, so $cu+v \in W$. (c) $\Rightarrow$ (a): taking $v = 0$ gives $cu \in W$, and $c = 1$ gives $u+v \in W$; so the operations of V give operations on W. The rules (V1), (V2), (V5)–(V8) hold in W because they hold in V; (V3) holds since $0 \in W$, and (V4) since $-w = (-1)w \in W$. □

So, to show that W is a subspace, check that it contains 0 and is closed under the two operations; to show that it is not, exhibit one failure.

Example 3.5. (a) $\{0\}$ and V itself are subspaces of V.

(b) In $\mathbb{R}^n$, $W = \{x : x_1 = 0\}$ is a subspace: $0 \in W$, and $(cx+y)_1 = cx_1+y_1 = 0$ for $x, y \in W$. For $n \geqslant 2$, $\{x : x_1 = 1+x_2\}$ is not, since it does not contain 0.

(c) In $\mathbb{R}^2$, $\{(x,y)' : 2x-5y = 0\}$ is a subspace: it contains 0, and $2(cx_1+x_2)-5(cy_1+y_2) = c(2x_1-5y_1)+(2x_2-5y_2)$ for $(x_1,y_1)'$ and $(x_2,y_2)'$ in it; but for $k \neq 0$ the set $\{(x,y)' : ax+by = k\}$ is not, since it does not contain 0.

(d) The symmetric $n \times n$ matrices form a subspace of $\mathbb{R}^{n\times n}$: $0' = 0$, and $(cA+B)' = cA'+B' = cA+B$ for symmetric A, B. So do the **skew-symmetric** matrices, those with $A' = -A$: $0' = -0$, and $(cA + B)' = cA' + B' = -(cA + B)$ if $A' = -A$ and $B' = -B$.

(e) For $A \in \mathbb{R}^{m\times n}$, the solution set $\{x \in \mathbb{R}^n : Ax = 0\}$ of a homogeneous system is a subspace of $\mathbb{R}^n$: $A0 = 0$, and $A(cu + v) = cAu + Av = 0$ if $Au = Av = 0$. For $b \neq 0$, the solutions of $Ax = b$ do not form a subspace ($A0 = 0 \neq b$).

(f) In statistics, the **centered** vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$, whose entries add up to 0, form a subspace: this is (e) with the $1 \times n$ matrix $A = \mathbf{1}'$. For every x, the deviation vector Cx of Example 1.14 lies in W_0, since $\mathbf{1}'Cx = \mathbf{1}'C'x = (C\mathbf{1})'x = 0$ (Proposition 1.13: $C' = C$, $C\mathbf{1} = 0$).

(g) The set $\{(x, y, z)' \in \mathbb{R}^3 : xy = 0\}$ is not a subspace: $(1, 0, 0)'$ and $(0, 1, 0)'$ lie in it, but their sum $(1, 1, 0)'$ does not. It is the union of the subspaces $\{x = 0\}$ and $\{y = 0\}$; a union of subspaces need not be a subspace.

Span

Definition 3.6. Let $v_1, \dots, v_k$ be vectors of a vector space V. Their **span** is the set of all their linear combinations,

$$\operatorname{span}\{v_1, \dots, v_k\} = \{c_1v_1 + \cdots + c_kv_k : \ c_1, \dots, c_k \in \mathbb{R}\},$$

and $\operatorname{span} \varnothing = \{0\}$. If $\operatorname{span}\{v_1, \dots, v_k\} = W$, we say that $v_1, \dots, v_k$ **span** W.

Theorem 3.7. $\operatorname{span}\{v_1, \dots, v_k\}$ is a subspace of V that contains $v_1, \dots, v_k$, and it is contained in every subspace that contains $v_1, \dots, v_k$: it is the smallest subspace containing them.

Proof. $0 = 0v_1 + \cdots + 0v_k$ lies in the span, and $c\sum_i a_iv_i + \sum_i b_iv_i = \sum_i(ca_i + b_i)v_i$; so the span is a subspace (Theorem 3.4(c)). It contains $v_j = 1v_j + \sum_{i\neq j} 0v_i$. A subspace that contains $v_1, \dots, v_k$ contains every c_iv_i and every sum of these, by closure. □

For vectors $v_1, \dots, v_k \in \mathbb{R}^m$, put $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{m\times k}$. Since $Vc = c_1v_1 + \cdots + c_kv_k$ (Theorem 1.4(b)),

$$\operatorname{span}\{v_1, \dots, v_k\} = \{Vc : \ c \in \mathbb{R}^k\}, \qquad b \in \operatorname{span}\{v_1, \dots, v_k\} \iff \text{the system } Vc = b \text{ is consistent}$$

So membership in a span is decided by row-reducing $[V \mid b]$ (Proposition 2.7, Theorem 2.22).

Example 3.8. Let $W = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^5$, where $v_1 = (1, 2, 0, 3, 0)'$, $v_2 = (0, 0, 1, 4, 0)'$ and $v_3 = (0, 0, 0, 0, 1)'$. Which $w = (w_1, \dots, w_5)'$ lie in W?

Solution. Row-reduce $[v_1 \ v_2 \ v_3 \mid w]$, keeping the w_i as letters:

$$\left(\begin{array}{ccc|c} 1&0&0&w_1\\ 2&0&0&w_2\\ 0&1&0&w_3\\ 3&4&0&w_4\\ 0&0&1&w_5 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-3R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{ccc|c} 1&0&0&w_1\\ 0&0&0&w_2-2w_1\\ 0&1&0&w_3\\ 0&4&0&w_4-3w_1\\ 0&0&1&w_5 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-4R_3} \left(\begin{array}{ccc|c} 1&0&0&w_1\\ 0&0&0&w_2-2w_1\\ 0&1&0&w_3\\ 0&0&0&w_4-3w_1-4w_3\\ 0&0&1&w_5 \end{array}\right)$$

$$\xrightarrow[R_3\leftrightarrow R_5]{R_2\leftrightarrow R_3} \left(\begin{array}{ccc|c} 1&0&0&w_1\\ 0&1&0&w_3\\ 0&0&1&w_5\\ 0&0&0&w_4-3w_1-4w_3\\ 0&0&0&w_2-2w_1 \end{array}\right)$$

By Proposition 2.7, $w \in W$ if and only if

$$w_2 = 2w_1, \qquad w_4 = 3w_1 + 4w_3,$$

and then $w = w_1v_1 + w_3v_2 + w_5v_3$. For example, $(-3, -6, 1, -5, 2)' \in W$, since $-6 = 2(-3)$ and $-5 = 3(-3) + 4 \cdot 1$; indeed it equals $-3v_1 + v_2 + 2v_3$. But $(2, 4, 6, 7, 8)' \notin W$, since $3 \cdot 2 + 4 \cdot 6 = 30 \neq 7$.

Example 3.9. $\operatorname{span}\{\mathbf{1}_n\} = \{c\mathbf{1}_n : c \in \mathbb{R}\}$ is the subspace of constant vectors. Every $x \in \mathbb{R}^n$ is the sum of a constant vector and a centered vector:

$$x = \bar{x}\mathbf{1} + Cx, \qquad \bar{x}\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}, \quad Cx \in W_0,$$

since $Cx = x - \bar{x}\mathbf{1}$ (Example 1.14) and $Cx \in W_0$ (Example 3.5(f)). For $x = (1, 3, 5, 7, 9)'$: $x = 5\mathbf{1} + (-4, -2, 0, 2, 4)'$.

Linear independence

Definition 3.10. Vectors $v_1, \ldots, v_k$ of a vector space are **linearly dependent** if $c_1v_1 + \cdots + c_kv_k = 0$ for some scalars $c_1, \ldots, c_k$, not all 0 (a **linear relation**). Otherwise they are **linearly independent**: $c_1v_1 + \cdots + c_kv_k = 0$ only for $c_1 = \cdots = c_k = 0$. The empty list is independent.

Proposition 3.11. Let $v_1, \ldots, v_k \in \mathbb{R}^m$ and $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{m\times k}$. Then $v_1, \ldots, v_k$ are linearly independent if and only if $Vc = 0$ has only the solution $c = 0$, that is, if and only if every column of the RREF of V is a pivot column (no free unknowns). In particular, more than m vectors of $\mathbb{R}^m$ are linearly dependent.

Proof. $Vc = c_1v_1 + \cdots + c_kv_k$, so the linear relations among the v_i are the solutions of $Vc = 0$. This system is consistent ($c = 0$), so by Theorem 2.23(b) $c = 0$ is its only solution exactly when there is no free unknown. If $k > m$, then $Vc = 0$ has a nontrivial solution (Theorem 2.11(a)). □

Proposition 3.12. Let $v_1, \ldots, v_k$ be vectors of a vector space.

(a) If some $v_i = 0$, or $v_i = v_j$ for some $i \neq j$, then $v_1, \ldots, v_k$ are dependent. A single vector v is independent if and only if $v \neq 0$.

(b) A sublist of an independent list is independent.

(c) For $k \geqslant 2$, $v_1, \ldots, v_k$ are dependent if and only if some v_j is a linear combination of the others. In particular, two vectors are dependent if and only if one of them is a multiple of the other.

Proof. (a) $1 \cdot 0 = 0$ and $v_i - v_j = 0$ are linear relations with a coefficient 1; and if $v \neq 0$, then $cv = 0$ forces $c = 0$. (b) A relation among some of the vectors is a relation among all of them, with coefficient 0 for the others. (c) If $v_j = \sum_{i\neq j} x_iv_i$, then $v_j - \sum_{i\neq j} x_iv_i = 0$ is a relation with coefficient 1 on v_j. Conversely, if $\sum_i x_iv_i = 0$ with $x_j \neq 0$, then $v_j = \sum_{i\neq j}(-x_i/x_j)v_i$. □

Example 3.13. Let a_1, a_2, a_3, a_4 be the columns of the matrix A of Example 2.6:

$$a_1 = \begin{pmatrix}1\\1\\2\end{pmatrix}, \quad a_2 = \begin{pmatrix}1\\1\\1\end{pmatrix}, \quad a_3 = \begin{pmatrix}1\\2\\5\end{pmatrix}, \quad a_4 = \begin{pmatrix}-2\\-3\\-4\end{pmatrix}$$

Four vectors of $\mathbb{R}^3$ are dependent (Proposition 3.11). A relation comes from the solutions $x = t(-2, 3, 1, 1)'$ of $Ax = 0$ found there:

$$-2a_1 + 3a_2 + a_3 + a_4 = 0, \qquad \text{so} \qquad a_4 = 2a_1 - 3a_2 - a_3$$

On the other hand a_1, a_2, a_3 are independent: the same row operations take $[a_1 \ a_2 \ a_3]$ to the first three columns of the RREF found there (row operations act on each column separately: Theorems 2.17(c) and 1.4(a)), which form I_3, so every column is a pivot column.

Bases

Definition 3.14. Vectors $v_1, \dots, v_k$ of a vector space V form a **basis** of a subspace W if they lie in W, are linearly independent, and span W.

- The standard unit vectors $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, the **standard basis**: $x = x_1e_1 + \cdots + x_ne_n$ for every $x \in \mathbb{R}^n$, and this is 0 only for $x = 0$.
- Let $E_{ij} \in \mathbb{R}^{m\times n}$ have (i, j) entry 1 and all other entries 0 (two indices; not the elementary matrices $E_1, E_2, \dots$ of Theorem 2.17(c)). The mn matrices E_{ij} form a basis of $\mathbb{R}^{m\times n}$: $A = \sum_i \sum_j a_{ij}E_{ij}$, and $\sum_i \sum_j c_{ij}E_{ij} = (c_{ij})$ is 0 only if every $c_{ij} = 0$.
- The empty list is a basis of $\{0\}$.

Theorem 3.15. If $v_1, \dots, v_k$ form a basis of W, every $w \in W$ can be written as $w = c_1v_1 + \cdots + c_kv_k$ in exactly one way; the numbers $c_1, \dots, c_k$ are the **coordinates** of w in this basis.

Proof. Such an expression exists because the v_i span W. If $\sum_i x_iv_i = \sum_i y_iv_i$, then $\sum_i(x_i - y_i)v_i = 0$, and independence gives $x_i = y_i$ for every i. □

Example 3.16. If $P \in \mathbb{R}^{n\times n}$ is invertible, its columns $p_1, \dots, p_n$ form a basis of $\mathbb{R}^n$, and the coordinates of $y \in \mathbb{R}^n$ are $x = P^{-1}y$. Indeed $Px = \sum_j x_jp_j$; $Px = 0$ only for $x = 0$ (Theorem 2.18), so the columns are independent; and $y = P(P^{-1}y)$, so they span. For example, let

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

Multiplying out gives $PQ = I_4$, so P is invertible and $P^{-1} = Q$ (Theorem 2.18). Hence the columns $p_1 = (1, 1, 0, 0)'$, $p_2 = (1, 2, 1, 0)'$, $p_3 = (0, 1, 2, 1)'$, $p_4 = (0, 0, 1, 2)'$ of P form a basis of $\mathbb{R}^4$. The coordinates of $\mathbf{1}_4$ are $Q\mathbf{1}_4 = (2, -1, 1, 0)'$: $\mathbf{1}_4 = 2p_1 - p_2 + p_3$.

Example 3.17. Let $t = (1, 2, 3, 4)'$ and $W = \operatorname{span}\{\mathbf{1}_4, t\}$, a subspace of $\mathbb{R}^4$. Any two of the vectors $\mathbf{1}_4$, t, $\mathbf{1}_4 + t = (2, 3, 4, 5)'$ form a basis of W.

- Each pair is independent: none of the three vectors is 0 or a multiple of another, since their first two entries are $(1, 1)$, $(1, 2)$, $(2, 3)$, no two of them proportional (Proposition 3.12(c)).
- Each pair spans W: the pair $\mathbf{1}_4, t$ by definition, and for $w = a\mathbf{1}_4 + bt \in W$,

$$w = (a - b)\,\mathbf{1}_4 + b\,(\mathbf{1}_4 + t) = (b - a)\,t + a\,(\mathbf{1}_4 + t)$$

All three vectors together also span W, but they are dependent: $\mathbf{1}_4 + t - (\mathbf{1}_4 + t) = 0$ is a linear relation with coefficients $1, 1, -1$. So they do not form a basis. A subspace has many bases.

Example 3.18. The symmetric 2×2 matrices (for instance 2×2 covariance matrices) form a subspace of $\mathbb{R}^{2\times 2}$ (Example 3.5(d)). The matrices E_{11}, $E_{12} + E_{21}$, E_{22} form a basis of it: they are symmetric, every symmetric matrix is

$$\begin{pmatrix} a & b \\ b & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

and this combination is 0 only if $a = b = d = 0$.

Example 3.19. In $W_0 = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$, the vectors $d_1 = (1, -1, 0, 0)'$, $d_2 = (0, 1, -1, 0)'$ and $d_3 = (0, 0, 1, -1)'$ form a basis. They lie in W_0. They span W_0: for $x \in W_0$, $x_4 = -x_1 - x_2 - x_3$, so

$$x_1d_1 + (x_1 + x_2)d_2 + (x_1 + x_2 + x_3)d_3 = \big(x_1,\ x_2,\ x_3,\ -x_1 - x_2 - x_3\big)' = x$$

They are independent: $c_1d_1 + c_2d_2 + c_3d_3 = (c_1,\ c_2 - c_1,\ c_3 - c_2,\ -c_3)' = 0$ forces $c_1 = 0$, then $c_2 = 0$, then $c_3 = 0$.

Sums and direct sums of subspaces

Definition 3.20. Let U and W be subspaces of a vector space V. Their **intersection** $U \cap W$ is the set of vectors that lie in both U and W, and their **sum** is

$$U + W = \{u + w : \ u \in U, \ w \in W\}$$

The sum is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$, that is, if 0 is the only vector that U and W have in common. Thus $V = U \oplus W$ means that $V = U + W$ and $U \cap W = \{0\}$.

Theorem 3.21. Let U and W be subspaces of a vector space V.

(a) $U \cap W$ and $U + W$ are subspaces of V.

(b) $U + W$ contains U and W, and it is contained in every subspace of V that contains $U \cup W$: it is the smallest subspace of V containing $U \cup W$. Moreover $U + W$ is the set of all linear combinations $c_1x_1 + \cdots + c_kx_k$ of finitely many vectors $x_1, \ldots, x_k \in U \cup W$ (this set is written $\operatorname{span}(U \cup W)$).

(c) If $U = \operatorname{span}\{u_1, \ldots, u_p\}$ and $W = \operatorname{span}\{w_1, \ldots, w_q\}$, then $U + W = \operatorname{span}\{u_1, \ldots, u_p, w_1, \ldots, w_q\}$.

Proof. (a)

- Both sets contain 0: $0 \in U$, $0 \in W$, and $0 = 0 + 0$.
- Let $x, y \in U \cap W$ and $c \in \mathbb{R}$. Then $cx + y \in U$ and $cx + y \in W$, since U and W are subspaces; so $cx + y \in U \cap W$.
- Let $x = u_1 + w_1$ and $y = u_2 + w_2$ with $u_1, u_2 \in U$, $w_1, w_2 \in W$, and let $c \in \mathbb{R}$. Then $cx + y = (cu_1 + u_2) + (cw_1 + w_2)$ with $cu_1 + u_2 \in U$ and $cw_1 + w_2 \in W$; so $cx + y \in U + W$.
- By Theorem 3.4(c), $U \cap W$ and $U + W$ are subspaces.

(b)

- Every $u \in U$ is $u = u + 0$, and every $w \in W$ is $w = 0 + w$; so U and W lie in $U + W$.
- If a subspace X of V contains $U \cup W$, then it contains u and w for all $u \in U$, $w \in W$, hence also $u + w$; so $U + W \subseteq X$.
- In a combination $c_1x_1 + \cdots + c_kx_k$ with all $x_i \in U \cup W$, the terms with $x_i \in U$ add up to a vector $u \in U$, and the other terms, whose x_i lie in W, add up to a vector $w \in W$ (an empty sum is 0). So the combination is $u + w \in U + W$. Conversely, $u + w = 1u + 1w$ is such a combination.

(c)

- If $u = \sum_i a_iu_i \in U$ and $w = \sum_j b_jw_j \in W$, then $u + w = \sum_i a_iu_i + \sum_j b_jw_j$ is a combination of $u_1, \ldots, u_p, w_1, \ldots, w_q$.
- Conversely, $\sum_i a_iu_i + \sum_j b_jw_j = u + w$ with $u = \sum_i a_iu_i \in U$ and $w = \sum_j b_jw_j \in W$. □

For matrices: let $A \in \mathbb{R}^{m\times p}$ have columns $a_1, \ldots, a_p$, let $B \in \mathbb{R}^{m\times q}$ have columns $b_1, \ldots, b_q$, and let $[A\ B] \in \mathbb{R}^{m\times(p+q)}$ be the matrix with columns $a_1, \ldots, a_p, b_1, \ldots, b_q$. By (c), the columns of $[A\ B]$ span the sum of the subspaces spanned by the columns of A and by the columns of B,

$$\operatorname{span}\{a_1, \ldots, a_p, b_1, \ldots, b_q\} = \operatorname{span}\{a_1, \ldots, a_p\} + \operatorname{span}\{b_1, \ldots, b_q\}$$

and, by Theorem 1.4(b), the vectors of this subspace are the vectors $Ax + By$ ($x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$).

Theorem 3.22. Let U and W be subspaces of a vector space V. Then $U \cap W = \{0\}$ if and only if every $x \in U + W$ can be written in exactly one way as $x = u + w$ with $u \in U$ and $w \in W$, that is: if $u + w = u_1 + w_1$ with $u, u_1 \in U$ and $w, w_1 \in W$, then $u = u_1$ and $w = w_1$.

Proof. • Let $U \cap W = \{0\}$, and let $u + w = u_1 + w_1$ as above. Then $u - u_1 = w_1 - w$. The left side lies in U and the right side in W, so this vector lies in $U \cap W = \{0\}$. Hence $u = u_1$ and $w = w_1$.

- Let $U \cap W \neq \{0\}$, and take $x \neq 0$ in $U \cap W$. Then $x = x + 0$ (with $x \in U$, $0 \in W$) and $x = 0 + x$ (with $0 \in U$, $x \in W$) are two different ways of writing $x \in U + W$. □

So, if $V = U \oplus W$, each $v \in V$ is $v = u + w$ for exactly one $u \in U$ and one $w \in W$ (the parts of v in U and in W).

Theorem 3.23. Let U and W be subspaces of a vector space V, let $u_1, \ldots, u_p$ form a basis of U, and let $w_1, \ldots, w_q$ form a basis of W. If $U \cap W = \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ form a basis of $U \oplus W$. If $U \cap W \neq \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly dependent. Hence the sum $U + W$ is direct if and only if $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly independent.

Proof. • The vectors $u_1, \ldots, u_p, w_1, \ldots, w_q$ lie in $U + W$ and span it, by Theorem 3.21(b),(c).

- Let $U \cap W = \{0\}$ and $\sum_i a_i u_i + \sum_j b_j w_j = 0$. Then $x = \sum_i a_i u_i = \sum_j (-b_j) w_j$ lies in U and in W, so $x = 0$. Hence $\sum_i a_i u_i = 0$ and $\sum_j b_j w_j = 0$, and since each list is independent, every a_i and every b_j is 0. So the vectors form a basis of $U + W = U \oplus W$.
- Let $U \cap W \neq \{0\}$, take $x \neq 0$ in $U \cap W$, and write $x = \sum_i a_i u_i = \sum_j b_j w_j$. Then $\sum_i a_i u_i + \sum_j (-b_j) w_j = 0$, and not every a_i is 0, since $x \neq 0$: a linear relation among $u_1, \ldots, u_p, w_1, \ldots, w_q$. □

Example 3.24. In $\mathbb{R}^4$ let $U = \{x : x_4 = 0\} = \operatorname{span}\{e_1, e_2, e_3\}$ and $W_0 = \{x : x_1 + x_2 + x_3 + x_4 = 0\} = \operatorname{span}\{d_1, d_2, d_3\}$, with $d_1 = (1, -1, 0, 0)'$, $d_2 = (0, 1, -1, 0)'$ and $d_3 = (0, 0, 1, -1)'$ (Example 3.19).

(a) $U + W_0 = \mathbb{R}^4$, but the sum is not direct.

- Every $x \in \mathbb{R}^4$ is $x = (x_1,\ x_2,\ x_3 + x_4,\ 0)' + (0,\ 0,\ -x_4,\ x_4)'$, with the first vector in U and the second, $-x_4 d_3$, in W_0. So $U + W_0 = \mathbb{R}^4$.
- $x \in U \cap W_0$ means $x_4 = 0$ and $x_1 + x_2 + x_3 = 0$, that is, $x = (a,\ b,\ -a-b,\ 0)' = a\, d_1 + (a+b)\, d_2$ for some $a, b \in \mathbb{R}$. So $U \cap W_0 = \operatorname{span}\{d_1, d_2\} \neq \{0\}$, and the sum is not direct.
- Accordingly (Theorem 3.22) the parts are not unique: for example

$$(1, 2, 3, 6)' = (1, 2, 9, 0)' + (0, 0, -6, 6)' = (7, 2, 3, 0)' + (-6, 0, 0, 6)',$$

  and the two parts in U differ by $(-6, 0, 6, 0)' = -6(d_1 + d_2) \in U \cap W_0$. Likewise $e_1, e_2, e_3, d_1, d_2, d_3$ are dependent, since $d_1 = e_1 - e_2$ (Theorem 3.23).

(b) $\mathbb{R}^4 = W_0 \oplus \operatorname{span}\{e_4\}$.

- $ce_4 = (0, 0, 0, c)'$ lies in W_0 only for $c = 0$; so $W_0 \cap \operatorname{span}\{e_4\} = \{0\}$.
- For $x \in \mathbb{R}^4$ and $c \in \mathbb{R}$, $x - ce_4 = (x_1,\ x_2,\ x_3,\ x_4 - c)'$ lies in W_0 exactly when $c = x_1 + x_2 + x_3 + x_4$. So every x is $x = (x_1,\ x_2,\ x_3,\ -x_1 - x_2 - x_3)' + (x_1 + x_2 + x_3 + x_4)\, e_4$ with the first vector in W_0, and in no other way. For example, $(1, 2, 3, 6)' = (1, 2, 3, -6)' + 12e_4$.
- By Theorem 3.23, d_1, d_2, d_3, e_4 form a basis of $\mathbb{R}^4$.

Example 3.25. The constant vectors and the centered vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$ (Example 3.5(f)) give a direct sum:

$$\mathbb{R}^n = \operatorname{span}\{\mathbf{1}_n\} \oplus W_0, \qquad x = \bar{x}\mathbf{1} + Cx$$

- Sum: every $x \in \mathbb{R}^n$ is $x = \bar{x}\mathbf{1} + Cx$ with $\bar{x}\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}$ and $Cx \in W_0$ (Example 3.9).
- Intersection: if $c\mathbf{1} \in W_0$, then $0 = \mathbf{1}'(c\mathbf{1}) = cn$ (Proposition 1.13), so $c = 0$.
- Hence $\bar{x}\mathbf{1}$ is the only constant vector whose difference from x is centered (Theorem 3.22). In matrix form, $x = \bar{J}_n x + C_n x$ with $\bar{J}_n x = \frac{1}{n}\mathbf{1}(\mathbf{1}'x) = \bar{x}\mathbf{1}$, and $\bar{J}_n + C_n = I_n$ (Definition 1.12). For $x = (1, 3, 5, 7, 9)'$ the two parts are $\bar{x}\mathbf{1} = 5\mathbf{1}$ and $Cx = (-4, -2, 0, 2, 4)'$.

- For $n = 4$, $\mathbf{1}_4$ together with the basis d_1, d_2, d_3 of W_0 (Example 3.19) is a basis of $\mathbb{R}^4$ (Theorem 3.23).
- The second subspace matters: also $\mathbb{R}^4 = W_0 \oplus \operatorname{span}\{e_4\}$ (Example 3.24(b)), and the parts of $x = (1, 2, 3, 6)'$ in W_0 differ: here $\bar{x} = 3$ and $Cx = (-2, -1, 0, 3)'$, there $(1, 2, 3, -6)'$.

Exercises

Exercise 3.1. Is the set of vectors $(x, y)' \in \mathbb{R}^2$ with $x \geqslant 0$ and $y \geqslant 0$ a subspace of $\mathbb{R}^2$?

Solution. No: $(1, 1)'$ lies in the set, but $(-1)(1, 1)' = (-1, -1)'$ does not; so the set is not closed under scalar multiplication (Theorem 3.4(b)).

Exercise 3.2. (a) In the vector space $\mathbb{R}^{4\times4}$, show that the set of lower triangular matrices is a subspace.

(b) In $\mathbb{R}^{4\times4}$, show that the matrices with trace 0 form a subspace, and so do the matrices A with $A\mathbf{1}_4 = 0$ (every row sum is 0). Do the invertible 4×4 matrices form a subspace?

Solution. (a) Let $A, B \in \mathbb{R}^{4\times4}$ be lower triangular, that is, $a_{ij} = b_{ij} = 0$ for $i < j$, and let $c \in \mathbb{R}$. For $i < j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0$ and $(cA)_{ij} = ca_{ij} = 0$; so $A + B$ and cA are lower triangular, and so is 0. By Theorem 3.4(b), the lower triangular matrices form a subspace.

(b) Let $A, B \in \mathbb{R}^{4\times4}$ and $c \in \mathbb{R}$; use Theorem 3.4(c).

- Trace: $\operatorname{tr}(0) = 0$, and if $\operatorname{tr}(A) = \operatorname{tr}(B) = 0$, then $\operatorname{tr}(cA+B) = c\operatorname{tr}(A)+\operatorname{tr}(B) = 0$ (Theorem 1.11(a)).
- Row sums: $0\mathbf{1}_4 = 0$, and if $A\mathbf{1}_4 = B\mathbf{1}_4 = 0$, then $(cA + B)\mathbf{1}_4 = cA\mathbf{1}_4 + B\mathbf{1}_4 = 0$ (Theorem 1.3).
- Invertible matrices: no. The zero matrix is not invertible, since $0e_1 = 0$ with $e_1 \neq 0$ (Theorem 2.18); so the set does not contain 0.

Exercise 3.3. Which of the following subsets of $\mathbb{R}^4$ are subspaces? (a) $\{x : x_1 + x_2 = x_3 + x_4\}$; (b) $\{x : x_1x_4 = x_2x_3\}$; (c) $\{x : x_1 + x_2 + x_3 + x_4 = 1\}$; (d) $\{x : x_1 \leqslant x_2\}$; (e) $\{x : x_1 = x_2 = x_3 = x_4\}$.

Solution. (a) Yes: it is the solution set of $Ax = 0$ for the 1×4 matrix $A = (1, 1, -1, -1)$, a subspace by Example 3.5(e).

(b) No: e_1 and e_4 lie in the set ($1 \cdot 0 = 0 \cdot 0$ and $0 \cdot 1 = 0 \cdot 0$), but $e_1 + e_4 = (1, 0, 0, 1)'$ does not, since $1 \cdot 1 \neq 0 \cdot 0$.

(c) No: 0 is not in the set, since $0 + 0 + 0 + 0 \neq 1$.

(d) No: $e_2 = (0, 1, 0, 0)'$ lies in the set, but $(-1)e_2 = (0, -1, 0, 0)'$ does not, since $0 > -1$.

(e) Yes: the set is $\{c\mathbf{1}_4 : c \in \mathbb{R}\} = \operatorname{span}\{\mathbf{1}_4\}$, a subspace by Theorem 3.7.

Exercise 3.4. Let $u_1 = (1, 2, 0, 1)'$, $u_2 = (0, 1, 1, 1)'$, and $v_1 = (1, 3, 1, 2)'$, $v_2 = (1, 4, 2, 3)'$, $v_3 = (3, 5, -1, 2)'$ in $\mathbb{R}^4$. Show that $\operatorname{span}\{v_1, v_2, v_3\} = \operatorname{span}\{u_1, u_2\}$, and find a linear relation among v_1, v_2, v_3.

Solution. Solving $au_1 + bu_2 = v_j$ from entries 1 and 3 (which give a and b directly) and checking the other entries gives the first three identities; the last two follow by solving the first two for u_1 and u_2:

$$v_1 = u_1 + u_2, \qquad v_2 = u_1 + 2u_2, \qquad v_3 = 3u_1 - u_2; \qquad u_1 = 2v_1 - v_2, \qquad u_2 = v_2 - v_1$$

- v_1, v_2, v_3 lie in the subspace $\operatorname{span}\{u_1, u_2\}$, so $\operatorname{span}\{v_1, v_2, v_3\} \subseteq \operatorname{span}\{u_1, u_2\}$ (Theorem 3.7).
- u_1, u_2 lie in the subspace $\operatorname{span}\{v_1, v_2, v_3\}$, so $\operatorname{span}\{u_1, u_2\} \subseteq \operatorname{span}\{v_1, v_2, v_3\}$ (Theorem 3.7).
- Substituting $u_1 = 2v_1-v_2$ and $u_2 = v_2-v_1$ into $v_3 = 3u_1-u_2$ gives $v_3 = 6v_1-3v_2-v_2+v_1 = 7v_1-4v_2$, so
$$7v_1 - 4v_2 - v_3 = 0$$
Check: $(7, 21, 7, 14)' - (4, 16, 8, 12)' - (3, 5, -1, 2)' = 0$.

Exercise 3.5. Let $W = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^4$, where $v_1 = (1,2,1,0)'$, $v_2 = (0,1,1,1)'$ and $v_3 = (1,3,3,2)'$.

(a) Which $w = (w_1, w_2, w_3, w_4)'$ lie in W?

(b) Show that $(3,6,4,1)' \in W$, and write it as a combination of v_1, v_2, v_3.

(c) Does $(1,1,1,1)'$ lie in W?

Solution. (a) With $V = [v_1\ v_2\ v_3]$, $w \in W$ if and only if $Vc = w$ is consistent (the paragraph after Theorem 3.7). Row-reduce $[V \mid w]$, keeping the w_i as letters:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 2 & 1 & 3 & w_2 \\ 1 & 1 & 3 & w_3 \\ 0 & 1 & 2 & w_4 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 1 & 2 & w_3 - w_1 \\ 0 & 1 & 2 & w_4 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 0 & 1 & w_1 - w_2 + w_3 \\ 0 & 0 & 1 & 2w_1 - w_2 + w_4 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 0 & 1 & w_1 - w_2 + w_3 \\ 0 & 0 & 0 & w_1 - w_3 + w_4 \end{array}\right)$$

The last matrix is an echelon form. By Theorem 2.22, $Vc = w$ is consistent if and only if its last row is not $(0\ 0\ 0 \mid d)$ with $d \neq 0$:

$$w \in W \iff w_1 - w_3 + w_4 = 0$$

Then every column of the left part is a pivot column, so the coefficients are unique (Theorem 2.23(b)); back substitution gives $c_3 = w_1 - w_2 + w_3$, $c_2 = (w_2 - 2w_1) - c_3 = -3w_1 + 2w_2 - w_3$ and $c_1 = w_1 - c_3 = w_2 - w_3$.

(b) For $w = (3,6,4,1)'$, $3 - 4 + 1 = 0$, so $w \in W$, with $c_3 = 3 - 6 + 4 = 1$, $c_2 = -9 + 12 - 4 = -1$ and $c_1 = 6 - 4 = 2$: $w = 2v_1 - v_2 + v_3$. Check: $(2,4,2,0)' - (0,1,1,1)' + (1,3,3,2)' = (3,6,4,1)'$.

(c) No: $w_1 - w_3 + w_4 = 1 - 1 + 1 = 1 \neq 0$.

Exercise 3.6. Let $t = (1,2,3,4)'$ and $W = \operatorname{span}\{\mathbf{1}_4, t\} \subseteq \mathbb{R}^4$.

(a) Show that $\mathbf{1}_4, t$ form a basis of W.

(b) Which $y = (y_1, y_2, y_3, y_4)'$ lie in W? Give the coordinates of such y.

(c) Find the coordinates of $(3,5,7,9)'$. Does $(1,2,4,8)'$ lie in W?

Solution. (a) The vectors lie in W and span it. If $a\mathbf{1}_4 + bt = (a+b,\ a+2b,\ a+3b,\ a+4b)' = 0$, the second entry minus the first gives $b = 0$, and then $a = 0$. So they are independent.

(b) Row-reduce $[\mathbf{1}_4\ t \mid y]$, keeping the y_i as letters:

$$\left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 1 & 2 & y_2 \\ 1 & 3 & y_3 \\ 1 & 4 & y_4 \end{array}\right) \xrightarrow[R_i \leftarrow R_i - R_1\ (i=2,3,4)]{} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 2 & y_3 - y_1 \\ 0 & 3 & y_4 - y_1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 0 & y_1 - 2y_2 + y_3 \\ 0 & 0 & 2y_1 - 3y_2 + y_4 \end{array}\right)$$

By Theorem 2.22,

$$y \in W \iff y_1 - 2y_2 + y_3 = 0 \ \text{ and } \ 2y_1 - 3y_2 + y_4 = 0$$

Then back substitution gives the coordinates $b = y_2 - y_1$ and $a = y_1 - b = 2y_1 - y_2$: $y = (2y_1 - y_2)\mathbf{1}_4 + (y_2 - y_1)t$.

(c) For $(3,5,7,9)'$: $3 - 10 + 7 = 0$ and $6 - 15 + 9 = 0$, so it lies in W, with $a = 1$, $b = 2$: $(3,5,7,9)' = \mathbf{1}_4 + 2t$. For $(1,2,4,8)'$: $1 - 4 + 4 = 1 \neq 0$, so it does not lie in W.

Exercise 3.7. Decide whether the following vectors are linearly independent.

(a) $(1,1,0,0)'$, $(0,1,1,0)'$ and $(0,0,1,1)'$ in $\mathbb{R}^4$.

(b) $h_1=(1,1,1,1)'$, $h_2=(1,-1,1,-1)'$, $h_3=(1,1,-1,-1)'$ and $h_4=(1,-1,-1,1)'$ in $\mathbb{R}^4$.

(c) $(1,1,0,0,0)'$, $(0,1,1,0,0)'$, $(0,0,1,1,0)'$, $(0,0,0,1,1)'$ and $(1,0,0,0,1)'$ in $\mathbb{R}^5$.

(d) $c_1=(1,1,0,0)'$, $c_2=(0,1,1,0)'$, $c_3=(0,0,1,1)'$ and $c_4=(1,0,0,1)'$ in $\mathbb{R}^4$.

Solution. (a) Independent: $a(1,1,0,0)'+b(0,1,1,0)'+c(0,0,1,1)'=(a,\ a+b,\ b+c,\ c)'=0$ gives $a=0$, $c=0$, and then $b=0$.

(b) Independent. If $\sum_i a_i h_i=0$, then

$$a_1+a_2+a_3+a_4=0,\quad a_1-a_2+a_3-a_4=0,\quad a_1+a_2-a_3-a_4=0,\quad a_1-a_2-a_3+a_4=0$$

The sum and the difference of the first two equations give $a_1+a_3=0$ and $a_2+a_4=0$; those of the last two give $a_1-a_3=0$ and $a_2-a_4=0$. Hence $a_1=a_2=a_3=a_4=0$.

(c) Independent. A combination with coefficients $a_1,\dots,a_5$ is $(a_1+a_5,\ a_1+a_2,\ a_2+a_3,\ a_3+a_4,\ a_4+a_5)'$. If it is 0, entries 2 to 5 give $a_2=-a_1$, $a_3=a_1$, $a_4=-a_1$, $a_5=a_1$, and then the first entry gives $2a_1=0$. So every $a_i=0$.

(d) Dependent. The same computation gives $a_1c_1+a_2c_2+a_3c_3+a_4c_4=(a_1+a_4,\ a_1+a_2,\ a_2+a_3,\ a_3+a_4)'$, which is 0 for $(a_1,a_2,a_3,a_4)=(1,-1,1,-1)$:

$$c_1-c_2+c_3-c_4=0$$

Exercise 3.8. Let $u_1=(1,2,3,1)'$, $u_2=(0,1,1,1)'$, $u_3=(1,2,3,0)'$ and $u_4=(0,2,2,1)'$.

(a) Are u_1,u_2,u_3 linearly independent?

(b) Are u_1,u_2,u_3,u_4 linearly independent? If not, find a linear relation among them, and write u_4 as a combination of u_1,u_2,u_3.

Solution. Row-reduce $V=[u_1\ u_2\ u_3\ u_4]$:

$$\begin{pmatrix}1&0&1&0\\2&1&2&2\\3&1&3&2\\1&1&0&1\end{pmatrix}\xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3-3R_1}\begin{pmatrix}1&0&1&0\\0&1&0&2\\0&1&0&2\\0&1&-1&1\end{pmatrix}\xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_2}\begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&0&0\\0&0&-1&-1\end{pmatrix}$$

$$\xrightarrow{R_3\leftrightarrow R_4}\begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&-1&-1\\0&0&0&0\end{pmatrix}\xrightarrow{R_3\leftarrow -R_3}\begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&1&1\\0&0&0&0\end{pmatrix}\xrightarrow{R_1\leftarrow R_1-R_3}\begin{pmatrix}1&0&0&-1\\0&1&0&2\\0&0&1&1\\0&0&0&0\end{pmatrix}$$

After the second step the entry of row 3 in column 3 is 0, so Step 2 interchanges rows 3 and 4. The last matrix is the RREF of V, with pivot columns $1,2,3$.

(a) Yes. Row operations act on each column separately (Theorem 2.17(c) and Theorem 1.4(a)), so the same operations take $[u_1\ u_2\ u_3]$ to the first three columns of the last matrix, whose columns are the standard unit vectors e_1,e_2,e_3 of $\mathbb{R}^4$. This is the RREF of $[u_1\ u_2\ u_3]$, and every column is a pivot column; so u_1,u_2,u_3 are independent (Proposition 3.11).

(b) No. Column 4 is not a pivot column, so $Vc=0$ has a free unknown and a solution $c\neq 0$ (Proposition 3.11). The RREF gives $c_1-c_4=0$, $c_2+2c_4=0$, $c_3+c_4=0$; with $c_4=t$, $c=t\,(1,-2,-1,1)'$. For $t=1$,

$$u_1-2u_2-u_3+u_4=0,\qquad\text{so}\qquad u_4=-u_1+2u_2+u_3$$

Check: $-(1,2,3,1)'+2(0,1,1,1)'+(1,2,3,0)'=(0,2,2,1)'=u_4$.

Exercise 3.9. For what values of the scalar k are the vectors

$$v_1 = \begin{pmatrix} k \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ k \\ 1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ k \\ 1 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ k \end{pmatrix}$$

linearly dependent? Give a linear relation in each such case.

Solution. Let $x_1v_1 + x_2v_2 + x_3v_3 + x_4v_4 = 0$ and $s = x_1 + x_2 + x_3 + x_4$. Entry i of the left side is $kx_i + \sum_{j\neq i} x_j = (k-1)x_i + s$, so the condition reads

$$(k-1)x_i + s = 0 \qquad (i = 1, 2, 3, 4)$$

Adding the four equations gives $(k-1)s + 4s = (k+3)s = 0$.

- $k \neq 1$ and $k \neq -3$: then $s = 0$, so $(k-1)x_i = 0$ and $x_i = 0$ for every i. The vectors are independent.
- $k = 1$: the equations read $s = 0$, which $x = (1, -1, 0, 0)'$ satisfies. Indeed $v_1 = v_2 = \mathbf{1}_4$, and $v_1 - v_2 = 0$.
- $k = -3$: the equations read $s = 4x_i$, which $x = \mathbf{1}_4$ satisfies ($s = 4$). Indeed $v_1 + v_2 + v_3 + v_4 = (-3+3)\mathbf{1}_4 = 0$.

So the vectors are linearly dependent exactly for $k = 1$ and $k = -3$.

Exercise 3.10. Show that the columns of the matrix $A = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & -5 & -3 & 1 \\ 5 & -1 & 0 & 2 \end{pmatrix}$ are linearly dependent.

How about the rows?

Solution. The columns satisfy, for example, $3\,\mathrm{col}_1(A) - \mathrm{col}_2(A) - 8\,\mathrm{col}_4(A) = 0$:

$$3\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} - 8\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3-3-0 \\ 3+5-8 \\ 15+1-16 \end{pmatrix} = 0$$

The coefficients $3, -1, 0, -8$ are not all 0, so the columns are dependent; equivalently, $Ax = 0$ for $x = (3, -1, 0, -8)' \neq 0$ (Theorem 1.4(b)). The rows satisfy $3\,\mathrm{row}_1(A) + 2\,\mathrm{row}_2(A) - \mathrm{row}_3(A) = 0$: $(3, 9, 6, 0) + (2, -10, -6, 2) - (5, -1, 0, 2) = (0, 0, 0, 0)$, that is, $y'A = 0$ for $y = (3, 2, -1)'$ (Theorem 1.4(c)). So the rows are dependent too.

Exercise 3.11. Let $W = \{x \in \mathbb{R}^5 : \ x_1 + x_2 + x_3 + x_4 + x_5 = 0, \ x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 0\}$.

(a) Find a basis of W.

(b) Show that $w = (0, 0, 1, -2, 1)' \in W$, and find its coordinates in this basis.

Solution. (a) W is the solution set of $Ax = 0$ with $A = \left(\begin{smallmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{smallmatrix}\right)$. Row-reduce:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix}$$

The free unknowns are x_3, x_4, x_5, and $x_1 = x_3 + 2x_4 + 3x_5$, $x_2 = -2x_3 - 3x_4 - 4x_5$ (Proposition 2.7(b)). So

$$W = \{x_3b_1 + x_4b_2 + x_5b_3 : \ x_3, x_4, x_5 \in \mathbb{R}\}, \qquad b_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \ b_2 = \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \ b_3 = \begin{pmatrix} 3 \\ -4 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

The b_i lie in W and span it. Entries $3,4,5$ of $c_1b_1+c_2b_2+c_3b_3$ are c_1,c_2,c_3, so the combination is 0 only for $c_1=c_2=c_3=0$. Hence b_1,b_2,b_3 form a basis of W.

(b) $0+0+1-2+1=0$ and $0+0+3-8+5=0$, so $w\in W$. By (a) its coordinates are its entries $3,4,5$, namely $(1,-2,1)$, and they are unique (Theorem 3.15): $w=b_1-2b_2+b_3=(1-4+3,\ -2+6-4,\ 1,\ -2,\ 1)'$.

Exercise 3.12. In $\mathbb{R}^4$ let $U=\operatorname{span}\{u_1,u_2\}$ and $W=\operatorname{span}\{w_1,w_2\}$, where

$$u_1=\begin{pmatrix}1\\1\\0\\0\end{pmatrix},\quad u_2=\begin{pmatrix}0\\0\\1\\1\end{pmatrix},\quad w_1=\begin{pmatrix}1\\0\\1\\0\end{pmatrix},\quad w_2=\begin{pmatrix}0\\1\\0\\1\end{pmatrix}$$

(a) Which $x\in\mathbb{R}^4$ lie in $U+W$? Find a basis of $U+W$.

(b) Find a basis of $U\cap W$. Is the sum $U+W$ direct?

(c) Find a vector of $U+W$ that is not in $U\cup W$, and write it in two ways as $u+w$ with $u\in U$, $w\in W$.

Solution. (a) By Theorem 3.21(c), $U+W=\operatorname{span}\{u_1,u_2,w_1,w_2\}$. Row-reduce $[u_1\ u_2\ w_1\ w_2\mid x]$:

$$\left(\begin{array}{cccc|c}1&0&1&0&x_1\\1&0&0&1&x_2\\0&1&1&0&x_3\\0&1&0&1&x_4\end{array}\right)\xrightarrow[R_4\leftarrow R_4-R_3]{R_2\leftarrow R_2-R_1}\left(\begin{array}{cccc|c}1&0&1&0&x_1\\0&0&-1&1&x_2-x_1\\0&1&1&0&x_3\\0&0&-1&1&x_4-x_3\end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3}\left(\begin{array}{cccc|c}1&0&1&0&x_1\\0&1&1&0&x_3\\0&0&-1&1&x_2-x_1\\0&0&-1&1&x_4-x_3\end{array}\right)\xrightarrow{R_4\leftarrow R_4-R_3}\left(\begin{array}{cccc|c}1&0&1&0&x_1\\0&1&1&0&x_3\\0&0&-1&1&x_2-x_1\\0&0&0&0&x_1-x_2-x_3+x_4\end{array}\right)$$

- By Theorem 2.22, $x\in U+W$ if and only if $x_1-x_2-x_3+x_4=0$.
- For $x=0$, the unknown c_4 is free; $c_4=1$ gives $c_3=1$, $c_2=-1$, $c_1=-1$, the relation $-u_1-u_2+w_1+w_2=0$. So $w_2=u_1+u_2-w_1\in\operatorname{span}\{u_1,u_2,w_1\}$, and $\operatorname{span}\{u_1,u_2,w_1\}$, a subspace containing u_1,u_2,w_1,w_2, contains $U+W$ (Theorem 3.7); the reverse inclusion holds since $u_1,u_2,w_1\in U+W$.
- u_1,u_2,w_1 are independent: $au_1+bu_2+cw_1=(a+c,\ a,\ b+c,\ b)'=0$ gives $a=b=0$, then $c=0$. So u_1,u_2,w_1 form a basis of $U+W$.

(b) The vectors of U are $au_1+bu_2=(a,a,b,b)'$, those of W are $cw_1+dw_2=(c,d,c,d)'$. These are equal if and only if $a=c$, $a=d$, $b=c$, $b=d$, that is, $a=b=c=d$. So $U\cap W=\{a\mathbf{1}_4:a\in\mathbb{R}\}=\operatorname{span}\{\mathbf{1}_4\}$, with basis $\mathbf{1}_4$ (Proposition 3.12(a)). Since $U\cap W\neq\{0\}$, the sum is not direct.

(c) Take $x=u_1+w_1=(2,1,1,0)'\in U+W$. It is not in U, whose vectors $(a,a,b,b)'$ have equal first and second entries, and not in W, whose vectors $(c,d,c,d)'$ have equal first and third entries. Adding $\mathbf{1}_4=u_1+u_2=w_1+w_2$ to the part in U and subtracting it from the part in W gives a second decomposition:

$$x=\underbrace{(1,1,0,0)'}_{u_1}+\underbrace{(1,0,1,0)'}_{w_1}=\underbrace{(2,2,1,1)'}_{2u_1+u_2}+\underbrace{(0,-1,0,-1)'}_{-w_2}$$

Exercise 3.13. In $\mathbb{R}^4$ let $U = \operatorname{span}\{\mathbf{1}_4, t\}$ with $t = (1,2,3,4)'$ (Exercise 3.6), and $W = \operatorname{span}\{s_1, s_2\}$ with $s_1 = (1,-2,1,0)'$, $s_2 = (0,1,-2,1)'$. Show that $\mathbb{R}^4 = U \oplus W$, and write $x = (2,1,2,5)'$ as $x = u + w$ with $u \in U$ and $w \in W$.

Solution. Let $M = [\mathbf{1}_4\ t\ s_1\ s_2]$ and row-reduce $[M \mid x]$:

$$\left(\begin{array}{cccc|c} 1&1&1&0&2\\ 1&2&-2&1&1\\ 1&3&1&-2&2\\ 1&4&0&1&5 \end{array}\right) \xrightarrow[R_i \leftarrow R_i - R_1\ (i=2,3,4)]{} \left(\begin{array}{cccc|c} 1&1&1&0&2\\ 0&1&-3&1&-1\\ 0&2&0&-2&0\\ 0&3&-1&1&3 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{cccc|c} 1&1&1&0&2\\ 0&1&-3&1&-1\\ 0&0&6&-4&2\\ 0&0&8&-2&6 \end{array}\right) \xrightarrow[R_4 \leftarrow \frac{1}{2}R_4]{R_3 \leftarrow \frac{1}{2}R_3} \left(\begin{array}{cccc|c} 1&1&1&0&2\\ 0&1&-3&1&-1\\ 0&0&3&-2&1\\ 0&0&4&-1&3 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1&1&1&0&2\\ 0&1&-3&1&-1\\ 0&0&3&-2&1\\ 0&0&1&1&2 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_3]{R_3 \leftrightarrow R_4} \left(\begin{array}{cccc|c} 1&1&1&0&2\\ 0&1&-3&1&-1\\ 0&0&1&1&2\\ 0&0&0&-5&-5 \end{array}\right)$$

- The left part is an echelon form with a pivot in every column, so the RREF of M is I_4, and $\mathbf{1}_4, t, s_1, s_2$ are independent (Proposition 3.11). By Theorem 2.18, M is invertible, and by Example 3.16 its columns form a basis of $\mathbb{R}^4$.
- $U + W = \operatorname{span}\{\mathbf{1}_4, t, s_1, s_2\} = \mathbb{R}^4$ (Theorem 3.21(c)).
- $\mathbf{1}_4, t$ form a basis of U (Exercise 3.6(a)), and s_1, s_2 form a basis of W, being a sublist of the independent list $\mathbf{1}_4, t, s_1, s_2$ (Proposition 3.12(b)). The combined list is independent, so the sum is direct (Theorem 3.23): $\mathbb{R}^4 = U \oplus W$.
- Back substitution gives $c_4 = 1$, $c_3 = 2 - c_4 = 1$, $c_2 = -1 + 3c_3 - c_4 = 1$, $c_1 = 2 - c_2 - c_3 = 0$. So

$$x = \underbrace{(1,2,3,4)'}_{u=t} + \underbrace{(1,-1,-1,1)'}_{w=s_1+s_2},$$

  and by Theorem 3.22 this is the only such decomposition.

Problems

Problem 3.1. Suppose that $v_1, \ldots, v_k$ span a vector space V but do not form a basis of V. Show that, for any $v \in V$, the representation of v as a linear combination of $v_1, \ldots, v_k$ is not unique.

Solution. Let $v \in V$. Since $v_1, \ldots, v_k$ span V, there are scalars $x_1, \ldots, x_k$ with $v = \sum_i x_i v_i$. Since $v_1, \ldots, v_k$ span V but do not form a basis, they are linearly dependent (Definition 3.14): there are scalars $z_1, \ldots, z_k$, not all 0, with $\sum_i z_i v_i = 0$. Let $y_i = x_i + z_i$. Then $\sum_i y_i v_i = \sum_i x_i v_i + \sum_i z_i v_i = v + 0 = v$, and $(y_1, \ldots, y_k) \neq (x_1, \ldots, x_k)$ because some $z_i \neq 0$. So v has two different representations. (Compare Theorem 3.15.)

Problem 3.2. Let U and W be subspaces of a vector space V. Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Solution.

- If $U \subseteq W$, then $U \cup W = W$, a subspace; likewise if $W \subseteq U$.
- Suppose that neither holds: there are $u \in U$ with $u \notin W$ and $w \in W$ with $w \notin U$. If $u + w \in U$, then $w = (u + w) - u \in U$ ✘; if $u + w \in W$, then $u = (u + w) - w \in W$ ✘. So $u + w \notin U \cup W$, although $u, w \in U \cup W$, and $U \cup W$ is not a subspace (Theorem 3.4(b)).

Problem 3.3. Let $v_1, \dots, v_k$ be linearly independent vectors of a vector space V, and let $v \in V$. Show that $v_1, \dots, v_k, v$ are linearly independent if and only if $v \notin \operatorname{span}\{v_1, \dots, v_k\}$.

Solution.

- If $v = \sum_i c_i v_i$, then $\sum_i c_i v_i + (-1)v = 0$ is a linear relation with coefficient -1 on v; so $v_1, \dots, v_k, v$ are dependent.
- If $v_1, \dots, v_k, v$ are dependent, there is a relation $\sum_i c_i v_i + cv = 0$ with not all coefficients 0. If $c = 0$, it is a relation among $v_1, \dots, v_k$ with not all $c_i = 0$ ✘. So $c \neq 0$, and $v = \sum_i (-c_i/c) v_i \in \operatorname{span}\{v_1, \dots, v_k\}$.

Problem 3.4. Let $v_1, \dots, v_k$ be vectors of a vector space, let $w = c_1 v_1 + \cdots + c_k v_k$ with $c_j \neq 0$, and let L' be the list $v_1, \dots, v_k$ with v_j replaced by w.

(a) Show that $\operatorname{span} L' = \operatorname{span}\{v_1, \dots, v_k\}$.

(b) Show that if $v_1, \dots, v_k$ are linearly independent, so is L'.

(c) Deduce that adding a multiple of one vector of a list to another vector of it, multiplying a vector by a nonzero scalar, or interchanging two vectors changes neither the span of the list nor its independence.

Solution. (a) The vectors of L' lie in the subspace $\operatorname{span}\{v_1, \dots, v_k\}$ (w is a combination of the v_i), so $\operatorname{span} L' \subseteq \operatorname{span}\{v_1, \dots, v_k\}$ (Theorem 3.7). Conversely,

$$v_j = c_j^{-1}\Big(w - \sum_{i \neq j} c_i v_i\Big) \in \operatorname{span} L',$$

and the other v_i lie in L'; so $\operatorname{span}\{v_1, \dots, v_k\} \subseteq \operatorname{span} L'$, again by Theorem 3.7.

(b) Let $aw + \sum_{i \neq j} a_i v_i = 0$. Substituting w,

$$ac_j v_j + \sum_{i \neq j} (ac_i + a_i) v_i = 0$$

By independence $ac_j = 0$, so $a = 0$ (as $c_j \neq 0$); then $a_i = ac_i + a_i = 0$ for $i \neq j$.

(c) Replacing v_j by $v_j + cv_l$ ($l \neq j$) is the case $c_j = 1$, $c_l = c$, other coefficients 0; replacing v_j by cv_j with $c \neq 0$ is the case $c_j = c$, other coefficients 0. Apply (a) and (b). An interchange only reorders the list, which changes neither the set of combinations nor the set of relations.

Problem 3.5. Let $v_1, \dots, v_k \in \mathbb{R}^m$ be nonzero, and let i_j be the position of the last nonzero entry of v_j. Show that if $i_1 < i_2 < \cdots < i_k$, then $v_1, \dots, v_k$ are linearly independent. Apply this to the vectors b_1, b_2, b_3 of Exercise 3.11.

Solution. Induction on k. For $k = 1$, $v_1 \neq 0$ is independent (Proposition 3.12(a)). Let $k \geqslant 2$ and $\sum_{j=1}^{k} c_j v_j = 0$. For $j < k$, the entries of v_j after position i_j are 0, and $i_j < i_k$; so entry i_k of the sum is $c_k (v_k)_{i_k}$. Hence $c_k (v_k)_{i_k} = 0$, and $c_k = 0$ as $(v_k)_{i_k} \neq 0$. Then $\sum_{j<k} c_j v_j = 0$, and $v_1, \dots, v_{k-1}$ satisfy the hypothesis, so $c_1 = \cdots = c_{k-1} = 0$ by induction.

For $b_1 = (1, -2, 1, 0, 0)'$, $b_2 = (2, -3, 0, 1, 0)'$, $b_3 = (3, -4, 0, 0, 1)'$ the last nonzero entries are in positions $3 < 4 < 5$, so b_1, b_2, b_3 are independent.

Problem 3.6. Let $A \in \mathbb{R}^{m\times n}$, and for a subspace W of $\mathbb{R}^n$ let $AW = \{Aw : \ w \in W\}$.

(a) Show that AW is a subspace of $\mathbb{R}^m$, and that if $v_1, \dots, v_k$ span W, then $Av_1, \dots, Av_k$ span AW.

(b) Suppose that $Ax = 0$ only for $x = 0$. Show that if $v_1, \dots, v_k \in \mathbb{R}^n$ are linearly independent, so are $Av_1, \dots, Av_k$.

(c) Let $P \in \mathbb{R}^{m\times m}$ be invertible and $a_1, \dots, a_k \in \mathbb{R}^m$. Show that $\sum_i c_i a_i = 0$ if and only if $\sum_i c_i P a_i = 0$. Deduce that if row operations take A to B, then the columns of A and the columns of B satisfy the same linear relations.

Solution. (a) $0 = A0 \in AW$. For $w, w_1 \in W$ and $c \in \mathbb{R}$, $cAw + Aw_1 = A(cw + w_1) \in AW$, since $cw + w_1 \in W$. So AW is a subspace (Theorem 3.4(c)). Every $w \in W$ is $w = \sum_i c_i v_i$, and then $Aw = \sum_i c_i Av_i$ (Theorem 1.3); conversely $\sum_i c_i Av_i = A\Big(\sum_i c_i v_i\Big) \in AW$. So $AW = \operatorname{span}\{Av_1, \dots, Av_k\}$.

(b) If $\sum_i c_i Av_i = 0$, then $A\Big(\sum_i c_i v_i\Big) = 0$, so $\sum_i c_i v_i = 0$, and every $c_i = 0$ by independence.

(c) $\sum_i c_i Pa_i = P\Big(\sum_i c_i a_i\Big)$. If $\sum_i c_i a_i = 0$, this is 0; conversely, if it is 0, multiplying by P^{-1} gives $\sum_i c_i a_i = 0$. If row operations take A to B, then $B = PA$ with P invertible (Theorem 2.17(c)), and $\operatorname{col}_j(B) = P \operatorname{col}_j(A)$ (Theorem 1.4(a)); apply the first part with $a_j = \operatorname{col}_j(A)$.

Problem 3.7. Let $n \geqslant 2$, $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$, and $d_i = e_i - e_{i+1}$ for $i = 1, \dots, n-1$.

(a) Show that $d_1, \dots, d_{n-1}$ form a basis of W_0, and that the coordinates of $x \in W_0$ are the partial sums $s_i = x_1 + \cdots + x_i$ $(i = 1, \dots, n-1)$.

(b) Write the deviation vector $(-4, -2, 0, 2, 4)'$ of Example 3.9 in this basis ($n = 5$).

Solution. (a)

- $\mathbf{1}'d_i = 1 - 1 = 0$, so every d_i lies in W_0.
- Span: let $x \in W_0$; then $s_n = x_1 + \cdots + x_n = 0$. Shifting the index in the second sum,

$$\sum_{i=1}^{n-1} s_i d_i = \sum_{i=1}^{n-1} s_i e_i - \sum_{i=2}^{n} s_{i-1} e_i = s_1 e_1 + \sum_{i=2}^{n-1} (s_i - s_{i-1}) e_i - s_{n-1} e_n$$

  Here $s_1 = x_1$, $s_i - s_{i-1} = x_i$, and $-s_{n-1} = x_n - s_n = x_n$; so the sum is x.
- Independence: $\sum_i c_i d_i = (c_1, \ c_2 - c_1, \ \dots, \ c_{n-1} - c_{n-2}, \ -c_{n-1})'$. If it is 0, the first entry gives $c_1 = 0$, and entry i gives $c_i = c_{i-1}$ for $i = 2, \dots, n-1$; so every $c_i = 0$.
- Hence $d_1, \dots, d_{n-1}$ form a basis of W_0, and by the span step and Theorem 3.15 the coordinates of x are $s_1, \dots, s_{n-1}$.

(b) The partial sums of $(-4, -2, 0, 2, 4)'$ are $-4, -6, -6, -4$ (and $s_5 = 0$):

$$(-4, -2, 0, 2, 4)' = -4d_1 - 6d_2 - 6d_3 - 4d_4$$

For $n = 4$ this is Example 3.19: $x = x_1 d_1 + (x_1 + x_2) d_2 + (x_1 + x_2 + x_3) d_3$.

Problem 3.8. Show that every $A \in \mathbb{R}^{n\times n}$ can be written in exactly one way as $A = B + C$ with B symmetric and C skew-symmetric, and conclude that $\mathbb{R}^{n\times n}$ is the direct sum of the subspace of symmetric matrices and the subspace of skew-symmetric matrices.

Solution. Both sets are subspaces of $\mathbb{R}^{n\times n}$ (Example 3.5(d)).

- Existence: let $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$. Then $B + C = A$, and, since $(A')' = A$ (Theorem 1.9), $B' = \frac{1}{2}(A' + A) = B$ and $C' = \frac{1}{2}(A' - A) = -C$.
- Uniqueness: let also $A = B_1 + C_1$ with B_1 symmetric and C_1 skew-symmetric. Then $B_1 - B = C - C_1$, where $B_1 - B$ is symmetric and $C - C_1$ is skew-symmetric. So $B_1 - B = (B_1 - B)' = (C - C_1)' = -(C - C_1) = -(B_1 - B)$, hence $2(B_1 - B) = 0$. Thus $B_1 = B$, and then $C_1 = A - B_1 = C$.
- So the sum of the two subspaces is all of $\mathbb{R}^{n\times n}$, and by Theorem 3.22 their intersection is $\{0\}$: $\mathbb{R}^{n\times n}$ is their direct sum.

For example, $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$ give

$$A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 1 & 2 & 4 & 0 \\ 2 & -2 & 5 & 1 \\ 0 & 2 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \\ 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -1 & 1 \\ -1 & 0 & 3 & -1 \\ 1 & -3 & 0 & -1 \\ -1 & 1 & 1 & 0 \end{pmatrix} = B + C$$

Problem 3.9. Let U, W, X be subspaces of a vector space V with $U \subseteq X$. Show that

$$X \cap (U + W) = U + (X \cap W)$$

Solution. Both sides are subspaces (Theorem 3.21(a)).

- $\supseteq$: let $u \in U$ and $y \in X \cap W$. Then $u \in X$ (as $U \subseteq X$) and $y \in X$, so $u + y \in X$; and $u + y \in U + W$ since $y \in W$.
- $\subseteq$: let $x \in X \cap (U + W)$, say $x = u + w$ with $u \in U$, $w \in W$. Then $w = x - u$ lies in X, since $x \in X$ and $u \in U \subseteq X$; so $w \in X \cap W$, and $x = u + w \in U + (X \cap W)$.

Problem 3.10. In $\mathbb{R}^2$ let $U = \operatorname{span}\{e_1\}$, $W = \operatorname{span}\{e_2\}$ and $X = \operatorname{span}\{e_1 + e_2\}$.

(a) Show that $U \cap W$, $U \cap X$ and $W \cap X$ are all $\{0\}$, but that 0 can be written as $u + w + x$ ($u \in U$, $w \in W$, $x \in X$) in more than one way.

(b) Show that $X \cap (U + W) \neq U + (X \cap W)$: the hypothesis $U \subseteq X$ of Problem 3.9 cannot be dropped.

Solution. (a) No two of the vectors e_1, e_2, $e_1 + e_2$ are multiples of each other: $ae_1 = be_2$ gives $(a, -b)' = 0$, $ae_1 = b(e_1 + e_2)$ gives $(a - b, -b)' = 0$, and $ae_2 = b(e_1 + e_2)$ gives $(-b, a - b)' = 0$; in each case $a = b = 0$. So each pairwise intersection is $\{0\}$. But

$$0 = 0 + 0 + 0 = e_1 + e_2 + \big(-(e_1 + e_2)\big)$$

are two different ways.

(b) $U + W = \operatorname{span}\{e_1, e_2\} = \mathbb{R}^2$ (Theorem 3.21(c)), so $X \cap (U + W) = X$. By (a), $X \cap W = \{0\}$, so $U + (X \cap W) = U$. And $X \neq U$, since $e_1 + e_2 \notin U$ by (a).

Chapter 4. Bases, dimension, and rank

Bases and dimension of subspaces of $\mathbb{R}^n$

Theorem 4.1. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows. The following are equivalent:

(i) the columns $a_1, \dots, a_n$ of A are linearly independent;

(ii) the system $Ax = 0$ has only the solution $x = 0$;

(iii) $r = n$, that is, every column of R is a pivot column.

In particular, if $n > m$, then any n vectors of $\mathbb{R}^m$ are linearly dependent.

We call this the **pivot test**.

Proof. • (i) $\Leftrightarrow$ (ii): since $Ax = x_1a_1+\cdots+x_na_n$ (Theorem 1.4(b)), the statement "$x_1a_1+\cdots+x_na_n = 0$ only for $x = 0$" says exactly that $Ax = 0$ only for $x = 0$.

- (ii) $\Leftrightarrow$ (iii): $Ax = 0$ and $Rx = 0$ have the same solutions (Theorem 2.3(b)). If $r < n$, there is a free unknown; giving it the value 1 and the other free unknowns the value 0 yields a solution $x \neq 0$ (Proposition 2.7(b)). If $r = n$, every unknown is a pivot unknown, there are no free unknowns, and the equations of $Rx = 0$ read $x_{k_i} = 0$ $(1 \leqslant i \leqslant n)$; so $x = 0$.
- If $n > m$, put the n vectors into a matrix $A \in \mathbb{R}^{m\times n}$. Its RREF has $r \leqslant m < n$ nonzero rows, so (iii) fails and the vectors are dependent. □

Remark 4.2. For the pivot test one may stop at an echelon form E of A (Definitions 2.4 and 2.5). Backward elimination adds multiples of a row i, whose leading 1 is in column k_i, to rows $l < i$, whose leading 1 is in a column $k_l < k_i$. Row i is 0 before column k_i, so row l changes only in columns $\geqslant k_i$ and keeps its leading 1 in column k_l. Hence the RREF of A reached from E has the same leading columns as E: the pivot columns of A, and their number r, can be read off from any echelon form of A. In particular, (iii) of Theorem 4.1 says that every column of E contains a leading 1. The same holds before the leading entries are scaled to 1: scaling a row does not move its leading entry.

Theorem 4.3. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and entries R_{ij}.

(a) The columns of A and those of R satisfy the same linear relations: for $x \in \mathbb{R}^n$, $x_1a_1+\cdots+x_na_n = 0$ if and only if $x_1 \operatorname{col}_1(R) + \cdots + x_n \operatorname{col}_n(R) = 0$.

(b) If $r \geqslant 1$, the pivot columns $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\operatorname{span}\{a_1, \dots, a_n\}$, and for every $j = 1, \dots, n$

$$a_j = R_{1j}a_{k_1} + R_{2j}a_{k_2} + \cdots + R_{rj}a_{k_r}$$

If $r = 0$, then $A = 0$ and $\operatorname{span}\{a_1, \dots, a_n\} = \{0\}$.

In particular, every finite list that spans a subspace W contains a basis of W.

Proof. (a) The two sums are Ax and Rx, and $Ax = 0$ if and only if $Rx = 0$ (Theorem 2.3(b)).

(b) If $r = 0$, then $R = 0$, and $A = P^{-1}R = 0$, where $R = PA$ with $P \in \mathbb{R}^{m\times m}$ invertible (Theorem 2.17(c)). Let $r \geqslant 1$.

- Columns of R: $\operatorname{col}_{k_i}(R) = e_i$, and rows $r+1, \dots, m$ of R are zero, so $\operatorname{col}_j(R) = \sum_{i=1}^r R_{ij}e_i = \sum_{i=1}^r R_{ij} \operatorname{col}_{k_i}(R)$ for every j.
- Relations: for a column $j \notin \{k_1, \dots, k_r\}$ let $x \in \mathbb{R}^n$ have $x_j = 1$, $x_{k_i} = -R_{ij}$ $(1 \leqslant i \leqslant r)$ and all other entries 0. By the first item, $Rx = \operatorname{col}_j(R) - \sum_i R_{ij} \operatorname{col}_{k_i}(R) = 0$; by (a), $Ax = a_j - \sum_i R_{ij}a_{k_i} = 0$. For $j = k_l$ the formula reads $a_{k_l} = a_{k_l}$, since R_{ik_l} is 1 for $i = l$ and 0 otherwise.

- Spanning: each a_j lies in the subspace $U = \operatorname{span}\{a_{k_1}, \dots, a_{k_r}\}$, so $\operatorname{span}\{a_1, \dots, a_n\} \subseteq U$ (Theorem 3.7); the reverse inclusion is clear.
- Independence: if $c_1 a_{k_1} + \cdots + c_r a_{k_r} = 0$, then by (a) $c_1 \operatorname{col}_{k_1}(R) + \cdots + c_r \operatorname{col}_{k_r}(R) = c_1 e_1 + \cdots + c_r e_r = 0$, so every $c_i = 0$.

For the last sentence, write the spanning vectors as the columns of a matrix and apply (b). □

By Theorem 4.3(a), a_j is a linear combination of $a_1, \dots, a_{j-1}$ if and only if $\operatorname{col}_j(R)$ is a linear combination of $\operatorname{col}_1(R), \dots, \operatorname{col}_{j-1}(R)$ (take $x_j = -1$ and $x_l = 0$ for $l > j$). In R this happens exactly when j is not a pivot column: $\operatorname{col}_{k_i}(R) = e_i$, while the columns before k_i are 0 in rows $i, \dots, m$; and a non-pivot column j equals $\sum_{k_i<j} R_{ij} \operatorname{col}_{k_i}(R)$. So the pivot columns of A are the columns that are not combinations of the earlier ones; they do not depend on the row operations used. This gives a second proof of Proposition 2.8: two RREFs R, S of A have the same pivot columns $k_1 < \cdots < k_r$, hence the same zero rows $r+1, \dots, m$; and for each j, both $R_{1j}, \dots, R_{rj}$ and $S_{1j}, \dots, S_{rj}$ are coefficients of a_j in the basis $a_{k_1}, \dots, a_{k_r}$ (Theorem 4.3(b)), so they are equal (Theorem 3.15). Thus $R = S$.

Theorem 4.4. Every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements. This number is the **dimension** of W, written $\dim W$. For example, $\dim \mathbb{R}^n = n$ (standard basis) and $\dim\{0\} = 0$ (the empty list is a basis of $\{0\}$).

Proof. • If W is spanned by $w_1, \dots, w_m$, then any $k > m$ vectors $v_1, \dots, v_k$ of W are linearly dependent. Indeed, write $v_j = a_{1j} w_1 + \cdots + a_{mj} w_m$ $(1 \leqslant j \leqslant k)$ and $A = (a_{ij}) \in \mathbb{R}^{m \times k}$. As $k > m$, the columns of A are dependent (Theorem 4.1): $Ax = 0$ for some $x \neq 0$. Then

$$x_1 v_1 + \cdots + x_k v_k = \sum_{j=1}^{k} x_j \sum_{i=1}^{m} a_{ij} w_i = \sum_{i=1}^{m} (Ax)_i w_i = 0$$

- Same size: if $v_1, \dots, v_k$ and $w_1, \dots, w_m$ are bases of W, each list spans W and is independent, so the first item gives $k \leqslant m$ and $m \leqslant k$.
- Existence: start with the empty list and, as long as the vectors chosen do not span W, adjoin a vector $v \in W$ outside their span. The list stays independent: in a relation $c_1 v_1 + \cdots + c_k v_k + cv = 0$, $c \neq 0$ would put v in the span of $v_1, \dots, v_k$, so $c = 0$, and then every $c_i = 0$. More than n vectors of $\mathbb{R}^n$ are dependent (Theorem 4.1), so the process stops after at most n steps, at a basis of W. □

By Theorem 4.3, $\dim \operatorname{span}\{a_1, \dots, a_n\} = r$, the number of pivots of the RREF of $[a_1 \ \cdots \ a_n]$.

Theorem 4.5. Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d$.

(a) If U is a subspace of W, then $\dim U \leqslant d$, and $\dim U = d$ only for $U = W$. In particular $d \leqslant n$, and $d = n$ only for $W = \mathbb{R}^n$.

(b) Every linearly independent list of vectors of W can be extended to a basis of W.

(c) More than d vectors of W are linearly dependent; fewer than d vectors do not span W.

(d) If $d \geqslant 1$, then d vectors of W are linearly independent $\Leftrightarrow$ they span W $\Leftrightarrow$ they form a basis of W.

Proof. If $d = 0$, then $W = \{0\}$, and (a)–(c) are immediate: every subspace U of W is $\{0\} = W$; every vector of W is 0, so a nonempty list of vectors of W is dependent (Proposition 3.12(a)); and the empty list, the only independent one, is a basis of W (Definition 3.14). Let $d \geqslant 1$, and let $w_1, \dots, w_d$ be a basis of W (Theorem 4.4). We prove (c) first.

(c)

- The vectors $w_1, \dots, w_d$ span W, so any $k > d$ vectors of W are linearly dependent, by the first step of the proof of Theorem 4.4 (with $m = d$).
- If $k < d$ vectors spanned W, the same step, applied to them, would make the $d > k$ vectors $w_1, \dots, w_d$ of W dependent ✘

(b)

- Let $v_1, \dots, v_k$ be linearly independent vectors of W. If they span W, they form a basis of W.
- Otherwise take $v \in W$ outside $\operatorname{span}\{v_1, \dots, v_k\}$. Then $v_1, \dots, v_k, v$ are independent: in a relation $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span, so $c = 0$, and then every $c_i = 0$.
- Repeat this. Each list obtained consists of independent vectors of W, so it has at most d vectors by (c); hence the process stops, at a basis of W that contains $v_1, \dots, v_k$.

(a)

- A basis of U (Theorem 4.4) consists of $\dim U$ independent vectors of W, so $\dim U \leqslant d$ by (c).
- If $U \neq W$, take $v \in W$ outside U. Adjoining v to a basis of U gives, as in (b), $\dim U + 1$ independent vectors of W, so $\dim U + 1 \leqslant d$ by (c). Hence $\dim U = d$ only for $U = W$.
- For the last sentence, apply this to W as a subspace of $\mathbb{R}^n$, which has dimension n (Theorem 4.4).

(d) Let $v_1, \dots, v_d \in W$. A basis of W is independent and spans W (Definition 3.14).

- Let $v_1, \dots, v_d$ be independent, and let $w \in W$. By (c), the $d+1$ vectors $w, v_1, \dots, v_d$ are dependent: $x_0w + x_1v_1 + \cdots + x_dv_d = 0$ with not all $x_i = 0$. Here $x_0 \neq 0$, since $v_1, \dots, v_d$ are independent; so $w = -(x_1/x_0)v_1 - \cdots - (x_d/x_0)v_d$. Thus $v_1, \dots, v_d$ span W, and they form a basis.
- Let $v_1, \dots, v_d$ span W. By Theorem 4.3, some of them form a basis of W, which has d vectors (Theorem 4.4); so it consists of all of $v_1, \dots, v_d$, and they form a basis. □

Transposition carries a subspace of $\mathbb{R}^{1\times n}$ onto a subspace of $\mathbb{R}^n$ and preserves linear combinations, hence independence, spanning and bases; so Theorems 4.4 and 4.5 also hold for subspaces of $\mathbb{R}^{1\times n}$, such as the row space $\mathcal{R}(A)$ below.

Example 4.6. Let $W = \operatorname{span}\{u_1, \dots, u_5\} \subseteq \mathbb{R}^4$, where

$$u_1 = \begin{pmatrix}1\\2\\1\\1\end{pmatrix}, \quad u_2 = \begin{pmatrix}1\\3\\2\\1\end{pmatrix}, \quad u_3 = \begin{pmatrix}1\\1\\0\\1\end{pmatrix}, \quad u_4 = \begin{pmatrix}1\\2\\2\\3\end{pmatrix}, \quad u_5 = \begin{pmatrix}1\\3\\1\\-1\end{pmatrix}$$

(a) Are $u_1, \dots, u_5$ linearly independent? (b) Find a basis of W among $u_1, \dots, u_5$, and $\dim W$. (c) Express the remaining vectors in this basis.

Solution. Row-reduce $A = [u_1 \ \cdots \ u_5]$:

$$\begin{pmatrix}1&1&1&1&1\\2&3&1&2&3\\1&2&0&2&1\\1&1&1&3&-1\end{pmatrix} \xrightarrow[R_3\leftarrow R_3-R_1,\ R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-2R_1} \begin{pmatrix}1&1&1&1&1\\0&1&-1&0&1\\0&1&-1&1&0\\0&0&0&2&-2\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-2R_3]{R_3\leftarrow R_3-R_2} \begin{pmatrix}1&1&1&1&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix}$$

$$\xrightarrow{R_1\leftarrow R_1-R_2-R_3} \begin{pmatrix}1&0&2&0&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix} = R$$

(In the second step, $R_4 \leftarrow R_4 - 2R_3$ uses the new row 3.) So $r = 3$, with pivot columns $k_1 = 1$, $k_2 = 2$, $k_3 = 4$.

(a) No: $r = 3 < 5$ (Theorem 4.1); indeed any five vectors of $\mathbb{R}^4$ are dependent.

(b) By Theorem 4.3, u_1, u_2, u_4 form a basis of W, and $\dim W = 3$. As $3 < 4$, $W \neq \mathbb{R}^4$ (Theorem 4.5(a)).

(c) Columns 3 and 5 of R give $u_3 = 2u_1 - u_2$ and $u_5 = u_1 + u_2 - u_4$ (Theorem 4.3(b)). Check: $2(1,2,1,1)' - (1,3,2,1)' = (1,1,0,1)' = u_3$ and $(1,2,1,1)' + (1,3,2,1)' - (1,2,2,3)' = (1,3,1,-1)' = u_5$. Equivalently, $2u_1 - u_2 - u_3 = 0$ and $u_1 + u_2 - u_4 - u_5 = 0$.

The four subspaces of a matrix

Definition 4.7. For $A \in \mathbb{R}^{m\times n}$,

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \mathcal{R}(A) = \{y'A : y \in \mathbb{R}^m\} \subseteq \mathbb{R}^{1\times n},$$
$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n, \qquad \mathcal{N}(A') = \{y \in \mathbb{R}^m : A'y = 0\} \subseteq \mathbb{R}^m$$

are the **column space**, **row space**, **null space** and **left null space** of A. The columns of A span $\mathcal{C}(A)$, the rows of A span $\mathcal{R}(A)$ (a space of row vectors), $\mathcal{N}(A)$ is the set of solutions of $Ax = 0$, and $y \in \mathcal{N}(A')$ if and only if $y'A = 0$ (transpose both sides). All four are subspaces (Theorem 3.7 for $\mathcal{C}(A)$ and $\mathcal{R}(A)$, Example 3.5(e) for $\mathcal{N}(A)$ and $\mathcal{N}(A')$).

Let R be the RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and other columns $f_1 < \cdots < f_{n-r}$, so that $x_{f_1}, \ldots, x_{f_{n-r}}$ are the free unknowns of $Ax = 0$ (Proposition 2.7(b)). For $j = 1, \ldots, n-r$, the **special solution** $s_j \in \mathbb{R}^n$ is the solution of $Ax = 0$ with free unknowns $x_{f_j} = 1$ and $x_{f_l} = 0$ $(l \neq j)$. By Proposition 2.7(b), its entries are

$$(s_j)_{f_l} = \delta_{jl} \quad (1 \leqslant l \leqslant n-r), \qquad (s_j)_{k_i} = -R_{if_j} \quad (1 \leqslant i \leqslant r)$$

Theorem 4.8. Let $A \in \mathbb{R}^{m\times n}$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$, where R is the RREF of A, with nonzero rows $\rho_1, \ldots, \rho_r$, pivot columns $k_1 < \cdots < k_r$ and free columns $f_1 < \cdots < f_{n-r}$. Then $P \in \mathbb{R}^{m\times m}$ is invertible, $PA = R$, and:

(a) if $r \geqslant 1$, the pivot columns $a_{k_1}, \ldots, a_{k_r}$ of A form a basis of $\mathcal{C}(A)$;

(b) if $r \geqslant 1$, the nonzero rows $\rho_1, \ldots, \rho_r$ of R form a basis of $\mathcal{R}(A)$;

(c) the special solutions $s_1, \ldots, s_{n-r}$ form a basis of $\mathcal{N}(A)$ (if $r = n$, then $\mathcal{N}(A) = \{0\}$, with the empty basis);

(d) the transposed rows $\mathrm{row}_{r+1}(P)', \ldots, \mathrm{row}_m(P)'$ form a basis of $\mathcal{N}(A')$ (if $r = m$, then $\mathcal{N}(A') = \{0\}$, with the empty basis).

Hence $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, $\dim \mathcal{N}(A) = n - r$ and $\dim \mathcal{N}(A') = m - r$ (if $r = 0$, then $A = 0$ and $\mathcal{C}(A) = \mathcal{R}(A) = \{0\}$).

Proof. The row operations multiply $[A \mid I_m]$ on the left by an invertible matrix E (Theorem 2.17(c)), and $E[A \mid I_m] = [EA \mid E]$, column by column (Theorem 1.4(a)). So $P = E$ is invertible and $PA = R$.

(a) $\mathcal{C}(A) = \mathrm{span}\{a_1, \ldots, a_n\}$; apply Theorem 4.3.

(b)

- Same row space: for $y \in \mathbb{R}^m$, $y'R = (y'P)A$ and $y'A = (y'P^{-1})R$, where $y'P = (P'y)'$ and $y'P^{-1} = ((P^{-1})'y)'$. So $\mathcal{R}(R) \subseteq \mathcal{R}(A)$ and $\mathcal{R}(A) \subseteq \mathcal{R}(R)$, that is, $\mathcal{R}(R) = \mathcal{R}(A)$.
- Spanning: the rows of R span $\mathcal{R}(R)$, and rows $r+1, \ldots, m$ are zero, so $\rho_1, \ldots, \rho_r$ span $\mathcal{R}(R) = \mathcal{R}(A)$.
- Independent: $\mathrm{col}_{k_i}(R) = e_i$, so entry k_i of $c_1\rho_1 + \cdots + c_r\rho_r$ is c_i; if this combination is 0, every $c_i = 0$.

(c) The solutions of $Ax = 0$ are those of $Rx = 0$ (Theorem 2.3(b)).

- Independent: entry f_l of $\sum_j c_j s_j$ is c_l, so $\sum_j c_j s_j = 0$ forces every $c_l = 0$.
- Spanning: let $t \in \mathcal{N}(A)$ and $\tilde{t} = \sum_j t_{f_j} s_j$. Then $\tilde{t} \in \mathcal{N}(A)$, and $\tilde{t}$ has the same free unknowns as t. A solution is determined by its free unknowns (Proposition 2.7(b)), so $t = \tilde{t} \in \operatorname{span}\{s_1, \dots, s_{n-r}\}$.
- If $r = n$, there are no free unknowns and $x = 0$ is the only solution.

(d)

- In $\mathcal{N}(A')$: for $i > r$, $\operatorname{row}_i(P)A = \operatorname{row}_i(PA) = \operatorname{row}_i(R) = 0$ (Theorem 1.4(c)), so $A' \operatorname{row}_i(P)' = (\operatorname{row}_i(P)A)' = 0$.
- Independent: if $c'P = c_1 \operatorname{row}_1(P) + \cdots + c_m \operatorname{row}_m(P) = 0$, then $c' = c'PP^{-1} = 0$; so the rows of P are independent, and so are rows $r+1, \dots, m$ (Proposition 3.12(b)).
- Spanning: let $A'y = 0$, so $y'A = 0$, and put $z' = y'P^{-1}$, so that $y' = z'P = \sum_{i=1}^m z_i \operatorname{row}_i(P)$. Then $0 = y'A = z'PA = z'R = z_1\rho_1 + \cdots + z_r\rho_r$, and (b) gives $z_1 = \cdots = z_r = 0$. So $y' = \sum_{i>r} z_i \operatorname{row}_i(P)$ (and $y = 0$ if $r = m$). □

Example 4.9. Find bases of the four subspaces of

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$$

Solution. Row-reduce $[A \mid I_3]$:

$$\left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 5 & 0 & 1 & 0 \\ 2 & 5 & 1 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & -2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 3 & -1 & 3 & -2 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) = [R \mid P]$$

So $r = 2$, with pivot columns $1, 2$ and free unknowns x_3, x_4. (The pivots are already visible in the echelon form after the second arrow; Remark 4.2.)

- $\mathcal{C}(A)$: $a_1 = (1, 1, 2)'$ and $a_2 = (2, 3, 5)'$ form a basis; $\dim \mathcal{C}(A) = 2$. Columns 3, 4 of R give
$$a_3 = 3a_1 - a_2, \qquad a_4 = -a_1 + 2a_2$$
Check: $3(1, 1, 2)' - (2, 3, 5)' = (1, 0, 1)'$ and $-(1, 1, 2)' + 2(2, 3, 5)' = (3, 5, 8)'$.
- $\mathcal{R}(A)$: $\rho_1 = (1, 0, 3, -1)$ and $\rho_2 = (0, 1, -1, 2)$ form a basis; $\dim \mathcal{R}(A) = 2$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 3x_3 - x_4 = 0$, $x_2 - x_3 + 2x_4 = 0$. The special solutions
$$s_1 = \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} \ (x_3 = 1,\ x_4 = 0), \qquad s_2 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix} \ (x_3 = 0,\ x_4 = 1)$$
form a basis of $\mathcal{N}(A)$; $\dim \mathcal{N}(A) = 2 = 4 - 2$. Here $As_1 = -3a_1 + a_2 + a_3 = 0$ is the relation $a_3 = 3a_1 - a_2$: each special solution is a column relation.
- $\mathcal{N}(A')$: row 3 of P gives the basis $y = (-1, -1, 1)'$; $\dim \mathcal{N}(A') = 1 = 3 - 2$. Indeed $y'A = -\operatorname{row}_1(A) - \operatorname{row}_2(A) + \operatorname{row}_3(A) = 0$, that is, $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$: each vector of $\mathcal{N}(A')$ is a row relation.
- $\mathcal{C}(R) \neq \mathcal{C}(A)$: every column of R has third entry 0, but a_1 has third entry 2.

Remark 4.10. Row operations do not change $\mathcal{R}(A)$ or $\mathcal{N}(A)$, so R may be used for these two spaces. They do change $\mathcal{C}(A)$ in general: a basis of $\mathcal{C}(A)$ consists of the pivot columns of A itself, not of R (Example 4.9).

Rank

Definition 4.11. By Theorem 4.8, $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, the number of pivots of the RREF of A (also when $r = 0$). This common number is the **rank** of A, written $\operatorname{rank} A$; by Remark 4.2 it is also the number of nonzero rows of any echelon form of A. A has **full column rank** if $\operatorname{rank} A = n$, and **full row rank** if $\operatorname{rank} A = m$. The number $\dim \mathcal{N}(A)$ is the **nullity** of A.

Theorem 4.12. For every $A \in \mathbb{R}^{m\times n}$,

$$\operatorname{rank} A + \dim \mathcal{N}(A) = n, \qquad \operatorname{rank} A + \dim \mathcal{N}(A') = m$$

Proof. Each of the n columns of R is either a pivot column (r of them) or a free column ($n - r$ of them), and by Theorem 4.8(c), $\dim \mathcal{N}(A) = n - r$, the number of free unknowns. Likewise $\dim \mathcal{N}(A') = m - r$, the number of zero rows of R (Theorem 4.8(d)). Since $\operatorname{rank} A = r$, both equalities follow. □

Theorem 4.13. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\operatorname{rank} A' = \operatorname{rank} A$, and $\operatorname{rank} A \leqslant \min\{m, n\}$.

(b) A has full column rank $\Leftrightarrow$ its columns are linearly independent $\Leftrightarrow \mathcal{N}(A) = \{0\}$.

(c) A has full row rank $\Leftrightarrow$ its rows are linearly independent $\Leftrightarrow \mathcal{N}(A') = \{0\} \Leftrightarrow \mathcal{C}(A) = \mathbb{R}^m$.

(d) If $m = n$, then A is invertible $\Leftrightarrow \operatorname{rank} A = n$.

Proof. (a) For $y \in \mathbb{R}^m$, $(y'A)' = A'y$, so transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$. It preserves linear combinations and is undone by transposing again, so it carries a basis of $\mathcal{R}(A)$ to a basis of $\mathcal{C}(A')$, and $\operatorname{rank} A' = \dim \mathcal{C}(A') = \dim \mathcal{R}(A) = \operatorname{rank} A$. Also $\operatorname{rank} A = r \leqslant m$, as R has m rows, and $r \leqslant n$, as $k_1 < \cdots < k_r$ lie in $\{1, \ldots, n\}$.

(b) Since $\operatorname{rank} A = r$, this is Theorem 4.1: (iii) $\Leftrightarrow$ (i) $\Leftrightarrow$ (ii).

(c)

- $\operatorname{rank} A = m$ if and only if $\dim \mathcal{N}(A') = m - \operatorname{rank} A = 0$ (Theorem 4.12), that is, $\mathcal{N}(A') = \{0\}$.
- $y'A = y_1 \operatorname{row}_1(A) + \cdots + y_m \operatorname{row}_m(A)$ (Theorem 1.4(c)), and $y'A = 0$ if and only if $A'y = 0$. So $\mathcal{N}(A') = \{0\}$ says exactly that the rows of A are independent.
- $\operatorname{rank} A = \dim \mathcal{C}(A)$; a subspace of $\mathbb{R}^m$ has dimension m only if it is $\mathbb{R}^m$ (Theorem 4.5(a)), and conversely $\dim \mathbb{R}^m = m$ (Theorem 4.4).

(d) A square matrix A is invertible if and only if $Ax = 0$ has only the solution $x = 0$ (Theorem 2.18), that is, $\mathcal{N}(A) = \{0\}$; by (b), this means $\operatorname{rank} A = n$. □

Corollary 4.14. Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and let $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$ be the augmented matrix.

(a) $Ax = b$ is consistent $\Leftrightarrow b \in \mathcal{C}(A) \Leftrightarrow \operatorname{rank}[A \mid b] = \operatorname{rank} A$.

(b) If $Ax = b$ is consistent, its solution is unique $\Leftrightarrow \operatorname{rank} A = n$.

Proof. (a)

- $Ax = x_1a_1 + \cdots + x_na_n$ (Theorem 1.4(b)), so $Ax = b$ has a solution if and only if b is a linear combination of the columns of A, that is, $b \in \mathcal{C}(A)$.
- $\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b])$: a combination of $a_1, \ldots, a_n$ is one of $a_1, \ldots, a_n, b$, with coefficient 0 on b.
- If $b = Ac$, then $x_1a_1 + \cdots + x_na_n + tb = A(x + tc) \in \mathcal{C}(A)$; so $\mathcal{C}([A \mid b]) = \mathcal{C}(A)$, and the ranks are equal.

- If the ranks are equal, the subspace $\mathcal{C}(A)$ of $\mathcal{C}([A \mid b])$ has the same dimension, so the two are equal (Theorem 4.5(a)), and $b \in \mathcal{C}([A \mid b]) = \mathcal{C}(A)$.

(b) By Theorem 2.23(a), the solutions are $x_p + z$, where x_p is one solution and z runs through the solutions of $Ax = 0$, that is, through $\mathcal{N}(A)$. So the solution is unique if and only if $\mathcal{N}(A) = \{0\}$, that is, $\operatorname{rank} A = n$ (Theorem 4.13(b)). □

In practice, consistency is read off from an echelon form of $[A \mid b]$ (Theorem 2.22).

Rank equalities and inequalities

Theorem 4.15. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

(a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$; and $\operatorname{rank}(AB) \leqslant \min\{\operatorname{rank} A, \operatorname{rank} B\}$.

(b) If A has full column rank, then $\mathcal{N}(AB) = \mathcal{N}(B)$ and $\operatorname{rank}(AB) = \operatorname{rank} B$.

(c) If B has full row rank, then $\mathcal{C}(AB) = \mathcal{C}(A)$ and $\operatorname{rank}(AB) = \operatorname{rank} A$.

(d) $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for all invertible $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$.

Proof. (a)

- $ABx = A(Bx) \in \mathcal{C}(A)$, and $Bx = 0$ gives $ABx = A0 = 0$.
- So $\operatorname{rank}(AB) = \dim \mathcal{C}(AB) \leqslant \dim \mathcal{C}(A) = \operatorname{rank} A$ (Theorem 4.5(a)).
- By this, $(AB)' = B'A'$ (Theorem 1.9) and Theorem 4.13(a), $\operatorname{rank}(AB) = \operatorname{rank}(B'A') \leqslant \operatorname{rank} B' = \operatorname{rank} B$.

(b)

- If $ABx = 0$, then $Bx \in \mathcal{N}(A) = \{0\}$ (Theorem 4.13(b)), so $x \in \mathcal{N}(B)$. With (a), $\mathcal{N}(AB) = \mathcal{N}(B)$.
- AB and B both have p columns, so by Theorem 4.12, $\operatorname{rank}(AB) = p - \dim \mathcal{N}(AB) = p - \dim \mathcal{N}(B) = \operatorname{rank} B$.

(c)

- $\mathcal{C}(B) = \mathbb{R}^n$ (Theorem 4.13(c)): every $x \in \mathbb{R}^n$ is Bw for some $w \in \mathbb{R}^p$, and then $Ax = (AB)w \in \mathcal{C}(AB)$.
- So $\mathcal{C}(A) \subseteq \mathcal{C}(AB)$; with (a), $\mathcal{C}(AB) = \mathcal{C}(A)$, and the ranks are equal.

(d) $Px = 0$ gives $x = P^{-1}Px = 0$, so P has full column rank (Theorem 4.13(b)); and $\operatorname{rank} Q = n$ (Theorem 4.13(d)), so Q has full row rank. By (b) and (c), $\operatorname{rank}(P(AQ)) = \operatorname{rank}(AQ) = \operatorname{rank} A$. □

Theorem 4.16. (a) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$, let $[A\ B] \in \mathbb{R}^{m\times(n+p)}$ have the columns of A followed by those of B. Then

$$\max\{\operatorname{rank} A, \operatorname{rank} B\} \leqslant \operatorname{rank}[A\ B] \leqslant \operatorname{rank} A + \operatorname{rank} B$$

(b) For $A \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{q\times n}$, let $\left(\begin{smallmatrix} A \\ C \end{smallmatrix}\right) \in \mathbb{R}^{(m+q)\times n}$ have the rows of A followed by those of C. Then

$$\max\{\operatorname{rank} A, \operatorname{rank} C\} \leqslant \operatorname{rank} \begin{pmatrix} A \\ C \end{pmatrix} \leqslant \operatorname{rank} A + \operatorname{rank} C$$

(c) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{q\times p}$, let $\operatorname{diag}(A, B) = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right) \in \mathbb{R}^{(m+q)\times(n+p)}$, with zero blocks of sizes $m \times p$ and $q \times n$. Then $\operatorname{rank}\operatorname{diag}(A, B) = \operatorname{rank} A + \operatorname{rank} B$.

Proof. (a)

- A combination of the columns of A, or of those of B, is a combination of the columns of $[A\ B]$ (the others with coefficient 0). So $\mathcal{C}(A)$ and $\mathcal{C}(B)$ are subspaces of $\mathcal{C}([A\ B])$, and Theorem 4.5(a) gives the lower bound.
- Let $r = \operatorname{rank} A$, $s = \operatorname{rank} B$, and take bases $v_1, \dots, v_r$ of $\mathcal{C}(A)$ and $w_1, \dots, w_s$ of $\mathcal{C}(B)$. Every column of $[A\ B]$ is a combination of these $r + s$ vectors of $\mathcal{C}([A\ B])$, so they span $\mathcal{C}([A\ B])$ (Theorem 3.7), and $\operatorname{rank}[A\ B] \leqslant r + s$ (Theorem 4.5(c)).

(b) Transposing turns the rows of $\binom{A}{C}$ into the columns of $[A'\ C']$, so $\binom{A}{C}' = [A'\ C']$. Apply (a) to A' and C', and use Theorem 4.13(a).

(c) Let $M = \operatorname{diag}(A, B)$, $r = \operatorname{rank} A$ and $s = \operatorname{rank} B$.

- Upper bound: $M = [M_1\ M_2]$ with $M_1 = \binom{A}{0}$ and $M_2 = \binom{0}{B}$. By (b), and as a zero matrix has rank 0, $r \leqslant \operatorname{rank} M_1 \leqslant r + 0$; so $\operatorname{rank} M_1 = r$, and likewise $\operatorname{rank} M_2 = s$. By (a), $\operatorname{rank} M \leqslant r + s$.
- Lower bound: let $a_{k_1}, \dots, a_{k_r}$ and $b_{l_1}, \dots, b_{l_s}$ be the pivot columns of A and of B, bases of $\mathcal{C}(A)$ and $\mathcal{C}(B)$ (Theorem 4.8(a)). The corresponding columns $\begin{pmatrix} a_{k_i} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ b_{l_j} \end{pmatrix}$ of M are independent: a combination of them is $\begin{pmatrix} \sum_i \alpha_i a_{k_i} \\ \sum_j \beta_j b_{l_j} \end{pmatrix}$, which is 0 only if all α_i and β_j are 0. These $r + s$ vectors lie in $\mathcal{C}(M)$, so $\operatorname{rank} M \geqslant r + s$ (Theorem 4.5(c)). □

Corollary 4.17. For $A, B \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(A + B) \leqslant \operatorname{rank} A + \operatorname{rank} B$.

Proof. For $x \in \mathbb{R}^n$, $(A + B)x = Ax + Bx$ is a combination of the columns of A and of B, so it lies in $\mathcal{C}([A\ B])$. Hence $\mathcal{C}(A + B) \subseteq \mathcal{C}([A\ B])$, so $\operatorname{rank}(A + B) \leqslant \operatorname{rank}[A\ B]$ (Theorem 4.5(a)), and $\operatorname{rank}[A\ B] \leqslant \operatorname{rank} A + \operatorname{rank} B$ (Theorem 4.16(a)). □

Theorem 4.18. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\mathcal{N}(A'A) = \mathcal{N}(A)$.

(b) $\operatorname{rank}(A'A) = \operatorname{rank} A = \operatorname{rank}(AA')$.

(c) $\mathcal{C}(A'A) = \mathcal{C}(A')$ and $\mathcal{C}(AA') = \mathcal{C}(A)$.

Proof. (a)

- If $Ax = 0$, then $A'Ax = A'0 = 0$.
- If $A'Ax = 0$, then $(Ax)'(Ax) = x'A'Ax = 0$. But $(Ax)'(Ax)$ is the sum of the squares of the entries of Ax (Theorem 1.9), so $Ax = 0$.

(b) $A'A$ and A both have n columns, so by (a) and Theorem 4.12, $\operatorname{rank}(A'A) = n - \dim \mathcal{N}(A'A) = n - \dim \mathcal{N}(A) = \operatorname{rank} A$. Applied to A' in place of A, this gives $\operatorname{rank}(AA') = \operatorname{rank} A' = \operatorname{rank} A$ (Theorem 4.13(a)).

(c) $\mathcal{C}(A'A) \subseteq \mathcal{C}(A')$ (Theorem 4.15(a)), and both have dimension $\operatorname{rank} A$, by (b) and Theorem 4.13(a); so they are equal (Theorem 4.5(a)). Likewise $\mathcal{C}(AA') \subseteq \mathcal{C}(A)$, both of dimension $\operatorname{rank} A$. □

Theorem 4.19 (Sylvester's inequality)**.** For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$,

$$\dim \mathcal{N}(AB) \leqslant \dim \mathcal{N}(A) + \dim \mathcal{N}(B), \qquad \text{equivalently} \qquad \operatorname{rank}(AB) \geqslant \operatorname{rank} A + \operatorname{rank} B - n$$

In particular, if $AB = 0$, then $\operatorname{rank} A + \operatorname{rank} B \leqslant n$.

Proof.
- Let $z_1, \dots, z_s$ be a basis of $\mathcal{N}(B)$. Since $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$ (Theorem 4.15(a)), it extends to a basis $z_1, \dots, z_s, w_1, \dots, w_t$ of $\mathcal{N}(AB)$ (Theorem 4.5(b)); so $\dim \mathcal{N}(AB) = s + t$.
- The vectors $Bw_1, \dots, Bw_t$ lie in $\mathcal{N}(A)$, since $A(Bw_j) = (AB)w_j = 0$.
- They are independent: if $c_1Bw_1 + \cdots + c_tBw_t = 0$, then $B(c_1w_1 + \cdots + c_tw_t) = 0$, so $c_1w_1 + \cdots + c_tw_t \in \mathcal{N}(B)$ equals some $d_1z_1 + \cdots + d_sz_s$; as $z_1, \dots, z_s, w_1, \dots, w_t$ are independent, every $c_j = 0$.

- Hence $t \leqslant \dim \mathcal{N}(A)$ (Theorem 4.5(c)), and $\dim \mathcal{N}(AB) = s + t \leqslant \dim \mathcal{N}(B) + \dim \mathcal{N}(A)$.
- By Theorem 4.12, $\dim \mathcal{N}(AB) = p - \operatorname{rank}(AB)$, $\dim \mathcal{N}(B) = p - \operatorname{rank} B$ and $\dim \mathcal{N}(A) = n - \operatorname{rank} A$. Substituting gives $p - \operatorname{rank}(AB) \leqslant (p - \operatorname{rank} B) + (n - \operatorname{rank} A)$, which is the rank form. If $AB = 0$, then $\operatorname{rank}(AB) = 0$. □

Example 4.20. Let $A = \operatorname{diag}(1, 1, 1, 0)$ and $B = \operatorname{diag}(0, 0, 1, 1)$ in $\mathbb{R}^{4\times 4}$. Their columns are $e_1, e_2, e_3, 0$ and $0, 0, e_3, e_4$, so $\mathcal{C}(A) = \operatorname{span}\{e_1, e_2, e_3\}$, $\mathcal{C}(B) = \operatorname{span}\{e_3, e_4\}$, $\operatorname{rank} A = 3$ and $\operatorname{rank} B = 2$.

- $AB = \operatorname{diag}(0, 0, 1, 0)$ (row i of AB is $a_{ii} \operatorname{row}_i(B)$, Theorem 1.4(c)) has rank 1: less than $\min\{3, 2\}$ (Theorem 4.15(a)), and equal to $\operatorname{rank} A + \operatorname{rank} B - 4 = 1$, so Sylvester's bound is attained (Theorem 4.19).
- $A + B = \operatorname{diag}(1, 1, 2, 1)$ has rank $4 < 3 + 2$ (Corollary 4.17); and $A + (-A) = 0$ has rank 0.
- $[A\ B] \in \mathbb{R}^{4\times 8}$ has the columns $e_1, e_2, e_3, 0, 0, 0, e_3, e_4$, so $\mathcal{C}([A\ B]) = \mathbb{R}^4$ and $\operatorname{rank}[A\ B] = 4$, strictly between $\max\{3, 2\}$ and $3 + 2$ (Theorem 4.16(a)).
- $\operatorname{diag}(A, B) = \operatorname{diag}(1, 1, 1, 0, 0, 0, 1, 1) \in \mathbb{R}^{8\times 8}$ has rank $5 = 3 + 2$ (Theorem 4.16(c)).
- $A'A = A^2 = A$ has rank $3 = \operatorname{rank} A$ (Theorem 4.18).

Theorem 4.21. Let $A \in \mathbb{R}^{m\times n}$ have rank r. If $r \geqslant 1$, a product $A = BC$ with $B \in \mathbb{R}^{m\times r}$ of full column rank and $C \in \mathbb{R}^{r\times n}$ of full row rank is a **full-rank factorization** of A.

(a) If $r \geqslant 1$, let R be the RREF of A, with pivot columns $k_1 < \cdots < k_r$; let $B = [a_{k_1}\ \cdots\ a_{k_r}] \in \mathbb{R}^{m\times r}$ consist of the pivot columns of A, and $C \in \mathbb{R}^{r\times n}$ of the r nonzero rows of R. Then $A = BC$ is a full-rank factorization. If $r = 0$, then $A = 0$, and there is nothing to factor.

(b) For every full-rank factorization $A = BC$, $\mathcal{C}(A) = \mathcal{C}(B)$ and $\mathcal{N}(A) = \mathcal{N}(C)$.

(c) For every full-rank factorization $A = BC$, the $r \times r$ matrices $B'B$ and CC' are invertible, and

$$L = (B'B)^{-1}B' \in \mathbb{R}^{r\times m}, \qquad K = C'(CC')^{-1} \in \mathbb{R}^{n\times r}$$

satisfy $LB = I_r$ and $CK = I_r$: B has a left inverse and C has a right inverse.

Proof. (a)

- Column j of BC is $B \operatorname{col}_j(C) = R_{1j}a_{k_1} + \cdots + R_{rj}a_{k_r}$ (Theorem 1.4(a),(b)), which is a_j by Theorem 4.3(b). So $BC = A$.
- The columns of B form a basis of $\mathcal{C}(A)$, so they are independent: B has full column rank (Theorem 4.13(b)).
- The rows of C form a basis of $\mathcal{R}(A)$ (Theorem 4.8(b)), so they are independent: C has full row rank (Theorem 4.13(c)).
- If $r = 0$, then $A = 0$ (Theorem 4.8).

(b) Theorem 4.15(b) gives $\mathcal{N}(BC) = \mathcal{N}(C)$, as B has full column rank, and Theorem 4.15(c) gives $\mathcal{C}(BC) = \mathcal{C}(B)$, as C has full row rank.

(c)

- $\operatorname{rank}(B'B) = \operatorname{rank} B = r$ and $\operatorname{rank}(CC') = \operatorname{rank} C = r$ (Theorem 4.18(b)), so the $r \times r$ matrices $B'B$ and CC' are invertible (Theorem 4.13(d)).
- $LB = (B'B)^{-1}(B'B) = I_r$ and $CK = (CC')(CC')^{-1} = I_r$. □

Example 4.22. For A of Example 4.9, Theorem 4.21(a) gives

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

- Columns 1, 2 of BC are a_1, a_2, and columns 3, 4 are $3a_1 - a_2 = a_3$ and $-a_1 + 2a_2 = a_4$ (Example 4.9).
- By Theorem 4.21(b), $\mathcal{C}(A) = \mathcal{C}(B) = \operatorname{span}\{a_1, a_2\}$ and $\mathcal{N}(A) = \mathcal{N}(C) = \operatorname{span}\{s_1, s_2\}$: the system $Cx = 0$ is $Rx = 0$ without its zero row.
- With $N = [s_1\ s_2] \in \mathbb{R}^{4\times 2}$, $AN = 0$ and $\operatorname{rank} N = 2$, and Sylvester's inequality $\operatorname{rank} A + \operatorname{rank} N \leqslant 4$ holds with equality: $2 + 2 = 4$.

Exercises

Exercise 4.1. Let $a_1, \dots, a_9$ be the columns of

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 5 \\ 0 & 1 & 1 & 2 & 2 & 3 & -1 & 3 & 3 \\ 1 & 1 & 2 & 3 & 5 & 5 & 1 & 0 & 4 \end{pmatrix}$$

Which of the following form a basis of $\mathbb{R}^3$? (a) a_1, a_2; (b) a_1, a_4, a_6, a_8; (c) a_2, a_4, a_7; (d) a_3, a_5, a_9.

Solution. (a), (b) No: $\dim \mathbb{R}^3 = 3$, so two vectors cannot span $\mathbb{R}^3$, and four vectors of $\mathbb{R}^3$ are dependent (Theorem 4.5(c)).

(c) Yes. Three vectors of $\mathbb{R}^3$ form a basis if and only if they are independent (Theorem 4.5(d)). Row-reduce $[a_2\ a_4\ a_7]$:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & -1 \\ 1 & 3 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 2 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 5 \end{pmatrix}$$
$$\xrightarrow[R_2 \leftarrow R_2 + 3R_3]{R_3 \leftarrow \frac{1}{5}R_3,\ R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

All three columns are pivot columns, so a_2, a_4, a_7 are independent (Theorem 4.1).

(d) No: the three vectors are dependent, since

$$7a_3 - 2a_5 - a_9 = 7\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - 2\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 - 2 - 5 \\ 7 - 4 - 3 \\ 14 - 10 - 4 \end{pmatrix} = 0$$

Exercise 4.2. Let $W = \mathcal{C}(A) \subseteq \mathbb{R}^4$ for

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 8 \\ 5 & 1 & -3 \\ -3 & -4 & -5 \end{pmatrix}$$

(a) Find a basis of W. (b) What is $\dim W$?

Solution. Let a_1, a_2, a_3 be the columns of A; they span W.

(a) a_1, a_2 are independent: otherwise, as $a_1 \neq 0$, $a_2 = ca_1$ for some c (Proposition 3.12(c)); the first entries give $c = 2$, and then the second entries give $3 = 2(-2)$ ✘. But

$$a_1 - 2a_2 + a_3 = \begin{pmatrix} 1 - 4 + 3 \\ -2 - 6 + 8 \\ 5 - 2 - 3 \\ -3 + 8 - 5 \end{pmatrix} = 0,$$

so $a_3 = 2a_2 - a_1$, and every combination $x_1a_1 + x_2a_2 + x_3a_3 = (x_1 - x_3)a_1 + (x_2 + 2x_3)a_2$ is a combination of a_1, a_2. Thus a_1, a_2 span W and form a basis of W. (Check by Theorem 4.3: the RREF of A has rows $(1, 0, -1)$, $(0, 1, 2)$ and two zero rows; its pivot columns are $1, 2$, and column 3 gives $a_3 = -a_1 + 2a_2$.)

(b) $\dim W = 2$.

Exercise 4.3. Let $u_1 = (1, -2, 1, -1)'$ and $u_2 = (0, 1, -1, 1)'$ in $\mathbb{R}^4$, and let $e_1, \dots, e_4$ be the standard unit vectors.

(a) Row-reduce $[u_1\ u_2\ e_1\ e_2\ e_3\ e_4] \in \mathbb{R}^{4\times 6}$, and find a basis of $\mathbb{R}^4$ that consists of u_1, u_2 and two of the standard unit vectors.

(b) Do u_1, u_2, e_1, e_2 form a basis of $\mathbb{R}^4$?

Solution. (a) Call the matrix $A = [u_1\ u_2\ e_1\ e_2\ e_3\ e_4]$; its columns span $\mathbb{R}^4$, since they include $e_1, \dots, e_4$.

$$\left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

The last matrix is an echelon form, with leading 1s in columns $1, 2, 3, 5$; these are the pivot columns of A (Remark 4.2). By Theorem 4.3(b), u_1, u_2, e_1, e_3 form a basis of the span of the columns, which is $\mathbb{R}^4$.

(b) No. Backward elimination finishes the RREF:

$$\xrightarrow{R_3 \leftarrow R_3 - R_4} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - 2R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cc|cccc} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

Column 4 of the RREF gives $e_2 = -u_1 - u_2 + e_1$ (Theorem 4.3(b)). So u_1, u_2, e_1, e_2 are linearly dependent and do not form a basis. Check: $-(1, -2, 1, -1)' - (0, 1, -1, 1)' + (1, 0, 0, 0)' = (0, 1, 0, 0)' = e_2$.

Exercise 4.4. Find the dimension of the space of solutions of each system, and a basis of it: (a) $x_1 - x_2 + x_3 - x_4 = 0$, for $x \in \mathbb{R}^4$; (b) $x_1 + x_2 + x_3 + x_4 + x_5 = 0$, $x_2 + x_4 = 0$, $x_1 - x_2 + x_3 - x_4 + x_5 = 0$, for $x \in \mathbb{R}^5$.

Solution. The space of solutions is $\mathcal{N}(A)$ for the coefficient matrix $A \in \mathbb{R}^{m\times n}$, of dimension $n - \operatorname{rank} A$ (Theorem 4.12), and the special solutions form a basis of it (Theorem 4.8(c)).

(a) $A = (1, -1, 1, -1)$ is a nonzero RREF with one row, so $\operatorname{rank} A = 1$ and the dimension is $4 - 1 = 3$. The free unknowns are x_2, x_3, x_4, and $x_1 = x_2 - x_3 + x_4$; the special solutions

$$s_1 = (1, 1, 0, 0)', \qquad s_2 = (-1, 0, 1, 0)', \qquad s_3 = (1, 0, 0, 1)'$$

form a basis.

(b) Row-reduce the coefficient matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & -2 & 0 & -2 & 0 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_2]{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank} A = 2$, and the dimension is $5 - 2 = 3$. The free unknowns are x_3, x_4, x_5, with $x_1 = -x_3 - x_5$ and $x_2 = -x_4$; the special solutions

$$s_1 = (-1, 0, 1, 0, 0)', \qquad s_2 = (0, -1, 0, 1, 0)', \qquad s_3 = (-1, 0, 0, 0, 1)'$$

form a basis.

Exercise 4.5. Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 1 & 2 & 1 & 0 & 1 \\ 2 & 4 & 1 & 1 & 0 \\ 1 & 2 & 2 & -1 & 3 \end{pmatrix}$$

(a) Find $\operatorname{rank} A$ and bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$, $\mathcal{N}(A)$ and $\mathcal{N}(A')$, and write the non-pivot columns of A as combinations of the pivot columns.

(b) For which $b \in \mathbb{R}^4$ is $Ax = b$ consistent? Give all solutions for $b = (1, 2, 3, 3)'$.

Solution. (a) Row-reduce $[A \mid I_4]$ (Theorem 4.8):

$$\left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 4 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 2 & -1 & 3 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1,\ R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -2 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & 4 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right) = [R \mid P]$$

Row 1 has entry 0 in column 3, so no backward step is needed. Thus $\operatorname{rank} A = 2$, with pivot columns $1, 3$ and free unknowns x_2, x_4, x_5.

- $\mathcal{C}(A)$: $a_1 = (1, 1, 2, 1)'$ and $a_3 = (0, 1, 1, 2)'$. Columns 2, 4, 5 of R give $a_2 = 2a_1$, $a_4 = a_1 - a_3$ and $a_5 = -a_1 + 2a_3$ (Theorem 4.3(b)). Check: $(1, 1, 2, 1)' - (0, 1, 1, 2)' = (1, 0, 1, -1)' = a_4$ and $-(1, 1, 2, 1)' + 2(0, 1, 1, 2)' = (-1, 1, 0, 3)' = a_5$.
- $\mathcal{R}(A)$: $\rho_1 = (1, 2, 0, 1, -1)$ and $\rho_2 = (0, 0, 1, -1, 2)$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 2x_2 + x_4 - x_5 = 0$, $x_3 - x_4 + 2x_5 = 0$. The special solutions

  $$s_1 = (-2, 1, 0, 0, 0)', \qquad s_2 = (-1, 0, 1, 1, 0)', \qquad s_3 = (1, 0, -2, 0, 1)'$$

  form a basis; $\dim \mathcal{N}(A) = 3 = 5 - 2$.
- $\mathcal{N}(A')$: rows 3, 4 of P give the basis $y_1 = (-1, -1, 1, 0)'$, $y_2 = (1, -2, 0, 1)'$; $\dim \mathcal{N}(A') = 2 = 4 - 2$. Indeed $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$ and $\operatorname{row}_4(A) = -\operatorname{row}_1(A) + 2\operatorname{row}_2(A)$.

(b) The row operations of (a) multiply on the left by the invertible matrix P of $[R \mid P]$ (Theorem 2.17(c)), so they take $[A \mid b]$ to $P[A \mid b] = [R \mid Pb]$ (Theorem 1.4(a)), where

$$Pb = (b_1,\ b_2 - b_1,\ b_3 - b_1 - b_2,\ b_4 + b_1 - 2b_2)'$$

By Proposition 2.7(a), $Ax = b$ is consistent if and only if $b_3 = b_1 + b_2$ and $b_4 = 2b_2 - b_1$, that is, $y_1'b = y_2'b = 0$. For $b = (1, 2, 3, 3)'$ both hold, and $Pb = (1, 1, 0, 0)'$. By Proposition 2.7(b), $x_1 = 1 - 2x_2 - x_4 + x_5$ and $x_3 = 1 + x_4 - 2x_5$, so the solutions are

$$x = (1, 0, 1, 0, 0)' + x_2 s_1 + x_4 s_2 + x_5 s_3 \qquad (x_2, x_4, x_5 \in \mathbb{R})$$

Check: $A(1, 0, 1, 0, 0)' = a_1 + a_3 = (1, 2, 3, 3)'$.

Exercise 4.6. Find $\operatorname{rank} A$, bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$ and $\mathcal{N}(A)$, and $\dim \mathcal{N}(A')$ for

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 2 \\ 0 & -2 & -2 & 3 \end{pmatrix},$$

and write the non-pivot column of A as a combination of the pivot columns.

Solution. Row-reduce A:

$$A \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -2 & -2 & 3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

After the second step column 3 has no nonzero entry in rows $3, 4$, and the next pivot column, 4, has its nonzero entry in row 4; so rows 3 and 4 are interchanged. The echelon form after the third arrow has three nonzero rows: $\operatorname{rank} A = 3$, with pivot columns $1, 2, 4$.

- $\mathcal{C}(A)$: $a_1 = (1,0,1,0)'$, $a_2 = (2,1,1,-2)'$, $a_4 = (1,-1,2,3)'$. Column 3 of R gives $a_3 = -a_1 + a_2$; check: $-(1,0,1,0)' + (2,1,1,-2)' = (1,1,0,-2)' = a_3$.
- $\mathcal{R}(A)$: $(1,0,-1,0)$, $(0,1,1,0)$, $(0,0,0,1)$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 - x_3 = 0$, $x_2 + x_3 = 0$, $x_4 = 0$; the special solution $s = (1,-1,1,0)'$ ($x_3 = 1$) is a basis, and $\dim \mathcal{N}(A) = 1 = 4 - 3$.
- $\dim \mathcal{N}(A') = 4 - \operatorname{rank} A = 1$ (Theorem 4.12).

Exercise 4.7. For $t \in \mathbb{R}$ let

$$A_t = \begin{pmatrix} t & 1 & 1 & 1 \\ 1 & t & 1 & 1 \\ 1 & 1 & t & 1 \\ 1 & 1 & 1 & t \end{pmatrix}$$

Find $\operatorname{rank} A_t$ for every t, and a basis of $\mathcal{N}(A_t)$ whenever A_t is not invertible.

Solution. Reverse the order of the rows ($R_1 \leftrightarrow R_4$, $R_2 \leftrightarrow R_3$), so that the first row is $(1,1,1,t)$, and eliminate:

$$\begin{pmatrix} 1 & 1 & 1 & t \\ 1 & 1 & t & 1 \\ 1 & t & 1 & 1 \\ t & 1 & 1 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1,\ R_4 \leftarrow R_4 - tR_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & 0 & t-1 & 1-t \\ 0 & t-1 & 0 & 1-t \\ 0 & 1-t & 1-t & 1-t^2 \end{pmatrix}$$

$$\xrightarrow[R_2 \leftrightarrow R_3]{R_4 \leftarrow R_4 + R_2 + R_3} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & t-1 & 1-t \\ 0 & 0 & 0 & -(t-1)(t+3) \end{pmatrix} = E_t$$

since $(1-t^2) + 2(1-t) = (1-t)(3+t)$. The pivot columns can be read off before the leading entries are scaled to 1 (Remark 4.2).

- $t \neq 1$, $t \neq -3$: all four diagonal entries of E_t are nonzero, so $\operatorname{rank} A_t = 4$ (Definition 4.11) and A_t is invertible (Theorem 4.13(d)).

- $t = 1$: rows 2, 3, 4 of E_1 are zero, so $\operatorname{rank} A_1 = 1$. Here $\mathcal{N}(A_1)$ is given by $x_1 + x_2 + x_3 + x_4 = 0$, with special solutions $(-1, 1, 0, 0)'$, $(-1, 0, 1, 0)'$, $(-1, 0, 0, 1)'$ as a basis; $\dim \mathcal{N}(A_1) = 3$.
- $t = -3$: E_{-3} has nonzero rows $(1, 1, 1, -3)$, $(0, -4, 0, 4)$, $(0, 0, -4, 4)$ and a zero row, so $\operatorname{rank} A_{-3} = 3$ and $\dim \mathcal{N}(A_{-3}) = 1$ (Theorem 4.12). Every row of A_t has sum $t + 3$, so $A_{-3}\mathbf{1}_4 = 0$; the nonzero vector $\mathbf{1}_4 = (1, 1, 1, 1)'$ is a basis of $\mathcal{N}(A_{-3})$ (Proposition 3.12(a), Theorem 4.5(d)).

Exercise 4.8. Let

$$A = \begin{pmatrix} 1 & -2 & 3 & -1 \\ 1 & -1 & 2 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 1 & 2 & -1 & 3 \end{pmatrix}$$

(each column is an arithmetic progression).

(a) Find the RREF R and $\operatorname{rank} A$, and the full-rank factorization $A = BC$ of Theorem 4.21(a).

(b) Compute $B'B$, CC' and the matrices L, K of Theorem 4.21(c), and verify $LB = I_2$ and $CK = I_2$.

(c) Find a basis of $\mathcal{N}(A)$.

Solution. (a) Subtract row 1 from the other rows; rows 2 to 5 become $k(0, 1, -1, 1)$ for $k = 1, 2, 3, 4$:

$$A \xrightarrow[}{R_i \leftarrow R_i - R_1 \ (i=2,\dots,5)} \begin{pmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & -2 & 2 \\ 0 & 3 & -3 & 3 \\ 0 & 4 & -4 & 4 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 + 2R_2]{R_i \leftarrow R_i - (i-1)R_2 \ (i=3,4,5)} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

So $\operatorname{rank} A = 2$, with pivot columns $1, 2$, and

$$B = \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

Columns 3, 4 of R say $a_3 = a_1 - a_2$ and $a_4 = a_1 + a_2$; check: $a_1 - a_2 = (3, 2, 1, 0, -1)'$ and $a_1 + a_2 = (-1, 0, 1, 2, 3)'$.

(b) With $a_1 = \mathbf{1}_5$ and $a_2 = (-2, -1, 0, 1, 2)'$,

$$B'B = \begin{pmatrix} a_1'a_1 & a_1'a_2 \\ a_2'a_1 & a_2'a_2 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 10 \end{pmatrix}, \qquad CC' = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

($\operatorname{row}_1(C) \operatorname{row}_2(C)' = 0 - 1 + 1 = 0$). Hence

$$L = \begin{pmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{10} \end{pmatrix} B' = \begin{pmatrix} \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{10} & 0 & \frac{1}{10} & \frac{1}{5} \end{pmatrix}, \qquad K = \frac{1}{3}C' = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Then $LB = \operatorname{diag}(\frac{1}{5}, \frac{1}{10})B'B = \operatorname{diag}(\frac{1}{5}, \frac{1}{10}) \operatorname{diag}(5, 10) = I_2$ and $CK = \frac{1}{3}CC' = I_2$.

(c) $\mathcal{N}(A) = \mathcal{N}(C)$ (Theorem 4.21(b)); $Cx = 0$ reads $x_1 = -x_3 - x_4$, $x_2 = x_3 - x_4$, and the special solutions $(-1, 1, 1, 0)'$ and $(-1, -1, 0, 1)'$ form a basis. They are the relations $a_3 = a_1 - a_2$ and $a_4 = a_1 + a_2$.

Exercise 4.9. For $A = BC$ of Exercise 4.8:

(a) compute $A'A$ from $A'A = C'(B'B)C$;

(b) row-reduce $A'A$, and check that $\operatorname{rank}(A'A) = \operatorname{rank} A$ and $\mathcal{N}(A'A) = \mathcal{N}(A)$;

(c) find $\operatorname{rank}(AA')$ without computing the 5×5 matrix AA'.

Solution. (a) $A'A = C'B'BC$ (Theorem 1.9). Let $\gamma_1 = (1,0,1,1)$ and $\gamma_2 = (0,1,-1,1)$ be the rows of C. The columns of $C'\operatorname{diag}(5,10)$ are $5\gamma_1'$ and $10\gamma_2'$, so the outer product expansion (Theorem 1.4(d)) gives

$$A'A = 5\gamma_1'\gamma_1 + 10\gamma_2'\gamma_2 = \begin{pmatrix} 5 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 \\ 5 & 0 & 5 & 5 \\ 5 & 0 & 5 & 5 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 10 & -10 & 10 \\ 0 & -10 & 10 & -10 \\ 0 & 10 & -10 & 10 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 5 & 5 \\ 0 & 10 & -10 & 10 \\ 5 & -10 & 15 & -5 \\ 5 & 10 & -5 & 15 \end{pmatrix}$$

(b) Scale $R_1 \leftarrow \frac{1}{5}R_1$, $R_2 \leftarrow \frac{1}{10}R_2$, giving rows $(1,0,1,1)$ and $(0,1,-1,1)$; then $R_3 \leftarrow R_3 - 5R_1 + 10R_2$ and $R_4 \leftarrow R_4 - 5R_1 - 10R_2$ give zero rows, since

$$(5,-10,15,-5) - (5,0,5,5) + (0,10,-10,10) = 0, \qquad (5,10,-5,15) - (5,0,5,5) - (0,10,-10,10) = 0$$

The RREF of $A'A$ is the matrix R of Exercise 4.8 without its last row. So $\operatorname{rank}(A'A) = 2 = \operatorname{rank} A$, and $A'Ax = 0$ and $Ax = 0$ both reduce to $Cx = 0$: $\mathcal{N}(A'A) = \mathcal{N}(A)$, as Theorem 4.18(a),(b) asserts.

(c) $\operatorname{rank}(AA') = \operatorname{rank} A = 2$ (Theorem 4.18(b)).

Exercise 4.10. Let $S \in \mathbb{R}^{4\times 4}$ have entries $s_{12} = s_{23} = s_{34} = 1$ and all other entries 0, so that $Se_1 = 0$ and $Se_j = e_{j-1}$ $(j = 2,3,4)$.

(a) Compute S^2, S^3, S^4, and $\operatorname{rank} S^k$ for $k = 0, \ldots, 4$.

(b) Show that Sylvester's inequality for the product $S \cdot S^k$ is an equality for $k = 1,2,3$.

(c) Compute $S'S$, SS' and $\operatorname{rank}(S + S')$.

Solution. (a) $\operatorname{col}_j(S^k) = S^k e_j$ (Theorem 1.4(b)), and applying S repeatedly gives $S^k e_j = e_{j-k}$ for $j > k$ and $S^k e_j = 0$ for $j \leqslant k$. So

$$S^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad S^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad S^4 = 0$$

The nonzero columns of S^k are $e_1, \ldots, e_{4-k}$, which are independent and span $\mathcal{C}(S^k)$; so $\operatorname{rank} S^k = 4 - k$ $(k = 0, \ldots, 4)$.

(b) Theorem 4.19 gives $\operatorname{rank}(S \cdot S^k) \geqslant \operatorname{rank} S + \operatorname{rank} S^k - 4 = 3 + (4-k) - 4 = 3 - k$, and by (a) $\operatorname{rank} S^{k+1} = 4 - (k+1) = 3 - k$.

(c) S' has entries 1 in positions $(2,1)$, $(3,2)$, $(4,3)$ and 0 elsewhere, so $S'e_j = e_{j+1}$ $(j = 1,2,3)$ and $S'e_4 = 0$. Hence $S'Se_1 = 0$, $S'Se_j = S'e_{j-1} = e_j$ $(j \geqslant 2)$, and $SS'e_j = Se_{j+1} = e_j$ $(j \leqslant 3)$, $SS'e_4 = 0$:

$$S'S = \operatorname{diag}(0,1,1,1), \qquad SS' = \operatorname{diag}(1,1,1,0),$$

both of rank $3 = \operatorname{rank} S$ (Theorem 4.18(b)). For $S + S'$:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[R_3 \leftrightarrow R_4]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

an echelon form with four nonzero rows; so $\operatorname{rank}(S + S') = 4$, below the bound $\operatorname{rank} S + \operatorname{rank} S' = 6$ of Corollary 4.17.

Problems

Problem 4.1. Let $A \in \mathbb{R}^{m\times n}$ have rank r. Show that A is the sum of r matrices of rank 1.

Solution. If $r = 0$, then $A = 0$, the empty sum. Let $r \geqslant 1$, and let $A = BC$ be the full-rank factorization of Theorem 4.21(a), where $B \in \mathbb{R}^{m\times r}$ has columns $b_1, \dots, b_r$ and $C \in \mathbb{R}^{r\times n}$. By the outer product expansion (Theorem 1.4(d)),

$$A = \sum_{j=1}^{r} A_j, \qquad A_j = b_j \operatorname{row}_j(C) \in \mathbb{R}^{m\times n}$$

- The columns of B are independent, and so are the rows of C (Theorem 4.13(b),(c)); hence every b_j and every $\operatorname{row}_j(C)$ is nonzero, since a list containing 0 is dependent (Proposition 3.12(a)).
- If the lth entry c_{jl} of $\operatorname{row}_j(C)$ is nonzero, then $\operatorname{col}_l(A_j) = c_{jl}b_j \neq 0$; so $A_j \neq 0$, and $\operatorname{rank} A_j \geqslant 1$.
- $\operatorname{rank} A_j \leqslant \operatorname{rank} b_j \leqslant 1$ (Theorems 4.15(a) and 4.13(a)), as b_j has one column.

So each A_j has rank 1.

Problem 4.2. Let $A \in \mathbb{R}^{m\times n}$. Show that $\operatorname{rank} A = 1$ if and only if $A = uv'$ for some nonzero $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$.

Solution.

- Let $\operatorname{rank} A = 1$, and let $A = BC$ be a full-rank factorization (Theorem 4.21(a)), with $B \in \mathbb{R}^{m\times 1}$ and $C \in \mathbb{R}^{1\times n}$. Put $u = B$ and $v = C'$, so $A = uv'$. The single column of B is independent (Theorem 4.13(b)), so $u \neq 0$; the single row of C is independent (Theorem 4.13(c)), so $v \neq 0$ (Proposition 3.12(a)).
- Let $A = uv'$ with $u_i \neq 0$ and $v_j \neq 0$ for some i, j. Then $\operatorname{rank} A \leqslant \operatorname{rank} u \leqslant 1$ (Theorems 4.15(a) and 4.13(a)). Also $a_{ij} = u_i v_j \neq 0$, so $A \neq 0$; as rank 0 forces $A = 0$ (Theorem 4.8), $\operatorname{rank} A = 1$.

Problem 4.3. (a) Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$. Show that $\mathcal{C}(A) \subseteq \mathcal{C}(B)$ if and only if $A = BX$ for some $X \in \mathbb{R}^{p\times n}$.

(b) Show that if $\mathcal{C}(A) \subseteq \mathcal{C}(B)$ and $\operatorname{rank} A = \operatorname{rank} B$, then $\mathcal{C}(A) = \mathcal{C}(B)$.

(c) Let $A \in \mathbb{R}^{m\times n}$ and $D \in \mathbb{R}^{q\times n}$. Show that $\mathcal{R}(A) \subseteq \mathcal{R}(D)$ if and only if $A = YD$ for some $Y \in \mathbb{R}^{m\times q}$.

Solution. (a) If $A = BX$, then $\mathcal{C}(A) \subseteq \mathcal{C}(B)$ by Theorem 4.15(a). Conversely, every column a_j of A lies in $\mathcal{C}(B)$, so $a_j = Bx_j$ for some $x_j \in \mathbb{R}^p$. With $X = [x_1 \ \cdots \ x_n]$, $\operatorname{col}_j(BX) = Bx_j = a_j$ (Theorem 1.4(a)), so $BX = A$.

(b) $\mathcal{C}(A)$ is a subspace of $\mathcal{C}(B)$ of the same dimension, so the two are equal (Theorem 4.5(a)).

(c) If $A = YD$, then $y'A = (y'Y)D = (Y'y)'D \in \mathcal{R}(D)$ for every $y \in \mathbb{R}^m$. Conversely, every row of A lies in $\mathcal{R}(D)$: $\operatorname{row}_i(A) = z_i'D$ for some $z_i \in \mathbb{R}^q$. Let Y have rows $z_1', \dots, z_m'$. Then $\operatorname{row}_i(YD) = \operatorname{row}_i(Y)D = z_i'D = \operatorname{row}_i(A)$ (Theorem 1.4(c)), so $YD = A$.

Problem 4.4. Show that $\operatorname{rank}(A'AB) = \operatorname{rank}(AB) = \operatorname{rank}(ABB')$ for all $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

Solution. Apply Theorem 4.18(b) to AB, with $(AB)' = B'A'$ (Theorem 1.9), and Theorem 4.15(a) twice:

$$\operatorname{rank}(AB) = \operatorname{rank}\Big((AB)'AB\Big) = \operatorname{rank}\Big(B'(A'AB)\Big) \leqslant \operatorname{rank}(A'AB) = \operatorname{rank}\Big(A'(AB)\Big) \leqslant \operatorname{rank}(AB),$$

$$\operatorname{rank}(AB) = \operatorname{rank}\Big(AB(AB)'\Big) = \operatorname{rank}\Big((ABB')A'\Big) \leqslant \operatorname{rank}(ABB') = \operatorname{rank}\Big((AB)B'\Big) \leqslant \operatorname{rank}(AB)$$

So equality holds throughout.

Problem 4.5. Let $A \in \mathbb{R}^{m\times n}$ with $m < n$. Show that there is no $B \in \mathbb{R}^{m\times n}$ with $B'A = I_n$.

Solution. If $B'A = I_n$, then by Theorems 4.13(a),(d) and 4.15(a),

$$n = \operatorname{rank} I_n = \operatorname{rank}(B'A) \leqslant \operatorname{rank} A \leqslant m < n \text{ ✘}$$

Problem 4.6. Let $A, B \in \mathbb{R}^{m\times n}$. Show that $\operatorname{rank}(-B) = \operatorname{rank} B$ and $\operatorname{rank}(A+B) \geqslant |\operatorname{rank} A - \operatorname{rank} B|$.

Solution. $\mathcal{C}(-B) = \{-Bx : x \in \mathbb{R}^n\} = \{B(-x) : x \in \mathbb{R}^n\} = \mathcal{C}(B)$, so $\operatorname{rank}(-B) = \operatorname{rank} B$. Since $A = (A+B) + (-B)$, Corollary 4.17 gives $\operatorname{rank} A \leqslant \operatorname{rank}(A+B) + \operatorname{rank} B$, that is, $\operatorname{rank}(A+B) \geqslant \operatorname{rank} A - \operatorname{rank} B$. Exchanging A and B gives $\operatorname{rank}(A+B) \geqslant \operatorname{rank} B - \operatorname{rank} A$.

Problem 4.7. Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^2 = I$. Show that $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) = n$.

Solution.

- $2I = (I-A) + (I+A)$, and $2I = \operatorname{diag}(2, \ldots, 2)$ is invertible (Definition 2.12). By Theorem 4.13(d) and Corollary 4.17, $n = \operatorname{rank}(2I) \leqslant \operatorname{rank}(I-A) + \operatorname{rank}(I+A)$.
- $(I-A)(I+A) = I + A - A - A^2 = I - A^2 = 0$ (Theorem 1.3), so $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) \leqslant n$ by Theorem 4.19.

Together, $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) = n$.

Problem 4.8. Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^2 = A$. Show that

(a) $\mathcal{C}(A) = \mathcal{N}(I-A)$;

(b) $\operatorname{rank} A + \operatorname{rank}(I-A) = n$;

(c) $(I-A)^2 = I - A$ and $\mathcal{C}(I-A) = \mathcal{N}(A)$.

Solution. (a) If $y = Ax$, then $(I-A)y = Ax - A^2x = 0$ (Theorem 1.3). If $(I-A)y = 0$, then $y = Ay \in \mathcal{C}(A)$.

(b) By Theorem 4.12 and (a), $\operatorname{rank}(I-A) = n - \dim \mathcal{N}(I-A) = n - \dim \mathcal{C}(A) = n - \operatorname{rank} A$.

(c) $(I-A)^2 = I - 2A + A^2 = I - A$. Applying (a) to the matrix $I-A$ gives $\mathcal{C}(I-A) = \mathcal{N}(I-(I-A)) = \mathcal{N}(A)$.

Problem 4.9. Let $A \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{m\times p}$, $B \in \mathbb{R}^{q\times p}$, and

$$M = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \in \mathbb{R}^{(m+q)\times(n+p)},$$

with a zero block of size $q \times n$.

(a) Show that $\operatorname{rank} M \geqslant \operatorname{rank} A + \operatorname{rank} B$.

(b) Show that the inequality can be strict, using the matrix S of Exercise 4.10 with blocks of size 2×2.

Solution. (a) Let $r = \operatorname{rank} A$ and $s = \operatorname{rank} B$, and let $a_{k_1}, \ldots, a_{k_r}$ and $b_{l_1}, \ldots, b_{l_s}$ be the pivot columns of A and of B, bases of $\mathcal{C}(A)$ and $\mathcal{C}(B)$ (Theorem 4.8(a); a list is empty if the rank is 0). Let c_l denote column l of C. Columns k_i and $n + l_j$ of M are $\binom{a_{k_i}}{0}$ and $\binom{c_{l_j}}{b_{l_j}}$. Suppose

$$\sum_{i=1}^{r} \alpha_i \begin{pmatrix} a_{k_i} \\ 0 \end{pmatrix} + \sum_{j=1}^{s} \beta_j \begin{pmatrix} c_{l_j} \\ b_{l_j} \end{pmatrix} = 0$$

The last q entries give $\sum_j \beta_j b_{l_j} = 0$, so every $\beta_j = 0$; then the first m entries give $\sum_i \alpha_i a_{k_i} = 0$, so every $\alpha_i = 0$. These $r + s$ independent vectors lie in $\mathcal{C}(M)$, so $\operatorname{rank} M \geqslant r + s$ (Theorem 4.5(c)).

(b) With $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$,

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} N & E \\ 0 & N \end{pmatrix},$$

and $\operatorname{rank} S = 3 > 1 + 1 = \operatorname{rank} N + \operatorname{rank} N$ (Exercise 4.10(a)). For $C = 0$, equality holds (Theorem 4.16(c)).

Problem 4.10 (Frobenius' inequality)**.** Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{p\times q}$.

(a) Show that $\operatorname{rank}(AB) + \operatorname{rank}(BC) \leqslant \operatorname{rank} B + \operatorname{rank}(ABC)$.

(b) Show that (a) with $B = I_n$ is the rank form of Sylvester's inequality (Theorem 4.19).

Solution. (a)

- $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$ (Theorem 4.15(a)). Extend a basis $z_1, \ldots, z_s$ of $\mathcal{N}(B)$ to a basis $z_1, \ldots, z_s$, $x_1, \ldots, x_t$ of $\mathcal{N}(AB)$ (Theorem 4.5(b)). By Theorem 4.12 (p columns), $t = (p - \operatorname{rank}(AB)) - (p - \operatorname{rank} B) = \operatorname{rank} B - \operatorname{rank}(AB)$.
- Likewise $\mathcal{N}(BC) \subseteq \mathcal{N}(ABC)$; extend a basis $v_1, \ldots, v_\sigma$ of $\mathcal{N}(BC)$ to a basis $v_1, \ldots, v_\sigma$, $w_1, \ldots, w_\tau$ of $\mathcal{N}(ABC)$. With q columns, $\tau = \operatorname{rank}(BC) - \operatorname{rank}(ABC)$.
- Each Cw_j lies in $\mathcal{N}(AB)$, as $AB(Cw_j) = ABCw_j = 0$; so $Cw_j = \sum_i d_i z_i + \sum_l \mu_l x_l$ for some numbers d_i, μ_l, and $BCw_j = \sum_l \mu_l Bx_l$, since $Bz_i = 0$. Thus $BCw_1, \ldots, BCw_\tau$ lie in $U = \operatorname{span}\{Bx_1, \ldots, Bx_t\}$, and $\dim U \leqslant t$, as the spanning list contains a basis of U (Theorem 4.3).
- $BCw_1, \ldots, BCw_\tau$ are independent: if $\sum_j c_j BCw_j = 0$, then $\sum_j c_j w_j \in \mathcal{N}(BC)$ is a combination of $v_1, \ldots, v_\sigma$, and since $v_1, \ldots, v_\sigma, w_1, \ldots, w_\tau$ are independent, every $c_j = 0$.
- By Theorem 4.5(c), $\tau \leqslant \dim U \leqslant t$, that is, $\operatorname{rank}(BC) - \operatorname{rank}(ABC) \leqslant \operatorname{rank} B - \operatorname{rank}(AB)$.

(b) For $B = I_n$ (so $p = n$), (a) reads $\operatorname{rank} A + \operatorname{rank} C \leqslant n + \operatorname{rank}(AC)$ (Theorems 1.3 and 4.13(d)), that is, $\operatorname{rank}(AC) \geqslant \operatorname{rank} A + \operatorname{rank} C - n$ for $A \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times q}$.

Chapter 5. Block matrices and determinants

Block matrices

Definition 5.1. Let $A \in \mathbb{R}^{m\times n}$. A **partition** of A splits the rows into consecutive groups of $m_1, \dots, m_r$ rows ($m_1 + \cdots + m_r = m$) and the columns into consecutive groups of $n_1, \dots, n_s$ columns ($n_1 + \cdots + n_s = n$). Put $M_0 = 0$, $M_i = m_1 + \cdots + m_i$ and $N_0 = 0$, $N_j = n_1 + \cdots + n_j$: row group i consists of the rows $M_{i-1} + 1, \dots, M_i$, and column group j of the columns $N_{j-1} + 1, \dots, N_j$. The **block** $A_{ij} \in \mathbb{R}^{m_i \times n_j}$ consists of the entries in row group i and column group j:

$$(A_{ij})_{xy} = a_{M_{i-1}+x,\, N_{j-1}+y} \qquad (1 \leqslant x \leqslant m_i,\ 1 \leqslant y \leqslant n_j)$$

We write

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1s} \\ \vdots & & \vdots \\ A_{r1} & \cdots & A_{rs} \end{pmatrix}$$

Lines between rows and between columns may mark the partition. The group sizes are arbitrary positive integers, so a block A_{ij} is square only when $m_i = n_j$. Two matrices of the same size, partitioned in the same way, are equal if and only if their corresponding blocks are equal (both statements say that all entries agree, Definition 1.1).

Definition 5.2. Let $A \in \mathbb{R}^{m\times n}$ be partitioned with row groups $m_1, \dots, m_r$ and column groups $n_1, \dots, n_s$.

(a) A partition of $B \in \mathbb{R}^{m\times n}$ is **conformal for addition** with that of A if it has the same row groups $m_1, \dots, m_r$ and the same column groups $n_1, \dots, n_s$.

(b) A partition of $B \in \mathbb{R}^{n\times p}$ is **conformal for multiplication** with that of A (in the product AB) if its row groups are $n_1, \dots, n_s$, the column groups of A. Its column groups $p_1, \dots, p_t$ ($p_1 + \cdots + p_t = p$) may be any.

Conformality for multiplication restricts only the column groups of A and the row groups of B; the row groups m_i of A and the column groups p_k of B are free. It means exactly that every block $A_{ij} \in \mathbb{R}^{m_i \times n_j}$ has as many columns as every block $B_{jk} \in \mathbb{R}^{n_j \times p_k}$ has rows, so that the product $A_{ij}B_{jk} \in \mathbb{R}^{m_i \times p_k}$ is defined (Definition 1.2).

Theorem 5.3. Let $A \in \mathbb{R}^{m\times n}$ be partitioned into blocks $A_{ij} \in \mathbb{R}^{m_i \times n_j}$ ($1 \leqslant i \leqslant r$, $1 \leqslant j \leqslant s$).

(a) If $B \in \mathbb{R}^{m\times n}$ is partitioned conformally for addition, into blocks $B_{ij} \in \mathbb{R}^{m_i \times n_j}$, then $A + B$ has the blocks $A_{ij} + B_{ij}$, and for $c \in \mathbb{R}$ the matrix cA has the blocks cA_{ij} (same partition).

(b) Partition $A' \in \mathbb{R}^{n\times m}$ with row groups $n_1, \dots, n_s$ and column groups $m_1, \dots, m_r$. Its (j, i) block is $A'_{ij} \in \mathbb{R}^{n_j \times m_i}$: transpose the array of blocks and also each block.

(c) If $B \in \mathbb{R}^{n\times p}$ is partitioned conformally for multiplication, into blocks $B_{jk} \in \mathbb{R}^{n_j \times p_k}$ ($1 \leqslant j \leqslant s$, $1 \leqslant k \leqslant t$), then $AB \in \mathbb{R}^{m\times p}$, partitioned with row groups $m_1, \dots, m_r$ and column groups $p_1, \dots, p_t$, has the blocks

$$(AB)_{ik} = \sum_{j=1}^{s} A_{ij}B_{jk} = A_{i1}B_{1k} + A_{i2}B_{2k} + \cdots + A_{is}B_{sk} \in \mathbb{R}^{m_i \times p_k},$$

where the term $A_{ij}B_{jk}$ is the product of an $m_i \times n_j$ and an $n_j \times p_k$ matrix.

So conformal block matrices are added and multiplied as if the blocks were numbers, but the order of the factors in each $A_{ij}B_{jk}$ must be kept. No group size has to equal another, and no block has to be square. For $r = s = t = 2$,

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

whenever A_{11} and A_{21} have as many columns as B_{11} and B_{12} have rows (the second groups then match as well, since $n_1 + n_2 = n$ is the number of columns of A and of rows of B). The cases $r = t = 1,\ s = 2$; $r = t = 2,\ s = 1$; and $r = s = 1,\ t = 2$ read

$$[A_1\ A_2]\begin{pmatrix}B_1\\B_2\end{pmatrix} = A_1B_1 + A_2B_2, \qquad \begin{pmatrix}A_1\\A_2\end{pmatrix}[B_1\ B_2] = \begin{pmatrix}A_1B_1 & A_1B_2\\A_2B_1 & A_2B_2\end{pmatrix}, \qquad A[B_1\ B_2] = [AB_1\ \ AB_2]$$

Proof. Let M_i and N_j be as in Definition 5.1. Each row index u of A lies in exactly one row group i, and then $u = M_{i-1} + x$ with $1 \leqslant x \leqslant m_i$; likewise for columns.

(a) Fix i, j and $1 \leqslant x \leqslant m_i$, $1 \leqslant y \leqslant n_j$, and put $u = M_{i-1} + x$, $v = N_{j-1} + y$. Since B has the same groups as A, entry (x, y) of B_{ij} is b_{uv}. By Definition 1.1, entry (x, y) of block (i, j) of $A + B$ is

$$(A + B)_{uv} = a_{uv} + b_{uv} = (A_{ij})_{xy} + (B_{ij})_{xy} = (A_{ij} + B_{ij})_{xy},$$

and in the same way $(cA)_{uv} = ca_{uv} = (cA_{ij})_{xy}$.

(b) Row group j of A' consists of the rows $N_{j-1} + 1, \ldots, N_j$ and column group i of the columns $M_{i-1} + 1, \ldots, M_i$. By Definition 1.8, entry (y, x) of the (j, i) block of A' is

$$(A')_{N_{j-1}+y,\, M_{i-1}+x} = a_{M_{i-1}+x,\, N_{j-1}+y} = (A_{ij})_{xy} = (A'_{ij})_{yx}$$

(c) For a row group i of A and a column group k of B, let $A_{i*} \in \mathbb{R}^{m_i \times n}$ consist of the rows of A in row group i, and $B_{*k} \in \mathbb{R}^{n \times p_k}$ of the columns of B in column group k. By Definition 5.1, the columns of A_{i*} in column group j are, in order, the columns of A_{ij}; since the row groups of B are $n_1, \ldots, n_s$ (conformality), the rows of B_{*k} in row group j are, in order, the rows of B_{jk}. In blocks,

$$A_{i*} = [A_{i1}\ \cdots\ A_{is}], \qquad B_{*k} = \begin{pmatrix}B_{1k}\\\vdots\\B_{sk}\end{pmatrix}$$

- Columns. By Theorem 1.4(a), each column of AB is A times the column of B with the same index. The columns of B in column group k are, in order, the columns of B_{*k}, so the columns of AB in column group k are, in order, A times the columns of B_{*k}, that is, the columns of the product AB_{*k} (again Theorem 1.4(a)).
- Rows. By Theorem 1.4(c), each row of AB_{*k} is the row of A with the same index times B_{*k}. The rows of A in row group i are, in order, the rows of A_{i*}, so the rows of AB_{*k} in row group i are, in order, the rows of $A_{i*}B_{*k}$ (again Theorem 1.4(c)). These are the entries of AB in row group i and column group k, that is, the block $(AB)_{ik}$ (Definition 5.1): $(AB)_{ik} = A_{i*}B_{*k} \in \mathbb{R}^{m_i \times p_k}$.
- Inner sum. By the outer product expansion (Theorem 1.4(d)),

$$A_{i*}B_{*k} = \sum_{l=1}^{n} \operatorname{col}_l(A_{i*}) \operatorname{row}_l(B_{*k})$$

The column groups of A cut $l = 1, \ldots, n$ into s consecutive stretches of $n_1, \ldots, n_s$ indices. In stretch j, the columns $\operatorname{col}_l(A_{i*})$ are $\operatorname{col}_1(A_{ij}), \ldots, \operatorname{col}_{n_j}(A_{ij})$ and the rows $\operatorname{row}_l(B_{*k})$ are $\operatorname{row}_1(B_{jk}), \ldots, \operatorname{row}_{n_j}(B_{jk})$, in this order (see above). So the terms of stretch j add up to

$$\sum_{y=1}^{n_j} \operatorname{col}_y(A_{ij}) \operatorname{row}_y(B_{jk}) = A_{ij}B_{jk},$$

the outer product expansion of the product of $A_{ij} \in \mathbb{R}^{m_i \times n_j}$ and $B_{jk} \in \mathbb{R}^{n_j \times p_k}$ (Theorem 1.4(d)). Adding the s stretches,

$$(AB)_{ik} = A_{i*}B_{*k} = \sum_{j=1}^{s} A_{ij}B_{jk}$$ □

In entries: let $u = M_{i-1} + x$ (row x of row group i) and let column v be the zth column of column group k of B. By Definition 1.2, $(AB)_{uv} = \sum_{l=1}^{n} a_{ul}b_{lv}$. Split the sum into the stretches of the column groups of A and write $l = N_{j-1} + y$ in stretch j; by Definition 5.1 (for A, and for B with row groups $n_1, \dots, n_s$),

$$\sum_{l=N_{j-1}+1}^{N_j} a_{ul}b_{lv} = \sum_{y=1}^{n_j} (A_{ij})_{xy}(B_{jk})_{yz} = (A_{ij}B_{jk})_{xz},$$

so the part of stretch j is the entry (x, z) of $A_{ij}B_{jk}$: this is (c), entry by entry.

Example 5.4. Partition $A \in \mathbb{R}^{5\times 6}$ with row groups $(2, 3)$ and column groups $(4, 2)$, and $B \in \mathbb{R}^{6\times 5}$ with row groups $(4, 2)$ and column groups $(3, 2)$:

$$A = \left(\begin{array}{cccc|cc} 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ \hline 1 & 1 & 1 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array}\right) = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \qquad B = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ \hline 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \end{array}\right) = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

The blocks are

$$A_{11} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 4}, \quad A_{12} = I_2, \quad A_{21} = \mathbf{1}_3\mathbf{1}_4' \in \mathbb{R}^{3\times 4}, \quad A_{22} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2},$$

$$B_{11} = \begin{pmatrix} I_3 \\ \mathbf{1}_3' \end{pmatrix} \in \mathbb{R}^{4\times 3}, \quad B_{12} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 2}, \quad B_{21} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 3}, \quad B_{22} = I_2$$

Only A_{12} and B_{22} are square.

(a) The column groups $(4, 2)$ of A are the row groups of B, so the partitions are conformal for multiplication (Definition 5.2). By Theorem 5.3(c), $AB \in \mathbb{R}^{5\times 5}$ has row groups $(2, 3)$, those of A, and column groups $(3, 2)$, those of B. Every term is a product of a block with n_j columns and a block with n_j rows:

$$\begin{aligned} (AB)_{11} &= A_{11}B_{11} + A_{12}B_{21}: && (2\times 4)(4\times 3) + (2\times 2)(2\times 3) && \text{is } 2\times 3, \\ (AB)_{12} &= A_{11}B_{12} + A_{12}B_{22}: && (2\times 4)(4\times 2) + (2\times 2)(2\times 2) && \text{is } 2\times 2, \\ (AB)_{21} &= A_{21}B_{11} + A_{22}B_{21}: && (3\times 4)(4\times 3) + (3\times 2)(2\times 3) && \text{is } 3\times 3, \\ (AB)_{22} &= A_{21}B_{12} + A_{22}B_{22}: && (3\times 4)(4\times 2) + (3\times 2)(2\times 2) && \text{is } 3\times 2 \end{aligned}$$

(b) The column sums of B_{11} and B_{12} are $\mathbf{1}_4'B_{11} = (2, 2, 2)$ and $\mathbf{1}_4'B_{12} = (2, 2)$, so $A_{21}B_{11} = (\mathbf{1}_3\mathbf{1}_4')B_{11} = \mathbf{1}_3(\mathbf{1}_4'B_{11})$ (Theorem 1.3) has all entries 2, and so has $A_{21}B_{12}$. With $A_{12} = B_{22} = I_2$,

$$\begin{aligned} (AB)_{11} &= A_{11}B_{11} + B_{21} = \begin{pmatrix} 2 & 2 & 2 \\ 0 & -2 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 0 & -1 & -1 \end{pmatrix}, \\ (AB)_{12} &= A_{11}B_{12} + I_2 = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} + I_2 = \begin{pmatrix} 3 & 2 \\ 2 & -1 \end{pmatrix}, \\ (AB)_{21} &= A_{21}B_{11} + A_{22}B_{21} = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 0 & -2 \\ 0 & 1 & -1 \\ 1 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{pmatrix}, \\ (AB)_{22} &= A_{21}B_{12} + A_{22} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \\ 3 & 3 \end{pmatrix}, \end{aligned}$$

so

$$AB = \left(\begin{array}{ccc|cc} 3 & 2 & 1 & 3 & 2 \\ 0 & -1 & -1 & 2 & -1 \\ \hline 4 & 2 & 0 & 4 & 2 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 0 & 3 & 3 \end{array}\right)$$

(c) Check of one entry. Entry $(4,2)$ of AB lies in row group 2 (row $u = 4 = M_1 + 2$, so $x = 2$) and column group 1 ($z = 2$). As in the paragraph after the proof of Theorem 5.3, the sum over $l = 1, \dots, 6$ splits at $N_1 = 4$:

$$(AB)_{42} = \underbrace{1\cdot 0 + 1\cdot 1 + 1\cdot 0 + 1\cdot 1}_{l=1,\dots,4} + \underbrace{0\cdot 0 + 1\cdot 1}_{l=5,6} = 2 + 1 = 3$$

The first part is row 2 of A_{21} times column 2 of B_{11}, entry $(2,2)$ of $A_{21}B_{11}$; the second is row 2 of A_{22} times column 2 of B_{21}, entry $(2,2)$ of $A_{22}B_{21}$.

(d) A partition of B that is not conformal. Give B the row groups $(3,3)$, with any column groups. The product AB is still defined, but $A_{11} \in \mathbb{R}^{2\times 4}$ has 4 columns while the new B_{11} has 3 rows, so $A_{11}B_{11}$ is not defined, and the blocks of AB cannot be computed from these blocks. Changing the column groups of B, say to $(1,4)$, keeps the partitions conformal.

(e) By Theorem 5.3(b), $A' \in \mathbb{R}^{6\times 5}$ with row groups $(4,2)$ and column groups $(2,3)$ is $A' = \left(\begin{smallmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{smallmatrix}\right)$, where $A'_{11} \in \mathbb{R}^{4\times 2}$, $A'_{21} = \mathbf{1}_4\mathbf{1}'_3 \in \mathbb{R}^{4\times 3}$, $A'_{12} = I_2$ and $A'_{22} \in \mathbb{R}^{2\times 3}$.

Definition 5.5. Let $A \in \mathbb{R}^{n\times n}$ be partitioned with the same groups $n_1, \dots, n_r$ for rows and columns. Then the **diagonal blocks** $A_{ii} \in \mathbb{R}^{n_i\times n_i}$ are square, while an off-diagonal block $A_{ij} \in \mathbb{R}^{n_i\times n_j}$ $(i \neq j)$ is rectangular when $n_i \neq n_j$. A is **block diagonal**, written $A = \operatorname{diag}(A_{11}, \dots, A_{rr})$, if $A_{ij} = 0$ for $i \neq j$; **upper block triangular** if $A_{ij} = 0$ for $i > j$; and **lower block triangular** if $A_{ij} = 0$ for $i < j$.

Proposition 5.6. Let $A, B \in \mathbb{R}^{n\times n}$ be partitioned as in Definition 5.5, with the same groups.

(a) If A and B are upper (lower) block triangular, so is AB, and its diagonal blocks are $A_{ii}B_{ii}$.

(b) $\operatorname{diag}(A_{11}, \dots, A_{rr}) \operatorname{diag}(B_{11}, \dots, B_{rr}) = \operatorname{diag}(A_{11}B_{11}, \dots, A_{rr}B_{rr})$. If every A_{ii} is invertible, then $\operatorname{diag}(A_{11}, \dots, A_{rr})^{-1} = \operatorname{diag}(A_{11}^{-1}, \dots, A_{rr}^{-1})$.

(c) If $A_{11} \in \mathbb{R}^{p\times p}$ and $A_{22} \in \mathbb{R}^{q\times q}$ are invertible and $A_{12} \in \mathbb{R}^{p\times q}$, then

$$\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}$$

In (c) only the diagonal blocks must be square; for $p \neq q$ the blocks A_{12} and $-A_{11}^{-1}A_{12}A_{22}^{-1}$ are $p \times q$.

Proof. (a)

- The column groups $n_1, \dots, n_r$ of A are the row groups of B, so by Theorem 5.3(c) the (i,k) block of AB is $\sum_{j=1}^{r} A_{ij}B_{jk} \in \mathbb{R}^{n_i\times n_k}$.
- Let A and B be upper block triangular. Then $A_{ij} = 0$ unless $i \leqslant j$, and $B_{jk} = 0$ unless $j \leqslant k$. So the term $A_{ij}B_{jk}$ has a zero factor, and is 0, unless $i \leqslant j \leqslant k$.
- If $i > k$, no j satisfies $i \leqslant j \leqslant k$, so $(AB)_{ik} = 0$: AB is upper block triangular. If $i = k$, only $j = i$ remains, so $(AB)_{ii} = A_{ii}B_{ii}$.
- For lower block triangular A, B, the term j can be nonzero only if $i \geqslant j \geqslant k$; the same argument applies with the inequalities reversed.

(b) A block diagonal matrix is upper and lower block triangular, so by (a) the product is upper and lower block triangular, that is, block diagonal, with diagonal blocks $A_{ii}B_{ii}$. With $B_{ii} = A_{ii}^{-1}$ this gives $\operatorname{diag}(A_{11}, \dots, A_{rr}) \operatorname{diag}(A_{11}^{-1}, \dots, A_{rr}^{-1}) = \operatorname{diag}(I_{n_1}, \dots, I_{n_r}) = I_n$ (both have entry 1 at every (u, u) and 0 elsewhere), and likewise in the other order; so the second matrix is the inverse of the first (Definition 2.12).

(c) Call the two matrices T and X; both have row and column groups (p, q), so the 2×2 block formula of Theorem 5.3(c) applies. The four blocks of TX are

$$(TX)_{11} = A_{11}A_{11}^{-1} + A_{12} \cdot 0 = I_p, \quad (TX)_{12} = A_{11}\left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) + A_{12}A_{22}^{-1} = 0,$$
$$(TX)_{21} = 0 \cdot A_{11}^{-1} + A_{22} \cdot 0 = 0, \quad (TX)_{22} = 0 \cdot \left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) + A_{22}A_{22}^{-1} = I_q,$$

and those of XT are

$$(XT)_{11} = A_{11}^{-1}A_{11} + \left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) \cdot 0 = I_p, \quad (XT)_{12} = A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12}A_{22}^{-1}A_{22} = 0,$$
$$(XT)_{21} = 0 \cdot A_{11} + A_{22}^{-1} \cdot 0 = 0, \quad (XT)_{22} = 0 \cdot A_{12} + A_{22}^{-1}A_{22} = I_q$$

So $TX = XT = I_{p+q}$, and $X = T^{-1}$ (Definition 2.12). □

Example 5.7. Find the inverse of

$$T = \left(\begin{array}{cc|ccc} 3 & 2 & 1 & 0 & 2 \\ 1 & 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right) = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix} \in \mathbb{R}^{5\times 5}$$

Solution. The groups are $(2, 3)$: $A_{11} = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right)$ is 2×2, $A_{22} = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ is 3×3, and $A_{12} = \left(\begin{smallmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ is 2×3. By Proposition 2.13 ($3 \cdot 1 - 2 \cdot 1 = 1$), $A_{11}^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ -1 & 3 \end{smallmatrix}\right)$. For $Y = \left(\begin{smallmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$,

$$A_{22}Y = \begin{pmatrix} 1 & -1+1 & 1-1 \\ 0 & 1 & -1+1 \\ 0 & 0 & 1 \end{pmatrix} = I_3,$$

so $A_{22}^{-1} = Y$ (Theorem 2.18). Then

$$A_{11}^{-1}A_{12} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -2 & 0 \\ 2 & 3 & 1 \end{pmatrix},$$

$$-A_{11}^{-1}A_{12}A_{22}^{-1} = -\begin{pmatrix} -1 & -2 & 0 \\ 2 & 3 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ -2 & -1 & 0 \end{pmatrix},$$

a 2×3 block. By Proposition 5.6(c) with $p = 2$, $q = 3$,

$$T^{-1} = \left(\begin{array}{cc|ccc} 1 & -2 & 1 & 1 & -1 \\ -1 & 3 & -2 & -1 & 0 \\ \hline 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

Block elimination and the Schur complement

Proposition 5.8. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{m\times p}$, $B \in \mathbb{R}^{m\times q}$, $C \in \mathbb{R}^{n\times p}$, $D \in \mathbb{R}^{n\times q}$, and let $V \in \mathbb{R}^{n\times m}$, $F \in \mathbb{R}^{m\times n}$, $H \in \mathbb{R}^{p\times q}$. Then

$$\begin{pmatrix} I_m & 0 \\ V & I_n \end{pmatrix} M = \begin{pmatrix} A & B \\ C+VA & D+VB \end{pmatrix}, \qquad \begin{pmatrix} I_m & F \\ 0 & I_n \end{pmatrix} M = \begin{pmatrix} A+FC & B+FD \\ C & D \end{pmatrix},$$

$$M \begin{pmatrix} I_p & H \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B+AH \\ C & D+CH \end{pmatrix}$$

These **block elementary matrices** are invertible: $\left(\begin{smallmatrix} I_m & 0 \\ V & I_n \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} I_m & 0 \\ -V & I_n \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} I_m & F \\ 0 & I_n \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} I_m & -F \\ 0 & I_n \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I_p & H \\ 0 & I_q \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} I_p & -H \\ 0 & I_q \end{smallmatrix}\right)$.

None of A, B, C, D, V, F, H needs to be square.

Proof. • M has row groups (m, n) and column groups (p, q). The matrices $\left(\begin{smallmatrix} I_m & 0 \\ V & I_n \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I_m & F \\ 0 & I_n \end{smallmatrix}\right)$ have column groups (m, n), and $\left(\begin{smallmatrix} I_p & H \\ 0 & I_q \end{smallmatrix}\right)$ has row groups (p, q); so all three products are conformal, and Theorem 5.3(c) applies.

- First product: block row 1 is $(I_mA + 0C,\ I_mB + 0D) = (A,\ B)$, and block row 2 is $(VA + I_nC,\ VB + I_nD) = (C + VA,\ D + VB)$, where $VA \in \mathbb{R}^{n\times p}$ and $VB \in \mathbb{R}^{n\times q}$ have the sizes of C and D.
- Second product: block row 1 is $(I_mA + FC,\ I_mB + FD) = (A + FC,\ B + FD)$, and block row 2 is $(0A + I_nC,\ 0B + I_nD) = (C,\ D)$.
- Third product: block column 1 is $(AI_p + B0,\ CI_p + D0) = (A,\ C)$, and block column 2 is $(AH + BI_q,\ CH + DI_q) = (B + AH,\ D + CH)$.
- Thus left multiplication by $\left(\begin{smallmatrix} I & 0 \\ V & I \end{smallmatrix}\right)$ adds V times the first block row to the second, the block analogue of the row operation $R_2 \leftarrow R_2 + cR_1$ of Definition 2.2; right multiplication by $\left(\begin{smallmatrix} I & H \\ 0 & I \end{smallmatrix}\right)$ adds the first block column times H to the second.
- Inverses. The first identity with $A = I_m$, $B = 0$, $C = V$, $D = I_n$ (so $p = m$, $q = n$) and with $-V$ in place of V gives

$$\begin{pmatrix} I_m & 0 \\ -V & I_n \end{pmatrix} \begin{pmatrix} I_m & 0 \\ V & I_n \end{pmatrix} = \begin{pmatrix} I_m & 0 \\ V - VI_m & I_n - V\cdot 0 \end{pmatrix} = I_{m+n},$$

  and with $C = -V$ and V itself it gives the product in the other order, $\left(\begin{smallmatrix} I_m & 0 \\ -V+V & I_n \end{smallmatrix}\right) = I_{m+n}$. In the same way the second identity gives $\left(\begin{smallmatrix} I_m & -F \\ 0 & I_n \end{smallmatrix}\right)\left(\begin{smallmatrix} I_m & F \\ 0 & I_n \end{smallmatrix}\right) = \left(\begin{smallmatrix} I_m & F-F \\ 0 & I_n \end{smallmatrix}\right) = I_{m+n}$ and the product in the other order; the third case is the second with p, q, H in place of m, n, F. □

Theorem 5.9. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$ invertible, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{n\times p}$, $D \in \mathbb{R}^{n\times q}$. Then

$$\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}, \qquad \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix},$$

and hence $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{smallmatrix}\right) M \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$. The matrix $D - CA^{-1}B \in \mathbb{R}^{n\times q}$ is the **Schur complement** of A in M.

Only A has to be square: B, C, D and M itself may be rectangular.

Proof. • Apply Proposition 5.8 with $m = p$. The choice $V = -CA^{-1} \in \mathbb{R}^{n\times p}$ makes the $(2,1)$ block $C + VA = C - CA^{-1}A = 0$, and then $D + VB = D - CA^{-1}B$. This is the first identity.

- The choice $H = -A^{-1}B \in \mathbb{R}^{p\times q}$ makes $B+AH = B-AA^{-1}B = 0$, and then $D+CH = D-CA^{-1}B$. This is the second identity.
- Put $S = D-CA^{-1}B$. Applying the column step to the result of the row step, by the third identity of Proposition 5.8 with C replaced by 0 and D by S, $\left(\begin{smallmatrix} A & B \\ 0 & S \end{smallmatrix}\right)\left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & B-AA^{-1}B \\ 0 & S-0\cdot A^{-1}B \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & S \end{smallmatrix}\right)$. □

In Theorem 5.9 the left factor adds $-CA^{-1}$ times the first block row of M to the second, and the right factor adds the first block column times $-A^{-1}B$ to the second (Proposition 5.8). When D is invertible, the matrix $A - BD^{-1}C$ is likewise called the **Schur complement** of D in M.

Let R_1, R_2 be the block rows of a matrix with row groups (p, q). The **block row operations** are $R_1 \leftarrow XR_1$ and $R_2 \leftarrow YR_2$ with $X \in \mathbb{R}^{p\times p}$ and $Y \in \mathbb{R}^{q\times q}$ invertible, $R_2 \leftarrow R_2 + VR_1$ with $V \in \mathbb{R}^{q\times p}$, and $R_1 \leftarrow R_1 + WR_2$ with $W \in \mathbb{R}^{p\times q}$. They are the left multiplications by $\operatorname{diag}(X, I_q)$, $\operatorname{diag}(I_p, Y)$, $\left(\begin{smallmatrix} I_p & 0 \\ V & I_q \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I_p & W \\ 0 & I_q \end{smallmatrix}\right)$ (Theorem 5.3(c) and Proposition 5.8), and these matrices are invertible (Propositions 5.6(b) and 5.8). **Block elimination** applies the steps of Gauss–Jordan elimination (Definition 2.5) to the block rows of $[M \mid I]$, with the inverse of a pivot block in place of the reciprocal of a pivot entry: the first pivot is a diagonal block of M, the second its Schur complement, and the result is $[I \mid M^{-1}]$, as in the computation of A^{-1} after Theorem 2.18.

Corollary 5.10 (Block elimination)**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A and its Schur complement $S = D-CA^{-1}B$ are invertible, then M is invertible, the block row operations $R_2 \leftarrow R_2 - CA^{-1}R_1$, then $R_1 \leftarrow A^{-1}R_1$ and $R_2 \leftarrow S^{-1}R_2$, then $R_1 \leftarrow R_1 - A^{-1}BR_2$ take $[M \mid I_{p+q}]$ to $[I_{p+q} \mid M^{-1}]$,

$$M^{-1} = \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix}\begin{pmatrix} A^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix}\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix},$$

and $M = \left(\begin{smallmatrix} I_p & 0 \\ CA^{-1} & I_q \end{smallmatrix}\right)\operatorname{diag}(A, S)\left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$.

(b) If D and its Schur complement $F = A-BD^{-1}C$ are invertible, then M is invertible, the block row operations $R_1 \leftarrow R_1 - BD^{-1}R_2$, then $R_1 \leftarrow F^{-1}R_1$ and $R_2 \leftarrow D^{-1}R_2$, then $R_2 \leftarrow R_2 - D^{-1}CR_1$ take $[M \mid I_{p+q}]$ to $[I_{p+q} \mid M^{-1}]$,

$$M^{-1} = \begin{pmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{pmatrix}\begin{pmatrix} F^{-1} & 0 \\ 0 & D^{-1} \end{pmatrix}\begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix},$$

and $M = \left(\begin{smallmatrix} I_p & BD^{-1} \\ 0 & I_q \end{smallmatrix}\right)\operatorname{diag}(F, D)\left(\begin{smallmatrix} I_p & 0 \\ D^{-1}C & I_q \end{smallmatrix}\right)$.

For $p \neq q$ the blocks B and C are rectangular.

Proof. (a)

- The three steps are the left multiplications by $E_1 = \left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{smallmatrix}\right)$, $E_2 = \operatorname{diag}(I_p, S^{-1})\operatorname{diag}(A^{-1}, I_q) = \operatorname{diag}(A^{-1}, S^{-1})$ (Proposition 5.6(b)) and $E_3 = \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$.
- By Theorem 5.9 (with $n = q$), $E_1M = \left(\begin{smallmatrix} A & B \\ 0 & S \end{smallmatrix}\right)$. By the block product formula (Theorem 5.3(c)), $E_2E_1M = \left(\begin{smallmatrix} A^{-1}A & A^{-1}B \\ S^{-1}\cdot 0 & S^{-1}S \end{smallmatrix}\right) = \left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$. By Proposition 5.8 (second identity, with upper right block $-A^{-1}B$), $E_3E_2E_1M = \left(\begin{smallmatrix} I_p & A^{-1}B-A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = I_{p+q}$.
- Put $E = E_3E_2E_1$, the displayed product. Then $EM = I_{p+q}$, so M is invertible with $M^{-1} = E$ (Theorem 2.18), and $E[M \mid I_{p+q}] = [EM \mid E] = [I_{p+q} \mid M^{-1}]$ (Theorem 1.4(a)).

- $E_1^{-1} = \left(\begin{smallmatrix} I_p & 0 \\ CA^{-1} & I_q \end{smallmatrix}\right)$ and $E_3^{-1} = \left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$ (Proposition 5.8), and $E_2^{-1} = \operatorname{diag}(A, S)$ (Proposition 5.6(b)). So $M = E^{-1} = E_1^{-1}E_2^{-1}E_3^{-1}$ (Theorem 2.15(a),(b)).

(b) The three steps are the left multiplications by $E_1' = \left(\begin{smallmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right)$, $E_2' = \operatorname{diag}(F^{-1}, D^{-1})$ (as in (a)) and $E_3' = \left(\begin{smallmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{smallmatrix}\right)$. By Proposition 5.8 and Theorem 5.3(c),

$$E_1'M = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} F & 0 \\ C & D \end{pmatrix}, \qquad E_2'E_1'M = \begin{pmatrix} I_p & 0 \\ D^{-1}C & I_q \end{pmatrix}$$

and $E_3'E_2'E_1'M = I_{p+q}$. The last two bullets of (a) apply to $E' = E_3'E_2'E_1'$, the displayed product, with $E_1'^{-1} = \left(\begin{smallmatrix} I_p & BD^{-1} \\ 0 & I_q \end{smallmatrix}\right)$, $E_2'^{-1} = \operatorname{diag}(F, D)$ and $E_3'^{-1} = \left(\begin{smallmatrix} I_p & 0 \\ D^{-1}C & I_q \end{smallmatrix}\right)$ (Propositions 5.8 and 5.6(b)). □

Example 5.11. Partition $M \in \mathbb{R}^{5\times 5}$ with groups $(2, 3)$ for rows and columns:

$$M = \left(\begin{array}{cc|ccc} 2 & 1 & 2 & 1 & 3 \\ 1 & 1 & 1 & 1 & 2 \\ \hline 2 & 1 & 3 & 2 & 3 \\ 1 & 1 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 & 7 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix}, \quad C = B', \quad D = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix}$$

Here A is 2×2, D is 3×3, B is 2×3 and C is 3×2. By Proposition 2.13 ($2 \cdot 1 - 1 \cdot 1 = 1$), $A^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Then

$$A^{-1}B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad CA^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

of sizes 2×3 and 3×2, and the Schur complement of A is the 3×3 matrix

$$S = D - CA^{-1}B = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

By Theorem 5.9,

$$\left(\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 1 \end{array}\right) M \left(\begin{array}{cc|ccc} 1 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & -1 & -1 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right) = \left(\begin{array}{cc|ccc} 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

All matrices here have integer entries, because A^{-1} does. Multiplying on the left and on the right by the inverses of the two outer factors (Proposition 5.8),

$$\begin{aligned} M &= \begin{pmatrix} I_2 & 0 \\ CA^{-1} & I_3 \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix}\begin{pmatrix} I_2 & A^{-1}B \\ 0 & I_3 \end{pmatrix} \\ &= \left(\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}\right)\left(\begin{array}{cc|ccc} 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)\left(\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right), \end{aligned}$$

so M is the product of a lower block triangular, a block diagonal and an upper block triangular factor, whose off-diagonal blocks $CA^{-1} \in \mathbb{R}^{3\times 2}$ and $A^{-1}B \in \mathbb{R}^{2\times 3}$ are rectangular. Since $A' = A$ and $C = B'$, $A^{-1}B = (A^{-1})'C' = (CA^{-1})'$ (Theorems 2.15(c) and 1.9), and the right factor is the transpose of the left (Theorem 5.3(b)).

Determinants: definition and computation

Definition 5.12. Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$. For $n \geqslant 2$ and $1 \leqslant i, j \leqslant n$, let $A(i|j) \in \mathbb{R}^{(n-1)\times(n-1)}$ be the matrix obtained from A by deleting row i and column j. The **determinant** $\det A$ is the number defined by $\det(a_{11}) = a_{11}$ if $n = 1$ and, recursively, for $n \geqslant 2$ by

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A(1|j) = a_{11} \det A(1|1) - a_{12} \det A(1|2) + \cdots + (-1)^{1+n} a_{1n} \det A(1|n)$$

The number $\det A(i|j)$ is the **minor** of a_{ij}, and $c_{ij} = (-1)^{i+j} \det A(i|j)$ is its **cofactor**. Thus $\det A = a_{11}c_{11} + a_{12}c_{12} + \cdots + a_{1n}c_{1n}$, the **expansion along the first row**. Many books write $|A|$ for $\det A$.

For $n = 2$, $A(1|1) = (a_{22})$ and $A(1|2) = (a_{21})$, so

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21},$$

the number $ad - bc$ whose nonvanishing decides invertibility in Proposition 2.13. The signs $(-1)^{i+j}$ form a checkerboard with $+$ on the diagonal, for $n = 3$ $\left(\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}\right)$.

Theorem 5.13. Let $A \in \mathbb{R}^{n\times n}$ with $n \geqslant 2$, and let c_{ij} be the cofactors of A. For every row i and every column j,

$$\det A = a_{i1}c_{i1} + a_{i2}c_{i2} + \cdots + a_{in}c_{in} = a_{1j}c_{1j} + a_{2j}c_{2j} + \cdots + a_{nj}c_{nj}$$

(**expansion along row** i and **along column** j).

Proof. Induction on n. For $n = 2$, $A(2|1) = (a_{12})$ and $A(2|2) = (a_{11})$, so $c_{11} = a_{22}$, $c_{12} = -a_{21}$, $c_{21} = -a_{12}$, $c_{22} = a_{11}$, and the expansions along row 1, row 2, column 1 and column 2 are

$$a_{11}a_{22} - a_{12}a_{21}, \qquad -a_{21}a_{12} + a_{22}a_{11}, \qquad a_{11}a_{22} - a_{21}a_{12}, \qquad -a_{12}a_{21} + a_{22}a_{11},$$

all equal to $\det A$. Let $n \geqslant 3$, and assume the theorem for all $(n-1) \times (n-1)$ matrices. For $i \geqslant 2$ and $j \neq k$, let $A(1, i|j, k) \in \mathbb{R}^{(n-2)\times(n-2)}$ be A with rows $1, i$ and columns j, k deleted.

- Positions in the minors. Let $i \geqslant 2$ and $j \neq k$. Deleting column j of A moves column k to position $\bar{k}$, where $\bar{k} = k$ if $k < j$ and $\bar{k} = k - 1$ if $k > j$; deleting column k moves column j to position $\bar{j}$, where $\bar{j} = j$ if $j < k$ and $\bar{j} = j - 1$ if $j > k$. Deleting row 1 moves row i to position $i - 1$; deleting row i leaves row 1 in position 1. So a_{ik} stands in row $i - 1$ and column $\bar{k}$ of $A(1|j)$; a_{ij} in row $i - 1$ and column $\bar{j}$ of $A(1|k)$; a_{1j} in row 1 and column $\bar{j}$ of $A(i|k)$; and a_{1k} in row 1 and column $\bar{k}$ of $A(i|j)$. Deleting the row and the column of that entry from the minor leaves, in all four cases, the rows of A other than $1, i$ and the columns other than j, k in their original order, that is, $A(1, i|j, k)$.
- Row $i \geqslant 2$ (row 1 is the definition). Expand $\det A$ by Definition 5.12, and each $\det A(1|j)$ along its row $i - 1$, whose entries are the a_{ik} with $k \neq j$ (induction hypothesis):

  $$\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A(1|j) = \sum_{j=1}^{n} \sum_{k\neq j} (-1)^{1+j} (-1)^{(i-1)+\bar{k}} a_{1j}a_{ik} \det A(1, i|j, k)$$

  In the expansion along row i, expand each $\det A(i|k)$ along its first row (Definition 5.12), whose entries are the a_{1j} with $j \neq k$, in columns $\bar{j}$:

  $$\sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det A(i|k) = \sum_{k=1}^{n} \sum_{j\neq k} (-1)^{i+k} (-1)^{1+\bar{j}} a_{ik}a_{1j} \det A(1, i|j, k)$$

  Both double sums run over all pairs $j \neq k$, and the term of the pair (j, k) has the factor $a_{1j}a_{ik} \det A(1, i|j, k)$ in both; only the signs remain to be compared. If $k < j$, then $\bar{k} = k$, $\bar{j} = j - 1$, and both signs are $(-1)^{i+j+k}$. If $k > j$, then $\bar{k} = k - 1$, $\bar{j} = j$, and both signs are $(-1)^{i+j+k+1}$. So the two sums are equal, and the expansion along row i equals $\det A$.

- Column j. With k as summation index, Definition 5.12 reads $\det A = \sum_{k=1}^{n}(-1)^{1+k}a_{1k}\det A(1|k)$, and its term $k=j$ is $(-1)^{1+j}a_{1j}\det A(1|j)$, the term $i=1$ of the expansion along column j. In the terms $i \geqslant 2$ of the expansion along column j, expand $\det A(i|j)$ along its first row (Definition 5.12), whose entries are the a_{1k} with $k \neq j$, in columns $\bar{k}$; in the terms $k \neq j$ of the definition, expand $\det A(1|k)$ along its column $\bar{j}$, which comes from column j of A and whose entries are the a_{ij} with $i \geqslant 2$, in rows $i-1$ (induction hypothesis). The minors are $A(1,i|j,k)$ in both cases (first bullet), so these two groups of terms become

  $$\sum_{i=2}^{n}\sum_{k\neq j}(-1)^{i+j}(-1)^{1+\bar{k}}a_{ij}a_{1k}\det A(1,i|j,k) \quad \text{and} \quad \sum_{k\neq j}\sum_{i=2}^{n}(-1)^{1+k}(-1)^{(i-1)+\bar{j}}a_{1k}a_{ij}\det A(1,i|j,k)$$

  Both run over all pairs (i,k) with $i \geqslant 2$ and $k \neq j$, with the factor $a_{ij}a_{1k}\det A(1,i|j,k)$ in both. If $k<j$, then $\bar{k}=k$, $\bar{j}=j-1$, and both signs are $(-1)^{i+j+k+1}$. If $k>j$, then $\bar{k}=k-1$, $\bar{j}=j$, and both signs are $(-1)^{i+j+k}$. So the two groups are equal, and the expansion along column j equals $\det A$. □

In practice one expands along a row or column with many zeros.

Example 5.14. Compute $\det A$ for $A = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$ by expansion along the first row, and check the value by expansion along the third column.

Solution. Row 1 has the nonzero entries $a_{11}=2$ and $a_{13}=1$, so $\det A = 2\det A(1|1) + 1\det A(1|3)$. Each 3×3 minor has a row with a single nonzero entry, and expanding along that row (Theorem 5.13),

$$\det A(1|1) = \det\begin{pmatrix} 3 & 0 & 2 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (-1)^{3+3}\cdot 1 \cdot \det\begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} = 6,$$

$$\det A(1|3) = \det\begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = (-1)^{2+2}\cdot 1 \cdot \det\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = -1,$$

so $\det A = 2\cdot 6 + 1\cdot(-1) = 11$. Column 3 has the nonzero entries $a_{13}=1$ and $a_{33}=2$, with cofactors $c_{13} = \det A(1|3) = -1$ and

$$c_{33} = (-1)^{3+3}\det A(3|3) = \det\begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 1 \end{pmatrix} = (-1)^{1+1}\cdot 2 \cdot \det\begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix} = 6$$

(expansion along row 1). By Theorem 5.13, $\det A = 1\cdot(-1) + 2\cdot 6 = 11$.

Theorem 5.15. Let $A \in \mathbb{R}^{n\times n}$.

(a) If A is upper or lower triangular, then $\det A = a_{11}a_{22}\cdots a_{nn}$. In particular $\det \operatorname{diag}(d_1,\ldots,d_n) = d_1d_2\cdots d_n$ and $\det I_n = 1$.

(b) $\det A' = \det A$.

Proof. Both parts are proved by induction on n; for $n=1$ there is nothing to prove. Let $n \geqslant 2$.

(a)

- If A is lower (upper) triangular, the entries of its first row (column) other than a_{11} are 0, so the definition (the expansion along column 1, Theorem 5.13) gives $\det A = a_{11}\det A(1|1)$.
- $A(1|1)$ is triangular of the same kind with diagonal entries $a_{22},\ldots,a_{nn}$, so $\det A(1|1) = a_{22}\cdots a_{nn}$ by induction.

(b)

- Row 1 of A' is $(a_{11}, a_{21}, \ldots, a_{n1})$, and $A'(1|j) = A(j|1)'$: both consist of the entries a_{lk} with $l \neq j$ and $k \neq 1$, arranged as transposes of each other.
- By the definition and the induction hypothesis, $\det A' = \sum_{j=1}^{n}(-1)^{1+j}a_{j1}\det A(j|1)' = \sum_{j=1}^{n}(-1)^{j+1}a_{j1}\det A(j|1)$, which is the expansion of $\det A$ along column 1 (Theorem 5.13). □

Theorem 5.16. Let $A \in \mathbb{R}^{n\times n}$ with $n \geqslant 2$, let $r \neq s$ be row indices, and let $c \in \mathbb{R}$.

(a) Multiplying row r of A by c multiplies $\det A$ by c. In particular, if A has a zero row, then $\det A = 0$.

(b) Interchanging rows r and s changes the sign of $\det A$.

(c) If two rows of A are equal, then $\det A = 0$.

(d) Adding c times row s to row r does not change $\det A$.

(e) For the elementary matrices of Definition 2.16, $\det D_r(c) = c$, $\det T_{rs}(c) = 1$ and $\det P_{rs} = -1$; and $\det(EA) = \det E \cdot \det A$ for every elementary matrix $E \in \mathbb{R}^{n\times n}$.

(f) Statements (a)–(d) hold with columns in place of rows.

Proof. (a) Let B be A with row r multiplied by c. Deleting row r gives $B(r|j) = A(r|j)$, so the expansion along row r (Theorem 5.13) gives $\det B = \sum_j (-1)^{r+j}ca_{rj}\det A(r|j) = c\det A$. A zero row is 0 times itself, so then $\det A = 0 \cdot \det A = 0$.

(b) Let B be A with rows r and s interchanged; induction on n.

- For $n = 2$, $\det \left(\begin{smallmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{smallmatrix}\right) = a_{21}a_{12} - a_{22}a_{11} = -\det A$.
- For $n \geqslant 3$, expand $\det A$ and $\det B$ along a row $i \notin \{r, s\}$, which is the same in both. Each $B(i|j)$ is $A(i|j)$ with two rows interchanged, so $\det B(i|j) = -\det A(i|j)$ by induction; hence $\det B = -\det A$.

(c) Interchanging the two equal rows does not change A, but by (b) it changes the sign of $\det A$. So $\det A = -\det A$, that is, $\det A = 0$.

(d) Let B be A with row r replaced by $\mathrm{row}_r(A) + c\,\mathrm{row}_s(A)$, and $\tilde{A}$ be A with row r replaced by $\mathrm{row}_s(A)$. Since $B(r|j) = \tilde{A}(r|j) = A(r|j)$, expanding along row r,

$$\det B = \sum_{j=1}^{n}(-1)^{r+j}(a_{rj} + ca_{sj})\det A(r|j) = \det A + c\det \tilde{A}$$

Rows r and s of $\tilde{A}$ are equal, so $\det \tilde{A} = 0$ by (c).

(e) $D_r(c)$ is diagonal and $T_{rs}(c)$ is triangular with diagonal entries 1, so $\det D_r(c) = c$ and $\det T_{rs}(c) = 1$ (Theorem 5.15(a)); P_{rs} is I_n with rows r, s interchanged, so $\det P_{rs} = -1$ by (b). EA is obtained from A by the row operation of E (Theorem 2.17(a)), so by (a), (d), (b) $\det(EA)$ is $c\det A$, $\det A$ or $-\det A$, that is, $\det E \cdot \det A$.

(f) The columns of A are the transposed rows of A', and $\det A = \det A'$ (Theorem 5.15(b)). An operation on the columns of A is the same operation on the rows of A'; so (a)–(d) for the rows of A' give (a)–(d) for the columns of A. □

Computing $\det A$ by row reduction. Forward elimination (Steps 1, 2 and 4 of Definition 2.5, without making the pivots 1) turns $A \in \mathbb{R}^{n\times n}$ into an upper triangular matrix U (the first nonzero entry of row i lies in column i or to its right), whose determinant is the product of its diagonal entries (Theorem 5.15(a)). By Theorem 5.16, each interchange of two rows changes the sign and adding a multiple of a row to another row changes nothing, so $\det A = (-1)^t u_{11}u_{22}\cdots u_{nn}$, where t is the number of interchanges; if a row is multiplied by $c \neq 0$ on the way, divide by c. Column operations may be used in the same way, and a row or column with a single nonzero entry can be expanded at once.

Example 5.17. Compute $\det A$ for $A = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix}$.

Solution. Since $a_{11} = 0$, begin with an interchange:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 2 & 2 & 4 \\ 0 & -2 & 1 & 3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -2 \end{pmatrix} = U$$

The interchange changes the sign and the additions change nothing, so $\det A = -\det U = -\big(1 \cdot 2 \cdot 1 \cdot (-2)\big) = 4$.

Example 5.18. Compute $\det A$ for $A = \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{pmatrix}$.

Solution.

$$A \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{4} R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 5R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 0 & -8 \end{pmatrix} = U$$

Only the second step changes the determinant: it multiplies it by $\frac{1}{4}$ (Theorem 5.16(a)). Hence $\det A = 4 \det U = 4 \cdot 1 \cdot 1 \cdot (-4) \cdot (-8) = 128$.

Main properties

Theorem 5.19. Let $A, B \in \mathbb{R}^{n \times n}$.

(a) A is invertible $\Leftrightarrow \det A \neq 0 \Leftrightarrow \operatorname{rank} A = n$.

(b) $\det(AB) = \det A \cdot \det B$.

(c) If A is invertible, then $\det A^{-1} = 1/\det A$. Moreover $\det(cA) = c^n \det A$ for $c \in \mathbb{R}$, and $\det A^k = (\det A)^k$ for $k = 1, 2, \ldots$

Proof. For $n = 1$, $\det(a) = a$: (a) holds since (a) is invertible if and only if $a \neq 0$, and (b), (c) are identities between numbers; let $n \geqslant 2$.

(a)

- Let $R = PA$ be the RREF of A, where $P = E_k \cdots E_1$ is a product of elementary matrices (Theorem 2.17(c)). By Theorem 5.16(e), applied k times, $\det R = \det E_k \cdots \det E_1 \cdot \det A$, and each $\det E_i$ is $c \neq 0$, 1 or -1.

- If A is invertible, then $R = I_n$ (Theorem 2.18), so $\det A \neq 0$. If not, then $\operatorname{rank} A < n$ (Theorem 4.13(d)), so R has fewer than n nonzero rows (Definition 4.11), that is, a zero row; then $\det R = 0$ (Theorem 5.16(a)) and $\det A = 0$. Finally, A is invertible if and only if $\operatorname{rank} A = n$ (Theorem 4.13(d)).

(b)

- If A is not invertible, neither is AB: an inverse C of AB would give $A(BC) = I_n$, and A would be invertible (Theorem 2.18). By (a), $\det(AB) = 0 = \det A \cdot \det B$.
- If A is invertible, then $A = F_1 \cdots F_k$ with elementary matrices F_i (Theorem 2.18). By Theorem 5.16(e), applied k times, $\det(AB) = \det F_1 \cdots \det F_k \cdot \det B$; for $B = I_n$ this reads $\det A = \det F_1 \cdots \det F_k$, so $\det(AB) = \det A \cdot \det B$.

(c) By (b), $\det A \cdot \det A^{-1} = \det(AA^{-1}) = \det I_n = 1$. Next, $cA = (cI_n)A$ and cI_n is diagonal, so $\det(cA) = c^n \det A$ by (b) and Theorem 5.15(a). Finally $\det A^{k+1} = \det(A^k A) = \det A^k \cdot \det A$ by (b), which gives the powers by induction on k. □

Theorem 5.20. Let $A \in \mathbb{R}^{n\times n}$ with $n \geqslant 2$, and let c_{ij} be the cofactors of A. The **adjugate** of A is the transpose of the matrix of cofactors: $\operatorname{adj} A = (c_{ij})'$, that is, $(\operatorname{adj} A)_{ij} = c_{ji}$. Then

$$A \operatorname{adj} A = (\det A)\, I_n$$

Hence, if $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$.

Proof.
- The (i,k) entry of $A \operatorname{adj} A$ is $\sum_{j=1}^n a_{ij}(\operatorname{adj} A)_{jk} = \sum_{j=1}^n a_{ij}c_{kj}$.
- For $k = i$, this is the expansion of $\det A$ along row i (Theorem 5.13).
- For $k \neq i$, let $\tilde{A}$ be A with row k replaced by row i. Since $\tilde{A}(k|j) = A(k|j)$, the cofactors of row k of $\tilde{A}$ are $c_{k1}, \ldots, c_{kn}$, and the sum is the expansion of $\det \tilde{A}$ along row k. As $\tilde{A}$ has two equal rows, the sum is 0 (Theorem 5.16(c)).
- If $\det A \neq 0$, then $A\Big(\frac{1}{\det A} \operatorname{adj} A\Big) = I_n$, so $\frac{1}{\det A} \operatorname{adj} A$ is the inverse of A (Theorem 2.18). □

For $n = 2$, $\operatorname{adj} \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, and the theorem gives again the formula of Proposition 2.13.

Theorem 5.21 (Cramer's rule)**.** Let $A \in \mathbb{R}^{n\times n}$ ($n \geqslant 2$) with $\det A \neq 0$, and let $b \in \mathbb{R}^n$. Then $Ax = b$ has exactly one solution x, and

$$x_j = \frac{\det A_j}{\det A} \qquad (1 \leqslant j \leqslant n),$$

where A_j is A with column j replaced by b.

Proof.
- By Theorem 5.20, A is invertible with $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$. If $Ax = b$, then $x = A^{-1}(Ax) = A^{-1}b$, so there is at most one solution; and $A(A^{-1}b) = b$, so $A^{-1}b$ is a solution. Hence the only solution is $x = \frac{1}{\det A}(\operatorname{adj} A)b$.
- Let c_{ij} be the cofactors of A. Entry j of $(\operatorname{adj} A)b$ is $\sum_{i=1}^n (\operatorname{adj} A)_{ji} b_i = \sum_{i=1}^n c_{ij} b_i$.
- A_j differs from A only in column j, so $A_j(i|j) = A(i|j)$, and the (i,j) entry of A_j is b_i. Expanding $\det A_j$ along column j (Theorem 5.13) gives

$$\det A_j = \sum_{i=1}^n b_i (-1)^{i+j} \det A_j(i|j) = \sum_{i=1}^n b_i (-1)^{i+j} \det A(i|j) = \sum_{i=1}^n c_{ij} b_i$$

Hence $x_j = \det A_j / \det A$. □

Remark 5.22. Let $A \in \mathbb{R}^{n\times n}$, $n \geqslant 2$, be symmetric. Deleting row j and column i of $A = A'$ gives the transpose of $A(i|j)$: $A(j|i) = A'(j|i) = A(i|j)'$. By Theorem 5.15(b), $\det A(j|i) = \det A(i|j)$, so the cofactors satisfy $c_{ji} = c_{ij}$.

Example 5.23. Find A^{-1} for $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$ by Theorem 5.20.

Solution. The cofactors are

$$\begin{aligned}
c_{11} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{12} &= -\det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, \\
c_{13} &= \det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{14} &= -\det \left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1, \\
c_{21} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{22} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, \\
c_{23} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{24} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = -1, \\
c_{31} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{32} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1, \\
c_{33} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{34} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1, \\
c_{41} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1, & c_{42} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 1, \\
c_{43} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{44} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1
\end{aligned}$$

(each 3×3 minor is triangular or has a row with a single nonzero entry). Along row 1, $\det A = 1 \cdot c_{11} + 1 \cdot c_{12} + 0 \cdot c_{13} + 0 \cdot c_{14} = 2$. Hence

$$A^{-1} = \frac{1}{2} \operatorname{adj} A = \frac{1}{2} \begin{pmatrix} c_{11} & c_{21} & c_{31} & c_{41} \\ c_{12} & c_{22} & c_{32} & c_{42} \\ c_{13} & c_{23} & c_{33} & c_{43} \\ c_{14} & c_{24} & c_{34} & c_{44} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

Check: $\operatorname{row}_4(A) \operatorname{adj} A = (-1, 0, 0, 1) \operatorname{adj} A = (-1+1,\ 1-1,\ -1+1,\ 1+1) = (0, 0, 0, 2)$, the last row of $2I_4$.

Example 5.24. For the matrix A of Example 5.23 and $b = (3, 1, 0, 0)'$, find x_3 in $Ax = b$ by Cramer's rule, and check it against $x = A^{-1}b$ computed with that example.

Solution. Here $\det A = 2$, and A_3 is A with column 3 replaced by b. Row 3 of A_3 has the single nonzero entry 1, at position $(3, 4)$; expanding along it and then along the last row of the 3×3 minor (Theorem 5.13),

$$\det A_3 = \det \begin{pmatrix} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix} = (-1)^{3+4} \det \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{pmatrix} = -(-1)(-1)^{3+1} \det \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} = -2$$

So $x_3 = -2/2 = -1$ by Theorem 5.21. With $\operatorname{adj} A$ of Example 5.23 and $b = 3e_1 + e_2$, $(\operatorname{adj} A)b = 3 \operatorname{col}_1(\operatorname{adj} A) + \operatorname{col}_2(\operatorname{adj} A)$, so

$$x = \frac{1}{2}(\operatorname{adj} A)b = \frac{1}{2}\big(3(1, 1, -1, 1)' + (-1, 1, 1, -1)'\big) = \frac{1}{2}(2, 4, -2, 2)' = (1, 2, -1, 1)',$$

whose third entry is -1; indeed $Ax = (1+2,\ 2-1,\ -1+1,\ -1+1)' = b$.

Determinants of block matrices and the Schur complement

Theorem 5.25. Let $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$. Then

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \cdot \det D = \det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$$

In particular $\det \operatorname{diag}(A, D) = \det A \cdot \det D$.

The diagonal blocks A and D are square, of possibly different sizes; for $p \neq q$ the off-diagonal blocks B and C are rectangular.

Proof. • The first row of $\left(\begin{smallmatrix} I_p & 0 \\ 0 & D \end{smallmatrix}\right)$ is $(1, 0, \ldots, 0)$, and deleting row 1 and column 1 leaves $\left(\begin{smallmatrix} I_{p-1} & 0 \\ 0 & D \end{smallmatrix}\right)$; expanding along the first row p times gives $\det \left(\begin{smallmatrix} I_p & 0 \\ 0 & D \end{smallmatrix}\right) = \det D$. Likewise, expanding $\left(\begin{smallmatrix} A & 0 \\ 0 & I_q \end{smallmatrix}\right)$ q times along its last row $(0, \ldots, 0, 1)$, whose entry 1 is a diagonal entry (sign $+1$), gives $\det A$.

- $\left(\begin{smallmatrix} I_p & B \\ 0 & I_q \end{smallmatrix}\right)$ is upper triangular with diagonal entries 1, so its determinant is 1 (Theorem 5.15(a)).
- By the block product formula (Theorem 5.3),

$$\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} A & B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$$

  and by Theorem 5.19(b) its determinant is $\det D \cdot 1 \cdot \det A$.

- The transpose of $\left(\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} A' & C' \\ 0 & D' \end{smallmatrix}\right)$ (Theorem 5.3(b)); by Theorem 5.15(b) and the case just proved, its determinant is $\det A' \cdot \det D' = \det A \cdot \det D$. Finally, $\operatorname{diag}(A, D)$ is the case $B = 0$. □

Theorem 5.26. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A is invertible, then $\det M = \det A \cdot \det(D - CA^{-1}B)$.

(b) If D is invertible, then $\det M = \det D \cdot \det(A - BD^{-1}C)$.

In particular, if A is invertible, then M is invertible if and only if its Schur complement $D - CA^{-1}B$ is invertible.

Proof. (a) By Theorem 5.9, $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A & B \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$. The left factor is lower triangular with diagonal entries 1, so its determinant is 1. By Theorems 5.19(b) and 5.25, $\det M = \det A \cdot \det(D - CA^{-1}B)$.

(b) By the block product formula, $\left(\begin{smallmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A-BD^{-1}C & B-BD^{-1}D \\ C & D \end{smallmatrix}\right) = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C & D \end{smallmatrix}\right)$. The left factor is upper triangular with diagonal entries 1; as in (a), $\det M = \det(A - BD^{-1}C) \cdot \det D$.

If A is invertible, then $\det A \neq 0$, so by (a) $\det M \neq 0$ if and only if $\det(D - CA^{-1}B) \neq 0$; apply Theorem 5.19(a) to M and to $D - CA^{-1}B$. □

Here A and D are square, of possibly different sizes p and q; $B \in \mathbb{R}^{p\times q}$ and $C \in \mathbb{R}^{q\times p}$ are rectangular when $p \neq q$. For $p = q = 1$ and $a \neq 0$, (a) reads $\det \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = a\left(d - \frac{cb}{a}\right) = ad - bc$.

Theorem 5.27. Let $A \in \mathbb{R}^{n\times n}$ be invertible and $u, v \in \mathbb{R}^n$. Then

$$\det(A + uv') = \det A \cdot (1 + v'A^{-1}u)$$

In particular $\det(I_n + uv') = 1 + v'u$. (This is the **matrix determinant lemma**.)

Proof. Apply Theorem 5.26 to $M = \left(\begin{smallmatrix} A & -u \\ v' & 1 \end{smallmatrix}\right) \in \mathbb{R}^{(n+1)\times(n+1)}$ ($p = n$, $q = 1$).

- By (a), with the invertible block A, whose Schur complement is the 1×1 matrix $1 - v'A^{-1}(-u)$: $\det M = \det A \cdot (1 + v'A^{-1}u)$.
- By (b), with the invertible block $D = (1)$: $\det M = \det(1)\cdot\det\Big(A - (-u)(1)^{-1}v'\Big) = \det(A + uv')$.
- Comparing gives the formula; for $A = I_n$, $A^{-1} = I_n$. □

Example 5.28. Let $n \geqslant 2$ and $\rho \in \mathbb{R}$. The **equicorrelation matrix** $R = (1-\rho)I_n + \rho J_n \in \mathbb{R}^{n\times n}$, with diagonal entries 1 and all other entries ρ. Find $\det R$, and the values of ρ for which R is invertible.

Solution.

- If $\rho = 1$, then $R = J_n$ has equal rows ($n \geqslant 2$), so $\det R = 0$ (Theorem 5.16(c)).
- Let $\rho \neq 1$. As $J_n = \mathbf{1}_n\mathbf{1}_n'$, $R = A + uv'$ with $A = (1-\rho)I_n$, $u = \rho\mathbf{1}_n$ and $v = \mathbf{1}_n$. Here $\det A = (1-\rho)^n$ (Theorem 5.15(a)), $A^{-1} = (1-\rho)^{-1}I_n$ and $v'A^{-1}u = \frac{\rho}{1-\rho}\mathbf{1}'\mathbf{1} = \frac{n\rho}{1-\rho}$ (Proposition 1.13). By Theorem 5.27,

$$\det R = (1-\rho)^n\Big(1 + \frac{n\rho}{1-\rho}\Big) = (1-\rho)^{n-1}(1-\rho+n\rho) = (1-\rho)^{n-1}\Big(1+(n-1)\rho\Big)$$

  This formula also gives 0 for $\rho = 1$.
- By Theorem 5.19(a), R is invertible if and only if $\det R \neq 0$, that is, $\rho \neq 1$ and $\rho \neq -1/(n-1)$. For instance, for $n = 3$ and $\rho = -\frac{1}{2}$ every row of R has sum 0, so $R\mathbf{1}_3 = 0$.

Rank and the Schur complement

Theorem 5.29. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$.

(a) If $A \in \mathbb{R}^{p\times p}$ is invertible, and $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{n\times p}$, $D \in \mathbb{R}^{n\times q}$, then $\operatorname{rank} M = p + \operatorname{rank}(D - CA^{-1}B)$.

(b) If $D \in \mathbb{R}^{q\times q}$ is invertible, and $A \in \mathbb{R}^{m\times p}$, $B \in \mathbb{R}^{m\times q}$, $C \in \mathbb{R}^{q\times p}$, then $\operatorname{rank} M = q + \operatorname{rank}(A - BD^{-1}C)$.

Proof. (a)

- By Theorem 5.9, $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{smallmatrix}\right) M \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$, and the two outer factors are invertible (Proposition 5.8).
- Invertible factors do not change the rank (Theorem 4.15(d)), and the rank of a block-diagonal matrix is the sum of the ranks of its blocks (Theorem 4.16(c)); so $\operatorname{rank} M = \operatorname{rank} A + \operatorname{rank}(D - CA^{-1}B)$, where $\operatorname{rank} A = p$ (Theorem 4.13(d)).

(b)

- By the block product formula, $\left(\begin{smallmatrix} I_m & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C & D \end{smallmatrix}\right)$ (as in the proof of Theorem 5.26(b)), and multiplying on the right by $\left(\begin{smallmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{smallmatrix}\right)$ gives $\left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C-DD^{-1}C & D \end{smallmatrix}\right) = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ 0 & D \end{smallmatrix}\right)$.
- Both outer factors are invertible (Proposition 5.8); as in (a), $\operatorname{rank} M = \operatorname{rank}(A - BD^{-1}C) + \operatorname{rank} D = \operatorname{rank}(A - BD^{-1}C) + q$. □

Example 5.30. Find the rank of

$$M = \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 \\ \hline 2 & 3 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 1 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbb{R}^{4\times 5}$$

Solution. Here $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has $\det A = 1$ and $A^{-1} = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (Proposition 2.13), $B \in \mathbb{R}^{2\times 3}$, $C \in \mathbb{R}^{2\times 2}$ and $D \in \mathbb{R}^{2\times 3}$. Then

$$CA^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad CA^{-1}B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix},$$

and the Schur complement of A is

$$D - CA^{-1}B = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \end{pmatrix},$$

a nonzero matrix with two equal rows, of rank 1. By Theorem 5.29(a), $\operatorname{rank} M = 2 + 1 = 3$. The equal rows of the Schur complement give a relation among the rows of M: by Theorem 5.9, the last two rows of $\left(\begin{smallmatrix} I_2 & 0 \\ -CA^{-1} & I_2 \end{smallmatrix}\right) M$ are those of $[0 \;\; D - CA^{-1}B]$, that is,

$$\operatorname{row}_3(M) - \operatorname{row}_1(M) - \operatorname{row}_2(M) = (0, 0, 1, -1, 0) = \operatorname{row}_4(M) - \operatorname{row}_2(M),$$

so $\operatorname{row}_3(M) = \operatorname{row}_1(M) + \operatorname{row}_4(M)$.

Exercises

Exercise 5.1. Let A and B be the 3×5 matrices and C the 5×4 matrix partitioned as

$$A = \left(\begin{array}{ccc|cc} 1 & 3 & -2 & 1 & 2 \\ 2 & 1 & 0 & -1 & 3 \\ \hline 0 & 0 & 1 & 2 & 1 \end{array}\right), \quad B = \left(\begin{array}{ccc|cc} 1 & -3 & -2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 0 \\ \hline 1 & 0 & 2 & 0 & 1 \end{array}\right), \quad C = \left(\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & 3 & 1 \\ \hline 3 & 1 & 0 & 2 \\ 2 & -1 & 1 & 1 \end{array}\right)$$

With blocks $A = (A_{st})$, $B = (B_{st})$, $C = (C_{st})$ $(s, t = 1, 2)$, show that

$$A + B = (A_{st} + B_{st}), \qquad A' = \begin{pmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{pmatrix}, \qquad AC = \begin{pmatrix} A_{11}C_{11} + A_{12}C_{21} & A_{11}C_{12} + A_{12}C_{22} \\ A_{21}C_{11} + A_{22}C_{21} & A_{21}C_{12} + A_{22}C_{22} \end{pmatrix}$$

Solution. Adding entrywise, $A + B = \left(\begin{array}{ccc|cc} 2 & 0 & -4 & 3 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ \hline 1 & 0 & 3 & 2 & 2 \end{array}\right)$, and each block is the sum of the corresponding blocks of A and B. Transposing,

$$A' = \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \\ \hline 1 & -1 & 2 \\ 2 & 3 & 1 \end{array}\right) = \begin{pmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{pmatrix}$$

Multiplying in the ordinary way, $AC = \left(\begin{array}{cc|cc} 10 & 5 & -2 & 3 \\ 5 & -2 & 7 & 3 \\ \hline 7 & 1 & 4 & 6 \end{array}\right)$. The blockwise products give the same four blocks:

$$A_{11}C_{11} + A_{12}C_{21} = \begin{pmatrix} 3 & 6 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 7 & -1 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 10 & 5 \\ 5 & -2 \end{pmatrix},$$

$$A_{11}C_{12} + A_{12}C_{22} = \begin{pmatrix} -4 & -1 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 7 & 3 \end{pmatrix},$$

$$A_{21}C_{11} + A_{22}C_{21} = (-1,\ 0) + (8,\ 1) = (7,\ 1),$$

$$A_{21}C_{12} + A_{22}C_{22} = (3,\ 1) + (1,\ 5) = (4,\ 6)$$

For instance, $A_{12}C_{21} = \left(\begin{smallmatrix} 1 & 2 \\ -1 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} 3 & 1 \\ 2 & -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 7 & -1 \\ 3 & -4 \end{smallmatrix}\right)$. This agrees with Theorem 5.3.

Exercise 5.2. Let

$$N = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ \hline 1 & 0 & 1 & 3 & 1 \\ 0 & 1 & 1 & 2 & 1 \end{array}\right) = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix},$$

with $A \in \mathbb{R}^{3\times 3}$, $C \in \mathbb{R}^{2\times 3}$ and $D \in \mathbb{R}^{2\times 2}$. Find N^{-1} by applying Proposition 5.6(c) to N', and check the answer by multiplication.

Solution. By Theorem 5.3(b), N' with groups $(3,2)$ is $N' = \left(\begin{smallmatrix} A' & C' \\ 0 & D' \end{smallmatrix}\right)$, upper block triangular, with

$$A' = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad C' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad D' = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$$

The matrix A' is the block A_{22} of Example 5.7, so $(A')^{-1} = \left(\begin{smallmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$; by Proposition 2.13 ($3\cdot 1 - 2\cdot 1 = 1$), $(D')^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ -1 & 3 \end{smallmatrix}\right)$. Then

$$(A')^{-1}C' = \begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad -(A')^{-1}C'(D')^{-1} = -\begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 1 & -2 \\ 0 & -1 \end{pmatrix},$$

so by Proposition 5.6(c) with $p = 3$, $q = 2$

$$(N')^{-1} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 4 \\ 0 & 1 & -1 & 1 & -2 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -1 & 3 \end{array}\right), \qquad N^{-1} = \left((N')^{-1}\right)' = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ \hline -2 & 1 & 0 & 1 & -1 \\ 4 & -2 & -1 & -2 & 3 \end{array}\right)$$

by Theorem 2.15(c). Check, with the blocks $A^{-1} = \left(\begin{smallmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{smallmatrix}\right)$, $R = \left(\begin{smallmatrix} -2 & 1 & 0 \\ 4 & -2 & -1 \end{smallmatrix}\right) \in \mathbb{R}^{2\times 3}$ and $D^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -2 & 3 \end{smallmatrix}\right)$ of N^{-1}: the $(2,1)$ block of NN^{-1} is

$$CA^{-1} + DR = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix} = 0,$$

the $(1,2)$ block is $A\cdot 0 + 0\cdot D^{-1} = 0$, and the diagonal blocks are $AA^{-1} = I_3$ and $DD^{-1} = I_2$. So $NN^{-1} = I_5$.

Exercise 5.3. Let

$$M = \left(\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 4 & 2 \\ 0 & 1 & 2 & 2 & 3 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with $A \in \mathbb{R}^{3\times 3}$, $B \in \mathbb{R}^{3\times 2}$, $C \in \mathbb{R}^{2\times 3}$, $D \in \mathbb{R}^{2\times 2}$. Find the Schur complement E of A and $\det M$. Then compute M^{-1} with Corollary 5.10(a).

Solution. Here $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ is the block A_{22} of Example 5.7, so $A^{-1} = \left(\begin{smallmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, and $\det A = 1$ (Theorem 5.15(a)). With $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 3 \end{smallmatrix}\right)$,

$$A^{-1}B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad CA^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix},$$

$$E = D - CA^{-1}B = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

So $\det M = \det A \cdot \det E = 1 \cdot 1 = 1$ (Theorem 5.26(a)), and $E^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ by Proposition 2.13.

- With $L = \left(\begin{smallmatrix} I_3 & 0 \\ -CA^{-1} & I_2 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} I_3 & -A^{-1}B \\ 0 & I_2 \end{smallmatrix}\right)$, Corollary 5.10(a) gives $M^{-1} = U \operatorname{diag}(A^{-1}, E^{-1}) L$.
- By the block product formula,

$$U \operatorname{diag}(A^{-1}, E^{-1}) = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{array}\right),$$

since $-A^{-1}BE^{-1} = -\left(\begin{smallmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -2 & 2 \\ 1 & -1 \\ 0 & -1 \end{smallmatrix}\right) \in \mathbb{R}^{3\times 2}$. Multiplying on the right by L adds to block column 1 the block column 2 times $-CA^{-1} \in \mathbb{R}^{2\times 3}$:

$$M^{-1} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{array}\right) \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \hline -1 & 0 & -1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \end{array}\right) = \left(\begin{array}{ccc|cc} 3 & -3 & 1 & -2 & 2 \\ -1 & 2 & -1 & 1 & -1 \\ 0 & 1 & 2 & 0 & -1 \\ \hline -1 & 1 & 0 & 1 & -1 \\ 1 & -2 & -1 & -1 & 2 \end{array}\right)$$

Check: $\operatorname{row}_1(M)M^{-1}$ is the sum of rows 1, 2, 4 of M^{-1}, namely $(3-1-1,\ -3+2+1,\ 1-1+0,\ -2+1+1,\ 2-1-1) = (1,0,0,0,0)$; and $\operatorname{row}_4(M)M^{-1} = (1,1,1,4,2)M^{-1} = (3-1+0-4+2,\ -3+2+1+4-4,\ 1-1+2+0-2,\ -2+1+0+4-2,\ 2-1-1-4+4) = (0,0,0,1,0)$.

Exercise 5.4. Compute, by row and column operations,

$$\det \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 1 & 0 & 3 \\ 3 & 1 & 2 & 3 \\ 1 & 0 & 1 & 3 \end{pmatrix} \qquad \text{and} \qquad \det \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 2 \\ -1 & -2 & 3 & 1 \\ 1 & 2 & 6 & 7 \end{pmatrix}$$

Solution. Call the two matrices A and B.

(A) Column 4 is 3 times $\mathbf{1}_4$. Taking the factor 3 out of column 4 (Theorem 5.16(a),(f)) and then subtracting row 1 from rows 2, 3, 4 (Theorem 5.16(d)),

$$\det A = 3 \det \begin{pmatrix} 1 & 2 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix} = 3 \det \begin{pmatrix} 1 & 2 & -1 & 1 \\ 1 & -1 & 1 & 0 \\ 2 & -1 & 3 & 0 \\ 0 & -2 & 2 & 0 \end{pmatrix}$$

The last matrix has the single nonzero entry 1 in column 4, at position $(1,4)$. Expanding along column 4 and then the 3×3 determinant along its first row,

$$\det A = 3 \cdot (-1)^{1+4} \det \begin{pmatrix} 1 & -1 & 1 \\ 2 & -1 & 3 \\ 0 & -2 & 2 \end{pmatrix} = -3\Big(1(-2+6) + 1(4-0) + 1(-4-0)\Big) = -3 \cdot 4 = -12$$

(B)

$$B \xrightarrow[R_3 \leftarrow R_3 + R_1,\ R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 6 & 6 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

No step changes the determinant, so $\det B = 1 \cdot 1 \cdot 3 \cdot 2 = 6$.

Exercise 5.5. Compute the determinant of

$$A = \begin{pmatrix} 0 & 4 & 0 & 5 \\ 1 & 0 & -1 & 2 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix}$$

(a) by repeated cofactor expansion: expand $\det A$ in terms of determinants of 3×3 matrices, these in terms of determinants of 2×2 matrices, and finally these in terms of determinants of 1×1 matrices; (b) by row reduction to triangular form.

Solution. (a)

- Column 1 has the single nonzero entry $a_{21} = 1$; expanding along column 1, $\det A = 1 \cdot (-1)^{2+1} \det \left(\begin{smallmatrix} 4 & 0 & 5 \\ 3 & 0 & -2 \\ 0 & -6 & 0 \end{smallmatrix}\right)$.
- Row 3 of this 3×3 matrix has the single nonzero entry -6, in column 2: $\det \left(\begin{smallmatrix} 4 & 0 & 5 \\ 3 & 0 & -2 \\ 0 & -6 & 0 \end{smallmatrix}\right) = (-6)(-1)^{3+2} \det \left(\begin{smallmatrix} 4 & 5 \\ 3 & -2 \end{smallmatrix}\right)$.
- Expanding the 2×2 determinant along its first row, $4(-1)^{1+1}\det(-2)+5(-1)^{1+2}\det(3) = -8-15 = -23$.
- Hence $\det A = (-1) \cdot 6 \cdot (-23) = 138$.

(b)

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 4 & 0 & 5 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & -23 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & -23 \end{pmatrix}$$

There are two interchanges, so $\det A = (-1)^2 \cdot 1 \cdot 1 \cdot (-6) \cdot (-23) = 138$.

Exercise 5.6. Let $\alpha \in \mathbb{R}$ and $A = \begin{pmatrix} 3 & \alpha & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 2 & 0 & 2 & 0 \\ 3 & 1 & 0 & -1 \end{pmatrix}$. For which α is $\det A = 0$? Find a nontrivial relation among the columns of A in that case.

Solution. Column 4 has the single nonzero entry $a_{44} = -1$. Expanding along column 4 and then along the first row,

$$\det A = (-1)(-1)^{4+4} \det \begin{pmatrix} 3 & \alpha & 1 \\ 0 & -1 & 1 \\ 2 & 0 & 2 \end{pmatrix} = -\Big(3(-2-0) - \alpha(0-2) + 1(0+2)\Big) = 4 - 2\alpha$$

So $\det A = 0$ exactly when $\alpha = 2$ (and A is invertible exactly when $\alpha \neq 2$, Theorem 5.19(a)). For $\alpha = 2$ the rank of A is less than 4, and indeed $x = (1, -1, -1, 2)'$ gives

$$Ax = (3 - 2 - 1 + 0,\ 0 + 1 - 1 + 0,\ 2 + 0 - 2 + 0,\ 3 - 1 + 0 - 2)' = 0,$$

that is, $a_1 - a_2 - a_3 + 2a_4 = 0$ for the columns a_j of A.

Exercise 5.7. Let $T = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$.

(a) Compute all cofactors of T.

(b) Compute $\det T$ by expansion along the second row, and check the answer by expansion along the fourth column.

(c) Use Theorem 5.20 to show that T^{-1} is the matrix K with entries $k_{ij} = \min(i, j)$.

(d) For $b = (1, 2, 3, 4)'$, find x_2 in $Tx = b$ by Cramer's rule, and check it with $x = Kb$.

Solution. (a) T is symmetric, so $c_{ji} = c_{ij}$ (Remark 5.22). The ten cofactors with $i \leqslant j$ are (each minor is expanded along a row or column with at most two nonzero entries)

$$c_{11} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = 2 \cdot 1 + 1 \cdot (-1) = 1, \quad c_{12} = -\det \begin{pmatrix} -1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = -(-1) = 1,$$

$$c_{13} = \det \begin{pmatrix} -1 & 2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = (-1)(-1)(1) = 1, \quad c_{14} = -\det \begin{pmatrix} -1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} = -(-1) = 1,$$

$$c_{22} = \det \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = 2(2 - 1) = 2, \quad c_{23} = -\det \begin{pmatrix} 2 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = -(-2) = 2,$$

$$c_{24} = \det \begin{pmatrix} 2 & -1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} = 2(-1)(-1) = 2, \quad c_{33} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 4 - 1 = 3,$$

$$c_{34} = -\det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix} = -(-3) = 3, \quad c_{44} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} = 2 \cdot 3 + 1 \cdot (-2) = 4$$

(the triangular minors by Theorem 5.15(a)). So $c_{ij} = \min(i, j)$ for all i, j.

(b) Along row 2: $\det T = (-1)c_{21} + 2c_{22} + (-1)c_{23} = -1 + 4 - 2 = 1$. Along column 4: $\det T = (-1)c_{34} + 1 \cdot c_{44} = -3 + 4 = 1$ (Theorem 5.13).

(c) Since $\det T = 1$, Theorem 5.20 gives $T^{-1} = \operatorname{adj} T = (c_{ij})'$, and $(c_{ij})' = (c_{ji}) = (\min(i, j)) = K$ by (a):

$$T^{-1} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Check: $\operatorname{row}_4(T)K = -(1, 2, 3, 3) + (1, 2, 3, 4) = (0, 0, 0, 1)$ and $\operatorname{row}_2(T)K = -(1, 1, 1, 1) + 2(1, 2, 2, 2) - (1, 2, 3, 3) = (0, 1, 0, 0)$.

(d) T_2 is T with column 2 replaced by b. Expanding along column 1 (Theorem 5.13),

$$\det T_2 = \det \begin{pmatrix} 2 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & 3 & 2 & -1 \\ 0 & 4 & -1 & 1 \end{pmatrix} = 2 \det \begin{pmatrix} 2 & -1 & 0 \\ 3 & 2 & -1 \\ 4 & -1 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & -1 \\ 4 & -1 & 1 \end{pmatrix} = 2 \cdot 9 + 1 = 19$$

and $\det T = 1$ by (b), so $x_2 = 19$ (Theorem 5.21). Check: $x_2 = \operatorname{row}_2(K)b = 1 + 4 + 6 + 8 = 19$; indeed $x = Kb = (10, 19, 26, 30)'$.

Exercise 5.8. Let $A, B \in \mathbb{R}^{4\times 4}$ with $\det A = 2$ and $\det B = -3$. Compute $\det(3A)$, $\det(A'B^{-1})$, $\det(A^2B')$, $\det(B^{-1}AB)$, $\det(-B)$ and $\det(\operatorname{adj} A)$.

Solution. By Theorems 5.19(b),(c) and 5.15(b) (B is invertible since $\det B \neq 0$):

$$\det(3A) = 3^4 \cdot 2 = 162, \qquad \det(A'B^{-1}) = \det A \cdot \frac{1}{\det B} = -\frac{2}{3},$$
$$\det(A^2B') = 2^2 \cdot (-3) = -12, \quad \det(B^{-1}AB) = \frac{1}{-3} \cdot 2 \cdot (-3) = 2,$$
$$\det(-B) = (-1)^4(-3) = -3$$

By Theorem 5.20, $A \operatorname{adj} A = 2I_4$, so $\det A \cdot \det(\operatorname{adj} A) = \det(2I_4) = 2^4$ and $\det(\operatorname{adj} A) = 16/2 = 8$.

Exercise 5.9. Let

$$M = \left(\begin{array}{cc|ccc} 3 & 1 & 4 & -1 & 7 \\ 5 & 2 & 2 & 9 & 3 \\ \hline 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

Compute $\det M$, $\det M^3$ and $\det(2M^{-1})$.

Solution. $M = \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$ with $A = \left(\begin{smallmatrix} 3 & 1 \\ 5 & 2 \end{smallmatrix}\right)$, $\det A = 6 - 5 = 1$, and $D = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, whose expansion along the first row gives $\det D = 2(4-1) - 1(2-0) = 4$. By Theorem 5.25, $\det M = 1 \cdot 4 = 4$; the block B plays no role. By Theorem 5.19(c), $\det M^3 = 4^3 = 64$ and $\det(2M^{-1}) = 2^5 \cdot \frac{1}{4} = 8$.

Exercise 5.10. Let

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad M = \begin{pmatrix} I_3 & B \\ C & I_2 \end{pmatrix} \in \mathbb{R}^{5\times 5}$$

(a) Compute $\det M$ with the Schur complement of I_3, and again with the Schur complement of I_2.

(b) Compute M^{-1} with Corollary 5.10(a).

Solution. (a) The Schur complement of I_3 is $S = I_2 - CI_3^{-1}B = I_2 - CB$ and that of I_2 is $I_3 - BC$, with

$$CB = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & -1 \\ -1 & -1 \end{pmatrix}, \qquad BC = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}, \quad I_3 - BC = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & -2 & 0 \end{pmatrix}$$

By Theorem 5.26(a), $\det M = \det I_3 \cdot \det S = 0 - 1 = -1$. Expanding $\det(I_3 - BC)$ along the first row gives $(-1)^{1+2}(-1)\det\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right) = 0 - 1 = -1$, so Theorem 5.26(b) gives again $\det M = \det I_2 \cdot \det(I_3 - BC) = -1$.

(b) The Schur complement $S = I_2 - CB$ of I_3 is invertible ($\det S = -1$), with $S^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)$ by the 2×2 formula of Proposition 2.13. Corollary 5.10(a) with $A = I_3$ gives $M^{-1} = \left(\begin{smallmatrix} I_3 & -B \\ 0 & I_2 \end{smallmatrix}\right) \operatorname{diag}(I_3, S^{-1}) \left(\begin{smallmatrix} I_3 & 0 \\ -C & I_2 \end{smallmatrix}\right)$. By the block product formula,

$$M^{-1} = \begin{pmatrix} I_3 & -BS^{-1} \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I_3 & 0 \\ -C & I_2 \end{pmatrix} = \begin{pmatrix} I_3 + BS^{-1}C & -BS^{-1} \\ -S^{-1}C & S^{-1} \end{pmatrix},$$

where

$$-BS^{-1} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad -S^{-1}C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad BS^{-1}C = -B(-S^{-1}C) = \begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$$

Hence

$$M^{-1} = \left(\begin{array}{ccc|cc} 2 & 0 & -1 & -1 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ \hline -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & -1 & 0 \end{array}\right)$$

Check: $\operatorname{row}_4(M)M^{-1} = (1,1,0,1,0)M^{-1} = (2-1-1,\ 0+0+0,\ -1+0+1,\ -1+1+1,\ 1+0-1) = (0,0,0,1,0)$.

Exercise 5.11. Use Theorem 5.27 to compute

$$\det\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 4 & 1 \\ 1 & 1 & 1 & 7 \end{pmatrix} \qquad \text{and} \qquad \det\begin{pmatrix} 2 & -1 & 1 & -1 \\ 2 & -1 & 2 & -2 \\ 3 & -3 & 4 & -3 \\ 4 & -4 & 4 & -3 \end{pmatrix}$$

Solution. The first matrix is $D + \mathbf{1}_4\mathbf{1}_4'$ with $D = \operatorname{diag}(1,2,3,6)$, $\det D = 36$ and $D^{-1} = \operatorname{diag}(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ (Proposition 5.6(b)). By Theorem 5.27,

$$\det(D + \mathbf{1}\mathbf{1}') = \det D \cdot (1 + \mathbf{1}'D^{-1}\mathbf{1}) = 36\Big(1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}\Big) = 36 \cdot 3 = 108$$

The second matrix is $I_4 + uv'$ with $u = (1,2,3,4)'$ and $v = (1,-1,1,-1)'$: its (i,j) entry is $\delta_{ij} + u_i v_j$, where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. Hence its determinant is $1 + v'u = 1 + (1 - 2 + 3 - 4) = -1$.

Exercise 5.12. For $t \in \mathbb{R}$ let

$$M = \left(\begin{array}{ccc|cc} 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & t & 2 \\ 0 & 1 & 1 & 2 & t+2 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with $A \in \mathbb{R}^{3\times 3}$, $B \in \mathbb{R}^{3\times 2}$, $C \in \mathbb{R}^{2\times 3}$, $D \in \mathbb{R}^{2\times 2}$. (a) Find $\det M$ with the Schur complement of A. (b) For which t is M invertible? (c) Find $\operatorname{rank} M$ for these exceptional values of t.

Solution. (a) Here $A = \operatorname{diag}(A_1, 1)$ with $A_1 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $C = B'$ and $D = \left(\begin{smallmatrix} t & 2 \\ 2 & t+2 \end{smallmatrix}\right)$. By Theorem 5.25, $\det A = \det A_1 \cdot 1 = 1$; by Propositions 2.13 and 5.6(b), $A^{-1} = \operatorname{diag}(A_1^{-1}, 1) = \left(\begin{smallmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. Then

$$CA^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}, \qquad CA^{-1}B = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix},$$

so the Schur complement is $S = D - CA^{-1}B = \left(\begin{smallmatrix} t-1 & 1 \\ 1 & t-1 \end{smallmatrix}\right)$. By Theorem 5.26(a), $\det M = \det A \cdot \det S = (t-1)^2 - 1 = t(t-2)$.

(b) By Theorem 5.19(a), M is invertible if and only if $t \neq 0$ and $t \neq 2$.

(c) For $t = 0$, $S = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; for $t = 2$, $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. In both cases the second row of S is a multiple of the first and $S \neq 0$, so $\operatorname{rank} S = 1$. By Theorem 5.29(a) with $p = 3$, $\operatorname{rank} M = 3 + 1 = 4$.

Problems

Problem 5.1. Let $A \in \mathbb{R}^{n\times n}$ be invertible and partitioned by columns as $A = [A_1\ A_2]$, where A_1 has k columns, and let A^{-1} be partitioned by rows into B_1 (k rows) and B_2. Show that

$$B_1A_1 = I, \quad B_1A_2 = 0, \quad B_2A_1 = 0, \quad B_2A_2 = I, \qquad A_1B_1 = I - A_2B_2, \quad A_2B_2 = I - A_1B_1$$

Solution. With the partition $(k, n-k)$ of I_n, Theorem 5.3(c) gives

$$\begin{pmatrix} B_1A_1 & B_1A_2 \\ B_2A_1 & B_2A_2 \end{pmatrix} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} [A_1\ A_2] = A^{-1}A = I_n = \begin{pmatrix} I_k & 0 \\ 0 & I_{n-k} \end{pmatrix},$$

and comparing blocks gives the first four identities. Also $A_1B_1 + A_2B_2 = [A_1\ A_2]\left(\begin{smallmatrix} B_1 \\ B_2 \end{smallmatrix}\right) = AA^{-1} = I_n$, which gives the last two.

Problem 5.2. Show that the matrix $Z = \begin{pmatrix} -I_m & B \\ C & I_n \end{pmatrix}$ ($B \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times m}$) satisfies $Z^2 = I_{m+n}$ if $B = 0$ or $C = 0$. Is this condition necessary?

Solution. By the 2×2 block formula (Theorem 5.3),

$$Z^2 = \begin{pmatrix} (-I)(-I) + BC & -B + B \\ -C + C & CB + I \end{pmatrix} = \begin{pmatrix} I_m + BC & 0 \\ 0 & I_n + CB \end{pmatrix},$$

so $Z^2 = I_{m+n}$ if and only if $BC = 0$ and $CB = 0$. This holds if $B = 0$ or $C = 0$. The condition is not necessary when $m, n \geqslant 2$: for $m = n = 2$ and $B = C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \neq 0$ we have $BC = CB = B^2 = \left(\begin{smallmatrix} 0\cdot0+1\cdot0 & 0\cdot1+1\cdot0 \\ 0\cdot0+0\cdot0 & 0\cdot1+0\cdot0 \end{smallmatrix}\right) = 0$; for larger m, n put this 2×2 matrix in the top left corner of zero matrices. If $m = 1$, then $CB \in \mathbb{R}^{n\times n}$ has entries c_ib_j, so $CB = 0$ forces $B = 0$ or $C = 0$; the case $n = 1$ is the same with BC.

Problem 5.3. Show that $Z_1 = \begin{pmatrix} I_m & B_1 \\ 0 & I_n \end{pmatrix}$ and $Z_2 = \begin{pmatrix} I_m & B_2 \\ 0 & I_n \end{pmatrix}$ ($B_1, B_2 \in \mathbb{R}^{m\times n}$) commute.

Solution. Write $Z_1 = I_{m+n} + X_1$ and $Z_2 = I_{m+n} + X_2$ with $X_i = \left(\begin{smallmatrix} 0 & B_i \\ 0 & 0 \end{smallmatrix}\right)$. By the block formula, $X_1X_2 = \left(\begin{smallmatrix} 0\cdot0+B_1\cdot0 & 0\cdot B_2+B_1\cdot0 \\ 0 & 0 \end{smallmatrix}\right) = 0$, and likewise $X_2X_1 = 0$. Hence

$$Z_1Z_2 = I + X_1 + X_2 + X_1X_2 = I + X_1 + X_2 = I + X_2 + X_1 + X_2X_1 = Z_2Z_1$$

Problem 5.4. If $A \in \mathbb{R}^{m\times m}$ and $D \in \mathbb{R}^{n\times n}$ are invertible and $C \in \mathbb{R}^{n\times m}$, show that

$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{pmatrix}$$

Solution. Look for the inverse in the form $\left(\begin{smallmatrix} A^{-1} & 0 \\ R & D^{-1} \end{smallmatrix}\right)$. By the block formula,

$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}\begin{pmatrix} A^{-1} & 0 \\ R & D^{-1} \end{pmatrix} = \begin{pmatrix} I_m & 0 \\ CA^{-1} + DR & I_n \end{pmatrix},$$

which is I if and only if $CA^{-1} + DR = 0$, that is, $R = -D^{-1}CA^{-1}$. With this R, the product in the other order is $\left(\begin{smallmatrix} A^{-1}A & 0 \\ RA+D^{-1}C & D^{-1}D \end{smallmatrix}\right) = \left(\begin{smallmatrix} I & 0 \\ -D^{-1}C+D^{-1}C & I \end{smallmatrix}\right) = I$.

Problem 5.5. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A and its Schur complement $E = D - CA^{-1}B$ are invertible, show that M is invertible and

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}$$

(b) If D and $F = A - BD^{-1}C$ are invertible, show that M is invertible and

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}$$

Solution. (a) By Corollary 5.10(a) (with E in place of S), M is invertible, and two applications of the block product formula (Theorem 5.3(c)) give

$$\begin{aligned} M^{-1} &= \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & E^{-1} \end{pmatrix} \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix} \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} \\ &= \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix} \end{aligned}$$

(b) By Corollary 5.10(b), M is invertible, and two applications of the block product formula give

$$\begin{aligned} M^{-1} &= \begin{pmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{pmatrix} \begin{pmatrix} F^{-1} & 0 \\ 0 & D^{-1} \end{pmatrix} \begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} F^{-1} & 0 \\ -D^{-1}CF^{-1} & D^{-1} \end{pmatrix} \begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix} \\ &= \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix} \end{aligned}$$

Problem 5.6. Let $K \in \mathbb{R}^{12\times 12}$ satisfy $K^5 = 3K$. What is the numerical value of $\det K$? Is there more than one possibility? If so, give all possible values.

Solution.

- By Theorem 5.19(c), $\det K^5 = (\det K)^5$ and $\det(3K) = 3^{12}\det K$. So $(\det K)^5 = 3^{12}\det K$, that is, $\det K \cdot \left((\det K)^4 - 3^{12}\right) = 0$.
- Hence $\det K = 0$ or $(\det K)^4 = 3^{12}$, that is, $\left((\det K)^2 - 3^6\right)\left((\det K)^2 + 3^6\right) = 0$. The second factor is positive, so $(\det K)^2 = 3^6$ and $\det K = \pm 3^3 = \pm 27$.
- All three values occur. Let t be the positive number with $t^4 = 3$, so $t^5 = 3t$. For $K = 0$, $\det K = 0$. For $K = tI_{12}$, $K^5 = t^5 I_{12} = 3K$ and $\det K = t^{12} = 3^3 = 27$. For $K = \operatorname{diag}(-t, t, \ldots, t)$, $K^5 = \operatorname{diag}(-t^5, t^5, \ldots, t^5) = 3K$ and $\det K = -t^{12} = -27$.

So $\det K$ is 0, 27 or -27.

Problem 5.7. Let $A \in \mathbb{R}^{n\times n}$ have integer entries.

(a) Show that if A is invertible and all entries of A^{-1} are integers, then $\det A = \pm 1$.

(b) Conversely, show that if $\det A = \pm 1$, then A is invertible and all entries of A^{-1} are integers.

Solution. (a)

- A matrix with integer entries has an integer determinant: for $n = 1$ it is the entry, and for $n \geqslant 2$ Definition 5.12 gives $\det A$ as a sum of products of entries and of determinants of $(n-1)\times(n-1)$ matrices with integer entries; use induction on n. So $d = \det A$ and $\det A^{-1}$ are integers.
- By Theorem 5.19(a),(c), $d \neq 0$ and $\det A^{-1} = 1/d$. So d and $1/d$ are nonzero integers.
- A nonzero integer has absolute value at least 1: $|d| \geqslant 1$ and $|1/d| \geqslant 1$, that is, $|d| \leqslant 1$. Hence $|d| = 1$, and $\det A = \pm 1$.

(b) For $n = 1$, $A = (\pm 1) = A^{-1}$. For $n \geqslant 2$, each cofactor $c_{ij} = (-1)^{i+j}\det A(i|j)$ is an integer by the first step of (a), applied to $A(i|j)$. As $\det A = \pm 1 \neq 0$, Theorem 5.20 gives $A^{-1} = \frac{1}{\det A}\operatorname{adj} A = \pm \operatorname{adj} A$, whose entries $\pm c_{ji}$ are integers.

Problem 5.8. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$. Show that $\det(I_m - AB) = \det(I_n - BA)$.

Solution. By the block product formula (Theorem 5.3),

$$\begin{pmatrix} I_m - AB & A \\ 0 & I_n \end{pmatrix}\begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m - AB + AB & A \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix},$$
$$\begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix}\begin{pmatrix} I_m & A \\ 0 & I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & BA + I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$$

So the two products are equal. By Theorems 5.19(b) and 5.25, the first has determinant $\det(I_m - AB)\det I_n \cdot \det I_m \det I_n = \det(I_m - AB)$, and the second $\det I_m \det I_n \cdot \det I_m \det(I_n - BA) = \det(I_n - BA)$.

Problem 5.9. Let $R = (1-\rho)I_n + \rho J_n$ be the equicorrelation matrix of Example 5.28, with $\rho \neq 1$ and $\rho \neq -1/(n-1)$. Show that

$$R^{-1} = \frac{1}{1-\rho}\Big(I_n - \frac{\rho}{1+(n-1)\rho}J_n\Big)$$

Solution. Let $\tau = \rho/(1+(n-1)\rho)$, so that $\tau(1-\rho+n\rho) = \rho$. Using $J_n^2 = nJ_n$ (Proposition 1.13),

$$\begin{aligned}\Big((1-\rho)I_n + \rho J_n\Big)(I_n - \tau J_n) &= (1-\rho)I_n + \Big(\rho - (1-\rho)\tau - n\rho\tau\Big)J_n \\ &= (1-\rho)I_n + \Big(\rho - \tau(1-\rho+n\rho)\Big)J_n = (1-\rho)I_n\end{aligned}$$

Dividing by $1-\rho \neq 0$ gives $R \cdot \frac{1}{1-\rho}(I_n - \tau J_n) = I_n$, so $\frac{1}{1-\rho}(I_n - \tau J_n) = R^{-1}$ (Theorem 2.18).

Problem 5.10. Let $A, B, C, D \in \mathbb{R}^{n\times n}$ with A invertible and $AC = CA$. Show that

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - CB)$$

Give $A, B, C, D \in \mathbb{R}^{2\times 2}$, with A invertible, for which the formula fails.

Solution.

- By Theorems 5.26(a) and 5.19(b),

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \cdot \det(D - CA^{-1}B) = \det\Big(A(D - CA^{-1}B)\Big) = \det(AD - ACA^{-1}B)$$

- From $AC = CA$, $ACA^{-1} = CAA^{-1} = C$, so the last determinant is $\det(AD - CB)$.
- Without $AC = CA$ the formula can fail. Take $A = \operatorname{diag}(1, 2)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $D = I_2$. Then $CA^{-1}B = \left(\begin{smallmatrix} 0 & \frac{1}{2} \\ 0 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{smallmatrix}\right)$, so by Theorem 5.26(a) the 4×4 determinant is $\det A \cdot \det \operatorname{diag}(\frac{1}{2}, 1) = 2 \cdot \frac{1}{2} = 1$, while $AD - CB = \operatorname{diag}(1, 2) - \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) = \operatorname{diag}(0, 2)$ has determinant 0.

Chapter 6. Orthogonality and projections

Inner products, lengths, and orthogonality

Definition 6.1. For $x, y \in \mathbb{R}^n$, the **inner product** of x and y is the number $x'y = x_1y_1 + \cdots + x_ny_n$; it equals $y'x$ (Theorem 1.9). The **length** (or norm) of x is $\|x\| = \sqrt{x'x} = (x_1^2 + \cdots + x_n^2)^{1/2}$, and $\|x - y\|$ is the **distance** between x and y. The vectors x and y are **orthogonal**, written $x \perp y$, if $x'y = 0$.

Since $x'x$ is a sum of squares, $\|x\| \geqslant 0$, and $\|x\| = 0$ only for $x = 0$; also $\|cx\| = |c|\,\|x\|$ for $c \in \mathbb{R}$. A vector of length 1 is a **unit vector**; for $x \neq 0$, $x/\|x\|$ is one. The zero vector is orthogonal to every vector, and it is the only vector orthogonal to itself.

Proposition 6.2. For $x, y \in \mathbb{R}^n$, $\|x + y\|^2 = \|x\|^2 + 2x'y + \|y\|^2$. In particular, if $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ (**Pythagoras**).

Proof. $(x + y)'(x + y) = x'x + x'y + y'x + y'y$, and $y'x = x'y$. □

Theorem 6.3. Let $x, y \in \mathbb{R}^n$.

(a) (**Cauchy–Schwarz inequality**) $|x'y| \leqslant \|x\|\,\|y\|$, with equality if and only if one of x, y is a multiple of the other.

(b) (**Triangle inequality**) $\|x + y\| \leqslant \|x\| + \|y\|$.

Proof. (a) If $x = 0$, both sides are 0, and $x = 0y$. Let $x \neq 0$.

- Put $c = x'y/\|x\|^2$ and $w = y - cx$. Then $x'w = x'y - c\,x'x = 0$, so $cx \perp w$.
- Pythagoras for $y = cx + w$ gives $\|y\|^2 = c^2\|x\|^2 + \|w\|^2 \geqslant c^2\|x\|^2 = (x'y)^2/\|x\|^2$. Multiply by $\|x\|^2$ and take square roots.
- Equality holds if and only if $w = 0$, that is, $y = cx$. Conversely, if $y = ax$, then $c = a$ and $w = 0$; if $x = ay$ with $x \neq 0$, then $a \neq 0$ and $y = a^{-1}x$.

(b) By Proposition 6.2 and (a), $\|x + y\|^2 \leqslant \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$, and both $\|x + y\|$ and $\|x\| + \|y\|$ are $\geqslant 0$. □

Definition 6.4. Let W be a subspace of $\mathbb{R}^n$. A vector $y \in \mathbb{R}^n$ is **orthogonal to** W, written $y \perp W$, if $y'w = 0$ for every $w \in W$. The **orthogonal complement** of W is $W^\perp = \{y \in \mathbb{R}^n : y \perp W\}$.

Theorem 6.5. Let W be a subspace of $\mathbb{R}^n$.

(a) If $w_1, \ldots, w_k$ span W and $A = [w_1 \ \cdots \ w_k] \in \mathbb{R}^{n\times k}$, then $y \perp W$ if and only if $y'w_1 = \cdots = y'w_k = 0$, that is, $A'y = 0$. So $W^\perp = \mathcal{N}(A')$, a subspace of $\mathbb{R}^n$.

(b) $W \cap W^\perp = \{0\}$.

(c) $\dim W + \dim W^\perp = n$ and $\mathbb{R}^n = W \oplus W^\perp$: every $y \in \mathbb{R}^n$ is $y = w + v$ with $w \in W$ and $v \in W^\perp$ in exactly one way.

Proof. (a) If $y \perp W$, then $y'w_j = 0$, as $w_j \in W$. Conversely, let all $y'w_j = 0$. Every $w \in W$ is $w = Ac$ for some $c \in \mathbb{R}^k$ (Definition 3.6 and Theorem 1.4(b)), and $y'w = (A'y)'c = 0$. The jth entry of $A'y$ is $w_j'y$ (Theorem 1.4(d)).

(b) If $y \in W \cap W^\perp$, then $y \perp y$, so $y = 0$.

(c) Let $d = \dim W$. If $d = 0$, then $W = \{0\}$ and $W^\perp = \mathbb{R}^n$. Let $d \geqslant 1$.

- Let $A \in \mathbb{R}^{n\times d}$ have a basis of W as its columns, so $\operatorname{rank} A = d$. By (a) and Theorem 4.12, $\dim W^\perp = \dim \mathcal{N}(A') = n - \operatorname{rank} A = n - d$.

- By (b), the sum $W + W^\perp$ is direct (Definition 3.20), and a basis of W together with a basis of $W^\perp$, $d + (n-d) = n$ vectors, is a basis of $W \oplus W^\perp$ (Theorem 3.23). So $W \oplus W^\perp$ is a subspace of $\mathbb{R}^n$ (Theorem 3.21(a)) of dimension n, hence equal to $\mathbb{R}^n$ (Theorem 4.5(a)). The decomposition $y = w + v$ is unique by Theorem 3.22. □

Corollary 6.6. For $A \in \mathbb{R}^{m\times n}$,

$$\mathcal{N}(A) = \mathcal{C}(A')^\perp, \quad \mathbb{R}^n = \mathcal{C}(A') \oplus \mathcal{N}(A); \qquad \mathcal{N}(A') = \mathcal{C}(A)^\perp, \quad \mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A')$$

So $Ax = 0$ if and only if x is orthogonal to every (transposed) row of A, and the four subspaces of A (Definition 4.7) are orthogonal in pairs: the transposed row space $\mathcal{C}(A')$ and the null space in $\mathbb{R}^n$, the column space and the left null space in $\mathbb{R}^m$.

Proof. The columns of A' span $\mathcal{C}(A')$, so Theorem 6.5(a) gives $\mathcal{C}(A')^\perp = \mathcal{N}((A')') = \mathcal{N}(A)$; applied to the columns of A, it gives $\mathcal{C}(A)^\perp = \mathcal{N}(A')$. The direct sums follow from Theorem 6.5(c). □

Example 6.7. For $A = \left(\begin{smallmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{smallmatrix}\right)$ of Example 4.9, the transposed nonzero rows $\rho_1' = (1,0,3,-1)'$, $\rho_2' = (0,1,-1,2)'$ of its RREF form a basis of $\mathcal{C}(A')$, and $s_1 = (-3,1,1,0)'$, $s_2 = (1,-2,0,1)'$ a basis of $\mathcal{N}(A)$. Indeed

$$\rho_1 s_1 = -3+0+3+0 = 0, \quad \rho_1 s_2 = 1+0+0-1 = 0, \quad \rho_2 s_1 = 0+1-1+0 = 0, \quad \rho_2 s_2 = 0-2+0+2 = 0,$$

and $2 + 2 = 4$. In $\mathbb{R}^3$, $y = (-1,-1,1)'$ is a basis of $\mathcal{N}(A')$, orthogonal to the basis $a_1 = (1,1,2)'$, $a_2 = (2,3,5)'$ of $\mathcal{C}(A)$: $-1-1+2 = 0$ and $-2-3+5 = 0$; and $2 + 1 = 3$.

Orthonormal bases, Gram–Schmidt, and QR

Proposition 6.8. Let $u_1, \dots, u_k \in \mathbb{R}^n$ be nonzero and pairwise orthogonal ($u_i'u_j = 0$ for $i \neq j$).

(a) If $y = c_1u_1 + \cdots + c_ku_k$, then $c_j = u_j'y/\|u_j\|^2$ for $j = 1, \dots, k$.

(b) $u_1, \dots, u_k$ are linearly independent.

Proof. (a) $u_j'y = \sum_i c_i u_j'u_i = c_j\|u_j\|^2$, and $\|u_j\|^2 > 0$. (b) If $\sum_i c_iu_i = 0$, then (a) with $y = 0$ gives every $c_j = 0$. □

Definition 6.9. Vectors $q_1, \dots, q_p \in \mathbb{R}^n$ are **orthonormal** if $q_i'q_j = 0$ for $i \neq j$ and $\|q_j\| = 1$ for every j. A matrix $Q = [q_1 \ \cdots \ q_p] \in \mathbb{R}^{n\times p}$ has **orthonormal columns** if $Q'Q = I_p$, since $(Q'Q)_{ij} = q_i'q_j$ (Theorem 1.4(d)). A square matrix $Q \in \mathbb{R}^{n\times n}$ with $Q'Q = I_n$ is an **orthogonal matrix**. A basis of a subspace consisting of orthonormal vectors is an **orthonormal basis**.

Proposition 6.10. Let $Q \in \mathbb{R}^{n\times p}$ satisfy $Q'Q = I_p$.

(a) $(Qx)'(Qy) = x'y$ and $\|Qx\| = \|x\|$ for all $x, y \in \mathbb{R}^p$: Q preserves inner products and lengths.

(b) If $p = n$, then Q is invertible, $Q^{-1} = Q'$ and $QQ' = I_n$: the rows of Q are orthonormal too.

Proof. (a) $(Qx)'(Qy) = x'Q'Qy = x'y$; take $y = x$. (b) Q' is a left inverse of the square matrix Q, so $Q^{-1} = Q'$ (Theorem 2.18), and $QQ' = QQ^{-1} = I_n$. □

Theorem 6.11 (Gram–Schmidt)**.** Let $a_1, \dots, a_p \in \mathbb{R}^n$ be linearly independent. Put $b_1 = a_1$ and, for $j = 2, \dots, p$,

$$b_j = a_j - t_{1j}b_1 - t_{2j}b_2 - \cdots - t_{j-1,j}b_{j-1}, \qquad t_{ij} = \frac{a_j'b_i}{\|b_i\|^2} \quad (i < j)$$

Then $b_1, \dots, b_p$ are nonzero and pairwise orthogonal, and $\operatorname{span}\{b_1, \dots, b_j\} = \operatorname{span}\{a_1, \dots, a_j\}$ for every j. Hence the vectors $q_j = b_j/\|b_j\|$ $(1 \leqslant j \leqslant p)$ form an orthonormal basis of $\operatorname{span}\{a_1, \dots, a_p\}$.

Proof. Induction on j: $b_1 = a_1 \neq 0$, since an independent list does not contain 0. Let $j \geqslant 2$, and let $b_1, \dots, b_{j-1}$ be nonzero, pairwise orthogonal, with span $W_{j-1} = \operatorname{span}\{a_1, \dots, a_{j-1}\}$; so b_j is defined.

- Orthogonal: for $l < j$, $b_l' b_j = b_l' a_j - \sum_{i<j} t_{ij} b_l' b_i = b_l' a_j - t_{lj}\|b_l\|^2 = 0$.
- Nonzero: otherwise $a_j = \sum_{i<j} t_{ij} b_i \in W_{j-1}$, and $a_1, \dots, a_j$ would be dependent (Proposition 3.12(c)) ✘
- Span: b_j is a combination of a_j and $b_1, \dots, b_{j-1} \in W_{j-1}$, and $a_j = b_j + \sum_{i<j} t_{ij} b_i$; so each of the lists $a_1, \dots, a_j$ and $b_1, \dots, b_j$ lies in the span of the other, and the two spans are equal (Theorem 3.7).

Finally, the q_j are pairwise orthogonal unit vectors, hence independent (Proposition 6.8), and they span $\operatorname{span}\{b_1, \dots, b_p\} = \operatorname{span}\{a_1, \dots, a_p\}$. □

Theorem 6.12 (QR decomposition)**.** Let $A = [a_1 \ \cdots \ a_p] \in \mathbb{R}^{n\times p}$ have full column rank p. Then $A = QR$, where $Q \in \mathbb{R}^{n\times p}$ has orthonormal columns ($Q'Q = I_p$) and $R \in \mathbb{R}^{p\times p}$ is upper triangular with positive diagonal entries; moreover $R = Q'A$.

Proof.
- The columns of A are independent (Theorem 4.13(b)). Apply Theorem 6.11 to them, and let $Q = [q_1 \ \cdots \ q_p]$.
- The construction gives $a_j = b_j + \sum_{i<j} t_{ij} b_i = \sum_{i\leqslant j} r_{ij} q_i$, where $r_{jj} = \|b_j\| > 0$ and $r_{ij} = t_{ij}\|b_i\|$ for $i < j$. Let $R = (r_{ij})$, with $r_{ij} = 0$ for $i > j$. Then $\operatorname{col}_j(QR) = Q \operatorname{col}_j(R) = \sum_{i\leqslant j} r_{ij} q_i = a_j$ (Theorem 1.4(a),(b)), so $A = QR$.
- $Q'Q = I_p$, as the q_j are orthonormal; hence $Q'A = Q'QR = R$. □

Example 6.13. Find the QR decomposition of $A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 0 & 2 & 4 \\ 2 & 2 & 5 \end{pmatrix}$.

Solution. Apply Gram–Schmidt (Theorem 6.11) to the columns a_1, a_2, a_3.

- $b_1 = a_1 = (1, 2, 0, 2)'$, $\|b_1\|^2 = 1 + 4 + 0 + 4 = 9$.
- $a_2' b_1 = 3 + 2 + 0 + 4 = 9$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (3, 1, 2, 2)' - (1, 2, 0, 2)' = (2, -1, 2, 0)'$, with $\|b_2\|^2 = 4 + 1 + 4 + 0 = 9$.
- $a_3' b_1 = 2 + 6 + 0 + 10 = 18$ and $a_3' b_2 = 4 - 3 + 8 + 0 = 9$, so $t_{13} = 2$, $t_{23} = 1$ and

$$b_3 = a_3 - 2b_1 - b_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \qquad \|b_3\|^2 = 4 + 0 + 4 + 1 = 9$$

- Check: $b_1' b_2 = 2 - 2 + 0 + 0 = 0$, $b_1' b_3 = -2 + 0 + 0 + 2 = 0$, $b_2' b_3 = -4 + 0 + 4 + 0 = 0$.
- All three lengths are 3, so $q_j = \frac{1}{3} b_j$, $r_{jj} = 3$, $r_{12} = 3t_{12} = 3$, $r_{13} = 3t_{13} = 6$ and $r_{23} = 3t_{23} = 3$:

$$Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 2 & 2 \\ 2 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 3 & 3 & 6 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}$$

As a check, $R = Q'A$; for instance $r_{23} = q_2' a_3 = \frac{1}{3}(4 - 3 + 8 + 0) = 3$.

Orthogonal projections

Definition 6.14. Let W be a subspace of $\mathbb{R}^n$ and $y \in \mathbb{R}^n$. By Theorem 6.5(c), $y = \hat{w} + (y - \hat{w})$ with $\hat{w} \in W$ and $y - \hat{w} \in W^\perp$ for exactly one vector $\hat{w}$, the **orthogonal projection** of y onto W. Thus $\hat{w}$ is the only vector of W with $y - \hat{w} \perp W$.

Theorem 6.15. Let $\hat{w}$ be the orthogonal projection of $y \in \mathbb{R}^n$ onto a subspace W of $\mathbb{R}^n$. Then

$$\|y - w\|^2 = \|y - \hat{w}\|^2 + \|\hat{w} - w\|^2 \qquad (w \in W)$$

Hence $\hat{w}$ is the vector of W closest to y: $\|y - w\| > \|y - \hat{w}\|$ for every $w \in W$ with $w \neq \hat{w}$.

Proof. $y - w = (y - \hat{w}) + (\hat{w} - w)$, where $\hat{w} - w \in W$ and $y - \hat{w} \perp W$; apply Pythagoras (Proposition 6.2). If $w \neq \hat{w}$, then $\|\hat{w} - w\|^2 > 0$. □

The number $\|y - \hat{w}\| = \min_{w \in W} \|y - w\|$ is the **distance** from y to W.

Idempotent matrices: rank equals trace

Theorem 6.16. Let $A \in \mathbb{R}^{n \times n}$ be idempotent, $A^2 = A$.

(a) $Av = v$ for every $v \in \mathcal{C}(A)$, and $\mathbb{R}^n = \mathcal{N}(A) \oplus \mathcal{C}(A)$: every $v \in \mathbb{R}^n$ is $v = (v - Av) + Av$ with $v - Av \in \mathcal{N}(A)$ and $Av \in \mathcal{C}(A)$.

(b) $I_n - A$ is idempotent.

(c) $\operatorname{rank} A = \operatorname{tr} A$.

(d) If moreover $A' = A$, then Av is the orthogonal projection of v onto $\mathcal{C}(A)$, for every $v \in \mathbb{R}^n$.

Proof. (a)

- If $v = Aw$, then $Av = A^2 w = Aw = v$. For every v, $A(v - Av) = Av - A^2 v = 0$.
- If $v \in \mathcal{N}(A) \cap \mathcal{C}(A)$, then $v = Av = 0$. So the sum $\mathcal{N}(A) + \mathcal{C}(A)$ is direct (Definition 3.20), and it is $\mathbb{R}^n$ by the displayed decomposition.

(b) $(I - A)^2 = I - 2A + A^2 = I - A$.

(c) If $A = 0$, then $\operatorname{rank} A = 0 = \operatorname{tr} A$. Let $r = \operatorname{rank} A \geqslant 1$.

- Take a full-rank factorization $A = BC$, with $B \in \mathbb{R}^{n \times r}$, $C \in \mathbb{R}^{r \times n}$, and $L \in \mathbb{R}^{r \times n}$, $K \in \mathbb{R}^{n \times r}$ such that $LB = I_r$ and $CK = I_r$ (Theorem 4.21(a),(c)).
- $A^2 = A$ reads $BCBC = BC$. Multiplying on the left by L and on the right by K, $L(BCBC)K = (LB)(CB)(CK) = CB$ and $L(BC)K = (LB)(CK) = I_r$; so $CB = I_r$.
- Hence $\operatorname{tr} A = \operatorname{tr}(BC) = \operatorname{tr}(CB) = \operatorname{tr} I_r = r$ (Theorem 1.11(b)).

(d) By Corollary 6.6, $\mathcal{N}(A) = \mathcal{C}(A')^\perp = \mathcal{C}(A)^\perp$. So in (a), $Av \in \mathcal{C}(A)$ and $v - Av \in \mathcal{C}(A)^\perp$: by Definition 6.14, Av is the orthogonal projection of v onto $\mathcal{C}(A)$. □

Projection matrices

Proposition 6.17. For every subspace W of $\mathbb{R}^n$, $(W^\perp)^\perp = W$.

Proof. If $w \in W$, then $y'w = 0$ for every $y \in W^\perp$, so $w \in (W^\perp)^\perp$; thus $W \subseteq (W^\perp)^\perp$. By Theorem 6.5(a),(c), applied to W and to the subspace $W^\perp$, $\dim(W^\perp)^\perp = n - \dim W^\perp = n - (n - \dim W) = \dim W$. Hence $(W^\perp)^\perp = W$ (Theorem 4.5(a)). □

Definition 6.18. Let W be a subspace of $\mathbb{R}^n$. A matrix $P \in \mathbb{R}^{n\times n}$ is a **projection matrix onto** W if Py is the orthogonal projection of y onto W for every $y \in \mathbb{R}^n$. Subspaces W_1, W_2 of $\mathbb{R}^n$ are **orthogonal**, written $W_1 \perp W_2$, if $w_1' w_2 = 0$ for all $w_1 \in W_1$ and $w_2 \in W_2$; their sum is then called an **orthogonal sum**.

Theorem 6.19. Let W be a subspace of $\mathbb{R}^n$.

(a) There is exactly one projection matrix onto W, written P_W. If $W = \{0\}$, then $P_W = 0$. If $d = \dim W \geqslant 1$ and the columns of $Q \in \mathbb{R}^{n\times d}$ form an orthonormal basis of W, then $P_W = QQ'$.

(b) P_W is symmetric and idempotent; $P_W w = w$ for $w \in W$ and $P_W v = 0$ for $v \in W^\perp$; $\mathcal{C}(P_W) = W$ and $\operatorname{rank} P_W = \operatorname{tr} P_W = \dim W$; $\|P_W y\|^2 = y' P_W y$ for $y \in \mathbb{R}^n$; and $I_n - P_W = P_{W^\perp}$.

(c) Conversely, every symmetric idempotent $P \in \mathbb{R}^{n\times n}$ is the projection matrix onto its column space: $P = P_{\mathcal{C}(P)}$.

Proof. (a)

- At most one: if P and $\tilde{P}$ are projection matrices onto W, then Pe_j and $\tilde{P}e_j$ are both the orthogonal projection of e_j, which is unique (Definition 6.14); so P and $\tilde{P}$ have the same columns (Theorem 1.4(b)).
- If $W = \{0\}$, then $W^\perp = \mathbb{R}^n$, and $y = 0 + y$ shows that the projection of every y is 0.
- If $d \geqslant 1$, Gram–Schmidt (Theorem 6.11) turns a basis of W (Theorem 4.4) into an orthonormal basis. If the columns of Q form any orthonormal basis of W, then $Q'Q = I_d$ (Definition 6.9), and for $y \in \mathbb{R}^n$, $QQ'y = Q(Q'y) \in W$ and $Q'(y - QQ'y) = Q'y - Q'y = 0$; so $y - QQ'y \perp W$ (Theorem 6.5(a)), and $QQ'y$ is the projection of y onto W (Definition 6.14). Hence $P_W = QQ'$.

(b)

- By (a), P_W is 0 or QQ'; both are symmetric (Theorem 1.9), and $(QQ')^2 = Q(Q'Q)Q' = QQ'$. Hence $\|P_W y\|^2 = y' P_W' P_W y = y' P_W^2 y = y' P_W y$.
- For $w \in W$, $w = w + 0$ with $0 \in W^\perp$; for $v \in W^\perp$, $v = 0 + v$ with $0 \in W$. By the uniqueness in Definition 6.14, $P_W w = w$ and $P_W v = 0$.
- Every $P_W y$ lies in W, and every $w \in W$ is $P_W w$; so $\mathcal{C}(P_W) = W$, and $\operatorname{rank} P_W = \dim W$ (Definition 4.11), which is $\operatorname{tr} P_W$ (Theorem 6.16(c)).
- For $y \in \mathbb{R}^n$, $(I_n - P_W)y = y - P_W y \in W^\perp$, and $y - (I_n - P_W)y = P_W y \in W = (W^\perp)^\perp$ (Proposition 6.17). By Definition 6.14, applied to the subspace $W^\perp$, $(I_n - P_W)y$ is the projection of y onto $W^\perp$; so $I_n - P_W = P_{W^\perp}$ by (a).

(c) By Theorem 6.16(d), Pv is the orthogonal projection of v onto $\mathcal{C}(P)$ for every $v \in \mathbb{R}^n$; so $P = P_{\mathcal{C}(P)}$ by (a). □

By (a), QQ' does not depend on the choice of the orthonormal basis of W.

Corollary 6.20. Let $A \in \mathbb{R}^{n\times p}$ have full column rank p, and let $A = QR$ as in Theorem 6.12. Then $\mathcal{C}(A) = \mathcal{C}(Q)$ and $P_{\mathcal{C}(A)} = QQ'$.

Proof.

- $\operatorname{col}_j(A) = Q \operatorname{col}_j(R) \in \mathcal{C}(Q)$ (Theorem 1.4(a),(b)), so $\mathcal{C}(A) \subseteq \mathcal{C}(Q)$.
- R is upper triangular with positive diagonal entries, so $\det R = r_{11} r_{22} \cdots r_{pp} \neq 0$ (Theorem 5.15(a)), and R is invertible (Theorem 5.19(a)). Then $Q = AR^{-1}$, and as in the first step $\mathcal{C}(Q) \subseteq \mathcal{C}(A)$.
- The columns of Q are orthonormal, hence linearly independent (Proposition 6.8(b)), and they span $\mathcal{C}(Q) = \mathcal{C}(A)$. So they form an orthonormal basis of $\mathcal{C}(A)$, and $P_{\mathcal{C}(A)} = QQ'$ by Theorem 6.19(a). □

Corollary 6.21. Let $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}_n' x = 0\}$.

(a) $P_{\operatorname{span}\{\mathbf{1}_n\}} = \bar{J}_n$ and $P_{W_0} = C_n$.

(b) $\operatorname{rank} C_n = \operatorname{tr} C_n = n - 1$.

Proof. (a) The vector $q = n^{-1/2}\mathbf{1}_n$ satisfies $q'q = n^{-1}\mathbf{1}'\mathbf{1} = 1$ (Proposition 1.13), so it is an orthonormal basis of $\operatorname{span}\{\mathbf{1}_n\}$, and $P_{\operatorname{span}\{\mathbf{1}_n\}} = qq' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$ (Theorem 6.19(a) and Definition 1.12). By Theorem 6.5(a), $W_0 = \mathcal{N}(\mathbf{1}') = \operatorname{span}\{\mathbf{1}_n\}^\perp$, so $P_{W_0} = I_n - \bar{J}_n = C_n$ (Theorem 6.19(b)).

(b) By (a) and Theorem 6.19(b), $\operatorname{rank} C_n = \operatorname{tr} C_n$. Every diagonal entry of $\bar{J}_n$ is $\frac{1}{n}$, so $\operatorname{tr} C_n = \operatorname{tr} I_n - \operatorname{tr} \bar{J}_n = n - 1$ (Theorem 1.11(a)). □

Example 6.22. For $y \in \mathbb{R}^n$, $\bar{J}_n y = \frac{1}{n}\mathbf{1}(\mathbf{1}'y) = \bar{y}\mathbf{1}$ and $C_n y = y - \bar{y}\mathbf{1}$. By Corollary 6.21(a), these are the orthogonal projections of y onto the constant vectors and onto the centered vectors $W_0 = \operatorname{span}\{\mathbf{1}\}^\perp$. So the decomposition $y = \bar{y}\mathbf{1} + C_n y$ along $\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \oplus W_0$ of Example 3.25 is the orthogonal one, and $\bar{y}\mathbf{1}$ is the constant vector closest to y (Theorem 6.15): $\sum_i (y_i - c)^2 > \sum_i (y_i - \bar{y})^2$ for every $c \neq \bar{y}$. For instance, $y = (9, 3, 5, 3)'$ has $\bar{y} = 5$ and $y = 5\,\mathbf{1}_4 + (4, -2, 0, -2)'$, with $\mathbf{1}_4'(4, -2, 0, -2)' = 0$. Likewise, for any nonzero $a \in \mathbb{R}^n$, $q = a/\|a\|$ is an orthonormal basis of $\operatorname{span}\{a\}$, so $P_{\operatorname{span}\{a\}} = qq' = \frac{aa'}{a'a}$, and the projection of y onto $\operatorname{span}\{a\}$ is $\frac{a'y}{a'a}\,a$ (Theorem 6.19(a)).

Example 6.23. Let $w_1 = \mathbf{1}_4$, $w_2 = (1, -1, 1, -1)'$, $w_3 = (1, 1, -1, -1)'$, $W = \operatorname{span}\{w_1, w_2, w_3\} \subseteq \mathbb{R}^4$ and $y = (9, 3, 5, 3)'$. Find P_W, the vector of W closest to y, and the distance from y to W.

Solution.

- $w_i'w_j = 0$ for $i \neq j$ and $w_j'w_j = 4$; for instance $w_2'w_3 = 1 - 1 - 1 + 1 = 0$. So the vectors $q_j = \frac{1}{2}w_j$ are orthonormal, hence linearly independent (Proposition 6.8(b)), and they form an orthonormal basis of W; $\dim W = 3$.
- With $Q = [q_1\ q_2\ q_3]$, Theorems 6.19(a) and 1.4(d) give

$$P_W = QQ' = \tfrac{1}{4}\big(w_1w_1' + w_2w_2' + w_3w_3'\big) = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix};$$

  for instance, the $(1, 4)$ entry is $\frac{1}{4}(1 - 1 - 1) = -\frac{1}{4}$. Also $\operatorname{tr} P_W = 3 = \dim W$ (Theorem 6.19(b)).
- $Q'y = \frac{1}{2}(w_1'y, w_2'y, w_3'y)' = \frac{1}{2}(20, 8, 4)' = (10, 4, 2)'$, so

$$P_W y = Q(Q'y) = 10q_1 + 4q_2 + 2q_3 = 5w_1 + 2w_2 + w_3 = \begin{pmatrix} 8 \\ 4 \\ 6 \\ 2 \end{pmatrix}, \qquad y - P_W y = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

  By Theorem 6.15, $(8, 4, 6, 2)'$ is the vector of W closest to y, at distance $\|y - P_W y\| = \sqrt{1 + 1 + 1 + 1} = 2$.
- Check: $h = (1, -1, -1, 1)'$ is orthogonal to w_1, w_2, w_3, so $h \in W^\perp$ (Theorem 6.5(a)); as $\dim W^\perp = 4 - 3 = 1$ (Theorem 6.5(c)) and $h \neq 0$, $W^\perp = \operatorname{span}\{h\}$ (Theorem 4.5(d)). By Theorem 6.19(b) and Example 6.22, $P_W = I_4 - P_{W^\perp} = I_4 - \frac{1}{4}hh'$, which is the matrix above. Pythagoras (Proposition 6.2): $\|y\|^2 = 81 + 9 + 25 + 9 = 124$ and $\|P_W y\|^2 + \|y - P_W y\|^2 = (64 + 16 + 36 + 4) + 4 = 124$.

Projections onto orthogonal sums and nested subspaces

Theorem 6.24. Let $W_1 \perp W_2$ be orthogonal subspaces of $\mathbb{R}^n$, and $P_1 = P_{W_1}$, $P_2 = P_{W_2}$. Then $W_1 \cap W_2 = \{0\}$, so the sum $W_1 \oplus W_2$ is direct, and

$$P_{W_1 \oplus W_2} = P_1 + P_2, \qquad P_1P_2 = P_2P_1 = 0, \qquad \|P_{W_1 \oplus W_2} y\|^2 = \|P_1 y\|^2 + \|P_2 y\|^2 \quad (y \in \mathbb{R}^n)$$

Proof. • If $w \in W_1 \cap W_2$, then $w'w = 0$, so $w = 0$; hence the sum is direct (Definition 3.20). Let $W = W_1 \oplus W_2$.

- $W_2 \subseteq W_1^\perp$, so $P_1 w_2 = 0$ for $w_2 \in W_2$ (Theorem 6.19(b)); as $P_2 y \in W_2$, $P_1 P_2 y = 0$ for every y, that is, $P_1 P_2 = 0$. Likewise $P_2 P_1 = 0$.
- Let $y \in \mathbb{R}^n$. Then $P_1 y + P_2 y \in W$, and $z = y - P_1 y - P_2 y$ satisfies $w_1' z = w_1'(y - P_1 y) - w_1' P_2 y = 0$ for $w_1 \in W_1$ (as $y - P_1 y \perp W_1$ and $P_2 y \in W_2$); likewise $w_2' z = 0$ for $w_2 \in W_2$. So $z \perp W$ (every $w \in W$ is $w_1 + w_2$, and $w'z = w_1'z + w_2'z = 0$), and $(P_1 + P_2)y$ is the projection of y onto W (Definition 6.14); hence $P_W = P_1 + P_2$ (Theorem 6.19(a)).
- $P_W y = P_1 y + P_2 y$ with $P_1 y \in W_1$ orthogonal to $P_2 y \in W_2$; apply Pythagoras (Proposition 6.2). □

Theorem 6.25. Let $U \subseteq W$ be subspaces of $\mathbb{R}^n$.

(a) $P_W P_U = P_U P_W = P_U$.

(b) W is the orthogonal sum of U and $W \cap U^\perp$; the projection matrix onto $W \cap U^\perp$ is $P_W - P_U$, and $\dim(W \cap U^\perp) = \dim W - \dim U$.

(c) For every $y \in \mathbb{R}^n$,

$$y = P_U y + (P_W - P_U)y + (I_n - P_W)y,$$

where the three terms lie in U, $W \cap U^\perp$ and $W^\perp$ and are pairwise orthogonal. Hence

$$\|y\|^2 = \|P_U y\|^2 + \|(P_W - P_U)y\|^2 + \|(I_n - P_W)y\|^2, \qquad \|P_W y\|^2 = \|P_U y\|^2 + \|(P_W - P_U)y\|^2$$

Proof. (a) For every y, $P_U y \in U \subseteq W$, so $P_W P_U y = P_U y$ (Theorem 6.19(b)): $P_W P_U = P_U$. As P_U and P_W are symmetric, $P_U P_W = P_U' P_W' = (P_W P_U)' = P_U' = P_U$.

(b) Let $V = W \cap U^\perp$, a subspace (Theorems 6.5(a) and 3.21(a)).

- $U \perp V$, since $V \subseteq U^\perp$; and $U + V \subseteq W$, since U and V lie in the subspace W.
- For $w \in W$, $w - P_U w$ lies in W (as $P_U w \in U \subseteq W$) and in $U^\perp$ (Definition 6.14); so $w = P_U w + (w - P_U w) \in U + V$. Hence $W = U + V$, an orthogonal sum.
- By Theorem 6.24, $P_W = P_U + P_V$, that is, $P_V = P_W - P_U$. By Theorems 6.19(b) and 1.11(a), $\dim V = \operatorname{tr} P_V = \operatorname{tr} P_W - \operatorname{tr} P_U = \dim W - \dim U$.

(c) The three terms add up to y, and they lie in U, V (by (b)) and $W^\perp$ (Theorem 6.19(b)). $U \perp V$, and U and V lie in W, so both are orthogonal to $W^\perp$. Pythagoras (Proposition 6.2) for $y = P_U y + \big((P_W - P_U)y + (I_n - P_W)y\big)$, and then for the second term, gives the first identity; for $P_W y = P_U y + (P_W - P_U)y$ it gives the second. □

Example 6.26. In Example 6.23, let $U = \operatorname{span}\{w_1\} = \operatorname{span}\{\mathbf{1}_4\} \subseteq W$, and $y = (9, 3, 5, 3)'$.

- $P_U = \bar{J}_4$ (Corollary 6.21(a)), and $P_U = q_1 q_1'$. By Theorem 6.25(b),

$$P_{W \cap U^\perp} = P_W - P_U = q_2 q_2' + q_3 q_3' = \frac{1}{4}\big(w_2 w_2' + w_3 w_3'\big) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

and $\dim(W \cap U^\perp) = 3 - 1 = 2$. Indeed $w_2, w_3 \in W$ are orthogonal to w_1, so they lie in $W \cap U^\perp$ (Theorem 6.5(a)), and as two linearly independent vectors of a subspace of dimension 2 they form a basis of it (Theorem 4.5(d)).

- $P_U y = \bar{y}\mathbf{1}_4 = 5\,\mathbf{1}_4$ (Example 6.22), $P_W y = (8, 4, 6, 2)'$ (Example 6.23), and Theorem 6.25(c) gives

$$\begin{pmatrix} 9 \\ 3 \\ 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ 5 \\ 5 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \\ 1 \\ -3 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \qquad 124 = 100 + 20 + 4,$$

with the middle term $(P_W - P_U)y = 2w_2 + w_3$. Also $\|P_W y\|^2 = 120 = 100 + 20$.

- Check: the middle term is orthogonal to $\mathbf{1}_4$ ($3 - 1 + 1 - 3 = 0$) and to $h = (1, -1, -1, 1)'$ ($3 + 1 - 1 - 3 = 0$); and $\|y - \bar{y}\mathbf{1}_4\|^2 = 16 + 4 + 0 + 4 = 24 = 20 + 4$, the squared lengths of the last two terms.

Exercises

Exercise 6.1. (a) Find all vectors of $\mathbb{R}^4$ that are orthogonal to both $\mathbf{1}_4$ and $(1, 2, 3, 4)'$.

(b) What is the dimension of the subspace of $\mathbb{R}^6$ consisting of the vectors orthogonal to both $(1, 1, -2, 3, 4, 5)'$ and $(0, 0, 1, 1, 0, 7)'$?

Solution. (a) With $A = [\mathbf{1}_4 \;\; (1, 2, 3, 4)']$, the vectors sought form $\mathcal{N}(A')$ (Theorem 6.5(a)). Row-reduce A':

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$

With $y_3 = s$ and $y_4 = t$ free, $y_1 = s + 2t$ and $y_2 = -2s - 3t$; so the vectors are

$$y = s \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix} \qquad (s, t \in \mathbb{R}),$$

a subspace of dimension $2 = 4 - 2$ (Theorem 6.5(c)). Check: $1 - 2 + 1 + 0 = 0$, $1 - 4 + 3 + 0 = 0$, $2 - 3 + 0 + 1 = 0$, $2 - 6 + 0 + 4 = 0$.

(b) Let a_1, a_2 be the two vectors and $A = [a_1 \;\; a_2] \in \mathbb{R}^{6\times 2}$. The subspace is $\operatorname{span}\{a_1, a_2\}^\perp = \mathcal{N}(A')$ (Theorem 6.5(a)). The vectors are independent: if $c_1 a_1 + c_2 a_2 = 0$, the first entry gives $c_1 = 0$, and then $c_2 a_2 = 0$ with $a_2 \neq 0$ gives $c_2 = 0$. So $\dim \operatorname{span}\{a_1, a_2\} = 2$, and the answer is $6 - 2 = 4$ (Theorem 6.5(c)).

Exercise 6.2. Let $Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$, with columns $q_1, \ldots, q_4$, and $y = (4, 2, 0, 6)'$.

(a) Show that Q is a symmetric orthogonal matrix, and find Q^{-1}.

(b) Find $c \in \mathbb{R}^4$ with $y = c_1 q_1 + \cdots + c_4 q_4$, and check that $\|c\| = \|y\|$.

(c) Find the vector of $\operatorname{span}\{q_1, q_2\}$ closest to y, and its distance from y.

Solution. (a) $Q' = Q$ by inspection. Each column of $2Q$ has squared length 4, and any two of them are orthogonal, for instance $(1, -1, 1, -1)'$ and $(1, 1, -1, -1)'$ have inner product $1 - 1 - 1 + 1 = 0$; so $Q'Q = I_4$ (Definition 6.9), and $Q^{-1} = Q' = Q$ (Proposition 6.10(b)).

(b) $y = Qc$ (Theorem 1.4(b)) gives $c = Q^{-1}y = Q'y$:

$$c = \tfrac{1}{2}\big(4 + 2 + 0 + 6,\; 4 - 2 + 0 - 6,\; 4 + 2 - 0 - 6,\; 4 - 2 - 0 + 6\big)' = (6, -2, 0, 4)'$$

(equivalently, $c_j = q_j' y$ by Proposition 6.8(a)). Then $\|c\|^2 = 36+4+0+16 = 56 = 16+4+0+36 = \|y\|^2$, as Proposition 6.10(a) says.

(c) q_1, q_2 form an orthonormal basis of $V = \operatorname{span}\{q_1, q_2\}$, so $P_V y = q_1(q_1' y) + q_2(q_2' y) = 6q_1 - 2q_2$ (Theorems 6.19(a) and 1.4(d)):

$$P_V y = 3\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 4 \end{pmatrix}, \qquad y - P_V y = \begin{pmatrix} 2 \\ -2 \\ -2 \\ 2 \end{pmatrix} = 4q_4$$

By Theorem 6.15, $(2,4,2,4)'$ is the closest vector, at distance $\|y - P_V y\| = \sqrt{16} = 4$; by (b), $y - P_V y = c_3 q_3 + c_4 q_4$ and $\|y - P_V y\|^2 = c_3^2 + c_4^2 = 16$.

Exercise 6.3. Let $A = [a_1\ a_2\ a_3] \in \mathbb{R}^{4\times 3}$ have the columns $a_1 = \mathbf{1}_4$, $a_2 = (2,0,2,0)'$ and $a_3 = (3,3,1,1)'$.

(a) Find the QR decomposition of A by the Gram–Schmidt process.

(b) Show that $Q'Q = I_3$ but $QQ' \neq I_4$, and that QQ' is the projection matrix P_W of Example 6.23.

Solution. (a) By Theorem 6.11:

- $b_1 = a_1$, $\|b_1\|^2 = 4$.
- $a_2' b_1 = 4$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (1,-1,1,-1)'$, with $\|b_2\|^2 = 4$.
- $a_3' b_1 = 8$ and $a_3' b_2 = 3 - 3 + 1 - 1 = 0$, so $t_{13} = 2$, $t_{23} = 0$ and $b_3 = a_3 - 2b_1 = (1,1,-1,-1)'$, with $\|b_3\|^2 = 4$.
- All three lengths are 2: $q_j = \frac{1}{2} b_j$, $r_{jj} = 2$, $r_{12} = 2t_{12} = 2$, $r_{13} = 2t_{13} = 4$ and $r_{23} = 0$:

$$Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \end{pmatrix}, \qquad R = \begin{pmatrix} 2 & 2 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Check: the third column of QR is $Q(4,0,2)' = \frac{1}{2}(4+2,\ 4+2,\ 4-2,\ 4-2)' = (3,3,1,1)' = a_3$.

(b) The columns of Q are orthonormal, so $Q'Q = I_3$. They are the vectors $q_j = \frac{1}{2} w_j$ of Example 6.23, so

$$QQ' = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix} = P_W \neq I_4;$$

for instance, its $(1,1)$ entry is $\frac{1}{4}(1+1+1) = \frac{3}{4}$. The rows of Q are not orthonormal. By Corollary 6.20, $\mathcal{C}(A) = \mathcal{C}(Q) = W$ and $QQ' = P_{\mathcal{C}(A)}$.

Exercise 6.4. Let $B = [a_1\ a_2\ a_3\ a_4] \in \mathbb{R}^{4\times 4}$, with a_1, a_2, a_3 of Exercise 6.3 and $a_4 = (4,0,0,0)'$.

(a) Continue the Gram–Schmidt process of Exercise 6.3(a) to find the QR decomposition of B. Which matrix of Exercise 6.2 is its Q?

(b) Solve $Bx = b$ for $b = (9,3,5,3)'$ by solving $Rx = Q'b$.

(c) Show that $|\det B| = \det R = 16$, without expanding $\det B$.

Solution. (a) $a_4' b_1 = 4$, $a_4' b_2 = 4$, $a_4' b_3 = 4$, so $t_{14} = t_{24} = t_{34} = 1$ and

$$b_4 = a_4 - b_1 - b_2 - b_3 = (4-1-1-1,\ 0-1+1-1,\ 0-1-1+1,\ 0-1+1+1)' = (1,-1,-1,1)',$$

with $\|b_4\|^2 = 4$. So $q_4 = \frac{1}{2} b_4$, $r_{44} = 2$ and $r_{14} = r_{24} = r_{34} = 2 \cdot 1 = 2$: $B = QR$ with

$$Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 2 & 2 & 4 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

the orthogonal matrix Q of Exercise 6.2.

(b) $Bx = b$ is $QRx = b$; multiplying by Q', and back by Q, it is equivalent to $Rx = Q'b$, as $Q'Q = QQ' = I_4$. Here $Q'b = \frac{1}{2}(20, 8, 4, 4)' = (10, 4, 2, 2)'$, and back substitution gives

$$2x_4 = 2, \quad 2x_3 + 2x_4 = 2, \quad 2x_2 + 2x_4 = 4, \quad 2x_1 + 2x_2 + 4x_3 + 2x_4 = 10 : \qquad x = (3, 1, 0, 1)'$$

Check: $3a_1 + a_2 + a_4 = (3 + 2 + 4,\ 3,\ 3 + 2,\ 3)' = (9, 3, 5, 3)'$.

(c) $\det B = \det Q \det R$ (Theorem 5.19(b)). By Theorem 5.15(b) and Theorem 5.19(b), $(\det Q)^2 = \det Q' \det Q = \det(Q'Q) = \det I_4 = 1$, so $\det Q = \pm 1$; and $\det R = 2^4 = 16$ (Theorem 5.15(a)). Hence $|\det B| = 16$.

Exercise 6.5. Let A be the matrix of Example 6.13, $W = \mathcal{C}(A)$ and $y = (0, 6, 3, 3)'$.

(a) Show that $h = (0, 2, 1, -2)'$ spans $W^\perp$.

(b) Compute $P_W = QQ'$, with Q of Example 6.13, and check that $P_W = I_4 - \frac{1}{9}hh'$.

(c) Find the vector of W closest to y and its distance from y, and write the closest vector as Ax by solving $Rx = Q'y$.

Solution. (a) By Example 6.13, $W = \operatorname{span}\{b_1, b_2, b_3\}$ with $b_1 = (1, 2, 0, 2)'$, $b_2 = (2, -1, 2, 0)'$, $b_3 = (-2, 0, 2, 1)'$ (Theorem 6.11). Now $b_1'h = 0 + 4 + 0 - 4 = 0$, $b_2'h = 0 - 2 + 2 + 0 = 0$ and $b_3'h = 0 + 0 + 2 - 2 = 0$, so $h \in W^\perp$ (Theorem 6.5(a)). As $\dim W = 3$, $\dim W^\perp = 1$ (Theorem 6.5(c)), and the nonzero vector h spans $W^\perp$ (Theorem 4.5(d)).

(b) By Corollary 6.20 and Theorem 1.4(d),

$$P_W = QQ' = \tfrac{1}{9}\big(b_1b_1' + b_2b_2' + b_3b_3'\big) = \frac{1}{9}\begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 5 & -2 & 4 \\ 0 & -2 & 8 & 2 \\ 0 & 4 & 2 & 5 \end{pmatrix};$$

for instance, the $(2, 3)$ entry is $\frac{1}{9}(0 - 2 + 0)$ and the $(3, 3)$ entry is $\frac{1}{9}(0 + 4 + 4)$. Since $h'h = 9$, $\frac{1}{9}hh'$ is the projection matrix onto $\operatorname{span}\{h\} = W^\perp$ (Example 6.22), and $P_W = I_4 - \frac{1}{9}hh'$ (Theorem 6.19(b)); entry by entry, $I_4 - \frac{1}{9}hh'$ is the matrix above, as hh' has rows 0, $(0, 4, 2, -4)$, $(0, 2, 1, -2)$, $(0, -4, -2, 4)$.

(c) $h'y = 0 + 12 + 3 - 6 = 9$, so $(I_4 - P_W)y = \frac{h'y}{h'h}h = h$ and $P_Wy = y - h = (0, 4, 2, 5)'$. This is the closest vector (Theorem 6.15), at distance $\|h\| = 3$. Next $Q'y = \frac{1}{3}(b_1'y, b_2'y, b_3'y)' = \frac{1}{3}(18, 0, 9)' = (6, 0, 3)'$, and with R of Example 6.13,

$$3x_3 = 3, \quad 3x_2 + 3x_3 = 0, \quad 3x_1 + 3x_2 + 6x_3 = 6 : \qquad x = (1, -1, 1)'$$

Then $Ax = QRx = QQ'y = P_Wy$ (Corollary 6.20); directly, $Ax = a_1 - a_2 + a_3 = (1 - 3 + 2,\ 2 - 1 + 3,\ 0 - 2 + 4,\ 2 - 2 + 5)' = (0, 4, 2, 5)'$.

Exercise 6.6. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{pmatrix}$, $W = \mathcal{C}(A)$ and $y = (2, 1, 1, 4, 8)'$.

(a) Find a basis of $W^\perp$.

(b) Find the QR decomposition of A, and compute $P_W = QQ'$. Check that $P_WA = A$ and $\operatorname{tr} P_W = 2$.

(c) Write $y = P_Wy + (I_5 - P_W)y$, the decomposition along $\mathbb{R}^5 = W \oplus W^\perp$, check that the two parts are orthogonal, and find the distance from y to W.

Solution. (a) $W^\perp = \mathcal{N}(A')$ (Theorem 6.5(a)). Row-reduce A':

$$\begin{pmatrix}1&1&1&2&3\\1&0&0&0&1\end{pmatrix} \xrightarrow{R_1\leftrightarrow R_2} \begin{pmatrix}1&0&0&0&1\\1&1&1&2&3\end{pmatrix} \xrightarrow{R_2\leftarrow R_2-R_1} \begin{pmatrix}1&0&0&0&1\\0&1&1&2&2\end{pmatrix}$$

With x_3, x_4, x_5 free, $x_1 = -x_5$ and $x_2 = -x_3 - 2x_4 - 2x_5$; so $v_1 = (0,-1,1,0,0)'$, $v_2 = (0,-2,0,1,0)'$, $v_3 = (-1,-2,0,0,1)'$ form a basis of $W^\perp$, of dimension $5-2=3$.

(b) By Theorem 6.11, $b_1 = a_1 = (1,1,1,2,3)'$, $\|b_1\|^2 = 1+1+1+4+9 = 16$; $a_2'b_1 = 1+3 = 4$, so $t_{12} = \frac{1}{4}$ and

$$b_2 = a_2 - \tfrac{1}{4}b_1 = \tfrac{1}{4}(3,-1,-1,-2,1)', \qquad \|b_2\|^2 = \tfrac{1}{16}(9+1+1+4+1) = 1$$

So $q_1 = \frac{1}{4}b_1$, $q_2 = b_2$, $r_{11} = 4$, $r_{12} = t_{12}\|b_1\| = 1$, $r_{22} = 1$ (Theorem 6.12):

$$Q = \frac{1}{4}\begin{pmatrix}1&3\\1&-1\\1&-1\\2&-2\\3&1\end{pmatrix}, \qquad R = \begin{pmatrix}4&1\\0&1\end{pmatrix}, \qquad P_W = QQ' = \frac{1}{8}\begin{pmatrix}5&-1&-1&-2&3\\-1&1&1&2&1\\-1&1&1&2&1\\-2&2&2&4&2\\3&1&1&2&5\end{pmatrix}$$

(Corollary 6.20); for instance, the $(1,5)$ entry is $\frac{1}{16}(1\cdot 3 + 3\cdot 1) = \frac{3}{8}$. Row by row, $P_W a_1 = \frac{1}{8}(5-1-1-4+9,\ -1+1+1+4+3,\ -1+1+1+4+3,\ -2+2+2+8+6,\ 3+1+1+4+15)' = (1,1,1,2,3)'$ and $P_W a_2 = \frac{1}{8}(5+3,\ -1+1,\ -1+1,\ -2+2,\ 3+5)' = (1,0,0,0,1)'$, so $P_W A = A$; and $\operatorname{tr} P_W = \frac{1}{8}(5+1+1+4+5) = 2 = \dim W$ (Theorem 6.19(b)).

(c) $Q'y = (\frac{1}{4}(2+1+1+8+24),\ \frac{1}{4}(6-1-1-8+8))' = (9,1)'$, so $P_W y = Q(Q'y) = 9q_1 + q_2 = \frac{1}{4}(9+3,\ 9-1,\ 9-1,\ 18-2,\ 27+1)' = (3,2,2,4,7)'$, and

$$\begin{pmatrix}2\\1\\1\\4\\8\end{pmatrix} = \begin{pmatrix}3\\2\\2\\4\\7\end{pmatrix} + \begin{pmatrix}-1\\-1\\-1\\0\\1\end{pmatrix}$$

The second part is $v_3 - v_1 \in W^\perp$, and $(P_W y)'(I_5 - P_W)y = -3-2-2+0+7 = 0$. The distance is $\|(-1,-1,-1,0,1)'\| = 2$ (Theorem 6.15).

Exercise 6.7. Let $E = \begin{pmatrix}1&1&-1&1\\1&0&1&0\\0&-1&2&-1\\-1&-1&1&-1\end{pmatrix}$ and $v = \mathbf{1}_4$.

(a) Show that E is idempotent but not symmetric.

(b) Find $\operatorname{rank} E$, $\operatorname{tr} E$, $\operatorname{rank}(I_4 - E)$ and $\operatorname{tr}(I_4 - E)$.

(c) Decompose v along $\mathbb{R}^4 = \mathcal{N}(E) \oplus \mathcal{C}(E)$. Are the two parts orthogonal?

Solution. Let $\rho_1, \dots, \rho_4$ be the rows of E; then $\rho_3 = \rho_2 - \rho_1$ and $\rho_4 = -\rho_1$.

(a) By Theorem 1.4(c), $\operatorname{row}_i(E^2) = \rho_i E = e_{i1}\rho_1 + e_{i2}\rho_2 + e_{i3}\rho_3 + e_{i4}\rho_4$. So $\rho_1 E = \rho_1 + \rho_2 - \rho_3 + \rho_4 = \rho_1 + \rho_2 - (\rho_2 - \rho_1) - \rho_1 = \rho_1$ and $\rho_2 E = \rho_1 + \rho_3 = \rho_2$; then $\rho_3 E = \rho_2 E - \rho_1 E = \rho_3$ and $\rho_4 E = -\rho_1 E = \rho_4$. Hence $E^2 = E$. It is not symmetric: $e_{13} = -1 \neq 0 = e_{31}$.

(b) $R_3 \leftarrow R_3 - R_2 + R_1$ and $R_4 \leftarrow R_4 + R_1$ make rows 3 and 4 zero, and then $R_2 \leftarrow R_2 - R_1$ gives an echelon form with the two nonzero rows $(1,1,-1,1)$ and $(0,-1,2,-1)$; so $\operatorname{rank} E = 2 = 1+0+2-1 = \operatorname{tr} E$. Next

$$I_4 - E = \begin{pmatrix}0&-1&1&-1\\-1&1&-1&0\\0&1&-1&1\\1&1&-1&2\end{pmatrix},$$

whose rows $\sigma_1, \dots, \sigma_4$ satisfy $\sigma_3 = -\sigma_1$ and $\sigma_4 = -2\sigma_1 - \sigma_2$. So $R_3 \leftarrow R_3 + R_1$, $R_4 \leftarrow R_4 + 2R_1 + R_2$ and $R_1 \leftrightarrow R_2$ give an echelon form with the two nonzero rows $\sigma_2 = (-1, 1, -1, 0)$ and $\sigma_1 = (0, -1, 1, -1)$, and $\operatorname{rank}(I_4 - E) = 2 = 0 + 1 - 1 + 2 = \operatorname{tr}(I_4 - E)$, as Theorem 6.16(b),(c) say.

(c) $Ev = (2, 2, 0, -2)' \in \mathcal{C}(E)$ (the row sums of E) and $v - Ev = (-1, -1, 1, 3)'$, which lies in $\mathcal{N}(E)$: $E(-1, -1, 1, 3)' = (-1 - 1 - 1 + 3,\ -1 + 0 + 1 + 0,\ 0 + 1 + 2 - 3,\ 1 + 1 + 1 - 3)' = 0$. So $v = (-1, -1, 1, 3)' + (2, 2, 0, -2)'$ (Theorem 6.16(a)). The parts are not orthogonal: $(v - Ev)'(Ev) = -2 - 2 + 0 - 6 = -10$. (By Theorem 6.16(d), they would be if E were symmetric.)

Exercise 6.8. Let $a_1 = (1, 2, 2, 0)'$, $a_2 = (3, 2, 1, 2)'$, $W = \operatorname{span}\{a_1, a_2\} \subseteq \mathbb{R}^4$ and $y = (4, 2, 5, 3)'$.

(a) Find an orthonormal basis q_1, q_2 of W by the Gram–Schmidt process.

(b) Compute P_W, and check that $\operatorname{tr} P_W = 2$.

(c) Find the vector of W closest to y, and write it as a combination of a_1 and a_2. Find the distance from y to W.

Solution. (a) $b_1 = a_1$, with $\|b_1\|^2 = 1 + 4 + 4 + 0 = 9$; $a_2' b_1 = 3 + 4 + 2 + 0 = 9$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (2, 0, -1, 2)'$, with $\|b_2\|^2 = 4 + 0 + 1 + 4 = 9$ (Theorem 6.11). Hence $q_1 = \frac{1}{3}(1, 2, 2, 0)'$ and $q_2 = \frac{1}{3}(2, 0, -1, 2)'$.

(b) With $Q = [q_1\ q_2]$, Theorems 6.19(a) and 1.4(d) give

$$P_W = QQ' = \frac{1}{9}\big(b_1 b_1' + b_2 b_2'\big) = \frac{1}{9}\left[\begin{pmatrix} 1 & 2 & 2 & 0 \\ 2 & 4 & 4 & 0 \\ 2 & 4 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 4 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \\ -2 & 0 & 1 & -2 \\ 4 & 0 & -2 & 4 \end{pmatrix}\right] = \frac{1}{9}\begin{pmatrix} 5 & 2 & 0 & 4 \\ 2 & 4 & 4 & 0 \\ 0 & 4 & 5 & -2 \\ 4 & 0 & -2 & 4 \end{pmatrix},$$

and $\operatorname{tr} P_W = \frac{1}{9}(5 + 4 + 5 + 4) = 2 = \dim W$ (Theorem 6.19(b)).

(c) The closest vector is the projection (Theorem 6.15):

$$P_W y = \frac{1}{9}\begin{pmatrix} 20 + 4 + 0 + 12 \\ 8 + 8 + 20 + 0 \\ 0 + 8 + 25 - 6 \\ 16 + 0 - 10 + 12 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 3 \\ 2 \end{pmatrix} = a_1 + a_2$$

The distance is $\|y - P_W y\| = \|(0, -2, 2, 1)'\| = \sqrt{0 + 4 + 4 + 1} = 3$. Check: $a_1'(y - P_W y) = 0 - 4 + 4 + 0 = 0$ and $a_2'(y - P_W y) = 0 - 4 + 2 + 2 = 0$.

Exercise 6.9. In $\mathbb{R}^4$ let $U = \operatorname{span}\{\mathbf{1}_4\}$, $W = \{x : x_1 + x_2 = x_3 + x_4\}$ and $y = (5, 1, 2, 0)'$.

(a) Show that $U \subseteq W$, and find $W^\perp$, P_W and P_U.

(b) Check that $P_W P_U = P_U$.

(c) Find an orthonormal basis of $W \cap U^\perp$, and check that the projection matrix onto $W \cap U^\perp$ is $P_W - P_U$.

(d) Write $y = P_U y + (P_W - P_U)y + (I_4 - P_W)y$, and check that the squared lengths of the three terms add up to $\|y\|^2$.

Solution. Let $h = (1, 1, -1, -1)'$, so that $W = \{x : h'x = 0\}$.

(a) $h'\mathbf{1}_4 = 0$, so $\mathbf{1}_4 \in W$ and $U \subseteq W$. By Theorem 6.5(a), $W = \mathcal{N}(h') = \operatorname{span}\{h\}^\perp$, so $W^\perp = \operatorname{span}\{h\}$ (Proposition 6.17). By Example 6.22 and Theorem 6.19(b), $P_W = I_4 - P_{\operatorname{span}\{h\}} = I_4 - \frac{1}{4}hh'$; and $P_U = \bar{J}_4$ (Corollary 6.21(a)):

$$P_W = \frac{1}{4}\begin{pmatrix} 3 & -1 & 1 & 1 \\ -1 & 3 & 1 & 1 \\ 1 & 1 & 3 & -1 \\ 1 & 1 & -1 & 3 \end{pmatrix}, \qquad P_U = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

(b) Every row of P_W has sum 1, that is, $P_W\mathbf{1} = \mathbf{1}$ (as $\mathbf{1} \in W$). So $P_W P_U = \frac{1}{4}(P_W\mathbf{1})\mathbf{1}' = \frac{1}{4}\mathbf{1}\mathbf{1}' = P_U$, as Theorem 6.25(a) says.

(c) $x \in W \cap U^\perp$ means $x_1 + x_2 = x_3 + x_4$ and $x_1 + x_2 + x_3 + x_4 = 0$, that is, $x_1 + x_2 = 0 = x_3 + x_4$. So $W \cap U^\perp = \{(s, -s, t, -t)' : s, t \in \mathbb{R}\}$, with the orthonormal basis $u_1 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)'$, $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, -1)'$. By Theorem 6.19(a),

$$P_{W\cap U^\perp} = u_1u_1' + u_2u_2' = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix},$$

which is $P_W - P_U$, as Theorem 6.25(b) says; for instance, row 1 of $P_W - P_U$ is $\frac{1}{4}(3-1,\ -1-1,\ 1-1,\ 1-1) = \frac{1}{2}(1, -1, 0, 0)$. The dimension of $W \cap U^\perp$ is $2 = 3 - 1$.

(d) $P_U y = \bar{y}\mathbf{1}_4 = 2\,\mathbf{1}_4$; $h'y = 5 + 1 - 2 - 0 = 4$, so $(I_4 - P_W)y = P_{W^\perp}y = \frac{h'y}{h'h}h = h$ and $P_W y = y - h = (4, 0, 3, 1)'$; and $(P_W - P_U)y = (2, -2, 1, -1)'$. So

$$\begin{pmatrix}5\\1\\2\\0\end{pmatrix} = \begin{pmatrix}2\\2\\2\\2\end{pmatrix} + \begin{pmatrix}2\\-2\\1\\-1\end{pmatrix} + \begin{pmatrix}1\\1\\-1\\-1\end{pmatrix}, \qquad 30 = 16 + 10 + 4,$$

as Theorem 6.25(c) says; also $\|P_W y\|^2 = 16 + 0 + 9 + 1 = 26 = 16 + 10$.

Exercise 6.10. Let $a_1 = \mathbf{1}_5$, $a_2 = (1, 2, 3, 4, 5)'$, $a_3 = (1, 4, 9, 16, 25)'$ and $a_4 = (1, 8, 27, 64, 125)'$, the values of $1, t, t^2, t^3$ at $t = 1, \ldots, 5$, and let $U_k = \operatorname{span}\{a_1, \ldots, a_k\}$ $(k = 1, \ldots, 4)$.

(a) Apply the Gram–Schmidt process to $a_1, \ldots, a_4$, and show that b_2, b_3 and $\frac{5}{6}b_4$ are

$$c_1 = (-2, -1, 0, 1, 2)', \qquad c_2 = (2, -1, -2, -1, 2)', \qquad c_3 = (-1, 2, 0, -2, 1)'$$

(b) Show that b_2, b_3, b_4 are the values at $t = 1, \ldots, 5$ of the polynomials $t - 3$, $(t-3)^2 - 2$ and $(t-3)^3 - \frac{17}{5}(t-3)$, and that $c_4 = (1, -4, 6, -4, 1)'$ spans $U_4^\perp$. (The vectors $c_1, \ldots, c_4$ are the tabulated coefficients of the orthogonal polynomials for five equally spaced levels.)

(c) For $y = (0, 7, 7, 5, 6)'$, compute the numbers $(c_j'y)^2/(c_j'c_j)$ $(j = 1, \ldots, 4)$ and check that they add up to $\sum_{i=1}^5 (y_i - \bar{y})^2$. Find the vector of U_3 closest to y, the polynomial of degree at most 2 whose values at $t = 1, \ldots, 5$ it lists, and the distance from y to U_3.

Solution. (a) By Theorem 6.11:

- $b_1 = \mathbf{1}_5$, $\|b_1\|^2 = 5$; $a_2'b_1 = 15$, so $t_{12} = 3$ and $b_2 = a_2 - 3b_1 = (-2, -1, 0, 1, 2)' = c_1$, with $\|b_2\|^2 = 10$.
- $a_3'b_1 = 55$ and $a_3'b_2 = -2 - 4 + 0 + 16 + 50 = 60$, so $t_{13} = 11$, $t_{23} = 6$ and $b_3 = a_3 - 11b_1 - 6b_2 = (1 - 11 + 12,\ 4 - 11 + 6,\ 9 - 11 + 0,\ 16 - 11 - 6,\ 25 - 11 - 12)' = c_2$, with $\|b_3\|^2 = 14$.
- $a_4'b_1 = 225$, $a_4'b_2 = -2 - 8 + 0 + 64 + 250 = 304$ and $a_4'b_3 = 2 - 8 - 54 - 64 + 250 = 126$, so $t_{14} = 45$, $t_{24} = \frac{152}{5}$, $t_{34} = 9$, and $b_4 = a_4 - 45b_1 - \frac{152}{5}b_2 - 9b_3 = \frac{1}{5}(-6, 12, 0, -12, 6)' = \frac{6}{5}c_3$.

(b) The values of $s = t - 3$ at $t = 1, \ldots, 5$ are $b_2 = (-2, -1, 0, 1, 2)'$; those of $s^2 - 2$ are $(4 - 2,\ 1 - 2,\ 0 - 2,\ 1 - 2,\ 4 - 2)' = b_3$; and those of $s^3 - \frac{17}{5}s$ are $(-8 + \frac{34}{5},\ -1 + \frac{17}{5},\ 0,\ 1 - \frac{17}{5},\ 8 - \frac{34}{5})' = \frac{1}{5}(-6, 12, 0, -12, 6)' = b_4$. Next $c_4'a_1 = 1 - 4 + 6 - 4 + 1 = 0$, $c_4'a_2 = 1 - 8 + 18 - 16 + 5 = 0$, $c_4'a_3 = 1 - 16 + 54 - 64 + 25 = 0$ and $c_4'a_4 = 1 - 32 + 162 - 256 + 125 = 0$, so $c_4 \in U_4^\perp$ (Theorem 6.5(a)). The vectors $b_1, \ldots, b_4$ are nonzero, pairwise orthogonal and span U_4 (Theorem 6.11), so they form a basis of U_4 (Proposition 6.8(b)); hence $\dim U_4^\perp = 5 - 4 = 1$ (Theorem 6.5(c)), and $c_4 \neq 0$ spans $U_4^\perp$ (Theorem 4.5(d)).

(c) $c_1'y = 0-7+0+5+12 = 10$, $c_2'y = 0-7-14-5+12 = -14$, $c_3'y = 0+14+0-10+6 = 10$ and $c_4'y = 0-28+42-20+6 = 0$; also $\bar{y} = \frac{1}{5}(0+7+7+5+6) = 5$. As $c_1'c_1 = 10$, $c_2'c_2 = 14$, $c_3'c_3 = 10$ and $c_4'c_4 = 70$, the four numbers are

$$\frac{10^2}{10} = 10, \qquad \frac{14^2}{14} = 14, \qquad \frac{10^2}{10} = 10, \qquad \frac{0^2}{70} = 0,$$

with sum $34 = 25+4+4+0+1 = \sum_i (y_i - 5)^2$. By (a), b_1, b_2, b_3 are nonzero, pairwise orthogonal and span U_3 (Theorem 6.11), so $b_j/\|b_j\|$ $(j = 1, 2, 3)$ form an orthonormal basis of U_3. By Theorems 6.15 and 6.19(a), the vector of U_3 closest to y is

$$P_{U_3}y = \sum_{j=1}^{3} \frac{b_j'y}{b_j'b_j}\, b_j = \frac{25}{5}\,\mathbf{1}_5 + \frac{10}{10}\,c_1 + \frac{-14}{14}\,c_2 = 5\,\mathbf{1}_5 + c_1 - c_2 = (1, 5, 7, 7, 5)',$$

and $y - P_{U_3}y = (-1, 2, 0, -2, 1)' = c_3$, so the distance from y to U_3 is $\sqrt{10}$. By (a) and (b), $P_{U_3}y = 5b_1 + b_2 - b_3$ lists the values at $t = 1, \ldots, 5$ of $5 + (t-3) - \Big((t-3)^2 - 2\Big) = 7 + (t-3) - (t-3)^2$; for instance $7 - 2 - 4 = 1$ at $t = 1$.

Problems

Problem 6.1. Show that $I_n - 2A$ is an orthogonal matrix for every symmetric idempotent $A \in \mathbb{R}^{n\times n}$. In particular, if $h \in \mathbb{R}^n$ and $h'h = 1$, then $I_n - 2hh'$ is orthogonal.

Solution. Using $A' = A$ and $A^2 = A$, $(I - 2A)'(I - 2A) = (I - 2A)^2 = I - 4A + 4A^2 = I$; so $I - 2A$ is orthogonal (Definition 6.9). The matrix hh' is symmetric and idempotent: $(hh')' = hh'$ and $(hh')(hh') = h(h'h)h' = hh'$.

Problem 6.2. Let $a \in \mathbb{R}^n$, $a \neq 0$, and $W = \{x \in \mathbb{R}^n : a'x = 0\}$. The distance from $y \in \mathbb{R}^n$ to a set $L \subseteq \mathbb{R}^n$ is the smallest value of $\|y - x\|$ over $x \in L$, if there is one (for a subspace, Theorem 6.15).

(a) Show that $W^\perp = \text{span}\{a\}$ and $P_W = I_n - \frac{aa'}{a'a}$, and that the distance from y to W is $|a'y|/\|a\|$, attained only at $x = y - \frac{a'y}{a'a}a$.

(b) Let $c \in \mathbb{R}$ and $L = \{x \in \mathbb{R}^n : a'x = c\}$. Show that the distance from y to L is $|a'y - c|/\|a\|$, attained only at $x = y - \frac{a'y-c}{a'a}a$.

(c) Find the distances from $y = (3, 1, 2, 4)'$ to the hyperplanes $x_1 + 2x_2 + 2x_3 + 4x_4 = 0$ and $x_1 + 2x_2 + 2x_3 + 4x_4 = 75$ in $\mathbb{R}^4$, and the closest points.

Solution. (a) $W = \mathcal{N}(a') = \text{span}\{a\}^\perp$ (Theorem 6.5(a)), so $W^\perp = (\text{span}\{a\}^\perp)^\perp = \text{span}\{a\}$ (Proposition 6.17). The projection matrix onto $\text{span}\{a\}$ is $\frac{aa'}{a'a}$ (Example 6.22), and $P_W = I_n - P_{W^\perp}$ (Theorem 6.19(b)); so $P_W = I_n - \frac{aa'}{a'a}$. By Theorem 6.15, $\|y - x\|$ $(x \in W)$ is smallest only at $x = P_W y = y - \frac{a'y}{a'a}a$, and the smallest value is

$$\|y - P_W y\| = \left\|\frac{a'y}{a'a}\,a\right\| = \frac{|a'y|}{a'a}\,\|a\| = \frac{|a'y|}{\|a\|}$$

(b) Put $x_0 = \frac{c}{a'a}a$, so that $a'x_0 = c$. For $x \in \mathbb{R}^n$, $a'x = c$ if and only if $a'(x - x_0) = 0$; so $x \in L$ if and only if $w = x - x_0 \in W$, and then $\|y - x\| = \|(y - x_0) - w\|$. By (a), applied to $y - x_0$, the smallest value is $|a'(y - x_0)|/\|a\| = |a'y - c|/\|a\|$, attained only at $w = (y - x_0) - \frac{a'y-c}{a'a}a$, that is, at $x = x_0 + w = y - \frac{a'y-c}{a'a}a$.

(c) Here $a = (1, 2, 2, 4)'$, $\|a\| = \sqrt{1+4+4+16} = 5$ and $a'y = 3+2+4+16 = 25$.

- Hyperplane through 0: the distance is $25/5 = 5$, and the closest point is $y - \frac{25}{25}a = (2, -1, 0, 0)'$ (check: $2 - 2 + 0 + 0 = 0$).

- Hyperplane $a'x = 75$: the distance is $|25 - 75|/5 = 10$, and the closest point is $y - \frac{25-75}{25}a = y + 2a = (5, 5, 6, 12)'$ (check: $5 + 10 + 12 + 48 = 75$).

Problem 6.3. Let $U_1, \ldots, U_4$ and $c_1, \ldots, c_4$ be as in Exercise 6.10. Show that $P_{U_k} - P_{U_{k-1}}$ is the projection matrix onto $\text{span}\{c_{k-1}\}$ $(k = 2, 3, 4)$, and that for every $y \in \mathbb{R}^5$

$$\sum_{i=1}^{5}(y_i - \bar{y})^2 = \frac{(c_1'y)^2}{10} + \frac{(c_2'y)^2}{14} + \frac{(c_3'y)^2}{10} + \frac{(c_4'y)^2}{70}$$

Solution.

- $U_{k-1} \subseteq U_k$, so $P_{U_k} - P_{U_{k-1}}$ is the projection matrix onto $V_k = U_k \cap U_{k-1}^{\perp}$, of dimension $k-(k-1) = 1$ (Theorem 6.25(b); $\dim U_k = k$, as $b_1, \ldots, b_k$ form a basis of U_k by the solution of Exercise 6.10(b)). By Exercise 6.10(a), $b_k \in \text{span}\{b_1, \ldots, b_k\} = U_k$, and b_k is orthogonal to $b_1, \ldots, b_{k-1}$, which span U_{k-1}; so $b_k \in V_k$ (Theorem 6.5(a)), and $V_k = \text{span}\{b_k\} = \text{span}\{c_{k-1}\}$ (Theorem 4.5(d)). Likewise $I_5 - P_{U_4} = P_{U_4^{\perp}}$ (Theorem 6.19(b)), with $U_4^{\perp} = \text{span}\{c_4\}$ by Exercise 6.10(b).

- By Example 6.22, $P_{U_1}y = \bar{y}\mathbf{1}_5$, and the projection of y onto $\text{span}\{c_j\}$ is $\frac{c_j'y}{c_j'c_j}c_j$, of squared length $(c_j'y)^2/(c_j'c_j)$. So

$$y - \bar{y}\mathbf{1}_5 = (P_{U_2} - P_{U_1})y + (P_{U_3} - P_{U_2})y + (P_{U_4} - P_{U_3})y + (I_5 - P_{U_4})y$$

  is a sum of multiples of $c_1, \ldots, c_4$, which are pairwise orthogonal (c_1, c_2, c_3 are multiples of b_2, b_3, b_4, and $c_4 \perp U_4$). Pythagoras (Proposition 6.2), applied three times, $c_1'c_1 = 10$, $c_2'c_2 = 14$, $c_3'c_3 = 10$ and $c_4'c_4 = 70$ give the identity.

Problem 6.4. Let U and W be subspaces of $\mathbb{R}^n$.

(a) Show that $(U + W)^{\perp} = U^{\perp} \cap W^{\perp}$.

(b) Show that $U \subseteq W$ if and only if $W^{\perp} \subseteq U^{\perp}$.

(c) Show that $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.

(d) Check (c) for the subspaces $U = \{x : x_4 = 0\}$ and $W_0 = \{x : x_1 + x_2 + x_3 + x_4 = 0\}$ of $\mathbb{R}^4$ of Example 3.24.

Solution. (a) U and W lie in $U+W$ (Theorem 3.21(b)), so a vector orthogonal to $U+W$ is orthogonal to U and to W. Conversely, if $y \in U^{\perp} \cap W^{\perp}$, then $y'(u + w) = y'u + y'w = 0$ for all $u \in U$ and $w \in W$, so $y \in (U + W)^{\perp}$.

(b) If $U \subseteq W$, a vector orthogonal to W is orthogonal to U: $W^{\perp} \subseteq U^{\perp}$. Conversely, if $W^{\perp} \subseteq U^{\perp}$, the first step, applied to the subspaces $W^{\perp} \subseteq U^{\perp}$, gives $(U^{\perp})^{\perp} \subseteq (W^{\perp})^{\perp}$, that is, $U \subseteq W$ (Proposition 6.17).

(c) $U^{\perp}$ and $W^{\perp}$ are subspaces (Theorem 6.5(a)), and so is $U^{\perp} + W^{\perp}$ (Theorem 3.21(a)). By (a), applied to $U^{\perp}$ and $W^{\perp}$, and by Proposition 6.17, applied to U, to W and then to $U^{\perp} + W^{\perp}$,

$$(U^{\perp} + W^{\perp})^{\perp} = (U^{\perp})^{\perp} \cap (W^{\perp})^{\perp} = U \cap W, \qquad \text{so} \qquad U^{\perp} + W^{\perp} = \left((U^{\perp} + W^{\perp})^{\perp}\right)^{\perp} = (U \cap W)^{\perp}$$

(d) $U = \text{span}\{e_4\}^{\perp}$ and $W_0 = \text{span}\{\mathbf{1}_4\}^{\perp}$ (Theorem 6.5(a)), so $U^{\perp} = \text{span}\{e_4\}$ and $W_0^{\perp} = \text{span}\{\mathbf{1}_4\}$ (Proposition 6.17), and $U^{\perp} + W_0^{\perp} = \text{span}\{e_4, \mathbf{1}_4\}$ (Theorem 3.21(c)). By Example 3.24, $U \cap W_0 = \text{span}\{d_1, d_2\}$ with $d_1 = (1, -1, 0, 0)'$ and $d_2 = (0, 1, -1, 0)'$, so $(U \cap W_0)^{\perp} = \{x : x_1 = x_2 = x_3\} = \text{span}\{(1, 1, 1, 0)', e_4\}$. This is $\text{span}\{e_4, \mathbf{1}_4\}$, since $\mathbf{1}_4 = (1, 1, 1, 0)' + e_4$ and $(1, 1, 1, 0)' = \mathbf{1}_4 - e_4$ (Theorem 3.7).

Problem 6.5. Let W be a subspace of $\mathbb{R}^n$ and $P = P_W = (p_{ij})$.

(a) Show that $\|Py\| \leqslant \|y\|$ for every $y \in \mathbb{R}^n$, with equality if and only if $y \in W$.

(b) Show that $0 \leqslant p_{ii} \leqslant 1$ for every i, and that $p_{11} + \cdots + p_{nn} = \dim W$.

(c) Show that $p_{ii} = 1$ if and only if $e_i \in W$, and that then column i of P is e_i and row i is e_i'. Show that $p_{ii} = 0$ if and only if $e_i \in W^\perp$, and that then row i and column i of P are zero.

(d) Find the diagonal entries of P_W in Example 6.23 and in Exercise 6.5, and illustrate (c).

Solution. (a) $y = Py + (I_n - P)y$, with $Py \in W$ and $(I_n - P)y = P_{W^\perp}y \in W^\perp$ (Theorem 6.19(b)); by Pythagoras (Proposition 6.2), $\|y\|^2 = \|Py\|^2 + \|(I_n - P)y\|^2 \geqslant \|Py\|^2$. Equality holds if and only if $(I_n - P)y = 0$, that is, $y = Py$. If $y = Py$, then $y \in \mathcal{C}(P) = W$; if $y \in W$, then $Py = y$ (Theorem 6.19(b)).

(b) $p_{ii} = e_i'Pe_i = \|Pe_i\|^2 \geqslant 0$ (Theorem 6.19(b)), and $\|Pe_i\|^2 \leqslant \|e_i\|^2 = 1$ by (a). The sum of the p_{ii} is $\operatorname{tr} P = \dim W$ (Theorem 6.19(b)).

(c) By (b) and (a), $p_{ii} = 1$ if and only if $\|Pe_i\| = \|e_i\|$, that is, $e_i \in W$. Then column i of P is $Pe_i = e_i$ (Theorem 1.4(b)), and row i is its transpose, as $P' = P$. Next, $p_{ii} = \|Pe_i\|^2 = 0$ if and only if $Pe_i = 0$. If $Pe_i = 0$, then $e_i = (I_n - P)e_i \in W^\perp$; if $e_i \in W^\perp$, then $Pe_i = 0$ (Theorem 6.19(b)). Then column i of P is $Pe_i = 0$, and so is row i.

(d) In Example 6.23: $\frac{3}{4}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}$, with sum $3 = \dim W$. In Exercise 6.5: $1, \frac{5}{9}, \frac{8}{9}, \frac{5}{9}$, with sum $3 = \dim W$. There $p_{11} = 1$, and indeed $e_1 \in W$, as $e_1'h = 0$ and $W = \operatorname{span}\{h\}^\perp$ (Proposition 6.17); and column 1 of P_W is e_1, row 1 is e_1'.

Problem 6.6. Let W_1, W_2 be subspaces of $\mathbb{R}^n$, $P_1 = P_{W_1}$ and $P_2 = P_{W_2}$. Show that the following are equivalent:

(i) $P_1 + P_2$ is idempotent;

(ii) $P_1P_2 = 0$;

(iii) $W_1 \perp W_2$.

Show that then $P_1 + P_2 = P_{W_1 \oplus W_2}$.

Solution.

- (i)⇒(ii): $(P_1 + P_2)^2 = P_1^2 + P_1P_2 + P_2P_1 + P_2^2 = P_1 + P_2 + P_1P_2 + P_2P_1$ (Theorem 6.19(b)), so (i) gives $P_1P_2 + P_2P_1 = 0$. Multiplying on the left and on the right by P_1 gives $2P_1P_2P_1 = 0$; multiplying on the left by P_1 only gives $P_1P_2 + P_1P_2P_1 = 0$. Hence $P_1P_2 = 0$.
- (ii)⇒(iii): for $w_1 \in W_1$ and $w_2 \in W_2$, $w_1 = P_1w_1$ and $w_2 = P_2w_2$ (Theorem 6.19(b)), so $w_1'w_2 = w_1'P_1'P_2w_2 = w_1'(P_1P_2)w_2 = 0$, as $P_1' = P_1$.
- (iii)⇒(i): by Theorem 6.24, $P_1 + P_2 = P_{W_1 \oplus W_2}$, which is idempotent (Theorem 6.19(b)). This also proves the last statement.

Problem 6.7. Let U, W be subspaces of $\mathbb{R}^n$.

(a) Show that P_UP_W is symmetric if and only if $P_UP_W = P_WP_U$, and that then $P_UP_W = P_{U \cap W}$.

(b) In $\mathbb{R}^4$, check (a) for $W_0 = \{x : \mathbf{1}'x = 0\}$ and the subspace W of Exercise 6.9, and find $P_{W_0 \cap W}$. Show that P_UP_V is not symmetric for $U = \operatorname{span}\{e_1\}$ and $V = \operatorname{span}\{e_1 + e_2\}$.

Solution. (a) $(P_UP_W)' = P_W'P_U' = P_WP_U$ (Theorems 1.9 and 6.19(b)), so P_UP_W is symmetric if and only if $P_UP_W = P_WP_U$. Let them commute, and put $P = P_UP_W$.

- $P^2 = P_UP_WP_UP_W = P_UP_UP_WP_W = P_UP_W = P$. So P is symmetric idempotent, and $P = P_{\mathcal{C}(P)}$ (Theorem 6.19(c)).

- $\mathcal{C}(P) = U \cap W$: for $y \in \mathbb{R}^n$, $Py = P_U(P_W y) \in U$ and $Py = P_W(P_U y) \in W$; conversely, for $x \in U \cap W$, $Px = P_U(P_W x) = P_U x = x$ (Theorem 6.19(b)), so $x \in \mathcal{C}(P)$. Hence $P = P_{U\cap W}$.

(b) $P_{W_0} = C_4 = I_4 - \frac{1}{4}\mathbf{1}\mathbf{1}'$ (Corollary 6.21(a)), and $P_W = I_4 - \frac{1}{4}hh'$ with $h = (1, 1, -1, -1)'$ (Exercise 6.9). As $\mathbf{1}'h = 0$,

$$P_{W_0}P_W = I_4 - \tfrac{1}{4}\mathbf{1}\mathbf{1}' - \tfrac{1}{4}hh' + \tfrac{1}{16}\mathbf{1}(\mathbf{1}'h)h' = I_4 - \tfrac{1}{4}\mathbf{1}\mathbf{1}' - \tfrac{1}{4}hh',$$

and $P_W P_{W_0}$ is the same matrix (exchange the roles of $\mathbf{1}$ and h). It is symmetric, and it is $P_{W_0\cap W}$ by (a); $W_0 \cap W = W \cap \operatorname{span}\{\mathbf{1}\}^\perp$, and the matrix is the $\frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$ of Exercise 6.9(c). For the second pair, $P_U = e_1 e_1'$ and $P_V = \frac{1}{2}(e_1 + e_2)(e_1 + e_2)'$ (Example 6.22), so $P_U P_V = \frac{1}{2}e_1(e_1 + e_2)'$, whose first row is $(\frac{1}{2}, \frac{1}{2}, 0, 0)$ and whose second row is 0: not symmetric. By (a), $P_U P_V \neq P_V P_U$.

Problem 6.8. For nonzero $u, v \in \mathbb{R}^n$, the **angle** between u and v is the $\theta \in [0, \pi]$ with $\cos\theta = u'v/(\|u\|\,\|v\|)$: this number lies in $[-1, 1]$ (Theorem 6.3(a)), and the cosine takes every value in $[-1, 1]$ exactly once on $[0, \pi]$ (an imported fact from trigonometry, used without proof, as are the values $\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, $\cos\frac{\pi}{2} = 0$, $\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$). Let $x, y \in \mathbb{R}^n$ be data vectors that are not constant, so that $u = C_n x \neq 0$ and $v = C_n y \neq 0$ (Example 1.14). The **sample correlation** of x and y is

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \, \sum_{i=1}^n (y_i - \bar{y})^2}}$$

(a) Show that $r = \cos\theta$ for the angle θ between u and v; hence $-1 \leqslant r \leqslant 1$.

(b) Show that $r = 0$ if and only if $u \perp v$; and that $|r| = 1$ if and only if $y_i = \alpha + \beta x_i$ $(i = 1, \ldots, n)$ for some $\alpha, \beta \in \mathbb{R}$ with $\beta \neq 0$, and then $r = 1$ if $\beta > 0$ and $r = -1$ if $\beta < 0$.

(c) Show that r does not change if x is replaced by $\alpha\mathbf{1}_n + \beta x$ with $\beta > 0$ (for instance, by a change of units).

(d) Compute the three sample correlations, and the corresponding angles, among the two variables of Example 1.15 and the data vector $y = (3, 1, 5, 3)'$.

Solution. (a) The entries of u and v are $x_i - \bar{x}$ and $y_i - \bar{y}$, so the numerator of r is $u'v$ and the denominator is $\|u\|\,\|v\|$. So $r = \cos\theta$, and $|r| \leqslant 1$ by the Cauchy–Schwarz inequality (Theorem 6.3(a)).

(b) $r = 0$ if and only if $u'v = 0$. By the equality case of Theorem 6.3(a), $|r| = 1$ if and only if one of u, v is a multiple of the other, that is, as both are nonzero, $v = \beta u$ for some $\beta \neq 0$; then $r = \beta u'u/(\|u\| \cdot |\beta|\,\|u\|) = \beta/|\beta|$. Finally, $v = \beta u$ means $y - \bar{y}\mathbf{1} = \beta(x - \bar{x}\mathbf{1})$, that is, $y = \alpha\mathbf{1} + \beta x$ with $\alpha = \bar{y} - \beta\bar{x}$; conversely, if $y = \alpha\mathbf{1} + \beta x$, then $v = C_n y = \alpha C_n\mathbf{1} + \beta C_n x = \beta u$, as $C_n\mathbf{1} = 0$ (Proposition 1.13).

(c) $C_n(\alpha\mathbf{1} + \beta x) = \beta u$, and $(\beta u)'v/(\|\beta u\|\,\|v\|) = \beta u'v/(|\beta|\,\|u\|\,\|v\|) = r$ for $\beta > 0$.

(d) By Example 1.15, the centered variables are $u_1 = (1, -1, 1, -1)'$ and $u_2 = (1, 1, -1, -1)'$, with $\|u_1\| = \|u_2\| = 2$ and $u_1'u_2 = 0$ (the off-diagonal entry of $X_0'CX_0 = 4I_2$). For y, $\bar{y} = 3$, $v = (0, -2, 2, 0)'$, $\|v\| = \sqrt{8} = 2\sqrt{2}$, $u_1'v = 0 + 2 + 2 + 0 = 4$ and $u_2'v = 0 - 2 - 2 + 0 = -4$. So the two variables have $r = 0$ $(\theta = \frac{\pi}{2})$; the first variable and y have $r = 4/(2 \cdot 2\sqrt{2}) = \frac{1}{\sqrt{2}}$ $(\theta = \frac{\pi}{4})$; and the second variable and y have $r = -\frac{1}{\sqrt{2}}$ $(\theta = \frac{3\pi}{4})$. Here $v = u_1 - u_2$ lies in the plane of u_1 and u_2, and $\frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$.

Problem 6.9. Let $A = [a_1 \ \cdots \ a_p] \in \mathbb{R}^{n\times p}$ have full column rank.

(a) Let $A = QR = \tilde{Q}\tilde{R}$, where $Q, \tilde{Q} \in \mathbb{R}^{n\times p}$ satisfy $Q'Q = \tilde{Q}'\tilde{Q} = I_p$ and $R, \tilde{R} \in \mathbb{R}^{p\times p}$ are upper triangular with positive diagonal entries. Show that $Q = \tilde{Q}$ and $R = \tilde{R}$: the QR decomposition of Theorem 6.12 is unique.

(b) Show that if the columns of A are pairwise orthogonal, then $Q = AD^{-1}$ and $R = D$, where $D = \operatorname{diag}(\|a_1\|, \dots, \|a_p\|)$.

Solution. (a) Let $q_j, \tilde{q}_j$ be the columns of $Q, \tilde{Q}$. By Theorem 1.4(a),(b), $a_j = \sum_{i\leqslant j} r_{ij}q_i = \sum_{i\leqslant j} \tilde{r}_{ij}\tilde{q}_i$. By induction on j, we show $\tilde{q}_j = q_j$ and $\tilde{r}_{ij} = r_{ij}$ for $i \leqslant j$.

- $j = 1$: $a_1 = r_{11}q_1$, so $\|a_1\| = r_{11}\|q_1\| = r_{11}$, as $r_{11} > 0$; likewise $\|a_1\| = \tilde{r}_{11}$. So $r_{11} = \tilde{r}_{11}$ and $q_1 = a_1/r_{11} = \tilde{q}_1$.
- Let $j \geqslant 2$, and let the claim hold for the columns before j. For $i < j$, orthonormality gives $q_i'a_j = \sum_{k\leqslant j} r_{kj}q_i'q_k = r_{ij}$, and likewise $\tilde{q}_i'a_j = \tilde{r}_{ij}$; as $\tilde{q}_i = q_i$, $\tilde{r}_{ij} = r_{ij}$.
- Then $r_{jj}q_j = a_j - \sum_{i<j} r_{ij}q_i = a_j - \sum_{i<j} \tilde{r}_{ij}\tilde{q}_i = \tilde{r}_{jj}\tilde{q}_j$. Taking lengths, $r_{jj} = \tilde{r}_{jj}$ (both are positive and $\|q_j\| = \|\tilde{q}_j\| = 1$), and then $q_j = \tilde{q}_j$.

(b) The columns of A are nonzero (they are linearly independent), so D is invertible with $D^{-1} = \operatorname{diag}(\|a_1\|^{-1}, \dots, \|a_p\|^{-1})$ (Proposition 5.6(b)). The columns $a_j/\|a_j\|$ of AD^{-1} (Theorem 1.4(a)) are orthonormal, so $(AD^{-1})'(AD^{-1}) = I_p$; D is upper triangular with positive diagonal entries; and $A = (AD^{-1})D$. By (a), $Q = AD^{-1}$ and $R = D$. For instance, the matrix $[w_1 \ w_2 \ w_3]$ of Example 6.23 has $Q = \frac{1}{2}[w_1 \ w_2 \ w_3]$ and $R = 2I_3$.

Problem 6.10. Let $A = [a_1 \ \cdots \ a_p] \in \mathbb{R}^{n\times p}$ have full column rank, with $A = QR$ as in Theorem 6.12.

(a) Show that $A'A = R'R$ and $\det(A'A) = (r_{11}r_{22}\cdots r_{pp})^2 > 0$.

(b) For $p = 2$, show that $\det(A'A) = \|a_1\|^2\|a_2\|^2 - (a_1'a_2)^2$, and deduce the strict inequality $|a_1'a_2| < \|a_1\|\,\|a_2\|$ for linearly independent a_1, a_2.

(c) Check (a) for the matrix of Example 6.13, and (b) for a_1, a_2 of Exercise 6.8.

Solution. (a) $A'A = R'Q'QR = R'R$, as $Q'Q = I_p$. By Theorems 5.19(b) and 5.15(a),(b), $\det(A'A) = \det R' \det R = (\det R)^2 = (r_{11}\cdots r_{pp})^2$, which is positive, as every $r_{jj} > 0$.

(b) $(A'A)_{ij} = a_i'a_j$ (Theorem 1.4(d)), so by the 2×2 case of Definition 5.12, $\det(A'A) = (a_1'a_1)(a_2'a_2) - (a_1'a_2)(a_2'a_1) = \|a_1\|^2\|a_2\|^2 - (a_1'a_2)^2$. If a_1, a_2 are linearly independent, then $A = [a_1 \ a_2]$ has full column rank 2, and $\det(A'A) > 0$ by (a); take square roots.

(c) For Example 6.13, the inner products of $a_1 = (1,2,0,2)'$, $a_2 = (3,1,2,2)'$, $a_3 = (2,3,4,5)'$ give

$$A'A = \begin{pmatrix} 9 & 9 & 18 \\ 9 & 18 & 27 \\ 18 & 27 & 54 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 3 & 3 & 0 \\ 6 & 3 & 3 \end{pmatrix} \begin{pmatrix} 3 & 3 & 6 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix} = R'R,$$

for instance $54 = 36+9+9$, and $\det(A'A) = (3\cdot 3\cdot 3)^2 = 729$. For Exercise 6.8, $\|a_1\|^2 = 9$, $\|a_2\|^2 = 18$, $a_1'a_2 = 9$, so $\det(A'A) = 162 - 81 = 81 = (3\cdot 3)^2$, with $r_{11} = \|b_1\| = 3$ and $r_{22} = \|b_2\| = 3$; and $9 < 3\sqrt{18}$.

Problem 6.11. Let $x \in \mathbb{R}^n$ with $x \neq \|x\|e_1$, let $v = x - \|x\|e_1$, and let $H_v = I_n - \frac{2}{v'v}vv'$, a **Householder reflection**.

(a) Show that H_v is symmetric and orthogonal, that $H_v v = -v$ and $H_v w = w$ for $w \perp v$, and that $H_v x = \|x\|e_1$.

(b) Compute H_v for $x = (1, 2, 2, 4)'$, and check that $H_v x = 5e_1$.

(c) Let $A = \left(\begin{smallmatrix} 1 & 3 \\ 2 & 1 \\ 2 & 0 \\ 4 & 0 \end{smallmatrix}\right)$, with H_v of (b). Compute $H_v A$. Find a 3×3 Householder reflection K such that $G = \mathrm{diag}(1, K)$ gives $GH_vA = \left(\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right)$ with $R \in \mathbb{R}^{2\times 2}$ upper triangular with positive diagonal entries. Deduce that $A = Q_1 R$, where Q_1 consists of the first two columns of the orthogonal matrix $H_v G$, and compare with Theorem 6.12.

Solution. (a)

- $v \neq 0$, and $h = v/\|v\|$ satisfies $h'h = 1$ and $hh' = \frac{1}{v'v}vv'$; so $H_v = I_n - 2hh'$ is symmetric (Theorem 1.9) and orthogonal (Problem 6.1).
- $H_v v = v - \frac{2}{v'v}v(v'v) = -v$, and $H_v w = w - \frac{2v'w}{v'v}v = w$ for $w \perp v$.
- $v'x = x'x - \|x\|x_1$ and $v'v = x'x - 2\|x\|x_1 + \|x\|^2 = 2\Big(\|x\|^2 - \|x\|x_1\Big) = 2v'x$; so $H_v x = x - \frac{2v'x}{v'v}v = x - v = \|x\|e_1$.

(b) $\|x\| = \sqrt{1+4+4+16} = 5$, $v = (-4, 2, 2, 4)'$, $v'v = 40$, and

$$H_v = I_4 - \frac{1}{20}\begin{pmatrix} 16 & -8 & -8 & -16 \\ -8 & 4 & 4 & 8 \\ -8 & 4 & 4 & 8 \\ -16 & 8 & 8 & 16 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 1 & 2 & 2 & 4 \\ 2 & 4 & -1 & -2 \\ 2 & -1 & 4 & -2 \\ 4 & -2 & -2 & 1 \end{pmatrix},$$

$$H_v x = \frac{1}{5}\begin{pmatrix} 1+4+4+16 \\ 2+8-2-8 \\ 2-2+8-8 \\ 4-4-4+4 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

(c)

- The first column of A is x, and $H_v(3, 1, 0, 0)' = \frac{1}{5}(3+2,\ 6+4,\ 6-1,\ 12-2)' = (1, 2, 1, 2)'$; so $H_v A = \left(\begin{smallmatrix} 5 & 1 \\ 0 & 2 \\ 0 & 1 \\ 0 & 2 \end{smallmatrix}\right)$.
- For $z = (2, 1, 2)'$, $\|z\| = 3$ and $w = z - 3e_1 = (-1, 1, 2)'$, of squared length 6; by (a) with $n = 3$,

$$K = I_3 - \frac{1}{3}\begin{pmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & -2 \\ 2 & -2 & -1 \end{pmatrix}, \qquad Kz = \frac{1}{3}\begin{pmatrix} 4+1+4 \\ 2+2-4 \\ 4-2-2 \end{pmatrix} = 3e_1$$

- $G = \mathrm{diag}(1, K)$ is symmetric, and $G'G = \mathrm{diag}(1, K'K) = I_4$ (Proposition 5.6(b)). Its first column is e_1, and $G(1, 2, 1, 2)' = (1, 3, 0, 0)'$; so $GH_vA = \left(\begin{smallmatrix} 5 & 1 \\ 0 & 3 \\ 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $R = \left(\begin{smallmatrix} 5 & 1 \\ 0 & 3 \end{smallmatrix}\right)$.
- $(H_vG)'(H_vG) = G'H_v'H_vG = I_4$, so H_vG is orthogonal; and $(H_vG)(GH_v) = H_vG^2H_v = H_v^2 = I_4$, as G and H_v are symmetric and orthogonal. Write $H_vG = [Q_1\ Q_2]$, with Q_2 its last two columns. Then $A = H_vG(GH_vA) = [Q_1\ Q_2]\left(\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right) = Q_1R$ (Theorem 5.3(c)), and the columns of Q_1 are $H_vGe_1 = H_ve_1 = \frac{1}{5}(1, 2, 2, 4)'$ and $H_vGe_2 = H_v(0, \frac{2}{3}, \frac{1}{3}, \frac{2}{3})' = \frac{1}{15}(14, 3, -2, -4)'$ (Theorem 1.4(a)).
- Gram–Schmidt (Theorem 6.12): $b_1 = x$, $\|b_1\| = 5$; $t_{12} = \frac{3+2}{25} = \frac{1}{5}$ and $b_2 = (3, 1, 0, 0)' - \frac{1}{5}(1, 2, 2, 4)' = \frac{1}{5}(14, 3, -2, -4)'$, with $\|b_2\|^2 = \frac{196+9+4+16}{25} = 9$. So $q_1 = \frac{1}{5}(1, 2, 2, 4)'$, $q_2 = \frac{1}{15}(14, 3, -2, -4)'$, $r_{11} = 5$, $r_{12} = t_{12}\|b_1\| = 1$ and $r_{22} = 3$: the same Q_1 and R, as Problem 6.9(a) requires.

Chapter 7. Eigenvalues and diagonalization

Eigenvalues and eigenvectors

Definition 7.1. Let $A \in \mathbb{R}^{n\times n}$. A number $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if $Av = \lambda v$ for some $v \in \mathbb{R}^n$ with $v \neq 0$; such a vector v is an **eigenvector** of A for λ. For $\lambda \in \mathbb{R}$, the **eigenspace** of A for λ is $E_\lambda(A) = \mathcal{N}(A - \lambda I_n)$, and $g_A(\lambda) = \dim E_\lambda(A)$ is the **geometric multiplicity** of λ.

In this course eigenvalues are real numbers. Since $Av - \lambda v = (A - \lambda I_n)v$ (Theorem 1.3), $E_\lambda(A) = \{v \in \mathbb{R}^n : Av = \lambda v\}$: it consists of 0 and the eigenvectors for λ, and it is a subspace of $\mathbb{R}^n$ (Definition 4.7). So λ is an eigenvalue of A if and only if $g_A(\lambda) \geqslant 1$. An eigenvector v spans a line that A maps into itself: $A(cv) = \lambda(cv)$ for every $c \in \mathbb{R}$.

Example 7.2. Let

$$A = \begin{pmatrix} 1 & 2 & -2 & -2 \\ -1 & 0 & -2 & 2 \\ 0 & -1 & -1 & 1 \\ -1 & 2 & -2 & 0 \end{pmatrix}, \qquad v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

Multiplying out,

$$Av_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \qquad Av_2 = \begin{pmatrix} -1 \\ -1 \\ -1 \\ -1 \end{pmatrix}, \qquad Av_3 = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix}, \qquad Av_4 = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 0 \end{pmatrix}$$

So $Av_1 = v_1$, $Av_2 = -v_2$, $Av_3 = 2v_3$ and $Av_4 = -2v_4$: the numbers 1, -1, 2 and -2 are eigenvalues of A, with eigenvectors v_1, v_2, v_3, v_4. The vector e_1 is not an eigenvector: $Ae_1 = (1, -1, 0, -1)'$ is not a multiple of e_1.

Theorem 7.3. Let $A \in \mathbb{R}^{n\times n}$ and $\lambda \in \mathbb{R}$. The following are equivalent:

(i) λ is an eigenvalue of A;

(ii) $A - \lambda I_n$ is singular (not invertible);

(iii) $\operatorname{rank}(A - \lambda I_n) < n$;

(iv) $\det(\lambda I_n - A) = 0$.

Moreover $g_A(\lambda) = n - \operatorname{rank}(A - \lambda I_n)$.

Proof.
- (i) $\Leftrightarrow$ (ii): λ is an eigenvalue if and only if $(A - \lambda I_n)v = 0$ for some $v \neq 0$, and this holds if and only if $A - \lambda I_n$ is not invertible (Theorem 2.18, (i) $\Leftrightarrow$ (iii)).
- (ii) $\Leftrightarrow$ (iii): Theorem 4.13(d).
- (ii) $\Leftrightarrow$ (iv): $A - \lambda I_n$ is singular if and only if $\det(A - \lambda I_n) = 0$ (Theorem 5.19(a)), and $\det(\lambda I_n - A) = \det\big((-1)(A - \lambda I_n)\big) = (-1)^n \det(A - \lambda I_n)$ (Theorem 5.19(c)).
- $g_A(\lambda) = \dim \mathcal{N}(A - \lambda I_n) = n - \operatorname{rank}(A - \lambda I_n)$ by Theorem 4.12. □

Remark 7.4 (Polynomials; imported facts)**.** A **polynomial** is a function $q(t) = c_0 + c_1 t + \cdots + c_m t^m$ ($t \in \mathbb{R}$) with real coefficients $c_0, \ldots, c_m$. If $c_m \neq 0$, then q has **degree** m and **leading coefficient** c_m; q has **degree at most** d if $c_k = 0$ for all $k > d$. A **root** of q is a number r with $q(r) = 0$. Sums and products of polynomials are polynomials, and the product of nonzero polynomials of degrees

m and l has degree $m+l$, since its leading coefficient is the product of theirs. If $q(r)=0$, then $q(t)=(t-r)q_1(t)$ for a polynomial q_1 (the **factor theorem**), since

$$q(t)=q(t)-q(r)=\sum_{k=1}^{m}c_k(t^k-r^k),\qquad t^k-r^k=(t-r)(t^{k-1}+t^{k-2}r+\cdots+r^{k-1})$$

The following facts are used without proof.

(A1) A nonzero polynomial of degree m has at most m roots. Hence two polynomials that agree at infinitely many points are equal: they have the same coefficients, and they agree at every $t\in\mathbb{R}$.

(A2) (Used only to describe non-real roots.) Every polynomial q of degree $m\geqslant 1$ can be written as $q(t)=c_m(t-z_1)\cdots(t-z_m)$ with complex numbers $z_1,\ldots,z_m$, the complex roots of q.

Definition 7.5. The **characteristic polynomial** of $A\in\mathbb{R}^{n\times n}$ is $p_A(t)=\det(tI_n-A)$ $(t\in\mathbb{R})$, and $p_A(t)=0$ is the **characteristic equation** of A.

For $A\in\mathbb{R}^{2\times 2}$,

$$p_A(t)=\det\begin{pmatrix}t-a_{11} & -a_{12}\\ -a_{21} & t-a_{22}\end{pmatrix}=t^2-(a_{11}+a_{22})t+(a_{11}a_{22}-a_{12}a_{21})=t^2-(\operatorname{tr}A)\,t+\det A$$

By Theorem 7.3, the eigenvalues of $A\in\mathbb{R}^{2\times 2}$ are the real roots of this quadratic.

Lemma 7.6. For every $t\in\mathbb{R}$ let $M(t)\in\mathbb{R}^{m\times m}$ have entries of the form $\alpha+\beta t$, where the numbers α,β do not depend on t. If at most d rows of $M(t)$ contain an entry with $\beta\neq 0$, then $\det M(t)$ is a polynomial in t of degree at most d.

Proof. Induction on m. For $m=1$, $\det M(t)=\alpha+\beta t$, with $\beta=0$ if $d=0$. Let $m\geqslant 2$. If no entry has $\beta\neq 0$, then $\det M(t)$ does not depend on t. Otherwise $d\geqslant 1$; let row i contain an entry with $\beta\neq 0$, and expand along row i (Theorem 5.13):

$$\det M(t)=\sum_{j=1}^{m}(-1)^{i+j}m_{ij}(t)\det M(t)(i|j)$$

Each $M(t)(i|j)\in\mathbb{R}^{(m-1)\times(m-1)}$ has entries of the same form, and its rows are the rows of $M(t)$ other than row i, each shortened by one entry; so at most $d-1$ of them contain an entry with $\beta\neq 0$. By the induction hypothesis, $\det M(t)(i|j)$ is a polynomial of degree at most $d-1$; multiplied by $m_{ij}(t)=\alpha+\beta t$, it has degree at most d, and so has the sum. □

Theorem 7.7. Let $A\in\mathbb{R}^{n\times n}$.

(a) p_A is a polynomial of degree n with leading coefficient 1; the coefficient of t^{n-1} is $-\operatorname{tr}A$, and the constant term is $(-1)^n\det A$:

$$p_A(t)=t^n-(\operatorname{tr}A)\,t^{n-1}+\cdots+(-1)^n\det A$$

(for $n=1$, $p_A(t)=t-a_{11}$, and the last two descriptions coincide).

(b) The eigenvalues of A are the real roots of p_A. Hence A has at most n eigenvalues.

(c) $p_{A'}=p_A$.

Proof. (a) Induction on n. For $n=1$, $p_A(t)=t-a_{11}$, and $-a_{11}=-\operatorname{tr}A=(-1)^1\det A$. Let $n\geqslant 2$. The first row of tI_n-A is $(t-a_{11},-a_{12},\ldots,-a_{1n})$, and $(tI_n-A)(1|1)=tI_{n-1}-A(1|1)$. By Definition 5.12,

$$p_A(t)=(t-a_{11})\,p_{A(1|1)}(t)+\sum_{j=2}^{n}(-1)^{1+j}(-a_{1j})\det\Big((tI_n-A)(1|j)\Big)$$

- For $j \geqslant 2$, the rows of $(tI_n - A)(1|j)$ are rows $2, \dots, n$ of $tI_n - A$ without their jth entry. Row i of $tI_n - A$ contains t only in column i, so the row that comes from row j is constant, and at most $n-2$ rows contain t. By Lemma 7.6, the sum over $j \geqslant 2$ is a polynomial of degree at most $n-2$.
- By the induction hypothesis, $p_{A(1|1)}(t) = t^{n-1} - \operatorname{tr}\Big(A(1|1)\Big)t^{n-2} + r(t)$ with r of degree at most $n-3$ ($r = 0$ if $n = 2$). Hence

$$(t - a_{11})\, p_{A(1|1)}(t) = t^n - \Big(a_{11} + \operatorname{tr} A(1|1)\Big)t^{n-1} + (\text{a polynomial of degree at most } n-2),$$

  and $a_{11} + \operatorname{tr} A(1|1) = \operatorname{tr} A$.
- So $p_A(t) = t^n - (\operatorname{tr} A)t^{n-1} + (\text{a polynomial of degree at most } n-2)$. Its constant term is $p_A(0) = \det(-A) = (-1)^n \det A$ (Theorem 5.19(c)).

(b) By Theorem 7.3, $\lambda \in \mathbb{R}$ is an eigenvalue if and only if $p_A(\lambda) = 0$. By (a) and (A1), p_A has at most n roots.

(c) $(tI_n - A)' = tI_n - A'$ (Theorem 1.9), so by Theorem 5.15(b), $p_{A'}(t) = \det\Big((tI_n - A)'\Big) = \det(tI_n - A) = p_A(t)$. □

By (b), the four eigenvalues $1, -1, 2, -2$ of the matrix $A \in \mathbb{R}^{4\times 4}$ of Example 7.2 are all its eigenvalues.

Definition 7.8. Let λ be an eigenvalue of $A \in \mathbb{R}^{n\times n}$. The **algebraic multiplicity** $a_A(\lambda)$ is the multiplicity of λ as a root of p_A: the largest integer m such that $p_A(t) = (t-\lambda)^m q(t)$ for all t, for some polynomial q. The characteristic polynomial p_A **splits** if

$$p_A(t) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_n) \qquad (t \in \mathbb{R})$$

for some $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, not necessarily distinct; then $\lambda_1, \dots, \lambda_n$ are **the eigenvalues of A listed with multiplicity**.

The largest m exists: if $p_A(t) = (t-\lambda)^m q(t)$, then $q \neq 0$, and by (A1) and the degree of a product (Remark 7.4), $n = m + \deg q \geqslant m$. For this largest m, $q(\lambda) \neq 0$: otherwise $q(t) = (t-\lambda)q_1(t)$ by the factor theorem, and $p_A(t) = (t-\lambda)^{m+1}q_1(t)$.

Proposition 7.9. Let $A \in \mathbb{R}^{n\times n}$, and let $p_A(t) = (t - \lambda_1)\cdots(t - \lambda_n)$ with $\lambda_1, \dots, \lambda_n \in \mathbb{R}$.

(a) The eigenvalues of A are the distinct numbers among $\lambda_1, \dots, \lambda_n$.

(b) Each eigenvalue λ occurs exactly $a_A(\lambda)$ times in the list $\lambda_1, \dots, \lambda_n$. Hence the sum of $a_A(\lambda)$ over the distinct eigenvalues λ of A is n.

Proof. (a) For $\mu \in \mathbb{R}$, $p_A(\mu) = (\mu - \lambda_1)\cdots(\mu - \lambda_n)$ is 0 if and only if some factor is 0; apply Theorem 7.7(b).

(b) Let λ occur c times. Then $p_A(t) = (t-\lambda)^c r(t)$, where $r(t)$ is the product of the factors $t - \lambda_i$ with $\lambda_i \neq \lambda$; so $r(\lambda) \neq 0$, and $a_A(\lambda) \geqslant c$. Suppose that $p_A(t) = (t-\lambda)^m q(t)$ with $m > c$. For $t \neq \lambda$, dividing by $(t-\lambda)^c$ gives $(t-\lambda)^{m-c}q(t) = r(t)$; by (A1) the two polynomials agree at every t, and at $t = \lambda$ this reads $0 = r(\lambda)$ ✘ So $a_A(\lambda) = c$. Summing over the distinct eigenvalues counts every index i once. □

Example 7.10. Find the eigenvalues and eigenspaces of

$$A = \begin{pmatrix} -1 & 2 & 3 & -1 \\ 2 & -1 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Solution. The matrix $tI_4 - A$ has the diagonal blocks $tI_2 - B$ and $tI_2 - D$ and the zero block 0 below them. By Theorem 5.25 and the formula for 2×2 matrices,

$$p_A(t) = p_B(t)\, p_D(t) = (t^2 + 2t - 3)(t^2 - 2t - 3) = (t+3)(t-1)(t-3)(t+1)$$

So p_A splits, the eigenvalues are -3, -1, 1 and 3 (Theorem 7.7(b)), and each has algebraic multiplicity 1 (Proposition 7.9). Multiplying out, $p_A(t) = (t^2-9)(t^2-1) = t^4 - 10t^2 + 9$: the coefficient 0 of t^3 is $-\operatorname{tr} A = -(-1-1+1+1)$, and the constant 9 is $(-1)^4 \det A$, as $\det A = \det B \det D = (-3)(-3) = 9$ (Theorems 5.25 and 7.7(a)). The eigenspaces are the null spaces of the following four matrices.

- $A + 3I_4 = \left(\begin{smallmatrix} 2 & 2 & 3 & -1 \\ 2 & 2 & 1 & 1 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 2 & 4 \end{smallmatrix}\right)$: rows 3 and 4 give $x_3 = x_4 = 0$, and then row 1 gives $x_1 + x_2 = 0$. So $E_{-3}(A) = \operatorname{span}\{(-1,1,0,0)'\}$.
- $A + I_4 = \left(\begin{smallmatrix} 0 & 2 & 3 & -1 \\ 2 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 \end{smallmatrix}\right)$: row 3 gives $x_3 = -x_4$; then row 2 gives $x_1 = 0$, and row 1 gives $2x_2 = 4x_4$. So $E_{-1}(A) = \operatorname{span}\{(0,2,-1,1)'\}$.
- $A - I_4 = \left(\begin{smallmatrix} -2 & 2 & 3 & -1 \\ 2 & -2 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{smallmatrix}\right)$: rows 3 and 4 give $x_4 = x_3 = 0$, and then row 1 gives $x_1 = x_2$. So $E_1(A) = \operatorname{span}\{(1,1,0,0)'\}$.
- $A - 3I_4 = \left(\begin{smallmatrix} -4 & 2 & 3 & -1 \\ 2 & -4 & 1 & 1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 2 & -2 \end{smallmatrix}\right)$: row 3 gives $x_3 = x_4$, and then rows 1 and 2 give $2x_1 - x_2 = x_4$ and $x_1 - 2x_2 = -x_4$, so $x_1 = x_2 = x_4$. So $E_3(A) = \operatorname{span}\{\mathbf{1}_4\}$; indeed every row of A adds up to 3.

Each geometric multiplicity is 1; accordingly each of the four matrices has rank 3 (Theorem 7.3). Check: $A(0,2,-1,1)' = (0,-2,1,-1)'$ and $A(-1,1,0,0)' = (3,-3,0,0)'$.

Example 7.11. (a) If $A \in \mathbb{R}^{n\times n}$ is upper or lower triangular, so is $tI_n - A$, with diagonal entries $t - a_{ii}$. By Theorem 5.15(a),

$$p_A(t) = (t - a_{11})(t - a_{22}) \cdots (t - a_{nn})$$

So p_A splits, the eigenvalues of A are its diagonal entries, and $a_A(\lambda)$ is the number of diagonal entries equal to λ (Proposition 7.9).

(b) Let $D = \operatorname{diag}(d_1, \ldots, d_n)$ and $\lambda \in \mathbb{R}$. The columns of $D - \lambda I_n$ are $(d_i - \lambda)e_i$. The nonzero ones span $\mathcal{C}(D - \lambda I_n)$ and are linearly independent (a combination $\sum_i c_i(d_i - \lambda)e_i$ has the entries $c_i(d_i - \lambda)$), so $\operatorname{rank}(D - \lambda I_n)$ is the number of i with $d_i \neq \lambda$. By Theorem 7.3 and (a), for every eigenvalue λ of D, $g_D(\lambda)$ and $a_D(\lambda)$ both equal the number of i with $d_i = \lambda$.

(c) For $N = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $p_N(t) = (t-1)^2$ and $a_N(1) = 2$; but $N - I_2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has rank 1, so $g_N(1) = 1$ and $E_1(N) = \operatorname{span}\{e_1\}$.

(d) The matrix $K = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ turns every vector of the plane by a right angle: $K(x_1, x_2)' = (-x_2, x_1)'$. It has no eigenvalue, since $p_K(t) = t^2 + 1 > 0$ for every $t \in \mathbb{R}$. By (A2), $p_K(t) = (t - \mathrm{i})(t + \mathrm{i})$ with the non-real roots $\pm\mathrm{i}$ ($\mathrm{i}^2 = -1$); non-real roots of a characteristic polynomial are not eigenvalues in this course, and p_K does not split.

Theorem 7.12. Let $A \in \mathbb{R}^{n\times n}$, $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}^n$ with $Av = \lambda v$. For a polynomial $q(t) = c_0 + c_1 t + \cdots + c_m t^m$ put $q(A) = c_0 I_n + c_1 A + \cdots + c_m A^m$.

(a) $A^k v = \lambda^k v$ for every $k \geqslant 0$ (with $\lambda^0 = 1$, also for $\lambda = 0$), and $q(A)v = q(\lambda)v$ for every polynomial q.

(b) A is invertible if and only if 0 is not an eigenvalue of A. In that case, if $v \neq 0$, then $\lambda \neq 0$ and $A^{-1}v = \lambda^{-1}v$.

Proof. (a) $A^0v = v = \lambda^0 v$, and if $A^kv = \lambda^k v$, then $A^{k+1}v = A(A^kv) = \lambda^k Av = \lambda^{k+1}v$. Hence $q(A)v = \sum_k c_k A^k v = \sum_k c_k \lambda^k v = q(\lambda)v$.

(b) 0 is an eigenvalue if and only if $A - 0I_n = A$ is singular (Theorem 7.3). If A is invertible and $v \neq 0$, then $\lambda \neq 0$, since 0 is not an eigenvalue; multiplying $Av = \lambda v$ by $\lambda^{-1}A^{-1}$ gives $\lambda^{-1}v = A^{-1}v$. □

So every eigenvector of A is an eigenvector of A^k, of $A + cI_n$ (for the eigenvalue $\lambda + c$), of $q(A)$, and of A^{-1} if A is invertible. In Example 7.2, v_1 and v_2 are eigenvectors of A^2 for the eigenvalue 1, and v_3 and v_4 for 4; the matrix A is invertible, and $v_1, \dots, v_4$ are eigenvectors of A^{-1} for 1, -1, $\frac{1}{2}$, $-\frac{1}{2}$ and of $A - 2I_4$ for -1, -3, 0, -4. But $q(A)$ may have further eigenvalues: for K of Example 7.11(d), $K^2 = -I_2$ has the eigenvalue -1, while K has none.

Definition 7.13. A matrix $B \in \mathbb{R}^{n\times n}$ is **similar** to $A \in \mathbb{R}^{n\times n}$ if $B = P^{-1}AP$ for some invertible $P \in \mathbb{R}^{n\times n}$. Then $A = (P^{-1})^{-1}BP^{-1}$, so A is similar to B as well, and A and B are called **similar**.

Theorem 7.14. Let $A \in \mathbb{R}^{n\times n}$, let $P \in \mathbb{R}^{n\times n}$ be invertible, and let $B = P^{-1}AP$.

(a) $p_B = p_A$. Hence A and B have the same eigenvalues, with the same algebraic multiplicities, and $\operatorname{tr} B = \operatorname{tr} A$, $\det B = \det A$.

(b) $\operatorname{rank} B = \operatorname{rank} A$, and for every $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}^n$: $v \in E_\lambda(A)$ if and only if $P^{-1}v \in E_\lambda(B)$. Moreover $g_B(\lambda) = g_A(\lambda)$.

Proof. (a) $P^{-1}(tI_n)P = tI_n$, so $tI_n - B = P^{-1}(tI_n - A)P$, and by Theorem 5.19(b),(c), $p_B(t) = \det P^{-1} \cdot p_A(t) \cdot \det P = p_A(t)$. The eigenvalues are the real roots of this polynomial (Theorem 7.7(b)), the algebraic multiplicities depend only on it, and so do the trace and the determinant (Theorem 7.7(a)).

(b) $\operatorname{rank} B = \operatorname{rank} A$ by Theorem 4.15(d). If $Av = \lambda v$, then $B(P^{-1}v) = P^{-1}Av = \lambda P^{-1}v$; if $B(P^{-1}v) = \lambda P^{-1}v$, then $Av = PBP^{-1}v = \lambda v$. Finally $B - \lambda I_n = P^{-1}(A - \lambda I_n)P$, so by Theorems 7.3 and 4.15(d), $g_B(\lambda) = n - \operatorname{rank}(B - \lambda I_n) = n - \operatorname{rank}(A - \lambda I_n) = g_A(\lambda)$. □

Matrices with the same characteristic polynomial need not be similar: $p_N = p_{I_2}$ for N of Example 7.11(c), but $P^{-1}I_2P = I_2 \neq N$ for every invertible P; accordingly $g_N(1) = 1 \neq 2 = g_{I_2}(1)$.

Proposition 7.15. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$.

(a) $t^n p_{AB}(t) = t^m p_{BA}(t)$ for every $t \in \mathbb{R}$.

(b) AB and BA have the same nonzero eigenvalues, with the same algebraic and the same geometric multiplicities. For $\lambda \neq 0$, $v \mapsto Bv$ maps $E_\lambda(AB)$ into $E_\lambda(BA)$, injectively: $Bv \neq 0$ for every nonzero $v \in E_\lambda(AB)$.

Proof. (a) For $t \in \mathbb{R}$ let $M = \begin{pmatrix} tI_m & A \\ B & I_n \end{pmatrix}$. By Theorem 5.26(b), with the invertible block I_n, $\det M = \det(tI_m - AB) = p_{AB}(t)$. For $t \neq 0$ the block tI_m is invertible, and Theorem 5.26(a) and Theorem 5.19(c) give

$$\det M = \det(tI_m)\det(I_n - t^{-1}BA) = t^m \det\Big(t^{-1}(tI_n - BA)\Big) = t^m t^{-n} p_{BA}(t)$$

So $t^n p_{AB}(t) = t^m p_{BA}(t)$ for $t \neq 0$, and by (A1) for every t.

(b) Let $\lambda \neq 0$.

- Eigenvalues: $\lambda^n \neq 0$ and $\lambda^m \neq 0$, so by (a), $p_{AB}(\lambda) = 0$ if and only if $p_{BA}(\lambda) = 0$.
- Algebraic multiplicities: let λ be an eigenvalue, $c = a_{BA}(\lambda)$, and $p_{BA}(t) = (t - \lambda)^c r(t)$ with $r(\lambda) \neq 0$ (the paragraph after Definition 7.8). If $p_{AB}(t) = (t - \lambda)^k q(t)$ with $k > c$, then by (a), $t^n(t - \lambda)^{k-c}q(t) = t^m r(t)$ for $t \neq \lambda$; by (A1) this holds also at $t = \lambda$, where it reads $0 = \lambda^m r(\lambda) \neq 0$ ✘ So $a_{AB}(\lambda) \leqslant a_{BA}(\lambda)$. Exchanging the roles of A and B leaves the identity (a) unchanged, so also $a_{BA}(\lambda) \leqslant a_{AB}(\lambda)$.

- Geometric multiplicities: if $ABv = \lambda v$, then $BA(Bv) = B(ABv) = \lambda Bv$, so $Bv \in E_\lambda(BA)$; and if $Bv = 0$, then $\lambda v = ABv = 0$, so $v = 0$. Let $v_1, \dots, v_g$ be a basis of $E_\lambda(AB)$. If $\sum_i c_i B v_i = 0$, then $B(\sum_i c_i v_i) = 0$, so $\sum_i c_i v_i = 0$ and every $c_i = 0$. Thus $E_\lambda(BA)$ contains g linearly independent vectors, and $g_{BA}(\lambda) \geqslant g_{AB}(\lambda)$ (Theorem 4.5(c)). Exchanging A and B gives the reverse inequality. □

Example 7.16. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \qquad AA' = \begin{pmatrix} 2 & 1 & 1 & -1 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix}$$

Expanding $\det(tI_3 - A'A)$ along the first row,

$$p_{A'A}(t) = (t-3)\Big((t-2)(t-3) - 1\Big) - (t-3) = (t-3)(t^2 - 5t + 4) = (t-1)(t-3)(t-4)$$

By Proposition 7.15(a) with A and $B = A'$ ($m = 4$, $n = 3$), $t^3 p_{AA'}(t) = t^4 p_{A'A}(t)$. So $p_{AA'}(t) = t(t-1)(t-3)(t-4)$ for $t \neq 0$, and by (A1) for every t. The vectors $(1,1,1)'$, $(-1,0,1)'$ and $(1,-2,1)'$ are eigenvectors of $A'A$ for 4, 3 and 1 (for instance $A'A(1,-2,1)' = (1,-2,1)'$). By part (b), applied to A' and A, the map $v \mapsto Av$ sends them to eigenvectors of AA' for the same eigenvalues:

$$A\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix}, \qquad A\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \\ 0 \end{pmatrix}$$

and indeed $AA'(-1,0,1,2)' = (-3,0,3,6)'$. The remaining eigenvalue 0 of AA' has $E_0(AA') = \mathcal{N}(AA') = \mathcal{N}(A')$ (Theorem 4.18(a) applied to A'). The system $A'x = 0$ reads $x_1 + x_2 - x_4 = 0$, $x_1 + x_3 = 0$, $x_2 + x_3 + x_4 = 0$, with the special solution $(1,0,-1,1)'$; so $E_0(AA') = \operatorname{span}\{(1,0,-1,1)'\}$.

Example 7.17. Let $n \geqslant 2$ and $\alpha, \beta \in \mathbb{R}$. Find the eigenvalues and eigenspaces of $\alpha I_n + \beta J_n$, in particular of J_n, of the centering matrix C_n and of the equicorrelation matrix $R = (1-\rho)I_n + \rho J_n$ of Example 5.28, and rederive $\det R$.

Solution. Let $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$, the centered vectors (Example 3.5(f)).

- $J_n\mathbf{1} = \mathbf{1}(\mathbf{1}'\mathbf{1}) = n\mathbf{1}$ (Proposition 1.13), and $J_n x = \mathbf{1}(\mathbf{1}'x) = 0$ for $x \in W_0$.
- $W_0 = \operatorname{span}\{\mathbf{1}\}^\perp$ has dimension $n-1$ (Theorem 6.5(a),(c)). Let $d_1, \dots, d_{n-1}$ be a basis of W_0. As $\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \oplus W_0$ (Example 3.25), the vectors $\mathbf{1}, d_1, \dots, d_{n-1}$ form a basis of $\mathbb{R}^n$ (Theorem 3.23), so $P = [\mathbf{1}\ d_1\ \cdots\ d_{n-1}]$ is invertible (Theorem 4.13(b),(d)).
- $J_n P = [n\mathbf{1}\ 0\ \cdots\ 0] = P\Lambda$ with $\Lambda = \operatorname{diag}(n, 0, \dots, 0)$, so $P^{-1}J_nP = \Lambda$ and
$$P^{-1}(\alpha I_n + \beta J_n)P = \alpha I_n + \beta\Lambda = \operatorname{diag}(\alpha + n\beta,\ \alpha,\ \dots,\ \alpha)$$
By Theorem 7.14(a) and Example 7.11(a), $p_{\alpha I_n + \beta J_n}(t) = (t - \alpha - n\beta)(t-\alpha)^{n-1}$.
- $(\alpha I_n + \beta J_n)\mathbf{1} = (\alpha + n\beta)\mathbf{1}$, and $(\alpha I_n + \beta J_n)x = \alpha x$ for $x \in W_0$. If $\beta \neq 0$, the two eigenvalues differ, and by Theorem 7.14(b) and Example 7.11(b) their geometric multiplicities are 1 and $n-1$; so $E_{\alpha+n\beta}(\alpha I_n + \beta J_n) = \operatorname{span}\{\mathbf{1}\}$ and $E_\alpha(\alpha I_n + \beta J_n) = W_0$ (each contains the subspace on the right, which has the same dimension; Theorem 4.5(a)).
- J_n ($\alpha = 0$, $\beta = 1$) has the eigenvalue n with eigenvector $\mathbf{1}$ and the eigenvalue 0 with eigenspace W_0: $p_{J_n}(t) = (t-n)t^{n-1}$. $C_n = I_n - \frac{1}{n}J_n$ ($\alpha = 1$, $\beta = -\frac{1}{n}$) has the eigenvalue 0 with eigenvector $\mathbf{1}$ and the eigenvalue 1 with eigenspace W_0: $p_{C_n}(t) = t(t-1)^{n-1}$. R ($\alpha = 1-\rho$, $\beta = \rho$) has the eigenvalues $1 + (n-1)\rho$, with eigenvector $\mathbf{1}$, and $1-\rho$, with eigenspace W_0 if $\rho \neq 0$.
- $\det R = \det(P^{-1}RP) = \Big(1 + (n-1)\rho\Big)(1-\rho)^{n-1}$ (Theorems 7.14(a) and 5.15(a)), the value found in Example 5.28. By Theorem 7.12(b), R is invertible if and only if neither eigenvalue is 0, that is, $\rho \neq 1$ and $\rho \neq -1/(n-1)$. For $n = 4$ and $\rho = \frac{1}{3}$, the eigenvalues of R listed with multiplicity are 2, $\frac{2}{3}$, $\frac{2}{3}$, $\frac{2}{3}$, and $\det R = 2 \cdot \frac{8}{27} = \frac{16}{27}$.

Diagonalization

Definition 7.18. A matrix $A \in \mathbb{R}^{n\times n}$ is **diagonalizable** if $A = PDP^{-1}$ for some invertible $P \in \mathbb{R}^{n\times n}$ and some diagonal $D \in \mathbb{R}^{n\times n}$, that is, if A is similar to a diagonal matrix.

Theorem 7.19. Let $A \in \mathbb{R}^{n\times n}$, let $P \in \mathbb{R}^{n\times n}$ have columns $p_1, \dots, p_n$, and let $D = \operatorname{diag}(d_1, \dots, d_n)$. Then P is invertible and $A = PDP^{-1}$ if and only if $p_1, \dots, p_n$ are linearly independent and $Ap_j = d_j p_j$ for $j = 1, \dots, n$. Hence A is diagonalizable if and only if $\mathbb{R}^n$ has a basis of eigenvectors of A; then each d_j is an eigenvalue of A, with eigenvector p_j.

Proof.
- $AP = [Ap_1 \ \cdots \ Ap_n]$ (Theorem 1.4(a)), and $PD = [d_1 p_1 \ \cdots \ d_n p_n]$, since right multiplication by a diagonal matrix scales the columns (Chapter 1). So $AP = PD$ if and only if $Ap_j = d_j p_j$ for every j.
- P is invertible if and only if its columns are linearly independent (Theorem 4.13(b),(d)), and then $A = PDP^{-1}$ if and only if $AP = PD$ (multiply on the right by P, respectively by P^{-1}).
- n vectors of $\mathbb{R}^n$ form a basis if and only if they are linearly independent (Theorem 4.5(d)). Linearly independent vectors are nonzero, so p_j is an eigenvector for d_j. □

If $A = PDP^{-1}$ and $c = P^{-1}x$ is the vector of coordinates of x in the basis $p_1, \dots, p_n$ (Example 3.16), then the coordinates of Ax are $P^{-1}Ax = DP^{-1}x = (d_1c_1, \dots, d_nc_n)'$: in the basis of eigenvectors, A multiplies the jth coordinate by d_j.

Theorem 7.20. Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of $A \in \mathbb{R}^{n\times n}$, and for each i let $v_{i1}, \dots, v_{im_i}$ be linearly independent vectors of $E_{\lambda_i}(A)$. Then the combined list $v_{11}, \dots, v_{1m_1}, \dots, v_{k1}, \dots, v_{km_k}$ is linearly independent. In particular, eigenvectors for distinct eigenvalues are linearly independent.

Proof.
- First, if $u_i \in E_{\lambda_i}(A)$ for $1 \leqslant i \leqslant k$ and $u_1 + \cdots + u_k = 0$, then every $u_i = 0$. Induction on k; for $k = 1$ the hypothesis reads $u_1 = 0$. For $k \geqslant 2$, apply $A - \lambda_k I_n$: as $(A - \lambda_k I_n)u_i = (\lambda_i - \lambda_k)u_i$,

$$(\lambda_1 - \lambda_k)u_1 + \cdots + (\lambda_{k-1} - \lambda_k)u_{k-1} = 0,$$

  with $(\lambda_i - \lambda_k)u_i \in E_{\lambda_i}(A)$. By the induction hypothesis $(\lambda_i - \lambda_k)u_i = 0$, so $u_i = 0$ for $i < k$ (as $\lambda_i \neq \lambda_k$), and then $u_k = 0$.
- Let $\sum_i \sum_j c_{ij}v_{ij} = 0$, and put $u_i = \sum_j c_{ij}v_{ij} \in E_{\lambda_i}(A)$. Then $\sum_i u_i = 0$, so every $u_i = 0$, and the independence of $v_{i1}, \dots, v_{im_i}$ gives $c_{ij} = 0$ for all i, j. □

Theorem 7.21. For every eigenvalue λ of $A \in \mathbb{R}^{n\times n}$, $1 \leqslant g_A(\lambda) \leqslant a_A(\lambda)$.

Proof. Let $g = g_A(\lambda)$; $g \geqslant 1$, as $E_\lambda(A)$ contains an eigenvector.

- Extend a basis $v_1, \dots, v_g$ of $E_\lambda(A)$ to a basis $v_1, \dots, v_n$ of $\mathbb{R}^n$ (Theorem 4.5(b)); then $P = [v_1 \ \cdots \ v_n]$ is invertible (Theorem 4.13(b),(d)).
- For $j \leqslant g$, $APe_j = Av_j = \lambda v_j = \lambda Pe_j$, so $P^{-1}APe_j = \lambda e_j$: the first g columns of $B = P^{-1}AP$ are $\lambda e_1, \dots, \lambda e_g$.
- If $g = n$, then $B = \lambda I_n$, and $p_A(t) = p_B(t) = (t - \lambda)^n$ (Theorem 7.14(a)). If $g < n$, then $B = \left(\begin{smallmatrix} \lambda I_g & B_{12} \\ 0 & B_{22} \end{smallmatrix}\right)$ with $B_{22} \in \mathbb{R}^{(n-g)\times(n-g)}$, and by Theorems 7.14(a), 5.25 and 5.15(a),

$$p_A(t) = p_B(t) = \det \begin{pmatrix} (t-\lambda)I_g & -B_{12} \\ 0 & tI_{n-g} - B_{22} \end{pmatrix} = (t-\lambda)^g \, p_{B_{22}}(t)$$

- In both cases $p_A(t) = (t-\lambda)^g q(t)$ for a polynomial q (Theorem 7.7(a)), so $a_A(\lambda) \geqslant g$ (Definition 7.8). □

Theorem 7.22. Let $A \in \mathbb{R}^{n\times n}$.

(a) A is diagonalizable if and only if p_A splits and $g_A(\lambda) = a_A(\lambda)$ for every eigenvalue λ of A.

(b) If A has n distinct eigenvalues, then A is diagonalizable.

Proof. (a) Let $A = PDP^{-1}$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. By Theorem 7.14 and Example 7.11(a),(b), $p_A(t) = p_D(t) = (t - d_1)\cdots(t - d_n)$ splits, and $g_A(\lambda) = g_D(\lambda) = a_D(\lambda) = a_A(\lambda)$ for every eigenvalue λ.

Conversely, let p_A split and $g_A(\lambda) = a_A(\lambda)$ for every eigenvalue λ. Let $\mu_1, \dots, \mu_s$ be the distinct eigenvalues, and choose a basis of each $E_{\mu_j}(A)$. Together these are $\sum_j g_A(\mu_j) = \sum_j a_A(\mu_j) = n$ vectors (Proposition 7.9(b)), linearly independent by Theorem 7.20, hence a basis of $\mathbb{R}^n$ consisting of eigenvectors (Theorem 4.5(d)). Apply Theorem 7.19.

(b) Eigenvectors $v_1, \dots, v_n$ for the n distinct eigenvalues are linearly independent (Theorem 7.20), hence a basis of $\mathbb{R}^n$ (Theorem 4.5(d)); apply Theorem 7.19. □

To diagonalize A: find the roots of p_A; if p_A does not split, A is not diagonalizable. For each eigenvalue λ, the special solutions of $(A - \lambda I_n)x = 0$ form a basis of $E_\lambda(A)$ (Theorem 4.8(c)), and $g_A(\lambda)$ is the number of free unknowns. If $g_A(\lambda) = a_A(\lambda)$ for every λ, put these bases into the columns of P and the corresponding eigenvalues, in the same order, on the diagonal of D; then $A = PDP^{-1}$. By Theorem 7.21, only eigenvalues with $a_A(\lambda) \geqslant 2$ need to be checked.

Example 7.23. (a) The matrix A of Example 7.10 has four distinct eigenvalues, so it is diagonalizable (Theorem 7.22(b)): $A = PDP^{-1}$ with $P = \begin{pmatrix} -1 & 0 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$, the eigenvectors found there, and $D = \operatorname{diag}(-3, -1, 1, 3)$.

(b) $N = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not diagonalizable, since $g_N(1) = 1 < 2 = a_N(1)$ (Example 7.11(c)): all eigenvectors of N are multiples of e_1, and they do not span $\mathbb{R}^2$.

(c) Show that 3 and 1 are eigenvalues of

$$B = \begin{pmatrix} 1 & 2 & -2 & 2 \\ -2 & 3 & -2 & 0 \\ 0 & -2 & 3 & -2 \\ 2 & -2 & 2 & 1 \end{pmatrix},$$

and diagonalize B.

Solution.

- In $B - 3I_4 = \begin{pmatrix} -2 & 2 & -2 & 2 \\ -2 & 0 & -2 & 0 \\ 0 & -2 & 0 & -2 \\ 2 & -2 & 2 & -2 \end{pmatrix}$, row 1 is row 2 minus row 3, row 4 is minus row 1, and rows 2 and 3 are linearly independent; so $\operatorname{rank}(B - 3I_4) = 2$ and $g_B(3) = 2$ (Theorem 7.3). Rows 2 and 3 give $x_1 + x_3 = 0$ and $x_2 + x_4 = 0$, with the special solutions $p_1 = (-1, 0, 1, 0)'$ and $p_2 = (0, -1, 0, 1)'$, a basis of $E_3(B)$.
- In $B - I_4 = \begin{pmatrix} 0 & 2 & -2 & 2 \\ -2 & 2 & -2 & 0 \\ 0 & -2 & 2 & -2 \\ 2 & -2 & 2 & 0 \end{pmatrix}$, rows 3 and 4 are minus rows 1 and 2, which are linearly independent; so $g_B(1) = 2$. Rows 1 and 2 give $x_2 = x_3 - x_4$ and $x_1 = x_2 - x_3 = -x_4$, with the special solutions $p_3 = (0, 1, 1, 0)'$ and $p_4 = (-1, -1, 0, 1)'$, a basis of $E_1(B)$.
- By Theorem 7.20, $p_1, \dots, p_4$ are linearly independent, so by Theorem 7.19, $B = PDP^{-1}$ with $D = \operatorname{diag}(3, 3, 1, 1)$ and

$$P = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & -1 & 1 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

(check $P^{-1}P = I_4$; for instance $\text{row}_1(P^{-1})p_4 = 1 - 1 = 0$).

- By Theorem 7.14(a) and Example 7.11(a), $p_B(t) = p_D(t) = (t-3)^2(t-1)^2$: 3 and 1 are the only eigenvalues, and $a_B(3) = g_B(3) = 2$, $a_B(1) = g_B(1) = 2$.

Proposition 7.24. Let $A = PDP^{-1} \in \mathbb{R}^{n\times n}$, where P is invertible with columns $p_1, \dots, p_n$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then $A^k = PD^kP^{-1}$ for $k \geqslant 0$, with $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$. For $x_0 \in \mathbb{R}^n$ and $c = P^{-1}x_0$,

$$A^k x_0 = c_1\lambda_1^k p_1 + c_2\lambda_2^k p_2 + \cdots + c_n\lambda_n^k p_n$$

Proof. $A^0 = I_n = PD^0P^{-1}$, and $A^{k+1} = A^kA = PD^kP^{-1}PDP^{-1} = PD^{k+1}P^{-1}$. A product of diagonal matrices is diagonal with the products of the diagonal entries (Chapter 1), so $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$. Then $A^kx_0 = PD^kc$ with $D^kc = (c_1\lambda_1^k, \dots, c_n\lambda_n^k)'$; apply Theorem 1.4(b). □

For B of Example 7.23(c), $B^k = P\,\text{diag}(3^k, 3^k, 1, 1)P^{-1} = 3^kE + (I_4 - E)$ with $E = P\,\text{diag}(1, 1, 0, 0)P^{-1}$, since $\text{diag}(3^k, 3^k, 1, 1) = 3^k\,\text{diag}(1, 1, 0, 0) + (I_4 - \text{diag}(1, 1, 0, 0))$ and $PP^{-1} = I_4$. Here $B = 3E + I_4 - E = I_4 + 2E$, so $E = \frac{1}{2}(B - I_4)$ and

$$B^k = I_4 + \tfrac{1}{2}(3^k - 1)(B - I_4), \qquad B^2 = I_4 + 4(B - I_4) = \begin{pmatrix} 1 & 8 & -8 & 8 \\ -8 & 9 & -8 & 0 \\ 0 & -8 & 9 & -8 \\ 8 & -8 & 8 & 1 \end{pmatrix}$$

Theorem 7.25. Let $A \in \mathbb{R}^{n\times n}$ be idempotent of rank r. Then A is diagonalizable, its eigenvalues lie in $\{0, 1\}$, $E_1(A) = \mathcal{C}(A)$, $E_0(A) = \mathcal{N}(A)$, and $p_A(t) = t^{n-r}(t-1)^r$. Hence $\text{tr}\,A = r$.

Proof.

- If $Av = \lambda v$ with $v \neq 0$, then $\lambda v = Av = A^2v = \lambda^2 v$ (Theorem 7.12(a)), so $\lambda^2 = \lambda$ and $\lambda \in \{0, 1\}$.
- $E_0(A) = \mathcal{N}(A - 0I_n) = \mathcal{N}(A)$. If $v \in \mathcal{C}(A)$, then $Av = v$ (Theorem 6.16(a)); if $Av = v$, then $v \in \mathcal{C}(A)$. So $E_1(A) = \mathcal{C}(A)$.
- $\mathbb{R}^n = \mathcal{N}(A) \oplus \mathcal{C}(A)$ (Theorem 6.16(a)). So a basis $p_1, \dots, p_r$ of $\mathcal{C}(A)$ together with a basis $p_{r+1}, \dots, p_n$ of $\mathcal{N}(A)$, which has dimension $n - r$ (Theorem 4.12), is a basis of $\mathbb{R}^n$ (Theorem 3.23) consisting of eigenvectors; for $r = 0$ or $r = n$, one of the two lists is empty (Definition 3.14). By Theorem 7.19, $A = PDP^{-1}$ with $P = [p_1 \ \cdots \ p_n]$ and $D = \text{diag}(1, \dots, 1, 0, \dots, 0)$ (r ones).
- By Theorem 7.14(a) and Example 7.11(a), $p_A(t) = p_D(t) = (t-1)^r t^{n-r}$ and $\text{tr}\,A = \text{tr}\,D = r$. □

This is a second proof of $\text{rank}\,A = \text{tr}\,A$ (Theorem 6.16(c)). For instance, a projection matrix P_W onto a subspace $W \subseteq \mathbb{R}^n$ of dimension r is idempotent of rank r (Theorem 6.19(b)), so $p_{P_W}(t) = t^{n-r}(t-1)^r$; and $p_{C_n}(t) = t(t-1)^{n-1}$, as $C_n = P_{W_0}$ has rank $n - 1$ (Corollary 6.21; compare Example 7.17).

Example 7.26. The matrix $E = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & 1 & -1 \end{pmatrix}$ of Exercise 6.7 is idempotent of rank 2, with pivot columns 1 and 2. So $p_1 = (1, 1, 0, -1)'$ and $p_2 = (1, 0, -1, -1)'$ form a basis of $\mathcal{C}(E) = E_1(E)$ (Theorems 4.8(a) and 7.25). The vectors $p_3 = (-1, 2, 1, 0)'$ and $p_4 = (0, -1, 0, 1)'$ satisfy $Ep_3 = Ep_4 = 0$ (multiply out), and they are linearly independent; as $\dim\mathcal{N}(E) = 4 - 2 = 2$ (Theorem 4.12), they form a basis of $\mathcal{N}(E) = E_0(E)$ (Theorem 4.5(d)). By Theorem 7.20, $p_1, \dots, p_4$ are linearly independent, so by Theorems 7.19 and 7.25,

$$E = P\,\text{diag}(1, 1, 0, 0)P^{-1}, \qquad P = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 2 & -1 \\ 0 & -1 & 1 & 0 \\ -1 & -1 & 0 & 1 \end{pmatrix}, \qquad p_E(t) = t^2(t-1)^2$$

Here $\det P = 1$, so P^{-1} has integer entries (Theorem 5.20).

Orthogonal triangularization

Lemma 7.27. Let $q_1, \dots, q_k \in \mathbb{R}^n$ be orthonormal, with $k < n$. Then there are $q_{k+1}, \dots, q_n \in \mathbb{R}^n$ such that $q_1, \dots, q_n$ is an orthonormal basis of $\mathbb{R}^n$; that is, there is $U \in \mathbb{R}^{n\times(n-k)}$ such that $[q_1 \ \cdots \ q_k \ U]$ is an orthogonal matrix.

Proof. The vectors $q_1, \dots, q_k$ are linearly independent (Proposition 6.8(b)); extend them to a basis $q_1, \dots, q_k, a_{k+1}, \dots, a_n$ of $\mathbb{R}^n$ (Theorem 4.5(b)), and apply the Gram–Schmidt process (Theorem 6.11) to this list. It leaves $q_1, \dots, q_k$ unchanged: if $b_i = q_i$ for $i < j \leqslant k$, then $t_{ij} = q_j' q_i / \|q_i\|^2 = 0$, so $b_j = q_j$, of length 1. Hence the resulting orthonormal basis of $\mathbb{R}^n$ begins with $q_1, \dots, q_k$. Let U have the other $n-k$ vectors as its columns; then $Q = [q_1 \ \cdots \ q_k \ U]$ satisfies $Q'Q = I_n$ (Definition 6.9). □

Theorem 7.28 (Schur triangularization)**.** Let $A \in \mathbb{R}^{n\times n}$ with $p_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$, $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. Then $A = QTQ'$, where $Q \in \mathbb{R}^{n\times n}$ is orthogonal and $T \in \mathbb{R}^{n\times n}$ is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$, in this order. As the factors of p_A may be listed in any order, the eigenvalues listed with multiplicity can be placed on the diagonal of T in any prescribed order.

Proof. Induction on n; for $n = 1$, $A = (\lambda_1) = 1 \cdot (\lambda_1) \cdot 1$. Let $n \geqslant 2$.

- λ_1 is an eigenvalue (Theorem 7.7(b)); let q_1 be an eigenvector for λ_1 of length 1 (divide any eigenvector by its length). By Lemma 7.27, $Q_1 = [q_1 \ U]$ is orthogonal for some $U \in \mathbb{R}^{n\times(n-1)}$; so $q_1'q_1 = 1$ and $U'q_1 = 0$.
- By the block product (Theorem 5.3) and $Aq_1 = \lambda_1 q_1$,

$$Q_1'AQ_1 = \begin{pmatrix} q_1'Aq_1 & q_1'AU \\ U'Aq_1 & U'AU \end{pmatrix} = \begin{pmatrix} \lambda_1 & b' \\ 0 & A_1 \end{pmatrix}, \qquad b' = q_1'AU, \quad A_1 = U'AU \in \mathbb{R}^{(n-1)\times(n-1)}$$

- $Q_1' = Q_1^{-1}$ (Proposition 6.10(b)), so by Theorems 7.14(a) and 5.25, $p_A(t) = (t-\lambda_1)p_{A_1}(t)$. Hence $p_{A_1}(t) = (t-\lambda_2)\cdots(t-\lambda_n)$ for $t \neq \lambda_1$, and by (A1) for every t.
- By the induction hypothesis, $A_1 = Q_2T_2Q_2'$ with Q_2 orthogonal and T_2 upper triangular with diagonal entries $\lambda_2, \dots, \lambda_n$. Let $Q = Q_1 \operatorname{diag}(1, Q_2)$. By Proposition 5.6(b), $Q'Q = \operatorname{diag}(1, Q_2')\, Q_1'Q_1 \operatorname{diag}(1, Q_2) = \operatorname{diag}(1, Q_2'Q_2) = I_n$, and

$$Q'AQ = \begin{pmatrix} 1 & 0 \\ 0 & Q_2' \end{pmatrix} \begin{pmatrix} \lambda_1 & b' \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & b'Q_2 \\ 0 & Q_2'A_1Q_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & b'Q_2 \\ 0 & T_2 \end{pmatrix},$$

  an upper triangular matrix T with diagonal entries $\lambda_1, \dots, \lambda_n$. As $QQ' = I_n$, $A = Q(Q'AQ)Q' = QTQ'$. □

Conversely, if $A = QTQ'$ with Q orthogonal and T upper triangular, then $p_A = p_T$ splits (Theorem 7.14(a), Example 7.11(a)). So such a factorization exists exactly when p_A splits; the matrix K of Example 7.11(d) has none.

Example 7.29. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad Q = \tfrac{1}{2}H$$

Every row of A adds up to 4, so $q_1 = \frac{1}{2}\mathbf{1}_4$, the first column of Q, is an eigenvector of length 1 for the eigenvalue 4. The columns of H are orthogonal and have squared length 4 (for instance, columns 2

and 3 give $1-1-1+1=0$), so $H'H = 4I_4$ and Q is orthogonal: $Q = [q_1\ U]$ is a completion of q_1 as in Lemma 7.27. As $H' = H$,

$$AH = \begin{pmatrix} 4 & 2 & 0 & -2 \\ 4 & 2 & -2 & 0 \\ 4 & -2 & 0 & 2 \\ 4 & -2 & -2 & 0 \end{pmatrix}, \qquad Q'AQ = \tfrac{1}{4}H(AH) = \begin{pmatrix} 4 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = T$$

In the notation of the proof of Theorem 7.28, $\lambda_1 = 4$, $b' = (0,-1,0)$, and $A_1 = U'AU$ is already upper triangular, so the induction stops after one step. Hence $A = QTQ'$, $p_A(t) = p_T(t) = (t-4)(t-2)(t-1)(t+1)$ (Theorem 7.14(a), Example 7.11(a)), $\operatorname{tr} T = 6 = \operatorname{tr} A$ and $\det T = -8 = \det A$. The matrix A has four distinct eigenvalues, so it is diagonalizable (Theorem 7.22(b)); but no orthogonal Q makes $Q'AQ$ diagonal: QDQ' is symmetric for every diagonal D, and A is not.

Corollary 7.30. Let $A \in \mathbb{R}^{n\times n}$ with $p_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$, $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$.

(a) $\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n$ and $\det A = \lambda_1\lambda_2\cdots\lambda_n$.

(b) For every polynomial q, $p_{q(A)}(t) = \big(t - q(\lambda_1)\big)\cdots\big(t - q(\lambda_n)\big)$: $p_{q(A)}$ splits with roots $q(\lambda_1), \ldots, q(\lambda_n)$. In particular $\operatorname{tr} A^k = \lambda_1^k + \cdots + \lambda_n^k$ for $k \geqslant 0$.

(c) If A is invertible, then $p_{A^{-1}}(t) = (t-\lambda_1^{-1})\cdots(t-\lambda_n^{-1})$.

Proof. Let $A = QTQ'$ as in Theorem 7.28; $Q' = Q^{-1}$, and T has the diagonal entries $\lambda_1, \ldots, \lambda_n$.

(a) $\operatorname{tr} A = \operatorname{tr}(TQ'Q) = \operatorname{tr} T$ (Theorem 1.11(b)) and $\det A = \det T$ (Theorem 7.14(a)); apply Theorem 5.15(a).

(b) $A^k = QT^kQ'$ for $k \geqslant 0$, by induction: $A^{k+1} = QT^kQ'QTQ' = QT^{k+1}Q'$. So $q(A) = Q\,q(T)\,Q'$. A product of upper triangular matrices is upper triangular, with the products of the diagonal entries on its diagonal (Chapter 1); so T^k and $q(T)$ are upper triangular with diagonal entries λ_i^k and $q(\lambda_i)$. By Theorem 7.14(a) and Example 7.11(a), $p_{q(A)}(t) = p_{q(T)}(t) = \prod_i \big(t - q(\lambda_i)\big)$. For $q(t) = t^k$, (a) applied to A^k gives $\operatorname{tr} A^k = \sum_i \lambda_i^k$.

(c) $\det A = \prod_i \lambda_i \neq 0$ (Theorem 5.19(a)), so every $\lambda_i \neq 0$, and $A^{-1} = QT^{-1}Q'$ (Theorem 2.15(b)). By induction on n, T^{-1} is upper triangular with diagonal entries $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$: for $n \geqslant 2$ write $T = \left(\begin{smallmatrix} T_1 & c \\ 0 & \lambda_n \end{smallmatrix}\right)$ with T_1 upper triangular with diagonal entries $\lambda_1, \ldots, \lambda_{n-1}$, invertible as $\det T_1 \neq 0$; by Proposition 5.6(c), $T^{-1} = \left(\begin{smallmatrix} T_1^{-1} & -T_1^{-1}c\lambda_n^{-1} \\ 0 & \lambda_n^{-1} \end{smallmatrix}\right)$. Conclude as in (b). □

For the matrix A of Example 7.10, with eigenvalues $-3, -1, 1, 3$,

$$A^2 = \begin{pmatrix} 5 & -4 & 0 & 8 \\ -4 & 5 & 8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 4 & 5 \end{pmatrix}, \qquad \operatorname{tr} A^2 = 20 = (-3)^2 + (-1)^2 + 1^2 + 3^2,$$

and A^{-1} has the eigenvalues $-\frac{1}{3}, -1, 1, \frac{1}{3}$ (part (c)). The hypothesis that p_A splits is needed: for K of Example 7.11(d), $K^2 = -I_2$ and $\operatorname{tr} K^2 = -2$, while a sum $\lambda_1^2 + \lambda_2^2$ of squares of real numbers is $\geqslant 0$.

Exercises

Exercise 7.1. Find the eigenvalues of

$$B = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ -1 & 0 & 1 & 3 \end{pmatrix}$$

and a basis of each eigenspace. Check that the coefficient of t^3 in $p_B(t)$ is $-\operatorname{tr} B$ and that the constant term is $\det B$.

Solution. Row 2 of $tI_4 - B$ is $(0,\ t-1,\ 0,\ 0)$. Expanding along it, and then the remaining 3×3 determinant (rows and columns $1, 3, 4$) along its second row $(0,\ t-3,\ 0)$ (Theorem 5.13),

$$p_B(t) = (t-1)\det \begin{pmatrix} t-3 & 0 & 1 \\ 0 & t-3 & 0 \\ 1 & -1 & t-3 \end{pmatrix} = (t-1)(t-3)\det \begin{pmatrix} t-3 & 1 \\ 1 & t-3 \end{pmatrix} = (t-1)(t-3)(t-2)(t-4)$$

Multiplying out, $p_B(t) = t^4 - 10t^3 + 35t^2 - 50t + 24$. Here $\operatorname{tr} B = 3+1+3+3 = 10$, and expanding $\det B$ in the same way, $\det B = 1 \cdot 3 \cdot (9-1) = 24$; so the coefficient of t^3 is $-\operatorname{tr} B$ and the constant term is $(-1)^4 \det B$, as Theorem 7.7(a) says. The eigenvalues are $1, 2, 3, 4$ (Theorem 7.7(b)).

- $B - I_4 = \left(\begin{smallmatrix} 2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 0 & 1 & 2 \end{smallmatrix}\right)$: row 3 gives $x_2 = -2x_3$, row 4 gives $x_1 = x_3 + 2x_4$, and then row 1 gives $3x_4 = 0$. So $E_1(B) = \operatorname{span}\{(1,-2,1,0)'\}$.
- $B - 2I_4 = \left(\begin{smallmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \end{smallmatrix}\right)$: rows 2 and 3 give $x_2 = x_3 = 0$, and row 1 gives $x_1 = x_4$. So $E_2(B) = \operatorname{span}\{(1,0,0,1)'\}$.
- $B - 3I_4 = \left(\begin{smallmatrix} 0 & 1 & 0 & -1 \\ 0 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{smallmatrix}\right)$: $x_2 = 0$, then $x_4 = 0$, and $x_1 = x_3$. So $E_3(B) = \operatorname{span}\{(1,0,1,0)'\}$.
- $B - 4I_4 = \left(\begin{smallmatrix} -1 & 1 & 0 & -1 \\ 0 & -3 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & -1 \end{smallmatrix}\right)$: $x_2 = 0$, then $x_3 = 0$, and $x_1 = -x_4$. So $E_4(B) = \operatorname{span}\{(-1,0,0,1)'\}$.

Check: $B(1,-2,1,0)' = (1,-2,1,0)'$ and $B(-1,0,0,1)' = (-4,0,0,4)'$. Also $1+2+3+4 = 10 = \operatorname{tr} B$ and $1 \cdot 2 \cdot 3 \cdot 4 = 24 = \det B$ (Corollary 7.30(a)).

Exercise 7.2. Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

(a) Compute BA and AB, find p_{BA}, and deduce p_{AB}.

(b) Find eigenvectors of AB for its nonzero eigenvalues from those of BA, and a basis of $E_0(AB)$. Diagonalize AB.

(c) Show that $\operatorname{tr}(AB)^k = 2^k + (-1)^k$ for $k \geqslant 1$.

Solution. (a)

$$BA = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \qquad AB = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix},$$

and $p_{BA}(t) = (t-1)t - 2 = t^2 - t - 2 = (t-2)(t+1)$. By Proposition 7.15(a) with $m = 4$, $n = 2$, $t^2 p_{AB}(t) = t^4 p_{BA}(t)$; so $p_{AB}(t) = t^2(t-2)(t+1)$ for $t \neq 0$, and by (A1) for every t.

(b)

- $BA(1,1)' = (2,2)'$ and $BA(1,-2)' = (-1,2)'$. By Proposition 7.15(b), applied to B and A, $v \mapsto Av$ maps $E_\lambda(BA)$ into $E_\lambda(AB)$ for $\lambda \neq 0$, with nonzero images:

$$v_1 = A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\2\\1\\0\end{pmatrix}, \qquad v_2 = A\begin{pmatrix}1\\-2\end{pmatrix} = \begin{pmatrix}1\\-1\\-2\\3\end{pmatrix},$$

  and indeed $ABv_1 = (2,4,2,0)' = 2v_1$ and $ABv_2 = (-1,1,2,-3)' = -v_2$.

- $Bx = 0$ reads $x_1 + x_3 = 0$, $x_2 + x_4 = 0$, with the special solutions $n_1 = (-1,0,1,0)'$ and $n_2 = (0,-1,0,1)'$. As $ABx = A(Bx)$, $\mathcal{N}(B) \subseteq \mathcal{N}(AB) = E_0(AB)$, so $g_{AB}(0) \geqslant 2$; and $g_{AB}(0) \leqslant a_{AB}(0) = 2$ (Theorem 7.21). So $\mathcal{N}(B) = E_0(AB)$ (Theorem 4.5(a)), with basis n_1, n_2.

- The eigenvalues 2 and -1 are simple, so $g_{AB} = a_{AB}$ for every eigenvalue (Theorem 7.21), p_{AB} splits, and by Theorem 7.22(a) and the recipe after it, $AB = QDQ^{-1}$ with $Q = [v_1\ v_2\ n_1\ n_2]$ and $D = \operatorname{diag}(2,-1,0,0)$.

(c) $p_{AB}(t) = (t-2)(t+1)(t-0)(t-0)$ splits, so by Corollary 7.30(b), $\operatorname{tr}(AB)^k = 2^k + (-1)^k + 0^k + 0^k = 2^k + (-1)^k$ for $k \geqslant 1$. For $k = 1$, $\operatorname{tr} AB = 1 + 1 + 0 - 1 = 1$.

Exercise 7.3. Let $A = \begin{pmatrix} 2 & -2 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 0 & 0 & 3 & 0 \\ -1 & -2 & 1 & 2 \end{pmatrix}$. Show that $\operatorname{rank}(A - 3I_4) = 1$, diagonalize A, find a formula for A^k ($k \geqslant 0$), and compute A^3.

Solution.

- $A - 3I_4 = uv'$ with $u = (1,1,0,1)'$ and $v' = (-1,-2,1,-1)$: rows 1, 2 and 4 equal v', and row 3 is 0. So $\operatorname{rank}(A - 3I_4) = 1$, $g_A(3) = 3$ (Theorem 7.3), and $a_A(3) \geqslant 3$ (Theorem 7.21).

- Hence $p_A(t) = (t-3)^3 q(t)$, where q has degree $4 - 3 = 1$ and leading coefficient 1 (Theorem 7.7(a) and Remark 7.4): $q(t) = t - \mu$. The coefficient of t^3 in $(t-3)^3(t-\mu)$ is $-(9+\mu)$, and it equals $-\operatorname{tr} A = -8$; so $\mu = -1$ and $p_A(t) = (t-3)^3(t+1)$.

- $E_3(A) = \mathcal{N}(uv') = \mathcal{N}(v')$, as $uv'x = (v'x)u$ and $u \neq 0$. The equation $-x_1 - 2x_2 + x_3 - x_4 = 0$, with x_2, x_3, x_4 free, has the special solutions $p_1 = (-2,1,0,0)'$, $p_2 = (1,0,1,0)'$ and $p_3 = (-1,0,0,1)'$. Next $Au = 3u + u(v'u) = 3u - 4u = -u$, so $p_4 = u$ spans $E_{-1}(A)$ (Theorem 7.21: $g_A(-1) \leqslant a_A(-1) = 1$).

- By Theorems 7.20 and 7.19, $A = PDP^{-1}$ with $P = [p_1\ p_2\ p_3\ p_4]$ and $D = \operatorname{diag}(3,3,3,-1)$. The row $w' = -\frac{1}{4}v' = \frac{1}{4}(1,2,-1,1)$ satisfies $w'p_j = 0$ for $j \leqslant 3$ (as $v'p_j = 0$) and $w'p_4 = -\frac{1}{4}v'u = 1$; so $w'P = e_4'$, and $w' = e_4'P^{-1} = \operatorname{row}_4(P^{-1})$.

By Proposition 7.24,

$$A^k = P\operatorname{diag}\big(3^k, 3^k, 3^k, (-1)^k\big)P^{-1} = 3^k M_1 + (-1)^k M_2,$$

with $M_1 = P\operatorname{diag}(1,1,1,0)P^{-1}$ and $M_2 = P\operatorname{diag}(0,0,0,1)P^{-1}$. By the outer product expansion (Theorem 1.4(d)), $M_2 = p_4 \operatorname{row}_4(P^{-1}) = -\frac{1}{4}uv'$, and $M_1 = I_4 - M_2$, as $M_1 + M_2 = PP^{-1}$. So

$$A^k = 3^k I_4 + \tfrac{1}{4}\big(3^k - (-1)^k\big)uv' \qquad (k \geqslant 0)$$

For $k = 1$ this gives back $A = 3I_4 + uv'$, and

$$A^3 = 27I_4 + 7uv' = \begin{pmatrix} 20 & -14 & 7 & -7 \\ -7 & 13 & 7 & -7 \\ 0 & 0 & 27 & 0 \\ -7 & -14 & 7 & 20 \end{pmatrix}$$

Exercise 7.4. Which of the following matrices are diagonalizable?

$$A_1 = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 0 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ -1 & 0 & 1 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Solution. (A_1) A_1 differs from the matrix B of Exercise 7.1 only in the $(2,2)$ entry. The same two expansions give $p_{A_1}(t) = (t-3)\cdot(t-3)(t-2)(t-4) = (t-3)^2(t-2)(t-4)$, so $a_{A_1}(3) = 2$. But $A_1 - 3I_4 = \left(\begin{smallmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{smallmatrix}\right)$ has rank 3: in a combination $a(0,1,0,-1) + b(0,1,0,0) + c(-1,0,1,0) = 0$ of its nonzero rows, the first entry gives $c = 0$, the fourth $a = 0$, and then $b = 0$. So $g_{A_1}(3) = 1 < 2$, and A_1 is not diagonalizable (Theorem 7.22(a)). Changing one entry of the diagonalizable B has destroyed diagonalizability.

(A_2) $p_{A_2}(t) = t^2 - 0\cdot t + (-1+2) = t^2 + 1$ has no real root, so A_2 has no eigenvalue and p_{A_2} does not split: A_2 is not diagonalizable (Theorem 7.22(a)).

(A_3) A_3 is upper triangular, so $p_{A_3}(t) = (t-1)^2(t-2)^2$ (Example 7.11(a)). In $A_3 - I_4 = \left(\begin{smallmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, rows 1 and 2 are linearly independent, row 3 is row 1 minus row 2, and row 4 is minus row 3; so $g_{A_3}(1) = 4 - 2 = 2$. In $A_3 - 2I_4 = \left(\begin{smallmatrix} -1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$, rows 1 and 2 are linearly independent and row 3 is minus row 2; so $g_{A_3}(2) = 2$. Both geometric multiplicities equal the algebraic ones, and A_3 is diagonalizable (Theorem 7.22(a)); for instance $(1,0,0,0)'$, $(0,-1,1,0)'$ span $E_1(A_3)$ and $(1,1,0,0)'$, $(-1,0,-1,1)'$ span $E_2(A_3)$.

Exercise 7.5. Let

$$A = \begin{pmatrix} 0 & 2 & -1 & 0 \\ 1 & 1 & -1 & 0 \\ -2 & 2 & 1 & 0 \\ 1 & -1 & 0 & -2 \end{pmatrix}, \qquad p_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad p_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad p_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad p_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

(a) Show that $p_1, \dots, p_4$ are eigenvectors of A, and find p_A, $\operatorname{tr} A$ and $\det A$.

(b) Diagonalize A and compute A^{-1}.

(c) For $x_0 = (3,2,2,1)'$, find $A^k x_0$ ($k \geqslant 0$), and check $A^3 x_0$ by computing Ax_0, A^2x_0, A^3x_0 in turn.

Solution. (a) Multiplying out, $Ap_1 = (2,2,0,0)' = 2p_1$, $Ap_2 = (1,1,1,0)' = p_2$, $Ap_3 = (-1,0,-1,-1)' = -p_3$ and $Ap_4 = \operatorname{col}_4(A) = (0,0,0,-2)' = -2p_4$. The eigenvalues $2, 1, -1, -2$ are distinct, so $A = PDP^{-1}$ with $P = [p_1\ p_2\ p_3\ p_4]$ and $D = \operatorname{diag}(2,1,-1,-2)$ (Theorems 7.20 and 7.19). By Theorem 7.14(a) and Example 7.11(a), $p_A(t) = (t-2)(t-1)(t+1)(t+2)$, $\operatorname{tr} A = 2+1-1-2 = 0$ and $\det A = 2\cdot 1\cdot(-1)\cdot(-2) = 4$. Directly, $\operatorname{tr} A = 0+1+1-2 = 0$.

(b)

$$P^{-1} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}$$

(check $P^{-1}P = I_4$). By Theorem 2.15(b), $A^{-1} = PD^{-1}P^{-1}$ with $D^{-1} = \operatorname{diag}(\frac{1}{2}, 1, -1, -\frac{1}{2})$:

$$A^{-1} = \frac{1}{2}\begin{pmatrix} -3 & 4 & 1 & 0 \\ -1 & 2 & 1 & 0 \\ -4 & 4 & 2 & 0 \\ -1 & 1 & 0 & -1 \end{pmatrix}$$

Check: $\mathrm{row}_1(A)\,\mathrm{col}_1(A^{-1}) = \frac{1}{2}(-2+4) = 1$ and $\mathrm{row}_1(A)\,\mathrm{col}_2(A^{-1}) = \frac{1}{2}(4-4) = 0$.
(c) $c = P^{-1}x_0 = (1,1,1,0)'$, that is, $x_0 = p_1 + p_2 + p_3$. By Proposition 7.24,

$$A^k x_0 = 2^k p_1 + p_2 + (-1)^k p_3 = \left(2^k + 1 + (-1)^k,\ 2^k + 1,\ 1 + (-1)^k,\ (-1)^k\right)'$$

For $k = 3$ this is $(8,9,0,-1)'$. Directly, $Ax_0 = (2,3,0,-1)'$, $A^2x_0 = A(2,3,0,-1)' = (6,5,2,1)'$ and $A^3x_0 = A(6,5,2,1)' = (8,9,0,-1)'$.

Exercise 7.6. Let B be the matrix of Example 7.23(c), and

$$F = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad G = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 2 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) Show that $p_B = p_F = p_G$.

(b) Diagonalize F, and find an invertible S with $F = S^{-1}BS$.

(c) Show that G is not similar to B.

Solution. (a) $p_B(t) = (t-3)^2(t-1)^2$ (Example 7.23(c)). F and G are upper triangular with diagonal entries $3,3,1,1$, so $p_F(t) = p_G(t) = (t-3)^2(t-1)^2$ (Example 7.11(a)).
(b)

- $F - 3I_4 = \left(\begin{smallmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{smallmatrix}\right)$: $(F-3I_4)x = 0$ gives $x_4 = 0$ and $x_3 = 0$, with x_1, x_2 free; so e_1, e_2 is a basis of $E_3(F)$.
- $F - I_4 = \left(\begin{smallmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$: $x_1 = 0$ and $x_2 = -x_3 + x_4$, with the special solutions $(0,-1,1,0)'$ and $(0,1,0,1)'$, a basis of $E_1(F)$.
- So $g_F(3) = a_F(3) = 2$ and $g_F(1) = a_F(1) = 2$, and $F = P_F D P_F^{-1}$ with $D = \mathrm{diag}(3,3,1,1)$ (Theorem 7.22(a) and the recipe after it), where

$$P_F = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad P_F^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  (check $P_F^{-1}P_F = I_4$).
- With P of Example 7.23(c), $B = PDP^{-1}$ with the same D, so $D = P^{-1}BP$ and $F = P_F P^{-1} B P P_F^{-1} = S^{-1}BS$ for $S = PP_F^{-1}$ (Theorem 2.15(b)):

$$S = PP_F^{-1} = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad BS = SF = \begin{pmatrix} -3 & 0 & 0 & -1 \\ 0 & -3 & -2 & 2 \\ 3 & 0 & 1 & 0 \\ 0 & 3 & 3 & -2 \end{pmatrix}$$

  The equality $BS = SF$ confirms $F = S^{-1}BS$.

(c) In $G - I_4 = \left(\begin{smallmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$, rows 1, 2, 3 are linearly independent (their leading entries lie in columns 1, 2, 4), so $\mathrm{rank}(G - I_4) = 3$ and $g_G(1) = 1$ (Theorem 7.3). As $g_B(1) = 2$, Theorem 7.14(b) shows that G is not similar to B; by (a), the characteristic polynomial does not detect this.

Exercise 7.7. Let B be the matrix of Example 7.23(c).

(a) Without computing these matrices, find the eigenvalues and eigenvectors of B^3, B^{-1}, $B + cI_4$ ($c \in \mathbb{R}$) and $q(B) = B^2 - 4B + 3I_4$, and the numbers $\operatorname{tr} B^3$, $\det B^3$, $\operatorname{tr} B^{-1}$, $\det B^{-1}$.

(b) Show that $q(B) = 0$. Deduce $B^{-1} = \frac{1}{3}(4I_4 - B)$ and $B^3 = 13B - 12I_4$, and check the values of $\operatorname{tr} B^{-1}$ and $\operatorname{tr} B^3$.

Solution. (a) B has the eigenvalue 3, with eigenvectors p_1, p_2, and the eigenvalue 1, with eigenvectors p_3, p_4, and $p_B(t) = (t-3)^2(t-1)^2$ splits. By Theorem 7.12, p_1, p_2 and p_3, p_4 are eigenvectors of B^3 for 27 and 1, of B^{-1} for $\frac{1}{3}$ and 1, of $B + cI_4$ for $3 + c$ and $1 + c$, and of $q(B)$ for $q(3) = 9 - 12 + 3 = 0$ and $q(1) = 0$. By Corollary 7.30(b),(c), these are the eigenvalues listed with multiplicity (each twice), and by (a) of that corollary $\operatorname{tr} B^3 = 56$, $\det B^3 = 729$, $\operatorname{tr} B^{-1} = \frac{8}{3}$, $\det B^{-1} = \frac{1}{9}$. For B^3, B^{-1} and $B + cI_4$, each of the two eigenvalues has algebraic multiplicity 2; as $p_1, \dots, p_4$ are linearly independent, Theorem 7.21 shows that its eigenspace is spanned by p_1, p_2, respectively p_3, p_4. For $q(B)$, the eigenspace for 0 contains $p_1, \dots, p_4$, so it is $\mathbb{R}^4$.

(b) $q(B)$ maps the basis $p_1, \dots, p_4$ of $\mathbb{R}^4$ to 0, so $q(B)x = 0$ for every $x \in \mathbb{R}^4$ and $q(B) = 0$ (multiplying out $(B - I_4)(B - 3I_4) = q(B)$ gives 0 as well). Hence $B(4I_4 - B) = 4B - B^2 = 3I_4$, and $B^{-1} = \frac{1}{3}(4I_4 - B)$ (Theorem 2.18):

$$B^{-1} = \frac{1}{3}\begin{pmatrix} 3 & -2 & 2 & -2 \\ 2 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ -2 & 2 & -2 & 3 \end{pmatrix}$$

Also $B^2 = 4B - 3I_4$, so $B^3 = 4B^2 - 3B = 4(4B - 3I_4) - 3B = 13B - 12I_4$. With $\operatorname{tr} B = 8$: $\operatorname{tr} B^{-1} = \frac{1}{3}(16 - 8) = \frac{8}{3}$ and $\operatorname{tr} B^3 = 104 - 48 = 56$.

Exercise 7.8. Let $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 2 \\ -1 & 1 & 2 & 1 \end{pmatrix}$, and let $Q = \frac{1}{2}H$ be the orthogonal matrix of Example 7.29.

(a) Show that $\frac{1}{2}\mathbf{1}_4$ is an eigenvector of A of length 1, and that $T = Q'AQ$ is upper triangular. Read off the eigenvalues of A.

(b) Use (a) to find $\operatorname{tr}(A^2 + I_4)$ and $\det(A^2 + I_4)$, and check the trace directly.

(c) Is A diagonalizable? Is there an orthogonal U with $U'AU$ diagonal?

Solution. (a) Every row of A adds up to 3, so $A\mathbf{1}_4 = 3\mathbf{1}_4$, and $\|\frac{1}{2}\mathbf{1}_4\| = 1$. As $H' = H$,

$$AH = \begin{pmatrix} 3 & 1 & 1 & -1 \\ 3 & 1 & -1 & 1 \\ 3 & -3 & 1 & 3 \\ 3 & -3 & -1 & -3 \end{pmatrix}, \qquad T = Q'AQ = \tfrac{1}{4}H(AH) = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

So $A = QTQ'$, and by Theorem 7.14(a) and Example 7.11(a) the eigenvalues of A are $3, 2, 1, -2$.

(b) As $Q'Q = QQ' = I_4$, $A^2 + I_4 = Q(T^2 + I_4)Q'$, with $T^2 + I_4 = \begin{pmatrix} 10 & -5 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 5 \end{pmatrix}$. By Theorem 7.14(a), $\operatorname{tr}(A^2 + I_4) = 10 + 5 + 2 + 5 = 22$ and $\det(A^2 + I_4) = 10 \cdot 5 \cdot 2 \cdot 5 = 500$, as Corollary 7.30(b) predicts ($q(t) = t^2 + 1$ has $q(3) = 10$, $q(2) = 5$, $q(1) = 2$, $q(-2) = 5$). Directly, $A^2 + I_4 = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ -1 & 1 & 7 & 3 \\ 1 & -1 & 3 & 7 \end{pmatrix}$, with trace 22.

(c) A has four distinct eigenvalues, so it is diagonalizable (Theorem 7.22(b)). If $U'AU = D$ with U orthogonal and D diagonal, then $A = UDU'$ would be symmetric; but $a_{14} = 0 \neq -1 = a_{41}$. So no such U exists.

Problems

Problem 7.1. Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^m = 0$ for some $m \geqslant 1$ (A is **nilpotent**).

(a) Show that 0 is an eigenvalue of A, and that it is the only one.

(b) Show that $p_A(t) = t^n$. [For $s \in \mathbb{R}$, $(I_n - sA)(I_n + sA + \cdots + s^{m-1}A^{m-1}) = I_n$.]

(c) Show that $\operatorname{tr} A = 0$, and that A is diagonalizable only if $A = 0$.

(d) Let $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix}$, $Y = \begin{pmatrix} -1 & -1 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ and

$$N = XY' = \begin{pmatrix} -1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \\ 0 & -2 & 1 & -1 \end{pmatrix}$$

Show that $N^2 = 0$, and find $a_N(0)$ and $g_N(0)$.

Solution. (a) $(\det A)^m = \det A^m = 0$ (Theorem 5.19(c)), so $\det A = 0$, A is singular, and 0 is an eigenvalue (Theorem 7.3). If $Av = \lambda v$ with $v \neq 0$, then $0 = A^m v = \lambda^m v$ (Theorem 7.12(a)), so $\lambda^m = 0$ and $\lambda = 0$.

(b)

- By Theorem 1.3, $(I_n - sA)(I_n + sA + \cdots + s^{m-1}A^{m-1}) = I_n - s^m A^m = I_n$, as in Problem 2.3. Put $f(s) = \det(I_n - sA)$ and $g(s) = \det(I_n + sA + \cdots + s^{m-1}A^{m-1})$; then $f(s)g(s) = 1$ for every s (Theorem 5.19(b)).
- f is a polynomial by Lemma 7.6, and so is g: by Definition 5.12, a determinant is a sum of products of entries and smaller determinants, so by induction on the size, a determinant whose entries are polynomials in s is a polynomial in s.
- By (A1), fg is the constant polynomial 1; so f and g are nonzero, and $\deg f + \deg g = 0$ (Remark 7.4). Hence f is constant: $f(s) = f(0) = \det I_n = 1$.
- For $t \neq 0$, $p_A(t) = \det\bigl(t(I_n - t^{-1}A)\bigr) = t^n f(t^{-1}) = t^n$ (Theorem 5.19(c)). By (A1), $p_A(t) = t^n$ for every t.

(c) By Theorem 7.7(a), $-\operatorname{tr} A$ is the coefficient of t^{n-1} in t^n, which is 0. If $A = PDP^{-1}$ with D diagonal, the diagonal entries of D are eigenvalues of A (Theorem 7.19), hence 0 by (a); so $D = 0$ and $A = 0$.

(d) $Y'X = \begin{pmatrix} -1+0+1 & 0-1+1 \\ -1+0+1 & 0+1-1 \end{pmatrix} = 0$, so $N^2 = X(Y'X)Y' = 0$. By (b), $p_N(t) = t^4$, so $a_N(0) = 4$ (Proposition 7.9). The rows $\rho_1, \ldots, \rho_4$ of N satisfy $\rho_3 = \rho_1 + \rho_2$ and $\rho_4 = \rho_1 - \rho_2$, and ρ_1, ρ_2 are linearly independent; so $\operatorname{rank} N = 2$ and $g_N(0) = 4 - 2 = 2$ (Theorem 7.3). A nonzero nilpotent matrix may thus have $g_N(0)$ larger than 1, but by (c) it is never diagonalizable.

Problem 7.2. Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^2 = I_n$.

(a) Show that $E = \frac{1}{2}(I_n + A)$ is idempotent and that $A = 2E - I_n$.

(b) Deduce that A is diagonalizable, that its eigenvalues lie in $\{1, -1\}$, that $E_1(A) = \mathcal{C}(I_n + A)$ and $E_{-1}(A) = \mathcal{C}(I_n - A)$, and that $g_A(1) + g_A(-1) = n$. Compare with Problem 4.7.

(c) Let $h \in \mathbb{R}^n$ with $h'h = 1$. Show that $G = I_n - 2hh'$ satisfies $G^2 = I_n$, and find its eigenvalues listed with multiplicity, $\operatorname{tr} G$ and $\det G$.

Solution. (a) $E^2 = \frac{1}{4}(I_n + 2A + A^2) = \frac{1}{4}(2I_n + 2A) = E$, and $2E - I_n = I_n + A - I_n = A$.

(b)

- By Theorem 7.25, $E = PDP^{-1}$ with P invertible and D diagonal, and the diagonal entries of D are eigenvalues of E (Theorem 7.19), so they lie in $\{0, 1\}$. Then $A = 2E - I_n = P(2D - I_n)P^{-1}$ with $2D - I_n$ diagonal, so A is diagonalizable. Its eigenvalues are the diagonal entries of $2D - I_n$ (Theorem 7.14(a), Example 7.11(a)), which lie in $\{1, -1\}$.
- $A - I_n = 2(E - I_n)$ and $A + I_n = 2E$, so $E_1(A) = E_1(E) = \mathcal{C}(E) = \mathcal{C}(I_n + A)$ and $E_{-1}(A) = \mathcal{N}(E)$ (Theorem 7.25). Moreover $\mathcal{N}(E) = \mathcal{C}(I_n - E)$: $E(I_n - E) = E - E^2 = 0$, and if $Ev = 0$, then $v = (I_n - E)v$. As $I_n - E = \frac{1}{2}(I_n - A)$, $E_{-1}(A) = \mathcal{C}(I_n - A)$.
- $g_A(1) + g_A(-1) = \dim \mathcal{C}(E) + \dim \mathcal{N}(E) = n$ (Theorem 4.12). In terms of ranks, $\operatorname{rank}(I_n + A) + \operatorname{rank}(I_n - A) = n$: the result of Problem 4.7.

(c) $G^2 = I_n - 4hh' + 4h(h'h)h' = I_n$. Here $I_n - G = 2hh'$, and $\mathcal{C}(hh') = \operatorname{span}\{h\}$ ($hh'x = (h'x)h$, and $hh'h = h$); so by (b), $E_{-1}(G) = \operatorname{span}\{h\}$, $g_G(-1) = 1$ and $g_G(1) = n - 1$. As G is diagonalizable, its eigenvalues listed with multiplicity are 1 ($n-1$ times) and -1 (Theorem 7.22(a)). By Corollary 7.30(a), $\operatorname{tr} G = n - 2$ (also $\operatorname{tr} G = n - 2h'h$) and $\det G = -1$.

Problem 7.3. Let $u, v \in \mathbb{R}^n$ ($n \geqslant 2$) and $A = I_n + uv'$.

(a) Show that $p_A(t) = (t-1)^{n-1}(t - 1 - v'u)$. [Theorem 5.27.]

(b) Show that $Ax = x$ whenever $v'x = 0$, and that $Au = (1 + v'u)u$.

(c) Show that A is diagonalizable if and only if $v'u \neq 0$ or $uv' = 0$.

(d) Deduce that $\det A = 1 + v'u$ and $\operatorname{tr} A = n + v'u$.

Solution. (a) For $t \neq 1$, $tI_n - A = (t-1)I_n + (-u)v'$ with $(t-1)I_n$ invertible, and by Theorem 5.27

$$p_A(t) = (t-1)^n\Big(1 - \frac{v'u}{t-1}\Big) = (t-1)^{n-1}(t - 1 - v'u)$$

By (A1), this holds for every t.

(b) $Ax = x + u(v'x) = x$ if $v'x = 0$, and $Au = u + u(v'u) = (1 + v'u)u$.

(c) If $uv' = 0$, then $A = I_n$ is diagonal. Let $uv' \neq 0$, so $u \neq 0$ and $v \neq 0$. Then $E_1(A) = \mathcal{N}(uv')$, and $uv'x = (v'x)u = 0$ if and only if $v'x = 0$; so $E_1(A) = \mathcal{N}(v')$, of dimension $n - 1$ (Theorem 4.12).

- If $v'u \neq 0$, then by (a) and Proposition 7.9, $a_A(1) = n - 1 = g_A(1)$, and the eigenvalue $1 + v'u \neq 1$ has $a_A(1 + v'u) = 1 = g_A(1 + v'u)$ (Theorem 7.21). So A is diagonalizable (Theorem 7.22(a)).
- If $v'u = 0$, then $p_A(t) = (t-1)^n$ and $a_A(1) = n > n - 1 = g_A(1)$: A is not diagonalizable.

For instance, $u = e_1$ and $v = e_2$ give the matrix N of Example 7.11(c).

(d) p_A splits by (a), so by Corollary 7.30(a), $\det A = 1^{n-1}(1+v'u) = 1+v'u$ and $\operatorname{tr} A = (n-1)+1+v'u = n + v'u$. This agrees with Theorem 5.27 and with $\operatorname{tr}(uv') = \operatorname{tr}(v'u) = v'u$ (Theorem 1.11(b)).

Problem 7.4. Let $n \geqslant 2$, $\alpha, \beta \in \mathbb{R}$, and $\Sigma = \alpha I_n + \beta J_n$, the **compound symmetric** matrix with all diagonal entries $\alpha + \beta$ and all other entries β.

(a) Show that $\Sigma = (\alpha + n\beta)\bar{J}_n + \alpha C_n$ and $\bar{J}_n C_n = C_n \bar{J}_n = 0$.

(b) Deduce that $\Sigma^k = (\alpha + n\beta)^k \bar{J}_n + \alpha^k C_n$ for $k \geqslant 0$.

(c) Show that Σ is invertible if and only if $\alpha \neq 0$ and $\alpha + n\beta \neq 0$, and that then

$$\Sigma^{-1} = \frac{1}{\alpha + n\beta}\bar{J}_n + \frac{1}{\alpha}C_n = \frac{1}{\alpha}\Big(I_n - \frac{\beta}{\alpha + n\beta}J_n\Big)$$

Compare with Problem 5.9.

(d) For $n = 4$ and $\alpha = \beta = 1$, give Σ, its eigenvalues and Σ^{-1}.

Solution. (a) $J_n = n\bar{J}_n$ and $\bar{J}_n + C_n = I_n$ (Definition 1.12), so $\Sigma = \alpha(\bar{J}_n + C_n) + n\beta\bar{J}_n = (\alpha + n\beta)\bar{J}_n + \alpha C_n$. By Proposition 1.13, $\bar{J}_n C_n = \bar{J}_n - \bar{J}_n^2 = 0$ and $C_n\bar{J}_n = \bar{J}_n - \bar{J}_n^2 = 0$.

(b) Induction on k: for $k = 0$, $\bar{J}_n + C_n = I_n$. Using $\bar{J}_n^2 = \bar{J}_n$, $C_n^2 = C_n$ and (a),

$$\Big((\alpha+n\beta)^k\bar{J}_n + \alpha^k C_n\Big)\Big((\alpha+n\beta)\bar{J}_n + \alpha C_n\Big) = (\alpha+n\beta)^{k+1}\bar{J}_n + \alpha^{k+1}C_n$$

(c) By Example 7.17, $p_\Sigma(t) = (t-\alpha-n\beta)(t-\alpha)^{n-1}$, so the eigenvalues of Σ are $\alpha+n\beta$ and α $(n-1 \geqslant 1)$. By Theorem 7.12(b), Σ is invertible if and only if both are nonzero. Then $M = (\alpha+n\beta)^{-1}\bar{J}_n + \alpha^{-1}C_n$ satisfies $\Sigma M = \bar{J}_n + C_n = I_n$, by the computation of (b), so $M = \Sigma^{-1}$ (Theorem 2.18). Moreover

$$\frac{1}{\alpha}\Big(I_n - \frac{\beta}{\alpha+n\beta}J_n\Big) = \frac{1}{\alpha}(\bar{J}_n + C_n) - \frac{n\beta}{\alpha(\alpha+n\beta)}\bar{J}_n = \frac{\alpha+n\beta-n\beta}{\alpha(\alpha+n\beta)}\bar{J}_n + \frac{1}{\alpha}C_n = M$$

For $\alpha = 1-\rho$ and $\beta = \rho$, $\alpha + n\beta = 1 + (n-1)\rho$, and this is the formula of Problem 5.9.

(d) $\Sigma = I_4 + J_4 = \left(\begin{smallmatrix} 2&1&1&1\\ 1&2&1&1\\ 1&1&2&1\\ 1&1&1&2 \end{smallmatrix}\right)$ has the eigenvalues 5 (eigenvector $\mathbf{1}_4$) and 1, 1, 1 (eigenspace W_0), and

$$\Sigma^{-1} = I_4 - \frac{1}{5}J_4 = \frac{1}{5}\begin{pmatrix} 4&-1&-1&-1\\ -1&4&-1&-1\\ -1&-1&4&-1\\ -1&-1&-1&4 \end{pmatrix}$$

Check: $\operatorname{row}_1(\Sigma)\Sigma^{-1} = \frac{1}{5}(8-3,\ -2+4-2,\ -2+4-2,\ -2+4-2) = (1,0,0,0)$.

Problem 7.5. (a) Let $T \in \mathbb{R}^{n\times n}$ be upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$, and $M_j = (T-\lambda_1 I_n)(T-\lambda_2 I_n)\cdots(T-\lambda_j I_n)$. Show by induction on j that the first j columns of M_j are zero; hence $M_n = 0$.

(b) Let $A \in \mathbb{R}^{n\times n}$ with $p_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$, $\lambda_i \in \mathbb{R}$. Show that $(A-\lambda_1 I_n)\cdots(A-\lambda_n I_n) = 0$, that is, $p_A(A) = 0$ (the **Cayley–Hamilton theorem** for a split p_A).

(c) For the matrix A of Example 7.2, show that $A^4 = 5A^2 - 4I_4$ and $A^{-1} = \frac{1}{4}(5A - A^3)$, and compute A^{-1}.

Solution. (a) Put $T_j = T - \lambda_j I_n$: it is upper triangular, and its (j,j) entry is 0. So for $i \leqslant j$, $\operatorname{col}_i(T_j)$ has nonzero entries only in rows $1, \dots, j-1$: for $i < j$ only in rows $1, \dots, i$, and for $i = j$ the diagonal entry is 0. For $j = 1$, $\operatorname{col}_1(M_1) = \operatorname{col}_1(T_1) = 0$. For $j \geqslant 2$, $M_j = M_{j-1}T_j$, and for $i \leqslant j$, $\operatorname{col}_i(M_j) = M_{j-1}\operatorname{col}_i(T_j)$ (Theorem 1.4(a)) is a combination of $M_{j-1}e_1, \dots, M_{j-1}e_{j-1}$, the first $j-1$ columns of M_{j-1}, which are 0 by the induction hypothesis. So the first j columns of M_j are 0, and $M_n = 0$.

(b) By Theorem 7.28, $A = QTQ'$ with Q orthogonal and T upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$. As $QQ' = I_n$, $A - \lambda_i I_n = Q(T-\lambda_i I_n)Q'$, and as $Q'Q = I_n$,

$$(A-\lambda_1 I_n)\cdots(A-\lambda_n I_n) = Q(T-\lambda_1 I_n)\cdots(T-\lambda_n I_n)Q' = QM_nQ' = 0$$

Multiplying out this product uses only the distributive law and $A^jA^l = A^{j+l}$, exactly as multiplying out $(t-\lambda_1)\cdots(t-\lambda_n)$; so the product is $p_A(A)$ in the sense of Theorem 7.12.

(c) The eigenvectors $v_1, \dots, v_4$ of Example 7.2 belong to distinct eigenvalues, so they are linearly independent (Theorem 7.20), and $A = PDP^{-1}$ with $P = [v_1\ \cdots\ v_4]$ and $D = \operatorname{diag}(1,-1,2,-2)$ (Theorem 7.19). By Theorem 7.14(a) and Example 7.11(a), $p_A(t) = (t-1)(t+1)(t-2)(t+2) = (t^2-1)(t^2-4) = t^4 - 5t^2 + 4$. By (b), $A^4 - 5A^2 + 4I_4 = 0$. Hence $A\cdot\frac{1}{4}(5A - A^3) = \frac{1}{4}(5A^2 - A^4) = I_4$, and $A^{-1} = \frac{1}{4}(5A - A^3)$ (Theorem 2.18). With

$$A^3 = \begin{pmatrix} 1&2&-2&-2\\ -7&-6&-2&14\\ 0&-7&-1&7\\ -7&2&-2&6 \end{pmatrix}, \qquad A^{-1} = \frac{1}{4}(5A - A^3) = \frac{1}{2}\begin{pmatrix} 2&4&-4&-4\\ 1&3&-4&-2\\ 0&1&-2&-1\\ 1&4&-4&-3 \end{pmatrix}$$

Check: $\operatorname{row}_1(A)\ \operatorname{col}_1(A^{-1}) = \frac{1}{2}(2+2+0-2) = 1$ and $\operatorname{row}_1(A)\ \operatorname{col}_2(A^{-1}) = \frac{1}{2}(4+6-2-8) = 0$.

Problem 7.6. A row vector $w' \neq 0$ with $w'A = \lambda w'$ is a **left eigenvector** of $A \in \mathbb{R}^{n\times n}$ for λ.

(a) Show that $w'A = \lambda w'$ if and only if $A'w = \lambda w$. Deduce that A and A' have the same eigenvalues, with the same algebraic and the same geometric multiplicities.

(b) Let $Av = \lambda v$ and $w'A = \mu w'$ with $\lambda \neq \mu$. Show that $w'v = 0$.

(c) Let $A = PDP^{-1}$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. Show that $\operatorname{row}_i(P^{-1})$ is a left eigenvector of A for d_i.

(d) For the matrix A of Example 7.2, find left eigenvectors for 1, -1, 2 and -2, and check (b).

Solution. (a) Transpose (Theorem 1.9): $(w'A)' = A'w$ and $(\lambda w')' = \lambda w$. By Theorem 7.7(c), $p_{A'} = p_A$, which gives the eigenvalues and the algebraic multiplicities. By Theorems 7.3 and 4.13(a), $g_{A'}(\lambda) = n - \operatorname{rank}\big((A - \lambda I_n)'\big) = n - \operatorname{rank}(A - \lambda I_n) = g_A(\lambda)$.

(b) $\mu w'v = (w'A)v = w'(Av) = \lambda w'v$, so $(\mu - \lambda)w'v = 0$ and $w'v = 0$.

(c) $A = PDP^{-1}$ gives $P^{-1}A = DP^{-1}$; row i reads $\operatorname{row}_i(P^{-1})A = d_i \operatorname{row}_i(P^{-1})$, as left multiplication by D scales the rows (Chapter 1). The rows of the invertible P^{-1} are linearly independent (Theorem 4.13(c),(d)), so none is 0.

(d) As in the solution of (c) of Problem 7.5, $A = PDP^{-1}$ with $P = [v_1\ v_2\ v_3\ v_4]$ and $D = \operatorname{diag}(1, -1, 2, -2)$. Here $\det P = 1$ and

$$P^{-1} = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 0 & -1 & 1 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

(check $P^{-1}P = I_4$). By (c), $(1, 1, -1, -1)$, $(0, -1, 1, 1)$, $(-1, 0, 0, 1)$ and $(0, 1, 0, -1)$ are left eigenvectors for 1, -1, 2 and -2; for instance $(-1, 0, 0, 1)A = -\operatorname{row}_1(A) + \operatorname{row}_4(A) = (-2, 0, 0, 2)$. The relations $\operatorname{row}_i(P^{-1})v_j = 0$ for $i \neq j$, the off-diagonal entries of $P^{-1}P = I_4$, are the statement of (b).

Problem 7.7. (a) Let $W \subseteq \mathbb{R}^n$ be a subspace of dimension r, $1 \leqslant r < n$, and $P = P_W = QQ'$ with $Q \in \mathbb{R}^{n\times r}$, $Q'Q = I_r$ (Theorem 6.19(a)). Apply Proposition 7.15 to Q and Q' to show $p_P(t) = t^{n-r}(t-1)^r$, and compare with Theorem 7.25.

(b) Show that $p_{I_n - P}(t) = t^r(t-1)^{n-r}$, and deduce $\operatorname{tr} P = r$, $\operatorname{tr}(I_n - P) = n - r$ and $\det P = \det(I_n - P) = 0$.

(c) Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$. Use Proposition 7.15(a) at $t = -1$ to show $\det(I_m + AB) = \det(I_n + BA)$ (compare Problem 5.8).

(d) For the projection matrix P_W of Example 6.23 ($n = 4$, $r = 3$), give the eigenvalues of P_W and of $I_4 - P_W$ listed with multiplicity, and check the traces.

Solution. (a) $Q \in \mathbb{R}^{n\times r}$ and $Q' \in \mathbb{R}^{r\times n}$ have the products $QQ' = P$ and $Q'Q = I_r$, and $p_{I_r}(t) = (t-1)^r$. By Proposition 7.15(a), $t^r p_P(t) = t^n(t-1)^r$; so $p_P(t) = t^{n-r}(t-1)^r$ for $t \neq 0$, and by (A1) for every t. This agrees with Theorem 7.25, as P is idempotent of rank r (Theorem 6.19(b)).

(b) $tI_n - (I_n - P) = (t-1)I_n + P = -\big((1-t)I_n - P\big)$, so by Theorem 5.19(c) and (a),

$$p_{I_n - P}(t) = (-1)^n p_P(1-t) = (-1)^n(1-t)^{n-r}(-t)^r = (-1)^{2n}(t-1)^{n-r}t^r = t^r(t-1)^{n-r}$$

Both polynomials split, and by Corollary 7.30(a), $\operatorname{tr} P = r$, $\operatorname{tr}(I_n - P) = n - r$, $\det P = 0$ (as $n - r \geqslant 1$) and $\det(I_n - P) = 0$ (as $r \geqslant 1$).

(c) By Theorem 5.19(c), $p_{AB}(-1) = \det(-I_m - AB) = (-1)^m \det(I_m + AB)$ and $p_{BA}(-1) = (-1)^n \det(I_n + BA)$. At $t = -1$, Proposition 7.15(a) reads $(-1)^n(-1)^m \det(I_m + AB) = (-1)^m(-1)^n \det(I_n + BA)$. Problem 5.8 is the same identity with $-A$ in place of A.

(d) By (a) and (b), the eigenvalues of P_W are $1, 1, 1, 0$ and those of $I_4 - P_W$ are $0, 0, 0, 1$. Indeed every diagonal entry of P_W is $\frac{3}{4}$, so $\operatorname{tr} P_W = 3$, and $\operatorname{tr}(I_4 - P_W) = 4 - 3 = 1$.

Chapter 8. The spectral theorem and quadratic forms

The spectral theorem

Remark 8.1 (Imported calculus facts: extrema)**.** The following facts are used without proof.

(C1) A continuous function on a nonempty, closed and bounded subset of $\mathbb{R}^n$ attains a maximum and a minimum on it. Polynomials in $x_1, \dots, x_n$ (sums of multiples of products of the x_i) are continuous on $\mathbb{R}^n$, and the **unit sphere** $\{x \in \mathbb{R}^n : x'x = 1\}$ is nonempty, closed and bounded; so a polynomial in $x_1, \dots, x_n$ attains a maximum and a minimum on the unit sphere.

(C2) If a differentiable function g on an open interval has a maximum at a point t_0 of the interval, then $g'(t_0) = 0$. Polynomials in one variable are differentiable, with the usual derivative.

Lemma 8.2. Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $S = \{x \in \mathbb{R}^n : x'x = 1\}$ be the unit sphere. The function $x \mapsto x'Ax$ attains a maximum on S. If $u \in S$ is a maximizer, then $Au = \lambda u$ with $\lambda = u'Au$; so u is a unit eigenvector of A for the eigenvalue λ.

Proof. • $x'Ax = \sum_i \sum_j a_{ij}x_ix_j$ is a polynomial in $x_1, \dots, x_n$, so it attains a maximum on S (Remark 8.1, (C1)).

- Let $u \in S$ be a maximizer, $\lambda = u'Au$, and fix $v \in \mathbb{R}^n$. For $t \in \mathbb{R}$ put $w = u + tv$ and $f(t) = w'Aw - \lambda\, w'w$. If $w \neq 0$, then $w/\|w\| \in S$, so $(w/\|w\|)'A(w/\|w\|) \leqslant \lambda$; multiplying by $w'w > 0$ gives $f(t) \leqslant 0$. If $w = 0$, then $f(t) = 0$. Hence $f(t) \leqslant 0 = f(0)$ for every $t \in \mathbb{R}$, as $f(0) = u'Au - \lambda u'u = \lambda - \lambda$.
- The 1×1 matrix $u'Av$ equals its transpose $v'A'u = v'Au$ (Theorem 1.9), and $u'u = 1$. Expanding,

$$f(t) = \lambda + 2t\, v'Au + t^2\, v'Av - \lambda\Big(1 + 2t\, v'u + t^2\, v'v\Big) = 2t\, v'(Au - \lambda u) + t^2\,(v'Av - \lambda\, v'v),$$

  a polynomial in t with a maximum at $t = 0$. By (C2), $0 = f'(0) = 2v'(Au - \lambda u)$.
- This holds for every $v \in \mathbb{R}^n$. For $v = Au - \lambda u$ it gives $\|Au - \lambda u\|^2 = 0$, so $Au = \lambda u$; and $u \neq 0$. □

Example 8.3. Let $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. For a unit vector $x \in \mathbb{R}^2$,

$$x'Ax = 3x_1^2 + 2x_1x_2 + 3x_2^2 = 3(x_1^2 + x_2^2) + 2x_1x_2 = 3 + 2x_1x_2$$

Since $0 \leqslant (x_1 - x_2)^2 = 1 - 2x_1x_2$ and $0 \leqslant (x_1 + x_2)^2 = 1 + 2x_1x_2$, the value lies between 2 and 4. The maximum 4 is attained exactly when $x_1 = x_2$, that is, at $u = \pm(1,1)'/\sqrt{2}$, and indeed $Au = \pm(4,4)'/\sqrt{2} = 4u$, as Lemma 8.2 says. The minimum 2 is attained exactly when $x_1 = -x_2$, at $v = \pm(-1,1)'/\sqrt{2}$, and $Av = 2v$. Here $p_A(t) = (t-3)^2 - 1 = (t-4)(t-2)$, so 4 and 2 are the eigenvalues of A (Theorem 7.7(b)), and $u \perp v$.

Theorem 8.4 (Spectral theorem)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric.

(a) There are an orthogonal $Q \in \mathbb{R}^{n\times n}$ and a diagonal $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with

$$A = Q\Lambda Q'$$

Equivalently $AQ = Q\Lambda$, that is, $Aq_i = \lambda_i q_i$ for the columns $q_1, \dots, q_n$ of Q; these columns form an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A.

(b) For every such factorization, $p_A(t) = (t - \lambda_1)\cdots(t - \lambda_n)$: p_A splits, and $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A listed with multiplicity. A is diagonalizable, and $g_A(\lambda) = a_A(\lambda)$ for every eigenvalue λ of A.

(c) Eigenvectors of A for distinct eigenvalues are orthogonal.

(d) The columns of Q can be ordered so that $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$.

Proof. (a) Induction on n. For $n = 1$, $A = (1)(a_{11})(1)$. Let $n \geqslant 2$, and let (a) hold for symmetric matrices of size $n-1$.

- By Lemma 8.2 there is a unit vector u with $Au = \lambda_1 u$, $\lambda_1 = u'Au$. By Lemma 7.27 (with $k = 1 < n$) there is $U \in \mathbb{R}^{n\times(n-1)}$ such that $Q_1 = [u\ U]$ is orthogonal; comparing blocks of $Q_1'Q_1 = I_n$ (Theorem 5.3(b),(c)), $u'u = 1$, $U'u = 0$ and $U'U = I_{n-1}$.
- $U'Au = \lambda_1 U'u = 0$ and $u'AU = (U'A'u)' = (U'Au)' = 0$. By the block product formula,

$$Q_1'AQ_1 = \begin{pmatrix} u'Au & u'AU \\ U'Au & U'AU \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & B \end{pmatrix}, \qquad B = U'AU \in \mathbb{R}^{(n-1)\times(n-1)}$$

- $B' = U'A'U = B$. By the induction hypothesis, $B = Q_2\Lambda_2 Q_2'$ with Q_2 orthogonal and Λ_2 diagonal; then $Q_2'BQ_2 = \Lambda_2$.
- Put $Q = Q_1 \operatorname{diag}(1, Q_2)$. By Proposition 5.6(b), $Q'Q = \operatorname{diag}(1, Q_2')Q_1'Q_1 \operatorname{diag}(1, Q_2) = \operatorname{diag}(1, Q_2'Q_2) = I_n$, so Q is orthogonal, and

$$Q'AQ = \begin{pmatrix} 1 & 0 \\ 0 & Q_2' \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & Q_2'BQ_2 \end{pmatrix} = \operatorname{diag}(\lambda_1, \Lambda_2) = \Lambda,$$

  a diagonal matrix.
- $QQ' = I_n$ (Proposition 6.10(b)), so $A = QQ'AQQ' = Q\Lambda Q'$. For orthogonal Q, $A = Q\Lambda Q'$ gives $AQ = Q\Lambda Q'Q = Q\Lambda$, and $AQ = Q\Lambda$ gives $A = AQQ' = Q\Lambda Q'$. Column i of $AQ = Q\Lambda$ reads $Aq_i = \lambda_i q_i$ (Theorem 1.4(a),(b)). The n orthonormal columns of Q are linearly independent (Proposition 6.8(b)), hence a basis of $\mathbb{R}^n$ (Theorem 4.5(d)).

(b) $Q^{-1} = Q'$ (Proposition 6.10(b)), so $\Lambda = Q'AQ = Q^{-1}AQ$: A and Λ are similar (Definition 7.13). By Theorems 7.14(a) and 5.15(a),

$$p_A(t) = p_\Lambda(t) = \det \operatorname{diag}(t - \lambda_1, \ldots, t - \lambda_n) = (t - \lambda_1) \cdots (t - \lambda_n)$$

So p_A splits (Definition 7.8), and its roots $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (Theorem 7.7(b)). Since $A = Q\Lambda Q^{-1}$, A is diagonalizable (Definition 7.18), and $g_A(\lambda) = a_A(\lambda)$ by Theorem 7.22(a).

(c) Let $Av = \lambda v$ and $Aw = \mu w$ with $\lambda \neq \mu$. As $A' = A$, $\lambda\, w'v = w'Av = (w'Av)' = v'Aw = \mu\, v'w = \mu\, w'v$. So $(\lambda - \mu)w'v = 0$, and $w'v = 0$.

(d) Order the indices as $i_1, \ldots, i_n$ with $\lambda_{i_1} \geqslant \cdots \geqslant \lambda_{i_n}$, and let $\tilde{Q} = [q_{i_1}\ \cdots\ q_{i_n}]$ and $\tilde{\Lambda} = \operatorname{diag}(\lambda_{i_1}, \ldots, \lambda_{i_n})$. The columns of $\tilde{Q}$ are orthonormal, so $\tilde{Q}$ is orthogonal; by (a), $A\tilde{Q} = \tilde{Q}\tilde{\Lambda}$ column by column, and hence $A = \tilde{Q}\tilde{\Lambda}\tilde{Q}'$. □

Corollary 8.5. Let $A = Q\Lambda Q' \in \mathbb{R}^{n\times n}$ with Q orthogonal, columns $q_1, \ldots, q_n$, and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, as in Theorem 8.4(a).

(a) $A = \lambda_1 q_1 q_1' + \lambda_2 q_2 q_2' + \cdots + \lambda_n q_n q_n'$ (the **spectral decomposition** of A).

(b) $\operatorname{rank} A$ is the number of indices i with $\lambda_i \neq 0$, and

$$\mathcal{C}(A) = \operatorname{span}\{q_i : \lambda_i \neq 0\}, \qquad \mathcal{N}(A) = \operatorname{span}\{q_i : \lambda_i = 0\}$$

(c) Let $\mu_1, \ldots, \mu_s$ be the distinct eigenvalues of A, and let $P_j = P_{E_{\mu_j}(A)}$ be the projection matrix onto $E_{\mu_j}(A)$ (Theorem 6.19(a)). The q_i with $\lambda_i = \mu_j$ form an orthonormal basis of $E_{\mu_j}(A)$, and

$$P_j = \sum_{i:\, \lambda_i = \mu_j} q_i q_i', \qquad A = \mu_1 P_1 + \cdots + \mu_s P_s, \qquad P_j P_l = 0 \ \ (j \neq l), \qquad P_1 + \cdots + P_s = I_n$$

Proof. (a) By the outer product expansion (Theorem 1.4(d)), $Q\Lambda Q' = \sum_i \operatorname{col}_i(Q\Lambda)\operatorname{row}_i(Q') = \sum_i \lambda_i q_i q_i'$, since $\operatorname{col}_i(Q\Lambda) = \lambda_i q_i$ and $\operatorname{row}_i(Q') = q_i'$.

(b)

- Q and Q' are invertible, so $\operatorname{rank} A = \operatorname{rank}\Lambda$ (Theorem 4.15(d)). The nonzero columns of Λ are the $\lambda_i e_i$ with $\lambda_i \neq 0$; they are independent and span $\mathcal{C}(\Lambda)$, so $\operatorname{rank}\Lambda$ is their number.
- By (a), $Ax = \sum_i \lambda_i (q_i' x) q_i \in \operatorname{span}\{q_i : \lambda_i \neq 0\}$; conversely $q_j = A(\lambda_j^{-1} q_j) \in \mathcal{C}(A)$ if $\lambda_j \neq 0$. So $\mathcal{C}(A) = \operatorname{span}\{q_i : \lambda_i \neq 0\}$ (Theorem 3.7).
- $Aq_i = \lambda_i q_i = 0$ if $\lambda_i = 0$, so $\operatorname{span}\{q_i : \lambda_i = 0\} \subseteq \mathcal{N}(A)$. Both have dimension $n - \operatorname{rank} A$: the first because these q_i are orthonormal, hence independent (Proposition 6.8(b)), the second by Theorem 4.12. So they are equal (Theorem 4.5(a)).

(c) Fix j, and let $\mathcal{I}_j = \{i : \lambda_i = \mu_j\}$.

- $A - \mu_j I_n = Q(\Lambda - \mu_j I_n)Q'$, as $QQ' = I_n$; this is a factorization of the same form, with the numbers $\lambda_i - \mu_j$. By (b), $E_{\mu_j}(A) = \mathcal{N}(A - \mu_j I_n) = \operatorname{span}\{q_i : i \in \mathcal{I}_j\}$, and these orthonormal q_i form a basis of it.
- Let Q_j have the columns q_i, $i \in \mathcal{I}_j$, an orthonormal basis of $E_{\mu_j}(A)$. By Theorem 6.19(a) and the outer product expansion (Theorem 1.4(d)), $P_j = Q_j Q_j' = \sum_{i\in\mathcal{I}_j} q_i q_i'$.
- Every i lies in exactly one $\mathcal{I}_j$, so by (a) $A = \sum_j \sum_{i\in\mathcal{I}_j} \lambda_i q_i q_i' = \sum_j \mu_j P_j$, and $\sum_j P_j = \sum_i q_i q_i' = QQ' = I_n$.
- For $j \neq l$, $P_j P_l = \sum_{i\in\mathcal{I}_j}\sum_{h\in\mathcal{I}_l} q_i (q_i' q_h) q_h' = 0$, since $q_i' q_h = 0$ for $i \neq h$. □

Example 8.6. Find an orthogonal Q with $Q'AQ$ diagonal, and the matrices P_j of Corollary 8.5(c), for

$$A = \begin{pmatrix} 11 & 2 & 6 & 6 \\ 2 & 11 & 6 & 6 \\ 6 & 6 & 11 & 2 \\ 6 & 6 & 2 & 11 \end{pmatrix}$$

Solution.

- In $tI_4 - A$, subtract row 2 from row 1 and row 4 from row 3; the new rows 1 and 3 are $(t-9)(1,-1,0,0)$ and $(t-9)(0,0,1,-1)$. Take the two factors $t-9$ out, and add column 1 to column 2 and column 3 to column 4 (Theorem 5.16(d),(a),(f)):

$$p_A(t) = (t-9)^2 \det \begin{pmatrix} 1 & -1 & 0 & 0 \\ -2 & t-11 & -6 & -6 \\ 0 & 0 & 1 & -1 \\ -6 & -6 & -2 & t-11 \end{pmatrix} = (t-9)^2 \det \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & t-13 & -6 & -12 \\ 0 & 0 & 1 & 0 \\ -6 & -12 & -2 & t-13 \end{pmatrix}$$

  Expanding along row 1 and then along the row $(0,1,0)$ (Theorem 5.13), $p_A(t) = (t-9)^2\big((t-13)^2 - 144\big) = (t-9)^2(t-25)(t-1)$. The eigenvalues are $\mu_1 = 25$, $\mu_2 = 9$ and $\mu_3 = 1$ (Theorem 7.7(b)), with $a_A(25) = a_A(1) = 1$ and $a_A(9) = 2$.
- The row sums of A are 25, so $A\mathbf{1}_4 = 25\,\mathbf{1}_4$; and $w = (1,1,-1,-1)'$ has $Aw = (11+2-12,\ 2+11-12,\ 12-11-2,\ 12-2-11)' = w$. By Theorem 8.4(b), $\dim E_{25}(A) = \dim E_1(A) = 1$, so $q_1 = \frac{1}{2}\mathbf{1}_4$ and $q_4 = \frac{1}{2}w$ are orthonormal bases of $E_{25}(A)$ and $E_1(A)$.
- $E_9(A)$: the RREF of $A - 9I_4 = \left(\begin{smallmatrix} 2 & 2 & 6 & 6 \\ 2 & 2 & 6 & 6 \\ 6 & 6 & 2 & 2 \\ 6 & 6 & 2 & 2 \end{smallmatrix}\right)$ has the rows $(1,1,0,0)$, $(0,0,1,1)$, $(0,0,0,0)$, $(0,0,0,0)$, so $E_9(A) = \{x : x_1 + x_2 = 0,\ x_3 + x_4 = 0\}$, of dimension 2. Its vectors $(1,-1,0,0)'$ and $(0,0,1,-1)'$ are orthogonal, so $q_2 = \frac{1}{\sqrt{2}}(1,-1,0,0)'$, $q_3 = \frac{1}{\sqrt{2}}(0,0,1,-1)'$ is an orthonormal basis of $E_9(A)$.

- $q_1, q_4 \perp q_2, q_3$ (Theorem 8.4(c)), and $q_1'q_4 = 0$. So $Q = [q_1\ q_2\ q_3\ q_4]$ is orthogonal and $Q'AQ = \Lambda = \operatorname{diag}(25, 9, 9, 1)$.
- By Corollary 8.5(c),

$$P_1 = q_1q_1' = \tfrac{1}{4}J_4, \qquad P_2 = q_2q_2' + q_3q_3' = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \qquad P_3 = q_4q_4' = \tfrac{1}{4}ww',$$

$P_1 + P_2 + P_3 = I_4$, and $A = 25P_1 + 9P_2 + P_3$; for instance the $(1,1)$, $(1,2)$ and $(1,3)$ entries are $\frac{25}{4} + \frac{9}{2} + \frac{1}{4} = 11$, $\frac{25}{4} - \frac{9}{2} + \frac{1}{4} = 2$ and $\frac{25}{4} - \frac{1}{4} = 6$.

The vectors $\frac{1}{2}(1,-1,1,-1)'$ and $\frac{1}{2}(1,-1,-1,1)'$ also form an orthonormal basis of $E_9(A)$; they give another Q, with entries $\pm\frac{1}{2}$, but the same

$$P_2 = \tfrac{1}{4}\Big((1,-1,1,-1)'(1,-1,1,-1) + (1,-1,-1,1)'(1,-1,-1,1)\Big)$$

Definition 8.7. Let $A = Q\Lambda Q' \in \mathbb{R}^{n\times n}$ be symmetric, as in Theorem 8.4(a), with distinct eigenvalues $\mu_1, \dots, \mu_s$ and $P_1, \dots, P_s$ as in Corollary 8.5(c). For a real function f defined at $\mu_1, \dots, \mu_s$, put

$$f(A) = Q\operatorname{diag}\Big(f(\lambda_1), \dots, f(\lambda_n)\Big)Q'$$

By the outer product expansion, as in Corollary 8.5(a),(c),

$$f(A) = \sum_{i=1}^{n} f(\lambda_i)q_iq_i' = \sum_{j=1}^{s} f(\mu_j) \sum_{i:\,\lambda_i=\mu_j} q_iq_i' = \sum_{j=1}^{s} f(\mu_j)P_j$$

The last sum is determined by A and f, so $f(A)$ does not depend on the choice of Q. Two cases agree with earlier notation. Since $Q'Q = QQ' = I_n$, induction on k gives $(Q\Lambda Q')^k = Q\Lambda^kQ'$ $(k = 0, 1, 2, \dots)$, so $f(t) = t^k$ gives $f(A) = A^k$. If every $\lambda_i \neq 0$, then $Q\Lambda^{-1}Q' \cdot Q\Lambda Q' = QQ' = I_n$, so $f(t) = 1/t$ gives $f(A) = Q\Lambda^{-1}Q' = A^{-1}$ (Theorem 2.18).

Proposition 8.8. Let $A = Q\Lambda Q'$ be symmetric as in Theorem 8.4(a), and let f, g be real functions defined at the eigenvalues of A.

(a) $f(A)$ is symmetric, $f(A)q_i = f(\lambda_i)q_i$, and $p_{f(A)}(t) = (t - f(\lambda_1)) \cdots (t - f(\lambda_n))$.

(b) $(f + g)(A) = f(A) + g(A)$, $(cf)(A) = cf(A)$ for $c \in \mathbb{R}$, and $(fg)(A) = f(A)g(A)$.

(c) For a polynomial $f = q$, $f(A)$ is the usual $q(A)$. If 0 is not an eigenvalue of A, then A is invertible, and $f(t) = 1/t$ gives $f(A) = A^{-1} = Q\Lambda^{-1}Q'$.

Proof. Write $D_f = \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n))$, so $f(A) = QD_fQ'$.

(a) $f(A)' = QD_f'Q' = f(A)$. Column i of $f(A)Q = QD_f$ gives $f(A)q_i = f(\lambda_i)q_i$. As $Q^{-1}f(A)Q = D_f$, $f(A)$ is similar to D_f, and $p_{f(A)} = p_{D_f}$ (Theorems 7.14(a) and 5.15(a)).

(b) $Q'Q = I_n$ and products of diagonal matrices are taken entrywise on the diagonal, so $f(A)g(A) = QD_fQ'QD_gQ' = QD_fD_gQ' = QD_{fg}Q' = (fg)(A)$; sums and multiples are handled in the same way.

(c) By Definition 8.7, $f(t) = t^k$ gives $f(A) = A^k$ $(k = 0, 1, 2, \dots)$; a polynomial is a sum of multiples of such powers, so (b) gives the first statement. The eigenvalues of A are $\lambda_1, \dots, \lambda_n$ (Theorem 8.4(b)); if none of them is 0, Definition 8.7 gives $A^{-1} = Q\Lambda^{-1}Q'$ for $f(t) = 1/t$. □

Example 8.9. For $A = 25P_1 + 9P_2 + P_3$ of Example 8.6, Definition 8.7 gives

$$A^k = 25^kP_1 + 9^kP_2 + P_3 \quad (k \geqslant 0), \qquad A^{-1} = \tfrac{1}{25}P_1 + \tfrac{1}{9}P_2 + P_3 = \frac{1}{225}\begin{pmatrix} 71 & 46 & -54 & -54 \\ 46 & 71 & -54 & -54 \\ -54 & -54 & 71 & 46 \\ -54 & -54 & 46 & 71 \end{pmatrix}$$

Check: row 1 of A times the first three columns of $225A^{-1}$ gives $781+92-324-324=225$, $506+142-324-324=0$ and $-594-108+426+276=0$; the other entries follow in the same way. The polynomial $f(t)=(t-25)(t-9)(t-1)$ vanishes at 25, 9 and 1, so $f(A)=\sum_j f(\mu_j)P_j=0$. As $f(t)=t^3-35t^2+259t-225$, Proposition 8.8(c) gives $f(A)=A^3-35A^2+259A-225I_4$, which is $(A-25I_4)(A-9I_4)(A-I_4)$ multiplied out; hence $(A-25I_4)(A-9I_4)(A-I_4)=0$.

Proposition 8.10. (a) Let $P\in\mathbb{R}^{n\times n}$ be symmetric and idempotent, with $\operatorname{rank}P=r\geqslant 1$. Then $P=Q_1Q_1'$ for some $Q_1\in\mathbb{R}^{n\times r}$ with $Q_1'Q_1=I_r$, and $\mathcal{C}(P)=\mathcal{C}(Q_1)$. Conversely, Q_1Q_1' is symmetric and idempotent of rank r for every $Q_1\in\mathbb{R}^{n\times r}$ with $Q_1'Q_1=I_r$.

Proof. • The eigenvalues of P lie in $\{0,1\}$ (Theorem 7.25).

- By Theorem 8.4(a),(b),(d), $P=Q\Lambda Q'$ with $\lambda_1\geqslant\cdots\geqslant\lambda_n$, the eigenvalues of P; so all λ_i lie in $\{0,1\}$. By Corollary 8.5(b), exactly r of them are nonzero, so $\lambda_1=\cdots=\lambda_r=1$ and $\lambda_{r+1}=\cdots=\lambda_n=0$.
- With $Q_1=[q_1\ \cdots\ q_r]$, Corollary 8.5(a),(b) give $P=\sum_{i\leqslant r}q_iq_i'=Q_1Q_1'$ (Theorem 1.4(d)) and $\mathcal{C}(P)=\operatorname{span}\{q_1,\ldots,q_r\}=\mathcal{C}(Q_1)$; and $Q_1'Q_1=I_r$.
- Conversely, $(Q_1Q_1')'=Q_1Q_1'$ and $(Q_1Q_1')^2=Q_1(Q_1'Q_1)Q_1'=Q_1Q_1'$, and by Theorems 6.16(c) and 1.11(b), $\operatorname{rank}(Q_1Q_1')=\operatorname{tr}(Q_1Q_1')=\operatorname{tr}(Q_1'Q_1)=\operatorname{tr}I_r=r$. □

In Example 8.6, P_2 is symmetric and idempotent of rank 2, and $P_2=Q_1Q_1'$ with $Q_1=[q_2\ q_3]$.

Quadratic forms and definiteness

Definition 8.11. Let $A\in\mathbb{R}^{n\times n}$ be symmetric. The function $x\mapsto x'Ax=\sum_{i=1}^n\sum_{j=1}^n a_{ij}x_ix_j$ $(x\in\mathbb{R}^n)$ is the **quadratic form** of A. The matrix A is

- **positive definite** (**PD**) if $x'Ax>0$ for every $x\neq 0$;
- **positive semidefinite** (**PSD**) if $x'Ax\geqslant 0$ for every x;
- **negative definite** (**negative semidefinite**) if $-A$ is PD (PSD);
- **indefinite** if $x'Ax>0$ for some x and $x'Ax<0$ for some other x.

For every $B\in\mathbb{R}^{n\times n}$, $x'Bx=(x'Bx)'=x'B'x$, so $x'Bx=x'\big(\frac12(B+B')\big)x$ with the symmetric matrix $\frac12(B+B')$; this is why quadratic forms are written with symmetric matrices.

Example 8.12. For $n=2$, $x'Ax=a_{11}x_1^2+2a_{12}x_1x_2+a_{22}x_2^2$: the coefficient of x_1x_2 is twice the off-diagonal entry.

(a) $2x_1^2+2x_1x_2+2x_2^2=x'A_1x$ with $A_1=\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$. Completing the square, it equals $2(x_1+\frac12x_2)^2+\frac32x_2^2$, which is positive unless $x_2=0$ and $x_1+\frac12x_2=0$, that is, unless $x=0$. So A_1 is PD.

(b) $x_1^2+4x_1x_2+x_2^2=x'A_2x$ with $A_2=\left(\begin{smallmatrix}1&2\\2&1\end{smallmatrix}\right)$ takes the value 6 at $(1,1)'$ and -2 at $(1,-1)'$: A_2 is indefinite. The nonsymmetric $B=\left(\begin{smallmatrix}1&3\\1&1\end{smallmatrix}\right)$ has the same quadratic form, since $\frac12(B+B')=A_2$.

Theorem 8.13. Let $A\in\mathbb{R}^{n\times n}$ be symmetric, $A=Q\Lambda Q'$ with $\lambda_1\geqslant\cdots\geqslant\lambda_n$ (Theorem 8.4(a),(d)).

(a) A is PD if and only if $\lambda_n>0$, that is, all eigenvalues of A are positive.

(b) A is PSD if and only if $\lambda_n\geqslant 0$, that is, all eigenvalues of A are $\geqslant 0$.

(c) $\lambda_n\,x'x\leqslant x'Ax\leqslant\lambda_1\,x'x$ for every $x\in\mathbb{R}^n$, with equality on the right for $x=q_1$ and on the left for $x=q_n$.

(d) A is negative definite if and only if $\lambda_1<0$, negative semidefinite if and only if $\lambda_1\leqslant 0$, and indefinite if and only if $\lambda_1>0>\lambda_n$.

Proof. (c) For $x \in \mathbb{R}^n$ put $y = Q'x$. Then $x = QQ'x = Qy$, $y'y = x'QQ'x = x'x$, and, as $Q'AQ = \Lambda$,

$$x'Ax = y'Q'AQy = y'\Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$$

Since $\lambda_n \leqslant \lambda_i \leqslant \lambda_1$ and $y_i^2 \geqslant 0$, this lies between $\lambda_n \sum_i y_i^2 = \lambda_n x'x$ and $\lambda_1 \sum_i y_i^2 = \lambda_1 x'x$. Moreover $q_1'Aq_1 = \lambda_1 q_1'q_1 = \lambda_1$ and $q_n'Aq_n = \lambda_n$.

(a) If $\lambda_n > 0$ and $x \neq 0$, then $x'Ax \geqslant \lambda_n x'x > 0$ by (c). If A is PD, then $\lambda_n = q_n'Aq_n > 0$. The eigenvalues of A are $\lambda_1, \ldots, \lambda_n$ (Theorem 8.4(b)).

(b) The same argument with $\geqslant$ in place of $>$.

(d) $-A = Q(-\Lambda)Q'$, so the eigenvalues of $-A$ are $-\lambda_1, \ldots, -\lambda_n$ (Theorem 8.4(b)); by (a) and (b), $-A$ is PD (PSD) if and only if $-\lambda_1 > 0$ ($\geqslant 0$). By (c), $x'Ax$ takes a positive value if $\lambda_1 > 0$ (at q_1) and none if $\lambda_1 \leqslant 0$; it takes a negative value if $\lambda_n < 0$ (at q_n) and none if $\lambda_n \geqslant 0$. □

For instance, A_1 and A_2 of Example 8.12 have the eigenvalues $3, 1$ and $3, -1$ (eigenvectors $(1,1)'$ and $(1,-1)'$).

Definition 8.14. Let $A \in \mathbb{R}^{n\times n}$, and let $\mathcal{I} = \{i_1 < \cdots < i_k\} \subseteq \{1, \ldots, n\}$ with $k \geqslant 1$. The **principal submatrix** $A_\mathcal{I} \in \mathbb{R}^{k\times k}$ keeps the rows and the columns of A with indices in $\mathcal{I}$: $(A_\mathcal{I})_{rs} = a_{i_r i_s}$. Its determinant is a **principal minor** of A. For $\mathcal{I} = \{1, \ldots, k\}$ we write $A_{[k]}$ (the kth **leading principal submatrix**), and $\det A_{[k]}$ is the kth **leading principal minor**. With $E_\mathcal{I} = [e_{i_1} \ \cdots \ e_{i_k}] \in \mathbb{R}^{n\times k}$,

$$A_\mathcal{I} = E_\mathcal{I}' A E_\mathcal{I},$$

since the (r,s) entry of $E_\mathcal{I}' A E_\mathcal{I}$ is $e_{i_r}' A e_{i_s} = a_{i_r i_s}$ (Theorem 1.4(b),(d)).

Corollary 8.15. Let $A \in \mathbb{R}^{n\times n}$ be symmetric.

(a) If A is PD, then A is invertible, $\det A > 0$, A^{-1} is PD, every principal submatrix of A is PD, and every diagonal entry of A is positive.

(b) If A is PSD, then $\det A \geqslant 0$, every principal submatrix of A is PSD (so all principal minors are $\geqslant 0$), and every diagonal entry of A is $\geqslant 0$.

(c) If A is PSD and invertible, then A is PD.

Proof. • By Theorem 8.4(b) and Corollary 7.30(a), $\det A = \lambda_1 \cdots \lambda_n$. If A is PD (PSD), every λ_i is > 0 ($\geqslant 0$) by Theorem 8.13(a),(b), so $\det A > 0$ ($\geqslant 0$); and $\det A > 0$ makes A invertible (Theorem 5.19(a)).

- Let A be PD. A^{-1} is symmetric (Theorem 2.15(c)). For $x \neq 0$, $y = A^{-1}x \neq 0$, as $x = Ay$; and $x'A^{-1}x = y'A'A^{-1}Ay = y'Ay > 0$.
- $A_\mathcal{I} = E_\mathcal{I}' A E_\mathcal{I}$ is symmetric. For $z \in \mathbb{R}^k$, $E_\mathcal{I} z = z_1 e_{i_1} + \cdots + z_k e_{i_k}$ is 0 only for $z = 0$, and $z'A_\mathcal{I} z = (E_\mathcal{I} z)' A (E_\mathcal{I} z)$. So $A_\mathcal{I}$ is PD (PSD) if A is; then $\det A_\mathcal{I} > 0$ ($\geqslant 0$) by the first item. The diagonal entry a_{ii} is the principal submatrix for $\mathcal{I} = \{i\}$, and $z'(a_{ii})z = a_{ii}z^2$.
- If A is PSD and invertible, then $\lambda_n \geqslant 0$, and 0 is not an eigenvalue (Theorem 7.12(b)); so $\lambda_n > 0$, and A is PD (Theorem 8.13(a)). □

Theorem 8.16. (a) For $B \in \mathbb{R}^{m\times n}$, the matrix $B'B$ (the **Gram matrix** of the columns $b_1, \ldots, b_n$ of B, with entries $b_i'b_j$) is PSD; it is PD if and only if $\operatorname{rank} B = n$.

(b) Let $A \in \mathbb{R}^{n\times n}$ be symmetric and $C \in \mathbb{R}^{n\times k}$. If A is PSD, then $C'AC$ is PSD. If A is PD and $\operatorname{rank} C = k$, then $C'AC$ is PD.

(c) If $A \in \mathbb{R}^{n\times n}$ is PSD of rank $r \geqslant 1$, then $A = B'B$ for some $B \in \mathbb{R}^{r\times n}$ of rank r.

Proof. (a) $B'B$ is symmetric (Theorem 1.9), and $x'B'Bx = (Bx)'(Bx) = \|Bx\|^2 \geqslant 0$, with equality if and only if $Bx = 0$. So $B'B$ is PD if and only if $\mathcal{N}(B) = \{0\}$, that is, $\operatorname{rank} B = n$ (Theorem 4.13(b)).

(b) $(C'AC)' = C'A'C = C'AC$, and $x'C'ACx = (Cx)'A(Cx) \geqslant 0$. If $\operatorname{rank} C = k$ and $x \neq 0$, then $Cx \neq 0$ (Theorem 4.13(b)), so $(Cx)'A(Cx) > 0$ when A is PD.

(c) Write $A = Q\Lambda Q'$ with $\lambda_1 \geqslant \cdots \geqslant \lambda_n \geqslant 0$ (Theorem 8.13(b)); exactly r of the λ_i are nonzero (Corollary 8.5(b)), so $\lambda_1 \geqslant \cdots \geqslant \lambda_r > 0$. Let $B \in \mathbb{R}^{r\times n}$ have the rows $\sqrt{\lambda_i}\, q_i'$ $(i \leqslant r)$. By the outer product expansion and Corollary 8.5(a), $B'B = \sum_{i\leqslant r}(\sqrt{\lambda_i}\, q_i)(\sqrt{\lambda_i}\, q_i') = \sum_{i\leqslant r} \lambda_i q_i q_i' = A$, and $\operatorname{rank} B = \operatorname{rank}(B'B) = \operatorname{rank} A = r$ (Theorem 4.18(b)). □

Proposition 8.17. Let $A \in \mathbb{R}^{n\times n}$ be PSD.

(a) For $x \in \mathbb{R}^n$, $x'Ax = 0$ if and only if $Ax = 0$.

(b) If $a_{ii} = 0$, then row i and column i of A are zero.

(c) $a_{ij}^2 \leqslant a_{ii}a_{jj}$ for all i, j.

Proof. (a) If $Ax = 0$, then $x'Ax = 0$. Conversely, let $x'Ax = 0$. If $A = 0$, then $Ax = 0$. Otherwise $A = B'B$ as in Theorem 8.16(c), $\|Bx\|^2 = x'Ax = 0$, so $Bx = 0$ and $Ax = B'Bx = 0$.

(b) $e_i'Ae_i = a_{ii} = 0$, so $Ae_i = 0$ by (a): column i is zero, and row i is its transpose, as $A' = A$.

(c) For $i = j$ there is nothing to prove. For $i < j$, the principal submatrix for $\{i, j\}$ is $\left(\begin{smallmatrix} a_{ii} & a_{ij} \\ a_{ij} & a_{jj} \end{smallmatrix}\right)$, and its determinant $a_{ii}a_{jj} - a_{ij}^2$ is $\geqslant 0$ (Corollary 8.15(b)); the case $i > j$ follows by symmetry. □

So $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ is not PSD ($2^2 > 1 \cdot 3$), and neither is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ ($a_{11} = 0$ but row 1 is not zero).

Lemma 8.18. Let $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12}' & A_{22} \end{pmatrix} \in \mathbb{R}^{n\times n}$ be symmetric, with $A_{11} \in \mathbb{R}^{p\times p}$ PD $(1 \leqslant p < n)$, and let $A_{22\cdot 1} = A_{22} - A_{12}'A_{11}^{-1}A_{12}$ be the Schur complement of A_{11}, a symmetric matrix. Then A is PD if and only if $A_{22\cdot 1}$ is PD, and A is PSD if and only if $A_{22\cdot 1}$ is PSD.

Proof. • A_{11} is invertible (Corollary 8.15(a)) and symmetric, so A_{11}^{-1} is symmetric (Theorem 2.15(c)) and $(A_{12}'A_{11}^{-1})' = A_{11}^{-1}A_{12}$. Comparing the $(2,2)$ blocks of $A' = A$ (Theorem 5.3(b)) gives $A_{22}' = A_{22}$, so $A_{22\cdot 1}' = A_{22} - A_{12}'A_{11}^{-1}A_{12} = A_{22\cdot 1}$. Let $L = \left(\begin{smallmatrix} I_p & 0 \\ -A_{12}'A_{11}^{-1} & I_{n-p} \end{smallmatrix}\right)$. Then $L' = \left(\begin{smallmatrix} I_p & -A_{11}^{-1}A_{12} \\ 0 & I_{n-p} \end{smallmatrix}\right)$, and by Theorem 5.9,

$$LAL' = \begin{pmatrix} A_{11} & 0 \\ 0 & A_{22\cdot 1} \end{pmatrix} =: D$$

- L is invertible (Proposition 5.8), so L' and $(L^{-1})'$ have rank n, and $A = L^{-1}D(L^{-1})'$.
- If A is PD (PSD), then $D = (L')'AL'$ is PD (PSD) by Theorem 8.16(b), and so is its principal submatrix $A_{22\cdot 1}$ (Corollary 8.15(a),(b)).
- Conversely, let $A_{22\cdot 1}$ be PD (PSD). For $x = (x_1', x_2')'$ with $x_1 \in \mathbb{R}^p$, $x'Dx = x_1'A_{11}x_1 + x_2'A_{22\cdot 1}x_2 \geqslant 0$; and if $x \neq 0$ and $A_{22\cdot 1}$ is PD, then $x_1 \neq 0$ or $x_2 \neq 0$, so one of the two terms is positive. Hence D is PD (PSD), and so is $A = ((L^{-1})')'D(L^{-1})'$ (Theorem 8.16(b)). □

Theorem 8.19 (Sylvester's criterion)**.** A symmetric $A \in \mathbb{R}^{n\times n}$ is PD if and only if all its leading principal minors $\det A_{[1]}, \ldots, \det A_{[n]}$ are positive.

Proof. If A is PD, every $A_{[k]}$ is PD, so $\det A_{[k]} > 0$ (Corollary 8.15(a)). Conversely, let all leading principal minors be positive; we use induction on n. For $n = 1$, $x'Ax = a_{11}x_1^2 > 0$ for $x \neq 0$. Let $n \geqslant 2$.

- $A_{[n-1]}$ is symmetric, and its leading principal minors are $\det A_{[1]}, \ldots, \det A_{[n-1]} > 0$; by the induction hypothesis, $A_{[n-1]}$ is PD.

- Write $A = \left(\begin{smallmatrix} A_{[n-1]} & b \\ b' & a_{nn} \end{smallmatrix}\right)$ and $s = a_{nn} - b'A_{[n-1]}^{-1}b$. By Theorem 5.26(a), $\det A = \det A_{[n-1]} \cdot s$, so $s = \det A_{[n]} / \det A_{[n-1]} > 0$.
- So the 1×1 Schur complement (s) is PD, and A is PD by Lemma 8.18. □

Remark 8.20. Leading principal minors $\geqslant 0$ do not make a symmetric matrix PSD: $A = \left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ has leading principal minors 0 and 0, but $e_2'Ae_2 = -1$. Its principal minor $\det A_{\{2\}} = -1$ is negative, so A is not PSD by Corollary 8.15(b).

Example 8.21. For which $a \in \mathbb{R}$ is $A = \begin{pmatrix} 1 & a & a & 0 \\ a & 1 & 0 & a \\ a & 0 & 1 & a \\ 0 & a & a & 1 \end{pmatrix}$ PD?

Solution.

- The first two leading principal minors are 1 and $1-a^2$. Expanding along the first row (Theorem 5.13), $\det A_{[3]} = (1-0) - a(a-0) + a(0-a) = 1 - 2a^2$.
- For $\det A$, subtract row 3 from row 2, which becomes $(0, 1, -1, 0)$, and then add column 2 to column 3 (Theorem 5.16(d),(f)):

$$\det A = \det \begin{pmatrix} 1 & a & 2a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & 1 & a \\ 0 & a & 2a & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 2a & 0 \\ a & 1 & a \\ 0 & 2a & 1 \end{pmatrix} = (1 - 2a^2) - 2a(a-0) = 1 - 4a^2,$$

  expanding along row 2 and then along the first row.
- By Theorem 8.19, A is PD if and only if $1 - a^2$, $1 - 2a^2$ and $1 - 4a^2$ are positive, that is, $a^2 < \frac{1}{4}$, or $-\frac{1}{2} < a < \frac{1}{2}$.

Check with eigenvalues: the orthogonal vectors $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$, $h_4 = (1,-1,-1,1)'$ satisfy

$$Ah_1 = (1+2a)h_1, \qquad Ah_2 = h_2, \qquad Ah_3 = h_3, \qquad Ah_4 = (1-2a)h_4$$

(for instance, row 1 of A times h_4 is $1 - a - a + 0$). So $A = Q\Lambda Q'$ with $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ orthogonal and $\Lambda = \operatorname{diag}(1+2a, 1, 1, 1-2a)$, and by Theorem 8.13(a) A is PD if and only if $1 - 2|a| > 0$, the same condition. For $a = \pm\frac{1}{2}$ the eigenvalues are $2, 1, 1, 0$: A is PSD but not PD; for $|a| > \frac{1}{2}$, A is indefinite (Theorem 8.13(d)).

Theorem 8.22 (Cholesky decomposition)**.** A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is PD if and only if $A = LL'$ for some lower triangular $L \in \mathbb{R}^{n\times n}$ with positive diagonal entries. This L is unique; it is the **Cholesky factor** of A.

Proof. • Let $A = LL'$ with L as stated. Then $\det L = l_{11} \cdots l_{nn} > 0$ (Theorem 5.15(a)), so L is invertible (Theorem 5.19(a)), $\operatorname{rank} L' = n$ (Theorem 4.13(a),(d)), and $A = (L')'L'$ is PD by Theorem 8.16(a).

- Existence for PD A, by induction on n. For $n = 1$, $a_{11} > 0$ (Corollary 8.15(a)) and $L = (\sqrt{a_{11}})$. Let $n \geqslant 2$, and write $A = \left(\begin{smallmatrix} A_{11} & b \\ b' & a \end{smallmatrix}\right)$ with $A_{11} = A_{[n-1]}$. A_{11} is PD (Corollary 8.15(a)), so $A_{11} = L_{11}L_{11}'$ with L_{11} lower triangular with positive diagonal entries (induction hypothesis), and L_{11} is invertible (first item). The Schur complement $s = a - b'A_{11}^{-1}b$ is positive (Lemma 8.18). Put $l = L_{11}^{-1}b$ and

$$L = \begin{pmatrix} L_{11} & 0 \\ l' & \sqrt{s} \end{pmatrix}, \qquad LL' = \begin{pmatrix} L_{11}L_{11}' & L_{11}l \\ l'L_{11}' & l'l + s \end{pmatrix} = \begin{pmatrix} A_{11} & b \\ b' & a \end{pmatrix} = A,$$

  since $L_{11}l = b$ and $l'l = b'(L_{11}^{-1})'L_{11}^{-1}b = b'(L_{11}L_{11}')^{-1}b = b'A_{11}^{-1}b$ (Theorem 2.15(b),(c)). L is lower triangular with positive diagonal entries.

- Uniqueness, by induction on n. For $n = 1$, $l_{11}^2 = a_{11}$ and $l_{11} > 0$ force $l_{11} = \sqrt{a_{11}}$. Let $n \geqslant 2$ and $A = MM'$ with $M = \left(\begin{smallmatrix} M_{11} & 0 \\ m' & \delta \end{smallmatrix}\right)$ lower triangular with positive diagonal entries. Comparing blocks of MM' and A: $M_{11}M_{11}' = A_{11}$, so $M_{11} = L_{11}$ (induction hypothesis); $M_{11}m = b$, so $m = L_{11}^{-1}b = l$; and $m'm + \delta^2 = a$, so $\delta^2 = a - l'l = s$ and $\delta = \sqrt{s}$. Hence $M = L$. □

Remark 8.23. For $i \geqslant j$, the (i,j) entry of LL' is $\sum_{k \leqslant j} l_{ik}l_{jk}$, since $l_{jk} = 0$ for $k > j$. Comparing with a_{ij} gives L column by column ($j = 1, \ldots, n$):

$$l_{jj} = \Big(a_{jj} - \sum_{k<j} l_{jk}^2\Big)^{1/2}, \qquad l_{ij} = \frac{1}{l_{jj}}\Big(a_{ij} - \sum_{k<j} l_{ik}l_{jk}\Big) \quad (i > j)$$

For a symmetric A, this succeeds with positive numbers under all square roots exactly when A is PD: if it succeeds, it produces L with $LL' = A$; if A is PD, the Cholesky factor satisfies these equations. Moreover $A_{[k]} = L_{[k]}L_{[k]}'$, as the first k rows of L are zero after column k; so $\det A_{[k]} = (l_{11} \cdots l_{kk})^2$.

Example 8.24. Find the Cholesky factor L of $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$, the matrix with entries $a_{ij} = \min(i,j)$, and use it to compute A^{-1}.

Solution. By Remark 8.23:

- column 1: $l_{11} = \sqrt{1} = 1$ and $l_{i1} = a_{i1}/1 = 1$ for $i = 2, 3, 4$;
- column 2: $l_{22} = \sqrt{2 - 1^2} = 1$ and $l_{i2} = (2 - 1 \cdot 1)/1 = 1$ for $i = 3, 4$;
- column 3: $l_{33} = \sqrt{3 - 1 - 1} = 1$ and $l_{43} = (3 - 1 - 1)/1 = 1$;
- column 4: $l_{44} = \sqrt{4 - 1 - 1 - 1} = 1$.

So L is the lower triangular matrix with all entries $l_{ij} = 1$ for $i \geqslant j$; indeed the (i,j) entry of LL' is $\mathrm{row}_i(L)\,\mathrm{row}_j(L)' = \min(i,j)$. Hence A is PD (Theorem 8.22), and all its leading principal minors equal 1 (Remark 8.23). The matrix $K = \left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{smallmatrix}\right)$ has $KL = I_4$ (row i of K subtracts row $i-1$ of L from row i), so $L^{-1} = K$ (Theorem 2.18) and, by Theorem 2.15(b),(c),

$$A^{-1} = (L')^{-1}L^{-1} = K'K = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Square roots, the Loewner order, and simultaneous diagonalization

Theorem 8.25. Let $A \in \mathbb{R}^{n\times n}$ be PSD, $A = Q\Lambda Q'$ as in Theorem 8.4(a); so $\lambda_i \geqslant 0$ for every i (Theorem 8.13(b)).

(a) There is exactly one PSD $B \in \mathbb{R}^{n\times n}$ with $B^2 = A$. It is the **square root** of A, written $A^{1/2}$, and

$$A^{1/2} = Q\Lambda^{1/2}Q', \qquad \Lambda^{1/2} = \mathrm{diag}\big(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}\big),$$

that is, $A^{1/2} = f(A)$ for $f(t) = \sqrt{t}$ (Definition 8.7).

(b) $A^{1/2}A = AA^{1/2}$, $\mathrm{rank}\, A^{1/2} = \mathrm{rank}\, A$, and $x'Ax = \|A^{1/2}x\|^2$ for $x \in \mathbb{R}^n$.

(c) If A is PD, then $A^{1/2}$ is PD, $A^{-1/2} := (A^{1/2})^{-1} = Q\Lambda^{-1/2}Q'$ with $\Lambda^{-1/2} = \mathrm{diag}(\lambda_1^{-1/2}, \ldots, \lambda_n^{-1/2})$, and

$$A^{-1/2}AA^{-1/2} = I_n, \qquad A^{-1/2}A^{-1/2} = A^{-1}$$

Proof. (a)

- $B_0 = Q\Lambda^{1/2}Q'$ is symmetric, it is PSD by Theorem 8.13(b), as its eigenvalues $\sqrt{\lambda_i}$ are $\geqslant 0$ (Theorem 8.4(b)), and $B_0^2 = Q\Lambda^{1/2}Q'Q\Lambda^{1/2}Q' = Q\Lambda Q' = A$.
- Let B be PSD with $B^2 = A$. By Theorem 8.4(a),(b), $B = PMP'$ with P orthogonal, columns $p_1, \dots, p_n$, and $M = \operatorname{diag}(\mu_1, \dots, \mu_n)$, all $\mu_i \geqslant 0$ (Theorem 8.13(b)); so $Bp_i = \mu_i p_i$ and $Ap_i = B^2 p_i = \mu_i^2 p_i$.
- Let $Ax = \alpha x$ with $\alpha \geqslant 0$, and write $x = \sum_i c_i p_i$ (the p_i form a basis). Then $\sum_i c_i \mu_i^2 p_i = Ax = \sum_i c_i \alpha p_i$, so $c_i(\mu_i^2 - \alpha) = 0$ for every i (independence). Thus $c_i = 0$ unless $\mu_i^2 = \alpha$, that is, unless $\mu_i = \sqrt{\alpha}$ (both are $\geqslant 0$); hence $Bx = \sum_i c_i \mu_i p_i = \sqrt{\alpha}\sum_i c_i p_i = \sqrt{\alpha}\, x$.
- For $x = q_j$ and $\alpha = \lambda_j$ this gives $Bq_j = \sqrt{\lambda_j}\, q_j = B_0 q_j$ (Proposition 8.8(a)). So $BQ = B_0 Q$, and $B = B_0$ since Q is invertible.

(b) Diagonal matrices commute, so $A^{1/2}A = Q\Lambda^{1/2}\Lambda Q' = Q\Lambda\Lambda^{1/2}Q' = AA^{1/2}$. The number of nonzero $\sqrt{\lambda_i}$ is the number of nonzero λ_i, so the ranks agree (Corollary 8.5(b)). Finally $x'Ax = x'A^{1/2}A^{1/2}x = (A^{1/2}x)'(A^{1/2}x)$, as $A^{1/2}$ is symmetric.

(c) If A is PD, every $\lambda_i > 0$, so every $\sqrt{\lambda_i} > 0$ and $A^{1/2}$ is PD (Theorem 8.13(a)). Let $F = Q\Lambda^{-1/2}Q'$, that is, $F = f(A)$ for $f(t) = t^{-1/2}$. By Proposition 8.8(b),(c),

$$FA^{1/2} = A^{1/2}F = Q\Lambda^{-1/2}\Lambda^{1/2}Q' = I_n, \qquad FA = Q\Lambda^{-1/2}\Lambda Q' = A^{1/2}, \qquad FF = Q\Lambda^{-1}Q' = A^{-1}$$

So $F = (A^{1/2})^{-1} = A^{-1/2}$, $A^{-1/2}AA^{-1/2} = A^{1/2}F = I_n$, and $A^{-1/2}A^{-1/2} = A^{-1}$. □

Example 8.26. For $A = 25P_1 + 9P_2 + P_3$ of Example 8.6, all eigenvalues are positive, so A is PD, and Theorem 8.25(a),(c) gives

$$A^{1/2} = 5P_1 + 3P_2 + P_3 = \begin{pmatrix} 3 & 0 & 1 & 1 \\ 0 & 3 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 3 \end{pmatrix}, \qquad A^{-1/2} = \tfrac{1}{5}P_1 + \tfrac{1}{3}P_2 + P_3 = \frac{1}{15}\begin{pmatrix} 7 & 2 & -3 & -3 \\ 2 & 7 & -3 & -3 \\ -3 & -3 & 7 & 2 \\ -3 & -3 & 2 & 7 \end{pmatrix}$$

(for instance the $(1,1)$ entry of $A^{1/2}$ is $\frac{5}{4} + \frac{3}{2} + \frac{1}{4} = 3$). Check: row 1 of $A^{1/2}$ times its columns gives $9+1+1 = 11$, $0+1+1 = 2$, $3+3 = 6$, $3+3 = 6$, the first row of A. The matrix

$$C = 5P_1 - 3P_2 + P_3 = \begin{pmatrix} 0 & 3 & 1 & 1 \\ 3 & 0 & 1 & 1 \\ 1 & 1 & 0 & 3 \\ 1 & 1 & 3 & 0 \end{pmatrix}$$

also satisfies $C^2 = 25P_1 + 9P_2 + P_3 = A$, because $P_j^2 = P_j$ and $P_jP_l = 0$ for $j \neq l$ (Corollary 8.5(c)); but C is not PSD, as its diagonal entry $c_{11} = 0$ sits in a nonzero row (Proposition 8.17(b)). Among the matrices whose square is A, Theorem 8.25(a) singles out the PSD one.

Definition 8.27. For symmetric $A, B \in \mathbb{R}^{n\times n}$ we write $A \geqslant B$ (or $B \leqslant A$) if $A - B$ is PSD, and $A > B$ (or $B < A$) if $A - B$ is PD. This is the **Loewner order**. In particular $A \geqslant 0$ means that A is PSD, and $A > 0$ that A is PD. It is not the entrywise comparison of matrices.

Proposition 8.28. Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric.

(a) $A \geqslant A$; if $A \geqslant B$ and $B \geqslant A$, then $A = B$; if $A \geqslant B$ and $B \geqslant D$ for a symmetric $D \in \mathbb{R}^{n\times n}$, then $A \geqslant D$.

(b) If $A \geqslant B$, then $C'AC \geqslant C'BC$ for every $C \in \mathbb{R}^{n\times k}$.

(c) If $A \geqslant B$, then $a_{ii} \geqslant b_{ii}$ for every i, and $\operatorname{tr} A \geqslant \operatorname{tr} B$.

(d) If $A \geqslant B > 0$, then $A > 0$ and $B^{-1} \geqslant A^{-1}$.

(e) If $A \geqslant B \geqslant 0$, then $\det A \geqslant \det B$.

Proof. (a) $A - A = 0$ is PSD. If $A - B$ and $B - A$ are PSD, then $F = A - B$ has $x'Fx \geqslant 0$ and $-x'Fx \geqslant 0$, so $x'Fx = 0$, for every x; by Proposition 8.17(a), $Fx = 0$ for every x, so $F = 0$. Finally $x'(A - D)x = x'(A - B)x + x'(B - D)x \geqslant 0$.

(b) $C'AC - C'BC = C'(A - B)C$ is PSD by Theorem 8.16(b).

(c) $a_{ii} - b_{ii} = e_i'(A - B)e_i \geqslant 0$; add over i.

(d)

- For $x \neq 0$, $x'Ax = x'(A - B)x + x'Bx \geqslant x'Bx > 0$; so $A > 0$.
- Let $G = B^{-1/2}$, symmetric and invertible (Theorem 8.25(c)). By (b), $M = GAG \geqslant GBG = I_n$. If $Mv = \theta v$ with $v'v = 1$, then $\theta = v'Mv \geqslant v'v = 1$. So $M = P\Theta P'$ (Theorem 8.4(a),(b)) with all $\theta_i \geqslant 1$; $M^{-1} = P\Theta^{-1}P'$ (Proposition 8.8(c)), and $I_n - M^{-1} = P(I_n - \Theta^{-1})P'$ has the eigenvalues $1 - 1/\theta_i \geqslant 0$, so $M^{-1} \leqslant I_n$ (Theorem 8.13(b)).
- $M^{-1} = G^{-1}A^{-1}G^{-1}$ (Theorem 2.15(b)). By (b) with $C = G$, $A^{-1} = GM^{-1}G \leqslant GI_nG = B^{-1}$ (Theorem 8.25(c)).

(e) By (a), $A \geqslant 0$, so $\det A \geqslant 0$ and $\det B \geqslant 0$ (Corollary 8.15(b)). If $\det B = 0$, we are done. Otherwise B is invertible, hence PD (Corollary 8.15(c)), and with G and M as in (d), all eigenvalues of M are $\geqslant 1$, so $\det M \geqslant 1$ (Theorem 8.4(b), Corollary 7.30(a)). By Theorem 5.19(b), $\det M = (\det G)^2 \det A$ and $(\det G)^2 \det B = \det(GBG) = \det I_n = 1$; hence $\det A = \det M \cdot \det B \geqslant \det B$. □

Example 8.29. (a) The Loewner order is not total: for $A = \operatorname{diag}(1, 0)$ and $B = \operatorname{diag}(0, 1)$, $A - B = \operatorname{diag}(1, -1)$ is indefinite, so neither $A \geqslant B$ nor $B \geqslant A$.

(b) It is not the entrywise order: $I_2 - \frac{1}{2}J_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ has the eigenvalues 0 and 1 (eigenvectors $(1, 1)'$ and $(1, -1)'$), so $I_2 \geqslant \frac{1}{2}J_2$, although the $(1, 2)$ entry of $\frac{1}{2}J_2$ is the larger one.

Theorem 8.30 (Simultaneous diagonalization)**.** Let $A \in \mathbb{R}^{n\times n}$ be PD and $B \in \mathbb{R}^{n\times n}$ symmetric.

(a) There is an invertible $P \in \mathbb{R}^{n\times n}$ with

$$P'AP = I_n, \qquad P'BP = \Theta = \operatorname{diag}(\theta_1, \ldots, \theta_n),$$

and P can be chosen so that $\theta_1 \geqslant \cdots \geqslant \theta_n$.

(b) For every such P, the columns p_i of P satisfy $Bp_i = \theta_i Ap_i$, and

$$\det(B - \theta A) = \det A \cdot (\theta_1 - \theta) \cdots (\theta_n - \theta) \qquad (\theta \in \mathbb{R})$$

So $\theta_1, \ldots, \theta_n$ are the roots of $\det(B - \theta A) = 0$, listed with multiplicity; they are the eigenvalues of $A^{-1/2}BA^{-1/2}$ listed with multiplicity.

(c) If $Bu = \theta Au$ and $Bv = \phi Av$ with $\theta \neq \phi$, then $u'Av = 0$.

Proof. (a) $C = A^{-1/2}BA^{-1/2}$ is symmetric, as $A^{-1/2}$ is (Theorem 8.25(c)). By Theorem 8.4(a),(d), $C = R\Theta R'$ with R orthogonal and $\theta_1 \geqslant \cdots \geqslant \theta_n$. Put $P = A^{-1/2}R$, invertible as a product of invertible matrices. Then $P'AP = R'A^{-1/2}AA^{-1/2}R = R'R = I_n$ and $P'BP = R'CR = \Theta$.

(b)

- $P'AP = I_n$ gives $AP = (P')^{-1}$, and then $P'BP = \Theta$ gives $BP = (P')^{-1}\Theta = AP\Theta$; column i reads $Bp_i = \theta_i Ap_i$.
- $P'(B - \theta A)P = \Theta - \theta I_n$, so $(\det P)^2 \det(B - \theta A) = \prod_i(\theta_i - \theta)$, and $(\det P)^2 \det A = \det(P'AP) = 1$ (Theorems 5.19(b) and 5.15(a),(b)). Multiplying the first equation by $\det A$ gives the formula.

- With C as in (a) and $(\det A^{-1/2})^2 \det A = \det I_n = 1$, $p_C(t) = \det\bigl(A^{-1/2}(tA-B)A^{-1/2}\bigr) = (\det A)^{-1}(-1)^n \det(B-tA) = \prod_i (t-\theta_i)$ (Theorem 5.19(b),(c)). So p_C splits with roots $\theta_1, \dots, \theta_n$.

(c) $\theta\, v'Au = v'Bu = (v'Bu)' = u'Bv = \phi\, u'Av$, and $v'Au = u'Av$; so $(\theta-\phi)u'Av = 0$. □

Example 8.31. Let $A = (\min(i,j)) \in \mathbb{R}^{4\times 4}$ be the matrix of Example 8.24, and

$$B = \begin{pmatrix} 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 6 & 6 \\ 3 & 5 & 6 & 5 \end{pmatrix},$$

the matrix with entries $b_{ij} = c_{\min(i,j)}$, where $(c_1, c_2, c_3, c_4) = (3, 5, 6, 5)$. Find P as in Theorem 8.30(a).

Solution.

- A is PD (Example 8.24). The (i,j) entry of $B - \theta A$ is $d_{\min(i,j)}$, with $d_m = c_m - \theta m$: $d_1 = 3-\theta$, $d_2 = 5-2\theta$, $d_3 = 6-3\theta$, $d_4 = 5-4\theta$.
- Subtract row 3 from row 4, then row 2 from row 3, then row 1 from row 2 (Theorem 5.16(d)). Row i minus row $i-1$ has the entry $d_j - d_j = 0$ for $j < i$ and $d_i - d_{i-1}$ for $j \geqslant i$, so the result is upper triangular, and (Theorem 5.15(a))

  $$\det(B-\theta A) = d_1(d_2-d_1)(d_3-d_2)(d_4-d_3) = (3-\theta)(2-\theta)(1-\theta)(-1-\theta)$$

  By Theorem 8.30(b), and as $\det A = 1$, $\theta_1 = 3$, $\theta_2 = 2$, $\theta_3 = 1$, $\theta_4 = -1$.
- Put $e_0 = 0$, $d_0 = 0$, and $p_k = e_k - e_{k-1}$. The ith entry of $(B-\theta A)p_k$ is $d_{\min(i,k)} - d_{\min(i,k-1)}$, which is 0 for $i < k$ and $d_k - d_{k-1} = (c_k - c_{k-1}) - \theta$ for $i \geqslant k$. As $c_k - c_{k-1} = \theta_k$ ($c_0 = 0$), $(B-\theta_k A)p_k = 0$, that is, $Bp_k = \theta_k A p_k$.
- Let $P = [p_1\ p_2\ p_3\ p_4]$. The columns of

  $$P = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad AP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

  give $P'AP = I_4$: the (j,k) entry $p_j' A p_k$ is entry j minus entry $j-1$ of column k of AP, which is 1 for $j = k$ and 0 otherwise. Then $P'BP = \operatorname{diag}(3, 2, 1, -1)$, since $(P'BP)_{jk} = p_j' B p_k = \theta_k p_j' A p_k$.

Here $P = K' = (L')^{-1}$ for the Cholesky factor L of A and $K = L^{-1}$ (Example 8.24, Theorem 2.15(c)), and $B = L \operatorname{diag}(3,2,1,-1) L'$. In the coordinates $y = P^{-1}x = L'x$, that is, $y_k = x_k + \cdots + x_4$, $x'Ax = y_1^2 + y_2^2 + y_3^2 + y_4^2$ and $x'Bx = 3y_1^2 + 2y_2^2 + y_3^2 - y_4^2$.

Covariance matrices of data

Definition 8.32. Let $n \geqslant 2$.

- For $u, v \in \mathbb{R}^n$, the **sample covariance** of u and v is

  $$s_{uv} = \frac{1}{n-1} \sum_{i=1}^{n} (u_i - \bar{u})(v_i - \bar{v}) = \frac{1}{n-1}\, u' C_n v,$$

  and s_{uu} is the **sample variance** of u, $\sqrt{s_{uu}}$ its **sample standard deviation**. If $s_{uu}, s_{vv} > 0$, $r_{uv} = s_{uv}/\sqrt{s_{uu} s_{vv}}$ is the sample correlation of u and v (Problem 6.8; the factors $\frac{1}{n-1}$ cancel).
- Let $X_0 \in \mathbb{R}^{n\times k}$ be a data matrix with ith row x_i' (so $x_i \in \mathbb{R}^k$ holds the k measurements on unit i) and mean vector $\bar{x} = \frac{1}{n} X_0' \mathbf{1}_n = \frac{1}{n} \sum_i x_i$. The **sample covariance matrix** of X_0 is

  $$S = \frac{1}{n-1} X_0' C_n X_0 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x})'$$

- If every $s_{jj} > 0$, let $D = \operatorname{diag}(s_{11}, \dots, s_{kk})$ and $D^{-1/2} = \operatorname{diag}(s_{11}^{-1/2}, \dots, s_{kk}^{-1/2})$. The **sample correlation matrix** is $R = D^{-1/2} S D^{-1/2}$, with entries $r_{jl} = s_{jl} / \sqrt{s_{jj} s_{ll}}$.

The two expressions agree. C_n is symmetric and idempotent with $C_n u = u - \bar{u}\mathbf{1}$ (Proposition 1.13, Example 1.14), so $u' C_n v = (C_n u)'(C_n v) = \sum_i (u_i - \bar{u})(v_i - \bar{v})$. Likewise $X_0' C_n X_0 = (C_n X_0)'(C_n X_0)$, where $C_n X_0 = X_0 - \mathbf{1}\bar{x}'$ has rows $(x_i - \bar{x})'$; by the outer product expansion (Theorem 1.4(d)) this is $\sum_i (x_i - \bar{x})(x_i - \bar{x})'$. The (j,l) entry of $X_0' C_n X_0$ is $\operatorname{col}_j(X_0)' C_n \operatorname{col}_l(X_0)$, so s_{jl} is the sample covariance of variables j and l, and s_{jj} the sample variance of variable j. The (j,l) entry of $R = D^{-1/2} S D^{-1/2}$ is $\sum_{a,b} (D^{-1/2})_{ja} s_{ab} (D^{-1/2})_{bl} = s_{jj}^{-1/2} s_{jl} s_{ll}^{-1/2} = r_{jl}$, as $D^{-1/2}$ is diagonal.

Proposition 8.33. Let $X_0 \in \mathbb{R}^{n \times k}$ ($n \geqslant 2$) have rows x_i' and sample covariance matrix S.

(a) For $a, b \in \mathbb{R}^k$, the sample variance of the n numbers $a'x_1, \dots, a'x_n$ is $a'Sa$, and the sample covariance of the numbers $a'x_i$ and $b'x_i$ ($i = 1, \dots, n$) is $a'Sb$.

(b) S is PSD; it is PD if and only if $\operatorname{rank}(C_n X_0) = k$, which requires $n \geqslant k + 1$.

(c) If every $s_{jj} > 0$, then R is PSD, $r_{jj} = 1$, and $|r_{jl}| \leqslant 1$ for all j, l.

Proof. (a) The numbers $a'x_i = x_i'a$ are the entries of $X_0 a$ (Theorem 1.4(c)). By Definition 8.32, their sample covariance with the entries of $X_0 b$ is $\frac{1}{n-1}(X_0 a)' C_n (X_0 b) = a'\Big(\frac{1}{n-1} X_0' C_n X_0\Big) b = a'Sb$; take $b = a$ for the variance.

(b) $S = \frac{1}{n-1}(C_n X_0)'(C_n X_0)$ with $\frac{1}{n-1} > 0$, so by Theorem 8.16(a) S is PSD, and PD if and only if $\operatorname{rank}(C_n X_0) = k$. As $\operatorname{rank}(C_n X_0) \leqslant \operatorname{rank} C_n = n - 1$ (Theorem 4.15(a), Corollary 6.21(b)), this needs $k \leqslant n - 1$.

(c) $D^{-1/2}$ is symmetric, so $R = (D^{-1/2})' S D^{-1/2}$ is PSD (Theorem 8.16(b)); $r_{jj} = s_{jj}/s_{jj} = 1$; and $r_{jl}^2 \leqslant r_{jj} r_{ll} = 1$ (Proposition 8.17(c)). □

Example 8.34. Five students have the scores $(9, 10, 11)$, $(5, 4, 5)$, $(3, 6, 5)$, $(3, 4, 7)$, $(5, 6, 7)$ on three tests: these are the rows x_i' of $X_0 \in \mathbb{R}^{5 \times 3}$. Find $\bar{x}$, S and R, the sample variances of the total score and of the difference between tests 1 and 2, and their sample covariance.

Solution. $\mathbf{1}' X_0 = (25,\ 30,\ 35)$, so $\bar{x} = (5, 6, 7)'$, and

$$C_5 X_0 = X_0 - \mathbf{1}\bar{x}' = \begin{pmatrix} 4 & 4 & 4 \\ 0 & -2 & -2 \\ -2 & 0 & -2 \\ -2 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad X_0' C_5 X_0 = (C_5 X_0)'(C_5 X_0) = \begin{pmatrix} 24 & 20 & 20 \\ 20 & 24 & 20 \\ 20 & 20 & 24 \end{pmatrix}$$

(for instance $24 = 16 + 4 + 4$ and $20 = 16 + 4$). Hence

$$S = \frac{1}{4} X_0' C_5 X_0 = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix} = I_3 + 5J_3, \qquad R = \tfrac{1}{6} S = \begin{pmatrix} 1 & \frac{5}{6} & \frac{5}{6} \\ \frac{5}{6} & 1 & \frac{5}{6} \\ \frac{5}{6} & \frac{5}{6} & 1 \end{pmatrix},$$

as $D = 6I_3$: the three tests are equicorrelated with $\rho = \frac{5}{6}$. The total score is $a'x_i$ with $a = \mathbf{1}_3$, so its sample variance is $a'Sa = \mathbf{1}'S\mathbf{1} = 48$, the sum of the entries of S (Proposition 8.33(a)); directly, the totals $30, 14, 14, 14, 18$ have mean 18 and squared deviations $144, 16, 16, 16, 0$, with sum $192 = 4 \cdot 48$. For the difference, $b = (1, -1, 0)'$: $b'Sb = 6 - 2 \cdot 5 + 6 = 2$; and $a'Sb = (16, 16, 16)b = 0$, as every column of S sums to 16: total and difference are uncorrelated. Rows 2, 3, 4 of $C_5 X_0$ form a matrix T with $\det T = 0 - (-2)(0 - 4) + (-2)(4 - 0) = -16$, so T is invertible (Theorem 5.19(a)) and $3 = \operatorname{rank} T \leqslant \operatorname{rank}(C_5 X_0) \leqslant 3$ ($T = [0\ I_3\ 0]\, C_5 X_0$, Theorem 4.15(a)). So S is PD (Proposition 8.33(b)).

Definition 8.35. Let $S \in \mathbb{R}^{k \times k}$ be PD. The **Mahalanobis distance** (with respect to S) between $x, y \in \mathbb{R}^k$ is

$$d_S(x, y) = \Big((x - y)' S^{-1} (x - y)\Big)^{1/2}$$

S^{-1} is PD (Corollary 8.15(a)), so $d_S(x,y) \geqslant 0$, with equality only for $x = y$. For $S = I_k$ this is the Euclidean distance $\|x-y\|$; for $S = \operatorname{diag}(s_{11},\dots,s_{kk})$, $d_S(x,y)^2 = \sum_j (x_j - y_j)^2/s_{jj}$ measures each coordinate in units of its standard deviation $\sqrt{s_{jj}}$.

Proposition 8.36. Let $X_0 \in \mathbb{R}^{n\times k}$ ($n \geqslant 2$) have rows x_i', mean vector $\bar{x}$ and a PD sample covariance matrix S.

(a) The vectors $z_i = S^{-1/2}(x_i - \bar{x})$ ($1 \leqslant i \leqslant n$) have mean vector 0 and sample covariance matrix I_k. (Passing from the x_i to the z_i is called **whitening** the data.)

(b) $d_S(x_i,\bar{x}) = \|z_i\|$ for every i, and $\sum_{i=1}^n d_S(x_i,\bar{x})^2 = (n-1)k$.

(c) Let $B \in \mathbb{R}^{k\times k}$ be invertible, $c \in \mathbb{R}^k$, and $y_i = Bx_i + c$. The y_i have mean vector $\bar{y} = B\bar{x}+c$ and the PD sample covariance matrix $S_y = BSB'$, and $d_{S_y}(y_i,y_l) = d_S(x_i,x_l)$ and $d_{S_y}(y_i,\bar{y}) = d_S(x_i,\bar{x})$ for all i,l.

(d) Let $S = Q\Lambda Q'$ as in Theorem 8.4(a),(d), so that $\lambda_1 \geqslant \cdots \geqslant \lambda_k > 0$ (Theorem 8.13(a)), and let $c > 0$. For $x \in \mathbb{R}^k$ put $w = Q'(x-\bar{x})$, so that $x = \bar{x} + w_1 q_1 + \cdots + w_k q_k$. Then

$$d_S(x,\bar{x})^2 = \frac{w_1^2}{\lambda_1} + \cdots + \frac{w_k^2}{\lambda_k}$$

So $\{x \in \mathbb{R}^k : d_S(x,\bar{x}) \leqslant c\}$ is an ellipse for $k = 2$ (an interval for $k = 1$, an ellipsoid for $k \geqslant 3$) with center $\bar{x}$ and axes along $q_1,\dots,q_k$: the point $\bar{x} + tq_j$ lies in it if and only if $|t| \leqslant c\sqrt{\lambda_j}$.

Proof. (a) $\frac{1}{n}\sum_i z_i = S^{-1/2}\Big(\frac{1}{n}\sum_i x_i - \bar{x}\Big) = 0$. As $S^{-1/2}$ is symmetric, the matrix with rows $z_i' = (x_i - \bar{x})'S^{-1/2}$ is $Z_0 = C_n X_0 S^{-1/2}$. Since C_n is symmetric and idempotent, $C_n Z_0 = Z_0$, and the sample covariance matrix of the z_i is

$$\frac{1}{n-1}Z_0'C_nZ_0 = \frac{1}{n-1}Z_0'Z_0 = S^{-1/2}\Big(\frac{1}{n-1}X_0'C_nX_0\Big)S^{-1/2} = S^{-1/2}SS^{-1/2} = I_k$$

(Theorem 8.25(c)).

(b) $\|z_i\|^2 = (x_i-\bar{x})'S^{-1/2}S^{-1/2}(x_i-\bar{x}) = (x_i-\bar{x})'S^{-1}(x_i-\bar{x})$ (Theorem 8.25(c)). The sum of the $\|z_i\|^2$ is the sum of the squares of all entries of Z_0, that is, $\operatorname{tr}(Z_0'Z_0) = \operatorname{tr}\big((n-1)I_k\big) = (n-1)k$ (Theorem 1.11(d)).

(c) The matrix with rows $y_i' = x_i'B' + c'$ is $Y_0 = X_0B' + \mathbf{1}c'$. So $\bar{y} = \frac{1}{n}Y_0'\mathbf{1} = B\bar{x} + c\cdot\frac{1}{n}\mathbf{1}'\mathbf{1} = B\bar{x}+c$; and $C_n\mathbf{1} = 0$ gives $C_nY_0 = C_nX_0B'$, so $S_y = \frac{1}{n-1}BX_0'C_nX_0B' = BSB'$, PD by Theorem 8.16(b). Now $S_y^{-1} = (B')^{-1}S^{-1}B^{-1}$ (Theorem 2.15(b)) and $y_i - y_l = B(x_i - x_l)$, so

$$(y_i-y_l)'S_y^{-1}(y_i-y_l) = (x_i-x_l)'B'(B')^{-1}S^{-1}B^{-1}B(x_i-x_l) = (x_i-x_l)'S^{-1}(x_i-x_l),$$

and in the same way with $y_i - \bar{y} = B(x_i-\bar{x})$.

(d) $QQ' = I_k$ gives $x - \bar{x} = Qw = \sum_j w_j q_j$, and $S^{-1} = Q\Lambda^{-1}Q'$ (Proposition 8.8(c)), so $d_S(x,\bar{x})^2 = (x-\bar{x})'Q\Lambda^{-1}Q'(x-\bar{x}) = w'\Lambda^{-1}w = \sum_j w_j^2/\lambda_j$. For $x = \bar{x}+tq_j$, $w = tQ'q_j = te_j$, so $d_S(x,\bar{x})^2 = t^2/\lambda_j$, which is $\leqslant c^2$ if and only if $|t| \leqslant c\sqrt{\lambda_j}$. □

In the coordinates w, the set in (d) is $\{w : \sum_j w_j^2/(c\sqrt{\lambda_j})^2 \leqslant 1\}$, the standard form of an ellipse or ellipsoid with semi-axes $c\sqrt{\lambda_1},\dots,c\sqrt{\lambda_k}$.

Example 8.37. For the scores of Example 8.34, find $S^{1/2}$, $S^{-1/2}$, the whitened data and the Mahalanobis distances of the students from $\bar{x}$.

Solution.

- Let $B = I_3 + J_3 = \left(\begin{smallmatrix} 2&1&1\\1&2&1\\1&1&2 \end{smallmatrix}\right)$. Its leading principal minors are $2, 3, 4$, so B is PD (Theorem 8.19); and $B^2 = I_3 + 2J_3 + J_3^2 = I_3 + 5J_3 = S$, as $J_3^2 = 3J_3$ (Proposition 1.13). By the uniqueness in Theorem 8.25(a), $S^{1/2} = B$. Next $(I_3 + J_3)(I_3 - \frac{1}{4}J_3) = I_3 + J_3 - \frac{1}{4}J_3 - \frac{3}{4}J_3 = I_3$, so

$$S^{-1/2} = B^{-1} = I_3 - \tfrac{1}{4}J_3 = \frac{1}{4}\begin{pmatrix} 3&-1&-1\\-1&3&-1\\-1&-1&3 \end{pmatrix}, \qquad S^{-1} = \left(S^{-1/2}\right)^2 = I_3 - \tfrac{5}{16}J_3$$

  (Theorem 8.25(c)).

- The whitened data $z_i = S^{-1/2}(x_i - \bar{x}) = (x_i - \bar{x}) - \frac{1}{4}\big(\mathbf{1}'(x_i - \bar{x})\big)\mathbf{1}$ are

$$(1,1,1)', \qquad (1,-1,-1)', \qquad (-1,1,-1)', \qquad (-1,-1,1)', \qquad (0,0,0)'$$

  (for student 2, $(0,-2,-2)' + \mathbf{1}$). They have mean 0 and sample covariance matrix I_3 (Proposition 8.36(a)); directly, the matrix Z_0 with rows z_i' has orthogonal columns of squared length 4, so $\frac{1}{4}Z_0'Z_0 = I_3$.

- By Proposition 8.36(b), $d_S(x_i, \bar{x})^2 = \|z_i\|^2$, which is $3, 3, 3, 3, 0$; the sum is $12 = (n-1)k$. The squared Euclidean distances from $\bar{x}$ are $48, 8, 8, 8, 0$. Student 1, four points above average on every test, follows the strong common pattern of the tests and is no more unusual than students 2, 3, 4, each average on one test and two points below on the other two.

- $S\mathbf{1} = 16\,\mathbf{1}$ and $Sv = v$ for every $v \perp \mathbf{1}$ (Example 7.17), so $\lambda_1 = 16$ and $\lambda_2 = \lambda_3 = 1$, with $q_1 = \frac{1}{\sqrt{3}}\mathbf{1}$. The ellipsoid $d_S(x, \bar{x}) \leqslant \sqrt{3}$ has the semi-axis $\sqrt{3}\cdot 4$ along q_1 and $\sqrt{3}\cdot 1$ in every direction orthogonal to q_1 (Proposition 8.36(d)). Student 1 lies at the end of the long axis: $x_1 = \bar{x} + 4\mathbf{1} = \bar{x} + 4\sqrt{3}\,q_1$.

Exercises

Exercise 8.1. Let $A = \begin{pmatrix} 5&3&2&0\\3&5&0&2\\2&0&5&3\\0&2&3&5 \end{pmatrix}$, and let $h_1, \dots, h_4$ be the vectors of Example 8.21.

(a) Show that $h_1, \dots, h_4$ are eigenvectors of A. Find an orthogonal Q and a diagonal Λ with $A = Q\Lambda Q'$ and decreasing diagonal entries, and write the spectral decomposition of A.

(b) Find $\operatorname{rank} A$ and bases of $\mathcal{C}(A)$ and $\mathcal{N}(A)$. Is A PD? PSD?

(c) Find the largest and the smallest value of $x'Ax$ on the unit sphere $x'x = 1$ of $\mathbb{R}^4$, and all points where they are attained.

Solution. (a) Multiplying the rows of A by $h_1, \dots, h_4$:

$$Ah_1 = \begin{pmatrix} 10\\10\\10\\10 \end{pmatrix} = 10h_1, \quad Ah_2 = \begin{pmatrix} 6\\6\\-6\\-6 \end{pmatrix} = 6h_2, \quad Ah_3 = \begin{pmatrix} 4\\-4\\4\\-4 \end{pmatrix} = 4h_3, \quad Ah_4 = \begin{pmatrix} 5-3-2\\3-5+2\\2-5+3\\-2-3+5 \end{pmatrix} = 0$$

The h_j are orthogonal with $\|h_j\| = 2$, so $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ is orthogonal, $AQ = Q\Lambda$ with $\Lambda = \operatorname{diag}(10, 6, 4, 0)$, and $A = Q\Lambda Q'$ (Theorem 8.4(a)). By Corollary 8.5(a), with $q_j = \frac{1}{2}h_j$,

$$A = 10q_1q_1' + 6q_2q_2' + 4q_3q_3' = \tfrac{5}{2}J_4 + \tfrac{3}{2}h_2h_2' + h_3h_3',$$

for instance the entries in row 1 are $\frac{5}{2} + \frac{3}{2} + 1 = 5$, $\frac{5}{2} + \frac{3}{2} - 1 = 3$, $\frac{5}{2} - \frac{3}{2} + 1 = 2$ and $\frac{5}{2} - \frac{3}{2} - 1 = 0$.

(b) By Corollary 8.5(b), $\operatorname{rank} A = 3$, $\mathcal{C}(A) = \operatorname{span}\{h_1, h_2, h_3\}$ and $\mathcal{N}(A) = \operatorname{span}\{h_4\}$; these vectors are independent (Proposition 6.8(b)), so they form bases. The eigenvalues $10, 6, 4, 0$ are $\geqslant 0$ and one is 0: A is PSD but not PD (Theorem 8.13(a),(b)).

(c) Put $y = Q'x$. Then $x = Qy$, $y'y = x'x = 1$ and $x'Ax = y'\Lambda y = 10y_1^2 + 6y_2^2 + 4y_3^2$. Hence $x'Ax = 10 - 4y_2^2 - 6y_3^2 - 10y_4^2 \leqslant 10$, with equality exactly when $y = \pm e_1$, that is, at $x = \pm q_1 = \pm\frac{1}{2}(1,1,1,1)'$; and $x'Ax \geqslant 0$, with equality exactly when $y = \pm e_4$, that is, at $x = \pm q_4 = \pm\frac{1}{2}(1,-1,-1,1)'$. The largest value is 10 and the smallest is 0 (Theorem 8.13(c)).

Exercise 8.2. Let $A = \begin{pmatrix} -1 & -2 & 2 & -2 \\ -2 & -1 & -2 & 2 \\ 2 & -2 & -1 & -2 \\ -2 & 2 & -2 & -1 \end{pmatrix}$.

(a) Show that $p_A(t) = (t-5)(t+3)^3$, find an orthonormal basis of each eigenspace, and an orthogonal Q with $Q'AQ$ diagonal.

(b) Find the projection matrices P_5 and P_{-3} onto $E_5(A)$ and $E_{-3}(A)$, and check that $A = 5P_5 - 3P_{-3}$.

(c) Use Definition 8.7 to compute A^{-1} and A^3.

Solution. (a) The rows of $tI_4 - A$ are $r_1 = (t+1, 2, -2, 2)$, $r_2 = (2, t+1, 2, -2)$, $r_3 = (-2, 2, t+1, 2)$ and $r_4 = (2, -2, 2, t+1)$. Replace r_1 by $r_1 - r_2 + r_3 - r_4 = (t-5)(1,-1,1,-1)$, take out $t-5$, and add column 1 to columns 2 and 4 and subtract it from column 3 (Theorem 5.16(d),(a),(f)):

$$p_A(t) = (t-5)\det\begin{pmatrix} 1 & -1 & 1 & -1 \\ 2 & t+1 & 2 & -2 \\ -2 & 2 & t+1 & 2 \\ 2 & -2 & 2 & t+1 \end{pmatrix} = (t-5)\det\begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & t+3 & 0 & 0 \\ -2 & 0 & t+3 & 0 \\ 2 & 0 & 0 & t+3 \end{pmatrix} = (t-5)(t+3)^3$$

(Theorem 5.15(a)).

- $v = (1,-1,1,-1)'$ has $Av = (-1+2+2+2,\ -2+1-2-2,\ 2+2-1+2,\ -2-2-2+1)' = 5v$, and $\dim E_5(A) = a_A(5) = 1$ (Theorem 8.4(b)); so $u_1 = \frac{1}{2}v$ is an orthonormal basis of $E_5(A)$.
- Every row of $A + 3I_4$ is $\pm 2v'$, so $E_{-3}(A) = \{x : x_1 - x_2 + x_3 - x_4 = 0\}$, of dimension $a_A(-3) = 3$ (Theorem 8.4(b)). The orthogonal vectors $(1,1,1,1)'$, $(1,1,-1,-1)'$, $(1,-1,-1,1)'$ lie in it, so $u_2 = \frac{1}{2}(1,1,1,1)'$, $u_3 = \frac{1}{2}(1,1,-1,-1)'$, $u_4 = \frac{1}{2}(1,-1,-1,1)'$ is an orthonormal basis of $E_{-3}(A)$ (Proposition 6.8(b), Theorem 4.5(d)).
- $Q = [u_1\ u_2\ u_3\ u_4]$ is orthogonal ($u_1 \perp u_2, u_3, u_4$ by Theorem 8.4(c)) and $Q'AQ = \operatorname{diag}(5,-3,-3,-3)$.

(b) By Corollary 8.5(c),

$$P_5 = u_1u_1' = \frac{1}{4}\begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}, \qquad P_{-3} = I_4 - P_5,$$

and $5P_5 - 3P_{-3} = 8P_5 - 3I_4 = 2vv' - 3I_4 = A$: the diagonal entries are $2 - 3 = -1$ and the others are $2v_iv_j = \pm 2$.

(c) With $f(t) = 1/t$ and $f(t) = t^3$ (Proposition 8.8(c)),

$$A^{-1} = \tfrac{1}{5}P_5 - \tfrac{1}{3}P_{-3} = \tfrac{8}{15}P_5 - \tfrac{1}{3}I_4 = \tfrac{1}{15}(2vv' - 5I_4) = \frac{1}{15}\begin{pmatrix} -3 & -2 & 2 & -2 \\ -2 & -3 & -2 & 2 \\ 2 & -2 & -3 & -2 \\ -2 & 2 & -2 & -3 \end{pmatrix},$$

$$A^3 = 125P_5 - 27P_{-3} = 152P_5 - 27I_4 = 38vv' - 27I_4 = \begin{pmatrix} 11 & -38 & 38 & -38 \\ -38 & 11 & -38 & 38 \\ 38 & -38 & 11 & -38 \\ -38 & 38 & -38 & 11 \end{pmatrix}$$

Check: row 1 of A times the first two columns of $15A^{-1}$ gives $3+4+4+4=15$ and $2+6-4-4=0$. (Thus $A^{-1}=\frac{1}{15}(A-2I_4)$, that is, $A^2-2A-15I_4=(A-5I_4)(A+3I_4)=0$.)

Exercise 8.3. Let $w=(1,-1,-1,1)'$ and $P=I_4-\frac{1}{4}ww'=\frac{1}{4}\begin{pmatrix}3&1&1&-1\\1&3&-1&1\\1&-1&3&1\\-1&1&1&3\end{pmatrix}$.

(a) Show that P is symmetric and idempotent, and find $\operatorname{rank} P$.

(b) Find Q_1 with orthonormal columns and $P=Q_1Q_1'$, as in Proposition 8.10(a).

(c) For $y=(2,0,0,2)'$, find the orthogonal projection $\hat{y}$ of y onto $\mathcal{C}(P)$, and check that $y-\hat{y}\perp\mathcal{C}(P)$.

Solution. (a) $P'=I_4-\frac{1}{4}(ww')'=P$, and with $w'w=4$, $P^2=I_4-\frac{1}{2}ww'+\frac{1}{16}w(w'w)w'=I_4-\frac{1}{4}ww'=P$. By Theorem 6.16(c), $\operatorname{rank}P=\operatorname{tr}P=\frac{1}{4}(3+3+3+3)=3$.

(b) With $h_1,\dots,h_4$ of Example 8.21, $Q=\frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ is orthogonal and $\frac{1}{2}h_4=\frac{1}{2}w$. By the outer product expansion (Theorem 1.4(d)), $I_4=QQ'=\frac{1}{4}\sum_j h_jh_j'$, so

$$P=I_4-\tfrac{1}{4}h_4h_4'=\tfrac{1}{4}(h_1h_1'+h_2h_2'+h_3h_3')=Q_1Q_1',\qquad Q_1=\frac{1}{2}\begin{pmatrix}1&1&1\\1&1&-1\\1&-1&1\\1&-1&-1\end{pmatrix},$$

and $Q_1'Q_1=I_3$.

(c) $P=P_{\mathcal{C}(P)}$ (Theorem 6.19(c)), so $\hat{y}=Py=y-\frac{1}{4}(w'y)w=(2,0,0,2)'-(1,-1,-1,1)'=(1,1,1,1)'$. Then $y-\hat{y}=w$, and $Q_1'w=0$, as $w=h_4$ is orthogonal to h_1,h_2,h_3; so $y-\hat{y}\perp\mathcal{C}(Q_1)=\mathcal{C}(P)$.

Exercise 8.4. Write each quadratic form as $x'Ax$ with A symmetric, and decide whether A is PD, PSD, negative definite, negative semidefinite or indefinite: (a) $x_1^2+6x_1x_2+x_2^2$; (b) $x_1^2+2x_2^2+x_3^2-2x_1x_2-2x_2x_3$; (c) $-2x_1^2-2x_2^2-2x_3^2+2x_1x_2+2x_2x_3$; (d) $2x_1^2+2x_2^2+2x_3^2+x_4^2-2x_1x_2-2x_2x_3-2x_3x_4$.

Solution. (a) $A=\left(\begin{smallmatrix}1&3\\3&1\end{smallmatrix}\right)$ gives 8 at $(1,1)'$ and $1-6+1=-4$ at $(1,-1)'$: indefinite. (The eigenvalues are 4 and -2.)

(b) $A=\left(\begin{smallmatrix}1&-1&0\\-1&2&-1\\0&-1&1\end{smallmatrix}\right)$. The form equals $(x_1-x_2)^2+(x_2-x_3)^2\geqslant 0$, and it is 0 at $x=(1,1,1)'\neq 0$: PSD but not PD. The leading principal minors $1,1,0$ do not decide this (Remark 8.20); the eigenvalues are $0,1,3$, with eigenvectors $(1,1,1)'$, $(1,0,-1)'$, $(1,-2,1)'$.

(c) $A=-\left(\begin{smallmatrix}2&-1&0\\-1&2&-1\\0&-1&2\end{smallmatrix}\right)$. The leading principal minors of $-A$ are 2, 3 and 4, so $-A$ is PD (Theorem 8.19) and A is negative definite.

(d) $A=\left(\begin{smallmatrix}2&-1&0&0\\-1&2&-1&0\\0&-1&2&-1\\0&0&-1&1\end{smallmatrix}\right)$, the matrix $A^{-1}=K'K$ of Example 8.24. So $x'Ax=\|Kx\|^2=x_1^2+(x_2-x_1)^2+(x_3-x_2)^2+(x_4-x_3)^2$, and A is PD (Theorem 8.16(a), as $\operatorname{rank}K=4$). Its leading principal minors are $2,3,4,1$.

Exercise 8.5. For $a\in\mathbb{R}$ let $A=\begin{pmatrix}4&2&2&a\\2&4&a&2\\2&a&4&2\\a&2&2&4\end{pmatrix}$, and let $h_1,\dots,h_4$ be the vectors of Example 8.21.

(a) Show that $h_1,\dots,h_4$ are eigenvectors of A, and find $\det A$.

(b) For which a is A PD? Answer with Theorem 8.19, and check with (a).

(c) For which a is A PSD? Find $\mathcal{N}(A)$ at the endpoints.

Solution. (a) The row sums are $8+a$, so $Ah_1 = (8+a)h_1$. Multiplying the rows of A by h_2, h_3, h_4,

$$Ah_2 = \begin{pmatrix} 4+2-2-a \\ 2+4-a-2 \\ 2+a-4-2 \\ a+2-2-4 \end{pmatrix} = (4-a)h_2, \quad Ah_3 = \begin{pmatrix} 4-2+2-a \\ 2-4+a-2 \\ 2-a+4-2 \\ a-2+2-4 \end{pmatrix} = (4-a)h_3,$$

and in the same way $Ah_4 = (4-2-2+a,\ 2-4-a+2,\ 2-a-4+2,\ a-2-2+4)' = ah_4$. So $A = Q\Lambda Q'$ with $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ and $\Lambda = \operatorname{diag}(8+a, 4-a, 4-a, a)$, and $\det A = a(a+8)(a-4)^2$ (Theorem 8.4(b), Corollary 7.30(a)).

(b) The leading principal minors are 4, $16-4=12$,

$$\det A_{[3]} = 4(16-a^2) - 2(8-2a) + 2(2a-8) = 32 + 8a - 4a^2 = -4(a-4)(a+2)$$

(expanding along the first row), and $\det A = a(a+8)(a-4)^2$. The third is positive exactly for $-2 < a < 4$, and the fourth exactly for $a \neq 4$ and $a(a+8) > 0$, that is, $a > 0$ or $a < -8$. By Theorem 8.19, A is PD if and only if $0 < a < 4$. By (a) and Theorem 8.13(a), A is PD if and only if $8+a > 0$, $4-a > 0$ and $a > 0$: the same condition.

(c) By (a) and Theorem 8.13(b), A is PSD if and only if $8+a \geqslant 0$, $4-a \geqslant 0$ and $a \geqslant 0$, that is, $0 \leqslant a \leqslant 4$. By Corollary 8.5(b), $\mathcal{N}(A) = \operatorname{span}\{h_4\}$ for $a = 0$ (eigenvalues $8, 4, 4, 0$) and $\mathcal{N}(A) = \operatorname{span}\{h_2, h_3\}$ for $a = 4$ (eigenvalues $12, 0, 0, 4$).

Exercise 8.6. Let $A = \begin{pmatrix} 9 & 3 & -6 & 3 \\ 3 & 5 & 0 & -1 \\ -6 & 0 & 9 & -1 \\ 3 & -1 & -1 & 4 \end{pmatrix}$. (a) Find the Cholesky factor L of A. (b) Find $\det A$. (c) Solve $Ax = b$ for $b = (9, 7, 2, 5)'$ by solving $Ly = b$ and then $L'x = y$.

Solution. (a) By Remark 8.23:

- column 1: $l_{11} = \sqrt{9} = 3$, $l_{21} = \frac{3}{3} = 1$, $l_{31} = -\frac{6}{3} = -2$, $l_{41} = \frac{3}{3} = 1$;
- column 2: $l_{22} = \sqrt{5-1} = 2$, $l_{32} = \big(0 - (-2)\cdot 1\big)/2 = 1$, $l_{42} = (-1-1\cdot 1)/2 = -1$;
- column 3: $l_{33} = \sqrt{9-4-1} = 2$, $l_{43} = \big(-1-1\cdot(-2)-(-1)\cdot 1\big)/2 = 1$;
- column 4: $l_{44} = \sqrt{4-1-1-1} = 1$.

So $L = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & 1 & 2 & 0 \\ 1 & -1 & 1 & 1 \end{pmatrix}$, and $LL' = A$; in particular A is PD (Theorem 8.22).

(b) $\det A = \det L \cdot \det L' = (3\cdot 2\cdot 2\cdot 1)^2 = 144$ (Theorems 5.19(b) and 5.15(a),(b)).

(c) $Ax = L(L'x)$, so x solves $Ax = b$ if $Ly = b$ and $L'x = y$. From the first equation downwards, $3y_1 = 9$, $y_1 + 2y_2 = 7$, $-2y_1 + y_2 + 2y_3 = 2$, $y_1 - y_2 + y_3 + y_4 = 5$ give $y = (3, 2, 3, 1)'$. Back substitution in $L'x = y$: $x_4 = 1$, $2x_3 + x_4 = 3$, $2x_2 + x_3 - x_4 = 2$, $3x_1 + x_2 - 2x_3 + x_4 = 3$ give $x = (1, 1, 1, 1)'$. Check: the row sums of A are $9, 7, 2, 5$. The solution is unique, as A is invertible.

Exercise 8.7. Let $A = I_4 + 6J_4$, the 4×4 matrix with diagonal entries 7 and all other entries 6.

(a) Find the eigenvalues of A, and compute $A^{1/2}$, $A^{-1/2}$ and A^{-1}. Check that $(A^{1/2})^2 = A$.

(b) Show that $C = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \end{pmatrix}$ also satisfies $C^2 = A$. Why does this not contradict Theorem 8.25(a)?

Solution. (a) By Example 7.17 (with $\alpha = 1$, $\beta = 6$), the eigenvalues are 25, with $E_{25}(A) = \operatorname{span}\{\mathbf{1}_4\}$, and 1, with $E_1(A) = W_0 = \{x : \mathbf{1}'x = 0\}$. The projection matrices onto these eigenspaces are $\bar{J}_4 = \frac{1}{4}J_4$ and $C_4 = I_4 - \frac{1}{4}J_4$ (Corollary 6.21(a)), so $A = 25\bar{J}_4 + C_4$ (Corollary 8.5(c)), and by Theorem 8.25(a),(c) and Definition 8.7

$$A^{1/2} = 5\bar{J}_4 + C_4 = I_4 + J_4, \qquad A^{-1/2} = \tfrac{1}{5}\bar{J}_4 + C_4 = I_4 - \tfrac{1}{5}J_4, \qquad A^{-1} = \tfrac{1}{25}\bar{J}_4 + C_4 = I_4 - \tfrac{6}{25}J_4$$

Indeed $(I_4 + J_4)^2 = I_4 + 2J_4 + J_4^2 = I_4 + 6J_4$, as $J_4^2 = 4J_4$ (Proposition 1.13).

(b) The $(1,1)$, $(1,2)$, $(1,3)$, $(1,4)$ entries of C^2 are $1+4+1+1 = 7$, $2+2+1+1 = 6$, $1+2+1+2 = 6$, $1 + 2 + 2 + 1 = 6$, and the other rows follow in the same way: $C^2 = A$. But C is not PSD: $(1,-1,0,0)C(1,-1,0,0)' = 1 - 4 + 1 = -2$. Theorem 8.25(a) asserts uniqueness only among PSD matrices. (With $h_1, \ldots, h_4$ of Example 8.21, $C = \frac{1}{4}(5h_1h_1' + h_2h_2' - h_3h_3' - h_4h_4')$, with the eigenvalues $5, 1, -1, -1$.)

Exercise 8.8. Let $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 4 & 0 & 0 \\ 4 & 7 & 3 & 0 \\ 0 & 3 & 5 & 2 \\ 0 & 0 & 2 & 3 \end{pmatrix}$.

(a) Show that A is PD, and find an invertible P with $P'AP = I_4$ and $P'BP$ diagonal.

(b) Show that $x'Ax \leqslant x'Bx \leqslant 4\,x'Ax$ for every $x \in \mathbb{R}^4$, that is, $A \leqslant B \leqslant 4A$.

Solution. (a) Expanding $\det A_{[k]}$ along its last row gives $\det A_{[k]} = 2\det A_{[k-1]} - \det A_{[k-2]}$ for $k = 3, 4$ (Theorem 5.13), so the leading principal minors are 1, $2 - 1 = 1$, $2 - 1 = 1$, $2 - 1 = 1$, and A is PD (Theorem 8.19). The rows of $B - \theta A$ are $(4-\theta,\ 4-\theta,\ 0,\ 0)$, $(4-\theta,\ 7-2\theta,\ 3-\theta,\ 0)$, $(0,\ 3-\theta,\ 5-2\theta,\ 2-\theta)$, $(0,\ 0,\ 2-\theta,\ 3-2\theta)$. Subtract row 1 from row 2, then the new row 2 from row 3, then the new row 3 from row 4 (Theorem 5.16(d)):

$$\det(B - \theta A) = \det \begin{pmatrix} 4-\theta & 4-\theta & 0 & 0 \\ 0 & 3-\theta & 3-\theta & 0 \\ 0 & 0 & 2-\theta & 2-\theta \\ 0 & 0 & 0 & 1-\theta \end{pmatrix} = (4-\theta)(3-\theta)(2-\theta)(1-\theta),$$

so $\theta_1 = 4$, $\theta_2 = 3$, $\theta_3 = 2$, $\theta_4 = 1$ (Theorem 8.30(b), as $\det A = 1$). The null spaces of $B - 4A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & -2 \\ 0 & 0 & -2 & -5 \end{pmatrix}$, $B - 3A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{pmatrix}$, $B - 2A = \begin{pmatrix} 2 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ and $B - A = \begin{pmatrix} 3 & 3 & 0 & 0 \\ 3 & 5 & 2 & 0 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ contain $p_1 = (1,0,0,0)'$, $p_2 = (-1,1,0,0)'$, $p_3 = (1,-1,1,0)'$ and $p_4 = (-1,1,-1,1)'$. Here $Ap_1 = (1,1,0,0)'$, $Ap_2 = (0,1,1,0)'$, $Ap_3 = (0,0,1,1)'$ and $Ap_4 = (0,0,0,1)'$, so $p_k'Ap_k = 1$, and $p_j'Ap_k = 0$ for $j \neq k$ (Theorem 8.30(c)). Hence

$$P = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

has $P'AP = I_4$ and $P'BP = \operatorname{diag}(4,3,2,1)$, since $(P'BP)_{jk} = p_j'Bp_k = \theta_k p_j'Ap_k$.

(b) $\det P = 1$, so every x is $x = Py$ with $y = P^{-1}x$; here P^{-1} is the upper triangular matrix with ones on the diagonal and on the superdiagonal (multiply out P with it), so $y_k = x_k + x_{k+1}$ for $k \leqslant 3$ and $y_4 = x_4$. Then $x'Ax = y'P'APy = y_1^2 + y_2^2 + y_3^2 + y_4^2$ and $x'Bx = y'P'BPy = 4y_1^2 + 3y_2^2 + 2y_3^2 + y_4^2$. Hence $x'Ax \leqslant x'Bx \leqslant 4\,x'Ax$.

Exercise 8.9. Three stocks each have correlation $\frac{2}{3}$ with a market index, and any two of the stocks have correlation r. The correlation matrix of (index, stock 1, stock 2, stock 3) is

$$R_r = \begin{pmatrix} 1 & \frac{2}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & 1 & r & r \\ \frac{2}{3} & r & 1 & r \\ \frac{2}{3} & r & r & 1 \end{pmatrix}$$

(a) Show that R_r is PSD if and only if $\frac{1}{6} \leqslant r \leqslant 1$, and PD if and only if $\frac{1}{6} < r < 1$.

(b) An analyst reports the sample correlations $\frac{2}{3}, \frac{2}{3}, \frac{2}{3}$ of three series with a fourth, and reports the three series as pairwise uncorrelated, all observed over the same periods. Explain why the report must contain an error.

Solution. (a) The block $A_{11} = (1)$ is PD, and its Schur complement is

$$A_{22\cdot 1} = \begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix} - \tfrac{4}{9} J_3 = (1-r) I_3 + \left(r - \tfrac{4}{9}\right) J_3$$

By Example 7.17, its eigenvalues are $(1-r) + 3\left(r - \frac{4}{9}\right) = 2r - \frac{1}{3}$ (eigenvector $\mathbf{1}_3$) and $1 - r$. By Lemma 8.18 and Theorem 8.13(a),(b), R_r is PSD if and only if $2r - \frac{1}{3} \geqslant 0$ and $1 - r \geqslant 0$, that is, $\frac{1}{6} \leqslant r \leqslant 1$; and PD if and only if both are positive, that is, $\frac{1}{6} < r < 1$.

(b) A sample correlation matrix is PSD (Proposition 8.33(c)), and $0 < \frac{1}{6}$, so by (a) the reported matrix R_0 is not; for instance $x = (-2, 1, 1, 1)'$ gives $x' R_0 x = 4 + 3 - 2 \cdot \frac{2}{3} \cdot 2 \cdot 3 = -1 < 0$. More generally, for any sample correlations r_{12}, r_{13}, r_{23} among the three series, the same x gives $0 \leqslant 7 - 8 + 2(r_{12} + r_{13} + r_{23})$: three series that each correlate $\frac{2}{3}$ with a fourth have pairwise correlations adding up to at least $\frac{1}{2}$.

Exercise 8.10. The monthly returns (in %) of three funds over five months are:

month	1	2	3	4	5
fund 1	10	2	8	4	6
fund 2	10	14	8	12	6
fund 3	10	8	8	8	6

Let $X_0 \in \mathbb{R}^{5 \times 3}$ have the months as rows.

(a) Compute the mean vector $\bar{x}$, the sample covariance matrix S and the sample correlation matrix R.

(b) The equally weighted portfolio has the returns $a' x_i$ with $a = \frac{1}{3}(1, 1, 1)'$. Compute their sample variance as $a' S a$, and directly from the five portfolio returns.

(c) Show that S is singular, find $a \neq 0$ with $a' S a = 0$, and explain this from the data. What is $\operatorname{rank}(C_5 X_0)$?

Solution. (a) $\bar{x} = \frac{1}{5}(30, 50, 40)' = (6, 10, 8)'$, and

$$C_5 X_0 = \begin{pmatrix} 4 & 0 & 2 \\ -4 & 4 & 0 \\ 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & -4 & -2 \end{pmatrix}, \qquad X_0' C_5 X_0 = \begin{pmatrix} 40 & -24 & 8 \\ -24 & 40 & 8 \\ 8 & 8 & 8 \end{pmatrix}, \qquad S = \frac{1}{4} X_0' C_5 X_0 = \begin{pmatrix} 10 & -6 & 2 \\ -6 & 10 & 2 \\ 2 & 2 & 2 \end{pmatrix}$$

With $D = \operatorname{diag}(10, 10, 2)$, $r_{12} = -\frac{6}{10} = -\frac{3}{5}$ and $r_{13} = r_{23} = \frac{2}{\sqrt{10 \cdot 2}} = \frac{1}{\sqrt{5}}$; so $R = \begin{pmatrix} 1 & -3/5 & 1/\sqrt{5} \\ -3/5 & 1 & 1/\sqrt{5} \\ 1/\sqrt{5} & 1/\sqrt{5} & 1 \end{pmatrix}$.

(b) $a'Sa = \frac{1}{9}\mathbf{1}'S\mathbf{1}$, and the entries of S add up to 18; so $a'Sa = 2$ (Proposition 8.33(a)). Directly, the portfolio returns are $10, 8, 8, 8, 6$, with mean 8 and squared deviations $4, 0, 0, 0, 4$; their sample variance is $\frac{8}{4} = 2$. Each of funds 1 and 2 alone has sample variance 10; their negative correlation makes the portfolio much less variable.

(c) In every month the return of fund 3 is the average of those of funds 1 and 2: $x_{i3} = \frac{1}{2}(x_{i1} + x_{i2})$. So for $a = (1, 1, -2)'$, every $a'x_i$ is 0; their sample variance $a'Sa$ is 0 (Proposition 8.33(a)), and $Sa = 0$ (Proposition 8.17(a)); directly, $S(1, 1, -2)' = (10 - 6 - 4,\ -6 + 10 - 4,\ 2 + 2 - 4)' = 0$. Thus S is singular and not PD, so $\operatorname{rank}(C_5X_0) < 3$ (Proposition 8.33(b)). The first two columns of C_5X_0 are not multiples of each other, so $\operatorname{rank}(C_5X_0) = 2$. (Fund 3 is a 50:50 mix of funds 1 and 2, and the combination "one unit each of funds 1 and 2, minus two units of fund 3" has the constant return 0.)

Exercise 8.11. Seven students take two quizzes (scores out of 10):

student	1	2	3	4	5	6	7
quiz 1	9	7	3	4	9	7	3
quiz 2	8	5	2	5	8	5	2

(a) Compute $\bar{x}$, S, S^{-1} and the sample correlation r_{12}.

(b) Compute the squared Mahalanobis distances $d_i^2 = d_S(x_i, \bar{x})^2$ of all students, and check $\sum_i d_i^2 = (n-1)k$.

(c) A screening rule flags a student with $d_i^2 > 3$ as unusual. Which student is flagged? Compare with the standardized scores $(x_{ij} - \bar{x}_j)/\sqrt{s_{jj}}$: is the flagged student more than one standard deviation from the average on either quiz?

Solution. (a) $\bar{x} = \frac{1}{7}(42, 35)' = (6, 5)'$. The rows of C_7X_0 are $(3, 3)$, $(1, 0)$, $(-3, -3)$, $(-2, 0)$, $(3, 3)$, $(1, 0)$, $(-3, -3)$, so

$$X_0'C_7X_0 = \begin{pmatrix} 42 & 36 \\ 36 & 36 \end{pmatrix}, \qquad S = \frac{1}{6}\begin{pmatrix} 42 & 36 \\ 36 & 36 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ 6 & 6 \end{pmatrix}$$

(for instance $42 = 9 + 1 + 9 + 4 + 9 + 1 + 9$). Then $\det S = 42 - 36 = 6$, and by the 2×2 formula of Proposition 2.13

$$S^{-1} = \frac{1}{6}\begin{pmatrix} 6 & -6 \\ -6 & 7 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & \frac{7}{6} \end{pmatrix}, \qquad r_{12} = \frac{6}{\sqrt{7 \cdot 6}} = \sqrt{\tfrac{6}{7}}$$

(b) For a deviation $(u, v) = (x_i - \bar{x})'$, $d_i^2 = u^2 - 2uv + \frac{7}{6}v^2 = (u - v)^2 + \frac{1}{6}v^2$. This gives

$$d_1^2 = d_3^2 = d_5^2 = d_7^2 = \tfrac{3}{2}, \qquad d_2^2 = d_6^2 = 1, \qquad d_4^2 = 4,$$

with sum $6 + 2 + 4 = 12 = 6 \cdot 2$, as Proposition 8.36(b) says.

(c) Only student 4 is flagged: $d_4^2 = 4 > 3$, while all other $d_i^2 \leqslant \frac{3}{2}$. The standardized scores of student 4 are $-2/\sqrt{7}$ and 0, and $2 < \sqrt{7}$: less than one standard deviation below average on quiz 1, exactly average on quiz 2. Students $1, 3, 5, 7$ have the standardized scores $\pm 3/\sqrt{7}$ and $\pm 3/\sqrt{6}$, more than one standard deviation from average on both quizzes ($3 > \sqrt{7}$, $3 > \sqrt{6}$), but $d_i^2 = \frac{3}{2}$. The quizzes are strongly correlated ($r_{12}^2 = \frac{6}{7}$); students $1, 3, 5, 7$ follow the common pattern, while student 4 is below average on quiz 1 without being below average on quiz 2. The Mahalanobis distance detects this; the coordinatewise scores do not.

Problems

Problem 8.1. Let $A, B \in \mathbb{R}^{n\times n}$ be PSD. Show that $\operatorname{tr}(AB) \geqslant 0$, with equality if and only if $AB = 0$.

Solution. Let $F = B^{1/2}A^{1/2}$ (Theorem 8.25(a)). Then $F'F = A^{1/2}B^{1/2}B^{1/2}A^{1/2} = A^{1/2}BA^{1/2}$, and by Theorem 1.11(b),(d),

$$\operatorname{tr}(AB) = \operatorname{tr}\Big(A^{1/2}(A^{1/2}B)\Big) = \operatorname{tr}\Big(A^{1/2}BA^{1/2}\Big) = \operatorname{tr}(F'F) = \sum_{i,j} f_{ij}^2 \geqslant 0$$

If $AB = 0$, then $\operatorname{tr}(AB) = 0$. Conversely, if $\operatorname{tr}(AB) = 0$, then $\operatorname{tr}(F'F) = 0$, so $F = 0$ (Theorem 1.11(d)); hence $BA = B^{1/2}FA^{1/2} = 0$, and $AB = A'B' = (BA)' = 0$.

Problem 8.2. (a) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Show that $A \geqslant B \geqslant 0$, but that $A^2 \geqslant B^2$ fails.

(b) Let $A, B \in \mathbb{R}^{n\times n}$ with $A \geqslant B \geqslant 0$. Show that $\operatorname{tr}(A^2) \geqslant \operatorname{tr}(B^2)$.

Solution. (a) $A - B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) = \mathbf{1}\mathbf{1}'$ and $B = e_1e_1'$ are PSD (Theorem 8.16(a)). But $A^2 = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 2 \end{smallmatrix}\right)$, $B^2 = B$, and $A^2 - B^2 = \left(\begin{smallmatrix} 4 & 3 \\ 3 & 2 \end{smallmatrix}\right)$ has determinant $-1 < 0$, so it is not PSD (Corollary 8.15(b)); for instance $x = (2, -3)'$ gives $x'(A^2 - B^2)x = 16 - 36 + 18 = -2$. Squaring does not preserve the Loewner order.

(b) By Theorem 1.11(a),(b), $\operatorname{tr}\Big((A-B)(A+B)\Big) = \operatorname{tr}(A^2)+\operatorname{tr}(AB)-\operatorname{tr}(BA)-\operatorname{tr}(B^2) = \operatorname{tr}(A^2)-\operatorname{tr}(B^2)$. Both $A - B$ and $A + B$ are PSD ($A \geqslant 0$ by Proposition 8.28(a), and a sum of PSD matrices is PSD), so this trace is $\geqslant 0$ by Problem 8.1. In (a), $\operatorname{tr}(A^2) = 7 \geqslant 1 = \operatorname{tr}(B^2)$.

Problem 8.3. Let $n \geqslant 2$, $\rho \in \mathbb{R}$, and let $R = (1 - \rho)I_n + \rho J_n$ be the equicorrelation matrix of Example 5.28.

(a) Show that $x'Rx = (1 - \rho)\|C_nx\|^2 + n\Big(1 + (n - 1)\rho\Big)\bar{x}^2$ for $x \in \mathbb{R}^n$, where $\bar{x} = \frac{1}{n}\mathbf{1}'x$.

(b) Show that R is PD if and only if $-\frac{1}{n-1} < \rho < 1$, and PSD if and only if $-\frac{1}{n-1} \leqslant \rho \leqslant 1$. Compare with the eigenvalues of R found in Example 7.17.

(c) Can n variables of a data matrix have all pairwise sample correlations equal to some $\rho < -\frac{1}{n-1}$?

Solution. (a) $x'J_nx = (\mathbf{1}'x)^2 = n^2\bar{x}^2$. Also $x = \bar{x}\mathbf{1} + C_nx$ (Example 1.14) with $\bar{x}\mathbf{1} \perp C_nx$ ($\mathbf{1}'C_nx = (C_n\mathbf{1})'x = 0$), so $x'x = n\bar{x}^2 + \|C_nx\|^2$ (Pythagoras, Proposition 6.2). Hence $x'Rx = (1 - \rho)\Big(n\bar{x}^2 + \|C_nx\|^2\Big) + \rho n^2\bar{x}^2 = (1 - \rho)\|C_nx\|^2 + n(1 - \rho + n\rho)\bar{x}^2$.

(b) Let $1 - \rho > 0$ and $1 + (n - 1)\rho > 0$. Both terms of (a) are $\geqslant 0$, and $x'Rx = 0$ forces $C_nx = 0$ and $\bar{x} = 0$, so $x = \bar{x}\mathbf{1} + C_nx = 0$: R is PD. Conversely, $x = \mathbf{1}$ gives $\bar{x} = 1$, $C_nx = 0$ and $x'Rx = n\Big(1 + (n - 1)\rho\Big)$, and $x = e_1 - e_2$ gives $\bar{x} = 0$, $C_nx = x$ and $x'Rx = 2(1 - \rho)$; so if R is PD, then $1 + (n - 1)\rho > 0$ and $1 - \rho > 0$, that is, $-\frac{1}{n-1} < \rho < 1$. The same argument with $\geqslant$ gives the PSD statement. By Example 7.17, the eigenvalues of R are $1 + (n - 1)\rho$ (eigenvector $\mathbf{1}$) and $1 - \rho$ (on W_0, multiplicity $n - 1$), so Theorem 8.13(a),(b) gives the same conditions.

(c) No: a sample correlation matrix is PSD (Proposition 8.33(c)), and by (b) the equicorrelation matrix with $\rho < -\frac{1}{n-1}$ is not. For three variables, even unequal pairwise sample correlations r_{12}, r_{13}, r_{23} cannot all be below $-\frac{1}{2}$: at $x = \mathbf{1}_3$, the quadratic form of their sample correlation matrix takes the value $3 + 2(r_{12} + r_{13} + r_{23}) < 3 - 3 = 0$.

Problem 8.4. (a) Let $A \in \mathbb{R}^{n\times n}$ be PD. Show that $\det A \leqslant a_{11}a_{22}\cdots a_{nn}$, with equality if and only if A is diagonal (**Hadamard's inequality**). Use Theorem 8.22.

(b) Deduce that $(\det B)^2 \leqslant \|b_1\|^2 \cdots \|b_n\|^2$ for every $B = [b_1 \ \cdots \ b_n] \in \mathbb{R}^{n\times n}$.

Solution. (a) Let $A = LL'$ with L the Cholesky factor. Then $\det A = (\det L)^2 = l_{11}^2 \cdots l_{nn}^2$ (Theorems 5.19(b) and 5.15(a)), and by Remark 8.23 $a_{jj} = \sum_{k\leqslant j} l_{jk}^2 \geqslant l_{jj}^2 > 0$. Multiplying these n inequalities between positive numbers gives $\det A \leqslant a_{11}\cdots a_{nn}$, with equality if and only if $a_{jj} = l_{jj}^2$ for every j, that is, $l_{jk} = 0$ for all $k < j$: L is diagonal. If L is diagonal, so is $A = LL'$; if A is diagonal, $\operatorname{diag}(\sqrt{a_{11}}, \ldots, \sqrt{a_{nn}})$ is a Cholesky factor of A, so L is diagonal by uniqueness.

(b) If $\det B = 0$, there is nothing to prove. Otherwise $\operatorname{rank} B = n$ (Theorem 5.19(a)), so $A = B'B$ is PD (Theorem 8.16(a)), with $a_{jj} = b_j'b_j = \|b_j\|^2$ and $\det A = \det B' \det B = (\det B)^2$ (Theorems 5.19(b) and 5.15(b)). Apply (a). For example, $B = [h_1 \ h_2 \ h_3 \ h_4]$, with the vectors of Example 8.21, has $B'B = 4I_4$, diagonal, and $(\det B)^2 = \det(B'B) = 4^4 = \|h_1\|^2 \cdots \|h_4\|^2$: equality.

Problem 8.5. Let W be a subspace of $\mathbb{R}^n$, and let $P = P_W = (p_{ij})$ be the projection matrix onto W (Theorem 6.19).

(a) Show that $0 \leqslant P \leqslant I_n$ in the Loewner order, and that $0 \leqslant p_{ii} \leqslant 1$ for every i.

(b) Show that $p_{ij}^2 \leqslant p_{ii}p_{jj}$ for all i, j, and that $p_{ii} = 0$ or $p_{ii} = 1$ implies $p_{ij} = 0$ for all $j \neq i$.

(c) Let $\mathbf{1}_n \in W$. Show that $p_{ii} \geqslant \frac{1}{n}$ for every i.

Solution. (a) P is symmetric and idempotent, and so is $I_n - P = P_{W^\perp}$ (Theorem 6.19(b)). Hence $P = P'P$ and $I_n - P = (I_n - P)'(I_n - P)$ are PSD (Theorem 8.16(a)), that is, $P \geqslant 0$ and $I_n \geqslant P$ (Definition 8.27). Their diagonal entries p_{ii} and $1 - p_{ii}$ are $\geqslant 0$ (Corollary 8.15(b)).

(b) $p_{ij}^2 \leqslant p_{ii}p_{jj}$ by Proposition 8.17(c). If $p_{ii} = 0$, row i of the PSD matrix P is zero (Proposition 8.17(b)). If $p_{ii} = 1$, the diagonal entry $1 - p_{ii}$ of the PSD matrix $I_n - P$ is 0, so row i of $I_n - P$ is zero: $-p_{ij} = 0$ for $j \neq i$.

(c) Let $U = \operatorname{span}\{\mathbf{1}_n\} \subseteq W$. Then $P_U = \bar{J}_n$ (Corollary 6.21(a)), and $P - \bar{J}_n$ is the projection matrix onto the subspace $W \cap U^\perp$ (Theorem 6.25(b)). By (a), applied to that subspace, its diagonal entries $p_{ii} - \frac{1}{n}$ are $\geqslant 0$. For example, P of Exercise 8.3 has $P\mathbf{1}_4 = \mathbf{1}_4 - \frac{1}{4}(w'\mathbf{1}_4)w = \mathbf{1}_4$, so $\mathbf{1}_4 \in \mathcal{C}(P)$, and indeed $p_{ii} = \frac{3}{4} \geqslant \frac{1}{4}$.

Problem 8.6. Let $A \in \mathbb{R}^{n\times n}$ be PD.

(a) Show that $(x'y)^2 \leqslant (x'Ax)(y'A^{-1}y)$ for all $x, y \in \mathbb{R}^n$, with equality if and only if one of x and $A^{-1}y$ is a multiple of the other.

(b) Let $y \neq 0$. Show that the largest value of $(x'y)^2/(x'Ax)$ over $x \neq 0$ is $y'A^{-1}y$, attained exactly at the vectors $x = cA^{-1}y$ with $c \neq 0$.

(c) Let $X_0 \in \mathbb{R}^{n\times k}$ have rows x_i', mean vector $\bar{x}$ and a PD sample covariance matrix S, and let $\mu \in \mathbb{R}^k$. For $a \neq 0$, the numbers $a'x_i$ have mean $a'\bar{x}$ and sample variance $a'Sa$ (Proposition 8.33(a)). Show that

$$\max_{a\neq 0} \frac{(a'\bar{x} - a'\mu)^2}{a'Sa} = d_S(\bar{x}, \mu)^2$$

Solution. (a) Let $u = A^{1/2}x$ and $v = A^{-1/2}y$ (Theorem 8.25). As $A^{1/2}$ is symmetric and $A^{1/2}A^{-1/2} = I_n$, $u'v = x'A^{1/2}A^{-1/2}y = x'y$; also $\|u\|^2 = x'Ax$ (Theorem 8.25(b)) and $\|v\|^2 = y'A^{-1/2}A^{-1/2}y = y'A^{-1}y$ (Theorem 8.25(c)). By the Cauchy–Schwarz inequality (Theorem 6.3(a)), $(x'y)^2 = (u'v)^2 \leqslant \|u\|^2\|v\|^2$, with equality if and only if one of u, v is a multiple of the other. Multiplying by $A^{-1/2}$, $u = cv$ holds if and only if $x = cA^{-1}y$, and $v = cu$ if and only if $A^{-1}y = cx$.

(b) For $x \neq 0$, $x'Ax > 0$, so by (a) the ratio is at most $y'A^{-1}y$. As $x \neq 0$ and $A^{-1}y \neq 0$, equality holds if and only if $x = cA^{-1}y$ with $c \neq 0$; and then $(x'y)^2/(x'Ax) = c^2(y'A^{-1}y)^2/\left(c^2y'A^{-1}AA^{-1}y\right) = y'A^{-1}y$.

(c) If $\bar{x} = \mu$, both sides are 0. Otherwise apply (b) with $A = S$, $y = \bar{x} - \mu$ and $x = a$: the maximum is $(\bar{x} - \mu)'S^{-1}(\bar{x} - \mu) = d_S(\bar{x}, \mu)^2$ (Definition 8.35), attained at $a = S^{-1}(\bar{x} - \mu)$. So the squared Mahalanobis distance is the largest squared standardized difference $(a'\bar{x} - a'\mu)^2/(a'Sa)$ over all linear combinations of the variables.

Problem 8.7. For $A, B \in \mathbb{R}^{n\times n}$, the **Hadamard product** $A \circ B \in \mathbb{R}^{n\times n}$ has the entries $(A \circ B)_{ij} = a_{ij}b_{ij}$.

(a) For $v \in \mathbb{R}^n$ let $D_v = \operatorname{diag}(v_1, \ldots, v_n)$. Show that $A \circ (vv') = D_v A D_v$.

(b) Let $A, B \in \mathbb{R}^{n\times n}$ be PSD. Show that $A \circ B$ is PSD (**Schur product theorem**).

(c) Let A be PD and B PSD with $b_{ii} > 0$ for every i. Show that $A \circ B$ is PD.

(d) Let R be the sample correlation matrix of Example 8.34. Show that the matrix $R \circ R$ of squared correlations is PD.

Solution. (a) The (i, j) entry of $D_v A D_v$ is $v_i a_{ij} v_j = a_{ij}(vv')_{ij}$ (Theorem 1.4(d)).

(b) Write $B = \sum_k \lambda_k q_k q_k'$ with Q orthogonal and all $\lambda_k \geqslant 0$ (Corollary 8.5(a), Theorem 8.13(b)). The Hadamard product is computed entrywise, so by (a)

$$A \circ B = \sum_k \lambda_k\, A \circ (q_k q_k') = \sum_k \lambda_k D_{q_k} A D_{q_k}, \qquad x'(A \circ B)x = \sum_k \lambda_k (D_{q_k}x)' A (D_{q_k}x) \geqslant 0,$$

and $A \circ B$ is symmetric, as $a_{ji}b_{ji} = a_{ij}b_{ij}$.

(c) Let $x'(A \circ B)x = 0$. Every term of the sum in (b) is $\geqslant 0$, so $(D_{q_k}x)'A(D_{q_k}x) = 0$, hence $D_{q_k}x = 0$ (A is PD), for every k with $\lambda_k > 0$; that is, $q_{ik}x_i = 0$ for all i and all such k, where q_{ik} is entry i of q_k. If $x_i \neq 0$, then $q_{ik} = 0$ for every k with $\lambda_k > 0$, and $b_{ii} = \sum_k \lambda_k q_{ik}^2 = 0$ (Corollary 8.5(a)), ✘. So $x = 0$, and $A \circ B$ is PD.

(d) S is PD (Example 8.34), so $R = D^{-1/2}SD^{-1/2}$ is PD (Theorem 8.16(b)) with unit diagonal; by (c), $R \circ R$ is PD. Directly, $R \circ R = \frac{1}{36}(11I_3 + 25J_3)$ has the eigenvalues $\frac{86}{36}$ and $\frac{11}{36}$ (Example 7.17).

Problem 8.8. (a) Let $Z \in \mathbb{R}^{n\times k}$ ($n \geqslant 2$) have column means 0 and sample covariance matrix I_k, let $\mu \in \mathbb{R}^k$ and $L \in \mathbb{R}^{k\times k}$. Show that $Y = \mathbf{1}_n\mu' + ZL'$ has mean vector μ and sample covariance matrix LL'.

(b) Let $X_0 \in \mathbb{R}^{n\times k}$ have rows x_i', mean vector $\bar{x}$ and a PD sample covariance matrix S with Cholesky factor L_S. Show that the vectors $z_i = L_S^{-1}(x_i - \bar{x})$ have mean vector 0 and sample covariance matrix I_k.

(c) Apply (b) to the data $(7, 5)$, $(3, 1)$, $(7, 3)$, $(3, 3)$, $(5, 3)$, and then (a), to produce five observations with mean vector $(10, 20)'$ and sample covariance matrix $\Sigma = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 5 \end{smallmatrix}\right)$.

Solution. (a) The mean vector of Y is $\frac{1}{n}Y'\mathbf{1} = \mu \cdot \frac{1}{n}\mathbf{1}'\mathbf{1} + L \cdot \frac{1}{n}Z'\mathbf{1} = \mu$. Since $C_n\mathbf{1} = 0$ and $C_nZ = Z$ (the columns of Z have mean 0, Example 1.14), $C_nY = ZL'$, and $\frac{1}{n-1}Z'Z = \frac{1}{n-1}Z'C_nZ = I_k$. Hence the sample covariance matrix of Y is $\frac{1}{n-1}(C_nY)'(C_nY) = L\Big(\frac{1}{n-1}Z'Z\Big)L' = LL'$.

(b) The matrix with rows $z_i' = (x_i - \bar{x})'(L_S^{-1})'$ is $Z = C_nX_0(L_S^{-1})'$. Then $Z'\mathbf{1} = L_S^{-1}X_0'C_n\mathbf{1} = 0$, $C_nZ = Z$, and, with $(L_S^{-1})' = (L_S')^{-1}$ (Theorem 2.15(c)),

$$\frac{1}{n-1}Z'C_nZ = L_S^{-1}\Big(\frac{1}{n-1}X_0'C_nX_0\Big)(L_S')^{-1} = L_S^{-1}L_SL_S'(L_S')^{-1} = I_k$$

(c) $\bar{x} = (5, 3)'$, the rows of C_5X_0 are $(2, 2)$, $(-2, -2)$, $(2, 0)$, $(-2, 0)$, $(0, 0)$, and $S = \frac{1}{4}\left(\begin{smallmatrix} 16 & 8 \\ 8 & 8 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 2 \end{smallmatrix}\right)$. By Remark 8.23, $L_S = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ ($l_{11} = 2$, $l_{21} = 2/2 = 1$, $l_{22} = \sqrt{2-1} = 1$), and $L_S^{-1} = \left(\begin{smallmatrix} 1/2 & 0 \\ -1/2 & 1 \end{smallmatrix}\right)$. So

$z_i = L_S^{-1}(x_i - \bar{x})$ gives $(1,1)'$, $(-1,-1)'$, $(1,-1)'$, $(-1,1)'$, $(0,0)'$; indeed $\frac{1}{4}Z'Z = \frac{1}{4}\left(\begin{smallmatrix} 4 & 0 \\ 0 & 4 \end{smallmatrix}\right) = I_2$. The Cholesky factor of Σ is $L = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 2 \end{smallmatrix}\right)$ ($l_{22} = \sqrt{5-1} = 2$). The rows $y_i' = (\mu + Lz_i)'$ of $Y = \mathbf{1}\mu' + ZL'$ are

$$(12, 23), \quad (8, 17), \quad (12, 19), \quad (8, 21), \quad (10, 20)$$

Check: the means are $\frac{50}{5} = 10$ and $\frac{100}{5} = 20$; the deviations $(2,3)$, $(-2,-3)$, $(2,-1)$, $(-2,1)$, $(0,0)$ give $\frac{1}{4}\left(\begin{smallmatrix} 16 & 8 \\ 8 & 20 \end{smallmatrix}\right) = \Sigma$.

Chapter 9. Matrix differentiation and variational principles

Differentials

Remark 9.1 (Imported calculus facts)**.** A matrix $X \in \mathbb{R}^{p\times q}$ is treated as a point of $\mathbb{R}^{pq}$, its entries listed column by column. A set $U \subseteq \mathbb{R}^n$ is **open** if every $x \in U$ has a ball $\{z : \|z-x\| < \delta\}$ $(\delta > 0)$ inside U. A ball contains the segment $\{(1-t)y + tz : 0 \leqslant t \leqslant 1\}$ between any two of its points: if $\|y-x\| < \delta$ and $\|z-x\| < \delta$, then $\|(1-t)y+tz-x\| \leqslant (1-t)\|y-x\| + t\|z-x\| < \delta$ by the triangle inequality (Theorem 6.3(b)). Let U be open and $f : U \to \mathbb{R}^m$. Then f is **differentiable at** $x \in U$ if there is a matrix $Df(x) \in \mathbb{R}^{m\times n}$, the **derivative** of f at x, with

$$f(x+h) = f(x) + Df(x)h + r(h), \qquad \|r(h)\|/\|h\| \to 0 \quad (h \to 0)$$

For $m = 1$, $\nabla f(x) = Df(x)' \in \mathbb{R}^n$ is the **gradient**. For an increment $\mathrm{d}x \in \mathbb{R}^n$, the **differential** of f at x is $\mathrm{d}f = Df(x)\,\mathrm{d}x$. A matrix-valued function $F = (f_{kl})$ is differentiable if its entries are, and $\mathrm{d}F = (\mathrm{d}f_{kl})$ has the shape of F. For $S \subseteq \mathbb{R}^n$ and $f : S \to \mathbb{R}$, a point $x^* \in S$ is a (**global**) **minimizer** of f on S if $f(x^*) \leqslant f(x)$ for all $x \in S$, and a **local minimizer** if this holds for all $x \in S$ in some ball around x^*; it is **strict** if the inequality is strict for $x \neq x^*$. (**Local**) **maximizers** are defined with $\geqslant$; a (**local**) **extremum** is a (local) minimizer or maximizer. If $S = U$ is open, a point x with $\nabla f(x) = 0$ is a **stationary point** of f. The following facts of multivariable calculus are used without proof.

(F1) $Df(x)$ is unique, and its (i,j) entry is the partial derivative $\partial f_i(x)/\partial x_j$. Hence, if f is differentiable at x and $\mathrm{d}f = B\,\mathrm{d}x$ for every $\mathrm{d}x$, then $Df(x) = B$ (**identification**). If all partial derivatives of f exist and are continuous on U (f is C^1), then f is differentiable on U. Polynomials in the coordinates, quotients of polynomials with nonzero denominator, exp, and log on $(0,\infty)$ are continuous with continuous partial derivatives of all orders, and so are their sums, products and compositions. A continuous function that is positive (or nonzero) at a point is positive (nonzero) on a ball around it.

(F2) (**Rules**) With all functions differentiable at x and A, C, α, β constant: a constant has differential 0, and

$$\mathrm{d}(\alpha F + \beta G) = \alpha\,\mathrm{d}F + \beta\,\mathrm{d}G, \qquad \mathrm{d}(AFC) = A(\mathrm{d}F)C, \qquad \mathrm{d}(F') = (\mathrm{d}F)', \qquad \mathrm{d}\operatorname{tr}F = \operatorname{tr}\mathrm{d}F,$$
$$\mathrm{d}(FG) = (\mathrm{d}F)G + F\,\mathrm{d}G \quad \text{(product rule)}$$

(**Chain rule**) If g is differentiable at $y = f(x)$, then $g \circ f$ is differentiable at x and $D(g\circ f)(x) = Dg(y)Df(x)$: the differential of $g(f(x))$ is $\mathrm{d}g = Dg(y)\,\mathrm{d}y$ with $\mathrm{d}y = \mathrm{d}f$.

(F3) If $f : U \to \mathbb{R}$ has continuous second partial derivatives (f is C^2), then the **Hessian** $Hf(x) = \left(\partial^2 f(x)/\partial x_i \partial x_j\right) \in \mathbb{R}^{n\times n}$ is symmetric, and ∇f is differentiable with $D(\nabla f)(x) = Hf(x)$. (**Taylor's formula**) If the segment from x to y lies in U, then $f(y) = f(x) + \nabla f(x)'(y-x) + \frac{1}{2}(y-x)'Hf(z)(y-x)$ for some z on the segment.

(F4) If $f : U \to \mathbb{R}$ is differentiable at a local extremum x, then $\nabla f(x) = 0$.

(F5) (**Lagrange multipliers**) Let $f, g_1, \ldots, g_k$ be C^1 on U, $g = (g_1, \ldots, g_k)'$ and $S = \{x \in U : g(x) = 0\}$. If x^* is a local extremum of f on S and $\operatorname{rank} Dg(x^*) = k$, then

$$\nabla f(x^*) = Dg(x^*)'\lambda = \lambda_1 \nabla g_1(x^*) + \cdots + \lambda_k \nabla g_k(x^*)$$

for some $\lambda \in \mathbb{R}^k$; the columns of $Dg(x^*)'$ are the gradients $\nabla g_i(x^*)$. A point $x \in S$ with $\operatorname{rank} Dg(x) = k$ and $\nabla f(x) = Dg(x)'\lambda$ for some $\lambda \in \mathbb{R}^k$ is a **stationary point of f on** S, and $\lambda_1, \ldots, \lambda_k$ are **Lagrange multipliers**.

(F6) (**Change of variables**) If F is a C^1 bijection between open sets $V, W \subseteq \mathbb{R}^N$ with C^1 inverse, then $\int_W g(y)\,\mathrm{d}y = \int_V g(F(x))\,|\det DF(x)|\,\mathrm{d}x$ for every integrable g; $|\det DF(x)|$ is the factor by which F scales volumes near x.

Theorem 9.2. Let $a \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times n}$ be constant. The functions $a'x$, Ax and $x'Bx$ ($x \in \mathbb{R}^n$) are differentiable everywhere, and

$$D(a'x) = a', \qquad D(Ax) = A, \qquad D(x'Bx) = x'(B + B')$$

In particular $D(x'Bx) = 2x'B$ and $\nabla(x'Bx) = 2Bx$ if B is symmetric.

Proof. $A(x + h) = Ax + Ah$ exactly, so $D(Ax) = A$ with remainder 0; $a'x$ is the case $A = a'$. By the product rule (Remark 9.1(F2)), $\mathrm{d}(x'Bx) = (\mathrm{d}x)'Bx + x'B\,\mathrm{d}x$, and the 1×1 matrix $(\mathrm{d}x)'Bx$ equals its transpose $x'B'\mathrm{d}x$ (Theorem 1.9). So $\mathrm{d}(x'Bx) = x'(B' + B)\,\mathrm{d}x$, and (F1) gives $D(x'Bx) = x'(B + B')$. □

Definition 9.3. The length of $X = (x_{ij}) \in \mathbb{R}^{p\times q}$ as a point of $\mathbb{R}^{pq}$ is its **Frobenius norm** $\|X\|_F = \left(\sum_{i,j} x_{ij}^2\right)^{1/2} = (\operatorname{tr} X'X)^{1/2}$ (Theorem 1.11(d)). Let $U \subseteq \mathbb{R}^{p\times q}$ be open and $f : U \to \mathbb{R}$ be differentiable at $X \in U$.

- The **matrix derivative** of f at X is the matrix of partial derivatives $\partial f/\partial X = (\partial f/\partial x_{ij}) \in \mathbb{R}^{p\times q}$; it has the shape of X.
- For an increment $\mathrm{d}X \in \mathbb{R}^{p\times q}$, the differential of f is

$$\mathrm{d}f = \sum_{i,j} \frac{\partial f}{\partial x_{ij}}\,\mathrm{d}x_{ij} = \operatorname{tr}\Big(\Big(\frac{\partial f}{\partial X}\Big)'\mathrm{d}X\Big),$$

  by (F1) and the identity $\operatorname{tr}(B'H) = \sum_{i,j} b_{ij}h_{ij}$ for $B, H \in \mathbb{R}^{p\times q}$ (the jth diagonal entry of $B'H$ is $\sum_i b_{ij}h_{ij}$).

Matrix derivatives are identified as follows: if f is differentiable at X and $\mathrm{d}f = \operatorname{tr}(B'\mathrm{d}X)$ for every $\mathrm{d}X \in \mathbb{R}^{p\times q}$, then $\partial f/\partial X = B$. Indeed, for $\mathrm{d}X = E_{ij}$ (Definition 3.14), $\mathrm{d}f = \partial f/\partial x_{ij}$ and $\operatorname{tr}(B'E_{ij}) = b_{ij}$.

Lemma 9.4. For $A \in \mathbb{R}^{m\times n}$ and $h \in \mathbb{R}^n$, $\|Ah\| \leqslant \|A\|_F\|h\|$.

Proof. The ith entry of Ah is $\operatorname{row}_i(A)h$, and $(\operatorname{row}_i(A)h)^2 \leqslant \|\operatorname{row}_i(A)'\|^2\|h\|^2$ by the Cauchy–Schwarz inequality (Theorem 6.3(a)). Summing over i gives $\|Ah\|^2 \leqslant \left(\sum_{i,j} a_{ij}^2\right)\|h\|^2$. □

Theorem 9.5. (a) Let $X \in \mathbb{R}^{p\times q}$. For constant $A \in \mathbb{R}^{q\times p}$, $\mathrm{d}\operatorname{tr}(AX) = \operatorname{tr}(A\,\mathrm{d}X)$, so $\partial\operatorname{tr}(AX)/\partial X = A'$. For constant $A \in \mathbb{R}^{p\times p}$, $\mathrm{d}\operatorname{tr}(X'AX) = \operatorname{tr}\big(X'(A + A')\,\mathrm{d}X\big)$, so $\partial\operatorname{tr}(X'AX)/\partial X = (A + A')X$.

(b) Let $n \geqslant 2$. The function $X \mapsto \det X$ is differentiable on $\mathbb{R}^{n\times n}$, and $\partial \det X/\partial x_{ij} = c_{ij}$, the cofactor of x_{ij} (Definition 5.12). Hence $\mathrm{d}\det X = \operatorname{tr}(\operatorname{adj}(X)\,\mathrm{d}X)$, and $\mathrm{d}\det X = \det X \cdot \operatorname{tr}(X^{-1}\mathrm{d}X)$ if X is invertible.

(c) On the open set $\{X \in \mathbb{R}^{n\times n} : \det X > 0\}$, $\mathrm{d}\log\det X = \operatorname{tr}(X^{-1}\mathrm{d}X)$, so $\partial\log\det X/\partial X = (X^{-1})'$.

(d) On the open set of invertible $X \in \mathbb{R}^{n\times n}$, $X \mapsto X^{-1}$ is differentiable and $\mathrm{d}(X^{-1}) = -X^{-1}(\mathrm{d}X)X^{-1}$. Hence, for constant $A \in \mathbb{R}^{n\times n}$, $\mathrm{d}\operatorname{tr}(AX^{-1}) = -\operatorname{tr}(X^{-1}AX^{-1}\mathrm{d}X)$ and $\partial\operatorname{tr}(AX^{-1})/\partial X = -(X^{-1}AX^{-1})'$.

Proof. By Definition 5.12 and induction on n, $\det X$ is a polynomial in the entries of X, hence continuous (Remark 9.1). So if $\det X_0 > 0$ (or $\det X_0 \neq 0$), then $\det X > 0$ ($\det X \neq 0$) for all X in a ball around X_0 (Remark 9.1): the sets in (c) and (d) are open.

(a) $\operatorname{tr}(A(X + H)) = \operatorname{tr}(AX) + \operatorname{tr}(AH)$ exactly, so $\mathrm{d}\operatorname{tr}(AX) = \operatorname{tr}(A\,\mathrm{d}X) = \operatorname{tr}((A')'\mathrm{d}X)$, and the identification after Definition 9.3 gives $\partial\operatorname{tr}(AX)/\partial X = A'$. By the product rule (Remark 9.1(F2)), $\mathrm{d}(X'AX) = (\mathrm{d}X)'AX + X'A\,\mathrm{d}X$ and $\mathrm{d}\operatorname{tr} = \operatorname{tr}\mathrm{d}$; and $\operatorname{tr}((\mathrm{d}X)'AX) = \operatorname{tr}\big(((\mathrm{d}X)'AX)'\big) = \operatorname{tr}(X'A'\mathrm{d}X)$ (Theorems 1.11(a) and 1.9). Hence $\mathrm{d}\operatorname{tr}(X'AX) = \operatorname{tr}(X'(A' + A)\,\mathrm{d}X) = \operatorname{tr}\big(((A + A')X)'\mathrm{d}X\big)$.

(b) The partial derivatives of the polynomial $\det X$ are polynomials, hence continuous, so det is differentiable (Remark 9.1(F1)). By the expansion along row i (Theorem 5.13), $\det X = \sum_k x_{ik}c_{ik}$, where the cofactors $c_{ik} = (-1)^{i+k}\det X(i|k)$ do not involve the entries of row i. So, as a function of x_{ij} with the other entries fixed, $\det X = x_{ij}c_{ij} + (\text{terms without } x_{ij})$, and $\partial \det X/\partial x_{ij} = c_{ij}$. Hence $\mathrm{d}\det X = \sum_{i,j} c_{ij}\,\mathrm{d}x_{ij} = \operatorname{tr}(C'\mathrm{d}X)$ with $C = (c_{ij})$, and $C' = \operatorname{adj} X$ (Theorem 5.20). If X is invertible, $\operatorname{adj} X = \det X \cdot X^{-1}$ by the same theorem.

(c) For $n \geqslant 2$, the chain rule (Remark 9.1(F2)) with $\mathrm{d}\log t = t^{-1}\mathrm{d}t$ and (b) gives $\mathrm{d}\log\det X = (\det X)^{-1}\mathrm{d}\det X = \operatorname{tr}(X^{-1}\mathrm{d}X)$; for $n = 1$ this is $\mathrm{d}\log x = x^{-1}\mathrm{d}x$. Since $\operatorname{tr}(X^{-1}\mathrm{d}X) = \operatorname{tr}\Big(((X^{-1})')'\mathrm{d}X\Big)$, the identification gives $\partial \log\det X/\partial X = (X^{-1})'$.

(d) By Theorem 5.20, the entries of $X^{-1} = \operatorname{adj}(X)/\det X$ are quotients of polynomials in the entries of X with nonzero denominator (for $n = 1$, $1/x$); their partial derivatives exist and are continuous, so $X \mapsto X^{-1}$ is differentiable (Remark 9.1(F1)). The product rule applied to the constant $XX^{-1} = I_n$ gives $(\mathrm{d}X)X^{-1} + X\,\mathrm{d}(X^{-1}) = 0$; multiplying by X^{-1} on the left, $\mathrm{d}(X^{-1}) = -X^{-1}(\mathrm{d}X)X^{-1}$. Then, by Theorem 1.11(c), $\mathrm{d}\operatorname{tr}(AX^{-1}) = \operatorname{tr}(A\,\mathrm{d}(X^{-1})) = -\operatorname{tr}(AX^{-1}(\mathrm{d}X)X^{-1}) = -\operatorname{tr}(X^{-1}AX^{-1}\mathrm{d}X)$, and the identification gives the matrix derivative. □

Example 9.6. Let $U = \{X \in \mathbb{R}^{4\times 4} : \det X > 0\}$ and $f(X) = \log\det X$ on U. Evaluate the matrix derivative of f at

$$X_0 = \Big(\min(i,j)\Big) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix},$$

and check it along a direction in which $f(X_0 + t\,\mathrm{d}X)$ is computed exactly.

Solution. By Theorem 9.5(c), U is open and $\partial f/\partial X = (X^{-1})'$ on U. Let $L \in \mathbb{R}^{4\times 4}$ have $l_{ik} = 1$ for $k \leqslant i$ and $l_{ik} = 0$ for $k > i$. Then $(LL')_{ij} = \sum_k l_{ik}l_{jk}$ counts the k with $k \leqslant i$ and $k \leqslant j$, so $LL' = X_0$. As L is lower triangular with unit diagonal, $\det L = \det L' = 1$ (Theorem 5.15), and $\det X_0 = 1$ (Theorem 5.19(b)). So $X_0 \in U$ and $f(X_0) = 0$. Let

$$T = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Column j of X_0T is $-c_{j-1} + 2c_j - c_{j+1}$ for $j \leqslant 3$ (with $c_0 = 0$) and $c_4 - c_3$, where c_j is column j of X_0; these are $e_1, \dots, e_4$. So $X_0T = I_4$ and $X_0^{-1} = T$ (Theorem 2.18), and, as $T' = T$,

$$\frac{\partial f}{\partial X}(X_0) = (X_0^{-1})' = T, \qquad \mathrm{d}f = \operatorname{tr}(T\,\mathrm{d}X) \text{ at } X_0$$

Check along the rank-one direction $\mathrm{d}X = uv'$ with $u = e_1 - e_2$ and $v = e_1$ (adding t to x_{11} and $-t$ to x_{21}): by the matrix determinant lemma (Theorem 5.27), $\det(X_0 + tuv') = \det X_0\,(1 + t\,v'X_0^{-1}u) = 1 + t\Big(2 - (-1)\Big) = 1 + 3t$. So $f(X_0 + tuv') = \log(1 + 3t)$ for $t > -\frac{1}{3}$, with derivative 3 at $t = 0$; and $\operatorname{tr}(Tuv') = v'Tu = t_{11} - t_{12} = 3$ (Theorem 1.11(b)).

Vectorization and Kronecker products

Definition 9.7. For $X \in \mathbb{R}^{p\times q}$ with columns $x_1, \dots, x_q$, the **vectorization** of X is $\operatorname{vec} X = (x_1', \dots, x_q')' \in \mathbb{R}^{pq}$. For $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{r\times s}$, the **Kronecker product** $A \otimes B \in \mathbb{R}^{mr\times ns}$ is the block matrix whose (i,j) block is $a_{ij}B$:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}$$

The point of $\mathbb{R}^{pq}$ attached to X in Remark 9.1 is $\operatorname{vec} X$, and $\|X\|_F = \|\operatorname{vec} X\|$. For $A, B \in \mathbb{R}^{p\times q}$, $\operatorname{tr}(A'B) = \sum_{i,j} a_{ij}b_{ij} = (\operatorname{vec} A)' \operatorname{vec} B$. The identity matrix satisfies $I_m \otimes I_r = I_{mr}$, and $I_p \otimes B = \operatorname{diag}(B, \ldots, B)$.

Theorem 9.8. Let the matrices have sizes for which the operations are defined.

(a) $(A \otimes B)(C \otimes D) = AC \otimes BD$.

(b) $(A\otimes B)' = A'\otimes B'$; $(\alpha A)\otimes B = A\otimes(\alpha B) = \alpha(A\otimes B)$ for $\alpha \in \mathbb{R}$; $(A_1+A_2)\otimes B = A_1\otimes B+A_2\otimes B$ and $A \otimes (B_1 + B_2) = A \otimes B_1 + A \otimes B_2$.

(c) If A and B are square and invertible, then $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$. If A and B are orthogonal (symmetric), so is $A \otimes B$.

(d) $\operatorname{vec}(AXB) = (B' \otimes A) \operatorname{vec} X$ for $A \in \mathbb{R}^{m\times p}$, $X \in \mathbb{R}^{p\times q}$ and $B \in \mathbb{R}^{q\times n}$.

(e) $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \cdot \operatorname{tr} B$ for square A and B.

(f) $\operatorname{rank}(A \otimes B) = \operatorname{rank} A \cdot \operatorname{rank} B$.

Proof. (a) Let $A \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{n\times k}$. By the block product formula (Theorem 5.3(c)), block (i, l) of the product is $\sum_j (a_{ij}B)(c_{jl}D) = \Big(\sum_j a_{ij}c_{jl}\Big)BD = (AC)_{il}BD$, block (i, l) of $AC \otimes BD$.

(b) By Theorem 5.3(b), block (j, i) of $(A \otimes B)'$ is $(a_{ij}B)' = a_{ij}B'$, and a_{ij} is the (j, i) entry of A'. The other identities hold block by block (Theorem 5.3(a)).

(c) By (a), $(A \otimes B)(A^{-1} \otimes B^{-1}) = I \otimes I = I$; apply Theorem 2.18. If $A'A = I$ and $B'B = I$, then $(A \otimes B)'(A \otimes B) = A'A \otimes B'B = I$ by (a) and (b). If $A' = A$ and $B' = B$, then $(A \otimes B)' = A \otimes B$ by (b).

(d) Let x_j be column j of X. Column k of AXB is $AX \operatorname{col}_k(B) = \sum_j b_{jk}Ax_j$ (Theorem 1.4). Block (k, j) of $B' \otimes A$ is $b_{jk}A$, so block k of $(B' \otimes A) \operatorname{vec} X$ is $\sum_j b_{jk}Ax_j$ as well.

(e) The diagonal blocks of $A \otimes B$ are $a_{ii}B$, so $\operatorname{tr}(A \otimes B) = \sum_i a_{ii} \operatorname{tr} B$.

(f) If $A = 0$ or $B = 0$, then $A \otimes B = 0$ and both sides are 0. Otherwise let $A = A_1A_2$ and $B = B_1B_2$ be full-rank factorizations (Theorem 4.21(a)), with $r = \operatorname{rank} A$ columns in A_1 and $s = \operatorname{rank} B$ columns in B_1.

- By (a), $A \otimes B = (A_1 \otimes B_1)(A_2 \otimes B_2)$, and by (a),(b), $(A_1 \otimes B_1)'(A_1 \otimes B_1) = A_1'A_1 \otimes B_1'B_1$ and $(A_2 \otimes B_2)(A_2 \otimes B_2)' = A_2A_2' \otimes B_2B_2'$.
- The $r \times r$ matrices $A_1'A_1$, A_2A_2' and the $s \times s$ matrices $B_1'B_1$, B_2B_2' are invertible (Theorem 4.21(c)), so by (c) both $rs \times rs$ Kronecker products above are invertible, of rank rs (Theorem 4.13(d)).
- By Theorem 4.18(b), $A_1 \otimes B_1$ has rank rs, its number of columns, and $A_2 \otimes B_2$ has rank rs, its number of rows. By Theorem 4.15(b), $\operatorname{rank}(A \otimes B) = \operatorname{rank}(A_2 \otimes B_2) = rs$. □

Theorem 9.9. Let $A \in \mathbb{R}^{p\times p}$ and $B \in \mathbb{R}^{q\times q}$.

(a) $\det(A \otimes B) = (\det A)^q(\det B)^p$.

(b) If $p_A(t) = \prod_{i=1}^{p}(t - \lambda_i)$ and $p_B(t) = \prod_{j=1}^{q}(t - \mu_j)$ with all $\lambda_i, \mu_j \in \mathbb{R}$, then

$$p_{A\otimes B}(t) = \prod_{i,j}(t - \lambda_i\mu_j), \qquad p_{A\otimes I_q + I_p\otimes B}(t) = \prod_{i,j}\Big(t - (\lambda_i + \mu_j)\Big)$$

(c) If $A = Q\Lambda Q'$ and $B = RMR'$ are spectral decompositions (Theorem 8.4) with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$, $M = \operatorname{diag}(\mu_1, \ldots, \mu_q)$, then $A \otimes B = (Q \otimes R)(\Lambda \otimes M)(Q \otimes R)'$ is a spectral decomposition: the columns $q_i \otimes r_j$ of $Q \otimes R$ are orthonormal eigenvectors of $A \otimes B$ for the eigenvalues $\lambda_i\mu_j$. If A and B are positive definite, so is $A \otimes B$.

Proof. (a)

- $I_p \otimes B = \operatorname{diag}(B, \dots, B)$, so $\det(I_p \otimes B) = (\det B)^p$ (Theorem 5.25 and induction on p). By Theorem 9.8(a), $A \otimes B = (A \otimes I_q)(I_p \otimes B)$.
- For $X \in \mathbb{R}^{q\times p}$, each entry of $\operatorname{vec}(X')$ is one entry of $\operatorname{vec} X$, and each entry of X occurs once; so $\operatorname{vec}(X') = K \operatorname{vec} X$ for a matrix $K \in \mathbb{R}^{pq\times pq}$ whose columns are $e_1, \dots, e_{pq}$ in some order, and $K'K = I_{pq}$. By Theorem 9.8(d), $(A \otimes I_q) \operatorname{vec} X = \operatorname{vec}(XA')$ and $(I_q \otimes A) \operatorname{vec} Y = \operatorname{vec}(AY)$ for $Y \in \mathbb{R}^{p\times q}$. Hence

$$K(A \otimes I_q) \operatorname{vec} X = \operatorname{vec}\Big((XA')'\Big) = \operatorname{vec}(AX') = (I_q \otimes A)K \operatorname{vec} X$$

  for every X, and $K(A \otimes I_q) = (I_q \otimes A)K$. As $(\det K)^2 = \det(K'K) = 1$ (Theorems 5.19(b) and 5.15(b)), taking determinants gives $\det(A \otimes I_q) = \det(I_q \otimes A) = (\det A)^q$. Multiply.

(b) By Schur triangularization (Theorem 7.28), $A = Q_1 T_1 Q_1'$ and $B = Q_2 T_2 Q_2'$ with Q_1, Q_2 orthogonal and T_1, T_2 upper triangular with diagonals $\lambda_1, \dots, \lambda_p$ and $\mu_1, \dots, \mu_q$. Let $P = Q_1 \otimes Q_2$, orthogonal by Theorem 9.8(c). By Theorem 9.8(a),(b), $A \otimes B = P(T_1 \otimes T_2)P'$ and $A \otimes I_q + I_p \otimes B = P(T_1 \otimes I_q + I_p \otimes T_2)P'$. In $T_1 \otimes T_2$ the blocks $(T_1)_{ij}T_2$ below the diagonal ($i > j$) are 0, and the diagonal blocks $\lambda_i T_2$ are upper triangular; so $T_1 \otimes T_2$ is upper triangular with diagonal entries $\lambda_i\mu_j$. Likewise $T_1 \otimes I_q + I_p \otimes T_2$ is upper triangular with diagonal blocks $\lambda_i I_q + T_2$, hence with diagonal entries $\lambda_i + \mu_j$. Similar matrices have the same characteristic polynomial (Theorem 7.14(a)), and $\det(tI - T)$ is the product of the diagonal entries of the triangular matrix $tI - T$ (Theorem 5.15(a)).

(c) $Q \otimes R$ is orthogonal (Theorem 9.8(c)), and $\Lambda \otimes M$ is diagonal with entries $\lambda_i\mu_j$; by Theorem 9.8(a),(b), $(Q \otimes R)(\Lambda \otimes M)(Q \otimes R)' = Q\Lambda Q' \otimes RMR' = A \otimes B$. Column $(i-1)q + j$ of $Q \otimes R$ has blocks $q_{ki}r_j$ ($k = 1, \dots, p$), that is, it is $q_i \otimes r_j$. If all λ_i and μ_j are positive, so are all $\lambda_i\mu_j$, and $A \otimes B$ is positive definite (Theorem 8.13(a)). □

Example 9.10. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. Find the eigenvalues, the determinant and the inverse of $A \otimes B$, and solve the matrix equation $AXB = C$ for $C = \left(\begin{smallmatrix} 11 & 1 \\ 13 & -1 \end{smallmatrix}\right)$.

Solution. A has the eigenvalues $3, 1$ with eigenvectors $(1, 1)'$, $(1, -1)'$, and B has the eigenvalues $4, 2$ with the same eigenvectors. By Theorem 9.9(c),

$$A \otimes B = \begin{pmatrix} 6 & 2 & 3 & 1 \\ 2 & 6 & 1 & 3 \\ 3 & 1 & 6 & 2 \\ 1 & 3 & 2 & 6 \end{pmatrix}$$

has the eigenvalues $12, 6, 4, 2$, with the orthogonal eigenvectors $(1, 1)' \otimes (1, 1)' = (1, 1, 1, 1)'$, $(1, 1)' \otimes (1, -1)' = (1, -1, 1, -1)'$, $(1, -1)' \otimes (1, 1)' = (1, 1, -1, -1)'$ and $(1, -1, -1, 1)'$. So $\det(A \otimes B) = 12 \cdot 6 \cdot 4 \cdot 2 = 576 = 3^2 \cdot 8^2 = (\det A)^2(\det B)^2$, as in Theorem 9.9(a). By Theorem 9.8(c), with $A^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and $B^{-1} = \frac{1}{8}\left(\begin{smallmatrix} 3 & -1 \\ -1 & 3 \end{smallmatrix}\right)$,

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1} = \frac{1}{24}\begin{pmatrix} 6 & -2 & -3 & 1 \\ -2 & 6 & 1 & -3 \\ -3 & 1 & 6 & -2 \\ 1 & -3 & -2 & 6 \end{pmatrix}$$

By Theorem 9.8(d), $AXB = C$ is the system $(B \otimes A) \operatorname{vec} X = \operatorname{vec} C$ of four equations, with $B' = B$ and the invertible matrix $B \otimes A$. Its solution $\operatorname{vec} X = (B^{-1} \otimes A^{-1}) \operatorname{vec} C = \operatorname{vec}(A^{-1}CB^{-1})$ needs only the two 2×2 inverses:

$$A^{-1}C = \frac{1}{3}\begin{pmatrix} 9 & 3 \\ 15 & -3 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}, \qquad X = A^{-1}CB^{-1} = \frac{1}{8}\begin{pmatrix} 8 & 0 \\ 16 & -8 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

Jacobian matrices of matrix maps

Definition 9.11. Let $U \subseteq \mathbb{R}^{p\times q}$ be open and $F : U \to \mathbb{R}^{r\times s}$ be differentiable at $X \in U$. The **Jacobian matrix** $J_F(X) \in \mathbb{R}^{rs\times pq}$ is the derivative of $\operatorname{vec} F$ as a function of $\operatorname{vec} X$:

$$\operatorname{vec}(\mathrm{d}F) = J_F(X)\operatorname{vec}(\mathrm{d}X)$$

If $rs = pq$, $\det J_F(X)$ is the **Jacobian determinant**. By (F1), if $\operatorname{vec}(\mathrm{d}F) = J\operatorname{vec}(\mathrm{d}X)$ for every $\mathrm{d}X$, then $J_F(X) = J$.

Theorem 9.12. (a) Let $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{q\times q}$ and $F(X) = AXB$ on $\mathbb{R}^{p\times q}$. Then $J_F(X) = B' \otimes A$ and $\det J_F(X) = (\det A)^q(\det B)^p$.

(b) Let $F(X) = X^{-1}$ on the open set of invertible $X \in \mathbb{R}^{n\times n}$. Then $J_F(X) = -(X^{-1})' \otimes X^{-1}$ and $\det J_F(X) = (-1)^n(\det X)^{-2n}$.

(c) Let $F(X) = X^2$ on $\mathbb{R}^{n\times n}$. Then $J_F(X) = X' \otimes I_n + I_n \otimes X$. If $p_X(t) = \prod_i (t - \lambda_i)$ with all $\lambda_i \in \mathbb{R}$, then $\det J_F(X) = \prod_{i,j}(\lambda_i + \lambda_j)$.

Proof. (a) $\mathrm{d}F = A(\mathrm{d}X)B$ (F2), so $\operatorname{vec}(\mathrm{d}F) = (B' \otimes A)\operatorname{vec}(\mathrm{d}X)$ (Theorem 9.8(d)). By Theorem 9.9(a) and $\det B' = \det B$ (Theorem 5.15(b)), $\det(B' \otimes A) = (\det B)^p(\det A)^q$.

(b) $\mathrm{d}F = -X^{-1}(\mathrm{d}X)X^{-1}$ (Theorem 9.5(d)); apply Theorem 9.8(d) with $A = -X^{-1}$, $B = X^{-1}$, and (b) of that theorem. Then $\det J_F(X) = (-1)^{n^2}\det\big((X^{-1})' \otimes X^{-1}\big) = (-1)^n(\det X)^{-n}(\det X)^{-n}$, by Theorems 5.19(c) and 9.9(a), as n^2 and n have the same parity.

(c) $\mathrm{d}F = (\mathrm{d}X)X + X\,\mathrm{d}X = I_n(\mathrm{d}X)X + X(\mathrm{d}X)I_n$, and Theorem 9.8(d) gives $J_F(X)$. $p_{X'} = p_X$, since $\det(tI - X') = \det(tI - X)$ (Theorem 5.15(b)). By Theorem 9.9(b) with $A = X'$ and $B = X$, the characteristic polynomial of $J_F(X)$ has the roots $\lambda_i + \lambda_j$, and the determinant is their product (Corollary 7.30(a)). □

Remark 9.13. By (F6), these determinants are the volume factors of the maps. For invertible A and B, multiplication $X \mapsto AXB$ is a bijection of $\mathbb{R}^{p\times q}$ and scales every volume by $|\det A|^q|\det B|^p$; for A and B of Example 9.10 this is $3^2 \cdot 8^2 = 576$. Inversion scales volumes near X by $|\det X|^{-2n}$. If p_X splits, by Theorem 9.12(c) the Jacobian of squaring is singular at X exactly when two eigenvalues, or one eigenvalue taken twice, add up to 0: for instance when 0 is an eigenvalue, or when 1 and -1 are; at a positive definite X all $\lambda_i + \lambda_j$ are positive.

Let $U \subseteq \mathbb{R}^{n\times n}$ be open and $F : U \to \mathbb{R}^{n\times n}$ differentiable. The **Newton step** for the equation $F(X) = 0$ from $X \in U$ replaces F near X by $F(X) + \mathrm{d}F$ and solves $F(X) + \mathrm{d}F = 0$ for the increment $\mathrm{d}X = H$: $X^{\text{new}} = X + H$, where $J_F(X)\operatorname{vec} H = -\operatorname{vec} F(X)$, provided $J_F(X)$ is invertible.

Example 9.14 (Newton–Schulz iteration)**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible, and $F(X) = X^{-1} - A$ on the open set of invertible X, so that $F(X) = 0$ exactly for $X = A^{-1}$.

(a) Show that the Newton step from X is $X^{\text{new}} = 2X - XAX$, and that $I_n - AX^{\text{new}} = (I_n - AX)^2$.

(b) Let $A = I_4 + N$ and $X_0 = I_4$, with N below. Compute the Newton iterates X_1 and X_2.

$$N = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution. (a) By Theorem 9.12(b), $J_F(X)$ has the nonzero determinant $(-1)^n(\det X)^{-2n}$, and $\mathrm{d}F = -X^{-1}HX^{-1}$ for the increment H. So the Newton step solves $-X^{-1}HX^{-1} = -(X^{-1} - A)$; multiplying by X on both sides, $H = X - XAX$, and $X^{\text{new}} = 2X - XAX$. Then $I_n - A(2X - XAX) =$

$I_n - 2AX + AXAX = (I_n - AX)^2$. By induction, the residuals $E_t = I_n - AX_t$ of Newton's method satisfy $E_t = E_0^{2^t}$: each step squares the residual, and uses only matrix products.

(b) $\det A = 1$ (Theorem 5.15(a)), $E_0 = I_4 - A = -N$, and $X_1 = 2I_4 - A = I_4 - N$, invertible with $\det X_1 = 1$. By (a), $E_1 = N^2$ and $E_2 = N^4$, where

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

So $AX_2 = I_4$ and $X_2 = A^{-1}$ (Theorem 2.18): two steps give the exact inverse. Explicitly $X_2 = X_1(2I_4 - AX_1) = (I_4 - N)(I_4 + N^2) = I_4 - N + N^2 - N^3$:

$$X_1 = \begin{pmatrix} 1 & -1 & -2 & -3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad X_2 = A^{-1} = \begin{pmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Hessians, convexity and Newton's method

Definition 9.15. Let $f : U \to \mathbb{R}$ be C^2 on the open set $U \subseteq \mathbb{R}^n$. The **Hessian** of f at $x \in U$ is the symmetric matrix $Hf(x) = \partial^2 f(x)/\partial x\, \partial x' = \left(\partial^2 f(x)/\partial x_i \partial x_j\right)$ of Remark 9.1(F3). The quadratic form $\mathrm{d}^2 f = (\mathrm{d}x)' Hf(x)\, \mathrm{d}x$ in the increment $\mathrm{d}x$ is the **second differential** of f at x. For f on an open subset of $\mathbb{R}^{p\times q}$, the Hessian is taken with respect to $\operatorname{vec} X$.

By (F3) and (F1), $Hf(x)$ is identified from $\mathrm{d}(\nabla f) = Hf(x)\, \mathrm{d}x$, and $\mathrm{d}^2 f$ is the differential of $\mathrm{d}f = (\mathrm{d}x)' \nabla f(x)$ with $\mathrm{d}x$ held fixed. For symmetric B, $\nabla(x'Bx) = 2Bx$ (Theorem 9.2), so $\mathrm{d}\nabla(x'Bx) = 2B\, \mathrm{d}x$ and the Hessian of $x'Bx$ is $2B$ at every x.

Theorem 9.16. Let f be C^2 on the open set $U \subseteq \mathbb{R}^n$, let $x \in U$, and let $Hf(x) = Q\Lambda Q'$ with $Q = [q_1 \ \cdots \ q_n]$ orthogonal and $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (Theorem 8.4(a),(d)).

(a) If f has a local extremum at x, then $\nabla f(x) = 0$.

(b) If $\nabla f(x) = 0$ and $\lambda_n > 0$ ($Hf(x)$ is positive definite), then f has a strict local minimum at x.

(c) If f has a local minimum at x, then $\lambda_n \geqslant 0$ ($Hf(x)$ is positive semidefinite).

(d) If $\nabla f(x) = 0$ and $\lambda_1 > 0 > \lambda_n$ ($Hf(x)$ is indefinite), then $f(x + tq_n) < f(x) < f(x + tq_1)$ for all small $t \neq 0$, and x is not a local extremum: it is a **saddle point**.

Applied to $-f$: if $\nabla f(x) = 0$ and $\lambda_1 < 0$, f has a strict local maximum at x; at a local maximum, $\lambda_1 \leqslant 0$.

Proof. (a) is Remark 9.1(F4).

- Let $\nabla f(x) = 0$ and $v \in \mathbb{R}^n$ with $v' Hf(x) v > 0$. The function $z \mapsto v' Hf(z) v$, a combination of the continuous second partial derivatives with constant coefficients $v_i v_j$, is continuous, so it is positive on a ball $\mathcal{B} \subseteq U$ around x (F1). For small $t \neq 0$, $x + tv \in \mathcal{B}$, so the segment from x to $x + tv$ lies in $\mathcal{B}$ (Remark 9.1), and by Taylor's formula (F3), $f(x + tv) - f(x) = \frac{1}{2} t^2 v' Hf(z_t) v > 0$ for some $z_t \in \mathcal{B}$. Likewise $v' Hf(x) v < 0$ gives $f(x + tv) < f(x)$ for all small $t \neq 0$.
- (c) At a local minimum $\nabla f(x) = 0$ by (a), and $f(x + tv) \geqslant f(x)$ for small t; by the first item $v' Hf(x) v < 0$ is impossible. So $Hf(x)$ is positive semidefinite, that is, $\lambda_n \geqslant 0$ (Theorem 8.13(b)).
- (d) $q_1' Hf(x) q_1 = \lambda_1 > 0$ and $q_n' Hf(x) q_n = \lambda_n < 0$; apply the first item. Every ball around x contains points $x + tq_1$ and $x + tq_n$.

- (b) Let $c = \lambda_n > 0$; then $h'Hf(x)h \geqslant c\|h\|^2$ for all h (Theorem 8.13(c)). There is a ball $\mathcal{B} \subseteq U$ around x, of radius δ, on which every entry of $Hf(z) - Hf(x)$ has absolute value less than $c/(2n)$ (F1, for each of the n^2 entries); so $\|Hf(z) - Hf(x)\|_F < c/2$ for $z \in \mathcal{B}$. By the Cauchy–Schwarz inequality and Lemma 9.4,

$$h'Hf(z)h = h'Hf(x)h + h'\Big(Hf(z) - Hf(x)\Big)h \geqslant c\|h\|^2 - \|h\|\,\Big\|\Big(Hf(z) - Hf(x)\Big)h\Big\| \geqslant \tfrac{c}{2}\|h\|^2$$

  For $y \in \mathcal{B}$, $y \neq x$, the segment from x to y lies in $\mathcal{B}$ (Remark 9.1), and by (F3), $f(y) = f(x) + \frac{1}{2}(y-x)'Hf(z)(y-x) \geqslant f(x) + \frac{c}{4}\|y-x\|^2 > f(x)$ for some $z \in \mathcal{B}$. □

The spectrum of $Hf(x)$ decides every case except $\lambda_n = 0 \leqslant \lambda_1$ or $\lambda_n \leqslant 0 = \lambda_1$: x_1^4, $-x_1^4$ and x_1^3 all have Hessian 0 at 0, and a minimum, a maximum and neither there.

Example 9.17. Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 > \cdots > \lambda_n > 0$ and orthonormal eigenvectors $q_1, \ldots, q_n$, and let $g(x) = \|A - \frac{1}{2}xx'\|_F^2$ on $\mathbb{R}^n$, the squared error of approximating A by the rank-one matrix $\frac{1}{2}xx'$. Find and classify all stationary points of g, and the minimum of g. Evaluate everything for

$$A = \begin{pmatrix} 15 & 5 & 10 & 2 \\ 5 & 15 & 2 & 10 \\ 10 & 2 & 15 & 5 \\ 2 & 10 & 5 & 15 \end{pmatrix}$$

Solution.

- By Theorem 1.11, $g(x) = \operatorname{tr}\Big((A - \frac{1}{2}xx')^2\Big) = \operatorname{tr} A^2 - x'Ax + \frac{1}{4}(x'x)^2$. By Theorem 9.2 and the product rule, $\mathrm{d}g = -2x'A\,\mathrm{d}x + (x'x)\,x'\mathrm{d}x$, so $\nabla g(x) = -2Ax + (x'x)x$. Differentiating again, $\mathrm{d}\nabla g = -2A\,\mathrm{d}x + 2x(x'\mathrm{d}x) + (x'x)\,\mathrm{d}x$, and

$$Hg(x) = -2A + (x'x)I_n + 2xx'$$

- $\nabla g(x) = 0$ means $Ax = \frac{1}{2}(x'x)x$: $x = 0$, or x is an eigenvector with $\frac{1}{2}\|x\|^2 = \lambda_k$. As the eigenvalues are distinct, the stationary points are 0 and $\pm x_k$ with $x_k = \sqrt{2\lambda_k}\,q_k$ $(k = 1, \ldots, n)$.
- At x_k, $x_k'x_k = 2\lambda_k$ and $Hg(x_k) = 2(\lambda_k I_n - A) + 4\lambda_k q_k q_k'$. So $Hg(x_k)q_j = 2(\lambda_k - \lambda_j)q_j$ for $j \neq k$ and $Hg(x_k)q_k = 4\lambda_k q_k$: the eigenvalues of the Hessian are read off from those of A. At $\pm x_1$ all are positive, a strict local minimum (Theorem 9.16(b)). At $\pm x_k$ with $k \geqslant 2$, exactly $k-1$ are negative and the others positive: a saddle point (Theorem 9.16(d)). At 0, $Hg(0) = -2A$ has all eigenvalues negative: a strict local maximum.
- By Theorem 8.13(c), $x'Ax \leqslant \lambda_1 s$ with $s = x'x$, so $g(x) \geqslant \operatorname{tr} A^2 - \lambda_1 s + \frac{1}{4}s^2 = \operatorname{tr} A^2 - \lambda_1^2 + (\frac{s}{2} - \lambda_1)^2 \geqslant \operatorname{tr} A^2 - \lambda_1^2$, with equality at $\pm x_1$. In general $g(\pm x_k) = \operatorname{tr} A^2 - 2\lambda_k^2 + \lambda_k^2 = \operatorname{tr} A^2 - \lambda_k^2$.

For the given A, the vectors $h_1 = (1,1,1,1)'$, $h_2 = (1,-1,1,-1)'$, $h_3 = (1,1,-1,-1)'$, $h_4 = (1,-1,-1,1)'$ are orthogonal of length 2, and $Ah_1 = 32h_1$, $Ah_2 = 18h_2$, $Ah_3 = 8h_3$, $Ah_4 = 2h_4$. With $q_k = \frac{1}{2}h_k$, $x_k = \sqrt{2\lambda_k}\,q_k$ gives the integer vectors $x_1 = 4h_1$, $x_2 = 3h_2$, $x_3 = 2h_3$, $x_4 = h_4$, and $\operatorname{tr} A^2 = 32^2 + 18^2 + 8^2 + 2^2 = 1416$ (Corollary 7.30(b)).

point	eigenvalues of Hg	type	g
$\pm 4h_1$	128, 28, 48, 60	strict local (and global) minimum	392
$\pm 3h_2$	72, −28, 20, 32	saddle	1092
$\pm 2h_3$	32, −48, −20, 12	saddle	1352
$\pm h_4$	8, −60, −32, −12	saddle	1412
0	−64, −36, −16, −4	strict local maximum	1416

The eigenvalues are listed in the order of the eigenvectors q_k first, then the other q_j by increasing j.

Definition 9.18. A set $C \subseteq \mathbb{R}^n$ is **convex** if $(1-t)x + ty \in C$ for all $x, y \in C$ and $0 \leqslant t \leqslant 1$. A function $f : C \to \mathbb{R}$ on a convex set C is **convex** if

$$f\Big((1-t)x + ty\Big) \leqslant (1-t)f(x) + tf(y) \qquad (x, y \in C,\ 0 \leqslant t \leqslant 1),$$

and **strictly convex** if the inequality is strict whenever $x \neq y$ and $0 < t < 1$. It is **concave** (**strictly concave**) if $-f$ is convex (strictly convex).

Every ball is convex (Remark 9.1); $\mathbb{R}^n$ and every open interval, such as $(0, \infty)$, are open and convex.

Theorem 9.19. Let f be C^2 on an open convex set $U \subseteq \mathbb{R}^n$.

(a) If $Hf(x)$ is positive semidefinite for every $x \in U$, then f is convex, $f(y) \geqslant f(x) + \nabla f(x)'(y-x)$ for all $x, y \in U$, and every stationary point of f is a global minimizer of f on U.

(b) If $Hf(x)$ is positive definite for every $x \in U$, then f is strictly convex, $f(y) > f(x) + \nabla f(x)'(y-x)$ for all $x \neq y$ in U, and f has at most one minimizer on U; a stationary point, if there is one, is the unique minimizer.

(c) If $Hf(x)$ is negative semidefinite (negative definite) for every $x \in U$, then f is concave (strictly concave), $f(y) \leqslant f(x) + \nabla f(x)'(y-x)$ for all $x, y \in U$, and every stationary point is a global maximizer (and there is at most one maximizer).

Proof. (a) Let $x, y \in U$. The segment from x to y lies in U, so by Remark 9.1(F3), $f(y) = f(x) + \nabla f(x)'(y-x) + \frac{1}{2}(y-x)'Hf(z)(y-x)$ for some $z \in U$, and the last term is $\geqslant 0$. This gives the **tangent inequality**. Next let $0 \leqslant t \leqslant 1$ and $z = (1-t)x + ty \in U$. The tangent inequality at z gives $f(x) \geqslant f(z) + \nabla f(z)'(x-z)$ and $f(y) \geqslant f(z) + \nabla f(z)'(y-z)$; multiplying by $1-t$ and t and adding,

$$(1-t)f(x) + tf(y) \geqslant f(z) + \nabla f(z)'\Big((1-t)x + ty - z\Big) = f(z)$$

So f is convex. If $\nabla f(x) = 0$, the tangent inequality reads $f(y) \geqslant f(x)$ for all $y \in U$.

(b) Now the last term in (F3) is > 0 for $y \neq x$, so the tangent inequality is strict. For $x \neq y$ and $0 < t < 1$, $z = (1-t)x + ty$ differs from x and from y ($z - x = t(y-x)$, $z - y = (1-t)(x-y)$), so both tangent inequalities in (a) are strict, and $(1-t)f(x) + tf(y) > f(z)$. If $x_1 \neq x_2$ were two minimizers, with $f(x_1) = f(x_2) = m$, then $z = \frac{1}{2}(x_1 + x_2) \in U$ would give $f(z) < \frac{1}{2}m + \frac{1}{2}m = m$ ✘ A stationary point is a minimizer by (a), hence the only one.

(c) Apply (a) and (b) to $-f$, whose gradient is $-\nabla f$ and whose Hessian is $-Hf$ (Remark 9.1(F2)); a minimizer of $-f$ is a maximizer of f. □

Strict convexity gives at most one minimizer, not existence: $f(x) = e^x$ on $\mathbb{R}$ has $f''(x) = e^x > 0$ and no minimizer.

Example 9.20 (Quadratic functions)**.** Let $A \in \mathbb{R}^{n\times n}$ be positive definite, $b \in \mathbb{R}^n$ and $q(x) = \frac{1}{2}x'Ax - b'x$ on $\mathbb{R}^n$.

(a) Show that $\hat{x} = A^{-1}b$ is the unique minimizer of q, and that

$$q(x) = q(\hat{x}) + \tfrac{1}{2}(x-\hat{x})'A(x-\hat{x}), \qquad q(\hat{x}) = -\tfrac{1}{2}b'A^{-1}b$$

Solution. (a) By Theorem 9.2, $\nabla q(x) = Ax - b$, so the Hessian of q is A at every x. By Theorem 9.19(b), q is strictly convex on $\mathbb{R}^n$, and its only stationary point $\hat{x} = A^{-1}b$ (A is invertible, Corollary 8.15(a)) is its unique minimizer. For $h \in \mathbb{R}^n$, as A is symmetric and $A\hat{x} = b$, $q(\hat{x}+h) = \frac{1}{2}\hat{x}'A\hat{x} + h'A\hat{x} + \frac{1}{2}h'Ah - b'\hat{x} - b'h = q(\hat{x}) + \frac{1}{2}h'Ah$; take $h = x - \hat{x}$. Finally $q(\hat{x}) = \frac{1}{2}b'\hat{x} - b'\hat{x} = -\frac{1}{2}b'A^{-1}b$.

Theorem 9.21. Let $\mathcal{P}_n$ be the set of positive definite $n\times n$ matrices; it is convex, as $x'((1-t)X + tY)x = (1-t)x'Xx + tx'Yx$.

(a) Let $X \in \mathcal{P}_n$ and $D \in \mathbb{R}^{n\times n}$ symmetric, $X_t = X + tD$, and $\mathcal{I} = \{t \in \mathbb{R} : X_t \in \mathcal{P}_n\}$. Then $\mathcal{I}$ is an open interval containing 0, $\varphi(t) = \log\det X_t$ is C^2 on $\mathcal{I}$, $\varphi'(t) = \operatorname{tr}(X_t^{-1}D)$, and

$$\varphi''(t) = -\operatorname{tr}(X_t^{-1}DX_t^{-1}D) = -(\operatorname{vec} D)'(X_t^{-1}\otimes X_t^{-1})\operatorname{vec} D = -\left\|X_t^{-1/2}DX_t^{-1/2}\right\|_F^2,$$

which is negative if $D \neq 0$. In differentials, for symmetric increments $\mathrm{d}X$: $\mathrm{d}^2\log\det X = -\operatorname{tr}(X^{-1}\mathrm{d}X\, X^{-1}\mathrm{d}X) = -(\operatorname{vec}\mathrm{d}X)'(X^{-1}\otimes X^{-1})\operatorname{vec}\mathrm{d}X \leqslant 0$.

(b) $\log\det$ is strictly concave on $\mathcal{P}_n$: for $X \neq Y$ in $\mathcal{P}_n$ and $0 < t < 1$, $\log\det\big((1-t)X + tY\big) > (1-t)\log\det X + t\log\det Y$.

(c) Let $S \in \mathcal{P}_n$. On $\mathcal{P}_n$, $g(X) = \log\det X - \operatorname{tr}(SX)$ has the unique maximizer $X = S^{-1}$, and the maximum is $-\log\det S - n$.

Proof. (a)

- If $s < u$ lie in $\mathcal{I}$ and $s \leqslant t \leqslant u$, then $X_t = \frac{u-t}{u-s}X_s + \frac{t-s}{u-s}X_u \in \mathcal{P}_n$ ($\mathcal{P}_n$ is convex); so $\mathcal{I}$ is an interval, and $0 \in \mathcal{I}$. Let $t_0 \in \mathcal{I}$. The leading principal minors of X_t are polynomials in t, positive at t_0 (Theorem 8.19), hence positive near t_0 (F1); so $\mathcal{I}$ is open.
- For $t \in \mathcal{I}$, $\det X_t > 0$ (Corollary 8.15(a)), and $\mathrm{d}X_t = D\,\mathrm{d}t$. By the chain rule and Theorem 9.5(c), $\varphi'(t) = \operatorname{tr}(X_t^{-1}D)$, whose entries are quotients of polynomials in t with denominator $\det X_t \neq 0$ (Theorem 5.20); by Theorem 9.5(d), $\mathrm{d}(X_t^{-1}) = -X_t^{-1}DX_t^{-1}\mathrm{d}t$, which gives φ'', and φ is C^2 (F1).
- As X_t^{-1} and D are symmetric, Theorem 9.8(d) gives $\operatorname{vec}(X_t^{-1}DX_t^{-1}) = (X_t^{-1}\otimes X_t^{-1})\operatorname{vec} D$, and $\operatorname{tr}(X_t^{-1}DX_t^{-1}D) = \operatorname{tr}\big(D'(X_t^{-1}DX_t^{-1})\big) = (\operatorname{vec} D)'(X_t^{-1}\otimes X_t^{-1})\operatorname{vec} D$ by the identity $\operatorname{tr}(A'B) = (\operatorname{vec} A)'\operatorname{vec} B$ after Definition 9.7.
- Let $Z = X_t^{-1/2}$, symmetric with $Z^2 = X_t^{-1}$ (Theorem 8.25(c)), and $M = ZDZ$, symmetric. By Theorem 1.11(c),(d), $\operatorname{tr}(X_t^{-1}DX_t^{-1}D) = \operatorname{tr}(ZZDZZD) = \operatorname{tr}(ZDZ\,ZDZ) = \operatorname{tr}(M'M) = \|M\|_F^2$. If $D \neq 0$, then $M \neq 0$, as Z is invertible, and $\varphi''(t) < 0$.

(b) Let $D = Y - X \neq 0$. For $0 \leqslant t \leqslant 1$, $X_t = (1-t)X + tY \in \mathcal{P}_n$, as $\mathcal{P}_n$ is convex; so $[0,1] \subseteq \mathcal{I}$. By (a), the 1×1 Hessian φ'' is negative definite on the open convex set $\mathcal{I}$, and φ is strictly concave on $\mathcal{I}$ (Theorem 9.19(c)). Hence $\varphi(t) = \varphi((1-t)\cdot 0 + t\cdot 1) > (1-t)\varphi(0) + t\varphi(1)$ for $0 < t < 1$, which is the claim.

(c) $S^{-1} \in \mathcal{P}_n$ (Corollary 8.15(a)). Let $Y \in \mathcal{P}_n$, $Y \neq S^{-1}$, $D = Y - S^{-1}$, and $\psi(t) = g(S^{-1} + tD) = \varphi(t) - \operatorname{tr}(I_n) - t\operatorname{tr}(SD)$ on $\mathcal{I}$, with φ as in (a) for $X = S^{-1}$. Then $\psi'(0) = \operatorname{tr}(SD) - \operatorname{tr}(SD) = 0$ and $\psi'' = \varphi'' < 0$ on $\mathcal{I}$. By Theorem 9.19(c), 0 is the unique maximizer of ψ on $\mathcal{I} \ni 1$, so $g(Y) = \psi(1) < \psi(0) = g(S^{-1}) = \log\det S^{-1} - n = -\log\det S - n$ (Theorem 5.19(c)). □

Corollary 9.22 (Hadamard's inequality)**.** If $X \in \mathbb{R}^{n\times n}$ is positive definite, then $\det X \leqslant x_{11}x_{22}\cdots x_{nn}$, with equality if and only if X is diagonal.

Proof. For $i = 1,\dots,n$ let $S_i = I_n - 2e_ie_i'$, the diagonal matrix with entry -1 in position i and 1 elsewhere; $\det S_i = -1$ (Theorem 5.15(a)). For symmetric Y, entry (k,l) of S_iYS_i is y_{kl} with its sign changed exactly when one of k,l equals i and the other does not. So $\frac{1}{2}(Y + S_iYS_i)$ is Y with the off-diagonal entries of row i and column i replaced by 0. Put $X^{(0)} = X$ and $X^{(i)} = \frac{1}{2}\big(X^{(i-1)} + S_iX^{(i-1)}S_i\big)$.

- By induction, $X^{(i)}$ is X with all off-diagonal entries in rows and columns $1,\dots,i$ replaced by 0; in particular $X^{(n)} = \operatorname{diag}(x_{11},\dots,x_{nn})$. If $X^{(i-1)} \in \mathcal{P}_n$, then $S_iX^{(i-1)}S_i \in \mathcal{P}_n$ (Theorem 8.16(b)), with the same determinant (Theorem 5.19(b)), and $X^{(i)} \in \mathcal{P}_n$ as $\mathcal{P}_n$ is convex.
- By Theorem 9.21(b) with $t = \frac{1}{2}$, $\log\det X^{(i)} \geqslant \frac{1}{2}\log\det X^{(i-1)} + \frac{1}{2}\log\det(S_iX^{(i-1)}S_i) = \log\det X^{(i-1)}$, strictly if $S_iX^{(i-1)}S_i \neq X^{(i-1)}$, and with equality ($X^{(i)} = X^{(i-1)}$) otherwise.

Hence $\log\det X \leqslant \log\det X^{(n)} = \log(x_{11}\cdots x_{nn})$. Equality forces $S_i X^{(i-1)} S_i = X^{(i-1)}$ for every i, that is, $x_{il} = 0$ for all $l > i$, the off-diagonal entries of row i of $X^{(i-1)}$ not yet replaced; so X is diagonal. Conversely, for diagonal X equality holds (Theorem 5.15(a)). □

This second proof of Hadamard's inequality uses only the strict concavity of $\log\det$; the first, Problem 8.4, derives it from the Cholesky decomposition.

Example 9.23. Let

$$A = \begin{pmatrix} 5 & 1 & 2 & 0 \\ 1 & 5 & 0 & 2 \\ 2 & 0 & 5 & 1 \\ 0 & 2 & 1 & 5 \end{pmatrix}$$

With $h_1, \ldots, h_4$ as in Example 9.17, $Ah_1 = 8h_1$, $Ah_2 = 6h_2$, $Ah_3 = 4h_3$, $Ah_4 = 2h_4$; so $A = Q\,\mathrm{diag}(8,6,4,2)Q'$ with $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ orthogonal, A is positive definite, and $\det A = 8{\cdot}6{\cdot}4{\cdot}2 = 384$ (Theorem 8.13(a), Corollary 7.30(a)). The steps of the proof of Corollary 9.22 give

$$X^{(1)} = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 2 \\ 0 & 0 & 5 & 1 \\ 0 & 2 & 1 & 5 \end{pmatrix}, \qquad X^{(2)} = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 1 & 5 \end{pmatrix}, \qquad X^{(3)} = X^{(4)} = 5I_4,$$

with $\det X^{(1)} = 5\cdot(5\cdot 24 - 0 + 2\cdot(0-10)) = 500$ (expansion along the first row and then the first row of the 3×3 block, Theorem 5.13), $\det X^{(2)} = 25\cdot 24 = 600$ and $\det X^{(3)} = 625$:

$$\det A = 384 < 500 < 600 < 625 = a_{11}a_{22}a_{33}a_{44}$$

The last step changes nothing, as row 4 has no off-diagonal entries left.

Definition 9.24. Let f be C^2 on an open set $U \subseteq \mathbb{R}^n$, and let $x \in U$ with $Hf(x)$ invertible. The **Newton step** from x is $x^{\mathrm{new}} = x + \Delta$, where Δ solves the linear system $Hf(x)\Delta = -\nabla f(x)$; that is, $x^{\mathrm{new}} = x - Hf(x)^{-1}\nabla f(x)$. **Newton's method** starts at a point $x_0 \in U$ and puts $x_{t+1} = x_t^{\mathrm{new}}$ $(t = 0, 1, 2, \ldots)$ as long as $x_t \in U$ and $Hf(x_t)$ is invertible.

As $D(\nabla f) = Hf$ (F3), this is the Newton step for the equation $\nabla f(x) = 0$: ∇f is replaced by $\nabla f(x) + Hf(x)\Delta$. Let $T(y) = f(x) + \nabla f(x)'(y-x) + \frac{1}{2}(y-x)'Hf(x)(y-x)$ $(y \in \mathbb{R}^n)$, the second-order Taylor polynomial of f at x. By Theorem 9.2 and the symmetry of $Hf(x)$, $\nabla T(y) = \nabla f(x) + Hf(x)(y-x)$, with Hessian $Hf(x)$: T has the same value, gradient and Hessian as f at x, and $\nabla T(y) = 0$ exactly at $y = x^{\mathrm{new}}$. The Newton step thus jumps to the stationary point of this quadratic approximation. If $Hf(x)$ is positive definite, x^{new} is the unique minimizer of T: with $z = y - x$, $T(y) = f(x) + \frac{1}{2}z'Hf(x)z - \big(-\nabla f(x)\big)'z$, and Example 9.20(a) with $A = Hf(x)$ and $b = -\nabla f(x)$ gives the unique minimizer $z = -Hf(x)^{-1}\nabla f(x) = x^{\mathrm{new}} - x$; the constant $f(x)$ does not change it. If $f(y) = \frac{1}{2}y'Ay - b'y$ with A positive definite, then $T = f$, and one Newton step from any x gives $x - A^{-1}(Ax - b) = A^{-1}b$, the minimizer.

Lagrange multipliers and the KKT system

Proposition 9.25 (Minimum-norm solution)**.** Let $A \in \mathbb{R}^{m\times n}$ have rank m, and let $b \in \mathbb{R}^m$. Then AA' is invertible, $x^* = A'(AA')^{-1}b$ solves $Ax = b$, and every solution x of $Ax = b$ satisfies

$$\|x\|^2 = \|x^*\|^2 + \|x - x^*\|^2$$

Hence x^* is the unique solution of minimum length.

Proof. $\operatorname{rank}(AA') = \operatorname{rank} A = m$ (Theorem 4.18(b)), so the $m \times m$ matrix AA' is invertible (Theorem 4.13(d)), and $Ax^* = AA'(AA')^{-1}b = b$. If $Ax = b$, then $A(x - x^*) = 0$, and, as $(AA')^{-1}$ is symmetric, $(x^*)'(x - x^*) = b'(AA')^{-1}A(x - x^*) = 0$. Pythagoras (Proposition 6.2) for $x = x^* + (x - x^*)$ gives the identity, and $\|x - x^*\|^2 > 0$ for $x \neq x^*$. □

Example 9.26 (Minimum-variance portfolio)**.** Wealth is split among k assets with weights $w = (w_1, \dots, w_k)'$, $\mathbf{1}'w = 1$ (negative weights, short sales, are allowed). The risk of the portfolio is $w'\Sigma w$ for a given positive definite $\Sigma \in \mathbb{R}^{k\times k}$, for instance a sample covariance matrix of asset returns. Find the w that minimizes $w'\Sigma w$ subject to $\mathbf{1}'w = 1$.

Solution. With $f(w) = w'\Sigma w$ and $g(w) = \mathbf{1}'w - 1$, $Dg = \mathbf{1}'$ has rank 1, so by Remark 9.1(F5) a minimizer satisfies $2\Sigma w = \lambda\mathbf{1}$, that is, $w = \frac{\lambda}{2}\Sigma^{-1}\mathbf{1}$; then $\mathbf{1}'w = 1$ gives $\frac{\lambda}{2} = 1/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$, where $\mathbf{1}'\Sigma^{-1}\mathbf{1} > 0$ because Σ^{-1} is positive definite (Corollary 8.15(a)). The candidate is

$$w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

It is the unique minimizer. Let $\mathbf{1}'w = 1$ and $d = w - w^*$, so $\mathbf{1}'d = 0$. Since $\Sigma w^* = \mathbf{1}/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$, $(w^*)'\Sigma d = \mathbf{1}'d/(\mathbf{1}'\Sigma^{-1}\mathbf{1}) = 0$, and

$$w'\Sigma w = (w^*)'\Sigma w^* + 2(w^*)'\Sigma d + d'\Sigma d = (w^*)'\Sigma w^* + d'\Sigma d \geqslant (w^*)'\Sigma w^*,$$

with equality only for $d = 0$, as Σ is positive definite. The minimum variance is $(w^*)'\Sigma w^* = 1/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$. For three assets with

$$\Sigma = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \qquad \Sigma^{-1} = \frac{1}{5}\begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(multiply to check), $\Sigma^{-1}\mathbf{1} = \frac{1}{5}(2, 1, 1)'$, $\mathbf{1}'\Sigma^{-1}\mathbf{1} = \frac{4}{5}$, and $w^* = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})'$, with variance $\frac{5}{4}$, less than the variance 2 of the least risky asset alone.

Theorem 9.27 (Quadratic minimization under linear constraints)**.** Let $A \in \mathbb{R}^{p\times p}$ be positive definite, $b \in \mathbb{R}^p$, $C \in \mathbb{R}^{m\times p}$ of rank m and $d \in \mathbb{R}^m$. Let $q(x) = \frac{1}{2}x'Ax - b'x$ and $\hat{x} = A^{-1}b$, its unconstrained minimizer (Example 9.20(a)). Then $V = CA^{-1}C'$ is positive definite, hence invertible, and with $\lambda = V^{-1}(C\hat{x} - d)$:

(a) the unique minimizer of q on $\{x \in \mathbb{R}^p : Cx = d\}$ is

$$x^* = \hat{x} - A^{-1}C'V^{-1}(C\hat{x} - d) = \hat{x} - A^{-1}C'\lambda$$

(b) $q(x) = q(x^*) + \frac{1}{2}(x - x^*)'A(x - x^*)$ for every x with $Cx = d$;

(c) $q(x^*) - q(\hat{x}) = \frac{1}{2}(C\hat{x} - d)'V^{-1}(C\hat{x} - d)$;

(d) (x^*, λ) is the unique solution of the $(p + m) \times (p + m)$ linear system

$$\begin{pmatrix} A & C' \\ C & 0 \end{pmatrix}\begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix}$$

Its first block row says $\nabla q(x^*) = Ax^* - b = C'(-\lambda)$: x^* is a stationary point of q on $\{x : Cx = d\}$ in the sense of Remark 9.1(F5), for $g(x) = Cx - d$ with $Dg(x) = C$, with multiplier vector $-\lambda$.

Proof. A^{-1} is positive definite (Corollary 8.15(a)) and $C' \in \mathbb{R}^{p\times m}$ has rank m (Theorem 4.13(a)), so $V = (C')'A^{-1}C'$ is positive definite (Theorem 8.16(b)), hence invertible (Corollary 8.15(a)).

(d) Let K be the block matrix. The Schur complement of A in K is $0 - CA^{-1}C' = -V$, so $\det K = \det A \cdot \det(-V) = (-1)^m \det A \det V \neq 0$ (Theorems 5.26(a) and 5.19(c)), and K is invertible (Theorem 5.19(a)): the system has exactly one solution. The pair (x^*, λ) is a solution: $Ax^* + C'\lambda = A\hat{x} - C'\lambda + C'\lambda = b$ and $Cx^* = C\hat{x} - V\lambda = C\hat{x} - (C\hat{x} - d) = d$.

(b) For $h \in \mathbb{R}^p$, as A is symmetric,

$$q(x^* + h) = \tfrac{1}{2}(x^*)'Ax^* + h'Ax^* + \tfrac{1}{2}h'Ah - b'x^* - b'h = q(x^*) + h'(Ax^* - b) + \tfrac{1}{2}h'Ah$$

By (d), $Ax^* - b = -C'\lambda$. If $Cx = d$, then $h = x - x^*$ satisfies $Ch = d - d = 0$, so $h'(Ax^* - b) = -(Ch)'\lambda = 0$, and $q(x) = q(x^*) + \tfrac{1}{2}h'Ah$.

(a) $Cx^* = d$ by (d). By (b), $q(x) \geqslant q(x^*)$ whenever $Cx = d$, with equality only if $(x-x^*)'A(x-x^*) = 0$, that is, $x = x^*$, as A is positive definite.

(c) By Example 9.20(a), $q(x^*) = q(\hat{x}) + \tfrac{1}{2}(x^* - \hat{x})'A(x^* - \hat{x})$, and $x^* - \hat{x} = -A^{-1}C'\lambda$. So $q(x^*) - q(\hat{x}) = \tfrac{1}{2}\lambda'CA^{-1}AA^{-1}C'\lambda = \tfrac{1}{2}\lambda'V\lambda = \tfrac{1}{2}(C\hat{x} - d)'V^{-1}(C\hat{x} - d)$, as V^{-1} is symmetric. □

For $A = I_p$ and $b = a$, $q(x) = \tfrac{1}{2}\|x - a\|^2 - \tfrac{1}{2}\|a\|^2$, and (a) gives the point of the set $\{x : Cx = d\}$ closest to a, namely $x^* = a - C'(CC')^{-1}(Ca - d)$; for $a = 0$ this is Proposition 9.25.

Example 9.28. Minimize

$$q(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2 - x_1x_2 - x_2x_3 - x_3x_4 - 2(x_1 + x_2 + x_3 + x_4)$$

subject to $x_1 + x_2 + x_3 + x_4 = 10$ and $x_1 - x_2 + x_3 - x_4 = 6$.

Solution. Here $q(x) = \tfrac{1}{2}x'Ax - b'x$ with $b = 2\mathbf{1}_4$ and

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}, \qquad A^{-1} = \frac{1}{5}\begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

(multiply to check $AA^{-1} = I_4$). The leading principal minors of A are $2, 3, 4, 5$, so A is positive definite (Theorem 8.19). The constraints are $Cx = d$ with

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad d = \begin{pmatrix} 10 \\ 6 \end{pmatrix},$$

and $\operatorname{rank} C = 2$ (the rows are not multiples of each other). The unconstrained minimizer is $\hat{x} = A^{-1}b = \tfrac{2}{5}(10, 15, 15, 10)' = (4, 6, 6, 4)'$, with $q(\hat{x}) = -\tfrac{1}{2}b'\hat{x} = -20$ (Example 9.20(a)). Next

$$A^{-1}C' = \frac{1}{5}\begin{pmatrix} 10 & 2 \\ 15 & -1 \\ 15 & 1 \\ 10 & -2 \end{pmatrix}, \qquad V = CA^{-1}C' = \begin{pmatrix} 10 & 0 \\ 0 & \frac{6}{5} \end{pmatrix}, \qquad C\hat{x} - d = \begin{pmatrix} 20 \\ 0 \end{pmatrix} - \begin{pmatrix} 10 \\ 6 \end{pmatrix} = \begin{pmatrix} 10 \\ -6 \end{pmatrix},$$

so $\lambda = V^{-1}(C\hat{x} - d) = (1, -5)'$. By Theorem 9.27(a),(c),

$$x^* = \begin{pmatrix} 4 \\ 6 \\ 6 \\ 4 \end{pmatrix} - \frac{1}{5}\begin{pmatrix} 10 & 2 \\ 15 & -1 \\ 15 & 1 \\ 10 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ -5 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 4 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 4 \\ 0 \end{pmatrix}, \qquad q(x^*) - q(\hat{x}) = \tfrac{1}{2}\big(10 \cdot 1 + (-6)(-5)\big) = 20,$$

so the constrained minimum is $q(x^*) = 0$. Checks: x^* satisfies both constraints; $q(x^*) = 16 + 4 + 16 - 8 - 8 - 20 = 0$; and $Ax^* - b = (6, -4, 6, -4)' - (2, 2, 2, 2)' = (4, -6, 4, -6)' = -C'\lambda$, as in Theorem 9.27(d).

Variational characterizations of eigenvalues

Theorem 9.29 (Rayleigh–Ritz)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $A = Q\Lambda Q'$ with $Q = [q_1 \ \cdots \ q_n]$ orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (Theorem 8.4(a),(d)). For $x \neq 0$ let $R_A(x) = x'Ax/x'x$, the **Rayleigh quotient**.

(a) $\lambda_n \leqslant R_A(x) \leqslant \lambda_1$ for every $x \neq 0$; $R_A(x) = \lambda_1$ if and only if $Ax = \lambda_1 x$, and $R_A(x) = \lambda_n$ if and only if $Ax = \lambda_n x$.

(b) For $1 \leqslant k \leqslant n$, $\max\{R_A(x) : x \neq 0,\ q_i'x = 0 \text{ for } i < k\} = \lambda_k$, attained at $x = q_k$ (for $k = 1$ the only condition is $x \neq 0$).

(c) The stationary points of $f(x) = x'Ax$ on the unit sphere $\{x : x'x = 1\}$ are exactly the unit eigenvectors of A, and $f(x) = \lambda$ at a unit eigenvector x for the eigenvalue λ. The maximum λ_1 and the minimum λ_n of f on the sphere are attained exactly at the unit eigenvectors for λ_1 and for λ_n.

Proof. For $x \in \mathbb{R}^n$ put $c = Q'x$, so $x = Qc = \sum_i c_i q_i$ with $c_i = q_i'x$ (Proposition 6.10(b)), $x'x = c'c$ (Proposition 6.10(a)) and $x'Ax = c'\Lambda c = \sum_i \lambda_i c_i^2$.

(a) $\lambda_1 x'x - x'Ax = \sum_i (\lambda_1 - \lambda_i)c_i^2 \geqslant 0$, so $R_A(x) \leqslant \lambda_1$, with equality if and only if $(\lambda_1 - \lambda_i)c_i = 0$ for every i. Since $(\lambda_1 I_n - A)x = Q(\lambda_1 I_n - \Lambda)c$ and Q is invertible, this happens if and only if $Ax = \lambda_1 x$. The case of λ_n is the same with $x'Ax - \lambda_n x'x = \sum_i (\lambda_i - \lambda_n)c_i^2$.

(b) The condition $q_i'x = 0$ for $i < k$ means $c_1 = \cdots = c_{k-1} = 0$. Then $x'Ax = \sum_{i\geqslant k} \lambda_i c_i^2 \leqslant \lambda_k \sum_{i\geqslant k} c_i^2 = \lambda_k x'x$. The vector q_k satisfies the condition, as the columns of Q are orthonormal, and $q_k'Aq_k = \lambda_k q_k'q_k = \lambda_k$.

(c) Take $U = \mathbb{R}^n$ and $g(x) = x'x - 1$ in Remark 9.1(F5). By Theorem 9.2, $Dg(x) = 2x'$, of rank 1 on the sphere, and $\nabla f(x) = 2Ax$. So a unit vector x is a stationary point of f on the sphere if and only if $2Ax = 2\lambda x$ for some $\lambda \in \mathbb{R}$, that is, x is an eigenvector; then $f(x) = x'(\lambda x) = \lambda$. On the sphere $f = R_A$, so the last statement is (a). □

For symmetric $A \in \mathbb{R}^{n\times n}$ we write $\lambda_1(A) \geqslant \cdots \geqslant \lambda_n(A)$ for its eigenvalues listed with multiplicity in decreasing order, that is, the diagonal entries of Λ in Theorem 9.29; they do not depend on the choice of Q, being the roots of p_A (Theorem 8.4(b)).

Lemma 9.30. Let W_1 and W_2 be subspaces of $\mathbb{R}^n$. Then $\dim(W_1 \cap W_2) \geqslant \dim W_1 + \dim W_2 - n$. In particular $W_1 \cap W_2 \neq \{0\}$ if $\dim W_1 + \dim W_2 > n$.

Proof. Let $d_1 = \dim W_1$ and $d_2 = \dim W_2$; $W_1 \cap W_2$ is a subspace (Theorem 3.21(a)). If $d_1 = 0$ or $d_2 = 0$, then $d_1 + d_2 - n \leqslant 0$ and there is nothing to prove. Otherwise let the columns of $A \in \mathbb{R}^{n\times d_1}$ and of $B \in \mathbb{R}^{n\times d_2}$ be bases of W_1 and of W_2, and let $M = [A \ \ -B] \in \mathbb{R}^{n\times(d_1+d_2)}$. By Theorems 4.12 and 4.13(a), $s = \dim \mathcal{N}(M) = d_1 + d_2 - \operatorname{rank} M \geqslant d_1 + d_2 - n$; let $s \geqslant 1$ (otherwise there is nothing to prove), and let $(a_1', b_1')', \dots, (a_s', b_s')'$ be a basis of $\mathcal{N}(M)$, with $a_j \in \mathbb{R}^{d_1}$, $b_j \in \mathbb{R}^{d_2}$. Then $Aa_j = Bb_j \in W_1 \cap W_2$. These s vectors are independent: if $\sum_j c_j Aa_j = 0$, then $\sum_j c_j a_j = 0$ (the columns of A are independent), next $B\sum_j c_j b_j = A\sum_j c_j a_j = 0$ gives $\sum_j c_j b_j = 0$, so $\sum_j c_j (a_j', b_j')' = 0$ and every $c_j = 0$. Hence $\dim(W_1 \cap W_2) \geqslant s$ (Theorem 4.5(c)). □

Theorem 9.31 (Courant–Fischer)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric. On every subspace $W \neq \{0\}$ of $\mathbb{R}^n$, $R_A(x)$ attains a minimum and a maximum over the nonzero $x \in W$, and for $1 \leqslant k \leqslant n$

$$\lambda_k(A) = \max_{\dim W = k} \ \min_{0\neq x\in W} R_A(x) = \min_{\dim W = n-k+1} \ \max_{0\neq x\in W} R_A(x),$$

where W runs over the subspaces of $\mathbb{R}^n$ of the given dimension; with $q_1, \dots, q_n$ as in Theorem 9.29, the outer maximum is attained at $\operatorname{span}\{q_1, \dots, q_k\}$ and the outer minimum at $\operatorname{span}\{q_k, \dots, q_n\}$.

Proof. Let $d = \dim W \geqslant 1$, and let the columns of $V \in \mathbb{R}^{n\times d}$ be an orthonormal basis of W (Theorem 6.11), so $V'V = I_d$. The nonzero $x \in W$ are the vectors Vc with $c \neq 0$, and $R_A(Vc) = c'(V'AV)c/c'c$; by Theorem 9.29(a) for the symmetric matrix $V'AV$, this quotient attains a minimum and a maximum. Write $\lambda_i = \lambda_i(A)$; for $x = \sum_i c_i q_i$, $x'Ax = \sum_i \lambda_i c_i^2$ as in the proof of Theorem 9.29.

- On $W_k = \operatorname{span}\{q_1, \ldots, q_k\}$, every $x = \sum_{i\leqslant k} c_i q_i$ has $x'Ax \geqslant \lambda_k \sum_{i\leqslant k} c_i^2 = \lambda_k x'x$, with equality at q_k; so $\min_{0\neq x\in W_k} R_A(x) = \lambda_k$. On $Z_k = \operatorname{span}\{q_k, \ldots, q_n\}$, every $x = \sum_{i\geqslant k} c_i q_i$ has $x'Ax \leqslant \lambda_k x'x$, with equality at q_k; so $\max_{0\neq x\in Z_k} R_A(x) = \lambda_k$.
- Let $\dim W = k$. Since $\dim W + \dim Z_k = n + 1 > n$, Lemma 9.30 gives $x \neq 0$ in $W \cap Z_k$, and $\min_{0\neq x\in W} R_A \leqslant R_A(x) \leqslant \lambda_k$. With the first item, the maximum over W of the minimum is λ_k, attained at W_k.
- Let $\dim W = n - k + 1$. Since $\dim W + \dim W_k = n + 1 > n$, there is $x \neq 0$ in $W \cap W_k$, and $\max_{0\neq x\in W} R_A \geqslant R_A(x) \geqslant \lambda_k$. With the first item, the minimum over W of the maximum is λ_k, attained at Z_k. □

Corollary 9.32 (Weyl's inequalities)**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric.

(a) If $A \leqslant B$ (Definition 8.27), then $\lambda_k(A) \leqslant \lambda_k(B)$ for $k = 1, \ldots, n$.

(b) $\lambda_k(A) + \lambda_n(B) \leqslant \lambda_k(A + B) \leqslant \lambda_k(A) + \lambda_1(B)$ for $k = 1, \ldots, n$.

Proof. (a) $B - A$ is positive semidefinite, so $R_A(x) \leqslant R_B(x)$ for $x \neq 0$. Let W_k be the subspace at which the maximum in Theorem 9.31 is attained for A. If x_0 minimizes R_B over the nonzero vectors of W_k, then by Theorem 9.31 for A and for B, $\lambda_k(A) = \min_{0\neq x\in W_k} R_A(x) \leqslant R_A(x_0) \leqslant R_B(x_0) = \min_{0\neq x\in W_k} R_B(x) \leqslant \lambda_k(B)$.

(b) By Theorem 9.29(a), $\lambda_n(B)x'x \leqslant x'Bx \leqslant \lambda_1(B)x'x$, that is, $A + \lambda_n(B)I_n \leqslant A + B \leqslant A + \lambda_1(B)I_n$. For $c \in \mathbb{R}$, $A + cI_n = Q(\Lambda + cI_n)Q'$, so $\lambda_k(A + cI_n) = \lambda_k(A) + c$. Apply (a). □

Corollary 9.33. In Theorem 9.27, the symmetric matrix $K = \left(\begin{smallmatrix} A & C' \\ C & 0 \end{smallmatrix}\right)$ has exactly p positive and m negative eigenvalues. Hence the stationary point (x^*, λ) of the Lagrangian $L(x, \mu) = q(x) + \mu'(Cx - d)$ on $\mathbb{R}^{p+m}$ is a saddle point.

Proof.
- Let $M = \left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_m \end{smallmatrix}\right)$, invertible with $\det M = 1$. As A is symmetric, $M' = \left(\begin{smallmatrix} I_p & -A^{-1}C' \\ 0 & I_m \end{smallmatrix}\right)$, and by Theorem 5.9, $MKM' = \operatorname{diag}(A, -V)$ with $V = CA^{-1}C'$ positive definite. So $w'MKM'w = u'Au - v'Vv$ for $w = (u', v')'$ with $u \in \mathbb{R}^p$.
- Let $W_+ = \{M'(u', 0')' : u \in \mathbb{R}^p\}$ and $W_- = \{M'(0', v')' : v \in \mathbb{R}^m\}$, of dimensions p and m (M' is invertible). For $0 \neq x = M'(u', 0')' \in W_+$, $x'Kx = (u', 0')MKM'(u', 0')' = u'Au > 0$, and for $0 \neq x \in W_-$, $x'Kx = -v'Vv < 0$. By Theorem 9.31 with $n = p + m$, $\lambda_p(K) \geqslant \min_{0\neq x\in W_+} R_K(x) > 0$ and, as $m = n - (p + 1) + 1$, $\lambda_{p+1}(K) \leqslant \max_{0\neq x\in W_-} R_K(x) < 0$.
- $\nabla L(x, \mu) = \big((Ax - b + C'\mu)', (Cx - d)'\big)'$ (Theorem 9.2), which vanishes exactly at the solution (x^*, λ) of the system in Theorem 9.27(d), and $HL = K$. By Theorem 9.16(d), (x^*, λ) is a saddle point. □

So the constrained minimum is not found by minimizing L over (x, μ): L is minimized in the p directions of W_+ and maximized in the m directions of W_-.

Theorem 9.34 (Ky Fan)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, $A = Q\Lambda Q'$ as in Theorem 9.29, and $1 \leqslant k \leqslant n$. For every $V \in \mathbb{R}^{n\times k}$ with $V'V = I_k$,

$$\lambda_{n-k+1} + \cdots + \lambda_n \leqslant \operatorname{tr}(V'AV) \leqslant \lambda_1 + \cdots + \lambda_k,$$

with equality on the right for $V = [q_1 \ \cdots \ q_k]$ and on the left for $V = [q_{n-k+1} \ \cdots \ q_n]$.

Proof. Let $w_i = \|V'q_i\|^2 = q_i'VV'q_i$. By Theorem 1.11(c), $\operatorname{tr}(V'AV) = \operatorname{tr}(V'Q\Lambda Q'V) = \sum_i \lambda_i q_i'VV'q_i = \sum_i \lambda_i w_i$, as $Q\Lambda Q' = \sum_i \lambda_i q_i q_i'$ (Corollary 8.5(a)).

- VV' is the projection matrix onto $\mathcal{C}(V)$ (Theorem 6.19(a)), so $w_i = \|VV'q_i\|^2 \leqslant \|q_i\|^2 = 1$ (Theorem 6.19(b) and Pythagoras, Proposition 6.2, for $q_i = VV'q_i + (I_n - VV')q_i$). Also $\sum_i w_i = \operatorname{tr}(V'QQ'V) = \operatorname{tr}(V'V) = k$.
- Hence

$$\sum_{i\leqslant k}\lambda_i - \sum_i \lambda_i w_i = \sum_{i\leqslant k}\lambda_i(1-w_i) - \sum_{i>k}\lambda_i w_i \geqslant \lambda_k\sum_{i\leqslant k}(1-w_i) - \lambda_k\sum_{i>k} w_i = \lambda_k\Big(k - \sum_i w_i\Big) = 0$$

  For $V = [q_1 \ \cdots \ q_k]$, $V'V = I_k$ and $\operatorname{tr}(V'AV) = \sum_{i\leqslant k} q_i'Aq_i = \sum_{i\leqslant k}\lambda_i$.
- The left inequality is the right one for $-A = Q(-\Lambda)Q'$, whose eigenvalues in decreasing order are $-\lambda_n \geqslant \cdots \geqslant -\lambda_1$. □

For $k = 1$ this is Theorem 9.29(a). If $A = S$ is the sample covariance matrix of a data matrix whose number of columns (variables) is the size n of A (Definition 8.32), then $\operatorname{tr}(V'SV) = \sum_j v_j'Sv_j$ is the sum of the sample variances of the k new variables $v_j'x_i$ (Proposition 8.33(a)); among all choices of k orthonormal directions $v_1, \ldots, v_k$, the eigenvectors $q_1, \ldots, q_k$ of S retain the largest total variance.

Example 9.35. For the matrix A of Example 9.17, with eigenvalues $32, 18, 8, 2$ and eigenvectors $h_1, \ldots, h_4$, find the maximum and the minimum of $\operatorname{tr}(V'AV)$ over $V \in \mathbb{R}^{4\times 2}$ with $V'V = I_2$.

Solution. By Theorem 9.34, the maximum is $32+18 = 50$, attained at $V = \frac{1}{2}[h_1 \ h_2]$, and the minimum is $8 + 2 = 10$, attained at $V = \frac{1}{2}[h_3 \ h_4]$. Other frames give intermediate values: $V = [e_1 \ e_2]$ gives $a_{11} + a_{22} = 30$, and $V = \frac{1}{\sqrt{2}}[e_1 + e_2 \ \ e_3 + e_4]$ gives $\frac{1}{2}(15+5+5+15) + \frac{1}{2}(15+5+5+15) = 40$. The frame $V = \frac{1}{\sqrt{2}}[e_1 + e_3 \ \ e_2 + e_4]$ gives $\frac{1}{2}(15+10+10+15)\cdot 2 = 50$: its columns span the same plane as $h_1 = (e_1 + e_3) + (e_2 + e_4)$ and $h_2 = (e_1 + e_3) - (e_2 + e_4)$, and $\operatorname{tr}(V'AV) = \operatorname{tr}(AVV')$ depends on V only through the projection matrix VV' onto that plane.

Theorem 9.36. Let $A \in \mathbb{R}^{n\times n}$ be symmetric and $B \in \mathbb{R}^{n\times n}$ positive definite. Let $P = [p_1 \ \cdots \ p_n]$ be invertible with $P'BP = I_n$ and $P'AP = \operatorname{diag}(\theta_1, \ldots, \theta_n)$, $\theta_1 \geqslant \cdots \geqslant \theta_n$; such a P exists by Theorem 8.30(a), applied to the positive definite matrix B and the symmetric matrix A. By Theorem 8.30(b), $\theta_1, \ldots, \theta_n$ are the roots of $\det(A - \theta B) = 0$, and $Ap_i = \theta_i Bp_i$.

(a) $x'Ax/x'Bx \leqslant \theta_1$ for every $x \neq 0$, with equality at $x = p_1$; so $\max_{x\neq 0} x'Ax/x'Bx = \theta_1$, the largest root of $\det(A - \theta B) = 0$.

(b) For $a, x \in \mathbb{R}^n$, $(a'x)^2 \leqslant (a'B^{-1}a)(x'Bx)$; if $a \neq 0$, equality holds if and only if x is a multiple of $B^{-1}a$. Hence $\max_{x\neq 0}(a'x)^2/x'Bx = a'B^{-1}a$, attained at $x = B^{-1}a$ when $a \neq 0$.

Proof. (a) For $x \neq 0$, $c = P^{-1}x \neq 0$, and $x = Pc$ gives $x'Bx = c'P'BPc = c'c$ and $x'Ax = c'P'APc = \sum_i \theta_i c_i^2 \leqslant \theta_1 c'c$. For $x = p_1$, $c = e_1$ and equality holds.

(b) Let $B^{1/2}$ be the positive semidefinite square root of B and $B^{-1/2} = (B^{1/2})^{-1}$; both are symmetric, and $B^{-1/2}B^{-1/2} = B^{-1}$ (Theorem 8.25(a),(c)). Then $a'x = (B^{-1/2}a)'(B^{1/2}x)$, and the Cauchy–Schwarz inequality (Theorem 6.3(a)) gives $(a'x)^2 \leqslant \|B^{-1/2}a\|^2\|B^{1/2}x\|^2 = (a'B^{-1}a)(x'Bx)$. If $a \neq 0$, then $B^{-1/2}a \neq 0$, and equality holds if and only if $B^{1/2}x = tB^{-1/2}a$ for some $t \in \mathbb{R}$, that is, $x = tB^{-1}a$. For $x \neq 0$, $x'Bx > 0$; dividing gives $(a'x)^2/x'Bx \leqslant a'B^{-1}a$, with equality at $x = B^{-1}a \neq 0$ if $a \neq 0$; if $a = 0$, both sides are 0 for every $x \neq 0$. □

Example 9.37 (Fisher's discriminant direction)**.** Observations of k variables fall into two groups. Group g ($g = 1, 2$) has $n_g \geqslant 2$ observations, with mean vector $\bar{x}_g$ and sample covariance matrix S_g (Definition 8.32). The **pooled covariance matrix** is $S_p = \big((n_1 - 1)S_1 + (n_2 - 1)S_2\big)/(n_1 + n_2 - 2)$; assume that S_p is positive definite and $d = \bar{x}_1 - \bar{x}_2 \neq 0$. For a direction $v \neq 0$, the values $v'x$ of the observations of group g have mean $v'\bar{x}_g$ and sample variance $v'S_gv$ (Proposition 8.33(a)); so

the difference of the group means is $v'd$, and the pooled sample variance $\big((n_1-1)v'S_1v+(n_2-1)v'S_2v\big)/(n_1+n_2-2)$ is $v'S_pv$. Fisher's criterion $F(v)=(v'd)^2/v'S_pv$ measures how well the groups separate along v. By Theorem 9.36(b), F is maximized exactly by the nonzero multiples of $S_p^{-1}d$, and the maximum is $d'S_p^{-1}d=d_{S_p}(\bar x_1,\bar x_2)^2$, the squared Mahalanobis distance between the group means (Definition 8.35).

For example, let group 1 consist of $(0,0)'$, $(1,2)'$, $(2,1)'$ and group 2 of $(2,0)'$, $(2,1)'$, $(5,2)'$. Then $\bar x_1=(1,1)'$, $\bar x_2=(3,1)'$, $d=(-2,0)'$, and the matrices of corrected sums of squares and products are $(n_1-1)S_1=\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$ and $(n_2-1)S_2=\left(\begin{smallmatrix}6&3\\3&2\end{smallmatrix}\right)$. So

$$S_p=\frac14\begin{pmatrix}8&4\\4&4\end{pmatrix}=\begin{pmatrix}2&1\\1&1\end{pmatrix},\qquad S_p^{-1}=\begin{pmatrix}1&-1\\-1&2\end{pmatrix},\qquad S_p^{-1}d=\begin{pmatrix}-2\\2\end{pmatrix},$$

and the best direction is $v=(1,-1)'$, with $F(v)=4=d'S_p^{-1}d$. The means differ only in the first variable, yet e_1 gives only $F(e_1)=4/2=2$: the second variable, correlated with the first within the groups, sharpens the separation. Along v, the values x_1-x_2 are $0,-1,1$ in group 1 and $2,1,3$ in group 2.

Exercises

Exercise 9.1. Let

$$A=\begin{pmatrix}1&1&1\\1&0&-1\end{pmatrix}\in\mathbb{R}^{2\times3},\qquad B=\begin{pmatrix}2&1\\-1&1\\1&-1\end{pmatrix}\in\mathbb{R}^{3\times2}$$

(a) Write out $A\otimes B\in\mathbb{R}^{6\times6}$ and find its rank.

(b) Compute AB and BA, show that $(A\otimes B)(B\otimes A)=AB\otimes BA$, and find the eigenvalues and the trace of this 6×6 matrix.

(c) Write the equation $AXB=C$ ($X\in\mathbb{R}^{3\times3}$, $C\in\mathbb{R}^{2\times2}$) as a linear system $M\operatorname{vec}X=\operatorname{vec}C$ with $M\in\mathbb{R}^{4\times9}$. Show that $\operatorname{rank}M=4$ and $MM'=B'B\otimes AA'$.

(d) Show that the solution of $AXB=C$ of least Frobenius norm is $X^*=A'(AA')^{-1}C(B'B)^{-1}B'$. Find X^* for $C=\operatorname{diag}(18,6)$.

Solution. (a) By Definition 9.7, the blocks of $A\otimes B$ are B,B,B in the first block row and $B,0,-B$ in the second:

$$A\otimes B=\begin{pmatrix}2&1&2&1&2&1\\-1&1&-1&1&-1&1\\1&-1&1&-1&1&-1\\2&1&0&0&-2&-1\\-1&1&0&0&1&-1\\1&-1&0&0&-1&1\end{pmatrix}$$

The two rows of A are not multiples of each other, nor are the two columns of B, so $\operatorname{rank}A=\operatorname{rank}B=2$, and $\operatorname{rank}(A\otimes B)=4$ by Theorem 9.8(f). Indeed rows 3 and 6 are the negatives of rows 2 and 5.

(b) Multiplying,

$$AB=\begin{pmatrix}2&1\\1&2\end{pmatrix},\qquad BA=\begin{pmatrix}3&2&1\\0&-1&-2\\0&1&2\end{pmatrix}$$

By Theorem 9.8(a), with the 2×3 and 3×2 factors A,B and the 3×2 and 2×3 factors B,A, $(A\otimes B)(B\otimes A)=AB\otimes BA$. Here $p_{AB}(t)=(t-2)^2-1=(t-1)(t-3)$ and, expanding along the

first column (Theorem 5.13), $p_{BA}(t) = (t-3)\big((t+1)(t-2)+2\big) = t(t-1)(t-3)$. By Theorem 9.9(b), the eigenvalues of $AB \otimes BA$ are the products $3\cdot 3, 3\cdot 1, 3\cdot 0, 1\cdot 3, 1\cdot 1, 1\cdot 0$, that is, $9, 3, 3, 1, 0, 0$, and the trace is $16 = \operatorname{tr}(AB)\operatorname{tr}(BA) = 4\cdot 4$ (Theorem 9.8(e)).

(c) By Theorem 9.8(d), $\operatorname{vec}(AXB) = (B' \otimes A)\operatorname{vec} X$, so $M = B' \otimes A \in \mathbb{R}^{4\times 9}$, and $\operatorname{rank} M = \operatorname{rank} B' \cdot \operatorname{rank} A = 2\cdot 2 = 4$ (Theorem 9.8(f), Theorem 4.13(a)). By Theorem 9.8(a),(b), $MM' = (B'\otimes A)(B\otimes A') = B'B \otimes AA'$. The rows of A and the columns of B are orthogonal, so $AA' = \operatorname{diag}(3,2)$, $B'B = \operatorname{diag}(6,3)$ and $MM' = \operatorname{diag}(18,12,9,6)$.

(d) M has rank 4, its number of rows, and $\|X\|_F = \|\operatorname{vec} X\|$. By Proposition 9.25, the solution of least norm is $\operatorname{vec} X^* = M'(MM')^{-1}\operatorname{vec} C$. By Theorem 9.8(c), $(MM')^{-1} = (B'B)^{-1} \otimes (AA')^{-1}$, with $(B'B)^{-1}$ symmetric, so $(MM')^{-1}\operatorname{vec} C = \operatorname{vec}\big((AA')^{-1}C(B'B)^{-1}\big)$; and $M' = B \otimes A'$ (Theorem 9.8(b)) maps $\operatorname{vec} Y$ to $\operatorname{vec}(A'YB')$ (Theorem 9.8(d)). Hence $X^* = A'(AA')^{-1}C(B'B)^{-1}B'$. For $C = \operatorname{diag}(18,6)$, $(AA')^{-1}C(B'B)^{-1} = \operatorname{diag}(\frac{1}{3},\frac{1}{2})\operatorname{diag}(18,6)\operatorname{diag}(\frac{1}{6},\frac{1}{3}) = I_2$, and

$$X^* = A'B' = (BA)' = \begin{pmatrix} 3 & 0 & 0 \\ 2 & -1 & 1 \\ 1 & -2 & 2 \end{pmatrix}, \qquad \|X^*\|_F^2 = 24$$

Check: $AX^*B = (AA')(B'B) = \operatorname{diag}(18,6)$. Other solutions are longer: $A(1,-2,1)' = 0$, so $X = X^* + (1,-2,1)'e_1'$ also solves $AXB = C$, and $\|X\|_F^2 = 24 + 6 = 30$ (Proposition 9.25).

Exercise 9.2. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 4 & -1 \\ 8 & 0 \end{smallmatrix}\right)$.

(a) Show that $AX + XB = C$ ($X \in \mathbb{R}^{2\times 2}$) is the linear system $(I_2 \otimes A + B' \otimes I_2)\operatorname{vec} X = \operatorname{vec} C$, and write out its matrix L.

(b) Find the eigenvalues of L from those of A and B, and conclude that the equation has exactly one solution for every C.

(c) Solve it.

Solution. (a) By Theorem 9.8(d), $\operatorname{vec}(AXI_2) = (I_2 \otimes A)\operatorname{vec} X$ and $\operatorname{vec}(I_2XB) = (B' \otimes I_2)\operatorname{vec} X$. Here

$$L = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 1 & 0 \\ 0 & 4 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

(b) B' has the eigenvalues $1, -1$ (the same as B) and A has $2, 3$. By Theorem 9.9(b) applied to $B' \otimes I_2 + I_2 \otimes A$, the eigenvalues of L are $1+2=3$, $1+3=4$, $-1+2=1$ and $-1+3=2$, the diagonal of the triangular matrix L. None is 0, so $\det L = 24 \neq 0$ and L is invertible (Theorem 5.19(a)).

(c) $\operatorname{vec} C = (4, 8, -1, 0)'$. Back substitution in $L\operatorname{vec} X = \operatorname{vec} C$ gives $x_{22} = 0$, $x_{12} = -1$, $x_{21} = 2$ and $3x_{11} = 4 - 2 + 1$, $x_{11} = 1$. So $X = \left(\begin{smallmatrix} 1 & -1 \\ 2 & 0 \end{smallmatrix}\right)$; check: $AX + XB = \left(\begin{smallmatrix} 4 & -2 \\ 6 & 0 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 1 \\ 2 & 0 \end{smallmatrix}\right) = C$.

Exercise 9.3. (a) Find the Jacobian determinant of $X \mapsto AXB$ on $\mathbb{R}^{2\times 3}$ for $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right)$.

(b) Find X_0^{-1} and the Jacobian determinant of $X \mapsto X^{-1}$ at $X_0 = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{smallmatrix}\right)$.

(c) Find the Jacobian matrix of $X \mapsto X'$ on $\mathbb{R}^{2\times 2}$ and its determinant.

(d) Find the Jacobian matrix of $X \mapsto X^2$ at $X_1 = \left(\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right)$, and show that it is singular. Compute $(X_1 + tE_{12})^2$ for $t \in \mathbb{R}$.

Solution. (a) $\det A = 2$ and $\det B = 2(4-1) - (2-1) + (1-2) = 4$. By Theorem 9.12(a) with $p = 2$, $q = 3$, the Jacobian matrix is $B' \otimes A \in \mathbb{R}^{6\times 6}$, with determinant $(\det A)^3(\det B)^2 = 8 \cdot 16 = 128$.

(b) $\det X_0 = 1\cdot 1 - 1\cdot(0-1) + 0 = 2$, and multiplying checks $X_0^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}$. By Theorem 9.12(b) with $n = 3$, the Jacobian determinant is $(-1)^3\, 2^{-6} = -\frac{1}{64}$.

(c) The map is linear, with $\operatorname{vec}(X') = (x_{11}, x_{12}, x_{21}, x_{22})'$ for $\operatorname{vec} X = (x_{11}, x_{21}, x_{12}, x_{22})'$. So the Jacobian matrix is the permutation matrix $K = [e_1\ e_3\ e_2\ e_4]$, obtained from I_4 by exchanging rows 2 and 3, and $\det K = -1$ (Theorem 5.16).

(d) By Theorem 9.12(c), the Jacobian matrix is

$$X_1' \otimes I_2 + I_2 \otimes X_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & -2 \end{pmatrix},$$

singular, as its second row is zero; X_1 is triangular with eigenvalues $1, -1$, and $1 + (-1) = 0$ (Remark 9.13). Indeed $(X_1 + tE_{12})^2 = \left(\begin{smallmatrix} 1 & 2+t \\ 0 & -1 \end{smallmatrix}\right)^2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ for every t: the whole line through X_1 in the direction E_{12} is mapped to I_2, and $\operatorname{vec} E_{12} = e_3$ spans part of the null space of the Jacobian matrix (its third column is zero).

Exercise 9.4. Let

$$X_0 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$$

(a) Show that $X_0^{-1} = \left(\begin{smallmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$, and find $\det X_0$, $\partial \det X/\partial X$ and $\partial \log\det X/\partial X$ at X_0.

(b) Find the differential of X^{-1} at X_0 for the increment $\mathrm{d}X = E_{41}$, and show that it equals vv'.

Solution. (a) Multiplying, X_0 times the given matrix is I_4, so it is X_0^{-1} (Theorem 2.18). X_0 is upper triangular with diagonal entries 1, so $\det X_0 = 1$ (Theorem 5.15(a)). By Theorem 9.5(b),(c),

$$\frac{\partial \det X}{\partial X}(X_0) = \frac{\partial \log\det X}{\partial X}(X_0) = (X_0^{-1})' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

(b) By Theorem 9.5(d), $\mathrm{d}(X^{-1}) = -X_0^{-1} e_4 e_1' X_0^{-1}$. Here $X_0^{-1} e_4 = (-1, 1, -1, 1)' = -v$ (the fourth column) and $e_1' X_0^{-1} = v'$ (the first row), so $\mathrm{d}(X^{-1}) = vv'$.

Exercise 9.5. Let

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 4 & -2 & 1 & 1 \end{pmatrix}$$

Run the Newton–Schulz iteration of Example 9.14 from $X_0 = I_4$: find the residuals $E_0 = I_4 - AX_0$, E_1, E_2 and the iterates X_1, X_2, and show that $X_2 = A^{-1}$.

Solution. Let $N = A - I_4$, strictly lower triangular. Then $E_0 = -N$, and by Example 9.14(a), $E_1 = N^2$ and $E_2 = N^4$, where

$$N^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ -5 & 3 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

So $AX_2 = I_4$, and $X_2 = A^{-1}$ (Theorem 2.18). The iterates are $X_1 = 2I_4 - A = I_4 - N$ and $X_2 = X_1(2I_4 - AX_1) = (I_4 - N)(I_4 + N^2) = I_4 - N + N^2 - N^3$:

$$X_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -4 & 2 & -1 & 1 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 7 & -3 & 1 & 0 \\ -15 & 5 & -1 & 1 \end{pmatrix}$$

Both are lower triangular with unit diagonal, hence invertible (Theorem 5.15(a)), as the iteration requires. Check one entry of $AX_2 = I_4$: row 4 of A times column 1 of X_2 is $4 + 4 + 7 - 15 = 0$.

Exercise 9.6. Let $M = 4I_4 + 3J_4$ and $f(x) = \frac{1}{4}(x'x)^2 - \frac{1}{2}x'Mx$ on $\mathbb{R}^4$.

(a) Find ∇f and Hf, and the eigenvalues and eigenvectors of M.

(b) Find all stationary points of f, and classify them by the eigenvalues of the Hessian. Treat $x = (1, 1, -1, -1)'$ explicitly.

(c) Find the global minimum of f.

Solution. (a) As in Example 9.17, $\nabla f(x) = (x'x)x - Mx$ and $Hf(x) = (x'x)I_4 + 2xx' - M$. Since $J_4\mathbf{1}_4 = 4\mathbf{1}_4$ and $J_4 v = 0$ for $v \perp \mathbf{1}_4$, $M\mathbf{1}_4 = 16\mathbf{1}_4$ and $Mv = 4v$ for every v in the 3-dimensional subspace $\mathbf{1}_4^{\perp}$.

(b) $\nabla f(x) = 0$ means $Mx = (x'x)x$: $x = 0$, or x is an eigenvector of M with $x'x$ equal to its eigenvalue. The stationary points are 0, $\pm 2\mathbf{1}_4$ ($x'x = 16$), and every $x \perp \mathbf{1}_4$ with $x'x = 4$, infinitely many.

- At $\pm 2\mathbf{1}_4$, $Hf = 16I_4 + 8J_4 - 4I_4 - 3J_4 = 12I_4 + 5J_4$, with eigenvalues 32 (on $\mathbf{1}_4$) and 12 (three times): a strict local minimum, with $f = 64 - 128 = -64$ (Theorem 9.16(b)).
- At $x \perp \mathbf{1}_4$ with $x'x = 4$, $Hf = 2xx' - 3J_4$. Then $Hf\,x = 8x$, $Hf\,\mathbf{1}_4 = -12\mathbf{1}_4$, and $Hf\,v = 0$ for $v \perp x, \mathbf{1}_4$: eigenvalues $8, -12, 0, 0$. As $8 > 0 > -12$, this is a saddle point (Theorem 9.16(d)), with $f = 4 - 8 = -4$; the zero eigenvalues belong to the directions along the sphere of stationary points. For $x = (1, 1, -1, -1)'$,
$$Hf(x) = \begin{pmatrix} -1 & -1 & -5 & -5 \\ -1 & -1 & -5 & -5 \\ -5 & -5 & -1 & -1 \\ -5 & -5 & -1 & -1 \end{pmatrix},$$
and $Hf(x)(1, -1, 0, 0)' = Hf(x)(0, 0, 1, -1)' = 0$.
- At 0, $Hf = -M$ has the eigenvalues $-16, -4, -4, -4$: a strict local maximum, $f(0) = 0$.

(c) By Theorem 8.13(c), $x'Mx \leqslant 16s$ with $s = x'x$, so $f(x) \geqslant \frac{1}{4}s^2 - 8s = \frac{1}{4}(s - 16)^2 - 64 \geqslant -64$. The global minimum is -64, attained at $\pm 2\mathbf{1}_4$.

Exercise 9.7. Let $S = A \otimes B$ be the matrix of Example 9.10, and $g(X) = \log\det X - \operatorname{tr}(SX)$ on $\mathcal{P}_4$.

(a) Maximize $g(cI_4)$ over $c > 0$.

(b) Find the maximizer and the maximum of g on $\mathcal{P}_4$, and compare with (a).

Solution. (a) $\operatorname{tr} S = 24$, so $g(cI_4) = 4\log c - 24c$, with derivative $4/c - 24$ and second derivative $-4/c^2 < 0$ on $(0, \infty)$. By Theorem 9.19(c), the stationary point $c = \frac{1}{6}$ is the maximizer, and $g(\frac{1}{6}I_4) = -4\log 6 - 4 = -2\log 36 - 4$.

(b) By Theorem 9.21(c), the maximizer is $S^{-1} = \frac{1}{24}\begin{pmatrix} 6 & -2 & -3 & 1 \\ -2 & 6 & 1 & -3 \\ -3 & 1 & 6 & -2 \\ 1 & -3 & -2 & 6 \end{pmatrix}$ (Example 9.10), and the maximum is $-\log\det S - 4 = -\log 576 - 4 = -2\log 24 - 4$, larger than the value $-2\log 36 - 4$ in (a).

Exercise 9.8. Let $a = (3,1,4,1)'$ and $L = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 1,\ x_1 + x_2 - x_3 - x_4 = 3\}$.

(a) Find the point x^* of L closest to a, and the distance from a to L, by Theorem 9.27 with $A = I_4$ and $b = a$.

(b) Check Theorem 9.27(b) for the point $x = (2,0,0,-1)'$ of L.

(c) Find the determinant of the 6×6 matrix $K = \left(\begin{smallmatrix} I_4 & C' \\ C & 0 \end{smallmatrix}\right)$ of Theorem 9.27(d), and the numbers of its positive and negative eigenvalues.

Solution. (a) With $A = I_4$ and $b = a$, $q(x) = \frac{1}{2}x'x - a'x = \frac{1}{2}\|x-a\|^2 - \frac{1}{2}\|a\|^2$, so minimizing q on L is minimizing the distance to a, and $\hat{x} = a$. Here

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \end{pmatrix}, \qquad d = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad V = CC' = 4I_2, \qquad Ca - d = \begin{pmatrix} 9 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \end{pmatrix},$$

$\operatorname{rank} C = 2$, and $\lambda = V^{-1}(Ca - d) = (2,-1)'$. By Theorem 9.27(a),

$$x^* = a - C'\lambda = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \\ -2 \end{pmatrix},$$

which satisfies $2 + 0 + 1 - 2 = 1$ and $2 + 0 - 1 + 2 = 3$. By Theorem 9.27(c), $\frac{1}{2}\|x^* - a\|^2 = q(x^*) - q(a) = \frac{1}{2}(Ca-d)'V^{-1}(Ca-d) = \frac{1}{2} \cdot \frac{64+16}{4} = 10$; so the distance is $\sqrt{20} = 2\sqrt{5}$, and directly $\|a - x^*\|^2 = 1 + 1 + 9 + 9 = 20$.

(b) $x = (2,0,0,-1)'$ lies in L ($2+0+0-1 = 1$, $2+0-0+1 = 3$). Here $q(x) = \frac{1}{2} \cdot 5 - (6-1) = -\frac{5}{2}$, $q(x^*) = \frac{1}{2} \cdot 9 - (6+4-2) = -\frac{7}{2}$, and $\frac{1}{2}\|x - x^*\|^2 = \frac{1}{2}\|(0,0,-1,1)'\|^2 = 1$; indeed $-\frac{5}{2} = -\frac{7}{2} + 1$.

(c) By Theorem 5.26(a), $\det K = \det I_4 \cdot \det(0 - CI_4^{-1}C') = \det(-4I_2) = 16$. By Corollary 9.33 with $p = 4$ and $m = 2$, K has four positive and two negative eigenvalues; their product 16 is positive, as it must be (Corollary 7.30(a)).

Exercise 9.9. Let A be the matrix of Example 9.23, and $h_1, \dots, h_4$ as in Example 9.17.

(a) Show that $A = I_2 \otimes B_0 + C_0 \otimes I_2$ with $B_0 = \left(\begin{smallmatrix} 5 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ and $C_0 = \left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$, and obtain the eigenvalues of A from those of B_0 and C_0.

(b) Show that every eigenvector of A is a multiple of one of $h_1, \dots, h_4$.

(c) Find all stationary points of $f(x) = x'Ax$ on the unit sphere of $\mathbb{R}^4$, and the maximum and the minimum of f on the sphere.

(d) Find the maximum of $R_A(x)$ over the nonzero x with $x_1 + x_2 + x_3 + x_4 = 0$, and a maximizer.

(e) Find the maximum of $\operatorname{tr}(V'AV)$ over $V \in \mathbb{R}^{4\times k}$ with $V'V = I_k$, for $k = 2$ and $k = 3$.

Solution. (a) $I_2 \otimes B_0 = \operatorname{diag}(B_0, B_0)$ and $C_0 \otimes I_2 = \left(\begin{smallmatrix} 0 & 2I_2 \\ 2I_2 & 0 \end{smallmatrix}\right)$; their sum is A. B_0 has the eigenvalues $6, 4$ and C_0 has $2, -2$ (eigenvectors $(1,1)'$, $(1,-1)'$ in both cases). By Theorem 9.9(b) applied to $C_0 \otimes I_2 + I_2 \otimes B_0$, the eigenvalues of A are $2+6 = 8$, $2+4 = 6$, $-2+6 = 4$ and $-2+4 = 2$.

(b) By Example 9.23, $Ah_1 = 8h_1$, $Ah_2 = 6h_2$, $Ah_3 = 4h_3$ and $Ah_4 = 2h_4$. The h_j are nonzero and pairwise orthogonal, hence independent (Proposition 6.8(b)), and a basis of $\mathbb{R}^4$ (Theorem 4.5(d)). For $x = \sum_j c_j h_j$, $Ax = 8c_1h_1 + 6c_2h_2 + 4c_3h_3 + 2c_4h_4$, so $Ax = \lambda x$ means $(8-\lambda)c_1 = (6-\lambda)c_2 = (4-\lambda)c_3 = (2-\lambda)c_4 = 0$. If $x \neq 0$, some $c_j \neq 0$, so $\lambda \in \{8,6,4,2\}$, and then the other three coefficients vanish: x is a multiple of one h_j.

(c) Each $\|h_j\| = 2$. By Theorem 9.29(c), the stationary points are the unit eigenvectors $\pm\frac{1}{2}h_1$ ($f = 8$), $\pm\frac{1}{2}h_2$ ($f = 6$), $\pm\frac{1}{2}h_3$ ($f = 4$) and $\pm\frac{1}{2}h_4$ ($f = 2$). The maximum of f on the sphere is 8 and the minimum is 2 (Theorem 9.29(a)).

(d) Let $q_j = \frac{1}{2}h_j$. The condition is $q_1'x = 0$. By Theorem 9.29(b) with $k = 2$ and $Q = [q_1\ q_2\ q_3\ q_4]$, the maximum is $\lambda_2 = 6$, attained at $x = q_2$ (or h_2).

(e) By Theorem 9.34, the maximum is $8+6 = 14$ for $k = 2$, attained at $V = [q_1\ q_2]$, and $8+6+4 = 18$ for $k = 3$, attained at $V = [q_1\ q_2\ q_3]$. For comparison, $V = [e_1\ e_2]$ gives $a_{11} + a_{22} = 10$.

Problems

Problem 9.1. (a) Show that $\operatorname{vec}(ab') = b \otimes a$ for $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$.

(b) Let $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{q\times q}$ and $f(X) = \operatorname{tr}(X'AXB)$ on $\mathbb{R}^{p\times q}$. Show that $f(X) = (\operatorname{vec} X)'(B' \otimes A)\operatorname{vec} X$, and find the Hessian of f. If A and B are positive definite and $C \in \mathbb{R}^{p\times q}$, show that $f(X) - 2\operatorname{tr}(C'X)$ has the unique minimizer $X = A^{-1}CB^{-1}$.

(c) Let $X \in \mathbb{R}^{n\times n}$ be positive definite with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n$, and $H \in \mathbb{R}^{n\times n}$ symmetric. Show that

$$\lambda_1^{-2}\|H\|_F^2 \leqslant \operatorname{tr}(X^{-1}HX^{-1}H) \leqslant \lambda_n^{-2}\|H\|_F^2,$$

two-sided bounds for the second differential of $\log\det$ in Theorem 9.21(a).

Solution. (a) Column j of ab' is $b_j a$, and block j of $b \otimes a$ is $b_j a$ (Definition 9.7).

(b) By the identity $\operatorname{tr}(P'Q) = (\operatorname{vec} P)'\operatorname{vec} Q$ after Definition 9.7 and Theorem 9.8(d), $f(X) = \operatorname{tr}\big(X'(AXB)\big) = (\operatorname{vec} X)'(B' \otimes A)\operatorname{vec} X$. With $x = \operatorname{vec} X$ and $G = B' \otimes A$, Theorem 9.2 gives $\nabla f = (G + G')x$ and the Hessian $G + G' = B' \otimes A + B \otimes A'$ (Theorem 9.8(b)), at every X. If A and B are positive definite, the Hessian is $2(B \otimes A)$, positive definite by Theorem 9.9(c), and $f(X) - 2\operatorname{tr}(C'X) = \frac{1}{2}x'\big(2(B \otimes A)\big)x - 2(\operatorname{vec} C)'x$ is a quadratic function as in Example 9.20(a). Its unique minimizer is $x = (B \otimes A)^{-1}\operatorname{vec} C = (B^{-1} \otimes A^{-1})\operatorname{vec} C = \operatorname{vec}(A^{-1}CB^{-1})$ (Theorem 9.8(c),(d), with B^{-1} symmetric).

(c) By Theorem 9.21(a) with $D = H$ and $t = 0$, $\operatorname{tr}(X^{-1}HX^{-1}H) = (\operatorname{vec} H)'(X^{-1} \otimes X^{-1})\operatorname{vec} H$. The eigenvalues of X^{-1} are λ_i^{-1} (with the eigenvectors of X), so by Theorem 9.9(c) those of $X^{-1} \otimes X^{-1}$ are $\lambda_i^{-1}\lambda_j^{-1}$, between λ_1^{-2} and λ_n^{-2}. Theorem 8.13(c) and $\|\operatorname{vec} H\| = \|H\|_F$ give the bounds.

Problem 9.2. Let $X \in \mathcal{P}_n$, D symmetric, and X_t, $\mathcal{I}$ as in Theorem 9.21(a).

(a) Show that $\psi(t) = \operatorname{tr}(X_t^{-1})$ has $\psi'(t) = -\operatorname{tr}(X_t^{-1}DX_t^{-1})$ and $\psi''(t) = 2\operatorname{tr}(X_t^{-1}DX_t^{-1}DX_t^{-1})$, and that $\psi''(t) > 0$ if $D \neq 0$.

(b) Conclude that $X \mapsto \operatorname{tr}(X^{-1})$ is strictly convex on $\mathcal{P}_n$.

(c) Check $\operatorname{tr}\big((\frac{1}{2}(X+Y))^{-1}\big) < \frac{1}{2}\big(\operatorname{tr} X^{-1} + \operatorname{tr} Y^{-1}\big)$ for $Y = I_4$ and $X = A \otimes B$ of Example 9.10.

Solution. (a) By Theorem 9.5(d) and the chain rule, $\mathrm{d}(X_t^{-1}) = -X_t^{-1}DX_t^{-1}\mathrm{d}t$, and $\mathrm{d}\operatorname{tr} = \operatorname{tr}\mathrm{d}$ (F2) gives ψ'. Differentiating $-X_t^{-1}DX_t^{-1}$ by the product rule gives $X_t^{-1}DX_t^{-1}DX_t^{-1} + X_t^{-1}DX_t^{-1}DX_t^{-1}$, and ψ'' follows. With the symmetric matrices $Z = X_t^{-1/2}$ (Theorem 8.25(c)) and $N = ZDZ^2$, $N'N = Z^2DZ\,ZDZ^2 = X_t^{-1}DX_t^{-1}DX_t^{-1}$, so $\psi''(t) = 2\|N\|_F^2$ (Theorem 1.11(d)). If $D \neq 0$, then $N \neq 0$, as Z is invertible.

(b) As in the proof of Theorem 9.21(b), with Theorem 9.19(b) in place of (c): for $X \neq Y$ in $\mathcal{P}_n$ and $D = Y - X$, ψ is strictly convex on $\mathcal{I} \supseteq [0,1]$, and $\operatorname{tr}\big(((1-t)X + tY)^{-1}\big) = \psi(t) < (1-t)\psi(0) + t\psi(1)$ for $0 < t < 1$.

(c) X has the eigenvalues $12, 6, 4, 2$, so $\operatorname{tr} X^{-1} = \frac{1}{12} + \frac{1}{6} + \frac{1}{4} + \frac{1}{2} = 1$ (Corollary 7.30(c)), and the right side is $\frac{1}{2}(1+4) = \frac{5}{2}$. With the same orthonormal eigenvectors, $\frac{1}{2}(X + I_4)$ has the eigenvalues $\frac{13}{2}, \frac{7}{2}, \frac{5}{2}, \frac{3}{2}$, so the left side is $2\big(\frac{1}{13} + \frac{1}{7} + \frac{1}{5} + \frac{1}{3}\big) = \frac{2056}{1365} < \frac{5}{2}$.

Problem 9.3. Let $Y \in \mathbb{R}^{n\times n}$ be positive definite, $Y = QMQ'$ with Q orthogonal and $M = \operatorname{diag}(\mu_1, \dots, \mu_n)$.

(a) Show that the linear map $H \mapsto YH + HY$ on $\mathbb{R}^{n\times n}$ has the matrix $Y \otimes I_n + I_n \otimes Y$ with respect to vec, with eigenvalues $\mu_i + \mu_j > 0$. Conclude that for every $C \in \mathbb{R}^{n\times n}$ the equation $YH + HY = C$ has exactly one solution, namely $H = QGQ'$ with $g_{ij} = (Q'CQ)_{ij}/(\mu_i + \mu_j)$, and that H is symmetric if C is.

(b) Let X be positive definite, D symmetric, and $Y(t) = (X + tD)^{1/2}$ for t in the interval $\mathcal{I}$ of Theorem 9.21(a). Show that if $t \mapsto Y(t)$ is differentiable at 0, then its derivative $Y'(0)$ is the solution H of $X^{1/2}H + HX^{1/2} = D$.

(c) Find this derivative for $X = \operatorname{diag}(1, 4, 9, 16)$ and $D = J_4$.

Solution. (a) By Theorem 9.8(d), $\operatorname{vec}(YHI_n) = (I_n \otimes Y)\operatorname{vec} H$ and $\operatorname{vec}(I_nHY) = (Y' \otimes I_n)\operatorname{vec} H = (Y \otimes I_n)\operatorname{vec} H$. By Theorem 9.9(b) with $A = B = Y$ (whose characteristic polynomial splits, Theorem 8.4(b)), the eigenvalues of $Y \otimes I_n + I_n \otimes Y$ are the $\mu_i + \mu_j$, all positive (Theorem 8.13(a)). So the matrix has nonzero determinant (Corollary 7.30(a)) and is invertible: the equation has exactly one solution for every C. For $H = QGQ'$, $YH + HY = Q(MG + GM)Q'$, and $(MG + GM)_{ij} = (\mu_i + \mu_j)g_{ij} = (Q'CQ)_{ij}$; so $YH + HY = QQ'CQQ' = C$. If $C' = C$, then $Q'CQ$ and G are symmetric, and so is H.

(b) $Y(t)^2 = X + tD$ for $t \in \mathcal{I}$ (Theorem 8.25(a)). If Y is differentiable at 0, the product rule (F2) at $t = 0$ gives $Y'(0)Y(0) + Y(0)Y'(0) = D$, with $Y(0) = X^{1/2}$; by (a) there is only one such matrix.

(c) $X^{1/2} = \operatorname{diag}(1, 2, 3, 4)$ (Theorem 8.25(a) with $Q = I_4$), so $\mu_i = i$, and $Y'(0) = \Big(1/(i + j)\Big)_{i,j=1}^{4}$:

$$Y'(0) = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \\ \frac{1}{5} & \frac{1}{6} & \frac{1}{7} & \frac{1}{8} \end{pmatrix}$$

Check: entry (i, j) of $X^{1/2}H + HX^{1/2}$ is $(i + j)/(i + j) = 1$.

Problem 9.4. Let $A \in \mathbb{R}^{n\times n}$ be positive definite, $A = Q\Lambda Q'$ with $\lambda_1 \geqslant \cdots \geqslant \lambda_n > 0$, let $0 < \alpha < 2/\lambda_1$, and let $X_0 = \alpha I_n$ and $X_{t+1} = 2X_t - X_tAX_t$ (Example 9.14).

(a) Show that $X_t = Q \operatorname{diag}\Big(\varphi_t(\lambda_1), \dots, \varphi_t(\lambda_n)\Big)Q'$ with $\varphi_t(\lambda) = \Big(1 - (1 - \alpha\lambda)^{2^t}\Big)/\lambda$, so that every X_t is positive definite.

(b) Show that $X_{t+1} - A^{-1} = -(X_t - A^{-1})A(X_t - A^{-1})$ and $\|X_t - A^{-1}\|_F^2 = \sum_i (1 - \alpha\lambda_i)^{2^{t+1}}/\lambda_i^2$, which tends to 0 as $t \to \infty$.

Solution. (a) $E_0 = I_n - \alpha A = Q \operatorname{diag}(1 - \alpha\lambda_i)Q'$. The identity $I_n - A(2X - XAX) = (I_n - AX)^2$ of Example 9.14(a) is algebraic, so $E_t = I_n - AX_t = E_0^{2^t} = Q \operatorname{diag}\Big((1 - \alpha\lambda_i)^{2^t}\Big)Q'$ (Proposition 7.24). Hence $X_t = A^{-1}(I_n - E_t) = Q\Lambda^{-1}Q'Q \operatorname{diag}\Big(1 - (1 - \alpha\lambda_i)^{2^t}\Big)Q'$, as claimed. Since $0 < \alpha\lambda_i < 2$, $|1 - \alpha\lambda_i| < 1$, so $0 \leqslant (1 - \alpha\lambda_i)^{2^t} < 1$ and $\varphi_t(\lambda_i) > 0$: X_t is positive definite (Theorem 8.13(a)).

(b) $-(X - A^{-1})A(X - A^{-1}) = -(XA - I_n)(X - A^{-1}) = -(XAX - 2X + A^{-1}) = 2X - XAX - A^{-1}$. By (a), $X_t - A^{-1} = -A^{-1}E_t = -Q \operatorname{diag}\Big((1 - \alpha\lambda_i)^{2^t}/\lambda_i\Big)Q'$, and $\|QDQ'\|_F^2 = \operatorname{tr}(QD^2Q') = \operatorname{tr} D^2$ for diagonal D (Theorem 1.11(b),(d)). As $|1 - \alpha\lambda_i| < 1$, each term tends to 0. The error is squared at each step, in the sense of the first identity.

Problem 9.5. (a) Let $M = \left(\begin{smallmatrix} A & B \\ B' & D \end{smallmatrix}\right)$ be positive definite, with square blocks A and D. Show that $\det M \leqslant \det A \cdot \det D$ (**Fischer's inequality**), with equality if and only if $B = 0$.

(b) Let M be positive definite and partitioned into k row and column groups, with square diagonal blocks $M_{11}, \dots, M_{kk}$. Show that $\det M \leqslant \det M_{11} \cdots \det M_{kk}$. For blocks of size 1 this is Corollary 9.22.

(c) Check (a) for the matrix A of Example 9.23, partitioned into 2×2 blocks.

Solution. (a) A and D are positive definite (Corollary 8.15(a)), and so is the Schur complement $S = D - B'A^{-1}B$ (Lemma 8.18); $\det M = \det A \cdot \det S$ (Theorem 5.26(a)). With the symmetric square root $A^{-1/2}$ (Theorem 8.25(c)), $D - S = B'A^{-1}B = (A^{-1/2}B)'(A^{-1/2}B)$ is positive semidefinite (Theorem 8.16(a)), so $D \geqslant S > 0$ and $\det D \geqslant \det S$ (Proposition 8.28(e)). This gives the inequality.

- If $B = 0$, then $S = D$ and equality holds.
- Let $\det D = \det S$. Put $R = S^{-1/2}DS^{-1/2}$; then $R - I = S^{-1/2}(D - S)S^{-1/2}$ is positive semidefinite (Theorem 8.16(b)), so the eigenvalues ρ_i of R satisfy $\rho_i \geqslant 1$ (Theorem 8.13(b), as $R - I$ has the eigenvalues $\rho_i - 1$), and $\prod_i \rho_i = \det R = \det D / \det S = 1$ (Corollary 7.30(a)). Hence every $\rho_i = 1$, $R = I$ (Theorem 8.4(a)), and $D = S$. Then $\|A^{-1/2}Bv\|^2 = v'(D - S)v = 0$ for every v, so $A^{-1/2}B = 0$ and $B = A^{1/2}A^{-1/2}B = 0$.

(b) Induction on k; for $k = 1$ there is nothing to prove. For $k \geqslant 2$, let A be the leading block formed by the first $k - 1$ groups and $D = M_{kk}$. A is positive definite (Corollary 8.15(a)) with diagonal blocks $M_{11}, \dots, M_{k-1,k-1}$. By (a) and the induction hypothesis, $\det M \leqslant \det A \cdot \det M_{kk} \leqslant \det M_{11} \cdots \det M_{k-1,k-1} \cdot \det M_{kk}$.

(c) $A_{11} = A_{22} = \left(\begin{smallmatrix} 5 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ have determinant 24, and $\det A = 384 < 576 = 24 \cdot 24$, with $A_{12} = 2I_2 \neq 0$.

Problem 9.6. Let $A, B \in \mathbb{R}^{n \times n}$ be symmetric and $1 \leqslant k \leqslant n$; $\lambda_i(\cdot)$ as after Theorem 9.29.

(a) Show that $a_{i_1 i_1} + \cdots + a_{i_k i_k} \leqslant \lambda_1(A) + \cdots + \lambda_k(A)$ for any distinct indices $i_1, \dots, i_k$, with equality for $k = n$. Check this for the matrix A of Example 9.17.

(b) Show that $\sum_{i \leqslant k} \lambda_i(A + B) \leqslant \sum_{i \leqslant k} \lambda_i(A) + \sum_{i \leqslant k} \lambda_i(B)$.

(c) Let $A = Q\Lambda Q'$ as in Theorem 9.29, with $\lambda_k > \lambda_{k+1}$ ($k < n$). Show that $V \in \mathbb{R}^{n \times k}$ with $V'V = I_k$ satisfies $\operatorname{tr}(V'AV) = \lambda_1 + \cdots + \lambda_k$ if and only if $\mathcal{C}(V) = \operatorname{span}\{q_1, \dots, q_k\}$.

Solution. (a) Apply Theorem 9.34 to $V = [e_{i_1} \ \cdots \ e_{i_k}]$, for which $V'V = I_k$ and $\operatorname{tr}(V'AV) = \sum_r a_{i_r i_r}$. For $k = n$, $\operatorname{tr} A = \sum_i \lambda_i(A)$ (Corollary 7.30(a)). For the matrix of Example 9.17, all diagonal entries are 15 and the eigenvalues are $32, 18, 8, 2$: $15 \leqslant 32$, $30 \leqslant 50$, $45 \leqslant 58$ and $60 = 60$.

(b) Let $V = [q_1 \ \cdots \ q_k]$ be built from orthonormal eigenvectors of $A + B$ as in Theorem 9.34. Then $\sum_{i \leqslant k} \lambda_i(A + B) = \operatorname{tr}(V'(A + B)V) = \operatorname{tr}(V'AV) + \operatorname{tr}(V'BV)$, and Theorem 9.34 for A and for B bounds the two terms.

(c) In the proof of Theorem 9.34, $w_i = \|V'q_i\|^2$ and $\sum_{i \leqslant k} \lambda_i - \operatorname{tr}(V'AV) = \sum_{i \leqslant k}(\lambda_i - \lambda_k)(1 - w_i) + \sum_{i > k}(\lambda_k - \lambda_i)w_i$, using $\sum_i w_i = k$; all terms are $\geqslant 0$.

- If equality holds, then $(\lambda_k - \lambda_i)w_i = 0$ for $i > k$, and $\lambda_k - \lambda_i \geqslant \lambda_k - \lambda_{k+1} > 0$; so $V'q_i = 0$ for $i > k$. The columns of V are then orthogonal to $q_{k+1}, \dots, q_n$, and as $x = \sum_i (q_i'x)q_i$ (Proposition 6.10(b)), they lie in $W = \operatorname{span}\{q_1, \dots, q_k\}$. Both $\mathcal{C}(V)$ and W have dimension k, so they are equal (Theorem 4.5).
- If $\mathcal{C}(V) = W$, then $VV' = P_W$ (Theorem 6.19(a)), so $w_i = q_i' P_W q_i$ is 1 for $i \leqslant k$ and 0 for $i > k$ (Theorem 6.19(b)), and $\operatorname{tr}(V'AV) = \sum_i \lambda_i w_i = \lambda_1 + \cdots + \lambda_k$.

Problem 9.7. Let $S \in \mathbb{R}^{n\times n}$ be positive definite, and $\varphi(X) = \log\det X + \operatorname{tr}(X^{-1}S)$ on the open set $U = \{X \in \mathbb{R}^{n\times n} : \det X > 0\}$.

(a) Show that $\partial\varphi/\partial X = (X^{-1} - X^{-1}SX^{-1})'$, which is 0 at $X = S$.

(b) Let Σ be positive definite and $C = S^{1/2}\Sigma^{-1}S^{1/2}$, with eigenvalues $\theta_1, \dots, \theta_n$. Show that C is positive definite and $\varphi(\Sigma) - \varphi(S) = \sum_i(\theta_i - 1 - \log\theta_i)$.

(c) Show that $\theta - 1 - \log\theta \geqslant 0$ for $\theta > 0$, with equality only for $\theta = 1$. Conclude that $\varphi(\Sigma) \geqslant \varphi(S)$ for every positive definite Σ, with equality only for $\Sigma = S$.

Solution. (a) By Theorem 9.5(c),(d) with $A = S$, $\mathrm{d}\varphi = \operatorname{tr}(X^{-1}\mathrm{d}X) - \operatorname{tr}(X^{-1}SX^{-1}\mathrm{d}X) = \operatorname{tr}\Big((X^{-1} - X^{-1}SX^{-1})\,\mathrm{d}X\Big)$, which gives $\partial\varphi/\partial X$. As $\det S > 0$ (Corollary 8.15(a)), $S \in U$, and at $X = S$, $X^{-1}SX^{-1} = S^{-1} = X^{-1}$: S is a stationary point of φ. This alone does not compare $\varphi(S)$ with $\varphi(\Sigma)$ for all positive definite Σ; (b) and (c) do.

(b) S is invertible (Corollary 8.15(a)), so $\operatorname{rank} S^{1/2} = \operatorname{rank} S = n$ (Theorem 8.25(b)) and the positive semidefinite matrix $S^{1/2}$ is invertible; also Σ^{-1} is positive definite (Corollary 8.15(a)); so $C = (S^{1/2})'\Sigma^{-1}S^{1/2}$ is positive definite (Theorem 8.16(b)). Write $C = Q\Theta Q'$ with Q orthogonal and $\Theta = \operatorname{diag}(\theta_1, \dots, \theta_n)$ (Theorem 8.4(a)); every $\theta_i > 0$ (Theorem 8.13(a)). Then $\operatorname{tr} C = \operatorname{tr}(\Theta Q'Q) = \sum_i \theta_i$ and $\det C = \det Q \det\Theta \det Q' = \prod_i \theta_i$, since $\det Q \det Q' = \det(QQ') = 1$ (Theorems 1.11(b), 5.19(b) and 5.15). On the other hand $\operatorname{tr} C = \operatorname{tr}(\Sigma^{-1}S^{1/2}S^{1/2}) = \operatorname{tr}(\Sigma^{-1}S)$ and $\det C = \det S/\det\Sigma$ (Theorem 5.19). With $\varphi(S) = \log\det S + n$,

$$\varphi(\Sigma) - \varphi(S) = \log\det\Sigma + \operatorname{tr} C - \log\det S - n = \operatorname{tr} C - \log\det C - n = \sum_{i=1}^{n}(\theta_i - 1 - \log\theta_i)$$

(c) $h(\theta) = \theta - 1 - \log\theta$ is C^2 on the open convex set $(0, \infty)$, with $h'(\theta) = 1 - 1/\theta$ and $h''(\theta) = 1/\theta^2 > 0$. By Theorem 9.19(b), its stationary point $\theta = 1$ is its unique minimizer, and $h(1) = 0$. So every term in (b) is $\geqslant 0$, and $\varphi(\Sigma) \geqslant \varphi(S)$. Equality forces every $\theta_i = 1$, that is, $C = QQ' = I_n$; then $\Sigma^{-1} = S^{-1/2}S^{-1/2} = S^{-1}$, and $\Sigma = S$.

Chapter 10. Singular value decomposition, generalized inverses, and principal components

The singular value decomposition

Theorem 10.1 (Singular value decomposition)**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$. Then

$$A = U\Sigma V',$$

where $U = [u_1 \ \cdots \ u_m] \in \mathbb{R}^{m\times m}$ and $V = [v_1 \ \cdots \ v_n] \in \mathbb{R}^{n\times n}$ are orthogonal, and $\Sigma \in \mathbb{R}^{m\times n}$ has the entries $\Sigma_{ii} = \sigma_i$ for $i \leqslant r$, where $\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_r > 0$, and all other entries 0. Equivalently, with $U_r = [u_1 \ \cdots \ u_r]$, $V_r = [v_1 \ \cdots \ v_r]$ and $D_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$,

$$A = U_r D_r V_r' = \sum_{i=1}^{r} \sigma_i u_i v_i' \qquad \text{(the } \textbf{compact form}\text{)}$$

Moreover $AV = U\Sigma$ and $A'U = V\Sigma'$, that is, $Av_i = \sigma_i u_i$ and $A'u_i = \sigma_i v_i$ for $i \leqslant r$, $Av_j = 0$ for $r < j \leqslant n$, and $A'u_j = 0$ for $r < j \leqslant m$. The numbers $\sigma_1, \ldots, \sigma_r$ are the **singular values** of A, the u_i and v_i are **left** and **right singular vectors**, and $A = U\Sigma V'$ is a **singular value decomposition** (SVD) of A.

Proof. • $A'A \in \mathbb{R}^{n\times n}$ is symmetric (Theorem 1.9). By Theorem 8.4(a),(d), $A'A = V\Lambda V'$ with $V = [v_1 \ \cdots \ v_n]$ orthogonal, $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ and $A'Av_i = \lambda_i v_i$.

- $\lambda_i = \lambda_i v_i' v_i = v_i' A'Av_i = \|Av_i\|^2 \geqslant 0$. The number of nonzero λ_i is $\operatorname{rank}(A'A)$ (Corollary 8.5(b)), which is r (Theorem 4.18(b)). As the λ_i are nonnegative and decreasing, $\lambda_1 \geqslant \cdots \geqslant \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$.
- Put $\sigma_i = \sqrt{\lambda_i}$ and $u_i = \sigma_i^{-1} Av_i$ for $i \leqslant r$. Then $u_i' u_j = (\sigma_i\sigma_j)^{-1} v_i' A'Av_j = (\sigma_i\sigma_j)^{-1}\lambda_j v_i' v_j$, which is 1 for $i = j$ and 0 for $i \neq j$: $u_1, \ldots, u_r$ are orthonormal. If $r < m$, extend them to an orthonormal basis $u_1, \ldots, u_m$ of $\mathbb{R}^m$ (Lemma 7.27). So $U = [u_1 \ \cdots \ u_m]$ is orthogonal.
- For $j > r$, $\|Av_j\|^2 = \lambda_j = 0$, so $Av_j = 0$. Hence column j of AV, which is Av_j, equals $\sigma_j u_j$ for $j \leqslant r$ and 0 for $j > r$; this is column j of $U\Sigma$ (Theorem 1.4(a),(b)). So $AV = U\Sigma$, and $A = AVV' = U\Sigma V'$, since $VV' = I_n$ (Proposition 6.10(b)).
- $A' = V\Sigma' U'$ (Theorem 1.9), so $A'U = V\Sigma'$; column j of $V\Sigma'$ is $\sigma_j v_j$ for $j \leqslant r$ and 0 for $j > r$.
- With $\Sigma = \left(\begin{smallmatrix} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $U = [U_r \ U_2]$ and $V = [V_r \ V_2]$ (blocks with no columns are absent), Theorem 5.3(c) gives $U\Sigma V' = [U_r D_r \ \ 0]\, V' = U_r D_r V_r'$, and the outer product expansion (Theorem 1.4(d)) gives $U_r D_r V_r' = \sum_{i\leqslant r} (\sigma_i u_i) v_i'$. □

For $A = 0$ the formula $A = U\Sigma V'$ holds with $\Sigma = 0$ and any orthogonal U, V.

Corollary 10.2. Let $A = U\Sigma V'$ be as in Theorem 10.1.

(a) The characteristic polynomials (Definition 7.5) are $p_{A'A}(t) = t^{n-r}\prod_{i=1}^{r}(t - \sigma_i^2)$ and $p_{AA'}(t) = t^{m-r}\prod_{i=1}^{r}(t - \sigma_i^2)$: the numbers $\sigma_1^2, \ldots, \sigma_r^2$ are the nonzero eigenvalues of $A'A$ and of AA', listed with multiplicity (Definition 7.8), and $A'Av_i = \sigma_i^2 v_i$, $AA'u_i = \sigma_i^2 u_i$ $(i \leqslant r)$. In particular the singular values are determined by A, whatever decomposition is used.

(b) $u_1, \ldots, u_r$ form an orthonormal basis of $\mathcal{C}(A)$, $u_{r+1}, \ldots, u_m$ one of $\mathcal{N}(A')$, $v_1, \ldots, v_r$ one of $\mathcal{C}(A')$, and $v_{r+1}, \ldots, v_n$ one of $\mathcal{N}(A)$ (the empty basis if $r = m$, respectively $r = n$).

(c) $A' = V\Sigma' U'$ is a singular value decomposition of A'; so A' has the same singular values as A.

(d) $\|Ax\|^2 = \sum_{i=1}^{r} \sigma_i^2 (v_i' x)^2$ for every $x \in \mathbb{R}^n$.

(e) If $A = \sum_{i=1}^{s} \tau_i x_i y_i'$ with $x_1, \dots, x_s \in \mathbb{R}^m$ orthonormal, $y_1, \dots, y_s \in \mathbb{R}^n$ orthonormal and $\tau_1 \geqslant \cdots \geqslant \tau_s > 0$, then $s = r$ and $\tau_i = \sigma_i$ for every i; completing the x_i and the y_i to orthogonal matrices gives a singular value decomposition with $u_i = x_i$ and $v_i = y_i$ $(i \leqslant r)$.

We put $\sigma_j(A) = \sigma_j$ for $j \leqslant r$ and $\sigma_j(A) = 0$ for $r < j \leqslant n$, and $\sigma_j(0) = 0$ for all j. By (a), $\sigma_j(A)^2 = \lambda_j(A'A)$ for $1 \leqslant j \leqslant n$, where $\lambda_1(A'A) \geqslant \cdots \geqslant \lambda_n(A'A)$ are the eigenvalues of $A'A$ listed with multiplicity, as defined after Theorem 9.29.

Proof. (a) $A'A = V\Sigma'U'U\Sigma V' = V(\Sigma'\Sigma)V'$ with $\Sigma'\Sigma = \operatorname{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0) \in \mathbb{R}^{n\times n}$. As $V^{-1} = V'$, $A'A$ is similar to $\Sigma'\Sigma$ (Definition 7.13), so $p_{A'A} = p_{\Sigma'\Sigma}$ (Theorem 7.14(a)), and $p_{\Sigma'\Sigma}(t) = \det(tI_n - \Sigma'\Sigma) = t^{n-r}\prod_i(t - \sigma_i^2)$ (Theorem 5.15(a)). Likewise $AA' = U(\Sigma\Sigma')U'$ with $\Sigma\Sigma' = \operatorname{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0) \in \mathbb{R}^{m\times m}$. By Theorem 10.1, $A'Av_i = \sigma_i A'u_i = \sigma_i^2 v_i$ and $AA'u_i = \sigma_i Av_i = \sigma_i^2 u_i$. So $\sigma_1 \geqslant \cdots \geqslant \sigma_r$ are the positive square roots of the nonzero roots of $p_{A'A}$, in decreasing order.

(b)

- $\mathcal{C}(A) = \mathcal{C}(U_r(D_rV_r')) \subseteq \mathcal{C}(U_r)$ (Theorem 4.15(a)), and $u_i = \sigma_i^{-1}Av_i \in \mathcal{C}(A)$ for $i \leqslant r$. So $u_1, \dots, u_r$ span $\mathcal{C}(A)$ (Theorem 3.7); being orthonormal, they are independent (Proposition 6.8(b)).
- $A'u_j = 0$ for $j > r$ (Theorem 10.1), so $u_{r+1}, \dots, u_m$ are $m - r$ orthonormal, hence independent, vectors of $\mathcal{N}(A')$, which has dimension $m - r$ (Theorem 4.12); so they form a basis of it (Theorem 4.5(d)); if $r = m$, then $\mathcal{N}(A') = \{0\}$ by Theorem 4.12, with the empty basis.
- The same two steps for $A' = V_rD_rU_r'$, with $v_i = \sigma_i^{-1}A'u_i$ and $Av_j = 0$ $(j > r)$, give the bases of $\mathcal{C}(A')$ and $\mathcal{N}(A)$ (the empty basis of $\mathcal{N}(A) = \{0\}$ if $r = n$).

(c) $A' = V\Sigma'U'$ (Theorem 1.9), where V, U are orthogonal, $\Sigma' \in \mathbb{R}^{n\times m}$ has $(\Sigma')_{ii} = \sigma_i$ for $i \leqslant r$ and all other entries 0, and $\operatorname{rank} A' = r$ (Theorem 4.13(a)). By (a), the singular values of A' are $\sigma_1, \dots, \sigma_r$.

(d) Put $c = V'x$, so $c_i = v_i'x$. Then $Ax = U\Sigma V'x = U\Sigma c$, and $\|Ax\| = \|\Sigma c\|$ (Proposition 6.10(a)), where $\Sigma c = (\sigma_1 c_1, \dots, \sigma_r c_r, 0, \dots, 0)'$.

(e) Orthonormal vectors are independent (Proposition 6.8(b)), so $s \leqslant m$ and $s \leqslant n$ (Theorem 4.5(c)). If $s < m$, extend $x_1, \dots, x_s$ to an orthonormal basis $x_1, \dots, x_m$ of $\mathbb{R}^m$ (Lemma 7.27), and likewise $y_1, \dots, y_n$. With the orthogonal matrices $X = [x_1 \ \cdots \ x_m]$ and $Y = [y_1 \ \cdots \ y_n]$, and $T \in \mathbb{R}^{m\times n}$ with $T_{ii} = \tau_i$ $(i \leqslant s)$ and all other entries 0, the outer product expansion gives $A = XTY'$. The columns $\tau_1 e_1, \dots, \tau_s e_s$ of T span $\mathcal{C}(T)$ and are independent, so $\operatorname{rank} A = \operatorname{rank} T = s$ (Theorem 4.15(d)). Thus $A = XTY'$ is a decomposition as in Theorem 10.1, and by (a) $\tau_i = \sigma_i$. □

Example 10.3. Find a singular value decomposition of

$$A = \begin{pmatrix} 6 & 0 & 0 \\ 4 & 4 & 6 \\ 5 & 2 & 6 \\ 2 & -4 & 0 \end{pmatrix}$$

Solution. Follow the construction in the proof of Theorem 10.1.

- The entries of $A'A$ are the inner products of the columns of A:

$$A'A = \begin{pmatrix} 81 & 18 & 54 \\ 18 & 36 & 36 \\ 54 & 36 & 72 \end{pmatrix} = 9B, \qquad B = \begin{pmatrix} 9 & 2 & 6 \\ 2 & 4 & 4 \\ 6 & 4 & 8 \end{pmatrix}$$

  For $w_1 = (2, 1, 2)'$, $w_2 = (-2, 2, 1)'$ and $w_3 = (1, 2, -2)'$, $Bw_1 = (32, 16, 32)' = 16w_1$, $Bw_2 = (-8, 8, 4)' = 4w_2$ and $Bw_3 = w_3$. So $A'Aw_k = \lambda_k w_k$ with $\lambda_1 = 144$, $\lambda_2 = 36$, $\lambda_3 = 9$. The w_k are orthogonal with length 3, so $v_k = \frac{1}{3}w_k$ gives an orthogonal $V = [v_1 \ v_2 \ v_3]$, and $\sigma_1 = 12$, $\sigma_2 = 6$, $\sigma_3 = 3$; in particular $\operatorname{rank} A = 3$.

- $u_k = \sigma_k^{-1} A v_k = (3\sigma_k)^{-1} A w_k$: $Aw_1 = (12, 24, 24, 0)'$, $Aw_2 = (-12, 6, 0, -12)'$ and $Aw_3 = (6, 0, -3, -6)'$ give

$$u_1 = \tfrac{1}{3}(1, 2, 2, 0)', \qquad u_2 = \tfrac{1}{3}(-2, 1, 0, -2)', \qquad u_3 = \tfrac{1}{3}(2, 0, -1, -2)'$$

- A vector x orthogonal to u_1, u_2, u_3 satisfies $x_1 + 2x_2 + 2x_3 = 0$, $-2x_1 + x_2 - 2x_4 = 0$ and $2x_1 - x_3 - 2x_4 = 0$, so $x = c\,(0, 2, -2, 1)'$; take $u_4 = \frac{1}{3}(0, 2, -2, 1)'$.

Hence $A = U\Sigma V'$ with

$$U = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 & 0 \\ 2 & 1 & 0 & 2 \\ 2 & 0 & -1 & -2 \\ 0 & -2 & -2 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \qquad V = \frac{1}{3}\begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix},$$

and the compact form is

$$A = 12u_1v_1' + 6u_2v_2' + 3u_3v_3' = \frac{4}{3}\begin{pmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 4 & 2 & 4 \\ 0 & 0 & 0 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 4 & -4 & -2 \\ -2 & 2 & 1 \\ 0 & 0 & 0 \\ 4 & -4 & -2 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 2 & 4 & -4 \\ 0 & 0 & 0 \\ -1 & -2 & 2 \\ -2 & -4 & 4 \end{pmatrix}$$

By Corollary 10.2(b), $\mathcal{C}(A) = \operatorname{span}\{u_1, u_2, u_3\}$, $\mathcal{N}(A') = \operatorname{span}\{(0, 2, -2, 1)'\}$ (indeed $A'(0, 2, -2, 1)' = 0$), $\mathcal{C}(A') = \mathbb{R}^3$ and $\mathcal{N}(A) = \{0\}$; by Corollary 10.2(a), the eigenvalues of the 4×4 matrix AA' are 144, 36, 9, 0.

Definition 10.4. For $A \in \mathbb{R}^{m\times n}$, $\|A\|_F = (\operatorname{tr} A'A)^{1/2} = \left(\sum_{i,j} a_{ij}^2\right)^{1/2}$ is the Frobenius norm of A (Definition 9.3), and the **spectral norm** of A is

$$\|A\|_2 = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|}$$

The maximum exists and equals $\sigma_1(A)$: if $A \neq 0$, Corollary 10.2(d) gives $\|Ax\|^2 \leqslant \sigma_1^2 \sum_{i=1}^n (v_i'x)^2 = \sigma_1^2 \|V'x\|^2 = \sigma_1^2\|x\|^2$ (Proposition 6.10(a)), with equality at $x = v_1$; and $\|0x\| = 0$.

Theorem 10.5. Let $A \in \mathbb{R}^{m\times n}$ and $\sigma_j = \sigma_j(A)$ $(1 \leqslant j \leqslant n)$.

(a) $\|A\|_F^2 = \sigma_1^2 + \cdots + \sigma_n^2$ and $\|A\|_2 = \sigma_1$. Both norms are unchanged when A is replaced by PAQ with $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ orthogonal.

(b) $\sigma_n\|x\| \leqslant \|Ax\| \leqslant \sigma_1\|x\|$ for every $x \in \mathbb{R}^n$, and $\sigma_n > 0$ if and only if $\operatorname{rank} A = n$. So if $\operatorname{rank} A = n$, then $\|Ax\| \geqslant \sigma_n\|x\|$, with equality at $x = v_n$ (the last column of V in Theorem 10.1).

(c) For $1 \leqslant k \leqslant n$, with W ranging over the subspaces of $\mathbb{R}^n$,

$$\sigma_k = \max_{\dim W = k} \min_{0\neq x\in W} \frac{\|Ax\|}{\|x\|} = \min_{\dim W = n-k+1} \max_{0\neq x\in W} \frac{\|Ax\|}{\|x\|}$$

(d) (**Weyl's inequality**) For $B \in \mathbb{R}^{m\times n}$ and $i, j \geqslant 1$ with $i+j-1 \leqslant n$, $\sigma_{i+j-1}(A+B) \leqslant \sigma_i(A) + \sigma_j(B)$.

Proof. Let $A = U\Sigma V'$ be a singular value decomposition (for $A = 0$, $V = I_n$). By Corollary 10.2(d) and the convention after it, for every $x \in \mathbb{R}^n$

$$\|Ax\|^2 = \sum_{j=1}^n \sigma_j^2 (v_j'x)^2, \qquad \sum_{j=1}^n (v_j'x)^2 = \|V'x\|^2 = \|x\|^2$$

(a) $\|A\|_F^2 = \operatorname{tr}(V\Sigma'\Sigma V') = \operatorname{tr}(\Sigma'\Sigma V'V) = \operatorname{tr}(\Sigma'\Sigma) = \sum_j \sigma_j^2$ (Theorem 1.11(b)), and $\|A\|_2 = \sigma_1$ by Definition 10.4. Next $\|PAQ\|_F^2 = \operatorname{tr}(Q'A'P'PAQ) = \operatorname{tr}(A'AQQ') = \|A\|_F^2$, and $\|PAQx\| = \|AQx\|$

(Proposition 6.10(a)); as x runs through the nonzero vectors of $\mathbb{R}^n$, so does Qx, with $\|Qx\| = \|x\|$. So the maxima defining $\|PAQ\|_2$ and $\|A\|_2$ coincide.

(b) $\sigma_n^2 \leqslant \sigma_j^2 \leqslant \sigma_1^2$ for every j, so the display gives $\sigma_n^2\|x\|^2 \leqslant \|Ax\|^2 \leqslant \sigma_1^2\|x\|^2$. Since $\operatorname{rank} A \leqslant n$, $\sigma_n > 0$ means $\operatorname{rank} A = n$. Then $\|Av_n\|^2 = \sigma_n^2$, as $v_j'v_n = 0$ for $j \neq n$.

(c) $\sigma_k^2 = \lambda_k(A'A)$ and $\|Ax\|^2/\|x\|^2 = x'(A'A)x/x'x$. By Theorem 9.31 for the symmetric matrix $A'A$, the two expressions in (c) with $\|Ax\|^2/\|x\|^2$ in place of $\|Ax\|/\|x\|$ equal σ_k^2. As $t \mapsto \sqrt{t}$ is increasing on $[0,\infty)$, the minimum (maximum) over x of $\|Ax\|/\|x\|$ is the square root of that of $\|Ax\|^2/\|x\|^2$, and likewise for the maximum (minimum) over W; so both expressions in (c) equal σ_k.

(d) Let $k = i + j - 1$, and let $w_1, \ldots, w_n$ be the columns of the matrix V of a singular value decomposition of B ($V = I_n$ if $B = 0$).

- On $S_1 = \operatorname{span}\{v_i, \ldots, v_n\}$, of dimension $n - i + 1$, $v_l'x = 0$ for $l < i$, so the display gives $\|Ax\|^2 = \sum_{l \geqslant i} \sigma_l(A)^2(v_l'x)^2 \leqslant \sigma_i(A)^2\|x\|^2$. Likewise $\|Bx\| \leqslant \sigma_j(B)\|x\|$ on $S_2 = \operatorname{span}\{w_j, \ldots, w_n\}$, of dimension $n - j + 1$.
- By Lemma 9.30, $\dim(S_1 \cap S_2) \geqslant (n - i + 1) + (n - j + 1) - n = n - k + 1 \geqslant 1$; let W be spanned by $n - k + 1$ vectors of a basis of $S_1 \cap S_2$.
- For $0 \neq x \in W$, the triangle inequality (Theorem 6.3(b)) gives $\|(A + B)x\| \leqslant \|Ax\| + \|Bx\| \leqslant (\sigma_i(A)+\sigma_j(B))\|x\|$. By (c) for $A+B$, $\sigma_k(A+B) \leqslant \max_{0\neq x\in W}\|(A+B)x\|/\|x\| \leqslant \sigma_i(A)+\sigma_j(B)$. □

In Example 10.3, $\|A\|_2 = 12$ and $\|A\|_F^2 = 36 + 68 + 65 + 20 = 189 = 12^2 + 6^2 + 3^2$ (the four terms are the squared lengths of the rows).

Definition 10.6. Let $A \in \mathbb{R}^{m\times n}$ have full column rank n. Its **condition number** is $\kappa(A) = \sigma_1(A)/\sigma_n(A) \geqslant 1$, where $\sigma_n(A) > 0$ by Theorem 10.5(b). With $A = U\Sigma V'$, $A'A = V\Sigma'U'U\Sigma V' = \sum_{j=1}^n \sigma_j^2 v_j v_j'$ has the singular values $\sigma_1(A)^2 \geqslant \cdots \geqslant \sigma_n(A)^2 > 0$ (Corollary 10.2(e)); hence $\kappa(A'A) = \kappa(A)^2$.

For A of full column rank n, $\sigma_n(A) = \min_{x\neq 0}\|Ax\|/\|x\|$ and $\sigma_1(A) = \max_{x\neq 0}\|Ax\|/\|x\|$ (Theorem 10.5(a),(b)). A large $\kappa(A)$ means that the combination Av_n of the columns of A, with unit coefficient vector v_n, is short compared with the longest one, Av_1: the columns are nearly linearly dependent. As rescaling a single column changes $\kappa(A)$, the columns are often scaled to equal length first. In Example 10.3, $\kappa(A) = 12/3 = 4$ and $\kappa(A'A) = 16$.

Proposition 10.7. If $A \in \mathbb{R}^{m\times n}$ has full column rank, then $\kappa(cA) = \kappa(A)$ for $c \neq 0$, and $\kappa(PAQ) = \kappa(A)$ for orthogonal $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$.

Proof. Both matrices have rank n (Theorem 4.15(d)). With $A = U\Sigma V'$, $cA = (\pm U)(|c|\Sigma)V'$ (with the sign of c) has the singular values $|c|\sigma_j$, and $PAQ = (PU)\Sigma(Q'V)'$, where PU and $Q'V$ are orthogonal ($(PU)'PU = U'P'PU = I_m$), has the singular values σ_j (Corollary 10.2(a)). □

Generalized inverses

Definition 10.8. Let $A \in \mathbb{R}^{m\times n}$. A matrix $G \in \mathbb{R}^{n\times m}$ with $AGA = A$ is a **generalized inverse** (**g-inverse**) of A; we write A^- for any such G.

Theorem 10.9. Let $A \in \mathbb{R}^{m\times n}$.

(a) Let $\operatorname{rank} A = r \geqslant 1$ and $A = U\Sigma V'$ as in Theorem 10.1. Then $G \in \mathbb{R}^{n\times m}$ is a g-inverse of A if and only if

$$G = V\begin{pmatrix} D_r^{-1} & K \\ L & N \end{pmatrix}U'$$

for some $K \in \mathbb{R}^{r\times(m-r)}$, $L \in \mathbb{R}^{(n-r)\times r}$ and $N \in \mathbb{R}^{(n-r)\times(m-r)}$ (blocks with no rows or no columns are absent). In particular every matrix has a g-inverse; for $A = 0$ every $G \in \mathbb{R}^{n\times m}$ is one.

(b) Let $\operatorname{rank} A = r \geqslant 1$ and $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{r\times r}$ invertible (blocks with no rows or no columns are absent). Then $G = \left(\begin{smallmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right) \in \mathbb{R}^{n\times m}$ is a g-inverse of A. More generally, if the submatrix M of A in rows $i_1 < \cdots < i_r$ and columns $j_1 < \cdots < j_r$ is invertible, the matrix $G \in \mathbb{R}^{n\times m}$ with entry $(M^{-1})_{st}$ in position (j_s, i_t) $(1 \leqslant s, t \leqslant r)$ and all other entries 0 is a g-inverse of A.

(c) If A is square and invertible, A^{-1} is its only g-inverse.

Proof. (a)

- Put $H = V'GU \in \mathbb{R}^{n\times m}$, so that $G = VHU'$. Then $AGA = U\Sigma V'VHU'U\Sigma V' = U\Sigma H\Sigma V'$. Multiplying by U' on the left and by V on the right, and back by U and V', shows that $AGA = A$ if and only if $\Sigma H\Sigma = \Sigma$.
- Partition $H = \left(\begin{smallmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{smallmatrix}\right)$ with $H_{11} \in \mathbb{R}^{r\times r}$, conformally with $\Sigma = \left(\begin{smallmatrix} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. By Theorem 5.3(c), $\Sigma H\Sigma = \left(\begin{smallmatrix} D_r H_{11} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. As D_r is invertible, $\Sigma H\Sigma = \Sigma$ if and only if $H_{11} = D_r^{-1}$, with $H_{12} = K$, $H_{21} = L$, $H_{22} = N$ arbitrary. For $A = 0$, $AGA = 0 = A$ for every G.

(b)

- If $r < m$ and $r < n$, Theorem 5.29(a) gives $r = \operatorname{rank} A = r + \operatorname{rank}(A_{22} - A_{21}A_{11}^{-1}A_{12})$, so $A_{22} = A_{21}A_{11}^{-1}A_{12}$. In every case the block product formula (Theorem 5.3) gives, with absent blocks omitted,

$$AG = \begin{pmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{pmatrix}, \qquad AGA = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{pmatrix} = A$$

- General position: let $P \in \mathbb{R}^{m\times m}$ have the rows $e'_{i_1}, \ldots, e'_{i_r}$ followed by the other e'_i, and $Q \in \mathbb{R}^{n\times n}$ the columns $e_{j_1}, \ldots, e_{j_r}$ followed by the other e_j. The rows of P and the columns of Q are orthonormal, so P and Q are orthogonal (Proposition 6.10(b), applied to P' and Q). Row s of PA is row i_s of A and column t of PAQ is column j_t of PA (Theorem 1.4), so PAQ has M as its leading $r \times r$ block, and $\operatorname{rank}(PAQ) = r$ (Theorem 4.15(d)).
- By the first item, $G_0 = \left(\begin{smallmatrix} M^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is a g-inverse of PAQ, so $G = QG_0P$ satisfies $AGA = P'(PAQ)G_0(PAQ)Q' = P'(PAQ)Q' = A$. By the outer product expansion, $QG_0P = \sum_{s,t\leqslant r}(M^{-1})_{st}(Qe_s)(P'e_t)' = \sum_{s,t\leqslant r}(M^{-1})_{st}e_{j_s}e'_{i_t}$, the matrix described.

(c) $A^{-1}AA^{-1} = A^{-1}$ shows that A^{-1} is a g-inverse. If $AGA = A$, then $G = A^{-1}(AGA)A^{-1} = A^{-1}AA^{-1} = A^{-1}$. □

So a g-inverse of a matrix A of rank $r \geqslant 1$ is found as follows: choose an invertible $r \times r$ submatrix M, replace it in A by $(M^{-1})'$ and all other entries by 0, and transpose. Such an M exists: the r pivot columns of A form a matrix $F \in \mathbb{R}^{m\times r}$ of rank r (Theorem 4.8(a)); F' has rank r (Theorem 4.13(a)), so r of its columns are independent (Theorem 4.8(a)), and the corresponding r rows of F form an $r \times r$ matrix with independent rows, which is invertible (Theorem 4.13(c),(d)).

Theorem 10.10. Let G be a g-inverse of $A \in \mathbb{R}^{m\times n}$.

(a) AG and GA are idempotent, and $\operatorname{rank}(AG) = \operatorname{rank}(GA) = \operatorname{rank} A \leqslant \operatorname{rank} G$.

(b) $\mathcal{C}(AG) = \mathcal{C}(A)$ and $\mathcal{N}(GA) = \mathcal{N}(A)$.

(c) G' is a g-inverse of A'.

Proof. (a) $(AG)^2 = (AGA)G = AG$ and $(GA)^2 = G(AGA) = GA$. By Theorem 4.15(a), $\operatorname{rank} A = \operatorname{rank}(AGA) \leqslant \operatorname{rank}(AG) \leqslant \operatorname{rank} A$, $\operatorname{rank} A = \operatorname{rank}(AGA) \leqslant \operatorname{rank}(GA) \leqslant \operatorname{rank} A$ and $\operatorname{rank}(GA) \leqslant \operatorname{rank} G$.

(b) $\mathcal{C}(AG) \subseteq \mathcal{C}(A) = \mathcal{C}(AGA) \subseteq \mathcal{C}(AG)$ (Theorem 4.15(a)). If $GAx = 0$, then $Ax = AGAx = 0$; and $Ax = 0$ gives $GAx = 0$.

(c) $A'G'A' = (AGA)' = A'$ (Theorem 1.9). □

Theorem 10.11. Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and let G be a g-inverse of A.

(a) $Ax = b$ is consistent if and only if $AGb = b$.

(b) If $Ax = b$ is consistent, its solutions are the vectors $x = Gb + (I_n - GA)z$, $z \in \mathbb{R}^n$.

(c) $\mathcal{N}(A) = \mathcal{C}(I_n - GA)$.

Proof. (a) If $AGb = b$, then $x = Gb$ is a solution. If $Ax_0 = b$, then $AGb = AGAx_0 = Ax_0 = b$.

(c) $A(I_n - GA)z = (A - AGA)z = 0$, so $\mathcal{C}(I_n - GA) \subseteq \mathcal{N}(A)$; and if $Az = 0$, then $z = (I_n - GA)z \in \mathcal{C}(I_n - GA)$.

(b) By (a), Gb is a solution. By Theorem 2.23(a) the solutions are $Gb + w$ with $w \in \mathcal{N}(A)$, and by (c) these w are the vectors $(I_n - GA)z$. □

Lemma 10.12. Let $A \in \mathbb{R}^{m\times n}$ and let G be a g-inverse of $A'A$.

(a) $AGA'A = A$ and $A'AGA' = A'$.

(b) GA' is a g-inverse of A.

Proof. (a)

- Each column of $E = AGA'A - A = A(GA'A - I_n)$ lies in $\mathcal{C}(A)$, and $A'E = A'AGA'A - A'A = 0$, so each column lies in $\mathcal{N}(A') = \mathcal{C}(A)^\perp$ (Corollary 6.6). As $\mathcal{C}(A) \cap \mathcal{C}(A)^\perp = \{0\}$ (Theorem 6.5(b)), $E = 0$.
- G' is a g-inverse of $(A'A)' = A'A$ (Theorem 10.10(c)), so the first identity with G' gives $AG'A'A = A$; transposing, $A'AGA' = A'$.

(b) $A(GA')A = AGA'A = A$ by (a). □

Proposition 10.13. Let $A = BC$ be a full-rank factorization of $A \in \mathbb{R}^{m\times n}$ of rank $r \geqslant 1$ (Theorem 4.21), and let $L \in \mathbb{R}^{r\times m}$ and $K \in \mathbb{R}^{n\times r}$ satisfy $LB = I_r$ and $CK = I_r$ (for instance $L = (B'B)^{-1}B'$ and $K = C'(CC')^{-1}$). Then $G = KL$ satisfies $AGA = A$ and $GAG = G$, and $\operatorname{rank} G = r$.

Proof. $AGA = B(CK)(LB)C = BC = A$ and $GAG = K(LB)(CK)L = KL = G$. By Theorem 10.10(a) and Theorem 4.15(a), $r = \operatorname{rank} A \leqslant \operatorname{rank} G = \operatorname{rank}(KL) \leqslant \operatorname{rank} K \leqslant r$, as K has r columns (Theorem 4.13(a)). □

Example 10.14. Let

$$A = \begin{pmatrix} 2 & 2 & 0 & 1 \\ 0 & -2 & 2 & 1 \\ 1 & 0 & 1 & 1 \\ 2 & 4 & -2 & 0 \end{pmatrix}, \qquad p_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 2 \end{pmatrix}, \quad p_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad q_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \quad q_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

Show that $A = p_1q_1' + p_2q_2'$ is a full-rank factorization of rank 2, find g-inverses of A from two invertible 2×2 submatrices, and use them to decide when $Ax = b$ is consistent and to solve it for $b = (2, 0, 1, 2)'$.

Solution.

- By the outer product expansion (Theorem 1.4(d)), $p_1q_1' + p_2q_2' = BC$ with $B = [p_1\ p_2] \in \mathbb{R}^{4\times 2}$ and $C \in \mathbb{R}^{2\times 4}$ with rows q_1', q_2', and

$$p_1q_1' + p_2q_2' = \begin{pmatrix} 1 & 2 & -1 & 0 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & -2 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = A$$

Since $p_1'p_2 = 0$ and $q_1'q_2 = 0$, the nonzero vectors p_1, p_2 are independent, and so are q_1, q_2 (Proposition 6.8(b)): B has full column rank and C full row rank (Theorem 4.13(b),(c)). By Theorem 4.15(b), $\operatorname{rank} A = \operatorname{rank} C = 2$, and $A = BC$ is a full-rank factorization. Also $B'B = \operatorname{diag}(6,3)$ and $CC' = \operatorname{diag}(6,3)$.

- The leading submatrix $M_1 = \left(\begin{smallmatrix} 2 & 2 \\ 0 & -2 \end{smallmatrix}\right)$ has $M_1^{-1} = \frac{1}{4}\left(\begin{smallmatrix} 2 & 2 \\ 0 & -2 \end{smallmatrix}\right)$. By Theorem 10.9(b),

$$G_1 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  is a g-inverse of A.

- The submatrix in rows $1,3$ and columns $3,4$ is $M_2 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with $M_2^{-1} = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. By the general rule of Theorem 10.9(b), the entries of M_2^{-1} go to the positions $(3,1)$, $(3,3)$, $(4,1)$, $(4,3)$:

$$G_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

- With the columns $a_1, \dots, a_4$ of A, $AG_2 = [-a_3 + a_4 \;\; 0 \;\; a_3 \;\; 0]$, so $AG_2b = b_1(1,-1,0,2)' + b_3(0,2,1,-2)' = (b_1,\ 2b_3 - b_1,\ b_3,\ 2b_1 - 2b_3)'$. By Theorem 10.11(a), $Ax = b$ is consistent if and only if $b_2 = 2b_3 - b_1$ and $b_4 = 2b_1 - 2b_3$.

- For $b = (2,0,1,2)'$ both conditions hold. The rows of G_2A are 0, 0, $\operatorname{row}_3(A) - \operatorname{row}_1(A) = (-1,-2,1,0)$ and $\operatorname{row}_1(A) = (2,2,0,1)$, so

$$G_2b = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix}, \qquad I_4 - G_2A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{pmatrix}$$

  By Theorem 10.11(b),(c), $\mathcal{N}(A) = \operatorname{span}\{(1,0,1,-2)', (0,1,2,-2)'\}$ and the solutions are

$$x = (0,0,-1,2)' + z_1(1,0,1,-2)' + z_2(0,1,2,-2)' \qquad (z_1, z_2 \in \mathbb{R})$$

  The solution $G_1b = (1,0,0,0)' = e_1$ ($z_1 = 1$, $z_2 = 0$) is visible directly: b is the first column of A.

The Moore–Penrose inverse

Theorem 10.15. Let $A \in \mathbb{R}^{m\times n}$.

(a) There is exactly one $G \in \mathbb{R}^{n\times m}$ with

$$AGA = A, \qquad GAG = G, \qquad (AG)' = AG, \qquad (GA)' = GA$$

(the **Penrose conditions**). It is the **Moore–Penrose inverse** of A, written A^+.

(b) If $\operatorname{rank} A = r \geqslant 1$ and $A = U\Sigma V'$ as in Theorem 10.1, then $A^+ = V\Sigma^+U' = \sum_{i=1}^{r} \sigma_i^{-1} v_i u_i'$, where $\Sigma^+ \in \mathbb{R}^{n\times m}$ has entries $(\Sigma^+)_{ii} = \sigma_i^{-1}$ for $i \leqslant r$ and all other entries 0. Also $0^+ = 0$.

(c) $A^+ = (A'A)^{-1}A'$ if $\operatorname{rank} A = n$; $A^+ = A'(AA')^{-1}$ if $\operatorname{rank} A = m$; and $A^+ = C'(CC')^{-1}(B'B)^{-1}B'$ for every full-rank factorization $A = BC$ (Theorem 4.21).

(d) $AA^+ = P_{\mathcal{C}(A)}$ and $A^+A = P_{\mathcal{C}(A')}$ (Theorem 6.19); if $\operatorname{rank} A = r \geqslant 1$ and $A = U\Sigma V'$ as in Theorem 10.1, then $AA^+ = U_rU_r'$ and $A^+A = V_rV_r'$. In particular AA^+ and A^+A are symmetric and idempotent, AA^+y is the orthogonal projection of $y \in \mathbb{R}^m$ onto $\mathcal{C}(A)$, and A^+Ax that of $x \in \mathbb{R}^n$ onto $\mathcal{C}(A')$.

(e) $(A^+)' = (A')^+$ and $(A^+)^+ = A$. If A is symmetric and $A = Q\Lambda Q'$ with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ (Theorem 8.4), then $A^+ = Q\Lambda^+Q'$, where $\Lambda^+ = \operatorname{diag}(\lambda_1^+, \ldots, \lambda_n^+)$, $\lambda^+ = 1/\lambda$ for $\lambda \neq 0$ and $0^+ = 0$.

Proof. (a), (b)

- Uniqueness: let G_1, G_2 satisfy the four conditions. Using them in turn,

$$\begin{aligned} G_1 &= G_1AG_1 = G_1(AG_1)' = G_1G_1'A' = G_1G_1'(AG_2A)' \\ &= G_1(AG_1)'(AG_2)' = G_1AG_1AG_2 = G_1AG_2, \\ G_2 &= G_2AG_2 = (G_2A)'G_2 = A'G_2'G_2 = (AG_1A)'G_2'G_2 \\ &= (G_1A)'(G_2A)'G_2 = G_1AG_2AG_2 = G_1AG_2, \end{aligned}$$

  so $G_1 = G_2$.

- Existence: for $A = 0$, $G = 0$ satisfies the conditions. Let $r \geqslant 1$ and $G = V\Sigma^+U'$. By the block product formula, $\Sigma\Sigma^+ \in \mathbb{R}^{m\times m}$ and $\Sigma^+\Sigma \in \mathbb{R}^{n\times n}$ are diagonal with diagonal entries 1 (r times) and 0; hence $\Sigma\Sigma^+\Sigma = \Sigma$ and $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$. Then $AGA = U\Sigma\Sigma^+\Sigma V' = A$, $GAG = V\Sigma^+\Sigma\Sigma^+U' = G$, and $AG = U(\Sigma\Sigma^+)U'$, $GA = V(\Sigma^+\Sigma)V'$ are symmetric. Finally $V\Sigma^+U' = V_rD_r^{-1}U_r' = \sum_{i\leqslant r}\sigma_i^{-1}v_iu_i'$, as in the proof of Theorem 10.1.

(c) In each case we check the four conditions and apply (a). If N is symmetric and invertible, then $(N^{-1})' = (N')^{-1} = N^{-1}$ (Theorem 2.15(c)), so $FN^{-1}F'$ is symmetric for every matrix F with as many columns as N (Theorem 1.9).

- $\operatorname{rank} A = n$: $\operatorname{rank}(A'A) = n$ (Theorem 4.18(b)), so the symmetric $n \times n$ matrix $A'A$ is invertible (Theorem 4.13(d)). For $G = (A'A)^{-1}A'$, $GA = I_n$, so $AGA = A$, $GAG = G$, GA is symmetric, and $AG = A(A'A)^{-1}A'$ is symmetric.

- $\operatorname{rank} A = m$: likewise AA' is invertible. For $G = A'(AA')^{-1}$, $AG = I_m$, so $AGA = A$, $GAG = G$, AG is symmetric, and $GA = A'(AA')^{-1}A$ is symmetric.

- $A = BC$ with $B \in \mathbb{R}^{m\times r}$, $C \in \mathbb{R}^{r\times n}$ of rank r: with $L = (B'B)^{-1}B'$ and $K = C'(CC')^{-1}$, $LB = CK = I_r$ (Theorem 4.21(c)). For $G = KL$, Proposition 10.13 gives $AGA = A$ and $GAG = G$, and $AG = B(CK)L = B(B'B)^{-1}B'$ and $GA = K(LB)C = C'(CC')^{-1}C$ are symmetric.

(d) For $A = 0$, $A^+ = 0$, and $AA^+ = 0 = P_{\{0\}} = P_{\mathcal{C}(A)}$, $A^+A = 0 = P_{\mathcal{C}(A')}$. Let $r \geqslant 1$. By (b) and $V'V = I_n$, $AA^+ = U\Sigma V'V\Sigma^+U' = U(\Sigma\Sigma^+)U'$, and $\Sigma\Sigma^+ = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$, so $AA^+ = U_rU_r'$ by the block product formula (Theorem 5.3(c)), with $U = [U_r\ U_2]$ as in the proof of Theorem 10.1. The columns of U_r form an orthonormal basis of $\mathcal{C}(A)$ (Corollary 10.2(b)), so $U_rU_r' = P_{\mathcal{C}(A)}$ (Theorem 6.19(a)). Likewise $A^+A = V(\Sigma^+\Sigma)V' = V_rV_r' = P_{\mathcal{C}(A')}$, as $v_1, \ldots, v_r$ form an orthonormal basis of $\mathcal{C}(A')$. The remaining statements follow from Theorem 6.19(a),(b).

(e) $G = (A^+)'$ satisfies the Penrose conditions for A': $A'GA' = (AA^+A)' = A'$, $GA'G = (A^+AA^+)' = G$, $A'G = (A^+A)' = A^+A$ and $GA' = (AA^+)' = AA^+$ are symmetric. The four conditions for A^+ are symmetric in A and A^+, so A satisfies them as the Moore–Penrose inverse of A^+. For symmetric $A = Q\Lambda Q'$ and $G = Q\Lambda^+Q'$: $\lambda\lambda^+\lambda = \lambda$ and $\lambda^+\lambda\lambda^+ = \lambda^+$ for every real λ, so $AGA = Q\Lambda\Lambda^+\Lambda Q' = A$, $GAG = G$, and $AG = GA = Q\Lambda\Lambda^+Q'$ is symmetric. □

Every Moore–Penrose inverse is a g-inverse, and by Theorem 10.9(c), $A^+ = A^{-1}$ for invertible A. In

Example 10.3, $\operatorname{rank} A = 3 = n$ and $A'A = 9B$; $B^{-1} = \frac{1}{16}\left(\begin{smallmatrix} 4 & 2 & -4 \\ 2 & 9 & -6 \\ -4 & -6 & 8 \end{smallmatrix}\right)$ (check: $BB^{-1} = I_3$), so (c) gives

$$A^+ = (A'A)^{-1}A' = \frac{1}{144}\begin{pmatrix} 4 & 2 & -4 \\ 2 & 9 & -6 \\ -4 & -6 & 8 \end{pmatrix}\begin{pmatrix} 6 & 4 & 5 & 2 \\ 0 & 4 & 2 & -4 \\ 0 & 6 & 6 & 0 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} 6 & 0 & 0 & 0 \\ 3 & 2 & -2 & -8 \\ -6 & 2 & 4 & 4 \end{pmatrix}$$

By (b), also $A^+ = \frac{1}{12}v_1u_1' + \frac{1}{6}v_2u_2' + \frac{1}{3}v_3u_3'$. By (d), $A^+A = I_3$ and $AA^+ = U_3U_3' = I_4 - u_4u_4' = \frac{1}{9}\left(\begin{smallmatrix} 9 & 0 & 0 & 0 \\ 0 & 5 & 4 & -2 \\ 0 & 4 & 5 & 2 \\ 0 & -2 & 2 & 8 \end{smallmatrix}\right)$, the projection matrix onto $\mathcal{C}(A) = \operatorname{span}\{u_4\}^\perp$.

Theorem 10.16. Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, and let $Ax = b$ be consistent.

(a) A^+b is a solution, and the solutions are the vectors $A^+b + (I_n - A^+A)z$, $z \in \mathbb{R}^n$.

(b) Every solution x satisfies $\|x\|^2 = \|A^+b\|^2 + \|x - A^+b\|^2$. So A^+b is the unique solution of minimum length, and the only solution in $\mathcal{C}(A')$.

Proof. (a) A^+ is a g-inverse of A (Theorem 10.15(a)); apply Theorem 10.11(a),(b) with $G = A^+$.
(b) Let $Ax = b$. Then $A(x - A^+b) = b - b = 0$, so $x - A^+b \in \mathcal{N}(A) = \mathcal{C}(A')^\perp$ (Corollary 6.6). Also $A^+b = A^+A(A^+b) \in \mathcal{C}(A^+A) = \mathcal{C}(A')$ (Theorem 10.15(d) and Theorem 6.19(b)). So $A^+b \perp x - A^+b$, and Pythagoras (Proposition 6.2) for $x = A^+b + (x - A^+b)$ gives the identity; hence $\|x\| > \|A^+b\|$ unless $x = A^+b$. If moreover $x \in \mathcal{C}(A')$, then $x - A^+b \in \mathcal{C}(A') \cap \mathcal{C}(A')^\perp = \{0\}$ (Theorem 6.5(b)). □

If $\operatorname{rank} A = m$, then $Ax = b$ is consistent for every b ($\mathcal{C}(A) = \mathbb{R}^m$, Theorem 4.13(c)), and $A^+b = A'(AA')^{-1}b$ (Theorem 10.15(c)) is the minimum-norm solution of Proposition 9.25.

Example 10.17. For the matrix $A = p_1q_1' + p_2q_2'$ of Example 10.14, find a compact singular value decomposition, A^+, AA^+ and A^+A, and the solution of minimum length of $Ax = (2, 0, 1, 2)'$.

Solution.

- $\|p_1\| = \|q_1\| = \sqrt{6}$ and $\|p_2\| = \|q_2\| = \sqrt{3}$, so
$$A = 6\,\frac{p_1}{\sqrt{6}}\Big(\frac{q_1}{\sqrt{6}}\Big)' + 3\,\frac{p_2}{\sqrt{3}}\Big(\frac{q_2}{\sqrt{3}}\Big)'$$
with $p_1 \perp p_2$ and $q_1 \perp q_2$. By Corollary 10.2(e), $\sigma_1 = 6$, $\sigma_2 = 3$, $u_1 = p_1/\sqrt{6}$, $v_1 = q_1/\sqrt{6}$, $u_2 = p_2/\sqrt{3}$, $v_2 = q_2/\sqrt{3}$.
- By Theorem 10.15(b),
$$A^+ = \frac{1}{6}v_1u_1' + \frac{1}{3}v_2u_2' = \frac{1}{36}q_1p_1' + \frac{1}{9}q_2p_2' = \frac{1}{36}\begin{pmatrix} 5 & 3 & 4 & 2 \\ 2 & -2 & 0 & 4 \\ 3 & 5 & 4 & -2 \\ 4 & 4 & 4 & 0 \end{pmatrix}$$
The full-rank factorization $A = BC$ of Example 10.14, with $B'B = CC' = \operatorname{diag}(6, 3)$, gives the same matrix by Theorem 10.15(c): $C'\operatorname{diag}(\frac{1}{6}, \frac{1}{3})\operatorname{diag}(\frac{1}{6}, \frac{1}{3})B' = \frac{1}{36}q_1p_1' + \frac{1}{9}q_2p_2'$.
- By Theorem 10.15(d),
$$AA^+ = \frac{1}{6}p_1p_1' + \frac{1}{3}p_2p_2' = \frac{1}{6}\begin{pmatrix} 3 & 1 & 2 & 2 \\ 1 & 3 & 2 & -2 \\ 2 & 2 & 2 & 0 \\ 2 & -2 & 0 & 4 \end{pmatrix}, \qquad A^+A = \frac{1}{6}q_1q_1' + \frac{1}{3}q_2q_2' = \frac{1}{6}\begin{pmatrix} 3 & 2 & 1 & 2 \\ 2 & 4 & -2 & 0 \\ 1 & -2 & 3 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}$$
are the projection matrices onto $\mathcal{C}(A)$ and $\mathcal{C}(A')$; both have trace $2 = \operatorname{rank} A$.
- The system is consistent (Example 10.14). By Theorem 10.16, its shortest solution is $x^+ = A^+(2, 0, 1, 2)' = \frac{1}{36}(18, 12, 6, 12)' = \frac{1}{6}(3, 2, 1, 2)'$, with $\|x^+\|^2 = \frac{18}{36} = \frac{1}{2}$. It is the first column of A^+A: the projection onto $\mathcal{C}(A')$ of the solution e_1. For the solutions e_1 and $(0, 0, -1, 2)'$ of Example 10.14, $\|e_1 - x^+\|^2 = \frac{9+4+1+4}{36} = \frac{1}{2}$ and $\|(0, 0, -1, 2)' - x^+\|^2 = \frac{9+4+49+100}{36} = \frac{9}{2}$, and indeed $1 = \frac{1}{2} + \frac{1}{2}$ and $5 = \frac{1}{2} + \frac{9}{2}$.

Low-rank approximation

Theorem 10.18 (Eckart–Young)**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$ and compact singular value decomposition $A = \sum_{i=1}^{r} \sigma_i u_i v_i'$, and for $0 \leqslant k < r$ let

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i' \qquad (A_0 = 0)$$

Then $\operatorname{rank} A_k = k$, and every $B \in \mathbb{R}^{m\times n}$ with $\operatorname{rank} B \leqslant k$ satisfies

(a) $\|A - B\|_2 \geqslant \sigma_{k+1} = \|A - A_k\|_2$;

(b) $\|A - B\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2 = \|A - A_k\|_F^2$.

Proof. • $A_0 = 0$ has rank 0. For $k \geqslant 1$, $A_k = \sum_{i\leqslant k} \sigma_i u_i v_i'$ is of the form of Corollary 10.2(e), so $\operatorname{rank} A_k = k$. Also $A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i'$ has the singular values $\sigma_{k+1} \geqslant \cdots \geqslant \sigma_r$ by the same corollary. By Theorem 10.5(a), $\|A - A_k\|_2 = \sigma_{k+1}$ and $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$.

• (a) $\dim \mathcal{N}(B) = n - \operatorname{rank} B \geqslant n - k$ (Theorem 4.12), and $W = \operatorname{span}\{v_1, \ldots, v_{k+1}\}$ has dimension $k+1$. By Lemma 9.30, $\dim(\mathcal{N}(B) \cap W) \geqslant (n-k) + (k+1) - n = 1$. Take $x \in \mathcal{N}(B) \cap W$ with $\|x\| = 1$; then $v_i' x = 0$ for $i > k+1$ and $\sum_{i\leqslant k+1} (v_i' x)^2 = 1$. By Corollary 10.2(d), $\|(A-B)x\|^2 = \|Ax\|^2 = \sum_{i\leqslant k+1} \sigma_i^2 (v_i' x)^2 \geqslant \sigma_{k+1}^2$, so $\|A - B\|_2 \geqslant \sigma_{k+1}$.

• (b) $\sigma_{k+1}(B) = 0$, as $\operatorname{rank} B \leqslant k$. By Weyl's inequality (Theorem 10.5(d)) for $A = (A-B) + B$, with $j = k+1$, $\sigma_{i+k}(A) \leqslant \sigma_i(A-B) + \sigma_{k+1}(B) = \sigma_i(A-B)$ for $1 \leqslant i \leqslant n-k$. By Theorem 10.5(a), $\|A - B\|_F^2 = \sum_{i=1}^{n} \sigma_i(A-B)^2 \geqslant \sum_{i=1}^{n-k} \sigma_{i+k}(A)^2 = \sum_{l=k+1}^{r} \sigma_l^2$. □

So A_k is a best approximation of A by matrices of rank at most k in both norms, and the fraction of the sum of squares $\|A\|_F^2 = \sum_{i,j} a_{ij}^2$ that it reproduces is $\|A_k\|_F^2 / \|A\|_F^2 = (\sigma_1^2 + \cdots + \sigma_k^2)/(\sigma_1^2 + \cdots + \sigma_r^2)$. For the matrix $A = p_1 q_1' + p_2 q_2'$ of Example 10.17, $A_1 = 6u_1 v_1' = p_1 q_1'$ and $A - A_1 = p_2 q_2'$:

$$A_1 = \begin{pmatrix} 1 & 2 & -1 & 0 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & -2 & 0 \end{pmatrix}, \qquad A - A_1 = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Every B of rank at most 1 has $\|A - B\|_2 \geqslant 3$ and $\|A - B\|_F^2 \geqslant 9$, and A_1 reproduces $36/45 = 4/5$ of $\|A\|_F^2 = 45$. By comparison, keeping the longest row of A, the fourth, and replacing the other rows by 0 gives a matrix of rank 1 with $\|A - B\|_F^2 = 45 - 24 = 21$.

Corollary 10.19. Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$. Then $\min\{\|A - B\|_2 : B \in \mathbb{R}^{m\times n}, \operatorname{rank} B < r\} = \sigma_r(A)$. In particular, if A has full column rank n, the smallest $\|E\|_2$ for which $A + E$ has linearly dependent columns is $\sigma_n(A)$, and $\sigma_n(A)/\|A\|_2 = 1/\kappa(A)$.

Proof. Theorem 10.18(a) with $k = r - 1$ gives the minimum, attained at $B = A_{r-1}$. If $\operatorname{rank} A = n$, the columns of $B = A + E$ are dependent if and only if $\operatorname{rank} B < n$ (Theorem 4.13(b)), and $\|E\|_2 = \|A - B\|_2$. Finally $\|A\|_2 = \sigma_1(A)$. □

So $1/\kappa(A)$ measures, relative to $\|A\|_2$, how far a matrix A of full column rank is from having linearly dependent columns. In Example 10.3, $\sigma_3 = 3 = \|A\|_2/4$: the matrix $E = -3u_3 v_3' = -\frac{1}{3}(2, 0, -1, -2)'(1, 2, -2)$, of spectral norm 3, makes the columns of $A + E = 12u_1 v_1' + 6u_2 v_2'$ dependent, and no E of smaller spectral norm does.

The orthogonal Procrustes problem

Theorem 10.20 (Orthogonal Procrustes)**.** Let $M \in \mathbb{R}^{n\times n}$, with singular value decomposition $M = U\Sigma V'$ (Theorem 10.1; for $M = 0$ take $U = V = I_n$), and $\sigma_j = \sigma_j(M)$.

(a) For every orthogonal $Q \in \mathbb{R}^{n\times n}$, $\operatorname{tr}(Q'M) \leqslant \sigma_1 + \cdots + \sigma_n$, with equality for $Q = UV'$.

(b) Let $B, C \in \mathbb{R}^{n\times k}$ and $M = CB'$. For every orthogonal $Q \in \mathbb{R}^{n\times n}$,

$$\|QB - C\|_F^2 \geqslant \|B\|_F^2 + \|C\|_F^2 - 2(\sigma_1 + \cdots + \sigma_n),$$

with equality for $Q = UV'$.

Proof. (a) $Z = V'Q'U$ is orthogonal, as $Z'Z = U'QVV'Q'U = I_n$. By Theorem 1.11(b), $\operatorname{tr}(Q'M) = \operatorname{tr}(Q'U\Sigma V') = \operatorname{tr}(V'Q'U\Sigma) = \operatorname{tr}(Z\Sigma) = \sum_j z_{jj}\sigma_j$, since Σ is diagonal with diagonal entries $\sigma_1, \ldots, \sigma_n$ (the convention after Corollary 10.2). Column j of Z is a unit vector, so $z_{jj} \leqslant 1$, and $\sigma_j \geqslant 0$; hence $\operatorname{tr}(Q'M) \leqslant \sum_j \sigma_j$. For $Q = UV'$, $Z = V'VU'U = I_n$, and equality holds.

(b) $(QB)'(QB) = B'Q'QB = B'B$, and $\operatorname{tr}(C'QB) = \operatorname{tr}\Big((C'QB)'\Big) = \operatorname{tr}(B'Q'C) = \operatorname{tr}(Q'CB')$ (Theorem 1.11(a),(b)). So

$$\|QB - C\|_F^2 = \operatorname{tr}\Big((QB - C)'(QB - C)\Big) = \operatorname{tr}(B'B) - 2\operatorname{tr}(Q'CB') + \operatorname{tr}(C'C),$$

and (a) applied to $M = CB'$ gives the bound and the case of equality. □

In (b) the columns of B and C are two configurations of k points in $\mathbb{R}^n$, and UV' is an orthogonal transformation (a rotation, possibly composed with a reflection) that moves the first configuration as close as possible to the second. For $B = I_n$, (b) says that UV' is an orthogonal matrix nearest to $C = M$ in the Frobenius norm.

Principal components

Definition 10.21. Let $X_0 \in \mathbb{R}^{n\times k}$ ($n \geqslant 2$) be a data matrix with rows $x_1', \ldots, x_n'$, mean vector $\bar{x}$, centered data matrix $X_c = C_n X_0$, whose row i is $(x_i - \bar{x})'$, and sample covariance matrix $S = X_c'X_c/(n-1)$ (Definition 8.32). Let $S = Q\Lambda Q'$ with $Q = [q_1 \ \cdots \ q_k]$ orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_k$ (Theorem 8.4(a),(d)); here $\lambda_k \geqslant 0$, since S is positive semidefinite (Proposition 8.33(b), Theorem 8.13(b)).

- q_j is the jth **loading vector**, and its entries are the **loadings** of the jth principal component;
- $Z = X_cQ \in \mathbb{R}^{n\times k}$ is the matrix of **principal component scores**: $z_{ij} = q_j'(x_i - \bar{x})$ is the score of observation i on the jth component, and column j of Z is the jth **principal component**;
- λ_j is the **variance** of the jth component;
- if $S \neq 0$, the **proportion of variance explained** by the first m components is $(\lambda_1 + \cdots + \lambda_m)/(\lambda_1 + \cdots + \lambda_k)$.

If λ_j is a simple root of p_S, its eigenspace has dimension 1 (Theorem 8.4(b)), and q_j is determined up to its sign; within an eigenspace of dimension at least 2, any orthonormal basis may serve as loading vectors.

For a subspace $W \neq \{0\}$ of $\mathbb{R}^k$ with orthonormal basis $w_1, \ldots, w_m$ (Theorem 6.11) and $Q_W = [w_1 \ \cdots \ w_m]$, the projection matrix onto W is $P_W = Q_W Q_W'$; it is symmetric and idempotent, of rank m (Theorem 6.19(a),(b)). For $W = \{0\}$, $P_W = 0$.

Theorem 10.22. In the setting of Definition 10.21:

(a) the columns of Z have sample mean 0 and sample covariance matrix Λ: the jth component has sample variance λ_j, and different components have sample covariance 0;

(b) the sample variance $a'Sa$ of the values $a'x_1, \dots, a'x_n$ is largest among unit vectors a at $a = q_1$, and, for $2 \leqslant j \leqslant k$, among unit vectors $a \perp q_1, \dots, q_{j-1}$ at $a = q_j$; the largest values are λ_1 and λ_j;

(c) $\operatorname{tr} S = s_{11} + \cdots + s_{kk} = \lambda_1 + \cdots + \lambda_k$ (the **total variance**);

(d) if $X_c \neq 0$ and $X_c = U\Sigma V'$ is a singular value decomposition, then $\lambda_j = \sigma_j(X_c)^2/(n-1)$ for every j, one may take $Q = V$, and then $Z = U\Sigma$;

(e) for $0 \leqslant m < k$, $W_m = \operatorname{span}\{q_1, \dots, q_m\}$ minimizes

$$\sum_{i=1}^{n} \|(x_i - \bar{x}) - P_W(x_i - \bar{x})\|^2$$

over all m-dimensional subspaces W of $\mathbb{R}^k$, with minimum $(n-1)(\lambda_{m+1} + \cdots + \lambda_k)$;

(f) for $1 \leqslant m < k$, with Z_m and Q_m the first m columns of Z and Q, $X_c P_{W_m} = Z_m Q_m'$: the projection of $x_i - \bar{x}$ onto W_m is $\sum_{j \leqslant m} z_{ij} q_j$. If $S \neq 0$, then $\|X_c - Z_m Q_m'\|_F^2 / \|X_c\|_F^2$ is 1 minus the proportion of variance explained by the first m components.

Proof. (a) $\mathbf{1}'Z = (C_n\mathbf{1})'X_0Q = 0$ (Proposition 1.13), so the mean vector of Z is 0. As $C_nZ = C_n^2X_0Q = Z$, the sample covariance matrix of Z (Definition 8.32) is $(n-1)^{-1}Z'C_nZ = (n-1)^{-1}Q'X_c'X_cQ = Q'SQ = \Lambda$.

(b) The sample variance is $a'Sa$ (Proposition 8.33(a)), which equals $a'Sa/a'a$ for unit a; apply Theorem 9.29(a),(b) to S and its orthonormal eigenvectors $q_1, \dots, q_k$.

(c) $\operatorname{tr} S = \operatorname{tr}(Q\Lambda Q') = \operatorname{tr}(\Lambda Q'Q) = \operatorname{tr}\Lambda$ (Theorem 1.11(b)).

(d) $S = (n-1)^{-1}V\Sigma'U'U\Sigma V' = V\Big(\Sigma'\Sigma/(n-1)\Big)V'$, where the diagonal matrix $\Sigma'\Sigma/(n-1) \in \mathbb{R}^{k\times k}$ has the decreasing diagonal entries $\sigma_j(X_c)^2/(n-1)$; so V may serve as Q. The λ_j, being the roots of p_S in decreasing order, are these diagonal entries for every choice of Q. Finally $Z = X_cV = U\Sigma V'V = U\Sigma$.

(e)

- Let W have dimension m. Row i of $X_c - X_cP_W$ is $(x_i-\bar{x})' - (x_i-\bar{x})'P_W = \Big((x_i-\bar{x}) - P_W(x_i-\bar{x})\Big)'$, as P_W is symmetric. So the sum to be minimized is $\|X_c - X_cP_W\|_F^2$ (Definition 10.4), and $\operatorname{rank}(X_cP_W) \leqslant \operatorname{rank} P_W = m$ (Theorem 4.15(a)).
- Let $r = \operatorname{rank} X_c$. By (d), $\lambda_j = 0$ for $j > r$ (if $X_c = 0$, then $S = 0$). If $m < r$, Theorem 10.18(b) and (d) give $\|X_c - X_cP_W\|_F^2 \geqslant \sum_{j=m+1}^{r} \sigma_j(X_c)^2 = (n-1)\sum_{j>m}\lambda_j$. If $m \geqslant r$, the right side is 0, and the inequality holds as the left side is nonnegative.
- For $W = W_m$, $P_{W_m} = Q_mQ_m'$, and $I_k - Q_mQ_m' = \sum_{j>m} q_jq_j'$ (from $QQ' = I_k$ and the outer product expansion) is symmetric and idempotent. By Theorem 1.11(b) and $Sq_j = \lambda_jq_j$,

$$\|X_c(I_k - Q_mQ_m')\|_F^2 = (n-1)\operatorname{tr}\Big(S(I_k - Q_mQ_m')\Big) = (n-1)\sum_{j>m} q_j'Sq_j = (n-1)\sum_{j>m}\lambda_j$$

(f) $X_cP_{W_m} = X_cQ_mQ_m' = Z_mQ_m'$, whose row i is $\sum_{j\leqslant m} z_{ij}q_j'$. By (c), $\|X_c\|_F^2 = \operatorname{tr}(X_c'X_c) = (n-1)\sum_j\lambda_j$, and by (e) $\|X_c - Z_mQ_m'\|_F^2 = (n-1)\sum_{j>m}\lambda_j$; so the ratio is $\sum_{j>m}\lambda_j / \sum_j \lambda_j$. □

Approximating each $x_i - \bar{x}$ by its projection onto W_m gives, by Theorem 10.22(f), $x_i \approx \bar{x} + \sum_{j\leqslant m} z_{ij}q_j$. In practice m is chosen where the λ_j become small compared with the leading ones.

Example 10.23. Three variables are measured on five units: $x_1 = (1,7,1)'$, $x_2 = (5,5,3)'$, $x_3 = (5,5,1)'$, $x_4 = (7,1,1)'$, $x_5 = (2,2,4)'$. Find the principal components, the proportions of variance explained, and the best approximation of the data by one component.

Solution.

- $\bar{x} = (4, 4, 2)'$, and the rows of X_c are $(-3, 3, -1)$, $(1, 1, 1)$, $(1, 1, -1)$, $(3, -3, -1)$, $(-2, -2, 2)$. So

$$X_c'X_c = \begin{pmatrix} 24 & -12 & -4 \\ -12 & 24 & -4 \\ -4 & -4 & 8 \end{pmatrix}, \qquad S = \frac{1}{4}X_c'X_c = \begin{pmatrix} 6 & -3 & -1 \\ -3 & 6 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

- $S(1, -1, 0)' = 9(1, -1, 0)'$, $S(1, 1, -1)' = 4(1, 1, -1)'$ and $S(1, 1, 2)' = (1, 1, 2)'$, and these three vectors are orthogonal. So $\lambda_1 = 9$, $\lambda_2 = 4$, $\lambda_3 = 1$, and

$$q_1 = \tfrac{1}{\sqrt{2}}(1, -1, 0)', \qquad q_2 = \tfrac{1}{\sqrt{3}}(1, 1, -1)', \qquad q_3 = \tfrac{1}{\sqrt{6}}(1, 1, 2)',$$

  each determined up to its sign, as the λ_j are simple roots of p_S. The total variance is $\operatorname{tr} S = 14$; the first component explains $9/14$ of it, the first two $13/14$. The first component contrasts the first two variables; the second compares their sum with the third.

- The scores $z_{ij} = q_j'(x_i - \bar{x})$ are

$$(z_{i1})_i = (-3\sqrt{2},\, 0,\, 0,\, 3\sqrt{2},\, 0), \qquad (z_{i2})_i = \tfrac{1}{\sqrt{3}}(1, 1, 3, 1, -6), \qquad (z_{i3})_i = \tfrac{1}{\sqrt{6}}(-2, 4, 0, -2, 0)$$

  Each has sum 0; $\sum_i z_{i1}^2 = 36 = 4\lambda_1$, $\sum_i z_{i2}^2 = 16 = 4\lambda_2$, $\sum_i z_{i3}^2 = 4 = 4\lambda_3$, and $\sum_i z_{i1}z_{i2} = \sum_i z_{i1}z_{i3} = \sum_i z_{i2}z_{i3} = 0$, as Theorem 10.22(a) says.

- With one component, $x_i \approx \bar{x} + z_{i1}q_1 = \bar{x} + \frac{1}{2}(x_{i1} - x_{i2})(1, -1, 0)'$, which gives $(1, 7, 2)'$, $(4, 4, 2)'$, $(4, 4, 2)'$, $(7, 1, 2)'$, $(4, 4, 2)'$. The squared distance from x_i to the nearest point $\bar{x} + P_W(x_i - \bar{x})$ of a line $\bar{x} + W$ through $\bar{x}$ is $\|(x_i - \bar{x}) - P_W(x_i - \bar{x})\|^2$ (Theorem 6.15). By Theorem 10.22(e), the line $\{\bar{x} + tq_1 : t \in \mathbb{R}\}$ minimizes the sum of these squared distances among all lines through $\bar{x}$. Here the errors are $(0, 0, -1)'$, $(1, 1, 1)'$, $(1, 1, -1)'$, $(0, 0, -1)'$, $(-2, -2, 2)'$, with squared lengths adding up to $1 + 3 + 3 + 1 + 12 = 20 = (n - 1)(\lambda_2 + \lambda_3)$, out of $\sum_i \|x_i - \bar{x}\|^2 = 56 = (n - 1)\operatorname{tr} S$.

- By Theorem 10.22(d), X_c has the singular values $\sqrt{4 \cdot 9} = 6$, $\sqrt{4 \cdot 4} = 4$ and $\sqrt{4 \cdot 1} = 2$.

Example 10.24. Principal components depend on the units of measurement: a variable with a large variance dominates q_1. Therefore the variables are often **standardized** first, $y_{ij} = (x_{ij} - \bar{x}_j)/\sqrt{s_{jj}}$ (all $s_{jj} > 0$). The standardized data matrix $Y = X_cD^{-1/2}$, $D = \operatorname{diag}(s_{11}, \ldots, s_{kk})$, is centered, and its sample covariance matrix is $(n - 1)^{-1}D^{-1/2}X_c'X_cD^{-1/2} = D^{-1/2}SD^{-1/2} = R$, the sample correlation matrix (Definition 8.32); so standardized principal components are computed from R, whose total variance is $\operatorname{tr} R = k$. For $k = 2$ and $r = r_{12}$, $R = \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$ has $R(1, 1)' = (1 + r)(1, 1)'$ and $R(1, -1)' = (1 - r)(1, -1)'$. If $r > 0$, then $\lambda_1 = 1 + r$, $q_1 = \frac{1}{\sqrt{2}}(1, 1)'$, $\lambda_2 = 1 - r$, $q_2 = \frac{1}{\sqrt{2}}(1, -1)'$; if $r < 0$, the two pairs are interchanged; if $r = 0$, then $R = I_2$ and every orthonormal basis gives loading vectors. The first component explains $(1 + |r|)/2$ of the total variance 2.

Exercises

Exercise 10.1. Let $A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \\ 3 & 3 \\ 1 & 1 \end{pmatrix}$.

(a) Find a singular value decomposition $A = U\Sigma V'$ with $U \in \mathbb{R}^{4\times 4}$, and the compact form.

(b) Give orthonormal bases of $\mathcal{C}(A)$, $\mathcal{N}(A')$, $\mathcal{C}(A')$ and $\mathcal{N}(A)$.

(c) Find $\|A\|_2$, $\|A\|_F$, $\kappa(A)$ and $\kappa(A'A)$.

(d) Find A^+ from the singular value decomposition, and check it with $(A'A)^{-1}A'$.

Solution. (a) $A'A = \left(\begin{smallmatrix} 20 & 16 \\ 16 & 20 \end{smallmatrix}\right)$, with $A'A(1,1)' = 36(1,1)'$ and $A'A(1,-1)' = 4(1,-1)'$. As in the proof of Theorem 10.1: $\sigma_1 = 6$, $\sigma_2 = 2$, $v_1 = \frac{1}{\sqrt{2}}(1,1)'$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)'$, $u_1 = \frac{1}{6}Av_1 = \frac{1}{6\sqrt{2}}(4,4,6,2)' = \frac{1}{3\sqrt{2}}(2,2,3,1)'$ and $u_2 = \frac{1}{2}Av_2 = \frac{1}{\sqrt{2}}(-1,1,0,0)'$. A vector y orthogonal to u_1 and u_2 satisfies $y_1 = y_2$ and $4y_1+3y_3+y_4 = 0$; the vectors $(0,0,1,-3)'$ and $(5,5,-6,-2)'$ satisfy both and are orthogonal, with lengths $\sqrt{10}$ and $3\sqrt{10}$. Take $u_3 = \frac{1}{\sqrt{10}}(0,0,1,-3)'$ and $u_4 = \frac{1}{3\sqrt{10}}(5,5,-6,-2)'$. Then $A = U\Sigma V'$ with $U = [u_1\ u_2\ u_3\ u_4]$, $\Sigma = \left(\begin{smallmatrix} 6 & 0 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $V = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and

$$A = 6u_1v_1' + 2u_2v_2' = \begin{pmatrix} 2 & 2 \\ 2 & 2 \\ 3 & 3 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

(b) By Corollary 10.2(b): u_1, u_2 for $\mathcal{C}(A)$; u_3, u_4 for $\mathcal{N}(A')$; v_1, v_2 for $\mathcal{C}(A') = \mathbb{R}^2$; the empty basis for $\mathcal{N}(A) = \{0\}$.

(c) $\|A\|_2 = 6$; $\|A\|_F^2 = 10 + 10 + 18 + 2 = 40 = 6^2 + 2^2$, so $\|A\|_F = 2\sqrt{10}$; $\kappa(A) = 6/2 = 3$ and $\kappa(A'A) = 9$ (Definition 10.6).

(d) By Theorem 10.15(b),

$$A^+ = \frac{1}{6}v_1u_1' + \frac{1}{2}v_2u_2' = \frac{1}{36}\begin{pmatrix} 2 & 2 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} -7 & 11 & 3 & 1 \\ 11 & -7 & 3 & 1 \end{pmatrix}$$

As $\operatorname{rank} A = 2$, Theorem 10.15(c) gives, with $(A'A)^{-1} = \frac{1}{144}\left(\begin{smallmatrix} 20 & -16 \\ -16 & 20 \end{smallmatrix}\right) = \frac{1}{36}\left(\begin{smallmatrix} 5 & -4 \\ -4 & 5 \end{smallmatrix}\right)$, $A^+ = \frac{1}{36}\left(\begin{smallmatrix} 5 & -4 \\ -4 & 5 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 3 & 3 & 1 \\ 3 & 1 & 3 & 1 \end{smallmatrix}\right) = \frac{1}{36}\left(\begin{smallmatrix} -7 & 11 & 3 & 1 \\ 11 & -7 & 3 & 1 \end{smallmatrix}\right)$.

Exercise 10.2. Let $A = \begin{pmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 4 & 2 & 4 \\ 8 & 4 & 8 \end{pmatrix}$.

(a) Write $A = uv'$ and find a compact singular value decomposition, $\|A\|_2$ and $\|A\|_F$.

(b) Find A^+, AA^+ and A^+A.

(c) Decide whether $Ax = b$ is consistent for $b = (3,6,6,12)'$ and for $b = e_1$. Find the solution of minimum length in the consistent case, and compare it with the solution $(0,3,0)'$.

Solution. (a) $A = uv'$ with $u = (1,2,2,4)'$, $\|u\| = 5$, and $v = (2,1,2)'$, $\|v\| = 3$. So $A = 15\,(u/5)(v/3)'$, and by Corollary 10.2(e) with $s = 1$, $\operatorname{rank} A = 1$, $\sigma_1 = 15$, $u_1 = u/5$, $v_1 = v/3$. By Theorem 10.5(a), $\|A\|_2 = \|A\|_F = 15$; indeed $\|A\|_F^2 = 9 + 36 + 36 + 144 = 225$.

(b) By Theorem 10.15(b) and (d),

$$A^+ = \frac{1}{15}v_1u_1' = \frac{1}{225}vu' = \frac{1}{225}\begin{pmatrix} 2 & 4 & 4 & 8 \\ 1 & 2 & 2 & 4 \\ 2 & 4 & 4 & 8 \end{pmatrix}, \qquad AA^+ = u_1u_1' = \frac{1}{25}uu', \qquad A^+A = v_1v_1' = \frac{1}{9}vv'$$

(c) A^+ is a g-inverse, so by Theorem 10.11(a), $Ax = b$ is consistent if and only if $AA^+b = \frac{u'b}{25}u$ equals b. For $b = (3,6,6,12)' = 3u$ it does; for $b = e_1$, $AA^+e_1 = \frac{1}{25}u \neq e_1$, and there is no solution. By Theorem 10.16, the shortest solution for $b = 3u$ is $x^+ = A^+b = \frac{u'b}{225}v = \frac{75}{225}v = \frac{1}{3}(2,1,2)'$, with $\|x^+\|^2 = 1$. The solution $x = (0,3,0)'$ ($Ax = 3\operatorname{col}_2(A) = 3u$) has $\|x\|^2 = 9 = 1 + \|\frac{1}{3}(-2,8,-2)'\|^2 = 1 + 8$.

Exercise 10.3. Four stores sell three products; the sales (in hundreds) are

$$B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 6 & 2 \\ 0 & 2 & 7 \\ 4 & 8 & 8 \end{pmatrix}$$

(row i: store i; column j: product j). Let $w_1 = (1,2,2)'$, $w_2 = (2,1,-2)'$ and $w_3 = (2,-2,1)'$.

(a) Compute Bw_1, Bw_2, Bw_3, and deduce the singular values and a compact singular value decomposition of B.

(b) Find the best approximation B_1 of B by a matrix of rank at most 1, the errors $\|B - B_1\|_2$ and $\|B - B_1\|_F$, and the fraction of $\|B\|_F^2$ that B_1 reproduces. Interpret B_1.

(c) Find the best approximation B_2 of rank at most 2 and its errors.

Solution. (a) $Bw_1 = (9,18,18,36)' = 9(1,2,2,4)'$, $Bw_2 = (12,6,-12,0)' = 6(2,1,-2,0)'$ and $Bw_3 = (6,-6,3,0)' = 3(2,-2,1,0)'$. The w_j are orthogonal with length 3, so the $v_j = \frac{1}{3}w_j$ form an orthogonal matrix V. The vectors $x_1 = \frac{1}{5}(1,2,2,4)'$, $x_2 = \frac{1}{3}(2,1,-2,0)'$, $x_3 = \frac{1}{3}(2,-2,1,0)'$ are orthonormal (lengths 1; inner products $\frac{1}{15}(2+2-4)$, $\frac{1}{15}(2-4+2)$, $\frac{1}{9}(4-2-2)$, all 0), and $Bv_1 = 15x_1$, $Bv_2 = 6x_2$, $Bv_3 = 3x_3$. Hence, by the outer product expansion,

$$B = BVV' = \sum_j (Bv_j)v_j' = 15x_1v_1' + 6x_2v_2' + 3x_3v_3',$$

and by Corollary 10.2(e) the singular values are $15, 6, 3$, with $u_j = x_j$.

(b) By Theorem 10.18,

$$B_1 = 15x_1v_1' = (1,2,2,4)'(1,2,2) = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \\ 4 & 8 & 8 \end{pmatrix}, \qquad B - B_1 = \begin{pmatrix} 4 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

$\|B - B_1\|_2 = 6$ and $\|B - B_1\|_F^2 = 36 + 9 = 45$, so $\|B - B_1\|_F = 3\sqrt{5}$; no matrix of rank at most 1 comes closer in either norm. As $\|B\|_F^2 = 225 + 36 + 9 = 270$ (the sum of the squared entries), B_1 reproduces $\frac{225}{270} = \frac{5}{6}$ of it. The entries of B_1 are products a_ib_j with $a = (1,2,2,4)'$ and $b = (1,2,2)'$: among all tables of the form "store effect times product effect" ab', B_1 has the smallest sum of squared differences from B; store 4 follows it exactly.

(c) By Theorem 10.18,

$$B_2 = B_1 + 6x_2v_2' = B_1 + \frac{2}{3}\begin{pmatrix} 2 \\ 1 \\ -2 \\ 0 \end{pmatrix}(2,1,-2) = \frac{1}{3}\begin{pmatrix} 11 & 10 & -2 \\ 10 & 14 & 8 \\ -2 & 8 & 20 \\ 12 & 24 & 24 \end{pmatrix},$$

with $B - B_2 = 3x_3v_3'$, $\|B - B_2\|_2 = 3$ and $\|B - B_2\|_F^2 = 9$; B_2 reproduces $\frac{261}{270} = \frac{29}{30}$ of $\|B\|_F^2$.

Exercise 10.4. Let

$$A = \begin{pmatrix} 1 & 4 & 0 & 2 \\ 4 & 1 & 2 & 0 \\ 0 & 2 & 1 & 4 \\ 2 & 0 & 4 & 1 \end{pmatrix}, \qquad h_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix},\ h_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix},\ h_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix},\ h_4 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

(a) Show that $Ah_1 = 7h_1$, $Ah_2 = -5h_2$, $Ah_3 = 3h_3$ and $Ah_4 = -h_4$.

(b) Find a singular value decomposition of A. Why is $U \neq V$?

(c) Find $\|A\|_2$, $\|A\|_F$ and $\kappa(A)$.

Solution. (a) $Ah_1 = (7,7,7,7)'$ (every row sum is 7), $Ah_2 = (-5,5,-5,5)'$, $Ah_3 = (3,3,-3,-3)'$ and $Ah_4 = (-1,1,1,-1)'$.

(b) The h_k are orthogonal with length 2, so $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ is orthogonal and $A = Q\Lambda Q'$, $\Lambda = \operatorname{diag}(7,-5,3,-1)$ (from $AQ = Q\Lambda$). Then $A'A = A^2 = Q\Lambda^2 Q'$, so $A'Av_k = \lambda_k^2 v_k$ for $v_k = \frac{1}{2}h_k$, with $\lambda_k^2 = 49, 25, 9, 1$ in decreasing order. As in the proof of Theorem 10.1, $\sigma_1 = 7$, $\sigma_2 = 5$, $\sigma_3 = 3$, $\sigma_4 = 1$, $V = Q$, and $u_k = \sigma_k^{-1} A v_k = (\lambda_k/|\lambda_k|)v_k$:

$$U = \frac{1}{2}[h_1\ \ -h_2\ \ h_3\ \ -h_4], \qquad \Sigma = \operatorname{diag}(7,5,3,1), \qquad V = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$$

The columns of U and V differ in sign exactly where the eigenvalue is negative.

(c) $\|A\|_2 = 7$; $\|A\|_F^2 = 4 \cdot 21 = 84 = 49 + 25 + 9 + 1$; $\operatorname{rank} A = 4$ and $\kappa(A) = 7$.

Exercise 10.5. Let A be the matrix of Exercise 10.4.

(a) Find the largest value of $\operatorname{tr}(Q'A)$ over the orthogonal $Q \in \mathbb{R}^{4\times 4}$, and an orthogonal matrix Q_0 at which it is attained. Compare with $Q = I_4$.

(b) Find an orthogonal matrix nearest to A in the Frobenius norm, and the distance. Compare with $\|I_4 - A\|_F$.

Solution. (a) By Theorem 10.20(a) and Exercise 10.4, the largest value is $7+5+3+1 = 16$, attained at

$$Q_0 = UV' = \sum_{k=1}^{4} u_k v_k' = \frac{1}{4}\big(h_1h_1' - h_2h_2' + h_3h_3' - h_4h_4'\big) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

(the (i,j) entry is $\frac{1}{4}$ times the sum of the products of the entries i and j of the h_k, with signs $+,-,+,-$). Indeed $\operatorname{tr}(Q_0'A) = a_{21} + a_{12} + a_{43} + a_{34} = 16$, whereas $\operatorname{tr}(I_4'A) = \operatorname{tr} A = 4$.

(b) Theorem 10.20(b) with $B = I_4$ and $C = A$ (so $CB' = A$) gives $\|Q - A\|_F^2 \geqslant \|I_4\|_F^2 + \|A\|_F^2 - 2\cdot 16 = 4 + 84 - 32 = 56$ for every orthogonal Q, with equality at $Q = Q_0$. So Q_0 is nearest, at distance $\sqrt{56} = 2\sqrt{14}$; directly, $Q_0 - A$ has four entries -1, four entries -3 and four entries -2, and $4 + 36 + 16 = 56$. By comparison, $\|I_4 - A\|_F^2 = 4 \cdot 16 + 4 \cdot 4 = 80$.

Exercise 10.6. Let $A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$.

(a) Show that $\operatorname{rank} A = 2$. Find a g-inverse G of A from its leading 2×2 submatrix, and verify $AGA = A$.

(b) Compute GA and $I_4 - GA$, and read off a basis of $\mathcal{N}(A)$.

(c) Use G to decide whether $Ax = b$ is consistent for $b = (2, 3, 5)'$ and for $c = (1, 0, 0)'$, and give all solutions when it is.

Solution. (a) $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$, and the first two rows are not proportional, so $\operatorname{rank} A = 2$ (Theorem 4.13(a),(c)). The leading submatrix $M = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 3 \end{smallmatrix}\right)$ has determinant 1 and $M^{-1} = \left(\begin{smallmatrix} 3 & -2 \\ -1 & 1 \end{smallmatrix}\right)$ (check: $MM^{-1} = I_2$). By Theorem 10.9(b),

$$G = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad AG = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

(column 1 of AG is $3a_1 - a_2 = (1, 0, 1)'$, column 2 is $-2a_1 + a_2 = (0, 1, 1)'$). The rows of AGA are $\operatorname{row}_1(A)$, $\operatorname{row}_2(A)$ and $\operatorname{row}_1(A) + \operatorname{row}_2(A) = \operatorname{row}_3(A)$; so $AGA = A$.

(b) The rows of GA are $3\operatorname{row}_1(A) - 2\operatorname{row}_2(A) = (1, 0, 3, -1)$, $-\operatorname{row}_1(A) + \operatorname{row}_2(A) = (0, 1, -1, 2)$ and two zero rows, so

$$I_4 - GA = \begin{pmatrix} 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

By Theorem 10.11(c), $\mathcal{N}(A)$ is spanned by the columns $(-3, 1, 1, 0)'$ and $(1, -2, 0, 1)'$, which are independent (entries 3 and 4); they form a basis.

(c) For every $b = (b_1, b_2, b_3)'$, $AGb = (b_1,\ b_2,\ b_1 + b_2)'$; so $Ax = b$ is consistent if and only if $b_3 = b_1 + b_2$ (Theorem 10.11(a)). For $b = (2, 3, 5)'$ it is, with $Gb = (0, 1, 0, 0)'$, and by Theorem 10.11(b) the solutions are

$$x = (0, 1, 0, 0)' + z_3(-3, 1, 1, 0)' + z_4(1, -2, 0, 1)' \qquad (z_3, z_4 \in \mathbb{R})$$

(check: $A(0, 1, 0, 0)' = a_2 = (2, 3, 5)'$). For $c = (1, 0, 0)'$, $AGc = (1, 0, 1)' \neq c$: no solution.

Exercise 10.7. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 1 & 2 & -1 \\ 1 & -2 & 3 \end{pmatrix}$.

(a) Compute $A'A$, show that $\operatorname{rank} A = 2$, and find a g-inverse G of $A'A$ from its leading 2×2 submatrix.

(b) Verify $AGA'A = A$, and write down the g-inverse GA' of A.

(c) Use GA' to decide whether $Ax = b$ is consistent for $b = (2, 2, 2, 2)'$ and for $b = e_1$. In the consistent case find all solutions and the one of minimum length.

Solution. (a) The columns are $a_1 = (1, 1, 1, 1)'$, $a_2 = (1, -1, 2, -2)'$ and $a_3 = (0, 2, -1, 3)' = a_1 - a_2$, and

$$A'A = \begin{pmatrix} 4 & 0 & 4 \\ 0 & 10 & -10 \\ 4 & -10 & 14 \end{pmatrix}$$

$a_1 \perp a_2$ are nonzero, hence independent (Proposition 6.8(b)), and $a_3 \in \operatorname{span}\{a_1, a_2\}$; so $\operatorname{rank} A = 2$, and $\operatorname{rank}(A'A) = 2$ (Theorem 4.18(b)). The leading submatrix $\operatorname{diag}(4, 10)$ is invertible, so $G = \operatorname{diag}(\frac{1}{4}, \frac{1}{10}, 0)$ is a g-inverse of $A'A$ (Theorem 10.9(b)).

(b) $GA'A = G(A'A) = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, so the columns of $A(GA'A)$ are a_1, a_2, $a_1 - a_2 = a_3$: $AGA'A = A$, as Lemma 10.12(a) says. By Lemma 10.12(b),

$$GA' = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{10} & -\frac{1}{10} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is a g-inverse of A.

(c) $A(GA')b = \frac{1}{4}(a_1'b)a_1 + \frac{1}{10}(a_2'b)a_2$. For $b = (2, 2, 2, 2)' = 2a_1$, $a_1'b = 8$ and $a_2'b = 0$, so $AGA'b = 2a_1 = b$: consistent. For $b = e_1$, $AGA'e_1 = \frac{1}{4}a_1 + \frac{1}{10}a_2 = \frac{1}{20}(7, 3, 9, 1)' \neq e_1$: inconsistent (Theorem 10.11(a)). For $b = 2a_1$, $GA'b = (2, 0, 0)'$ and $I_3 - GA'A = \left(\begin{smallmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, so the solutions are $x = (2, 0, 0)' + z(-1, 1, 1)'$, $z \in \mathbb{R}$ (Theorem 10.11(b)). By Theorem 10.16(b), the shortest one lies in $\mathcal{C}(A') = \mathcal{N}(A)^\perp$ (Corollary 6.6), where $\mathcal{N}(A) = \operatorname{span}\{(-1, 1, 1)'\}$: $(2-z)(-1)+z+z = 0$ gives $z = \frac{2}{3}$ and $x^+ = \frac{1}{3}(4, 2, 2)'$, with $\|x^+\|^2 = \frac{8}{3}$; the solution $(2, 0, 0)'$ has $\|(2, 0, 0)'\|^2 = 4 = \frac{8}{3} + \frac{4}{3}$.

Exercise 10.8. Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}, \qquad f_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},\ f_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},\ g_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix},\ g_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

(a) Show that $A = f_1 g_1' + f_2 g_2'$, and find $\operatorname{rank} A$, the singular values and a compact singular value decomposition.

(b) Find A^+ and AA^+.

(c) Find the condition on $b \in \mathbb{R}^3$ for $Ax = b$ to be consistent. For $b = (8, 4, 6)'$ find the solution of minimum length, and compare it with the solution $(6, 2, 0, 0, 0)'$.

Solution. (a) $f_1 g_1'$ has the three rows $(1, 0, 1, 0, 1)$, and $f_2 g_2'$ the rows $(0, 1, 0, 1, 0)$, $(0, -1, 0, -1, 0)$, 0; their sum is A. As $f_1 \perp f_2$, $g_1 \perp g_2$, $\|f_1\|\|g_1\| = \sqrt{3}\sqrt{3} = 3$ and $\|f_2\|\|g_2\| = \sqrt{2}\sqrt{2} = 2$,

$$A = 3\,\frac{f_1}{\sqrt{3}}\Big(\frac{g_1}{\sqrt{3}}\Big)' + 2\,\frac{f_2}{\sqrt{2}}\Big(\frac{g_2}{\sqrt{2}}\Big)'$$

is of the form of Corollary 10.2(e): $\operatorname{rank} A = 2$, $\sigma_1 = 3$, $\sigma_2 = 2$, $u_1 = f_1/\sqrt{3}$, $v_1 = g_1/\sqrt{3}$, $u_2 = f_2/\sqrt{2}$, $v_2 = g_2/\sqrt{2}$.

(b) By Theorem 10.15(b),(d),

$$A^+ = \frac{1}{3}v_1u_1' + \frac{1}{2}v_2u_2' = \frac{1}{9}g_1f_1' + \frac{1}{4}g_2f_2' = \frac{1}{36}\begin{pmatrix} 4 & 4 & 4 \\ 9 & -9 & 0 \\ 4 & 4 & 4 \\ 9 & -9 & 0 \\ 4 & 4 & 4 \end{pmatrix}$$

and $AA^+ = \frac{1}{3}f_1f_1' + \frac{1}{2}f_2f_2' = \frac{1}{6}\left(\begin{smallmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{smallmatrix}\right)$.

(c) $Ax = b$ is consistent if and only if $b \in \mathcal{C}(A) = \operatorname{span}\{f_1, f_2\}$ (Corollary 10.2(b)). The vector $(1, 1, -2)'$ is orthogonal to f_1 and f_2, so it lies in $\mathcal{C}(A)^\perp$, which has dimension $3-2 = 1$ (Theorem 6.5(c)); so $\mathcal{C}(A)^\perp = \operatorname{span}\{(1, 1, -2)'\}$, and $\mathcal{C}(A) = \{b : b_1 + b_2 - 2b_3 = 0\}$ (Proposition 6.17). For $b = (8, 4, 6)'$, $8 + 4 = 12$: consistent. By Theorem 10.16, $x^+ = A^+b = \frac{1}{36}(72, 36, 72, 36, 72)' = (2, 1, 2, 1, 2)'$, with $\|x^+\|^2 = 14$. The solution $x = (6, 2, 0, 0, 0)'$ ($Ax = 6a_1 + 2a_2 = (8, 4, 6)'$) has $\|x\|^2 = 40 = 14 + \|(4, 1, -2, -1, -2)'\|^2 = 14 + 26$.

Exercise 10.9. Three variables are measured on five units: $(5,7,1)$, $(5,3,5)$, $(1,5,7)$, $(3,4,4)$, $(1,1,3)$.

(a) Find $\bar{x}$, the centered data matrix X_c and S.

(b) Show that $(1,1,-1)'$, $(0,1,1)'$ and $(2,-1,1)'$ are eigenvectors of S, and give the loading vectors, the variances of the components and the proportions of variance explained by the first component and by the first two.

(c) Compute the scores, and check Theorem 10.22(a) for them.

(d) Find the sum over the units of the squared errors of the approximation by two components, and the singular values of X_c.

Solution. (a) $\bar{x} = (3,4,4)'$; the rows of X_c are $(2,3,-3)$, $(2,-1,1)$, $(-2,1,3)$, $(0,0,0)$, $(-2,-3,-1)$, and

$$X_c'X_c = \begin{pmatrix} 16 & 8 & -8 \\ 8 & 20 & -4 \\ -8 & -4 & 20 \end{pmatrix}, \qquad S = \frac{1}{4}X_c'X_c = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 5 & -1 \\ -2 & -1 & 5 \end{pmatrix}$$

(b) $S(1,1,-1)' = (8,8,-8)'$, $S(0,1,1)' = (0,4,4)'$ and $S(2,-1,1)' = (4,-2,2)'$. The three vectors are orthogonal, so $\lambda_1 = 8$, $\lambda_2 = 4$, $\lambda_3 = 2$, with $q_1 = \frac{1}{\sqrt{3}}(1,1,-1)'$, $q_2 = \frac{1}{\sqrt{2}}(0,1,1)'$, $q_3 = \frac{1}{\sqrt{6}}(2,-1,1)'$. The total variance is $\operatorname{tr} S = 14$; the first component explains $8/14 = 4/7$, the first two $12/14 = 6/7$.

(c) $z_{ij} = q_j'(x_i - \bar{x})$:

$$(z_{i1})_i = \tfrac{1}{\sqrt{3}}(8,0,-4,0,-4), \qquad (z_{i2})_i = \tfrac{1}{\sqrt{2}}(0,0,4,0,-4), \qquad (z_{i3})_i = \tfrac{1}{\sqrt{6}}(-2,6,-2,0,-2)$$

Each has sum 0; the sums of squares are $\frac{96}{3} = 32 = 4\lambda_1$, $\frac{32}{2} = 16 = 4\lambda_2$ and $\frac{48}{6} = 8 = 4\lambda_3$; and the sums of products of two different columns are $\frac{1}{\sqrt{6}}(-16+16) = 0$, $\frac{1}{\sqrt{18}}(-16+8+8) = 0$ and $\frac{1}{\sqrt{12}}(-8+8) = 0$.

(d) By Theorem 10.22(e),(f), the sum is $(n-1)\lambda_3 = 8 = \sum_i z_{i3}^2$, out of $(n-1)\operatorname{tr} S = 56$. By Theorem 10.22(d), the singular values of X_c are $\sqrt{4\cdot 8} = 4\sqrt{2}$, $\sqrt{4\cdot 4} = 4$ and $\sqrt{4\cdot 2} = 2\sqrt{2}$.

Problems

Problem 10.1. Verify the Penrose conditions to show:

(a) $\operatorname{diag}(d_1,\dots,d_n)^+ = \operatorname{diag}(d_1^+,\dots,d_n^+)$, where $d^+ = 1/d$ for $d \neq 0$ and $0^+ = 0$;

(b) $P^+ = P$ for every symmetric idempotent P;

(c) $A^+ = A'$ if $A'A = I$;

(d) $(uv')^+ = vu'/(\|u\|^2\|v\|^2)$ for nonzero $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$;

(e) $(AB)^+ \neq B^+A^+$ for $A = (1 \;\; 1)$ and $B = (1 \;\; 0)'$.

Solution. (a) For $D = \operatorname{diag}(d_i)$ and $G = \operatorname{diag}(d_i^+)$, $DG = GD = \operatorname{diag}(d_i d_i^+)$ is symmetric, and $dd^+d = d$, $d^+dd^+ = d^+$ for every real d; so $DGD = D$ and $GDG = G$.

(b) With $G = P$: $PPP = P$, and $PP = P$ is symmetric.

(c) With $G = A'$: $AA'A = A$, $A'AA' = A'$, and AA' and $A'A = I$ are symmetric.

(d) Let $c = \|u\|^2\|v\|^2$ and $G = vu'/c$. Then $AG = uv'vu'/c = uu'/\|u\|^2$ and $GA = vu'uv'/c = vv'/\|v\|^2$ are symmetric, $AGA = uu'uv'/\|u\|^2 = uv'$, and $GAG = vv'vu'/(\|v\|^2 c) = G$.

(e) $AB = (1)$, so $(AB)^+ = 1$. By (d) and (c), $A^+ = \frac{1}{2}(1,1)'$ and $B^+ = (1 \;\; 0)$, so $B^+A^+ = \frac{1}{2} \neq 1$.

Problem 10.2. Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ listed with multiplicity.

(a) Show that the singular values of A are the numbers $|\lambda_i|$ with $\lambda_i \neq 0$, in decreasing order, and that $\|A\|_2 = \max_i |\lambda_i|$.

(b) Show that if A is positive semidefinite, a spectral decomposition $A = Q\Lambda Q'$ with $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ is a singular value decomposition.

(c) Show that $\|A\|_2 = \max_{x\neq 0} |x'Ax|/x'x$.

Solution. (a) By Theorem 8.4, $A = Q\Lambda Q'$ with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_i)$. Then $A'A = A^2 = Q\Lambda^2 Q'$ is similar to Λ^2, so its eigenvalues listed with multiplicity are $\lambda_1^2, \dots, \lambda_n^2$ (Theorems 7.14(a) and 5.15(a)). By Corollary 10.2(a), the singular values are the square roots $\sqrt{\lambda_i^2} = |\lambda_i|$ of the nonzero ones, in decreasing order, and $\|A\|_2 = \sigma_1(A) = \max_i |\lambda_i|$ (Theorem 10.5(a)).

(b) $\lambda_n \geqslant 0$ (Theorem 8.13(b)), so with $r = \operatorname{rank} A$, the number of nonzero λ_i (Corollary 8.5(b)), $\lambda_1 \geqslant \cdots \geqslant \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$, and $A = Q\Lambda Q'$ is of the form of Theorem 10.1 with $U = V = Q$ and $\Sigma = \Lambda$.

(c) Order the eigenvalues as $\lambda_1 \geqslant \cdots \geqslant \lambda_n$. By Theorem 9.29(a), $\lambda_n \leqslant x'Ax/x'x \leqslant \lambda_1$ for $x \neq 0$, with the values λ_1 and λ_n attained at eigenvectors. So the largest value of $|x'Ax|/x'x$ is $\max\{|\lambda_1|, |\lambda_n|\} = \max_i |\lambda_i|$, which is $\|A\|_2$ by (a).

Problem 10.3. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. Show that:

(a) $\|AB\|_2 \leqslant \|A\|_2 \|B\|_2$;

(b) $\|AB\|_F \leqslant \|A\|_2 \|B\|_F$;

(c) $\|A\|_2 \leqslant \|A\|_F \leqslant \sqrt{\operatorname{rank} A}\, \|A\|_2$, with equality on the left if and only if $\operatorname{rank} A \leqslant 1$, and on the right if and only if all nonzero singular values of A are equal;

(d) $\|A'\|_2 = \|A\|_2$ and $\|A'A\|_2 = \|A\|_2^2$.

Solution. (a) By Theorem 10.5(a),(b), $\|ABx\| \leqslant \|A\|_2 \|Bx\| \leqslant \|A\|_2 \|B\|_2 \|x\|$ for every x; divide by $\|x\|$ ($x \neq 0$) and take the maximum.

(b) Let $b_1, \dots, b_p$ be the columns of B. Column j of AB is Ab_j (Theorem 1.4(a)), and $\|M\|_F^2$ is the sum of the squared lengths of the columns of M. So $\|AB\|_F^2 = \sum_j \|Ab_j\|^2 \leqslant \|A\|_2^2 \sum_j \|b_j\|^2 = \|A\|_2^2 \|B\|_F^2$.

(c) Let $r = \operatorname{rank} A$ and $\sigma_i = \sigma_i(A)$. By Theorem 10.5(a), $\|A\|_F^2 = \sum_{i\leqslant r} \sigma_i^2$, which is at least $\sigma_1^2 = \|A\|_2^2$, with equality if and only if $\sigma_2 = \cdots = \sigma_r = 0$, that is, $r \leqslant 1$; and at most $r\sigma_1^2$, with equality if and only if $\sigma_i = \sigma_1$ for all $i \leqslant r$. The left inequality also follows from Lemma 9.4: $\|Ax\|/\|x\| \leqslant \|A\|_F$ for every $x \neq 0$.

(d) A' has the singular values of A (Corollary 10.2(c)). $A'A$ is positive semidefinite (Theorem 8.16(a)) with the nonzero eigenvalues $\sigma_i(A)^2$ (Corollary 10.2(a)), so by Problem 10.2(b) its largest singular value is $\sigma_1(A)^2$ (also for $A = 0$).

Problem 10.4. Let $A \in \mathbb{R}^{n\times n}$.

(a) Show that $A = WP$ with W orthogonal and $P = (A'A)^{1/2}$ (Theorem 8.25), the **polar decomposition** of A; with $A = U\Sigma V'$, one may take $W = UV'$.

(b) Show that if A is invertible and $A = W_1 P_1$ with W_1 orthogonal and P_1 positive semidefinite, then $P_1 = (A'A)^{1/2}$ and $W_1 = AP_1^{-1}$.

(c) Find the polar decomposition of the matrix A of Exercise 10.4.

Solution. (a) If $A = 0$, take $W = I_n$ and $P = 0$. Otherwise let $A = U\Sigma V'$ (Theorem 10.1); Σ is square, diagonal, with nonnegative entries. $W = UV'$ is orthogonal, since $(UV')'UV' = VU'UV' = I_n$. $P = V\Sigma V'$ is symmetric, positive semidefinite ($x'Px = \sum_i \Sigma_{ii}(v_i'x)^2 \geqslant 0$), and

$P^2 = V\Sigma^2V' = V\Sigma'U'U\Sigma V' = A'A$; so $P = (A'A)^{1/2}$ by the uniqueness in Theorem 8.25. Finally $WP = UV'V\Sigma V' = U\Sigma V' = A$.

(b) $A'A = P_1W_1'W_1P_1 = P_1^2$, so $P_1 = (A'A)^{1/2}$ by the uniqueness in Theorem 8.25. As $(\det P_1)^2 = \det(A'A) = (\det A)^2 \neq 0$ (Theorems 5.19 and 5.15(b)), P_1 is invertible, and $W_1 = AP_1^{-1}$.

(c) With the singular value decomposition of Exercise 10.4, $W = UV'$ is the permutation matrix Q_0 of Exercise 10.5, and

$$P = V\Sigma V' = \frac{1}{4}\big(7h_1h_1' + 5h_2h_2' + 3h_3h_3' + h_4h_4'\big) = \begin{pmatrix} 4 & 1 & 2 & 0 \\ 1 & 4 & 0 & 2 \\ 2 & 0 & 4 & 1 \\ 0 & 2 & 1 & 4 \end{pmatrix}$$

Check: WP interchanges rows $1, 2$ and rows $3, 4$ of P, which gives A. Thus A stretches along the orthogonal directions $h_1, \ldots, h_4$ by the factors $7, 5, 3, 1$ and then permutes the coordinates.

Problem 10.5. Let $A \in \mathbb{R}^{m\times n}$.

(a) Show that $A'H$ is a g-inverse of A for every g-inverse H of AA'.

(b) Show that $A^+ = (A'A)^+A' = A'(AA')^+$.

(c) Show that $(A'A)^+ = A^+(A^+)'$.

Solution. (a) H' is a g-inverse of $(AA')' = AA'$ (Theorem 10.10(c)). Lemma 10.12(a) for A' in place of A (so that $(A')'A' = AA'$) and the g-inverse H' gives $A'H'AA' = A'$; transposing, $AA'HA = A$, that is, $A(A'H)A = A$.

(b) For $A = 0$ all three matrices are 0. Let $A = U\Sigma V'$ have rank $r \geqslant 1$. $A'A = V(\Sigma'\Sigma)V'$ with $\Sigma'\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)$, so by Theorem 10.15(e), $(A'A)^+ = V(\Sigma'\Sigma)^+V'$ with $(\Sigma'\Sigma)^+ = \operatorname{diag}(\sigma_1^{-2}, \ldots, \sigma_r^{-2}, 0, \ldots, 0)$. Hence $(A'A)^+A' = V(\Sigma'\Sigma)^+\Sigma'U' = V\Sigma^+U' = A^+$ (Theorem 10.15(b)), as the (i,i) entry of $(\Sigma'\Sigma)^+\Sigma' \in \mathbb{R}^{n\times m}$ is $\sigma_i^{-2}\sigma_i = \sigma_i^{-1}$ for $i \leqslant r$ and all its other entries are 0. Likewise $AA' = U(\Sigma\Sigma')U'$, $(AA')^+ = U(\Sigma\Sigma')^+U'$ and $A'(AA')^+ = V\Sigma'(\Sigma\Sigma')^+U' = V\Sigma^+U'$.

(c) $A^+(A^+)' = V\Sigma^+U'U(\Sigma^+)'V' = V\Sigma^+(\Sigma^+)'V' = V(\Sigma'\Sigma)^+V' = (A'A)^+$, since $\Sigma^+(\Sigma^+)' = \operatorname{diag}(\sigma_1^{-2}, \ldots, \sigma_r^{-2}, 0, \ldots, 0) \in \mathbb{R}^{n\times n}$.

Problem 10.6. Let G be a g-inverse of $A \in \mathbb{R}^{m\times n}$.

(a) Show that GAG is a g-inverse of A and $\operatorname{rank}(GAG) = \operatorname{rank} A$.

(b) Show that $GAG = G$ if and only if $\operatorname{rank} G = \operatorname{rank} A$.

(c) For $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, show that $G \in \mathbb{R}^{2\times 2}$ is a g-inverse of A if and only if the entries of G add up to 1. Give one of rank 2, and check (b) for A^+.

Solution. (a) $A(GAG)A = (AGA)GA = AGA = A$. By Theorem 4.15(a) and Theorem 10.10(a), $\operatorname{rank} A = \operatorname{rank}\big(A(GAG)A\big) \leqslant \operatorname{rank}(GAG) \leqslant \operatorname{rank}(GA) = \operatorname{rank} A$.

(b) If $GAG = G$, then $\operatorname{rank} G = \operatorname{rank} A$ by (a). Conversely, let $\operatorname{rank} G = \operatorname{rank} A$. Then $\mathcal{C}(GA) \subseteq \mathcal{C}(G)$ (Theorem 4.15(a)) and $\operatorname{rank}(GA) = \operatorname{rank} A = \operatorname{rank} G$ (Theorem 10.10(a)), so $\mathcal{C}(GA) = \mathcal{C}(G)$ (Theorem 4.5(a)). Hence every column of G is GA times some vector, that is, $G = GAW$ for some $W \in \mathbb{R}^{n\times m}$ (Theorem 1.4(a)), and $GAG = (GA)^2W = GAW = G$, as GA is idempotent.

(c) $A = \mathbf{1}_2\mathbf{1}_2'$, so $AGA = \mathbf{1}_2(\mathbf{1}_2'G\mathbf{1}_2)\mathbf{1}_2' = (\mathbf{1}_2'G\mathbf{1}_2)A$, where $\mathbf{1}_2'G\mathbf{1}_2$ is the sum of the entries of G. So $AGA = A$ if and only if this sum is 1. For instance $G = \frac{1}{2}I_2$ is a g-inverse of rank $2 \neq 1 = \operatorname{rank} A$, and by (b) $GAG \neq G$ (indeed $GAG = \frac{1}{4}A$). By Problem 10.1(d), $A^+ = \mathbf{1}\mathbf{1}'/4 = \frac{1}{4}A$: its entries add up to 1, it has rank 1, and $A^+AA^+ = A^+$, as (b) requires.

Problem 10.7. Let $k \geqslant 2$ variables have the sample covariance matrix $S = s^2\big((1-\rho)I_k + \rho J_k\big)$ with $s > 0$ and $0 < \rho < 1$ (equal variances, equal correlations ρ).

(a) Show that $\lambda_1 = s^2(1 + (k-1)\rho)$ with $q_1 = k^{-1/2}\mathbf{1}_k$, and $\lambda_2 = \cdots = \lambda_k = s^2(1-\rho)$, where $q_2, \ldots, q_k$ can be any orthonormal basis of $W_0 = \{x \in \mathbb{R}^k : \mathbf{1}'x = 0\}$.

(b) Show that the first score of observation i is $\sqrt{k}$ times the average of its k centered values, and that the first component explains $(1 + (k-1)\rho)/k$ of the total variance.

(c) For $k = 4$ test scores with $\rho = \frac{1}{2}$, compute this proportion. Does the second loading vector have a meaning of its own?

Solution. (a) $S\mathbf{1} = s^2\big((1-\rho)\mathbf{1} + \rho k\mathbf{1}\big) = s^2(1 + (k-1)\rho)\mathbf{1}$, and for $x \in W_0$, $J_k x = \mathbf{1}(\mathbf{1}'x) = 0$, so $Sx = s^2(1-\rho)x$ (compare Example 7.17). Extend $q_1 = k^{-1/2}\mathbf{1}$ to an orthonormal basis $q_1, \ldots, q_k$ of $\mathbb{R}^k$ (Lemma 7.27); then $q_2, \ldots, q_k \perp \mathbf{1}$ lie in W_0, and $SQ = Q\Lambda$, so $S = Q\Lambda Q'$ with $\Lambda = \operatorname{diag}\big(s^2(1 + (k-1)\rho), s^2(1-\rho), \ldots, s^2(1-\rho)\big)$, in decreasing order since $k\rho > 0$. The same holds for every orthonormal basis $q_2, \ldots, q_k$ of W_0, which has dimension $k - 1$ ($W_0 = \operatorname{span}\{\mathbf{1}\}^{\perp}$, Theorem 6.5(c)).

(b) $z_{i1} = k^{-1/2}\mathbf{1}'(x_i - \bar{x}) = \sqrt{k} \cdot \frac{1}{k}\sum_j (x_{ij} - \bar{x}_j)$. The total variance is $\operatorname{tr} S = ks^2$ (Theorem 10.22(c)), so the proportion is $s^2(1 + (k-1)\rho)/(ks^2) = (1 + (k-1)\rho)/k$.

(c) $(1 + \frac{3}{2})/4 = \frac{5}{8}$. Each further component explains $(1-\rho)/k = \frac{1}{8}$. Since any orthonormal basis of W_0 may serve as q_2, q_3, q_4, the individual vector q_2 has no meaning of its own; only the space W_0 of contrasts (coefficient vectors with sum 0) is determined.

Chapter 11. Random vectors and the multivariate normal distribution

Random vectors

Definition 11.1. A **random vector** $x = (x_1, \dots, x_n)'$ is a vector whose entries are random variables, and a **random matrix** $X = (x_{ij})$ is a matrix whose entries are random variables. Expectations are taken entrywise:

$$\mathrm{E}\, x = (\mathrm{E}\, x_1, \dots, \mathrm{E}\, x_n)', \qquad \mathrm{E}\, X = (\mathrm{E}\, x_{ij})$$

Let $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ be random vectors with means $\mu = \mathrm{E}\, x$ and $\nu = \mathrm{E}\, y$.

- The **covariance matrix** of x is $\operatorname{var}(x) = \mathrm{E}(x-\mu)(x-\mu)' \in \mathbb{R}^{n\times n}$. Its (i,j) entry is $\operatorname{cov}(x_i, x_j) = \mathrm{E}(x_i - \mu_i)(x_j - \mu_j)$, and its diagonal entries are the variances $\operatorname{var}(x_i)$, whose square roots are the **standard deviations**.
- The **cross-covariance matrix** of x and y is $\operatorname{cov}(x, y) = \mathrm{E}(x-\mu)(y-\nu)' \in \mathbb{R}^{n\times m}$, with (i,j) entry $\operatorname{cov}(x_i, y_j)$; so $\operatorname{var}(x) = \operatorname{cov}(x, x)$. The vectors x and y are **uncorrelated** if $\operatorname{cov}(x, y) = 0$.
- If every $\operatorname{var}(x_i) > 0$, the **correlation matrix** of x is

  $$\operatorname{corr}(x) = D^{-1/2} \operatorname{var}(x) D^{-1/2}, \qquad D = \operatorname{diag}\Big(\operatorname{var}(x_1), \dots, \operatorname{var}(x_n)\Big),$$

  where $D^{-1/2}$ is the diagonal matrix with diagonal entries $\operatorname{var}(x_i)^{-1/2}$. Its (i,j) entry is the correlation $\operatorname{corr}(x_i, x_j) = \operatorname{cov}(x_i, x_j)/\sqrt{\operatorname{var}(x_i)\operatorname{var}(x_j)}$, and its diagonal entries are 1.

Remark 11.2 (Imported probability facts)**.** Random variables, probabilities $\mathrm{P}(\cdot)$, expectations and independence are used as in a first course in probability. The **distribution** of a random vector $x \in \mathbb{R}^n$ is the assignment $A \mapsto \mathrm{P}(x \in A)$ $(A \subseteq \mathbb{R}^n)$; probabilities, expectations $\mathrm{E}\, g(x)$, and the distribution of $g(x)$ depend only on the distribution of x. We write $x \sim \mathcal{D}$ if x has the distribution $\mathcal{D}$, and $u \perp\!\!\!\perp v$ if u and v are independent. The vector x has the **density** f if $\mathrm{P}(x \in A) = \int_A f(u)\, du$ for all (reasonable) sets $A \subseteq \mathbb{R}^n$. Its **moment generating function** (MGF) is $M_x(t) = \mathrm{E}\, e^{t'x}$ $(t \in \mathbb{R}^n)$, where this expectation is finite. The **conditional distribution** of a random vector y given $x = a$ describes y when x is known to equal a; its mean and covariance matrix are the **conditional mean** $\mathrm{E}(y \mid x = a)$ and the **conditional covariance matrix** $\operatorname{var}(y \mid x = a)$. The following facts are used without proof.

(P1) Every random variable whose mean or variance is used has a finite variance. Expectation is linear, $\mathrm{E}(aX + bY) = a\,\mathrm{E}\, X + b\,\mathrm{E}\, Y$ $(a, b \in \mathbb{R})$; $\mathrm{E}\, c = c$ for a constant c; and $\mathrm{E}\, X \geqslant 0$ if $X \geqslant 0$. If X takes only the values $a_1, \dots, a_m$, with probabilities $p_1, \dots, p_m$, then $\mathrm{E}\, g(X) = g(a_1)p_1 + \cdots + g(a_m)p_m$ for every function g; in particular, a random variable that equals 1 with probability p and 0 otherwise has expectation p. If $u_1, \dots, u_m$ are independent random vectors and $g_1, \dots, g_m$ are functions, then $g_1(u_1), \dots, g_m(u_m)$ are independent, and if they are random variables with finite means, $\mathrm{E}\big(g_1(u_1) \cdots g_m(u_m)\big) = \mathrm{E}\, g_1(u_1) \cdots \mathrm{E}\, g_m(u_m)$. A random variable Y has $\operatorname{var}(Y) = 0$ if and only if $Y = \mathrm{E}\, Y$ with probability 1.

(P2) (*Uniqueness.*) If the MGFs of two random vectors are finite and equal on a set $\{t : \|t\| < \delta\}$ $(\delta > 0)$, the two vectors have the same distribution.

(P3) (*Independence.*) Let $u_1 \in \mathbb{R}^{n_1}, \dots, u_m \in \mathbb{R}^{n_m}$ have MGFs that are finite near 0, and let $u = (u_1', \dots, u_m')'$. Then $u_1, \dots, u_m$ are independent if and only if $M_u(t) = M_{u_1}(t_1) \cdots M_{u_m}(t_m)$ for all $t = (t_1', \dots, t_m')'$ near 0.

(P4) (*Densities.*) If $z \in \mathbb{R}^n$ has density f, $B \in \mathbb{R}^{n\times n}$ is invertible and $\mu \in \mathbb{R}^n$, then $x = Bz + \mu$ has density $f\big(B^{-1}(x-\mu)\big)/|\det B|$. If $u_1, \dots, u_m$ are independent with densities $f_1, \dots, f_m$, then $(u_1', \dots, u_m')'$ has density $f_1(u_1) \cdots f_m(u_m)$.

(P5) (*Normal and derived distributions.*) The **standard normal distribution** $N(0,1)$ has density $\varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}$, where e is the base of the natural logarithm log and $2 < e < 3$, mean 0, variance 1 and MGF $e^{t^2/2}$ ($t \in \mathbb{R}$). Let $z_1, \dots, z_r$ be independent with distribution $N(0,1)$. Then $z_1^2 + \cdots + z_r^2$ has the **chi-square distribution** χ_r^2 with r **degrees of freedom** ($r \geqslant 1$); if $z \sim N(0,1)$ and $v \sim \chi_\nu^2$ are independent, $z/\sqrt{v/\nu}$ has the t **distribution** t_ν (ν degrees of freedom); if $u \sim \chi_a^2$ and $v \sim \chi_b^2$ are independent, $(u/a)/(v/b)$ has the F **distribution** $F_{a,b}$ (a and b degrees of freedom). For $v \sim \chi_r^2$, $\mathrm{E}\, v = r$ and $\mathrm{E}\, e^{tv} = (1-2t)^{-r/2}$ ($t < \frac{1}{2}$); for $v \sim \chi_2^2$, $\mathrm{P}(v \leqslant c) = 1 - e^{-c/2}$ ($c \geqslant 0$). These distributions have continuous distribution functions, strictly increasing on $\mathbb{R}$ ($N(0,1)$, t_ν) or on $[0,\infty)$ (χ_r^2, $F_{a,b}$). So each single number has probability 0, and for $0 < \tau < 1$ there is exactly one τ**-quantile** c with $\mathrm{P}(V \leqslant c) = \tau$, written z_τ, $\chi_{r,\tau}^2$, $t_{\nu,\tau}$, $F_{a,b,\tau}$ for V with distribution $N(0,1)$, χ_r^2, t_ν, $F_{a,b}$.

(P6) (*Conditioning.*) If w and x_2 are independent random vectors and g is a function, then for every a the conditional distribution of $w + g(x_2)$ given $x_2 = a$ is the distribution of $w + g(a)$.

Theorem 11.3. Let $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ be random vectors with means μ and ν, let X be a random matrix, and let A, B, C, F, b be constant matrices and vectors of sizes for which the expressions below are defined.

(a) $\mathrm{E}(Ax + b) = A\mu + b$ and $\mathrm{E}(CXF) = C(\mathrm{E}\, X)F$.

(b) $\mathrm{var}(Ax + b) = A\,\mathrm{var}(x)A'$ and $\mathrm{cov}(Ax, By) = A\,\mathrm{cov}(x,y)B'$.

(c) $\mathrm{var}(x) = \mathrm{E}(xx') - \mu\mu'$.

(d) If $m = n$, then $\mathrm{var}(x + y) = \mathrm{var}(x) + \mathrm{var}(y) + \mathrm{cov}(x,y) + \mathrm{cov}(y,x)$.

(e) $\mathrm{cov}(y,x) = \mathrm{cov}(x,y)'$, and $\mathrm{var}(x)$ is symmetric. If x and y are independent, they are uncorrelated.

Proof. (a) Entry i of $Ax + b$ is $\sum_j a_{ij}x_j + b_i$, with expectation $\sum_j a_{ij}\mu_j + b_i$ by (P1). Likewise the (i,l) entry of CXF is $\sum_j \sum_k c_{ij}x_{jk}f_{kl}$, with expectation $\sum_j \sum_k c_{ij}\,\mathrm{E}(x_{jk})f_{kl}$, the (i,l) entry of $C(\mathrm{E}\, X)F$.

(b) By (a), $Ax + b - \mathrm{E}(Ax + b) = A(x - \mu)$ and $By - \mathrm{E}(By) = B(y - \nu)$. Applying (a) to the random matrix $(x-\mu)(y-\nu)'$,

$$\mathrm{cov}(Ax, By) = \mathrm{E}\Big(A(x-\mu)(y-\nu)'B'\Big) = A\ \mathrm{E}\Big((x-\mu)(y-\nu)'\Big)B' = A\,\mathrm{cov}(x,y)B',$$

and in the same way $\mathrm{var}(Ax + b) = \mathrm{E}\Big(A(x-\mu)(x-\mu)'A'\Big) = A\,\mathrm{var}(x)A'$.

(c) $(x-\mu)(x-\mu)' = xx' - x\mu' - \mu x' + \mu\mu'$, and by (a) its expectation is $\mathrm{E}(xx') - \mu\mu' - \mu\mu' + \mu\mu'$.

(d) $x + y - (\mu + \nu) = (x - \mu) + (y - \nu)$. Expand $\Big((x-\mu) + (y-\nu)\Big)\Big((x-\mu) + (y-\nu)\Big)'$ into four terms and take expectations.

(e) $\Big((x-\mu)(y-\nu)'\Big)' = (y-\nu)(x-\mu)'$, and transposition commutes with taking expectations entrywise; for $y = x$ this gives $\mathrm{var}(x)' = \mathrm{var}(x)$. If $x \perp\!\!\!\perp y$, then $x_i - \mu_i$ and $y_j - \nu_j$ are independent with means 0 (P1), so $\mathrm{cov}(x_i, y_j) = \mathrm{E}(x_i - \mu_i)\ \mathrm{E}(y_j - \nu_j) = 0$. □

Theorem 11.4. Let $x \in \mathbb{R}^n$ be a random vector with mean μ and covariance matrix $\Sigma = \mathrm{var}(x)$.

(a) $\mathrm{var}(a'x) = a'\Sigma a$ for every $a \in \mathbb{R}^n$, and Σ is positive semidefinite.

(b) Σ is singular if and only if $a'x = a'\mu$ with probability 1 for some $a \neq 0$.

Proof. (a) Theorem 11.3(b) with $A = a'$ gives $\mathrm{var}(a'x) = a'\Sigma a$. This number is $\mathrm{E}(a'x - a'\mu)^2 \geqslant 0$ (P1), and Σ is symmetric (Theorem 11.3(e)); so Σ is positive semidefinite (Definition 8.11).

(b) Σ is singular if and only if $\Sigma a = 0$ for some $a \neq 0$ (Theorem 2.18). As Σ is positive semidefinite, $\Sigma a = 0$ if and only if $a'\Sigma a = 0$ (Proposition 8.17(a)). By (a) and (P1), $a'\Sigma a = \mathrm{var}(a'x) = 0$ if and only if $a'x = \mathrm{E}(a'x) = a'\mu$ with probability 1. □

Example 11.5. Let u_1, u_2 be uncorrelated random variables with variance 1, and $x = (u_1,\ u_2,\ u_1{+}u_2)'$. Then $x = Bu$ with $u = (u_1, u_2)'$, $\operatorname{var}(u) = I_2$ and $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, so by Theorem 11.3(b)

$$\operatorname{var}(x) = BB' = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

This matrix is singular: $\operatorname{var}(x)a = 0$ for $a = (1, 1, -1)'$. Correspondingly $a'x = u_1 + u_2 - (u_1 + u_2) = 0$, as Theorem 11.4(b) says. Likewise, the shares of a budget spent on k items add up to 1, so their covariance matrix is singular.

Theorem 11.6. Let $x \in \mathbb{R}^n$ have mean μ and covariance matrix Σ, and let $A \in \mathbb{R}^{n\times n}$. Then

$$\mathrm{E}(x'Ax) = \operatorname{tr}(A\Sigma) + \mu' A\mu$$

Proof. The 1×1 matrix $x'Ax$ is its own trace, and $\operatorname{tr}(x'Ax) = \operatorname{tr}(Axx')$ (Theorem 1.11(b)). A trace is a sum of entries, so by (P1) and Theorem 11.3(a),(c),

$$\mathrm{E}(x'Ax) = \operatorname{tr}\Big(A\ \mathrm{E}(xx')\Big) = \operatorname{tr}\Big(A(\Sigma + \mu\mu')\Big) = \operatorname{tr}(A\Sigma) + \operatorname{tr}(A\mu\mu') = \operatorname{tr}(A\Sigma) + \mu' A\mu,$$

where $\operatorname{tr}(A\mu\mu') = \operatorname{tr}(\mu' A\mu)$ by Theorem 1.11(b). □

Definition 11.7. Let observations be modeled as a random vector y whose distribution depends on unknown parameters. An **estimator** of a parameter θ is a random variable $T = g(y)$, where the function g does not involve unknown parameters; its value at the observed y is an **estimate** of θ. The estimator T is **unbiased** if $\mathrm{E}\,T = \theta$ for all values of the parameters.

Example 11.8. Let $y \in \mathbb{R}^n$ have mean μ and covariance matrix $\sigma^2 I_n$, and let $P \in \mathbb{R}^{n\times n}$ be symmetric and idempotent of rank r. Then $\operatorname{tr} P = \operatorname{rank} P = r$ (Theorem 6.16(c)), and Theorem 11.6 gives

$$\mathrm{E}(y'Py) = \operatorname{tr}(P\sigma^2 I_n) + \mu' P\mu = \sigma^2 r + \mu' P\mu$$

- Let $n \geqslant 2$ and $P = C_n$, which is symmetric and idempotent of rank $n - 1$ (Proposition 1.13, Corollary 6.21(b)). By Example 1.14, $y'C_n y = \sum_i (y_i - \bar{y})^2$ and $\mu' C_n \mu = \sum_i (\mu_i - \bar{\mu})^2$, where $\bar{\mu} = \frac{1}{n}\mathbf{1}'\mu$. So
$$\mathrm{E}\sum_{i=1}^{n}(y_i - \bar{y})^2 = (n-1)\sigma^2 + \sum_{i=1}^{n}(\mu_i - \bar{\mu})^2$$
- If the observations have a common mean, $\mu = \mu_0\mathbf{1}$, then $C_n\mu = 0$ and the last sum is 0. So $s^2 = \frac{1}{n-1}\sum_i (y_i - \bar{y})^2$ is an unbiased estimator of σ^2: $\mathrm{E}\,s^2 = \sigma^2$ for all values of μ_0 and σ^2. The estimator $\frac{1}{n}\sum_i (y_i - \bar{y})^2$ has expectation $\frac{n-1}{n}\sigma^2$, which is smaller than σ^2 when $\sigma^2 > 0$; so it is not unbiased.
- If the means drift, s^2 overestimates σ^2 on average. For $n = 4$ and $\mu = (1, 3, 5, 7)'$, $\bar{\mu} = 4$ and $\sum_i (\mu_i - \bar{\mu})^2 = 9 + 1 + 1 + 9 = 20$, so $\mathrm{E}\,s^2 = \sigma^2 + \frac{20}{3}$.

Proposition 11.9. Let $x_1, \ldots, x_n \in \mathbb{R}^k$ ($n \geqslant 2$) be random vectors with common mean μ and common covariance matrix Σ, uncorrelated in pairs: $\operatorname{cov}(x_i, x_j) = 0$ for $i \neq j$ (for instance, independent; Theorem 11.3(e)). Let

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad S = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})',$$

the mean vector and the sample covariance matrix (Definition 8.32) of the random data matrix with rows $x_1', \ldots, x_n'$. Then

$$\mathrm{E}\,\bar{x} = \mu, \qquad \operatorname{var}(\bar{x}) = \frac{1}{n}\Sigma, \qquad \mathrm{E}\,S = \Sigma$$

Proof. • $\mathrm{E}\,\bar{x} = \frac{1}{n}\sum_i \mathrm{E}\,x_i = \mu$ (Theorem 11.3(a)).

- Let $u = (x_1', \dots, x_n')' \in \mathbb{R}^{nk}$. The (i,j) block of $\mathrm{var}(u)$ is $\mathrm{cov}(x_i, x_j)$, so $\mathrm{var}(u) = \mathrm{diag}(\Sigma, \dots, \Sigma)$. With $G = [I_k \ \cdots \ I_k] \in \mathbb{R}^{k \times nk}$, $\bar{x} = \frac{1}{n}Gu$; by Theorem 11.3(b) and the block product (Theorem 5.3(c)), $\mathrm{var}(\bar{x}) = \frac{1}{n^2} G\,\mathrm{diag}(\Sigma, \dots, \Sigma)G' = \frac{1}{n^2}\, n\Sigma$.
- Expanding, with $\sum_i x_i = n\bar{x}$, $\sum_i (x_i - \bar{x})(x_i - \bar{x})' = \sum_i x_i x_i' - n\bar{x}\bar{x}'$. By Theorem 11.3(c), $\mathrm{E}(x_i x_i') = \Sigma + \mu\mu'$ and $\mathrm{E}(\bar{x}\bar{x}') = \frac{1}{n}\Sigma + \mu\mu'$. So this sum has expectation $n(\Sigma + \mu\mu') - n(\frac{1}{n}\Sigma + \mu\mu') = (n-1)\Sigma$, and $\mathrm{E}\,S = \Sigma$. □

Theorem 11.10 (Best linear predictor)**.** Let y be a random variable and $x \in \mathbb{R}^k$ a random vector, with $\mathrm{E}\,y = \mu_y$, $\mathrm{E}\,x = \mu_x$, $\mathrm{var}(y) = \sigma_{yy}$, $\mathrm{cov}(x, y) = \sigma_{xy} \in \mathbb{R}^k$, $\sigma_{yx} = \sigma_{xy}'$, and $\mathrm{var}(x) = \Sigma_{xx}$ positive definite. Put

$$b^* = \Sigma_{xx}^{-1}\sigma_{xy}, \qquad a^* = \mu_y - b^{*\prime}\mu_x$$

Then for all $a \in \mathbb{R}$ and $b \in \mathbb{R}^k$,

$$\mathrm{E}(y - a - b'x)^2 = \left(\sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}\right) + (b - b^*)'\Sigma_{xx}(b - b^*) + (\mu_y - a - b'\mu_x)^2$$

So the **mean squared error** $\mathrm{E}(y - a - b'x)^2$ is smallest exactly for $a = a^*$, $b = b^*$, with minimum $\sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}$: the Schur complement of Σ_{xx} in $\mathrm{var}\big((x', y)'\big) = \left(\begin{smallmatrix} \Sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{smallmatrix}\right)$ (Theorem 5.9). The random variable $a^* + b^{*\prime}x$ is the **best linear predictor** of y from x, and the prediction error $u = y - a^* - b^{*\prime}x$ has mean 0 and is uncorrelated with x.

Proof. • Let $v = y - a - b'x$. By Theorem 11.3(c) for $n = 1$, $\mathrm{E}\,v^2 = \mathrm{var}(v) + (\mathrm{E}\,v)^2$, and $\mathrm{E}\,v = \mu_y - a - b'\mu_x$.

- $v = (1, -b')(y, x')' - a$, and $\mathrm{var}\big((y, x')'\big)$ has the blocks σ_{yy}, σ_{yx}, σ_{xy}, Σ_{xx}; by Theorem 11.3(b), $\mathrm{var}(v) = \sigma_{yy} - 2b'\sigma_{xy} + b'\Sigma_{xx}b$.
- As $\Sigma_{xx}b^* = \sigma_{xy}$, $(b - b^*)'\Sigma_{xx}(b - b^*) = b'\Sigma_{xx}b - 2b'\sigma_{xy} + \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}$; substituting gives the identity.
- Σ_{xx} is positive definite, so the middle term is $\geqslant 0$, with equality only for $b = b^*$; the last term is $\geqslant 0$, with equality only for $a = \mu_y - b'\mu_x$. Hence the minimum is attained exactly at $b = b^*$, $a = a^*$.
- As $a^* = \mu_y - b^{*\prime}\mu_x$, $u = y - a^* - b^{*\prime}x = (y - \mu_y) - b^{*\prime}(x - \mu_x)$, so $\mathrm{E}\,u = 0$ (P1). With $b^{*\prime}(x - \mu_x) = (x - \mu_x)'b^*$, $(x - \mu_x)u = (x - \mu_x)(y - \mu_y) - (x - \mu_x)(x - \mu_x)'b^*$, and by (P1) and Theorem 11.3(a), $\mathrm{cov}(x, u) = \mathrm{E}\big((x - \mu_x)u\big) = \sigma_{xy} - \Sigma_{xx}b^* = 0$. □

Example 11.11. Let $(y, x_1, x_2, x_3)'$ have mean $(8, 1, 2, 3)'$ and covariance matrix

$$\begin{pmatrix} \sigma_{yy} & \sigma_{yx} \\ \sigma_{xy} & \Sigma_{xx} \end{pmatrix} = \left(\begin{array}{c|ccc} 12 & 3 & 4 & 3 \\ \hline 3 & 2 & 1 & 0 \\ 4 & 1 & 2 & 1 \\ 3 & 0 & 1 & 2 \end{array}\right),$$

which is positive definite (leading principal minors 12, 15, 10, 8; Theorem 8.19); so is Σ_{xx} (Corollary 8.15(a)), and Σ_{xx} is invertible. So $b^* = \Sigma_{xx}^{-1}\sigma_{xy}$ is the unique solution of $\Sigma_{xx}b = \sigma_{xy}$. As

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix},$$

$b^* = (1, 1, 1)'$ and $a^* = 8 - (1 + 2 + 3) = 2$. By Theorem 11.10 the best linear predictor of y is $2 + x_1 + x_2 + x_3$, with mean squared error $12 - (3, 4, 3)(1, 1, 1)' = 2$. With fewer predictors the minimum is larger:

- with x_j alone it is $\sigma_{yy} - \sigma_{x_jy}^2/\sigma_{x_jx_j}$, that is, $12 - \frac{9}{2} = \frac{15}{2}$, $12 - \frac{16}{2} = 4$ and $12 - \frac{9}{2} = \frac{15}{2}$ for $j = 1, 2, 3$;
- with $(x_1, x_3)'$, whose covariance matrix is $2I_2$, it is $12 - (3, 3)\frac{1}{2}I_2(3, 3)' = 3$;
- with no predictor, $\mathrm{E}(y - a)^2 = \mathrm{var}(y) + (\mu_y - a)^2$ is smallest for $a = \mu_y = 8$, with value 12.

So x_2 is the best single predictor, but the pair $(x_1, x_3)'$, whose members are worse one at a time, does better than x_2.

The multivariate normal distribution

Definition 11.12. Let $\mu \in \mathbb{R}^n$, and let $\Sigma \in \mathbb{R}^{n\times n}$ be symmetric and positive semidefinite. A random vector $x \in \mathbb{R}^n$ has the **multivariate normal distribution** $N_n(\mu, \Sigma)$, written $x \sim N_n(\mu, \Sigma)$, if

$$\mathrm{E}\, e^{t'x} = \exp\Big(t'\mu + \tfrac{1}{2}t'\Sigma t\Big) \qquad \text{for all } t \in \mathbb{R}^n$$

For $n = 1$ we write $N(\mu, \sigma^2)$ ($\sigma^2 \geqslant 0$), and for every n we also write $N(\mu, \Sigma)$ when n is clear from the context. Random vectors $x_1, \ldots, x_m$ are **jointly normal** if $(x_1', \ldots, x_m')'$ is multivariate normal.

For $\mu = 0$ and $\sigma^2 = 1$ the MGF is $e^{t^2/2}$. So by (P2) and (P5), a random variable has the distribution $N(0, 1)$ of Definition 11.12 if and only if it has the standard normal distribution of (P5). The definition uses the MGF rather than a density because Σ is allowed to be singular.

Theorem 11.13. Let $z = (z_1, \ldots, z_k)'$ have independent components with distribution $N(0, 1)$, and let $\mu \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n\times k}$. Then $\mathrm{E}\, e^{s'z} = e^{s's/2}$ for all $s \in \mathbb{R}^k$, and

$$\mu + Bz \sim N_n(\mu, BB')$$

Hence $N_n(\mu, \Sigma)$ exists for every symmetric positive semidefinite Σ (take $B = \Sigma^{1/2}$, Theorem 8.25, or, if Σ is positive definite, its Cholesky factor, Theorem 8.22), and every $x \sim N_n(\mu, \Sigma)$ has $\mathrm{E}\, x = \mu$ and $\mathrm{var}(x) = \Sigma$.

Proof. • By (P1) and (P5), $\mathrm{E}\, e^{s'z} = \mathrm{E} \prod_j e^{s_jz_j} = \prod_j \mathrm{E}\, e^{s_jz_j} = \prod_j e^{s_j^2/2} = e^{s's/2}$: the factors $e^{s_jz_j}$ are functions of the independent z_j, with finite means.

- For $t \in \mathbb{R}^n$, $t'(\mu + Bz) = t'\mu + (B't)'z$, so by (P1) $\mathrm{E}\, e^{t'(\mu+Bz)} = e^{t'\mu}\ \mathrm{E}\, e^{(B't)'z} = e^{t'\mu + t'BB't/2}$. Also BB' is symmetric and positive semidefinite (Theorem 8.16(a)). So $\mu + Bz \sim N_n(\mu, BB')$.
- $\Sigma^{1/2}$ is symmetric with $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$ (Theorem 8.25), and a Cholesky factor L of a positive definite Σ satisfies $LL' = \Sigma$. Either choice of B gives $BB' = \Sigma$.
- By (P5) and (P1), $\mathrm{E}\, z = 0$, $\mathrm{var}(z_j) = 1$, and $\mathrm{cov}(z_i, z_j) = \mathrm{E}(z_iz_j) - \mathrm{E}\, z_i\ \mathrm{E}\, z_j = 0$ for $i \neq j$; so $\mathrm{var}(z) = I_k$, and Theorem 11.3(a),(b) gives $\mathrm{E}(\mu + Bz) = \mu$ and $\mathrm{var}(\mu + Bz) = BB'$.
- If $x \sim N_n(\mu, \Sigma)$ and $BB' = \Sigma$, then x and $\mu + Bz$ have the same MGF, finite on $\mathbb{R}^n$. By (P2) they have the same distribution, hence the same mean μ and covariance matrix Σ. □

Theorem 11.14. Let $x \sim N_n(\mu, \Sigma)$.

(a) For $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, $Ax + b \sim N_m(A\mu + b, A\Sigma A')$.

(b) Every subvector of x is normal, with the corresponding entries of μ and the corresponding rows and columns of Σ; and $a'x \sim N(a'\mu, a'\Sigma a)$ for every $a \in \mathbb{R}^n$.

(c) Conversely, if $y \in \mathbb{R}^n$ is a random vector such that $a'y$ has a normal distribution for every $a \in \mathbb{R}^n$, then $y \sim N_n\Big(\mathrm{E}\, y, \mathrm{var}(y)\Big)$.

Proof. (a) For $t \in \mathbb{R}^m$, $t'(Ax+b) = (A't)'x + t'b$, so by (P1)

$$\mathrm{E}\, e^{t'(Ax+b)} = e^{t'b} \exp\Big((A't)'\mu + \tfrac{1}{2}(A't)'\Sigma(A't)\Big) = \exp\Big(t'(A\mu+b) + \tfrac{1}{2}t'(A\Sigma A')t\Big),$$

and $A\Sigma A'$ is symmetric and positive semidefinite (Theorem 8.16(b)).

(b) The subvector $(x_{i_1}, \dots, x_{i_m})'$ is Ex, where E has the rows $e'_{i_1}, \dots, e'_{i_m}$; the sth entry of $E\mu$ is μ_{i_s}, and the (s,l) entry of $E\Sigma E'$ is $e'_{i_s}\Sigma e_{i_l} = \sigma_{i_s i_l}$. Apply (a). For $a'x$ take $A = a'$.

(c) Each $y_i = e'_i y$ is normal, hence has a finite variance (Theorem 11.13); let $\mu = \mathrm{E}\, y$ and $\Sigma = \mathrm{var}(y)$. For $t \in \mathbb{R}^n$, $t'y$ is normal with mean $t'\mu$ and variance $t'\Sigma t$ (Theorems 11.3(a) and 11.4(a)); as the parameters of a normal distribution are its mean and variance (Theorem 11.13), $t'y \sim N(t'\mu, t'\Sigma t)$. Evaluating its MGF at 1, $\mathrm{E}\, e^{t'y} = \exp(t'\mu + \frac{1}{2}t'\Sigma t)$. This holds for every t, and Σ is symmetric and positive semidefinite (Theorem 11.4(a)); so $y \sim N_n(\mu, \Sigma)$. □

Theorem 11.15. Let $x \sim N_n(\mu, \Sigma)$ with Σ positive definite. Then x has the density

$$f(u) = \frac{1}{(2\pi)^{n/2}(\det\Sigma)^{1/2}} \exp\Big(-\frac{1}{2}(u-\mu)'\Sigma^{-1}(u-\mu)\Big) \qquad (u \in \mathbb{R}^n)$$

Let $\Sigma = Q\Lambda Q'$ with $Q = [q_1 \ \cdots \ q_n]$ orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ (Theorem 8.4(a)). For $c > 0$, the density is constant on the ellipsoid $\{u : (u-\mu)'\Sigma^{-1}(u-\mu) = c\}$, whose center is μ, whose axes point along $q_1, \dots, q_n$, and whose semi-axes are $\sqrt{c\lambda_1}, \dots, \sqrt{c\lambda_n}$.

Proof.
- By Theorem 8.22, $\Sigma = LL'$ with L lower triangular with positive diagonal entries. So $\det L > 0$ (Theorem 5.15(a)) and $\det\Sigma = \det L \det L' = (\det L)^2$ (Theorems 5.19(b) and 5.15(b)): L is invertible (Theorem 5.19(a)) and $\det L = (\det\Sigma)^{1/2}$.
- Let $z \in \mathbb{R}^n$ have independent $N(0,1)$ components. By (P4) and (P5), z has the density $g(v) = \prod_j \varphi(v_j) = (2\pi)^{-n/2}e^{-v'v/2}$ $(v \in \mathbb{R}^n)$, and $y = Lz + \mu$ has, at u, the density $g(w)/\det L = (2\pi)^{-n/2}e^{-w'w/2}/\det L$ with $w = L^{-1}(u-\mu)$.
- $w'w = (u-\mu)'(L^{-1})'L^{-1}(u-\mu) = (u-\mu)'(LL')^{-1}(u-\mu) = (u-\mu)'\Sigma^{-1}(u-\mu)$ (Theorem 2.15(b),(c)). So y has the density f.
- $y \sim N_n(\mu, \Sigma)$ (Theorem 11.13), so x has the distribution of y (P2), and with it the density f.
- The λ_j are positive (Theorem 8.13(a)), and $(u-\mu)'\Sigma^{-1}(u-\mu) = d_\Sigma(u,\mu)^2$ (Definition 8.35). Part (d) of Proposition 8.36 uses only that S is positive definite; with $S = \Sigma$, $\bar{x} = \mu$ and $\sqrt{c}$ in place of c, it describes the set where this equals c. □

Example 11.16. Let $n = 2$, $\sigma_1, \sigma_2 > 0$, $-1 < \rho < 1$, and $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$. Then $\sigma_1^2 > 0$ and $\det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2) > 0$, so Σ is positive definite (Theorem 8.19), and $x \sim N_2(\mu, \Sigma)$ has $\mathrm{var}(x_j) = \sigma_j^2$ and $\mathrm{corr}(x_1, x_2) = \rho$ (Theorem 11.13). By Proposition 2.13,

$$\Sigma^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}$$

With $v_j = (u_j - \mu_j)/\sigma_j$, Theorem 11.15 gives the **bivariate normal density**

$$f(u_1, u_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\Big(-\frac{v_1^2 - 2\rho v_1 v_2 + v_2^2}{2(1-\rho^2)}\Big)$$

For instance, for $\mu = 0$, $\sigma_1 = \sigma_2 = 1$ and $\rho = \frac{1}{2}$, $f(u) = \frac{1}{\pi\sqrt{3}} \exp\Big(-\frac{2}{3}(u_1^2 - u_1u_2 + u_2^2)\Big)$. This Σ has the eigenvalues $\frac{3}{2}$, $\frac{1}{2}$ with eigenvectors $(1,1)'$, $(1,-1)'$, so the contours of f are ellipses with the long axis on the line $u_1 = u_2$ and semi-axes in the ratio $\sqrt{3} : 1$.

Theorem 11.17. (a) Let $x = (x_1', x_2')' \sim N_n(\mu, \Sigma)$, with $\mu = (\mu_1', \mu_2')'$ and $\Sigma = \left(\begin{smallmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{smallmatrix}\right)$ partitioned conformally. Then $x_1 \perp\!\!\!\perp x_2$ if and only if $\Sigma_{12} = 0$.

(b) Let $x \sim N_n(\mu, \Sigma)$, $A \in \mathbb{R}^{p\times n}$ and $B \in \mathbb{R}^{q\times n}$. Then $Ax \perp\!\!\!\perp Bx$ if and only if $A\Sigma B' = 0$.

(c) The components of $x \sim N_n(\mu, D)$ with D diagonal are independent. More generally, if $x = (x_1', \dots, x_m')' \sim N_n(\mu, \Sigma)$ and the blocks $\Sigma_{ij} = \operatorname{cov}(x_i, x_j)$ are 0 for $i \neq j$, then $x_1, \dots, x_m$ are independent.

(d) Conversely, if $x_1, \dots, x_m$ are independent and $x_i \sim N_{n_i}(\mu_i, \Sigma_i)$, then $(x_1', \dots, x_m')' \sim N_n\big(\mu, \operatorname{diag}(\Sigma_1, \dots, \Sigma_m)\big)$, with $\mu = (\mu_1', \dots, \mu_m')'$ and $n = n_1 + \cdots + n_m$.

Proof. (c) Partition $\mu = (\mu_1', \dots, \mu_m')'$ and $t = (t_1', \dots, t_m')'$ in the same way as x. By the block product (Theorem 5.3(c)), $t'\mu = \sum_i t_i'\mu_i$ and $t'\Sigma t = \sum_i \sum_j t_i'\Sigma_{ij}t_j = \sum_i t_i'\Sigma_{ii}t_i$. So

$$M_x(t) = \prod_{i=1}^m \exp\Big(t_i'\mu_i + \tfrac{1}{2}t_i'\Sigma_{ii}t_i\Big) = \prod_{i=1}^m M_{x_i}(t_i),$$

since $x_i \sim N(\mu_i, \Sigma_{ii})$ (Theorem 11.14(b)). All these MGFs are finite, and (P3) gives the independence. For $N_n(\mu, D)$ take blocks of size 1.

(a) If $\Sigma_{12} = 0$, apply (c) with $m = 2$. If $x_1 \perp\!\!\!\perp x_2$, then $\Sigma_{12} = \operatorname{cov}(x_1, x_2) = 0$ (Theorems 11.13 and 11.3(e)).

(b) By Theorem 11.14(a), $y = \left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) x$ is normal with covariance matrix $\left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) \Sigma \begin{bmatrix} A' & B' \end{bmatrix} = \left(\begin{smallmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{smallmatrix}\right)$, and its two blocks are Ax and Bx. Apply (a) to y.

(d) By (P1), $\mathrm{E}\, e^{t'x} = \mathrm{E} \prod_i e^{t_i'x_i} = \prod_i \mathrm{E}\, e^{t_i'x_i} = \exp\Big(\sum_i (t_i'\mu_i + \frac{1}{2}t_i'\Sigma_i t_i)\Big)$, which is $\exp\Big(t'\mu + \frac{1}{2}t' \operatorname{diag}(\Sigma_1, \dots, \Sigma_m)t\Big)$. The matrix $\operatorname{diag}(\Sigma_1, \dots, \Sigma_m)$ is symmetric, and positive semidefinite since $t' \operatorname{diag}(\Sigma_1, \dots, \Sigma_m)t = \sum_i t_i'\Sigma_i t_i \geqslant 0$. □

Corollary 11.18. Let $z \sim N_n(0, \sigma^2 I_n)$, and let $Q \in \mathbb{R}^{n\times n}$ be orthogonal. Then $Qz \sim N_n(0, \sigma^2 I_n)$. More generally, if $Q_1 \in \mathbb{R}^{n\times r}$ has orthonormal columns, then $Q_1'z \sim N_r(0, \sigma^2 I_r)$, and its r entries are independent with distribution $N(0, \sigma^2)$.

Proof. By Theorem 11.14(a), $Qz \sim N_n(0, \sigma^2 QQ')$ and $Q_1'z \sim N_r(0, \sigma^2 Q_1'Q_1)$, where $QQ' = I_n$ (Proposition 6.10(b)) and $Q_1'Q_1 = I_r$. The entries of $Q_1'z$ are independent by Theorem 11.17(c), and each is $N(0, \sigma^2)$ by Theorem 11.14(b). □

Remark 11.19. Joint normality cannot be dropped from Theorem 11.17(a).

- *Uncorrelated, not independent.* Let u take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $v = u^2$. Then $\mathrm{E}\, u = 0$ and $\mathrm{E}\, u^3 = 0$ (P1), so $\operatorname{cov}(u, v) = \mathrm{E}(uv) - \mathrm{E}\, u\, \mathrm{E}\, v = \mathrm{E}\, u^3 = 0$. But $\mathrm{P}(u = 0,\ v = 1) = 0 \neq \frac{1}{3} \cdot \frac{2}{3} = \mathrm{P}(u = 0)\, \mathrm{P}(v = 1)$, so u and v are not independent.
- *Normal and uncorrelated, not jointly normal.* Let $x \sim N(0, 1)$, let s be independent of x with $\mathrm{P}(s = 1) = \mathrm{P}(s = -1) = \frac{1}{2}$, and put $y = sx$. Let $\iota_\pm$ be 1 if $s = \pm 1$ and 0 otherwise, so that $\mathrm{E}\, \iota_\pm = \frac{1}{2}$ and $e^{ty} = \iota_+ e^{tx} + \iota_- e^{-tx}$. By (P1) and (P5), $\mathrm{E}\, e^{ty} = \frac{1}{2}e^{t^2/2} + \frac{1}{2}e^{t^2/2} = e^{t^2/2}$, so $y \sim N(0, 1)$; and $\operatorname{cov}(x, y) = \mathrm{E}(sx^2) = \mathrm{E}\, s\ \mathrm{E}(x^2) = 0$. If x and y were jointly normal, then $x + y$ would be normal with mean 0 and variance 2 (Theorems 11.14(b), 11.3(d)), with MGF e^{t^2}. But $x + y = (1 + s)x$, and in the same way $\mathrm{E}\, e^{t(x+y)} = \frac{1}{2}e^{2t^2} + \frac{1}{2}$, which at $t = 1$ is $\frac{1}{2}e^2 + \frac{1}{2} \neq e$. So x and y are each normal and uncorrelated, but not jointly normal, and by Theorem 11.17(d) they are not independent.

Theorem 11.20. Let $x = (x_1', x_2')' \sim N_n(\mu, \Sigma)$, partitioned as in Theorem 11.17(a), with Σ_{22} positive definite, and put

$$B = \Sigma_{12}\Sigma_{22}^{-1}, \qquad \Sigma_{11\cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

(a) $w = x_1 - Bx_2$ is independent of x_2, and $w \sim N(\mu_1 - B\mu_2, \Sigma_{11\cdot 2})$.

(b) The conditional distribution of x_1 given $x_2 = a$ is $N(\mu_1 + B(a-\mu_2), \Sigma_{11\cdot 2})$. So the conditional mean $\mathrm{E}(x_1 \mid x_2 = a) = \mu_1 + B(a-\mu_2)$ is linear in a, and the conditional covariance matrix $\mathrm{var}(x_1 \mid x_2 = a) = \Sigma_{11\cdot 2}$ does not depend on a.

Proof. (a) $w = [I \;\; -B]\,x$ and $x_2 = [0 \;\; I]\,x$. By Theorem 11.14(a), w is normal with mean $\mu_1 - B\mu_2$. By the block product, with $\Sigma_{21} = \Sigma_{12}'$ and Σ_{22}^{-1} symmetric,

$$[I \;\; -B]\,\Sigma\,[0 \;\; I]' = \Sigma_{12} - B\Sigma_{22} = 0,$$
$$[I \;\; -B]\,\Sigma\,[I \;\; -B]' = \Sigma_{11} - B\Sigma_{21} - \Sigma_{12}B' + B\Sigma_{22}B' = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

The first identity and Theorem 11.17(b) give $w \perp\!\!\!\perp x_2$; the second is the covariance matrix of w.

(b) $x_1 = w + Bx_2$ with $w \perp\!\!\!\perp x_2$. By (P6), the conditional distribution of x_1 given $x_2 = a$ is the distribution of $w + Ba$, which is $N(\mu_1 - B\mu_2 + Ba, \Sigma_{11\cdot 2})$ by Theorem 11.14(a). □

For a single variable $x_1 = y$ and $x_2 = x \in \mathbb{R}^k$, $B = \sigma_{yx}\Sigma_{xx}^{-1} = b^{*\prime}$ and $\Sigma_{11\cdot 2} = \sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}$ in the notation of Theorem 11.10: under joint normality the conditional mean $\mathrm{E}(y \mid x = a) = a^* + b^{*\prime}a$ is the best linear predictor, and the conditional variance is its mean squared error.

Example 11.21. Let $(y, x_1, x_2, x_3)'$ be normal with the mean and the covariance matrix of Example 11.11.

- Given $x = (x_1, x_2, x_3)' = a$: $B = b^{*\prime} = (1, 1, 1)$ and $\Sigma_{11\cdot 2} = 12 - 10 = 2$, so $y \mid x = a \sim N\big(8 + (a_1 - 1) + (a_2 - 2) + (a_3 - 3),\ 2\big) = N(2 + a_1 + a_2 + a_3,\ 2)$. For instance, given $x = (2, 3, 5)'$, $y \sim N(12, 2)$. Equivalently, by Theorem 11.20(a), $y = 2 + x_1 + x_2 + x_3 + u$ with $u = y - 2 - x_1 - x_2 - x_3 \sim N(0, 2)$ independent of x.
- Given $y = c$: $B = \sigma_{xy}/\sigma_{yy} = \frac{1}{12}(3, 4, 3)'$ and

$$\Sigma_{xx} - \frac{1}{12}\sigma_{xy}\sigma_{yx} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} - \frac{1}{12}\begin{pmatrix} 9 & 12 & 9 \\ 12 & 16 & 12 \\ 9 & 12 & 9 \end{pmatrix} = \begin{pmatrix} 5/4 & 0 & -3/4 \\ 0 & 2/3 & 0 \\ -3/4 & 0 & 5/4 \end{pmatrix},$$

  so $x \mid y = c \sim N_3\Big((1, 2, 3)' + \frac{c-8}{12}(3, 4, 3)',\ \Sigma_{xx} - \frac{1}{12}\sigma_{xy}\sigma_{yx}\Big)$.
- The variables x_1 and x_3 are uncorrelated, hence independent (Theorems 11.14(b) and 11.17(a)); for each value of y their conditional covariance is $-\frac{3}{4}$. The variables x_1 and x_2 have covariance 1, but for each value of y their conditional covariance is 0, so under the conditional distribution given $y = c$ they are independent (Theorems 11.14(b) and 11.17(c), applied to the subvector $(x_1, x_2)'$ of a random vector with that distribution).

Example 11.22. Let $(x, y)'$ be bivariate normal with means μ_x, μ_y, standard deviations $\sigma_x, \sigma_y > 0$ and correlation ρ. Theorem 11.20 with $x_1 = y$, $x_2 = x$ and $B = \rho\sigma_x\sigma_y/\sigma_x^2$ gives

$$y \mid x = a \;\sim\; N\Big(\mu_y + \rho\frac{\sigma_y}{\sigma_x}(a - \mu_x),\ \sigma_y^2(1 - \rho^2)\Big)$$

The line $a \mapsto \mu_y + \rho(\sigma_y/\sigma_x)(a - \mu_x)$ is the **line of conditional means** of y given x, and knowing x multiplies the variance of y by $1 - \rho^2$. In standard units,

$$\frac{\mathrm{E}(y \mid x = a) - \mu_y}{\sigma_y} = \rho\,\frac{a - \mu_x}{\sigma_x}$$

If $|\rho| < 1$ and $a \neq \mu_x$, the conditional mean of y lies fewer standard deviations from μ_y than a lies from μ_x. Galton observed this in the heights of parents and their adult children and called it **regression to the mean**. It holds in both directions: exchanging the roles of x and y gives $(\mathrm{E}(x \mid y = c) - \mu_x)/\sigma_x = \rho(c - \mu_y)/\sigma_y$.

Quadratic forms in normal vectors

Theorem 11.23. Let $z \sim N_n(0, I_n)$, and let $A \in \mathbb{R}^{n\times n}$ be symmetric and idempotent of rank $r \geqslant 1$. Then $z'Az \sim \chi^2_r$.

Proof. • By Proposition 8.10(a), $A = Q_1Q_1'$ for some $Q_1 \in \mathbb{R}^{n\times r}$ with $Q_1'Q_1 = I_r$.

• By Corollary 11.18, $w = Q_1'z$ has independent entries $w_1, \ldots, w_r$ with distribution $N(0,1)$. So $z'Az = (Q_1'z)'(Q_1'z) = w_1^2 + \cdots + w_r^2 \sim \chi^2_r$ by (P5). □

Corollary 11.24. (a) If $x \sim N_n(0, \sigma^2 I_n)$ with $\sigma > 0$, and $P \in \mathbb{R}^{n\times n}$ is symmetric and idempotent of rank $r \geqslant 1$, then $x'Px/\sigma^2 \sim \chi^2_r$.

(b) If $x \sim N_n(\mu, \Sigma)$ with Σ positive definite, then $(x-\mu)'\Sigma^{-1}(x-\mu) \sim \chi^2_n$.

Proof. (a) $z = x/\sigma \sim N_n(0, I_n)$ (Theorem 11.14(a)), and $x'Px/\sigma^2 = z'Pz$. Apply Theorem 11.23.

(b) $\Sigma^{-1/2}$ is symmetric, $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_n$ and $\Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1}$ (Theorem 8.25(c)). So $z = \Sigma^{-1/2}(x-\mu) \sim N_n(0, I_n)$ (Theorem 11.14(a)), and $(x-\mu)'\Sigma^{-1}(x-\mu) = z'I_nz \sim \chi^2_n$ by Theorem 11.23. □

In terms of the Mahalanobis distance d_Σ (Definition 8.35), (b) says that $d_\Sigma(x,\mu)^2 \sim \chi^2_n$. Hence the ellipsoid $\{u : (u-\mu)'\Sigma^{-1}(u-\mu) \leqslant \chi^2_{n,1-\alpha}\}$ of Theorem 11.15 contains x with probability $1-\alpha$. For $n = 2$, (P5) gives $\mathrm{P}\big(d_\Sigma(x,\mu)^2 \leqslant c\big) = 1 - e^{-c/2}$; this equals $1-\alpha$ for $c = 2\log(1/\alpha)$, so $\chi^2_{2,1-\alpha} = 2\log(1/\alpha)$, for instance $\chi^2_{2,1-\alpha} = 2\log 20$ for $\alpha = \frac{1}{20}$.

Theorem 11.25. Let $z \sim N_n(0, \sigma^2 I_n)$ with $\sigma > 0$.

(a) For $A \in \mathbb{R}^{p\times n}$ and $B \in \mathbb{R}^{q\times n}$, $Az \perp\!\!\!\perp Bz$ if and only if $AB' = 0$.

(b) If $P_1, P_2 \in \mathbb{R}^{n\times n}$ are symmetric and idempotent and $P_1P_2 = 0$, then $z'P_1z \perp\!\!\!\perp z'P_2z$.

(c) If $P \in \mathbb{R}^{n\times n}$ is symmetric and idempotent and $B \in \mathbb{R}^{q\times n}$ satisfies $BP = 0$, then $Bz \perp\!\!\!\perp z'Pz$.

Proof. (a) By Theorem 11.17(b), $Az \perp\!\!\!\perp Bz$ if and only if $A(\sigma^2 I_n)B' = \sigma^2 AB' = 0$.

(b) $z'P_iz = z'P_i'P_iz = \|P_iz\|^2$ is a function of P_iz, and $P_1z \perp\!\!\!\perp P_2z$ by (a), as $P_1P_2' = P_1P_2 = 0$. Apply (P1).

(c) $z'Pz = \|Pz\|^2$, and $Bz \perp\!\!\!\perp Pz$ by (a), as $BP' = BP = 0$. Apply (P1). □

Theorem 11.26 (Cochran)**.** Let $A_1, \ldots, A_k \in \mathbb{R}^{n\times n}$ be symmetric with $A_1 + \cdots + A_k = I_n$, let $r_i = \operatorname{rank} A_i \geqslant 1$, and suppose that $r_1 + \cdots + r_k = n$.

(a) Every A_i is idempotent, and $A_iA_j = 0$ for $i \neq j$.

(b) If $z \sim N_n(0, \sigma^2 I_n)$ with $\sigma > 0$, then $z'A_1z/\sigma^2, \ldots, z'A_kz/\sigma^2$ are independent, and $z'A_iz/\sigma^2 \sim \chi^2_{r_i}$.

Proof. (a)

• For each i, Corollary 8.5(a),(b) and Theorem 1.4(d) give $A_i = U_i\Lambda_iU_i'$, where Λ_i is the diagonal matrix of the r_i nonzero eigenvalues of A_i (listed with multiplicity) and the columns of $U_i \in \mathbb{R}^{n\times r_i}$ are corresponding orthonormal eigenvectors of A_i; so $U_i'U_i = I_{r_i}$, and Λ_i is invertible.

• Let $U = [U_1 \ \cdots \ U_k]$, an $n \times n$ matrix since $\sum_i r_i = n$, and $\Lambda = \operatorname{diag}(\Lambda_1, \ldots, \Lambda_k)$. By the block product, $I_n = \sum_i A_i = U\Lambda U'$. So $n = \operatorname{rank} I_n \leqslant \operatorname{rank} U$ (Theorem 4.15(a)), and U is invertible (Theorem 4.13(d)).

• The matrix $G = U'U$ is invertible (Theorem 2.15(b),(c)). Multiplying $I_n = U\Lambda U'$ on the left by U' and on the right by U gives $G = G\Lambda G$; multiplying by G^{-1} on the left gives $I_n = \Lambda G$, so $G = \Lambda^{-1} = \operatorname{diag}(\Lambda_1^{-1}, \ldots, \Lambda_k^{-1})$ (Theorem 2.18).

- The (i,j) block of G is $U_i'U_j$, and $U_i'U_i = I_{r_i}$. Comparing blocks, $\Lambda_i^{-1} = I_{r_i}$ and $U_i'U_j = 0$ for $i \neq j$. Hence $A_i = U_iU_i'$, $A_i^2 = U_i(U_i'U_i)U_i' = A_i$, and $A_iA_j = U_i(U_i'U_j)U_j' = 0$ for $i \neq j$.

(b) By (a), $U'U = G = I_n$: U is orthogonal, and $w = U'z \sim N_n(0, \sigma^2 I_n)$ (Corollary 11.18). Its blocks $w_i = U_i'z \in \mathbb{R}^{r_i}$ have zero cross-covariances, so they are independent (Theorem 11.17(c)), and $w_i \sim N_{r_i}(0, \sigma^2 I_{r_i})$ (Theorem 11.14(b)). Now $z'A_iz = z'U_iU_i'z = \|w_i\|^2$, and $\|w_i\|^2/\sigma^2 \sim \chi^2_{r_i}$ (Corollary 11.24(a) with $P = I_{r_i}$). These are functions of the independent w_i, hence independent (P1). □

Theorem 11.27. Let $x_1, \dots, x_n$ be independent with distribution $N(\mu, \sigma^2)$, where $n \geqslant 2$ and $\sigma > 0$, and let $\bar{x} = \frac{1}{n}\sum_i x_i$ and $s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$. Then

$$\bar{x} \sim N\Big(\mu, \frac{\sigma^2}{n}\Big), \qquad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \bar{x} \perp\!\!\!\perp s^2, \qquad \frac{\sqrt{n}(\bar{x}-\mu)}{s} \sim t_{n-1}$$

Proof. • By Theorem 11.17(d), $x = (x_1, \dots, x_n)' \sim N_n(\mu\mathbf{1}, \sigma^2 I_n)$. So $\varepsilon = x - \mu\mathbf{1} \sim N_n(0, \sigma^2 I_n)$ and $\bar{x} = \frac{1}{n}\mathbf{1}'x \sim N(\mu, \sigma^2/n)$ (Theorem 11.14(a),(b)).

- $(n-1)s^2 = x'C_nx$ (Example 1.14), and $C_n\mathbf{1} = 0$, so $(n-1)s^2 = \varepsilon' C_n \varepsilon$. As C_n is symmetric and idempotent of rank $n - 1 \geqslant 1$ (Proposition 1.13, Corollary 6.21(b)), Corollary 11.24(a) gives $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$.
- $\bar{x} - \mu = \frac{1}{n}\mathbf{1}'\varepsilon$, and $\frac{1}{n}\mathbf{1}'C_n = \frac{1}{n}(C_n\mathbf{1})' = 0$. By Theorem 11.25(c), $\bar{x} - \mu \perp\!\!\!\perp \varepsilon' C_n \varepsilon$, hence $\bar{x} \perp\!\!\!\perp s^2$ (P1).
- So $u = \sqrt{n}(\bar{x} - \mu)/\sigma \sim N(0,1)$ and $v = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ are independent (P1), and $u/\sqrt{v/(n-1)} = \sqrt{n}(\bar{x}-\mu)/s \sim t_{n-1}$ by (P5). (Here $s > 0$ with probability 1, since $\mathrm{P}(v = 0) = 0$ by (P5).) □

Example 11.28. A **confidence interval** with level $1-\alpha$ $(0 < \alpha < 1)$ for a parameter θ is an interval $[L, U]$ whose endpoints are computed from the observations and satisfy $\mathrm{P}(L \leqslant \theta \leqslant U) = 1 - \alpha$ for all values of the parameters. In Theorem 11.27, let $u = \sqrt{n}(\bar{x} - \mu)/\sigma \sim N(0,1)$ and $v = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$, independent by Theorem 11.27 and (P1), and let $T = \sqrt{n}(\bar{x} - \mu)/s = u/\sqrt{v/(n-1)}$ and $c = t_{n-1,1-\alpha/2}$.

- $-T$ also has the distribution t_{n-1}: $-u \sim N(0,1)$, since $\mathrm{E}\, e^{\lambda(-u)} = e^{(-\lambda)^2/2} = e^{\lambda^2/2}$ for all $\lambda \in \mathbb{R}$, and $-u \perp\!\!\!\perp v$ (P1), so $-T = (-u)/\sqrt{v/(n-1)} \sim t_{n-1}$ by (P5).
- Here $c > 0$: $\mathrm{P}(T \leqslant 0) = \mathrm{P}(-T \leqslant 0) = \mathrm{P}(T \geqslant 0)$, and these two numbers add up to $1 + \mathrm{P}(T = 0) = 1$ (P5), so $\mathrm{P}(T \leqslant 0) = \frac{1}{2} < 1 - \frac{\alpha}{2} = \mathrm{P}(T \leqslant c)$, whereas $c \leqslant 0$ would give $\{T \leqslant c\} \subseteq \{T \leqslant 0\}$. Hence $\{T > c\}$ and $\{T < -c\}$ are disjoint. By (P5), $\mathrm{P}(T > c) = 1 - \mathrm{P}(T \leqslant c) = \frac{\alpha}{2}$, and $\mathrm{P}(T < -c) = \mathrm{P}(-T > c) = \frac{\alpha}{2}$. So $\mathrm{P}(-c \leqslant T \leqslant c) = 1 - \alpha$.
- $-c \leqslant T \leqslant c$ if and only if $\bar{x} - cs/\sqrt{n} \leqslant \mu \leqslant \bar{x} + cs/\sqrt{n}$. So $\big[\bar{x} - t_{n-1,1-\alpha/2}\, s/\sqrt{n},\ \bar{x} + t_{n-1,1-\alpha/2}\, s/\sqrt{n}\big]$ is a confidence interval for μ with level $1 - \alpha$.
- Let $c_1 = \chi^2_{n-1,\alpha/2}$ and $c_2 = \chi^2_{n-1,1-\alpha/2}$. By (P5), $\mathrm{P}(c_1 \leqslant v \leqslant c_2) = \mathrm{P}(v \leqslant c_2) - \mathrm{P}(v < c_1) = (1 - \frac{\alpha}{2}) - \frac{\alpha}{2} = 1 - \alpha$, since $\mathrm{P}(v = c_1) = 0$. As $c_1 \leqslant v \leqslant c_2$ if and only if $(n-1)s^2/c_2 \leqslant \sigma^2 \leqslant (n-1)s^2/c_1$, $\big[(n-1)s^2/\chi^2_{n-1,1-\alpha/2},\ (n-1)s^2/\chi^2_{n-1,\alpha/2}\big]$ is a confidence interval for σ^2 with level $1 - \alpha$.

For the nine measurements $5, 7, 9, 9, 10, 11, 11, 13, 15$, modeled as independent $N(\mu, \sigma^2)$: $\bar{x} = 10$, $\sum_i (x_i - \bar{x})^2 = 25 + 9 + 1 + 1 + 0 + 1 + 1 + 9 + 25 = 72$, $s^2 = 72/8 = 9$ and $s = 3$. So $s/\sqrt{n} = 1$, and the interval with level $1 - \alpha$ is $\big[10 - t_{8,1-\alpha/2},\ 10 + t_{8,1-\alpha/2}\big]$.

Example 11.29. Two dimensions of each of $n = 25$ machined parts are measured; the measurement vectors $x_1, \dots, x_n$ are modeled as independent $N_2(\mu, \Sigma)$, with $\Sigma = \left(\begin{smallmatrix} 4 & 3 \\ 3 & 9 \end{smallmatrix}\right)$ known from long experience and μ unknown. So the standard deviations are 2 and 3, and the correlation is $\frac{1}{2}$.

- By Theorem 11.17(d), $u = (x_1', \dots, x_n')'$ is normal with covariance matrix $\operatorname{diag}(\Sigma, \dots, \Sigma)$, and $\bar{x} = \frac{1}{n} G u$ with $G = [I_2 \ \cdots \ I_2]$. As in the proof of Proposition 11.9, Theorem 11.14(a) gives $\bar{x} \sim N_2(\mu, \frac{1}{n}\Sigma)$. The standard deviations of $\bar{x}_1$ and $\bar{x}_2$ are $\frac{2}{5}$ and $\frac{3}{5}$.
- Σ is positive definite (minors 4, 27; Theorem 8.19). By Corollary 11.24(b), with $(\frac{1}{n}\Sigma)^{-1} = n\Sigma^{-1}$, $q(\mu) = n(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu) \sim \chi^2_2$, so $\mathrm{P}\big(q(\mu) \leqslant \chi^2_{2,1-\alpha}\big) = 1 - \alpha$. Thus the ellipse $\{m \in \mathbb{R}^2 : n(\bar{x} - m)'\Sigma^{-1}(\bar{x} - m) \leqslant 2\log(1/\alpha)\}$, computed from the observations, contains μ with probability $1 - \alpha$ for every μ: a **confidence region** with level $1 - \alpha$.
- Let $\alpha = \frac{1}{20}$, and compare two possible target values m_1, m_2 with $\bar{x} - m_1 = (\frac{4}{5}, \frac{6}{5})'$ and $\bar{x} - m_2 = (\frac{4}{5}, -\frac{6}{5})'$. Each coordinate of $\bar{x}$ is two of its standard deviations away from the corresponding coordinate of m_1, and also of m_2. With $\Sigma^{-1} = \frac{1}{27}\left(\begin{smallmatrix} 9 & -3 \\ -3 & 4 \end{smallmatrix}\right)$ and $d = 5(\bar{x} - m)$, $q(m) = 25(\bar{x} - m)'\Sigma^{-1}(\bar{x} - m) = d'\Sigma^{-1}d = \frac{1}{27}(9d_1^2 - 6d_1d_2 + 4d_2^2)$, where $d = (4, 6)'$ for m_1 and $d = (4, -6)'$ for m_2:

$$q(m_1) = \frac{144 - 144 + 144}{27} = \frac{16}{3}, \qquad q(m_2) = \frac{144 + 144 + 144}{27} = 16$$

  As $e < 3$, $e^8 < 3^8 = 6561 < 8000 = 20^3$, so $e^{8/3} < 20$ and $q(m_1) = \frac{16}{3} < 2\log 20$: m_1 lies inside the confidence ellipse. As $e > 2$, $e^8 > 256 > 20$, so $q(m_2) = 16 > 2\log 20$: m_2 lies outside. The two measurements are positively correlated, so deviations of the same sign are common and deviations of opposite signs are unusual; only the joint region takes this into account.

Maximum likelihood for the multivariate normal

Definition 11.30. Let observations be modeled as a random vector x with a density f_θ that depends on a parameter θ ranging over a set Θ. For the observed value of x, the function $L(\theta) = f_\theta(x)$ $(\theta \in \Theta)$ is the **likelihood**, $\ell(\theta) = \log L(\theta)$ is the **log-likelihood**, and a $\hat{\theta} \in \Theta$ with $L(\hat{\theta}) \geqslant L(\theta)$ for all $\theta \in \Theta$ is a **maximum likelihood estimate** (MLE) of θ.

Theorem 11.31. Let $x_1, \dots, x_n \in \mathbb{R}^p$ be observations modeled as independent with distribution $N_p(\mu, \Sigma)$, where $\mu \in \mathbb{R}^p$ and the symmetric positive definite Σ are unknown. Let

$$\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i, \qquad S_n = \frac{1}{n}\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})',$$

and suppose that S_n is positive definite (this requires $n \geqslant p + 1$, Proposition 8.33(b)).

(a) The log-likelihood is

$$\ell(\mu, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\Big(\log\det\Sigma + \operatorname{tr}(\Sigma^{-1}S_n) + (\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu)\Big)$$

(b) $\ell(\mu, \Sigma) \leqslant \ell(\bar{x}, S_n) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log\det S_n - \frac{np}{2}$, with equality only for $\mu = \bar{x}$ and $\Sigma = S_n$. As log is strictly increasing, $(\bar{x}, S_n)$ is the unique MLE of (μ, Σ).

(c) Regarded as random vectors, $\bar{x}$ and S_n satisfy $\mathrm{E}\,\bar{x} = \mu$ and $\mathrm{E}\,S_n = \frac{n-1}{n}\Sigma$: the MLE of Σ is not unbiased, while $S = \frac{n}{n-1}S_n$ is.

Proof. (a)

- By Theorem 11.15 and (P4), the stacked vector $(x_1', \dots, x_n')'$ has the density $\prod_i f(x_i)$, with f the $N_p(\mu, \Sigma)$ density; so

$$\ell(\mu, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log\det\Sigma - \frac{1}{2}\sum_{i=1}^n (x_i - \mu)'\Sigma^{-1}(x_i - \mu)$$

- Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$. As $\sum_i (x_i - \bar{x}) = 0$, the cross terms add up to 0:

$$\sum_i (x_i - \mu)'\Sigma^{-1}(x_i - \mu) = \sum_i (x_i - \bar{x})'\Sigma^{-1}(x_i - \bar{x}) + n(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu)$$

- $(x_i - \bar{x})'\Sigma^{-1}(x_i - \bar{x}) = \operatorname{tr}\Big(\Sigma^{-1}(x_i - \bar{x})(x_i - \bar{x})'\Big)$ (Theorem 1.11(b)), and the sum over i is $\operatorname{tr}(\Sigma^{-1} n S_n)$.

(b)

- Σ^{-1} is positive definite (Corollary 8.15(a)), so the last term in (a) is $\geqslant 0$, with equality only for $\mu = \bar{x}$. It remains to show that $g(\Sigma) = \log\det\Sigma + \operatorname{tr}(\Sigma^{-1} S_n) \geqslant \log\det S_n + p$, with equality only for $\Sigma = S_n$.
- This is Problem 9.7(b),(c), with p in place of n and the positive definite $S = S_n$: its function $\varphi(X) = \log\det X + \operatorname{tr}(X^{-1} S_n)$ has $\varphi(\Sigma) = g(\Sigma)$ and $\varphi(S_n) = \log\det S_n + \operatorname{tr} I_p = \log\det S_n + p$.

(c) The x_i are independent, hence uncorrelated (Theorem 11.3(e)), with mean μ and covariance matrix Σ (Theorem 11.13). By Proposition 11.9, $\mathrm{E}\,\bar{x} = \mu$ and $\mathrm{E}\,S_n = \frac{n-1}{n}\,\mathrm{E}\,S = \frac{n-1}{n}\Sigma$. □

Remark 11.32. On the open set $U = \{X \in \mathbb{R}^{p\times p} : \det X > 0\}$, the function $g(X) = \log\det X + \operatorname{tr}(X^{-1} S_n)$ has $\partial g/\partial X = (X^{-1} - X^{-1} S_n X^{-1})'$ (Problem 9.7(a) with $S = S_n$). As $X^{-1} - X^{-1} S_n X^{-1} = X^{-1}(X - S_n)X^{-1}$, this derivative is 0 only at $X = S_n$. A stationary point alone does not show that S_n minimizes g over the positive definite matrices; the proof of Theorem 11.31(b) does. Likewise, by Theorem 11.31(a), ℓ depends on μ only through

$$-\frac{n}{2}(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu) = -\frac{n}{2}\Big(\mu'\Sigma^{-1}\mu - 2\bar{x}'\Sigma^{-1}\mu + \bar{x}'\Sigma^{-1}\bar{x}\Big),$$

whose gradient with respect to μ is $n\Sigma^{-1}(\bar{x} - \mu)$ (Theorem 9.2); it vanishes only at $\mu = \bar{x}$.

Example 11.33. Model the four observations $(3,5,6)$, $(3,3,2)$, $(1,3,4)$, $(1,1,4)$ as independent from $N_3(\mu, \Sigma)$ ($n = 4$, $p = 3$). Then $\bar{x} = (2,3,4)'$, and the deviations $x_i - \bar{x}$ are the rows of

$$D = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 0 & -2 \\ -1 & 0 & 0 \\ -1 & -2 & 0 \end{pmatrix}, \qquad nS_n = D'D = \begin{pmatrix} 4 & 4 & 0 \\ 4 & 8 & 4 \\ 0 & 4 & 8 \end{pmatrix}, \qquad S_n = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Here $S_n = LL'$ with $L = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right)$, so S_n is positive definite (Theorem 8.22) and $\det S_n = (\det L)^2 = 1$. By Theorem 11.31, the MLE is $\hat{\mu} = (2,3,4)'$, $\hat{\Sigma} = S_n$, and the largest value of the log-likelihood is $-6\log(2\pi) - 2\log 1 - 6 = -6(1 + \log 2\pi)$. The unbiased estimate of Σ is $S = \frac{4}{3}S_n$. For $p = 1$, Theorem 11.31 says that for independent observations from $N(\mu, \sigma^2)$ the MLE is $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \frac{n-1}{n}s^2$, when the x_i are not all equal.

Exercises

Exercise 11.1. Let $x \in \mathbb{R}^4$ have $\mathrm{E}\,x = \mu = (1,0,2,-1)'$ and

$$\operatorname{var}(x) = \Sigma = \begin{pmatrix} 9 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 9 & 5 \\ 3 & 1 & 5 & 4 \end{pmatrix}$$

(a) Show that $\Sigma = LL'$ for $L = \left(\begin{smallmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{smallmatrix}\right)$, and deduce that no linear combination $a'x$ with $a \neq 0$ is constant with probability 1.

(b) Find $\mathrm{E}\,y$ and $\mathrm{var}(y)$ for $y = (x_1 - x_2,\ x_3 - x_4 - 3)'$.

(c) Find $\mathrm{cov}(x_1 + x_3,\ x_2 - x_4)$ and $\mathrm{corr}(x)$.

Solution. (a) The (i,j) entry of LL' is $\mathrm{row}_i(L)\,\mathrm{row}_j(L)'$ (Theorem 1.4(d)). With the rows $(3,0,0,0)$, $(0,1,0,0)$, $(1,2,2,0)$, $(1,1,1,1)$ this gives $9,0,3,3$ in row 1; $1,2,1$ at $(2,2)$, $(2,3)$, $(2,4)$; $1+4+4=9$ and $1+2+2=5$ at $(3,3)$, $(3,4)$; and 4 at $(4,4)$. So $LL' = \Sigma$. As L is lower triangular with positive diagonal entries, Σ is positive definite (Theorem 8.22), hence invertible (Corollary 8.15(a)), and by Theorem 11.4(b) no $a'x$ with $a \neq 0$ is constant with probability 1.

(b) $y = Ax + b$ with $A = \left(\begin{smallmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{smallmatrix}\right)$ and $b = (0,-3)'$. By Theorem 11.3(a),(b), $\mathrm{E}\,y = A\mu + b = (1-0,\ 2+1-3)' = (1,0)'$, and the rows of $A\Sigma$ are $\mathrm{row}_1(\Sigma) - \mathrm{row}_2(\Sigma)$ and $\mathrm{row}_3(\Sigma) - \mathrm{row}_4(\Sigma)$:

$$A\Sigma = \begin{pmatrix} 9 & -1 & 1 & 2 \\ 0 & 1 & 4 & 1 \end{pmatrix}, \qquad \mathrm{var}(y) = A\Sigma A' = \begin{pmatrix} 10 & -1 \\ -1 & 3 \end{pmatrix}$$

(c) $\Sigma(0,1,0,-1)' = \mathrm{col}_2(\Sigma) - \mathrm{col}_4(\Sigma) = (-3,0,-3,-3)'$, so by Theorem 11.3(b) $\mathrm{cov}(x_1+x_3,\ x_2-x_4) = (1,0,1,0)(-3,0,-3,-3)' = -6$. The standard deviations are $3,1,3,2$, so $D^{-1/2} = \mathrm{diag}(\frac{1}{3}, 1, \frac{1}{3}, \frac{1}{2})$ and

$$\mathrm{corr}(x) = D^{-1/2}\Sigma D^{-1/2} = \begin{pmatrix} 1 & 0 & 1/3 & 1/2 \\ 0 & 1 & 2/3 & 1/2 \\ 1/3 & 2/3 & 1 & 5/6 \\ 1/2 & 1/2 & 5/6 & 1 \end{pmatrix}$$

Exercise 11.2. Let x be as in Exercise 11.1.

(a) Use Theorem 11.6 to find $\mathrm{E}(x_1x_3 + x_2^2)$ and $\mathrm{E}\|x\|^2$, and check the first value directly from the means, variances and covariances.

(b) Find the best linear predictor of x_1 from $(x_2, x_3, x_4)'$ and its mean squared error. Compare with the best linear predictors from x_2 alone and from x_4 alone.

(c) Check directly that the prediction error u of (b) has mean 0, that $\mathrm{cov}(u, x_2) = \mathrm{cov}(u, x_3) = \mathrm{cov}(u, x_4) = 0$, and that $\mathrm{var}(u)$ is the mean squared error.

Solution. (a) $x_1x_3 + x_2^2 = x'Ax$ with $A = e_1e_3' + e_2e_2'$. The rows of $A\Sigma$ are $\mathrm{row}_3(\Sigma)$, $\mathrm{row}_2(\Sigma)$, 0, 0, so $\mathrm{tr}(A\Sigma) = \sigma_{31} + \sigma_{22} = 3 + 1 = 4$; and $\mu'A\mu = \mu_1\mu_3 + \mu_2^2 = 2$. Hence $\mathrm{E}(x_1x_3 + x_2^2) = 4 + 2 = 6$. Directly, by Theorem 11.3(c), $\mathrm{E}(x_1x_3) = \mathrm{cov}(x_1, x_3) + \mu_1\mu_3 = 3 + 2 = 5$ and $\mathrm{E}\,x_2^2 = \mathrm{var}(x_2) + \mu_2^2 = 1$, with sum 6. With $A = I_4$, $\mathrm{E}\|x\|^2 = \mathrm{tr}\,\Sigma + \mu'\mu = 23 + 6 = 29$.

(b) In Theorem 11.10 take $y = x_1$ and $(x_2, x_3, x_4)'$ as the predictors:

$$\Sigma_{xx} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 9 & 5 \\ 1 & 5 & 4 \end{pmatrix}, \qquad \sigma_{xy} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}, \qquad \Sigma_{xx} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$$

Σ_{xx} is a principal submatrix of the positive definite Σ, so it is positive definite and invertible (Corollary 8.15(a)), and $b^* = \Sigma_{xx}^{-1}\sigma_{xy} = (-1,0,1)'$. Then $a^* = \mu_1 - b^{*\prime}(\mu_2, \mu_3, \mu_4)' = 1 - (0+0-1) = 2$, the best linear predictor is $2 - x_2 + x_4$, and its mean squared error is $9 - (0,3,3)(-1,0,1)' = 6$. From x_4 alone the minimum is $9 - 3^2/4 = \frac{27}{4}$. As $\mathrm{cov}(x_1, x_2) = 0$, the best linear predictor from x_2 alone is the constant $\mu_1 = 1$, with mean squared error $\mathrm{var}(x_1) = 9$: alone, x_2 is useless, but together with x_4 it receives the coefficient -1.

(c) $u = x_1 - 2 + x_2 - x_4 = c'x - 2$ with $c = (1,1,0,-1)'$. Then $\mathrm{E}\,u = c'\mu - 2 = 1 + 0 + 0 + 1 - 2 = 0$, and $\Sigma c = \mathrm{col}_1(\Sigma) + \mathrm{col}_2(\Sigma) - \mathrm{col}_4(\Sigma) = (6,0,0,0)'$. By Theorem 11.3(b), $\mathrm{cov}(u, x_j) = c'\Sigma e_j$ is entry j of Σc, which is 0 for $j = 2,3,4$, and $\mathrm{var}(u) = c'\Sigma c = 6$.

Exercise 11.3. Let $x \sim N_4(\mu, \Sigma)$ with $\mu = (1, 2, 0, -1)'$ and

$$\Sigma = \begin{pmatrix} 3 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

(a) Find the distributions of $y = (x_1 - x_2,\ x_3 - x_4)'$ and of $(x_1, x_4)'$.

(b) Which of the six pairs $\{x_i, x_j\}$ are independent? Is x_1 independent of $(x_3, x_4)'$?

(c) Find c such that $x_1 - cx_3$ and x_3 are independent.

(d) Find the conditional distribution of $(x_1, x_2)'$ given $(x_3, x_4)' = (a_3, a_4)'$, and check that $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11\cdot 2}$.

Solution. (a) $y = Ax$ with $A = \left(\begin{smallmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{smallmatrix}\right)$. The rows of $A\Sigma$ are $\mathrm{row}_1(\Sigma) - \mathrm{row}_2(\Sigma) = (2, -2, 0, -1)$ and $\mathrm{row}_3(\Sigma) - \mathrm{row}_4(\Sigma) = (1, 0, 1, 0)$, so by Theorem 11.14(a)

$$y \sim N_2\left(\begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}\right), \qquad \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\right),$$

the second by Theorem 11.14(b).

(b) Each pair forms a normal subvector of x (Theorem 11.14(b)), so Theorem 11.17(a) applies to it: $x_1 \perp\!\!\!\perp x_4$, as $\sigma_{14} = 0$, and the other five pairs are not independent, as their covariances are not 0. The vector $(x_1, x_3, x_4)'$ is normal, and the cross-covariance of x_1 and $(x_3, x_4)'$ is $(\sigma_{13}, \sigma_{14}) = (1, 0) \neq 0$; so x_1 is not independent of $(x_3, x_4)'$.

(c) By Theorem 11.17(b) with $A = (1, 0, -c, 0)$ and $B = e_3'$, $A\Sigma B' = \sigma_{13} - c\sigma_{33} = 1 - 2c$, which is 0 exactly for $c = \frac{1}{2}$.

(d) In Theorem 11.20, the first block is $(x_1, x_2)'$ and the second $(x_3, x_4)'$:

$$\Sigma_{22} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Sigma_{22}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \Sigma_{12} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad B = \Sigma_{12}\Sigma_{22}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

with Σ_{22}^{-1} from Proposition 2.13 ($\det \Sigma_{22} = 1$), and

$$\Sigma_{11\cdot 2} = \Sigma_{11} - B\Sigma_{21} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

The conditional mean is $(1, 2)' + B(a_3 - 0,\ a_4 + 1)' = (a_3 - a_4,\ a_4 + 3)'$, so $(x_1, x_2)' \mid (x_3, x_4)' = (a_3, a_4)' \sim N_2\big((a_3 - a_4,\ a_4 + 3)',\ \Sigma_{11\cdot 2}\big)$. Check: the leading principal minors of Σ are 3, 8, 12 and $\det \Sigma = 3$, and $\det \Sigma_{22} \det \Sigma_{11\cdot 2} = 1 \cdot 3$ (Theorem 5.26(b)).

Exercise 11.4. A random vector $x \in \mathbb{R}^2$ has the density

$$f(u) = \frac{1}{2\pi} \exp\Big(-\frac{1}{2}\big(u_1^2 - 2u_1u_2 + 2u_2^2 - 2u_2 + 1\big)\Big)$$

Show that x is bivariate normal, and find μ, Σ, $\mathrm{corr}(x_1, x_2)$ and the conditional distribution of x_1 given $x_2 = a$.

Solution. Let $P = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and $\mu = (1, 1)'$. Then $u'Pu = u_1^2 - 2u_1u_2 + 2u_2^2$, $P\mu = (0, 1)'$ and $\mu' P \mu = 1$, so

$$(u - \mu)'P(u - \mu) = u'Pu - 2(P\mu)'u + \mu' P \mu = u_1^2 - 2u_1u_2 + 2u_2^2 - 2u_2 + 1$$

The matrix $\Sigma = P^{-1} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (Proposition 2.13) is positive definite (minors 2, 1; Theorem 8.19), and $\det \Sigma = 1$. So f is the $N_2(\mu, \Sigma)$ density of Theorem 11.15. Hence $\mathrm{P}(x \in A) = \int_A f = \mathrm{P}(y \in A)$ for $y \sim N_2(\mu, \Sigma)$ and all A: $x \sim N_2\big((1, 1)', \Sigma\big)$. Then $\mathrm{corr}(x_1, x_2) = 1/\sqrt{2 \cdot 1} = 1/\sqrt{2}$. By Theorem 11.20, $B = \sigma_{12}/\sigma_{22} = 1$ and $\Sigma_{11\cdot 2} = 2 - 1 = 1$, so $x_1 \mid x_2 = a \sim N(1 + (a - 1),\ 1) = N(a, 1)$.

Exercise 11.5. A program produces independent $N(0,1)$ numbers.

(a) Let $L \in \mathbb{R}^{4\times 4}$ have $l_{ij} = 1$ for $j \leqslant i$ and $l_{ij} = 0$ for $j > i$. Show that L is the Cholesky factor of $\Sigma = (\min(i,j))_{i,j\leqslant 4}$, and that $x = \mu + Lz \sim N_4(\mu, \Sigma)$ when z consists of four of the numbers. Describe $x_k - \mu_k$ in terms of z.

(b) Compute x for $\mu = (1, 0, -1, 0)'$ and $z = (1, -2, 2, 1)'$.

Solution. (a) L is lower triangular with positive diagonal entries. The (i,j) entry of LL' is $\operatorname{row}_i(L)\operatorname{row}_j(L)'$ (Theorem 1.4(d)); $\operatorname{row}_i(L)$ has ones in positions $1,\dots,i$, so this product counts the positions $k \leqslant \min(i,j)$ and equals $\min(i,j)$. So

$$LL' = \Sigma = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix},$$

and by Theorem 8.22 L is the Cholesky factor of Σ; Theorem 11.13 gives $\mu + Lz \sim N_4(\mu, LL')$. Here $x_k - \mu_k = z_1 + \cdots + z_k$: the partial sums of the numbers, a random walk with $N(0,1)$ steps.
(b) $Lz = (1,\ 1-2,\ 1-2+2,\ 1-2+2+1)' = (1, -1, 1, 2)'$, so $x = (2, -1, 0, 2)'$.

Exercise 11.6. Let $x \sim N_4(\mu, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 4 & 1 & 2 & 1 \\ 1 & 4 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

The correlation computed from the conditional covariance matrix of $(x_1, x_2)'$ given some of the other variables is the **partial correlation** of x_1 and x_2 given those variables.

(a) Show that Σ is positive definite. Find the conditional distribution of $(x_1, x_2)'$ given $(x_3, x_4)' = (a_3, a_4)'$, and compare the partial correlation of x_1 and x_2 given x_3, x_4 with $\operatorname{corr}(x_1, x_2)$.

(b) Find the partial correlation of x_1 and x_2 given x_3 alone, and given x_4 alone.

(c) Let $u = (x_1, x_2)' - B(x_3, x_4)'$, with B as in (a). Compute $\operatorname{var}(u)$ from Theorem 11.3(b), and compare with (a).

Solution. (a) The leading principal minors of Σ are 4, 15, 6 and $\det\Sigma = 3$, all positive (Theorem 8.19). In Theorem 11.20 with first block $(x_1, x_2)'$ and second block $(x_3, x_4)'$: $\Sigma_{22} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $\Sigma_{22}^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ (Proposition 2.13), $\Sigma_{12} = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$B = \Sigma_{12}\Sigma_{22}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \Sigma_{11\cdot 2} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

So $(x_1, x_2)' \mid (x_3, x_4)' = (a_3, a_4)' \sim N_2\big((\mu_1, \mu_2)' + (a_3 - \mu_3)(1,1)',\ \Sigma_{11\cdot 2}\big)$. The partial correlation is $-1/\sqrt{2\cdot 2} = -\frac{1}{2}$, while $\operatorname{corr}(x_1, x_2) = 1/\sqrt{4\cdot 4} = \frac{1}{4}$: the sign changes.
(b) Given x_3: $\Sigma_{11} - \frac{1}{2}(2,2)'(2,2) = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, partial correlation $-\frac{1}{2}$. Given x_4: $\Sigma_{11} - (1,1)'(1,1) = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, partial correlation 0.
(c) $u = (x_1 - x_3,\ x_2 - x_3)' = Kx$ with $K = \left(\begin{smallmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{smallmatrix}\right)$. The rows of $K\Sigma$ are $(2, -1, 0, 0)$ and $(-1, 2, 0, 0)$, so $\operatorname{var}(u) = K\Sigma K' = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right) = \Sigma_{11\cdot 2}$. This used only the covariance matrix: u_1 and u_2 are the prediction errors of the best linear predictors of x_1 and x_2 from $(x_3, x_4)'$ (Theorem 11.10), and the partial correlation is their correlation.

Exercise 11.7. Let $y \sim N_4(\mu, \sigma^2 I_4)$ with $\sigma > 0$, and

$$P = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

(a) Show that P and $I_4 - P$ are symmetric and idempotent of rank 2, that $y'Py = \frac{1}{2}(y_1 + y_2)^2 + \frac{1}{2}(y_3 + y_4)^2$, and that $y'(I_4 - P)y = \frac{1}{2}(y_1 - y_2)^2 + \frac{1}{2}(y_3 - y_4)^2$.

(b) Find $\mathrm{E}(y'Py)$ and $\mathrm{E}\big(y'(I_4 - P)y\big)$ for $\mu = (1, 3, 5, 7)'$.

(c) Let $\mu = (m_1, m_1, m_2, m_2)'$. Show that $y'(I_4 - P)y/\sigma^2 \sim \chi^2_2$, that $y'(I_4 - P)y$ is independent of Py, and that $\frac{1}{2}y'(I_4 - P)y$ is an unbiased estimator of σ^2.

Solution. (a) $P = \frac{1}{2}\operatorname{diag}(J_2, J_2)$ is symmetric, and $J_2^2 = 2J_2$ (Proposition 1.13), so $P^2 = \frac{1}{4}\operatorname{diag}(J_2^2, J_2^2) = \frac{1}{2}\operatorname{diag}(J_2, J_2) = P$ (Theorem 5.3(c)). By Theorem 6.16(b),(c), $I_4 - P$ is idempotent, $\operatorname{rank} P = \operatorname{tr} P = 2$ and $\operatorname{rank}(I_4 - P) = \operatorname{tr}(I_4 - P) = 2$. As $Py = \frac{1}{2}(y_1 + y_2,\ y_1 + y_2,\ y_3 + y_4,\ y_3 + y_4)'$, $y'Py = \frac{1}{2}(y_1+y_2)^2+\frac{1}{2}(y_3+y_4)^2$; and $y'(I_4-P)y = y'y-y'Py$, where $y_1^2+y_2^2-\frac{1}{2}(y_1+y_2)^2 = \frac{1}{2}(y_1-y_2)^2$, and likewise for y_3, y_4.

(b) By Example 11.8 with $r = 2$, $\mathrm{E}(y'Py) = 2\sigma^2 + \mu'P\mu = 2\sigma^2 + \frac{1}{2}\cdot 4^2 + \frac{1}{2}\cdot 12^2 = 2\sigma^2 + 80$ and $\mathrm{E}\big(y'(I_4 - P)y\big) = 2\sigma^2 + \frac{1}{2}\cdot 2^2 + \frac{1}{2}\cdot 2^2 = 2\sigma^2 + 4$.

(c) Let $\varepsilon = y - \mu \sim N_4(0, \sigma^2 I_4)$ (Theorem 11.14(a)). Here $(I_4 - P)\mu = 0$, so $(I_4 - P)y = (I_4 - P)\varepsilon$ and, as $I_4 - P$ is symmetric and idempotent, $y'(I_4 - P)y = \|(I_4 - P)y\|^2 = \varepsilon'(I_4 - P)\varepsilon$. Corollary 11.24(a) gives $y'(I_4 - P)y/\sigma^2 \sim \chi^2_2$. As $P(I_4 - P) = P - P^2 = 0$, Theorem 11.25(c) gives $P\varepsilon \perp\!\!\!\perp \varepsilon'(I_4 - P)\varepsilon$, and $Py = P\mu + P\varepsilon$ is a function of $P\varepsilon$ (P1). $\mathrm{E}\big(y'(I_4 - P)y\big) = 2\sigma^2 + 0$ (Example 11.8), so $\frac{1}{2}y'(I_4 - P)y$ is unbiased for σ^2 for all values of m_1, m_2, σ^2 (Definition 11.7).

Exercise 11.8. Let $z \sim N_4(0, \sigma^2 I_4)$ with $\sigma > 0$, and

$$q_1 = \tfrac{1}{4}(z_1 + z_2 + z_3 + z_4)^2, \quad q_2 = \tfrac{1}{2}(z_1 - z_2)^2, \quad q_3 = \tfrac{1}{6}(z_1 + z_2 - 2z_3)^2, \quad q_4 = \tfrac{1}{12}(z_1 + z_2 + z_3 - 3z_4)^2$$

(a) Write $q_i = z'A_i z$ with A_i symmetric of rank 1, and show that $A_1 + A_2 + A_3 + A_4 = I_4$.

(b) Show that $q_1/\sigma^2, \ldots, q_4/\sigma^2$ are independent with distribution χ^2_1.

(c) Show that $q_2 + q_3 + q_4 = \sum_i (z_i - \bar{z})^2$, that it is independent of $\bar{z}$, and that $(q_2 + q_3 + q_4)/\sigma^2 \sim \chi^2_3$.

Solution. (a) Let $a_1 = \mathbf{1}_4$, $a_2 = (1, -1, 0, 0)'$, $a_3 = (1, 1, -2, 0)'$ and $a_4 = (1, 1, 1, -3)'$, with $a_i'a_i = 4, 2, 6, 12$. Then $q_i = (a_i'z)^2/(a_i'a_i) = z'A_i z$ for $A_i = a_i a_i'/(a_i'a_i)$, which is symmetric of rank 1 (its columns are multiples of $a_i \neq 0$, not all 0). The six products $a_i'a_j$ $(i < j)$ are 0: for instance $a_3'a_4 = 1 + 1 - 2 + 0 = 0$. So the vectors $u_i = a_i/\sqrt{a_i'a_i}$ are orthonormal, $U = [u_1\ \cdots\ u_4]$ satisfies $U'U = I_4$, and $UU' = I_4$ (Proposition 6.10(b)). By the outer product expansion (Theorem 1.4(d)), $UU' = \sum_i u_i u_i' = \sum_i A_i$.

(b) The ranks add up to 4; apply Theorem 11.26(b).

(c) $A_2 + A_3 + A_4 = I_4 - A_1 = I_4 - \bar{J}_4 = C_4$, so $q_2 + q_3 + q_4 = z'C_4 z = \sum_i (z_i - \bar{z})^2$ (Example 1.14). As $\bar{z} = \frac{1}{4}\mathbf{1}'z$ and $\frac{1}{4}\mathbf{1}'C_4 = 0$, Theorem 11.25(c) gives $\bar{z} \perp\!\!\!\perp z'C_4 z$. C_4 is symmetric and idempotent of rank 3 (Proposition 1.13, Corollary 6.21(b)), so $(q_2 + q_3 + q_4)/\sigma^2 \sim \chi^2_3$ by Corollary 11.24(a).

Exercise 11.9. Let $x \sim N_2(\mu, \Sigma)$ with $\mu = (1,1)'$ and $\Sigma = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 5 \end{smallmatrix}\right)$, and let $d^2 = (x-\mu)'\Sigma^{-1}(x-\mu)$.

(a) Show that $\mathrm{P}(d^2 \leqslant c) = 1 - e^{-c/2}$ $(c \geqslant 0)$, and find c with $\mathrm{P}(d^2 \leqslant c) = 1-\alpha$.

(b) Describe the ellipse $E_\alpha = \{u : (u-\mu)'\Sigma^{-1}(u-\mu) \leqslant c\}$ for this c.

(c) Which of the points $(4,-2)$ and $(5,5)$ lie in E_α for $\alpha = \frac{1}{20}$? For each, find the probability that d^2 exceeds its value of $(u-\mu)'\Sigma^{-1}(u-\mu)$, and compare with the Euclidean distances from μ.

Solution. (a) Σ is positive definite (minors 5, 16; Theorem 8.19), so $d^2 \sim \chi^2_2$ (Corollary 11.24(b)), and $\mathrm{P}(d^2 \leqslant c) = 1 - e^{-c/2}$ by (P5). This is $1-\alpha$ for $c = 2\log(1/\alpha)$.

(b) $\Sigma(1,1)' = 8(1,1)'$ and $\Sigma(1,-1)' = 2(1,-1)'$: the eigenvalues are 8 and 2, with unit eigenvectors $\frac{1}{\sqrt2}(1,1)'$ and $\frac{1}{\sqrt2}(1,-1)'$. By Theorem 11.15, E_α has center $(1,1)$, semi-axis $\sqrt{8c} = 4\sqrt{\log(1/\alpha)}$ along $(1,1)'$, and semi-axis $\sqrt{2c} = 2\sqrt{\log(1/\alpha)}$ along $(1,-1)'$.

(c) $\det\Sigma = 16$ and $\Sigma^{-1} = \frac{1}{16}\left(\begin{smallmatrix} 5 & -3 \\ -3 & 5 \end{smallmatrix}\right)$. For $(4,-2)$, $u-\mu = (3,-3)'$ and $(u-\mu)'\Sigma^{-1}(u-\mu) = \frac{1}{16}(45+54+45) = 9$. As $e > 2$, $e^{9/2} > 2^{9/2} = 16\sqrt2 > 20$ (since $512 > 400$), so $9 > 2\log 20$: the point lies outside $E_{1/20}$, and $\mathrm{P}(d^2 > 9) = e^{-9/2}$. For $(5,5)$, $u-\mu = (4,4)'$ and the value is $\frac{1}{16}(80-96+80) = 4$. As $e < 3$, $e^2 < 9 < 20$, so $4 < 2\log 20$: the point lies inside, and $\mathrm{P}(d^2 > 4) = e^{-2}$. The Euclidean distances are $3\sqrt2$ and $4\sqrt2$: the point farther from μ is the less unusual one, because it lies in the direction $(1,1)'$ of the large variance.

Exercise 11.10. Three measurements $4, 7, 10$ are modeled as independent $N(\mu, \sigma^2)$.

(a) Write down $\bar x$, $(n-1)s^2$ and the interval of Example 11.28 for σ^2.

(b) Use (P5) to show that $\chi^2_{2,\tau} = -2\log(1-\tau)$ $(0 < \tau < 1)$, and write the interval of (a) in closed form. Does it contain 1 for $\alpha = \frac{1}{20}$?

(c) Write down the interval of Example 11.28 for μ.

Solution. (a) Here $n = 3$, $\bar x = 7$ and $(n-1)s^2 = 9 + 0 + 9 = 18$. By Example 11.28 the interval is $\Big[18/\chi^2_{2,1-\alpha/2},\ 18/\chi^2_{2,\alpha/2}\Big]$.

(b) By (P5), $\mathrm{P}(v \leqslant c) = 1 - e^{-c/2}$ for $v \sim \chi^2_2$, and $1 - e^{-c/2} = \tau$ exactly for $c = -2\log(1-\tau)$. So $\chi^2_{2,1-\alpha/2} = 2\log(2/\alpha)$ and $\chi^2_{2,\alpha/2} = 2\log\big(2/(2-\alpha)\big)$, so the interval is

$$\left[\frac{9}{\log(2/\alpha)},\ \frac{9}{\log\big(2/(2-\alpha)\big)}\right], \qquad \text{for } \alpha = \tfrac{1}{20}: \quad \left[\frac{9}{\log 40},\ \frac{9}{\log(40/39)}\right]$$

As $e > 2$, $e^9 > 512 > 40$, so $\log 40 < 9$ and the lower endpoint exceeds 1: the interval does not contain 1. By Problem 9.7(c) with $\theta = \frac{40}{39}$, $\log\frac{40}{39} < \frac{40}{39} - 1 = \frac{1}{39}$, so the upper endpoint exceeds $9 \cdot 39 = 351$. Three observations say little about σ^2.

(c) $s = 3$ and $s/\sqrt3 = \sqrt3$, so the interval is $\Big[7 - \sqrt3\, t_{2,1-\alpha/2},\ 7 + \sqrt3\, t_{2,1-\alpha/2}\Big]$.

Exercise 11.11. The five observations $(5,3)$, $(4,4)$, $(3,5)$, $(2,1)$, $(1,2)$ are modeled as independent from $N_2(\mu, \Sigma)$.

(a) Find the MLE $(\hat\mu, \hat\Sigma)$, the largest value of the log-likelihood, and the correlation computed from $\hat\Sigma$.

(b) Compute $\ell(\bar x, S)$ for the unbiased estimate S, and check that it is smaller than $\ell(\bar x, S_n)$.

(c) Find the eigenvalues θ_j of $C = S_n^{1/2} S^{-1} S_n^{1/2}$, and check the identity $\varphi(S) - \varphi(S_n) = \sum_j(\theta_j - 1 - \log\theta_j)$ of Problem 9.7(b) for $\varphi(X) = \log\det X + \mathrm{tr}(X^{-1}S_n)$.

Solution. (a) $\bar{x} = (3,3)'$, and the deviations $x_i - \bar{x}$ are $(2,0)$, $(1,1)$, $(0,2)$, $(-1,-2)$, $(-2,-1)$. So

$$nS_n = \sum_i (x_i - \bar{x})(x_i - \bar{x})' = \begin{pmatrix} 10 & 5 \\ 5 & 10 \end{pmatrix}, \qquad S_n = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det S_n = 3 > 0$$

(for instance $5 = 0+1+0+2+2$). As S_n is positive definite (Theorem 8.19), Theorem 11.31 gives $\hat{\mu} = (3,3)'$, $\hat{\Sigma} = S_n$, and the largest log-likelihood $-5\log(2\pi) - \frac{5}{2}\log 3 - 5$. The correlation computed from $\hat{\Sigma}$ is $1/\sqrt{2 \cdot 2} = \frac{1}{2}$.

(b) $S = \frac{5}{4}S_n$, $\det S = \frac{25}{16} \cdot 3 = \frac{75}{16}$, and $\operatorname{tr}(S^{-1}S_n) = \operatorname{tr}(\frac{4}{5}I_2) = \frac{8}{5}$. By Theorem 11.31(a), $\ell(\bar{x}, S) = -5\log(2\pi) - \frac{5}{2}\Big(\log\frac{75}{16} + \frac{8}{5}\Big)$. Hence

$$\ell(\bar{x}, S_n) - \ell(\bar{x}, S) = \frac{5}{2}\Big(\log\frac{75}{16} + \frac{8}{5} - \log 3 - 2\Big) = \frac{5}{2}\Big(\log\frac{25}{16} - \frac{2}{5}\Big) = 5\log\frac{5}{4} - 1$$

This is positive: $(\frac{5}{4})^5 = \frac{3125}{1024} > 3 > e$, so $5\log\frac{5}{4} > 1$.

(c) $S^{-1} = \frac{4}{5}S_n^{-1}$ and $S_n^{-1} = S_n^{-1/2}S_n^{-1/2}$ with $S_n^{-1/2} = (S_n^{1/2})^{-1}$ (Theorem 8.25(c)), so $C = \frac{4}{5}S_n^{1/2}S_n^{-1}S_n^{1/2} = \frac{4}{5}I_2$ and $\theta_1 = \theta_2 = \frac{4}{5}$. Then $\sum_j(\theta_j - 1 - \log\theta_j) = 2\log\frac{5}{4} - \frac{2}{5}$, and $\varphi(S) - \varphi(S_n) = \log\frac{75}{16} + \frac{8}{5} - \log 3 - 2 = 2\log\frac{5}{4} - \frac{2}{5}$, the same number; it is $\frac{2}{5}\Big(\ell(\bar{x}, S_n) - \ell(\bar{x}, S)\Big)$, as Theorem 11.31(a) predicts.

Exercise 11.12. Galton related the heights of adult children to the *mid-parent height*, the average height of the two parents. Model the mid-parent height x and the child's height y, in centimeters, as bivariate normal with means $\mu_x = \mu_y = 170$, standard deviations $\sigma_x = 4$, $\sigma_y = 5$ and correlation $\rho = \frac{3}{5}$.

(a) Find the conditional distribution of a child's height given mid-parent height 178, and given 162.

(b) Express the conditional means of (a) in standard units, and explain the regression to the mean.

(c) Find an interval $[L, U]$, computed from a, with $\mathrm{P}(L \leqslant y \leqslant U) = 1 - \alpha$ under the conditional distribution of y given $x = a$, for $a = 178$.

(d) Find $\mathrm{E}(x \mid y = 180)$. Do tall children have parents as extreme as themselves?

Solution. (a) By Example 11.22, the line of conditional means of y given x has slope $\rho\sigma_y/\sigma_x = \frac{3}{5} \cdot \frac{5}{4} = \frac{3}{4}$, and the conditional variance is $\sigma_y^2(1-\rho^2) = 25 \cdot \frac{16}{25} = 16$. The conditional means are $170 + \frac{3}{4}(178 - 170) = 176$ and $170 + \frac{3}{4}(162 - 170) = 164$. So $y \mid x = 178 \sim N(176, 16)$ and $y \mid x = 162 \sim N(164, 16)$.

(b) 178 is $8/4 = 2$ standard deviations above μ_x, and 176 is $6/5$ standard deviations above μ_y: $\frac{6}{5} = \frac{3}{5} \cdot 2$. Likewise 162 is 2 standard deviations below μ_x, and 164 is $\frac{6}{5}$ below μ_y. The children of tall parents are tall, but less extreme than their parents in standard units, and the children of short parents are short, but less so; this is the regression to the mean of Example 11.22, since $\rho < 1$.

(c) Given $x = 178$, $y \sim N(176, 16)$, so $w = (y - 176)/4 \sim N(0,1)$ (Theorem 11.14(a)). Let $c = z_{1-\alpha/2}$. As in Example 11.28, with w in place of T ($-w \sim N(0,1)$), $\mathrm{P}(-c \leqslant w \leqslant c) = 1 - \alpha$. Hence $\Big[176 - 4z_{1-\alpha/2},\ 176 + 4z_{1-\alpha/2}\Big]$ is a prediction interval with level $1 - \alpha$.

(d) Exchanging the roles of x and y (Example 11.22), the slope is $\rho\sigma_x/\sigma_y = \frac{3}{5} \cdot \frac{4}{5} = \frac{12}{25}$, and $\mathrm{E}(x \mid y = 180) = 170 + \frac{12}{25} \cdot 10 = 170 + \frac{24}{5}$. The child is 2 standard deviations above μ_y, but the expected mid-parent height only $\frac{24}{5}/4 = \frac{6}{5}$ above μ_x. So regression to the mean also works backwards: it is a consequence of $|\rho| < 1$, not a force acting from one generation to the next.

Problems

Problem 11.1. (a) Let $x \in \mathbb{R}^n$ have mean μ and positive definite covariance matrix Σ (x need not be normal). Show that $\mathrm{E}\big((x-\mu)'\Sigma^{-1}(x-\mu)\big) = n$.

(b) Let $x_1, \dots, x_n \in \mathbb{R}^k$ satisfy the hypotheses of Proposition 11.9, with sample covariance matrix $S = (s_{jl})$. Apply Example 11.8 to the numbers $a'x_1, \dots, a'x_n$ for suitable $a \in \mathbb{R}^k$ to give a second proof of $\mathrm{E}\, S = \Sigma$.

Solution. (a) $x - \mu$ has mean 0 and covariance matrix Σ (Theorem 11.3(a),(b)). By Theorem 11.6 with $A = \Sigma^{-1}$, $\mathrm{E}\big((x-\mu)'\Sigma^{-1}(x-\mu)\big) = \mathrm{tr}(\Sigma^{-1}\Sigma) + 0 = \mathrm{tr}\, I_n = n$.

(b) For $a \in \mathbb{R}^k$ let $y = (a'x_1, \dots, a'x_n)'$. By Theorem 11.3(a),(b), $\mathrm{E}(a'x_i) = a'\mu$, $\mathrm{var}(a'x_i) = a'\Sigma a$, and $\mathrm{cov}(a'x_i, a'x_j) = a'\,\mathrm{cov}(x_i, x_j)a = 0$ for $i \neq j$; so $\mathrm{E}\, y = (a'\mu)\mathbf{1}$ and $\mathrm{var}(y) = (a'\Sigma a)I_n$. Its sum of squares about the mean is $\sum_i (a'x_i - a'\bar{x})^2 = a'\big(\sum_i (x_i - \bar{x})(x_i - \bar{x})'\big)a = (n-1)a'Sa$. By the common-mean case of Example 11.8, $(n-1)\,\mathrm{E}(a'Sa) = (n-1)a'\Sigma a$, that is, $a'(\mathrm{E}\, S)a = a'\Sigma a$ (Theorem 11.3(a)). With $a = e_j$ this gives $\mathrm{E}\, s_{jj} = \sigma_{jj}$. With $a = e_j + e_l$ and $a = e_j - e_l$ ($j \neq l$), as S and Σ are symmetric, $\mathrm{E}\, s_{jj} \pm 2\,\mathrm{E}\, s_{jl} + \mathrm{E}\, s_{ll} = \sigma_{jj} \pm 2\sigma_{jl} + \sigma_{ll}$; subtracting the two equations gives $\mathrm{E}\, s_{jl} = \sigma_{jl}$.

Problem 11.2. Let $x \sim N_n(\mu, \Sigma)$ with $\mathrm{rank}\, \Sigma = r$, $1 \leqslant r < n$.

(a) Show that $\Sigma = CC'$ for some $C \in \mathbb{R}^{n \times r}$ of rank r, and that $w = (C'C)^{-1}C'(x-\mu) \sim N_r(0, I_r)$.

(b) Let $d = x - \mu - Cw$. Show that $\mathrm{E}\, d = 0$ and $\mathrm{var}(d) = 0$, and deduce that $x = \mu + Cw$ with probability 1. (Consider $Y = \|d\|$.)

(c) For x as in Example 11.5 with $u \sim N_2(0, I_2)$, find C and w.

Solution. (a) By Theorem 8.16(c), $\Sigma = B'B$ with $B \in \mathbb{R}^{r \times n}$ of rank r; take $C = B'$, of rank r (Theorem 4.13(a)). Then $C'C \in \mathbb{R}^{r \times r}$ has rank r (Theorem 4.18(b)), so it is invertible. With $K = (C'C)^{-1}C'$, $w = K(x-\mu)$ is normal with mean 0 and covariance matrix $K\Sigma K' = (C'C)^{-1}C'CC'C(C'C)^{-1} = I_r$ (Theorem 11.14(a)).

(b) $d = (I_n - CK)(x-\mu)$, so $\mathrm{E}\, d = 0$ and $\mathrm{var}(d) = (I_n - CK)CC'(I_n - CK)' = 0$, since $(I_n - CK)C = C - C(C'C)^{-1}C'C = 0$. By Theorem 11.6, $\mathrm{E}\, Y^2 = \mathrm{E}(d'd) = \mathrm{tr}\,\mathrm{var}(d) = 0$. So $\mathrm{var}(Y) = \mathrm{E}\, Y^2 - (\mathrm{E}\, Y)^2 = -(\mathrm{E}\, Y)^2 \leqslant 0$; as a variance is $\geqslant 0$ (P1), $\mathrm{var}(Y) = 0$ and $\mathrm{E}\, Y = 0$. By (P1), $Y = \mathrm{E}\, Y = 0$ with probability 1, that is, $d = 0$ and $x = \mu + Cw$ with probability 1.

(c) Here $\mu = 0$ and $\Sigma = BB'$ with $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ of rank 2, so $C = B$ will do. Then $C'C = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $(C'C)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, and $C'x = (x_1 + x_3,\ x_2 + x_3)' = (2u_1 + u_2,\ u_1 + 2u_2)'$, so $w = \frac{1}{3}\big(2(2u_1 + u_2) - (u_1 + 2u_2),\ -(2u_1 + u_2) + 2(u_1 + 2u_2)\big)' = (u_1, u_2)' = u$, and $x = Cu$.

Problem 11.3. Let $x \sim N_n(\mu, \Sigma)$ with $n \geqslant 3$ and Σ positive definite, and let $\Omega = \Sigma^{-1} = (\omega_{ij})$, the **precision matrix**. Partition $x = (x'_{(1)}, x'_{(2)})'$ with $x_{(1)} = (x_1, x_2)'$ and $x_{(2)} = (x_3, \dots, x_n)'$, and partition Σ and Ω conformally.

(a) Show that $\Omega_{11} = \Sigma_{11\cdot 2}^{-1}$, with $\Sigma_{11\cdot 2}$ as in Theorem 11.20.

(b) Deduce that the partial correlation of x_1 and x_2 given $x_3, \dots, x_n$ is $-\omega_{12}/\sqrt{\omega_{11}\omega_{22}}$. So x_1 and x_2 are conditionally uncorrelated given all the other variables if and only if $\omega_{12} = 0$.

(c) Show that $\mathrm{var}(x_1 \mid x_2, \dots, x_n) = 1/\omega_{11}$.

(d) Check that $3\Sigma^{-1} = \left(\begin{smallmatrix} 2 & 1 & -3 & 0 \\ 1 & 2 & -3 & 0 \\ -3 & -3 & 9 & -3 \\ 0 & 0 & -3 & 6 \end{smallmatrix}\right)$ for the matrix Σ of Exercise 11.6, and compare (b) with that exercise. Which pairs are conditionally uncorrelated given the other two variables?

Solution. (a) Σ_{22} is positive definite (Corollary 8.15(a)), and $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11\cdot 2}$ (Theorem 5.26(b)) with $\det \Sigma > 0$, so $\Sigma_{11\cdot 2}$ is invertible. By Problem 5.5(b) with $D = \Sigma_{22}$ and $F = \Sigma_{11\cdot 2}$, the leading 2×2 block of Σ^{-1} is F^{-1}.

(b) Ω is positive definite (Corollary 8.15(a)), so its principal submatrix Ω_{11} is positive definite, $\omega_{11}, \omega_{22} > 0$ and $\det \Omega_{11} = \omega_{11}\omega_{22} - \omega_{12}^2 > 0$. By (a) and Proposition 2.13,

$$\Sigma_{11\cdot 2} = \Omega_{11}^{-1} = \frac{1}{\omega_{11}\omega_{22} - \omega_{12}^2} \begin{pmatrix} \omega_{22} & -\omega_{12} \\ -\omega_{12} & \omega_{11} \end{pmatrix},$$

and the correlation computed from it is $-\omega_{12}/\sqrt{\omega_{22}\omega_{11}}$. It is 0 exactly when $\omega_{12} = 0$.

(c) Repeat (a) with the partition $x_{(1)} = x_1$, $x_{(2)} = (x_2, \dots, x_n)'$: the $(1,1)$ entry ω_{11} of Ω is $\Sigma_{11\cdot 2}^{-1}$, so the conditional variance $\Sigma_{11\cdot 2}$ (Theorem 11.20(b)) is $1/\omega_{11}$.

(d) Multiplying, $\Sigma \cdot \frac{1}{3}\begin{pmatrix} 2 & 1 & -3 & 0 \\ 1 & 2 & -3 & 0 \\ -3 & -3 & 9 & -3 \\ 0 & 0 & -3 & 6 \end{pmatrix} = I_4$; for instance row 1 of Σ times column 1 is $\frac{1}{3}(8+1-6+0) = 1$, and times column 3 is $\frac{1}{3}(-12 - 3 + 18 - 3) = 0$. So $\Omega = \Sigma^{-1}$ (Theorem 2.18), and (b) gives the partial correlation $-1/\sqrt{2 \cdot 2} = -\frac{1}{2}$ of x_1 and x_2 given x_3, x_4, as in Exercise 11.6(a). The zero entries $\omega_{14} = \omega_{24} = 0$ show, by (b) applied after renumbering the variables, that x_1, x_4 and x_2, x_4 are conditionally uncorrelated given the other two variables; the other four pairs are not.

Problem 11.4. Let $x_1, \dots, x_n$ ($n \geqslant 2$) be independent with distribution $N_p(\mu, \Sigma)$, and let $\bar{x}$ and S be as in Proposition 11.9.

(a) Show that $\operatorname{cov}(\bar{x},\ x_i - \bar{x}) = 0$ for every i.

(b) Show that $\bar{x} \sim N_p(\mu, \frac{1}{n}\Sigma)$ and that $\bar{x}$ and S are independent.

(c) Let Σ be positive definite. Show that $n(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu) \sim \chi^2_p$, independently of S.

Solution. (a) Let $u = (x_1', \dots, x_n')'$. By Theorem 11.17(d), u is normal with covariance matrix $V = \operatorname{diag}(\Sigma, \dots, \Sigma)$. Let $G = [I_p \ \cdots \ I_p] \in \mathbb{R}^{p \times np}$ and let $E_i \in \mathbb{R}^{p \times np}$ have the block I_p in position i and zero blocks elsewhere; so $\bar{x} = \frac{1}{n}Gu$ and $x_i = E_i u$. By the block product (Theorem 5.3(c)), $GVE_i' = \Sigma$ and $GVG' = n\Sigma$. By Theorem 11.3(b),

$$\operatorname{cov}\Big(\bar{x},\ x_i - \bar{x}\Big) = \frac{1}{n}GV\Big(E_i - \frac{1}{n}G\Big)' = \frac{1}{n}\Sigma - \frac{1}{n^2}n\Sigma = 0$$

(b) $\bar{x} \sim N_p(\mu, \frac{1}{n^2}GVG') = N_p(\mu, \frac{1}{n}\Sigma)$ by Theorem 11.14(a). Let K have the block rows $E_1 - \frac{1}{n}G, \dots, E_n - \frac{1}{n}G$, so that $Ku = (x_1' - \bar{x}', \dots, x_n' - \bar{x}')'$. By (a), the cross covariance $\frac{1}{n}GVK'$ has zero blocks, so $\bar{x} \perp\!\!\!\perp Ku$ by Theorem 11.17(b). As $S = \frac{1}{n-1}\sum_i (x_i - \bar{x})(x_i - \bar{x})'$ is a function of Ku, $\bar{x} \perp\!\!\!\perp S$ (P1).

(c) $\frac{1}{n}\Sigma$ is positive definite with inverse $n\Sigma^{-1}$, so Corollary 11.24(b) applied to $\bar{x} \sim N_p(\mu, \frac{1}{n}\Sigma)$ gives the distribution χ^2_p. The statistic is a function of $\bar{x}$, hence independent of S by (b) and (P1).

Problem 11.5. Let $n \geqslant 2$, $\sigma > 0$, $-1/(n-1) \leqslant \rho < 1$, $R = (1-\rho)I_n + \rho J_n$, and $x \sim N_n(\mu\mathbf{1}, \sigma^2 R)$: equicorrelated observations.

(a) Show that $\sigma^2 R$ is positive semidefinite.

(b) Find the distributions of $\bar{x}$ and of $C_n x$, and show that $\bar{x} \perp\!\!\!\perp C_n x$; hence $\bar{x} \perp\!\!\!\perp s^2$, where $s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$.

(c) Show that $(n-1)s^2/\Big(\sigma^2(1-\rho)\Big) \sim \chi^2_{n-1}$.

(d) Let $\rho > -1/(n-1)$. Show that $T = \sqrt{n}(\bar{x} - \mu)/s$ has the distribution of κV, where $V \sim t_{n-1}$ and $\kappa = \sqrt{(1 + (n-1)\rho)/(1-\rho)}$. For $n = 9$ and $\rho = \frac{1}{4}$, show that the interval of Example 11.28 with level $1 - \alpha$ contains μ with probability less than $1 - \alpha$.

Solution. (a) By Example 7.17, the eigenvalues of R, listed with multiplicity, are $1+(n-1)\rho$ and $1-\rho$ ($n-1$ times); both are $\geqslant 0$ here. So R, and with it $\sigma^2 R$, is positive semidefinite (Theorem 8.13(b)).

(b) Each row of R has sum $1+(n-1)\rho$, so $R\mathbf{1} = (1+(n-1)\rho)\mathbf{1}$ and $\mathbf{1}'R = (1+(n-1)\rho)\mathbf{1}'$. By Theorem 11.14(a),(b), $\bar{x} = \frac{1}{n}\mathbf{1}'x \sim N\Big(\mu, \frac{\sigma^2}{n^2}\mathbf{1}'R\mathbf{1}\Big) = N\Big(\mu, \frac{\sigma^2}{n}(1+(n-1)\rho)\Big)$, and $C_n x \sim N_n(0, \sigma^2 C_n R C_n)$, where $C_n\mu\mathbf{1} = 0$ and $C_n R C_n = (1-\rho)C_n^2 + \rho\, C_n\mathbf{1}\mathbf{1}'C_n = (1-\rho)C_n$. By Theorem 11.17(b) with $A = \frac{1}{n}\mathbf{1}'$ and $B = C_n$, $A(\sigma^2 R)B' = \frac{\sigma^2}{n}(1+(n-1)\rho)\mathbf{1}'C_n = 0$, so $\bar{x} \perp\!\!\!\perp C_n x$. As $s^2 = \|C_n x\|^2/(n-1)$ is a function of $C_n x$, $\bar{x} \perp\!\!\!\perp s^2$ (P1).

(c) Let $z \sim N_n(0, I_n)$. Then $\sigma\sqrt{1-\rho}\, C_n z \sim N_n\Big(0, \sigma^2(1-\rho)C_n\Big)$, the distribution of $C_n x$. The distribution of a function of a random vector depends only on the distribution of the vector (Remark 11.2), so $(n-1)s^2/(\sigma^2(1-\rho)) = \|C_n x\|^2/(\sigma^2(1-\rho))$ has the distribution of $\|C_n z\|^2 = z'C_n z$, which is χ^2_{n-1} (Theorem 11.23).

(d) Let $u = \sqrt{n}(\bar{x}-\mu)/\Big(\sigma\sqrt{1+(n-1)\rho}\Big) \sim N(0,1)$ and $v = (n-1)s^2/(\sigma^2(1-\rho)) \sim \chi^2_{n-1}$; they are independent by (b) and (P1). Then

$$\frac{u}{\sqrt{v/(n-1)}} = \frac{\sqrt{n}(\bar{x}-\mu)}{\sigma\sqrt{1+(n-1)\rho}} \cdot \frac{\sigma\sqrt{1-\rho}}{s} = \frac{T}{\kappa},$$

so $T = \kappa V$ with $V = u/\sqrt{v/(n-1)} \sim t_{n-1}$ (P5). For $n = 9$ and $\rho = \frac{1}{4}$, $\kappa^2 = (1+2)/\frac{3}{4} = 4$ and $\kappa = 2$. With $c = t_{8,1-\alpha/2} > 0$ (Example 11.28), the interval contains μ exactly when $|T| \leqslant c$, that is, $|V| \leqslant \frac{c}{2}$. As in Example 11.28, $-V \sim t_8$, so $\mathrm{P}(V < -\frac{c}{2}) = \mathrm{P}(V > \frac{c}{2})$ and $\mathrm{P}(|V| \leqslant \frac{c}{2}) = 2\,\mathrm{P}(V \leqslant \frac{c}{2}) - 1$. The distribution function of t_8 is strictly increasing (P5) and $\frac{c}{2} < c$, so $\mathrm{P}(V \leqslant \frac{c}{2}) < 1 - \frac{\alpha}{2}$ and the probability is less than $2(1-\frac{\alpha}{2}) - 1 = 1-\alpha$: with positively correlated observations the interval of Example 11.28 is too short.

Problem 11.6. Let $x_1, \ldots, x_m$ be $N(\mu_1, \sigma^2)$ and $y_1, \ldots, y_n$ be $N(\mu_2, \sigma^2)$, all $m+n$ independent, with $m, n \geqslant 2$ and $\sigma > 0$, and let $s_p^2 = \Big(\sum_i (x_i - \bar{x})^2 + \sum_j (y_j - \bar{y})^2\Big)/(m+n-2)$.

(a) Show that $T = \dfrac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)}{s_p\sqrt{1/m + 1/n}} \sim t_{m+n-2}$.

(b) Deduce a confidence interval with level $1-\alpha$ for $\mu_1 - \mu_2$.

(c) Four plants given a new fertilizer have heights $12, 15, 13, 16$, and five control plants $10, 9, 12, 11, 8$. Compute T for $\mu_1 = \mu_2$ and the interval of (b).

Solution. (a)

- By Theorem 11.17(d), $v = (x_1, \ldots, x_m, y_1, \ldots, y_n)'$ is normal with mean $(\mu_1\mathbf{1}'_m, \mu_2\mathbf{1}'_n)'$ and covariance matrix $\sigma^2 I_{m+n}$; so $\varepsilon = v - \mathrm{E}\, v \sim N_{m+n}(0, \sigma^2 I_{m+n})$.
- $\bar{x} - \bar{y} - (\mu_1 - \mu_2) = a'\varepsilon$ with $a = (\frac{1}{m}\mathbf{1}'_m, -\frac{1}{n}\mathbf{1}'_n)'$, $a'a = \frac{1}{m} + \frac{1}{n}$.
- As in the proof of Theorem 11.27, $(m+n-2)s_p^2 = \varepsilon' P \varepsilon$ with $P = \mathrm{diag}(C_m, C_n)$. P is symmetric and idempotent (Proposition 5.6(b)), of rank $(m-1)+(n-1)$ (Theorem 4.16(c), Corollary 6.21(b)).
- $a'P = (\frac{1}{m}\mathbf{1}'_m C_m,\ -\frac{1}{n}\mathbf{1}'_n C_n) = 0$, so $a'\varepsilon \perp\!\!\!\perp \varepsilon' P \varepsilon$ (Theorem 11.25(c)). Moreover $u = a'\varepsilon/(\sigma\sqrt{a'a}) \sim N(0,1)$ and $w = \varepsilon' P \varepsilon/\sigma^2 \sim \chi^2_{m+n-2}$ (Theorem 11.14(b), Corollary 11.24(a)), and $T = u/\sqrt{w/(m+n-2)} \sim t_{m+n-2}$ by (P1) and (P5).

(b) Exactly as in Example 11.28, with $c = t_{m+n-2,1-\alpha/2}$, $\mathrm{P}(-c \leqslant T \leqslant c) = 1-\alpha$, so $\bar{x} - \bar{y} \pm c\, s_p\sqrt{1/m + 1/n}$ is a confidence interval with level $1-\alpha$ for $\mu_1 - \mu_2$.

(c) $\bar{x} = 14$, $\sum_i (x_i - \bar{x})^2 = 4 + 1 + 1 + 4 = 10$; $\bar{y} = 10$, $\sum_j (y_j - \bar{y})^2 = 0 + 1 + 4 + 1 + 4 = 10$. So $s_p^2 = \frac{20}{7}$, $s_p^2(\frac{1}{4} + \frac{1}{5}) = \frac{20}{7} \cdot \frac{9}{20} = \frac{9}{7}$, and $s_p\sqrt{1/m + 1/n} = 3/\sqrt{7}$. For $\mu_1 = \mu_2$, $T = 4\sqrt{7}/3$. The interval is $\Big[4 - \frac{3}{\sqrt{7}}t_{7,1-\alpha/2},\ 4 + \frac{3}{\sqrt{7}}t_{7,1-\alpha/2}\Big]$; it contains 0 exactly when $t_{7,1-\alpha/2} \geqslant 4\sqrt{7}/3$.

Problem 11.7. Let $z \sim N_n(0, I_n)$, and let $A \in \mathbb{R}^{n\times n}$ be symmetric with $z'Az \sim \chi^2_r$ for some $r \geqslant 1$. Show that A is idempotent of rank r: the converse of Theorem 11.23. (Compare MGFs, and use Remark 7.4.)

Solution.

- By Theorem 8.4(a), $A = Q\Lambda Q'$ with Q orthogonal; by Corollary 11.18, $w = Q'z$ has independent $N(0,1)$ entries, and $z'Az = w'\Lambda w = \sum_i \lambda_i w_i^2$.
- Each $w_i^2 \sim \chi^2_1$, with MGF $(1-2s)^{-1/2}$ for $s < \frac{1}{2}$ (P5). Choose $\delta > 0$ with $\delta < \frac{1}{2}$ and $2|\lambda_i|\delta < 1$ for all i. For $|t| < \delta$, the factors $e^{t\lambda_i w_i^2}$ are functions of the independent w_i with finite means, so by (P1)
$$\mathrm{E}\, e^{t\, z'Az} = \prod_{i=1}^{n} \mathrm{E}\, e^{(t\lambda_i)w_i^2} = \prod_{i=1}^{n} (1-2\lambda_i t)^{-1/2}$$
- As $z'Az \sim \chi^2_r$, the left side is also $(1-2t)^{-r/2}$. Both sides are positive; raising them to the power -2 gives $\prod_i (1-2\lambda_i t) = (1-2t)^r$ for $|t| < \delta$. Two polynomials that agree at infinitely many points are equal (Remark 7.4, (A1)), so this holds for all t.
- If $\lambda_i \neq 0$, then $t = 1/(2\lambda_i)$ is a root of the left side, hence of the right side, which vanishes only at $t = \frac{1}{2}$; so $\lambda_i = 1$. Thus every $\lambda_i \in \{0, 1\}$, and if m of them equal 1, then $(1-2t)^m = (1-2t)^r$; comparing degrees, $m = r$.
- So $\Lambda^2 = \Lambda$, $A^2 = Q\Lambda Q'Q\Lambda Q' = Q\Lambda^2 Q' = A$, and $\operatorname{rank} A = r$, the number of nonzero λ_i (Corollary 8.5(b)).

Problem 11.8. In $N \geqslant 2$ independent trials, each trial results in exactly one of $k \geqslant 2$ categories, category j with probability $p_j > 0$ ($p_1 + \cdots + p_k = 1$). Let $y_i \in \mathbb{R}^k$ be the indicator vector of trial i ($y_i = e_j$ if trial i results in category j), $c = y_1 + \cdots + y_N$ the vector of counts, $p = (p_1, \ldots, p_k)'$ and $D = \operatorname{diag}(p_1, \ldots, p_k)$.

(a) Show that $\mathrm{E}\, c = Np$ and $\operatorname{var}(c) = NV$ with $V = D - pp'$.

(b) Show that $V\mathbf{1} = 0$, $\mathcal{N}(V) = \operatorname{span}\{\mathbf{1}\}$ and $\operatorname{rank} V = k - 1$. Explain $V\mathbf{1} = 0$ by Theorem 11.4.

(c) Show that D^{-1} is a generalized inverse of V (Definition 10.8).

(d) Let $u = (c - Np)/\sqrt{N}$. Show that **Pearson's statistic** $X^2 = \sum_j (c_j - Np_j)^2/(Np_j)$ equals $u'D^{-1}u$, and that $u'Gu = X^2$ for every generalized inverse G of V.

(e) Suppose, as an approximation for large N, that $u \sim N_k(0, V)$. Show that then $X^2 \sim \chi^2_{k-1}$.

Solution. (a) Entry j of y_i is 1 with probability p_j and 0 otherwise, so $\mathrm{E}\, y_i = p$ (P1). As y_i is a unit vector, $y_i y_i' = \operatorname{diag}(y_i)$, so $\mathrm{E}(y_i y_i') = D$ and $\operatorname{var}(y_i) = D - pp' = V$ (Theorem 11.3(c)). The y_i are independent, hence uncorrelated (Theorem 11.3(e)). By Proposition 11.9 applied to $\bar{y} = c/N$, $\mathrm{E}\, c = N\, \mathrm{E}\, \bar{y} = Np$ and $\operatorname{var}(c) = N^2 \operatorname{var}(\bar{y}) = N^2 V/N = NV$.

(b) $V\mathbf{1} = p - p(p'\mathbf{1}) = p - p = 0$. If $Vx = 0$, then $Dx = p(p'x)$, so $x_j = p'x$ for every j (divide by $p_j > 0$), and $x = (p'x)\mathbf{1}$. Hence $\mathcal{N}(V) = \operatorname{span}\{\mathbf{1}\}$ and $\operatorname{rank} V = k - 1$ (Theorem 4.12). Every trial falls in exactly one category, so $\mathbf{1}'c = N = \mathbf{1}'\, \mathrm{E}\, c$ always and $\operatorname{var}(\mathbf{1}'c) = 0$ (P1). By Theorem 11.4(a), $\mathbf{1}' \operatorname{var}(c)\mathbf{1} = 0$ and $\operatorname{var}(c)$ is positive semidefinite, hence $\operatorname{var}(c)\mathbf{1} = 0$ (Proposition 8.17(a)); as $\operatorname{var}(c) = NV$, this is $V\mathbf{1} = 0$. So $\operatorname{var}(c)$ is singular, as Theorem 11.4(b) with $a = \mathbf{1}$ says.

(c) As $p'D^{-1} = \mathbf{1}'$, $VD^{-1} = I_k - p\mathbf{1}'$; as $\mathbf{1}'D = p'$ and $\mathbf{1}'p = 1$,
$$VD^{-1}V = (I_k - p\mathbf{1}')(D - pp') = D - pp' - pp' + p(\mathbf{1}'p)p' = D - pp' = V$$

(d) $u'D^{-1}u = \sum_j u_j^2/p_j = \sum_j (c_j - Np_j)^2/(Np_j) = X^2$. Next $\mathbf{1}'u = (\mathbf{1}'c - N)/\sqrt{N} = 0$, so $u \in W = \operatorname{span}\{\mathbf{1}\}^\perp$, a subspace of dimension $k-1$ (Theorem 6.5(c)). It contains $\mathcal{C}(V)$, since $\mathbf{1}'Vw = (V\mathbf{1})'w = 0$

for every $w \in \mathbb{R}^k$, and $\dim \mathcal{C}(V) = \operatorname{rank} V = k-1$ by (b); so $\mathcal{C}(V) = W$ (Theorem 4.5(a)), and $u = Vw$ for some $w \in \mathbb{R}^k$. For every generalized inverse G of V, $u'Gu = w'VGVw = w'Vw$, the same number for all G; by (c) it equals $u'D^{-1}u = X^2$.

(e) Let $q = (\sqrt{p_1}, \dots, \sqrt{p_k})'$, so $q'q = 1$, and $D^{-1/2} = \operatorname{diag}(p_1^{-1/2}, \dots, p_k^{-1/2})$. Then $D^{-1/2}VD^{-1/2} = I_k - qq' = P$, symmetric and idempotent ($P^2 = I_k - 2qq' + q(q'q)q' = P$) of rank $\operatorname{tr} P = k-1$ (Theorem 6.16(c)). So $v = D^{-1/2}u \sim N_k(0, P)$ (Theorem 11.14(a)). For $z \sim N_k(0, I_k)$, $Pz \sim N_k(0, PP') = N_k(0, P)$ as well; hence $X^2 = v'v$ has the distribution of $(Pz)'(Pz) = z'Pz$, which is χ^2_{k-1} (Theorem 11.23).

Problem 11.9. In an experiment of Mendel's type, $N = 160$ pea seeds are classified as round yellow, round green, wrinkled yellow and wrinkled green, with counts $c = (93,\ 27,\ 31,\ 9)'$. Mendel's theory predicts the probabilities $p = \frac{1}{16}(9, 3, 3, 1)'$.

(a) Compute the expected counts Np_j, the contributions $(c_j - Np_j)^2/(Np_j)$ and Pearson's X^2 (Problem 11.8).

(b) Under the approximation of Problem 11.8(e), $X^2 \sim \chi^2_3$. Show that then $\mathrm{P}(X^2 > \frac{8}{15}) > \frac{1}{2}$, and comment on the fit of the theory. (Compare $z_1^2 + z_2^2 + z_3^2$ with $z_1^2 + z_2^2$.)

(c) With V as in Problem 11.8, let V_{11} be its leading 3×3 block and $D_3 = \operatorname{diag}(p_1, p_2, p_3)$. Show that $V_{11}^{-1} = D_3^{-1} + \frac{1}{p_4}\mathbf{1}_3\mathbf{1}_3'$, so that $G = \left(\begin{smallmatrix} V_{11}^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is a generalized inverse of V (Theorem 10.9(b)); and check directly that $u'Gu = X^2$ for these data.

Solution. (a) The expected counts are $160 \cdot \frac{9}{16} = 90$, $160 \cdot \frac{3}{16} = 30$ (twice) and $160 \cdot \frac{1}{16} = 10$, and $c - Np = (3, -3, 1, -1)'$. The contributions are

$$\frac{9}{90} = \frac{1}{10}, \qquad \frac{9}{30} = \frac{3}{10}, \qquad \frac{1}{30}, \qquad \frac{1}{10},$$

and $X^2 = \frac{3+9+1+3}{30} = \frac{8}{15}$.

(b) Let z_1, z_2, z_3 be independent $N(0,1)$. By (P5), $v_3 = z_1^2 + z_2^2 + z_3^2 \sim \chi^2_3$ and $v_2 = z_1^2 + z_2^2 \sim \chi^2_2$, and $v_3 \geqslant v_2$. So the event $\{v_2 > c\}$ is contained in $\{v_3 > c\}$, and by (P5), $\mathrm{P}(v_3 > c) \geqslant \mathrm{P}(v_2 > c) = e^{-c/2}$. For $c = \frac{8}{15}$, $e^{-4/15} > \frac{1}{2}$, since $e^4 < 3^4 = 81 < 2^{15}$. So under the theory a value of X^2 larger than the observed one has probability more than $\frac{1}{2}$: the counts agree well with Mendel's $9:3:3:1$ law.

(c) The leading block is $V_{11} = D_3 - p_{(3)}p_{(3)}'$ with $p_{(3)} = (p_1, p_2, p_3)'$. As $p_{(3)}'D_3^{-1} = \mathbf{1}_3'$, $D_3\mathbf{1}_3 = p_{(3)}$ and $\mathbf{1}_3'p_{(3)} = p_1 + p_2 + p_3 = 1 - p_4$,

$$\Big(D_3 - p_{(3)}p_{(3)}'\Big)\Big(D_3^{-1} + \frac{1}{p_4}\mathbf{1}_3\mathbf{1}_3'\Big) = I_3 - p_{(3)}\mathbf{1}_3' + \frac{1}{p_4}\Big(p_{(3)} - p_{(3)}(1-p_4)\Big)\mathbf{1}_3' = I_3,$$

so V_{11} is invertible with the stated inverse (Theorem 2.18). As $\operatorname{rank} V = 3$ (Problem 11.8(b)), G is a generalized inverse of V (Theorem 10.9(b)). With u as in Problem 11.8(d) and $u_{(3)} = (u_1, u_2, u_3)'$,

$$u'Gu = u_{(3)}'V_{11}^{-1}u_{(3)} = \sum_{j\leqslant 3}\frac{u_j^2}{p_j} + \frac{1}{p_4}\Big(\sum_{j\leqslant 3}u_j\Big)^2 = \sum_{j=1}^{4}\frac{u_j^2}{p_j} = X^2,$$

since $\sum_{j\leqslant 3} u_j = -u_4$. For these data, $\sum_{j\leqslant 3}(c_j - Np_j) = 3 - 3 + 1 = 1 = -(c_4 - Np_4)$, and $u'Gu = \frac{1}{10} + \frac{3}{10} + \frac{1}{30} + \frac{1}{160\cdot\frac{1}{16}} = \frac{8}{15}$, as Problem 11.8(d) predicts.

Chapter 12. Linear models and generalized linear models

Least squares

In the **linear regression model**, the response vector $y = (y_1, \dots, y_n)'$ holds n observations, and x_{ij} is the value of the jth explanatory variable on observation i ($1 \leqslant j \leqslant k$):

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i \quad (1 \leqslant i \leqslant n), \qquad \text{that is,} \qquad y = X\beta + \varepsilon$$

The **design matrix** $X = [\mathbf{1}_n \; X_0] \in \mathbb{R}^{n\times p}$, $p = k+1$, has the data matrix $X_0 = (x_{ij})$ (Example 1.15) as its last k columns and ith row $x_i' = (1, x_{i1}, \dots, x_{ik})$; $\beta = (\beta_0, \dots, \beta_k)'$, the coefficient β_0 of $\mathbf{1}_n$ is the **intercept**, and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)'$ is the vector of errors. For any $X \in \mathbb{R}^{n\times p}$ with rows x_i' and any $y \in \mathbb{R}^n$, a **least-squares estimate** of β is a vector $\hat\beta \in \mathbb{R}^p$ that minimizes

$$S(\beta) = \|y - X\beta\|^2 = \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \qquad (\beta \in \mathbb{R}^p);$$

then $\hat y = X\hat\beta$ is the vector of **fitted values**, $e = y - \hat y$ the vector of **residuals**, and $\mathrm{SSE} = S(\hat\beta) = e'e$, the minimum of S, the **error sum of squares**. The p equations $X'X\beta = X'y$ are the **normal equations**. The results below hold for every $X \in \mathbb{R}^{n\times p}$, with or without the column $\mathbf{1}_n$, unless an intercept is assumed.

Theorem 12.1. Let $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$.

(a) The normal equations $X'X\beta = X'y$ (unknown $\beta \in \mathbb{R}^p$) are consistent.

(b) $X'X$ is invertible if and only if X has full column rank, $\operatorname{rank} X = p$. Then the normal equations have exactly one solution, $\hat\beta = (X'X)^{-1}X'y$.

(c) If $\operatorname{rank} X < p$ and $\hat\beta$ is one solution, the solutions are the vectors $\hat\beta + z$ with $z \in \mathcal{N}(X)$; there are infinitely many, and all of them give the same vector $X\hat\beta$.

Proof. (a) $X'y \in \mathcal{C}(X') = \mathcal{C}(X'X)$ (Theorem 4.18(c)), so the system is consistent (Corollary 4.14(a)).
(b) $X'X \in \mathbb{R}^{p\times p}$ is invertible if and only if $\operatorname{rank}(X'X) = p$ (Theorem 4.13(d)), that is, $\operatorname{rank} X = p$ (Theorem 4.18(b)). Then multiplying $X'X\beta = X'y$ by $(X'X)^{-1}$ gives $\beta = (X'X)^{-1}X'y$, and this vector is a solution.
(c) By Theorem 2.23(a), the solutions are $\hat\beta + z$ with $z \in \mathcal{N}(X'X) = \mathcal{N}(X)$ (Theorem 4.18(a)), and $X(\hat\beta + z) = X\hat\beta + Xz = X\hat\beta$. As $\operatorname{rank} X < p$, some $z \neq 0$ lies in $\mathcal{N}(X)$ (Theorem 4.13(b)), and the solutions $\hat\beta + tz$ ($t \in \mathbb{R}$) are pairwise distinct, since $t_1 z = t_2 z$ forces $t_1 = t_2$. □

Theorem 12.2. Let $X \in \mathbb{R}^{n\times p}$ have full column rank p, so that $X'X$ is invertible (Theorem 12.1(b)). Put

$$H = X(X'X)^{-1}X' \in \mathbb{R}^{n\times n} \;\; (\text{the } \textbf{hat matrix}), \qquad M = I_n - H \;\; (\text{the } \textbf{residual maker})$$

(a) For every $y \in \mathbb{R}^n$, Hy is the orthogonal projection of y onto $\mathcal{C}(X)$, and $y = Hy + My$ is the decomposition of y along $\mathbb{R}^n = \mathcal{C}(X) \oplus \mathcal{C}(X)^\perp$. So $H = P_{\mathcal{C}(X)}$ and $M = P_{\mathcal{C}(X)^\perp}$.

(b) $H' = H = H^2$ and $M' = M = M^2$: both are symmetric and idempotent.

(c) $HX = X$, $MX = 0$ and $HM = MH = 0$.

(d) $\operatorname{rank} H = \operatorname{tr} H = p$ and $\operatorname{rank} M = \operatorname{tr} M = n - p$.

(e) If $X = QR$ as in Theorem 6.12, then $X'X = R'R$ and $H = QQ'$.

(f) The diagonal entry h_{ii} of $H = (h_{ij})$ is the **leverage** of observation i. Then $\sum_i h_{ii} = p$, $0 \leqslant h_{ii} \leqslant 1$, $h_{ij}^2 \leqslant h_{ii}h_{jj}$, $h_{ii} = 1$ implies $h_{ij} = 0$ for $j \neq i$, and $h_{ii} \geqslant \frac{1}{n}$ for every i if $\mathbf{1}_n \in \mathcal{C}(X)$.

Proof. (a) $Hy = X\big((X'X)^{-1}X'y\big) \in \mathcal{C}(X)$, and $X'(y - Hy) = X'y - (X'X)(X'X)^{-1}X'y = 0$, so $My = y - Hy \in \mathcal{C}(X)^\perp$ (Theorem 6.5(a)). By Definition 6.14, Hy is the projection; so $H = P_{\mathcal{C}(X)}$ (Theorem 6.19(a)), and $M = I_n - P_{\mathcal{C}(X)} = P_{\mathcal{C}(X)^\perp}$ (Theorem 6.19(b)).

(b) Projection matrices are symmetric and idempotent (Theorem 6.19(b)); apply (a).

(c) $HX = X(X'X)^{-1}X'X = X$, $MX = X - HX = 0$, and $HM = H - H^2 = 0 = MH$ by (b).

(d) By Theorem 6.19(b), $\operatorname{rank} H = \operatorname{tr} H = \dim \mathcal{C}(X) = \operatorname{rank} X = p$ and $\operatorname{rank} M = \operatorname{tr} M = \dim \mathcal{C}(X)^\perp = n - p$ (Theorem 6.5(c)).

(e) $X'X = R'Q'QR = R'R$, as $Q'Q = I_p$. By Corollary 6.20, $P_{\mathcal{C}(X)} = QQ'$; so $H = QQ'$ by (a).

(f) $\sum_i h_{ii} = \operatorname{tr} H = p$ by (d). As $H = P_{\mathcal{C}(X)}$, the other statements are Problem 8.5(a)–(c) with $W = \mathcal{C}(X)$. □

Theorem 12.3. Let $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$.

(a) $\hat\beta$ is a least-squares estimate if and only if $X'X\hat\beta = X'y$. For every solution $\hat\beta$ of the normal equations,
$$S(\beta) = S(\hat\beta) + \|X(\hat\beta - \beta)\|^2 \qquad (\beta \in \mathbb{R}^p)$$

(b) All least-squares estimates give the same vector of fitted values $\hat y = X\hat\beta$, the orthogonal projection of y onto $\mathcal{C}(X)$; the vector of residuals $e = y - \hat y$ lies in $\mathcal{C}(X)^\perp$.

(c) If $\operatorname{rank} X = p$, the least-squares estimate is unique, $\hat\beta = (X'X)^{-1}X'y$, and $\hat y = Hy$, $e = My$ and $\mathrm{SSE} = y'My$.

Proof. (a)

- Let $X'X\hat\beta = X'y$. Then $X'(y - X\hat\beta) = 0$, so $y - X\hat\beta \perp \mathcal{C}(X)$ (Theorem 6.5(a)); as $X\hat\beta \in \mathcal{C}(X)$, it is the orthogonal projection of y onto $\mathcal{C}(X)$ (Definition 6.14).
- For $\beta \in \mathbb{R}^p$, $X\beta \in \mathcal{C}(X)$, and Theorem 6.15 gives $\|y - X\beta\|^2 = \|y - X\hat\beta\|^2 + \|X\hat\beta - X\beta\|^2$, the identity. Hence $S(\beta) \geqslant S(\hat\beta)$: $\hat\beta$ is a least-squares estimate.
- Solutions exist (Theorem 12.1(a)). If β^* is a least-squares estimate and $\hat\beta$ a solution, then $S(\beta^*) \leqslant S(\hat\beta)$, so the identity gives $\|X(\hat\beta - \beta^*)\|^2 = 0$, that is, $X\beta^* = X\hat\beta$; then $X'X\beta^* = X'X\hat\beta = X'y$.

(b) By the last step, every least-squares estimate β^* gives $X\beta^* = X\hat\beta$, the projection found in the first step; and $e = y - X\hat\beta \perp \mathcal{C}(X)$.

(c) By Theorem 12.1(b) and (a), $(X'X)^{-1}X'y$ is the only least-squares estimate. Then $\hat y = X(X'X)^{-1}X'y = Hy$, $e = y - Hy = My$, and $\mathrm{SSE} = e'e = y'M'My = y'My$ (Theorem 12.2(b)). □

The formula $\hat\beta = (X'X)^{-1}X'y$ needs $\operatorname{rank} X = p$; since $\operatorname{rank} X \leqslant n$ (Theorem 4.13(a)), this requires at least p observations.

Proposition 12.4. Let $\hat\beta$ be a least-squares estimate, $\hat y = X\hat\beta$ and $e = y - \hat y$.

(a) $X'e = 0$: the residuals are orthogonal to every column of X.

(b) If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\mathbf{1}'e = e_1 + \cdots + e_n = 0$; hence $\hat y$ and y have the same mean $\bar y$.

(c) $\hat y'e = 0$.

Proof. (a) This is $X'(y - X\hat\beta) = 0$, the normal equations (Theorem 12.3(a)). (b) If $\mathbf{1} = Xc$, then $\mathbf{1}'e = c'X'e = 0$ by (a), and $\mathbf{1}'\hat y = \mathbf{1}'y - \mathbf{1}'e = \mathbf{1}'y$. (c) $\hat y'e = \hat\beta'X'e = 0$. □

Proposition 12.5. Let $X = [\mathbf{1}_n\ X_0]$ with $X_0 \in \mathbb{R}^{n\times k}$, let $\bar x = \frac{1}{n}X_0'\mathbf{1}_n$ be the mean vector of X_0, and let $S_c = X_0'C_nX_0$.

(a) $X'X = \begin{pmatrix} n & n\bar{x}' \\ n\bar{x} & X_0'X_0 \end{pmatrix}$, the Schur complement of the entry n in $X'X$ is S_c, and $\det(X'X) = n \det S_c$. Hence X has full column rank $p = k+1$ if and only if S_c is invertible.

(b) If S_c is invertible, then

$$(X'X)^{-1} = \begin{pmatrix} \frac{1}{n} + \bar{x}'S_c^{-1}\bar{x} & -\bar{x}'S_c^{-1} \\ -S_c^{-1}\bar{x} & S_c^{-1} \end{pmatrix}$$

Proof. (a) By the block product formula (Theorem 5.3(c)), $X'X = \left(\begin{smallmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'X_0 \\ X_0'\mathbf{1} & X_0'X_0 \end{smallmatrix}\right)$, with $\mathbf{1}'\mathbf{1} = n$ and $X_0'\mathbf{1} = n\bar{x}$. The Schur complement of $A = (n)$ is $X_0'X_0 - (n\bar{x})\frac{1}{n}(n\bar{x}') = X_0'X_0 - n\bar{x}\bar{x}' = X_0'C_nX_0$ (Example 1.15). Theorem 5.26(a) gives $\det(X'X) = n \det S_c$, so $X'X$ is invertible if and only if S_c is (Theorem 5.19(a)); and $X'X$ is invertible if and only if $\operatorname{rank} X = p$ (Theorem 12.1(b)).

(b) Apply Problem 5.5(a) with $A = (n)$, $B = n\bar{x}'$, $C = n\bar{x}$, $D = X_0'X_0$ and $E = S_c$: here $A^{-1}B = \bar{x}'$ and $CA^{-1} = \bar{x}$, so $A^{-1} + A^{-1}BE^{-1}CA^{-1} = \frac{1}{n} + \bar{x}'S_c^{-1}\bar{x}$, $-A^{-1}BE^{-1} = -\bar{x}'S_c^{-1}$ and $-E^{-1}CA^{-1} = -S_c^{-1}\bar{x}$. □

For $k = 1$ and $X_0 = x$, $S_c = x'C_nx = \sum_i (x_i - \bar{x})^2$ (Example 1.14), so the last diagonal entry of $(X'X)^{-1}$ is $1/\sum_i (x_i - \bar{x})^2$.

Example 12.6. Four units have the data matrix X_0 of Example 1.15 and the responses $y = (9, 3, 5, 3)'$. The entries of $X'X$ and $X'y$ are inner products of columns (Theorems 1.4(d) and 1.9):

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}$$

- By Example 1.15, $\bar{x} = (1, 2)'$ and $S_c = 4I_2$, which is invertible. So $\operatorname{rank} X = 3$, and Proposition 12.5(b), with $\frac{1}{4} + \bar{x}'S_c^{-1}\bar{x} = \frac{1}{4} + \frac{1+4}{4} = \frac{3}{2}$ and $\bar{x}'S_c^{-1} = (\frac{1}{4}, \frac{1}{2})$, gives

$$(X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}, \qquad \hat{\beta} = (X'X)^{-1}X'y = \frac{1}{4}\begin{pmatrix} 120 - 28 - 88 \\ -20 + 28 \\ -40 + 44 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

- The fitted values are $\hat{y} = X\hat{\beta} = (8, 4, 6, 2)'$, the residuals $e = (1, -1, -1, 1)'$, and $\mathrm{SSE} = 4$. As $\mathrm{SSE} > 0$, the system $X\beta = y$ has no solution. As Proposition 12.4 says, $X'e = (0,\ 2-2,\ 3-3-1+1)' = 0$, $\mathbf{1}'e = 0$ and $\hat{y}'e = 8 - 4 - 6 + 2 = 0$.

- Gram–Schmidt on the columns gives $b_1 = \mathbf{1}_4$, $b_2 = (2, 0, 2, 0)' - b_1 = (1, -1, 1, -1)'$ and $b_3 = (3, 3, 1, 1)' - 2b_1 - 0\,b_2 = (1, 1, -1, -1)'$, each of length 2. So $X = QR$ with $Q = \frac{1}{2}[b_1\ b_2\ b_3]$ and $R = Q'X$ (Theorem 6.12), and Theorem 12.2(e) and the outer product expansion (Theorem 1.4(d)) give

$$R = \begin{pmatrix} 2 & 2 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad H = QQ' = \frac{1}{4}\big(J_4 + b_2b_2' + b_3b_3'\big) = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix}$$

  $R'R = X'X$ and $Hy = (8, 4, 6, 2)'$. Here $M = I_4 - H = \frac{1}{4}ee'$, every leverage is $h_{ii} = \frac{3}{4} \geqslant \frac{1}{4}$, $\operatorname{tr} H = 3$ and $\operatorname{tr} M = 1$.

- By the identity of Theorem 12.3(a), for instance $\beta = (1, 2, 2)'$ gives $S(\beta) = 4 + \|X(0, 0, -1)'\|^2 = 4 + (9 + 9 + 1 + 1) = 24$.

Corollary 12.7. Let $k = 1$ (**simple regression**): $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, where $x_i = x_{i1}$ is a number, and $X = [\mathbf{1}_n \; x]$ with $x = (x_1, \dots, x_n)'$. Put

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2, \qquad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$

(expand, using $\sum_i x_i = n\bar{x}$ and $\sum_i y_i = n\bar{y}$, as in Example 1.14).

(a) X has full column rank if and only if x is not constant, that is, $S_{xx} > 0$.

(b) If $S_{xx} > 0$, then $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$: the fitted line $\hat{\beta}_0 + \hat{\beta}_1 x$ passes through the point $(\bar{x}, \bar{y})$.

Proof. (a) Here $S_c = S_{xx}$ (Proposition 12.5 and the remark after it), so X has full column rank if and only if $S_{xx} \neq 0$; a sum of squares is 0 exactly when every $x_i - \bar{x}$ is 0.

(b) By Proposition 12.5(b), row 2 of $(X'X)^{-1}$ is $(-\bar{x}/S_{xx},\ 1/S_{xx})$, and $X'y = (n\bar{y},\ \sum_i x_i y_i)'$; so $\hat{\beta}_1 = (\sum_i x_i y_i - n\bar{x}\bar{y})/S_{xx} = S_{xy}/S_{xx}$ (Theorem 12.1(b)). The first normal equation, $n\hat{\beta}_0 + n\bar{x}\hat{\beta}_1 = n\bar{y}$, gives $\hat{\beta}_0 + \hat{\beta}_1\bar{x} = \bar{y}$. □

Rank in regression

Theorem 12.8. Let $X \in \mathbb{R}^{n\times p}$ have any rank, let G be a generalized inverse of $X'X$ (Definition 10.8), and let $y \in \mathbb{R}^n$.

(a) $P_X = XGX'$ does not depend on the choice of G, and $P_X = XX^+ = P_{\mathcal{C}(X)}$. So P_X is symmetric and idempotent, $P_X X = X$, and $\operatorname{rank} P_X = \operatorname{rank} X$.

(b) $GX'y$ solves the normal equations. All solutions, that is, all least-squares estimates, are $GX'y + (I_p - GX'X)z$ ($z \in \mathbb{R}^p$), and every one of them gives $X\hat{\beta} = P_X y$.

Proof. By Lemma 10.12(a), $XGX'X = X$ and $X'XGX' = X'$.

- (a) For $y \in \mathbb{R}^n$, $XGX'y = X(GX'y) \in \mathcal{C}(X)$ and $X'(y - XGX'y) = X'y - X'y = 0$, so $y - XGX'y \perp \mathcal{C}(X)$ (Theorem 6.5(a)), and $XGX'y$ is the orthogonal projection of y onto $\mathcal{C}(X)$ (Definition 6.14). Hence $XGX' = P_{\mathcal{C}(X)}$ for every G (Theorem 6.19(a)), which is XX^+ (Theorem 10.15(d)). By Theorem 6.19(b), P_X is symmetric and idempotent, $P_X x = x$ for every column x of X, and $\operatorname{rank} P_X = \dim \mathcal{C}(X) = \operatorname{rank} X$.
- (b) $X'X(GX'y) = X'y$. The normal equations are consistent (Theorem 12.1(a)), so Theorem 10.11(b) with $A = X'X$ and $b = X'y$ gives the solutions, which are the least-squares estimates (Theorem 12.3(a)); and $X\Big(GX'y + (I_p - GX'X)z\Big) = P_X y + (X - XGX'X)z = P_X y$. □

If $\operatorname{rank} X = p$, then $G = (X'X)^{-1}$ is the only generalized inverse of $X'X$ (Theorem 10.9(c)), and $P_X = H$.

Proposition 12.9. Let $X \in \mathbb{R}^{n\times p}$ with $p \geqslant 2$. Then X has full column rank if and only if no column of X is a linear combination of the other columns. If some column is such a combination (**exact collinearity**), then $Xz = 0$ for some $z \neq 0$, $X'X$ is singular, and β and $\beta + tz$ give the same $X\beta$ for every $t \in \mathbb{R}$.

Proof. X has full column rank if and only if its columns are independent (Theorem 4.13(b)), and $p \geqslant 2$ vectors are dependent if and only if one of them is a linear combination of the others (Proposition 3.12(c)). In that case $Xz = 0$ for some $z \neq 0$ (Theorem 4.13(b)), $X'X$ is singular (Theorem 12.1(b)), and $X(\beta + tz) = X\beta$. □

For example, if a regression uses the scores x_{i1}, x_{i2} of two tests and also their total $x_{i3} = x_{i1} + x_{i2}$, then column 4 of X is the sum of columns 2 and 3, and $X(0, 1, 1, -1)' = 0$.

Theorem 12.10. Let $n_1, \dots, n_g \geqslant 1$ and $n = n_1 + \cdots + n_g$. In the **one-way layout**, n observations fall into g groups, listed group by group (first the n_1 observations of group 1, then the n_2 of group 2, and so on); G_l is the set of observations in group l, and the **indicator** $d_l \in \mathbb{R}^n$ of group l has entry 1 in the rows $i \in G_l$ and 0 elsewhere. The model $y_i = \mu + \alpha_l + \varepsilon_i$ $(i \in G_l)$ reads $y = X\beta + \varepsilon$ with $\beta = (\mu, \alpha_1, \dots, \alpha_g)'$ and

$$X = [\mathbf{1}_n \; d_1 \; \cdots \; d_g] \in \mathbb{R}^{n \times (g+1)}$$

Let $X_1 = [d_1 \; \cdots \; d_g] \in \mathbb{R}^{n \times g}$ (intercept dropped).

(a) $X_1'X_1 = \operatorname{diag}(n_1, \dots, n_g)$, $X_1'y$ holds the g group totals, and X_1 has full column rank. Moreover $X = X_1K$ with $K = [\mathbf{1}_g \; I_g] \in \mathbb{R}^{g \times (g+1)}$ of full row rank.

(b) $\operatorname{rank} X = g$ and $\mathcal{N}(X) = \operatorname{span}\{(-1, 1, \dots, 1)'\}$: X does not have full column rank.

(c) $\mathcal{C}(X_1) = \mathcal{C}(X)$.

(d) $P_{\mathcal{C}(X)} = \sum_l n_l^{-1} d_l d_l' = \operatorname{diag}(\bar{J}_{n_1}, \dots, \bar{J}_{n_g})$, and $P_{\mathcal{C}(X)}y = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g$, where $\bar{y}_l = n_l^{-1} \sum_{i \in G_l} y_i$ is the **group mean**: the fitted value of each observation is the mean of its group.

(e) With the **grand mean** $\bar{y}$ of all observations, $y_i = \bar{y} + (\bar{y}_l - \bar{y}) + (y_i - \bar{y}_l)$ $(i \in G_l)$ splits y into three orthogonal vectors, in subspaces of dimensions 1, $g - 1$ and $n - g$, and

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{l=1}^{g} n_l (\bar{y}_l - \bar{y})^2 + \sum_{l=1}^{g} \sum_{i \in G_l} (y_i - \bar{y}_l)^2$$

Proof. (a) No observation lies in two groups, so $d_l'd_m = 0$ for $l \neq m$, and $d_l'd_l = n_l$; also $d_l'y$ is the total of group l. These are the entries of $X_1'X_1$ and $X_1'y$ (Theorems 1.4(d) and 1.9). As $X_1'X_1$ is invertible, $\operatorname{rank} X_1 = g$ (Theorem 12.1(b)). Next $X_1\mathbf{1}_g = d_1 + \cdots + d_g = \mathbf{1}_n$, so $X_1K = [X_1\mathbf{1}_g \;\; X_1] = X$ (Theorem 5.3(c)); the columns of K include those of I_g, so $\mathcal{C}(K) = \mathbb{R}^g$, and K has full row rank (Theorem 4.13(c)).

(b) By (a) and Theorem 4.15(b), $\operatorname{rank} X = \operatorname{rank} K = g$ and $\mathcal{N}(X) = \mathcal{N}(K)$. Now $K(c_0, c_1, \dots, c_g)' = c_0\mathbf{1}_g + (c_1, \dots, c_g)'$ is 0 if and only if $c_l = -c_0$ for every l, that is, $(c_0, c_1, \dots, c_g)' = -c_0(-1, 1, \dots, 1)'$. As $g < g + 1$, X does not have full column rank.

(c) K has full row rank by (a), so $\mathcal{C}(X) = \mathcal{C}(X_1K) = \mathcal{C}(X_1)$ (Theorem 4.15(c)).

(d) By (a), (c) and Theorem 12.2(a), $P_{\mathcal{C}(X)} = P_{\mathcal{C}(X_1)} = X_1(X_1'X_1)^{-1}X_1'$, which is $\sum_l n_l^{-1} d_l d_l'$ by the outer product expansion (Theorem 1.4(d)). The (i, j) entry of $d_l d_l'$ is 1 if $i, j \in G_l$ and 0 otherwise, which gives the block diagonal form; and $n_l^{-1} d_l d_l' y = n_l^{-1}(d_l'y) d_l = \bar{y}_l d_l$.

(e) Let $U = \operatorname{span}\{\mathbf{1}_n\} \subseteq W = \mathcal{C}(X)$. Then $P_U y = \bar{y}\mathbf{1}_n$ (Corollary 6.21(a)), and $P_W y$ is given by (d). By Theorem 6.25(b),(c), $y = P_U y + (P_W - P_U)y + (I_n - P_W)y$, with entries $\bar{y}$, $\bar{y}_l - \bar{y}$ and $y_i - \bar{y}_l$ in row $i \in G_l$, is a sum of three orthogonal vectors in subspaces of dimensions 1, $g - 1$ and $n - g$ (Theorem 6.5(c)), and the squared lengths add up. Subtract $\|P_U y\|^2 = n\bar{y}^2$ and use $\|y\|^2 - n\bar{y}^2 = \sum_i (y_i - \bar{y})^2$ (Example 1.14). □

Using the intercept together with all g indicators (the **dummy variable trap**) thus makes $X'X$ singular; dropping the intercept gives a design of full column rank with the same column space, hence the same fitted values.

Example 12.11. Six plants of three types have weights $5, 7, 9$ (type 1), $3, 5$ (type 2) and 1 (type 3): the one-way layout with $g = 3$, $n_1 = 3$, $n_2 = 2$, $n_3 = 1$ and $y = (5, 7, 9, 3, 5, 1)'$. The entries of $X'X$

are $\mathbf{1}'\mathbf{1} = 6$, $\mathbf{1}'d_l = d_l'd_l = n_l$ and $d_l'd_m = 0$ $(l \neq m)$, as in the proof of Theorem 12.10(a):

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'X = \left(\begin{array}{c|ccc} 6 & 3 & 2 & 1 \\ \hline 3 & 3 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right), \qquad X'y = \begin{pmatrix} 30 \\ 21 \\ 8 \\ 1 \end{pmatrix}$$

- The lower right block $D = \operatorname{diag}(3,2,1)$ is invertible, and its Schur complement is $6 - (3,2,1)D^{-1}(3,2,1)' = 6 - (3+2+1) = 0$. So $\operatorname{rank}(X'X) = 3 + 0 = 3$ (Theorem 5.29(b)) and $\det(X'X) = 0$ (Theorem 5.26(b)), in line with $\operatorname{rank} X = 3$ (Theorem 12.10(b)).
- By Theorem 10.9(b), $G = \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$ is a generalized inverse of $X'X$. The rows of $GX'X$ are 0, $(1,1,0,0)$, $(1,0,1,0)$, $(1,0,0,1)$, so $I_4 - GX'X$ has first column $(1,-1,-1,-1)'$ and all other columns 0. By Theorem 12.8(b) the least-squares estimates are

  $$GX'y + t(-1,1,1,1)' = (-t,\ 7+t,\ 4+t,\ 1+t)' \qquad (t \in \mathbb{R}),$$

  and all give $\hat{y} = (7,7,7,4,4,1)'$, the group means (Theorem 12.10(d)).
- Intercept dropped: $X_1'X_1 = \operatorname{diag}(3,2,1)$ and $X_1'y = (21,8,1)'$, so $\hat{\beta} = (7,4,1)'$, the three group means; this is the estimate above with $t = 0$, without its first entry.
- The grand mean is $\bar{y} = 5$, and Theorem 12.10(e) reads

  $$\begin{pmatrix} 5 \\ 7 \\ 9 \\ 3 \\ 5 \\ 1 \end{pmatrix} = 5\,\mathbf{1}_6 + \begin{pmatrix} 2 \\ 2 \\ 2 \\ -1 \\ -1 \\ -4 \end{pmatrix} + \begin{pmatrix} -2 \\ 0 \\ 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \qquad \sum_i (y_i - 5)^2 = 40 = 30 + 10$$

  with $30 = 3 \cdot 4 + 2 \cdot 1 + 16$ and $10 = 4 + 0 + 4 + 1 + 1 + 0$.

Sums of squares and R^2

Definition 12.12. For a least-squares fit, with fitted values $\hat{y}$ and residuals $e = y - \hat{y}$, the **total** and **regression sums of squares** are

$$\mathrm{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2, \qquad \mathrm{SSR} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2,$$

and $\mathrm{SSE} = e'e$. If $\mathrm{SST} > 0$, the **coefficient of determination** is $R^2 = \mathrm{SSR}/\mathrm{SST}$.

Theorem 12.13. If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$, where $\mathrm{SST} = \|C_n y\|^2 = y'C_n y$ and $\mathrm{SSR} = \|C_n \hat{y}\|^2$. Hence, if $\mathrm{SST} > 0$, then $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ and $0 \leqslant R^2 \leqslant 1$.

Proof. • For $v \in \mathbb{R}^n$, $C_n v = v - \bar{v}\mathbf{1}$, so $\|C_n v\|^2 = \sum_i (v_i - \bar{v})^2$, and $\|C_n v\|^2 = v'C_n'C_n v = v'C_n v$ (Proposition 1.13). With $v = y$ this is SST; with $v = \hat{y}$, whose mean is $\bar{y}$ (Proposition 12.4(b)), it is SSR.

- $C_n e = e - \bar{e}\mathbf{1} = e$, as $\bar{e} = 0$ (Proposition 12.4(b)). So $C_n y = C_n \hat{y} + C_n e = C_n \hat{y} + e$.
- $(C_n \hat{y})'e = \hat{y}'C_n e = \hat{y}'e = 0$ (Proposition 12.4(c)). By Pythagoras (Proposition 6.2), $\|C_n y\|^2 = \|C_n \hat{y}\|^2 + \|e\|^2$, that is, $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$.

- Dividing by $\mathrm{SST} > 0$ gives $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$; and $0 \leqslant \mathrm{SSR} \leqslant \mathrm{SST}$, as $\mathrm{SSE} \geqslant 0$. □

This is Theorem 6.25(c) for $U = \operatorname{span}\{\mathbf{1}_n\} \subseteq W = \mathcal{C}(X)$: $P_U y = \bar{y}\mathbf{1}_n$, $P_W y = \hat{y}$, and $\|y\|^2 - n\bar{y}^2 = \mathrm{SST}$; Theorem 12.10(e) is the one-way case. For the design $X = \mathbf{1}_n$ alone, $\hat{y} = \bar{y}\mathbf{1}$, so $\mathrm{SSR} = 0$; R^2 is the fraction of SST that the other columns of X explain. In Example 12.6, $\bar{y} = 5$, $C_4 y = (4, -2, 0, -2)'$ and $C_4\hat{y} = (3, -1, 1, -3)'$, so $\mathrm{SST} = 24$, $\mathrm{SSR} = 20$, $24 = 20 + 4$ and $R^2 = \frac{5}{6}$.

The linear model and the Gauss–Markov theorem

Definition 12.14. The **linear model** is $y = X\beta + \varepsilon$, where the matrix $X \in \mathbb{R}^{n\times p}$ (with or without the column $\mathbf{1}_n$) is known (nonrandom), $\beta \in \mathbb{R}^p$ and $\sigma^2 > 0$ are unknown parameters, and the errors form a random vector with

$$\mathrm{E}\,\varepsilon = 0, \qquad \operatorname{var}\varepsilon = \sigma^2 I_n$$

(Definition 11.1): the errors $\varepsilon_1, \ldots, \varepsilon_n$ have mean 0 and common variance σ^2, and they are uncorrelated.

- The response $y = X\beta + I_n\varepsilon$ is a random vector with $\mathrm{E}\,y = X\beta$ and $\operatorname{var} y = \sigma^2 I_n$ (Theorem 11.3(a),(b)).
- Estimators, estimates and unbiased estimators are those of Definition 11.7, for the parameters β and σ^2: an estimator is computed from y and X alone, and an estimator $\hat{\tau}$ of a function τ of the parameters (such as $c'\beta$ or σ^2) is unbiased if $\mathrm{E}\,\hat{\tau} = \tau$ for every $\beta \in \mathbb{R}^p$ and every $\sigma^2 > 0$. An estimator is **linear** if it equals Ay for a fixed matrix A.
- When $\operatorname{rank} X = p$, the least-squares estimator $\hat{\beta} = (X'X)^{-1}X'y$, the fitted values $\hat{y} = Hy$, the residuals $e = My$ and $\mathrm{SSE} = y'My$ (Theorem 12.3(c)) are functions of the random vector y. As before, x_i' denotes row i of X.

Lemma 12.15. Let y be a random vector with $\mathrm{E}\,y = X\beta$ for an unknown $\beta \in \mathbb{R}^p$, and let $A \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{m\times p}$. Then Ay is an unbiased estimator of $C\beta$ if and only if $AX = C$. In particular, for $a \in \mathbb{R}^n$ and $c \in \mathbb{R}^p$, $a'y$ is an unbiased estimator of $c'\beta$ if and only if $X'a = c$.

Proof. $\mathrm{E}(Ay) = A\,\mathrm{E}\,y = AX\beta$ (Theorem 11.3(a)). If $AX = C$, this is $C\beta$. Conversely, if $AX\beta = C\beta$ for every β, then $\beta = e_j$ gives $\operatorname{col}_j(AX) = \operatorname{col}_j(C)$ for each j, so $AX = C$. The second statement is the case $A = a'$, $C = c'$, since $a'X = c'$ means $X'a = c$. □

Theorem 12.16. In the linear model let $\operatorname{rank} X = p$, and put $A = (X'X)^{-1}X'$, so that $\hat{\beta} = Ay$.

(a) $\hat{\beta} = \beta + A\varepsilon$, $\mathrm{E}\,\hat{\beta} = \beta$ and $\operatorname{var}\hat{\beta} = \sigma^2(X'X)^{-1}$.

(b) $e = M\varepsilon$, $\mathrm{E}\,e = 0$ and $\operatorname{var} e = \sigma^2 M$, so $\operatorname{var} e_i = \sigma^2(1 - h_{ii})$; $\mathrm{E}\,\hat{y} = X\beta$ and $\operatorname{var}\hat{y} = \sigma^2 H$; and $\operatorname{cov}(\hat{\beta}, e) = 0$, $\operatorname{cov}(\hat{y}, e) = 0$.

(c) If $n > p$, then $\mathrm{E}\,\mathrm{SSE} = \sigma^2(n - p)$, and $s^2 = \mathrm{SSE}/(n - p)$ is an unbiased estimator of σ^2.

Only $\mathrm{E}\,\varepsilon = 0$ and $\operatorname{var}\varepsilon = \sigma^2 I_n$ are used; no distribution is assumed.

Proof. • (a) $\hat{\beta} = A(X\beta + \varepsilon) = (X'X)^{-1}X'X\beta + A\varepsilon = \beta + A\varepsilon$. By Theorem 11.3(a),(b), $\mathrm{E}\,\hat{\beta} = \beta + A\,\mathrm{E}\,\varepsilon = \beta$ and $\operatorname{var}\hat{\beta} = A(\sigma^2 I_n)A' = \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$, as $(X'X)^{-1}$ is symmetric (Theorem 2.15(c)).

- (b) $MX = 0$ and $HX = X$ (Theorem 12.2(c)), so $e = My = MX\beta + M\varepsilon = M\varepsilon$ and $\hat{y} = Hy = X\beta + H\varepsilon$. As H and M are symmetric and idempotent (Theorem 12.2(b)), Theorem 11.3(a),(b) gives $\mathrm{E}\,e = 0$, $\operatorname{var} e = M(\sigma^2 I_n)M' = \sigma^2 M$, $\mathrm{E}\,\hat{y} = X\beta$ and $\operatorname{var}\hat{y} = \sigma^2 H$. The ith diagonal entry of $M = I_n - H$ is $1 - h_{ii}$.
- The deviations from the means are $\hat{\beta} - \mathrm{E}\,\hat{\beta} = A\varepsilon$, $\hat{y} - \mathrm{E}\,\hat{y} = H\varepsilon$ and $e - \mathrm{E}\,e = M\varepsilon$, so by Definition 11.1 and Theorem 11.3(b), $\operatorname{cov}(\hat{\beta}, e) = A(\sigma^2 I_n)M' = \sigma^2(X'X)^{-1}(MX)' = 0$ and $\operatorname{cov}(\hat{y}, e) = \sigma^2 HM = 0$ (Theorem 12.2(c)).

- (c) SSE $= y'My$, where M is symmetric and idempotent of rank $n-p$ (Theorem 12.2(b),(d)) and $MX\beta = 0$. Example 11.8 with $P = M$ and $\mu = X\beta$ gives E SSE $= \sigma^2(n-p) + \beta'X'MX\beta = \sigma^2(n-p)$. □

Definition 12.17. Let y be a random vector with $\mathrm{E}\, y = X\beta$, and let $c \in \mathbb{R}^p$. A linear unbiased estimator $a'y$ of $c'\beta$ is a **best linear unbiased estimator** (**BLUE**) of $c'\beta$ if $\operatorname{var}(a'y) \leqslant \operatorname{var}(b'y)$ for every linear unbiased estimator $b'y$ of $c'\beta$ and all values of the parameters.

Theorem 12.18 (Gauss–Markov)**.** In the linear model let $\operatorname{rank} X = p$.

(a) Let $A \in \mathbb{R}^{p\times n}$ satisfy $AX = I_p$, so that Ay is a linear unbiased estimator of β (Lemma 12.15), and put $\Delta = A - (X'X)^{-1}X'$. Then

$$\operatorname{var}(Ay) = \operatorname{var}\hat\beta + \sigma^2\Delta\Delta'$$

so $\operatorname{var}(Ay) \geqslant \operatorname{var}\hat\beta$ in the Loewner order (Definition 8.27), with equality only if $A = (X'X)^{-1}X'$. In particular $\operatorname{var}((Ay)_j) \geqslant \operatorname{var}\hat\beta_j$ for every j.

(b) Let $c \in \mathbb{R}^p$. Every linear unbiased estimator $a'y$ of $c'\beta$ satisfies

$$\operatorname{var}(a'y) = \operatorname{var}(c'\hat\beta) + \sigma^2\|a - X(X'X)^{-1}c\|^2$$

Hence $c'\hat\beta$ is a BLUE of $c'\beta$ (Definition 12.17), and the only one.

Only $\mathrm{E}\,\varepsilon = 0$ and $\operatorname{var}\varepsilon = \sigma^2 I_n$ are used.

Proof. (a)

- $\Delta X = AX - (X'X)^{-1}X'X = I_p - I_p = 0$.
- $\operatorname{var}(Ay) = A(\sigma^2 I_n)A' = \sigma^2 AA'$ (Theorem 11.3(b)), and, as $(X'X)^{-1}$ is symmetric,

$$\begin{aligned} AA' &= \big((X'X)^{-1}X' + \Delta\big)\big(X(X'X)^{-1} + \Delta'\big) \\ &= (X'X)^{-1} + (X'X)^{-1}(\Delta X)' + \Delta X(X'X)^{-1} + \Delta\Delta' = (X'X)^{-1} + \Delta\Delta' \end{aligned}$$

  With $\operatorname{var}\hat\beta = \sigma^2(X'X)^{-1}$ (Theorem 12.16(a)) this is the identity.
- $\Delta\Delta' = (\Delta')'\Delta'$ is positive semidefinite (Theorem 8.16(a)), so $\operatorname{var}(Ay) \geqslant \operatorname{var}\hat\beta$, and the diagonal entries compare in the same way (Proposition 8.28(c)). If $\operatorname{var}(Ay) = \operatorname{var}\hat\beta$, then $\Delta\Delta' = 0$, so $\operatorname{tr}(\Delta\Delta') = 0$ and $\Delta = 0$ (Theorem 1.11(d)).

(b)

- By Lemma 12.15, $X'a = c$. Put $a_0 = X(X'X)^{-1}c$ and $d = a - a_0$. Then $a_0'y = c'(X'X)^{-1}X'y = c'\hat\beta$, and $X'd = c - X'X(X'X)^{-1}c = 0$, so $a_0'd = c'(X'X)^{-1}X'd = 0$.
- By Theorem 11.3(b) and Pythagoras (Proposition 6.2), $\operatorname{var}(a'y) = \sigma^2 a'a = \sigma^2(a_0'a_0 + d'd) = \operatorname{var}(c'\hat\beta) + \sigma^2\|d\|^2$, the identity.
- So $\operatorname{var}(a'y) \geqslant \operatorname{var}(c'\hat\beta)$ for every linear unbiased $a'y$: $c'\hat\beta$ is a BLUE. If $a'y$ is also a BLUE, then $\operatorname{var}(a'y) \leqslant \operatorname{var}(c'\hat\beta)$, so $\|d\|^2 = 0$, $a = a_0$ and $a'y = c'\hat\beta$. □

Definition 12.19. Let $X \in \mathbb{R}^{n\times p}$ and $c \in \mathbb{R}^p$. The linear function $c'\beta$ of $\beta \in \mathbb{R}^p$ is **estimable** if for every $y \in \mathbb{R}^n$ the number $c'\hat\beta$ is the same for all solutions $\hat\beta$ of the normal equations $X'X\beta = X'y$.

Theorem 12.20. Let $X \in \mathbb{R}^{n\times p}$, $c \in \mathbb{R}^p$, let G be a generalized inverse of $X'X$, and $P_X = XGX'$. The following are equivalent:

(i) $c'\beta$ is estimable;

(ii) in the linear model, $c'\beta$ has a linear unbiased estimator;

(iii) $c \in \mathcal{C}(X')$, that is, $c = X'a$ for some $a \in \mathbb{R}^n$;

(iv) $c \perp \mathcal{N}(X)$;

(v) $c'GX'X = c'$.

If they hold, then, with a as in (iii), $c'\hat\beta = c'GX'y = a'P_Xy$ for every solution $\hat\beta$ of $X'X\beta = X'y$, and $c'Gc = a'P_Xa$ does not depend on the choice of G.

Proof. • (ii) $\Leftrightarrow$ (iii): by Lemma 12.15, $a'y$ is unbiased for $c'\beta$ exactly when $X'a = c$.

- (iii) $\Rightarrow$ (i) and the formulas: if $c = X'a$, every solution has $c'\hat\beta = a'X\hat\beta = a'P_Xy$ (Theorem 12.8(b)), and $GX'y$ is one of them; also $c'Gc = a'XGX'a = a'P_Xa$, with P_X independent of G (Theorem 12.8(a)).
- (i) $\Rightarrow$ (iv): take $y = 0$. The solutions of $X'X\beta = 0$ form $\mathcal{N}(X'X) = \mathcal{N}(X)$ (Theorem 4.18(a)), and $\beta = 0$ is one of them; so $c'z = c'0 = 0$ for every $z \in \mathcal{N}(X)$.
- (iv) $\Rightarrow$ (iii): write $c = X'a + z$ with $z \in \mathcal{N}(X)$ ($\mathbb{R}^p = \mathcal{C}(X') \oplus \mathcal{N}(X)$, Corollary 6.6). Then $0 = c'z = a'Xz + z'z = z'z$, so $z = 0$.
- (iii) $\Leftrightarrow$ (v): if $c = X'a$, then $c'GX'X = a'XGX'X = a'X = c'$ (Lemma 10.12(a)); conversely, $c' = c'GX'X$ gives $c = X'(XG'c)$. □

Thus $c'\beta$ is estimable exactly when c' is a linear combination of the rows of X; in particular every $x_i'\beta$ is estimable, with estimate the fitted value $x_i'\hat\beta$. If $\operatorname{rank} X = p$, then $\mathcal{N}(X) = \{0\}$ and every $c'\beta$ is estimable.

Proposition 12.21. In the linear model let $X \in \mathbb{R}^{n\times p}$ have any rank, let G be a generalized inverse of $X'X$, $P_X = XGX'$, let $\hat\beta$ be any solution of the normal equations, and let $c'\beta$ be estimable. Every linear unbiased estimator $b'y$ of $c'\beta$ satisfies $P_Xb = XGc$, $c'\hat\beta = (P_Xb)'y$ and

$$\operatorname{var}(b'y) = \operatorname{var}(c'\hat\beta) + \sigma^2\|b - P_Xb\|^2, \qquad \operatorname{var}(c'\hat\beta) = \sigma^2 c'Gc$$

Hence $c'\hat\beta$ is the only BLUE of $c'\beta$; it is the same for every solution $\hat\beta$.

Proof. • $X'b = c$ (Lemma 12.15), so $P_Xb = XGX'b = XGc$. By Theorem 12.20 with $a = b$, $c'\hat\beta = b'P_Xy = (P_Xb)'y$ for every solution $\hat\beta$, as P_X is symmetric (Theorem 12.8(a)).

- P_X is symmetric and idempotent, so $(P_Xb)'(b - P_Xb) = b'(P_X - P_X^2)b = 0$, and Pythagoras gives $\|b\|^2 = \|P_Xb\|^2 + \|b - P_Xb\|^2$. Multiplying by σ^2 gives the identity (Theorem 11.3(b)), and $\operatorname{var}(c'\hat\beta) = \sigma^2\|P_Xb\|^2 = \sigma^2 b'P_Xb = \sigma^2 b'XGX'b = \sigma^2 c'Gc$.
- $c'\hat\beta = (P_Xb)'y$ is linear, and unbiased: $P_XX = X$ (Theorem 12.8(a)), so $X'(P_Xb) = (P_XX)'b = X'b = c$. By the identity it is a BLUE, and a BLUE $b'y$ has $b = P_Xb$, that is, $b'y = c'\hat\beta$. □

Example 12.22. In the one-way layout of Example 12.11 ($\beta = (\mu, \alpha_1, \alpha_2, \alpha_3)'$, $\mathcal{N}(X) = \operatorname{span}\{(-1,1,1,1)'\}$):

- By Theorem 12.20(iv), $c'\beta$ is estimable if and only if $-c_1 + c_2 + c_3 + c_4 = 0$. So $\mu + \alpha_l$ and $\alpha_l - \alpha_m$ are estimable, while μ and each single α_l are not; by Theorem 12.20(ii), no linear estimator is unbiased for α_1.
- With $G = \operatorname{diag}(0, \frac13, \frac12, 1)$ and $\hat\beta = GX'y = (0, 7, 4, 1)'$, Proposition 12.21 gives the BLUE of $\alpha_1 - \alpha_2$ ($c = (0, 1, -1, 0)'$): $c'\hat\beta = 7 - 4 = 3$, the difference $\bar y_1 - \bar y_2$ of the group means, with variance $\sigma^2 c'Gc = \sigma^2(\frac13 + \frac12) = \frac56\sigma^2$. The BLUE of $\mu + \alpha_3$ is $\bar y_3 = 1$, with variance σ^2.

- Every least-squares estimate $(-t,\ 7+t,\ 4+t,\ 1+t)'$ gives $\hat{\alpha}_1 - \hat{\alpha}_2 = 3$, whereas $\hat{\alpha}_1 = 7+t$ depends on t.

Proposition 12.23. Let $X \in \mathbb{R}^{n\times p}$ have rank p, let $W \in \mathbb{R}^{n\times n}$ be positive definite, and let $y \in \mathbb{R}^n$. Then $X'WX$ is positive definite, and the unique minimizer of $Q(\beta) = (y - X\beta)'W(y - X\beta)$ $(\beta \in \mathbb{R}^p)$ is the **weighted least-squares estimate**

$$\hat{\beta}_W = (X'WX)^{-1}X'Wy$$

For $W = \operatorname{diag}(w_1, \ldots, w_n)$ with all $w_i > 0$, $Q(\beta) = \sum_i w_i(y_i - x_i'\beta)^2$.

Proof. $A = X'WX$ is positive definite (Theorem 8.16(b)). As $W' = W$, $Q(\beta) = y'Wy - 2\beta'X'Wy + \beta'X'WX\beta = y'Wy + 2q(\beta)$ with $q(\beta) = \frac{1}{2}\beta'A\beta - b'\beta$ and $b = X'Wy$. By Example 9.20(a), the unique minimizer of q, hence of Q, is $A^{-1}b = \hat{\beta}_W$. A diagonal W with positive diagonal entries is positive definite, as $v'Wv = \sum_i w_i v_i^2 > 0$ for $v \neq 0$, and $Q(\beta) = \sum_i w_i u_i^2$ for $u = y - X\beta$. □

Theorem 12.24 (Aitken)**.** Let $y = X\beta + \varepsilon$, where $X \in \mathbb{R}^{n\times p}$ is known with $\operatorname{rank} X = p$, $\mathrm{E}\,\varepsilon = 0$ and $\operatorname{var}\varepsilon = \sigma^2 V$ with $V \in \mathbb{R}^{n\times n}$ known and positive definite, and β and $\sigma^2 > 0$ are unknown. Then $X'V^{-1}X$ is invertible. Put

$$\tilde{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}y \qquad \text{(the } \textbf{generalized least-squares estimator}\text{)}$$

(a) $\tilde{\beta}$ is the unique minimizer of $(y - X\beta)'V^{-1}(y - X\beta)$, $\mathrm{E}\,\tilde{\beta} = \beta$ and $\operatorname{var}\tilde{\beta} = \sigma^2(X'V^{-1}X)^{-1}$.

(b) For every $c \in \mathbb{R}^p$, $c'\tilde{\beta}$ is the only BLUE of $c'\beta$.

(c) $\hat{\beta} = (X'X)^{-1}X'y$ is unbiased, and $\operatorname{var}\hat{\beta} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1} \geqslant \operatorname{var}\tilde{\beta}$.

Proof.
- Let $T = V^{-1/2}$, so that $TVT = I_n$ and $T^2 = V^{-1}$ (Theorem 8.25(c)). Also $T = (V^{1/2})^{-1}$ is symmetric, as $V^{1/2}$ is positive definite, hence symmetric (Theorem 2.15(c)).
- Put $y^* = Ty$, $X^* = TX$, $\varepsilon^* = T\varepsilon$. Then $y^* = X^*\beta + \varepsilon^*$, $\mathrm{E}\,\varepsilon^* = 0$ and $\operatorname{var}\varepsilon^* = T(\sigma^2 V)T' = \sigma^2 I_n$ (Theorem 11.3(a),(b)), and $\operatorname{rank} X^* = \operatorname{rank} X = p$ (Theorem 4.15(d)): the transformed model is a linear model with full column rank.
- $X^{*\prime}X^* = X'T^2X = X'V^{-1}X$, which is therefore invertible (Theorem 12.1(b)), and $X^{*\prime}y^* = X'V^{-1}y$. So $\tilde{\beta}$ is the least-squares estimator of the transformed model.
- (a) $\|y^* - X^*\beta\|^2 = (y - X\beta)'T^2(y - X\beta) = (y - X\beta)'V^{-1}(y - X\beta)$, whose unique minimizer is $\tilde{\beta}$ (Theorem 12.3(c)). Theorem 12.16(a) for the transformed model gives $\mathrm{E}\,\tilde{\beta} = \beta$ and $\operatorname{var}\tilde{\beta} = \sigma^2(X^{*\prime}X^*)^{-1}$.
- (b) For $a \in \mathbb{R}^n$, $a'y = (T^{-1}a)'y^*$, as T^{-1} is symmetric. So the linear estimators of the two descriptions are the same random variables, with the same means and variances, and the linear unbiased estimators of $c'\beta$ are the same. Theorem 12.18(b) for the transformed model gives the claim.
- (c) $\hat{\beta} = Ay$ with $A = (X'X)^{-1}X'$ and $AX = I_p$, so $\hat{\beta}$ is unbiased (Lemma 12.15) and $\operatorname{var}\hat{\beta} = A(\sigma^2 V)A'$ (Theorem 11.3(b)). Also $\hat{\beta} = (AT^{-1})y^*$ with $(AT^{-1})X^* = AX = I_p$, so Theorem 12.18(a) for the transformed model gives $\operatorname{var}\hat{\beta} \geqslant \operatorname{var}\tilde{\beta}$. □

If the errors are uncorrelated with $\operatorname{var}\varepsilon_i = \sigma^2 v_i$ and known $v_i > 0$ (**heteroscedastic** errors), then $V = \operatorname{diag}(v_1, \ldots, v_n)$ and $V^{-1} = \operatorname{diag}(1/v_1, \ldots, 1/v_n)$, and $\tilde{\beta}$ is the weighted least-squares estimate with weights $w_i = 1/v_i$ (Proposition 12.23).

Inference in the normal linear model

Definition 12.25. The **normal linear model** is the linear model of Definition 12.14 with $\varepsilon \sim N_n(0, \sigma^2 I_n)$. Then $y \sim N_n(X\beta, \sigma^2 I_n)$ (Theorem 11.14(a)), so $y_1, \ldots, y_n$ are independent with $y_i \sim N(x_i'\beta, \sigma^2)$ (Theorems 11.17(c) and 11.14(b)). In general, let $\alpha \in (0,1)$, and let the distribution of a random vector y depend on unknown parameters ϑ.

- The likelihood, the log-likelihood and the maximum likelihood estimates (MLE) of ϑ are those of Definition 11.30, with the density $f(y;\vartheta)$ of y; for a discrete y, $f(y;\vartheta)$ is instead the probability that y takes the observed value.
- Confidence intervals of level $1-\alpha$ for real functions of the parameters, such as $c'\beta$, are as in Example 11.28. For a further random variable y_0 (a future observation), an interval $[L, U]$ whose endpoints are computed from y is a **prediction interval** of level $1-\alpha$ for y_0 if $\mathrm{P}(L \leqslant y_0 \leqslant U) = 1-\alpha$ for all values of the parameters.
- A **hypothesis** H_0 is a condition on the parameters, such as $R\beta = r$. A **test** of H_0 is given by a **rejection region** $\mathcal{R}$, a set of possible values of y: H_0 is rejected if the observed y lies in $\mathcal{R}$. The test has **level** α if $\mathrm{P}(y \in \mathcal{R}) = \alpha$ for all parameter values that satisfy H_0. Usually $\mathcal{R}$ is described through a **test statistic** T computed from y, as $\{T > c\}$ or $\{|T| > c\}$. When H_0 is rejected at level α, the departure from H_0 is called **significant** at level α.
- The quantiles z_τ, $t_{\nu,\tau}$, $F_{a,b,\tau}$ and $\chi^2_{r,\tau}$ $(0 < \tau < 1)$, and the facts (P1)–(P6), are those of Remark 11.2.

Lemma 12.26. Let $T \sim N(0,1)$ and $c = z_{1-\alpha/2}$, or $T \sim t_\nu$ and $c = t_{\nu,1-\alpha/2}$. Then $-T$ has the same distribution as T, and $\mathrm{P}(|T| \leqslant c) = 1-\alpha$.

Proof. If $T \sim N(0,1)$, then $-T \sim N(0,1)$ by Theorem 11.14(a). If $T \sim t_\nu$, only the distribution of T matters (Remark 11.2), so take $T = u/\sqrt{v/\nu}$ with $u \sim N(0,1)$ and $v \sim \chi^2_\nu$ independent (P5); then $-T \sim t_\nu$ as in the first bullet of Example 11.28, with ν in place of $n-1$. In both cases the argument of the second bullet of that example gives $\mathrm{P}(|T| \leqslant c) = 1-\alpha$. □

Lemma 12.27. $\log t \leqslant t-1$ for every $t > 0$, with equality only for $t = 1$ (log is the natural logarithm).

Proof. This is Problem 9.7(c), with t in place of θ. □

Theorem 12.28. In the normal linear model let $\operatorname{rank} X = p$ and $n > p$.

(a) $\hat\beta \sim N_p(\beta, \sigma^2(X'X)^{-1})$; hence $c'\hat\beta \sim N(c'\beta, \sigma^2 c'(X'X)^{-1}c)$ for $c \in \mathbb{R}^p$.

(b) $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$; in particular $\mathrm{P}(\mathrm{SSE} = 0) = 0$.

(c) $\hat\beta$ and SSE are independent.

(d) The log-likelihood is $\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - X\beta\|^2$ ($\beta \in \mathbb{R}^p$, $\sigma^2 > 0$). If $\mathrm{SSE} > 0$, its only maximizer is $(\hat\beta, \mathrm{SSE}/n)$.

Proof.
- (a) $\hat\beta = \beta + A\varepsilon$ with $A = (X'X)^{-1}X'$ (Theorem 12.16(a)), so Theorem 11.14(a) gives $\hat\beta \sim N_p(\beta, A(\sigma^2 I_n)A') = N_p(\beta, \sigma^2(X'X)^{-1})$, and then, with the $1 \times p$ matrix c', $c'\hat\beta \sim N(c'\beta, \sigma^2 c'(X'X)^{-1}c)$.
- (b) M is symmetric and idempotent of rank $n - p \geqslant 1$ (Theorem 12.2(b),(d)), and $e = M\varepsilon$ (Theorem 12.16(b)), so $\mathrm{SSE} = e'e = \varepsilon'M'M\varepsilon = \varepsilon'M\varepsilon$. Apply Corollary 11.24(a) to $\varepsilon \sim N_n(0, \sigma^2 I_n)$. Each single number, 0 in particular, has probability 0 under χ^2_{n-p} (Remark 11.2(P5)).
- (c) $AM = (X'X)^{-1}(MX)' = 0$, so $A\varepsilon$ and $\varepsilon'M\varepsilon$ are independent (Theorem 11.25(c)). Hence so are $\hat\beta = \beta + A\varepsilon$ and $\mathrm{SSE} = \varepsilon'M\varepsilon$ (Remark 11.2(P1)).

- (d) $\sigma^2 I_n$ is positive definite with determinant σ^{2n} (Theorem 5.15(a)) and inverse $\sigma^{-2} I_n$, so by Theorem 11.15 the density of y is $(2\pi\sigma^2)^{-n/2} \exp\bigl(-\|y - X\beta\|^2/(2\sigma^2)\bigr)$, whose logarithm is ℓ.
- For fixed σ^2, $\|y - X\beta\|^2 = \mathrm{SSE} + \|X(\hat\beta - \beta)\|^2$ (Theorem 12.3(a)), and $X(\hat\beta - \beta) \neq 0$ for $\beta \neq \hat\beta$ (Theorem 4.13(b)). So $\ell(\beta, \sigma^2) < \ell(\hat\beta, \sigma^2)$ for $\beta \neq \hat\beta$.
- Put $\hat v = \mathrm{SSE}/n > 0$ and $t = \hat v/\sigma^2$. Then

$$\ell(\hat\beta, \sigma^2) - \ell(\hat\beta, \hat v) = -\frac{n}{2} \log \frac{\sigma^2}{\hat v} - \frac{n\hat v}{2\sigma^2} + \frac{n}{2} = \frac{n}{2}(\log t - t + 1) \leqslant 0,$$

  with equality only for $t = 1$ (Lemma 12.27). So $\ell(\beta, \sigma^2) \leqslant \ell(\hat\beta, \sigma^2) \leqslant \ell(\hat\beta, \hat v)$, with equality throughout only for $\beta = \hat\beta$ and $\sigma^2 = \hat v$. □

By Theorem 12.16(c), $\mathrm{E}(\mathrm{SSE}/n) = \sigma^2(n-p)/n < \sigma^2$: the maximum likelihood estimator of σ^2 is biased, and the unbiased s^2 is used instead.

Theorem 12.29. In the normal linear model let $\operatorname{rank} X = p$ and $n > p$, let $c \in \mathbb{R}^p$ be nonzero, put $t^* = t_{n-p,1-\alpha/2}$, and let $\mathrm{SE}(c'\hat\beta) = s\sqrt{c'(X'X)^{-1}c}$, the **standard error** of $c'\hat\beta$, where $s = \sqrt{s^2}$ (Theorem 12.16(c)); by Theorem 12.28(b), $s > 0$ with probability 1.

(a) $T = (c'\hat\beta - c'\beta)/\mathrm{SE}(c'\hat\beta) \sim t_{n-p}$.

(b) $\bigl[c'\hat\beta - t^* \mathrm{SE}(c'\hat\beta),\ c'\hat\beta + t^* \mathrm{SE}(c'\hat\beta)\bigr]$ is a confidence interval of level $1-\alpha$ for $c'\beta$. For $r_0 \in \mathbb{R}$, the test of $H_0\colon c'\beta = r_0$ that rejects when $|T_0| > t^*$, where $T_0 = (c'\hat\beta - r_0)/\mathrm{SE}(c'\hat\beta)$, has level α; it rejects exactly when r_0 lies outside this interval.

(c) Let $x_0 \in \mathbb{R}^p$, and let $y_0 = x_0'\beta + \varepsilon_0$ be a future observation with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of ε. Put $w_0 = s\sqrt{1 + x_0'(X'X)^{-1}x_0}$. Then $(y_0 - x_0'\hat\beta)/w_0 \sim t_{n-p}$, and $[x_0'\hat\beta - t^* w_0,\ x_0'\hat\beta + t^* w_0]$ is a prediction interval of level $1-\alpha$ for y_0.

Proof.
- (a) $(X'X)^{-1}$ is positive definite (Theorem 8.16(a), Corollary 8.15(a)), so $c'(X'X)^{-1}c > 0$. By Theorem 12.28(a) and Theorem 11.14(a), $z = (c'\hat\beta - c'\beta)/(\sigma\sqrt{c'(X'X)^{-1}c}) \sim N(0,1)$. Next $v = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$ (Theorem 12.28(b)), and z and v are independent, as functions of $\hat\beta$ and of SSE (Theorem 12.28(c), (P1)). Since $s/\sigma = \sqrt{v/(n-p)}$, $T = z/\sqrt{v/(n-p)} \sim t_{n-p}$ (P5).
- (b) By (a) and Lemma 12.26, $\mathrm{P}(|T| \leqslant t^*) = 1 - \alpha$, and $|T| \leqslant t^*$ holds exactly when $c'\beta$ lies in the interval. If H_0 holds, then $T_0 = T$, so $\mathrm{P}(|T_0| > t^*) = 1 - (1-\alpha) = \alpha$. Finally $|T_0| > t^*$ means $|c'\hat\beta - r_0| > t^* \mathrm{SE}(c'\hat\beta)$, that is, r_0 lies outside the interval.
- (c) As $\varepsilon \sim N_n(0, \sigma^2 I_n)$ and $\varepsilon_0 \sim N(0, \sigma^2)$ are independent, $\tilde\varepsilon = (\varepsilon', \varepsilon_0)' \sim N_{n+1}(0, \sigma^2 I_{n+1})$ (Theorem 11.17(d)).
- With $A = (X'X)^{-1}X'$, $y_0 - x_0'\hat\beta = \varepsilon_0 - x_0'A\varepsilon = b'\tilde\varepsilon$, where $b = (-x_0'A,\ 1)'$ has $b'b = x_0'AA'x_0 + 1 = 1 + x_0'(X'X)^{-1}x_0$. So $y_0 - x_0'\hat\beta \sim N\bigl(0, \sigma^2(1 + x_0'(X'X)^{-1}x_0)\bigr)$ (Theorem 11.14(b)).
- $\mathrm{SSE} = \varepsilon' M \varepsilon = \tilde\varepsilon' P \tilde\varepsilon$ with $P = \operatorname{diag}(M, 0)$ symmetric and idempotent, and $b'P = (-x_0'AM,\ 0) = 0$. By Theorem 11.25(c), $y_0 - x_0'\hat\beta$ and SSE are independent. As in (a), $(y_0 - x_0'\hat\beta)/w_0 \sim t_{n-p}$, and as in (b), $\mathrm{P}(|y_0 - x_0'\hat\beta| \leqslant t^* w_0) = 1 - \alpha$. □

For $x_0 \neq 0$, the confidence interval of (b) for $c = x_0$ has half-width $t^* s\sqrt{x_0'(X'X)^{-1}x_0}$; the prediction interval of (c) is wider, because the new error ε_0 adds σ^2 to the variance.

Theorem 12.30. Let $X \in \mathbb{R}^{n\times p}$ have rank p, let $R \in \mathbb{R}^{q\times p}$ have rank q, and let $r \in \mathbb{R}^q$, $y \in \mathbb{R}^n$ and $\hat\beta = (X'X)^{-1}X'y$. Then $V = R(X'X)^{-1}R'$ is positive definite, and:

(a) the unique minimizer of $S(\beta) = \|y - X\beta\|^2$ subject to $R\beta = r$ is the **constrained least-squares estimate**

$$\hat{\beta}_R = \hat{\beta} - (X'X)^{-1}R'V^{-1}(R\hat{\beta} - r);$$

(b) for every β with $R\beta = r$, $\|y - X\beta\|^2 = \|y - X\hat{\beta}_R\|^2 + \|X(\hat{\beta}_R - \beta)\|^2$;

(c) $\mathrm{SSE}_R - \mathrm{SSE} = (R\hat{\beta} - r)'V^{-1}(R\hat{\beta} - r)$, where $\mathrm{SSE}_R = \|y - X\hat{\beta}_R\|^2$ and $\mathrm{SSE} = \|y - X\hat{\beta}\|^2$.

Proof. $A = X'X$ is positive definite (Theorem 8.16(a)). With $b = X'y$,

$$S(\beta) = y'y - 2\beta'X'y + \beta'X'X\beta = y'y + 2q(\beta), \qquad q(\beta) = \tfrac{1}{2}\beta'A\beta - b'\beta,$$

and $A^{-1}b = \hat{\beta}$. Apply Theorem 9.27 with $C = R$ and $d = r$: there $V = CA^{-1}C'$ is positive definite, and the point x^* is $\hat{\beta}_R$. (a) S and q have the same minimizers on $\{\beta : R\beta = r\}$; apply Theorem 9.27(a). (b) By Theorem 9.27(b), $S(\beta) - S(\hat{\beta}_R) = 2\Big(q(\beta) - q(\hat{\beta}_R)\Big) = (\beta - \hat{\beta}_R)'X'X(\beta - \hat{\beta}_R) = \|X(\beta - \hat{\beta}_R)\|^2$. (c) By Theorem 9.27(c), $\mathrm{SSE}_R - \mathrm{SSE} = S(\hat{\beta}_R) - S(\hat{\beta}) = 2\Big(q(\hat{\beta}_R) - q(\hat{\beta})\Big) = (R\hat{\beta} - r)'V^{-1}(R\hat{\beta} - r)$. □

Theorem 12.31. In the normal linear model let $\operatorname{rank} X = p$ and $n > p$, let $R \in \mathbb{R}^{q\times p}$ have rank q, let $r \in \mathbb{R}^q$, and let $\hat{\beta}_R$ and SSE_R be as in Theorem 12.30.

(a) For every y with $\mathrm{SSE} > 0$,

$$F = \frac{(R\hat{\beta} - r)'\Big(R(X'X)^{-1}R'\Big)^{-1}(R\hat{\beta} - r)/q}{s^2} = \frac{(\mathrm{SSE}_R - \mathrm{SSE})/q}{\mathrm{SSE}/(n-p)}$$

(b) If $H_0\colon R\beta = r$ holds, then F, which is defined with probability 1 (Theorem 12.28(b)), has the distribution $F_{q,n-p}$; so the test that rejects H_0 when $F > F_{q,n-p,1-\alpha}$ has level α.

(c) Let H_0 set q given coefficients to 0: $r = 0$, and R consists of the corresponding rows of I_p, which are linearly independent, so $\operatorname{rank} R = q$ (Theorem 4.13(c)). Let X_R be X without the corresponding columns. Then SSE_R is the error sum of squares of the least-squares fit with design X_R. If $\mathbf{1}_n \in \mathcal{C}(X_R)$ and $\mathrm{SSE} > 0$, then, with R^2 and R_0^2 the coefficients of determination of the fits with designs X and X_R,

$$F = \frac{(R^2 - R_0^2)/q}{(1 - R^2)/(n-p)}$$

In particular, for the **overall F test** of $H_0\colon \beta_1 = \cdots = \beta_k = 0$, where $X = [\mathbf{1}_n\ X_0]$ and $p = k + 1 \geqslant 2$, $\mathrm{SSE}_R = \mathrm{SST}$, $\mathrm{SSE}_R - \mathrm{SSE} = \mathrm{SSR}$ and

$$F = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)}$$

(d) If $q = 1$ and $R = c'$, then $F = T_0^2$, where T_0 is the statistic of Theorem 12.29(b) for $r_0 = r$.

Part (a) and the identities in (c) and (d) are algebraic: they hold for every y with $\mathrm{SSE} > 0$.

Proof. • (a) $R(X'X)^{-1}R'$ is positive definite, hence invertible (Theorem 12.30, Corollary 8.15(a)); by Theorem 12.30(c) the numerator quadratic form is $\mathrm{SSE}_R - \mathrm{SSE}$, and $s^2 = \mathrm{SSE}/(n-p)$.

• (b) Under H_0, $R\hat{\beta} \sim N_q(r, \Sigma_R)$ with $\Sigma_R = \sigma^2 R(X'X)^{-1}R'$ positive definite (Theorems 12.28(a) and 11.14(a)). By Corollary 11.24(b), $u = (R\hat{\beta} - r)'\Sigma_R^{-1}(R\hat{\beta} - r) \sim \chi^2_q$. As a function of $\hat{\beta}$, u is independent of $v = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$ (Theorem 12.28(b),(c), (P1)), and $F = (u/q)/(v/(n-p))$. So $F \sim F_{q,n-p}$ (P5), and $\mathrm{P}(F > F_{q,n-p,1-\alpha}) = 1 - (1-\alpha) = \alpha$.

- (c) Let γ collect the coefficients that H_0 does not set to 0. The β with $R\beta = 0$ give exactly the vectors $X\beta = X_R\gamma$, γ arbitrary. By Theorem 12.30(a), SSE_R is the minimum of $\|y - X\beta\|^2$ over these β, that is, the minimum of $\|y - X_R\gamma\|^2$: the error sum of squares of the design X_R. If $\mathbf{1}_n \in \mathcal{C}(X_R)$, then also $\mathbf{1}_n \in \mathcal{C}(X)$, $\mathrm{SST} \geqslant \mathrm{SSE} > 0$, and Theorem 12.13 gives $\mathrm{SSE} = \mathrm{SST}(1 - R^2)$ and $\mathrm{SSE}_R = \mathrm{SST}(1 - R_0^2)$; substitute into (a) and cancel SST.
- For the overall test, $X_R = \mathbf{1}_n$, whose fitted values are $\bar{y}\mathbf{1}_n$ (Example 6.22). So $\mathrm{SSE}_R = \mathrm{SST}$, $R_0^2 = 0$ and $\mathrm{SSE}_R - \mathrm{SSE} = \mathrm{SSR}$ (Theorem 12.13).
- (d) For $R = c'$ the numerator of F is $(c'\hat{\beta} - r)^2/(c'(X'X)^{-1}c)$, so $F = T_0^2$. □

Example 12.32. Eight plots receive a fertilizer at two levels, coded $x_{i1} = -1, 1$, and irrigation at two levels, $x_{i2} = -1, 1$, two plots for each combination; the yields are

x_{i1}	-1	-1	-1	-1	1	1	1	1
x_{i2}	-1	-1	1	1	-1	-1	1	1
y_i	9	3	9	7	11	13	11	17

In the normal linear model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ ($n = 8$, $p = 3$), find the confidence interval of level $1 - \alpha$ for β_1, the tests of $\beta_2 = 0$ and of $\beta_1 = \beta_2 = 0$ at level α, and the prediction interval of level $1 - \alpha$ for a plot with $x_1 = x_2 = 1$. Put $t^* = t_{5,1-\alpha/2}$.

Solution.

- The three columns of X are orthogonal with squared length 8, so $X'X = 8I_3$, $X'y = (80, 24, 8)'$ and $\hat{\beta} = (10, 3, 1)'$. The fitted values are $6, 8, 12, 14$ for the four combinations, the residuals $e = (3, -3, 1, -1, -1, 1, -3, 3)'$, $\mathrm{SSE} = 40$ and $s^2 = 40/5 = 8$. With $\bar{y} = 10$, $\mathrm{SSR} = 2(16 + 4 + 4 + 16) = 80$, $\mathrm{SST} = 120$ and $R^2 = \frac{2}{3}$.
- $\mathrm{SE}(\hat{\beta}_j) = s\sqrt{1/8} = 1$ for every j. By Theorem 12.29(b) the interval for β_1 is $[3 - t^*,\ 3 + t^*]$, and the test of $\beta_2 = 0$ has $T_0 = 1$: it rejects if and only if $t^* < 1$; equivalently $F = T_0^2 = 1$ (Theorem 12.31(d)).
- The overall F statistic is $F = (80/2)/(40/5) = 5 = \frac{(2/3)/2}{(1/3)/5}$ (Theorem 12.31(c)); the test rejects if and only if $F_{2,5,1-\alpha} < 5$.
- For $x_0 = (1, 1, 1)'$, $x_0'\hat{\beta} = 14$ and $x_0'(X'X)^{-1}x_0 = \frac{3}{8}$, so $w_0 = s\sqrt{1 + \frac{3}{8}} = \sqrt{11}$: the prediction interval is $[14 - t^*\sqrt{11},\ 14 + t^*\sqrt{11}]$ (Theorem 12.29(c)), while the confidence interval for $x_0'\beta$ is $[14 - t^*\sqrt{3},\ 14 + t^*\sqrt{3}]$.

Exponential families and generalized linear models

Definition 12.33. Let $\Theta \subseteq \mathbb{R}$ be an open interval, $\phi > 0$, and $b\colon \Theta \to \mathbb{R}$ twice continuously differentiable. A family of distributions of a real random variable y, indexed by $\theta \in \Theta$, is a **one-parameter exponential family** with **natural parameter** θ, **dispersion** ϕ and **cumulant function** b if y has a density, or probabilities, of the form

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

for a function c that does not involve θ.

Remark 12.34 (Imported facts on series)**.** For integers $0 \leqslant k \leqslant m$, $\binom{m}{k} = \frac{m!}{k!\,(m-k)!}$, with $0! = 1$. The following facts are used without proof.

(S1) (*Binomial theorem.*) $(a + b)^m = \sum_{k=0}^{m} \binom{m}{k} a^k b^{m-k}$ for $a, b \in \mathbb{R}$ and every integer $m \geqslant 0$.

(S2) (*Exponential series.*) $\sum_{k=0}^{\infty} \mu^k/k! = e^{\mu}$ for every $\mu \in \mathbb{R}$.

(S3) If a random variable Y takes only the values $0, 1, 2, \ldots$, then $\mathrm{E}\, g(Y) = \sum_{k=0}^{\infty} g(k)\, \mathrm{P}(Y = k)$ for every function g for which this series converges absolutely.

Proposition 12.35. Each of the following families is a one-parameter exponential family (Definition 12.33) with $\Theta = \mathbb{R}$, and in each case

$$\mathrm{E}\, y = b'(\theta), \qquad \operatorname{var} y = \phi\, b''(\theta) > 0$$

Here $\operatorname{logit} \pi = \log\bigl(\pi/(1-\pi)\bigr)$ for $0 < \pi < 1$, and $\operatorname{expit}(t) = e^t/(1+e^t)$ $(t \in \mathbb{R})$ is the **logistic function**, the inverse of logit.

(a) **Binomial**: $y \sim \mathrm{Bin}(m, \pi)$ ($m \geqslant 1$ an integer, $0 < \pi < 1$), that is, $\mathrm{P}(y = k) = \binom{m}{k}\pi^k(1-\pi)^{m-k}$ for $k = 0, \ldots, m$ (these add up to 1 by Remark 12.34(S1)); for $m = 1$ (**Bernoulli**), $\mathrm{P}(y = 1) = \pi = 1 - \mathrm{P}(y = 0)$. Here $\theta = \operatorname{logit} \pi$, $\phi = 1$, $b(\theta) = m \log(1 + e^\theta)$, $c(k, \phi) = \log \binom{m}{k}$, and $\mathrm{E}\, y = m\pi$, $\operatorname{var} y = m\pi(1-\pi)$.

(b) **Poisson**: $y \sim \mathrm{Poisson}(\mu)$ $(\mu > 0)$, that is, $\mathrm{P}(y = k) = e^{-\mu}\mu^k/k!$ for $k = 0, 1, 2, \ldots$ (these add up to 1 by Remark 12.34(S2)). Here $\theta = \log \mu$, $\phi = 1$, $b(\theta) = e^\theta$, $c(k, \phi) = -\log k!$, and $\mathrm{E}\, y = \operatorname{var} y = \mu$.

(c) **Normal**: $y \sim N(\mu, \sigma^2)$ $(\sigma^2 > 0)$, with density $(2\pi\sigma^2)^{-1/2} \exp\bigl(-(y-\mu)^2/(2\sigma^2)\bigr)$ (Theorem 11.15). Here $\theta = \mu$, $\phi = \sigma^2$, $b(\theta) = \theta^2/2$, $c(y, \phi) = -y^2/(2\phi) - \frac{1}{2}\log(2\pi\phi)$, and $\mathrm{E}\, y = \mu$, $\operatorname{var} y = \sigma^2$.

Proof. Direct computation gives $1 - \operatorname{expit}(t) = 1/(1+e^t)$, $\operatorname{expit}' = \operatorname{expit}(1 - \operatorname{expit})$, $\operatorname{logit} \operatorname{expit}(t) = t$ and $\operatorname{expit}(\operatorname{logit} \pi) = \pi$.

- (a) With $\theta = \operatorname{logit} \pi$, $\pi = \operatorname{expit}(\theta)$ and $1 - \pi = 1/(1+e^\theta)$, so $\pi^k(1-\pi)^{m-k} = \bigl(\pi/(1-\pi)\bigr)^k(1-\pi)^m = e^{k\theta}(1+e^\theta)^{-m}$: this is the stated form. Then $b'(\theta) = m \operatorname{expit}(\theta) = m\pi$ and $b''(\theta) = m \operatorname{expit}(\theta)(1 - \operatorname{expit}(\theta)) = m\pi(1-\pi) > 0$.
- Since $k\binom{m}{k} = m\binom{m-1}{k-1}$ for $k \geqslant 1$, Remark 11.2(P1) and the binomial theorem (Remark 12.34(S1)) give

$$\mathrm{E}\, y = \sum_{k=1}^{m} k \binom{m}{k} \pi^k (1-\pi)^{m-k} = m\pi \sum_{j=0}^{m-1} \binom{m-1}{j} \pi^j (1-\pi)^{m-1-j} = m\pi(\pi + 1 - \pi)^{m-1} = m\pi$$

  In the same way, $k(k-1)\binom{m}{k} = m(m-1)\binom{m-2}{k-2}$ for $k \geqslant 2$ gives $\mathrm{E}\, y(y-1) = m(m-1)\pi^2$ (both sides are 0 if $m = 1$). By Theorem 11.3(c), $\operatorname{var} y = \mathrm{E}\, y(y-1) + \mathrm{E}\, y - (\mathrm{E}\, y)^2 = m(m-1)\pi^2 + m\pi - m^2\pi^2 = m\pi(1-\pi)$.
- (b) $e^{-\mu}\mu^k/k! = \exp\{k \log \mu - \mu - \log k!\}$, the stated form with $\theta = \log \mu$ and $b(\theta) = e^\theta$; so $b'(\theta) = b''(\theta) = e^\theta = \mu > 0$. The series below converge by Remark 12.34(S2), and (S3) gives

$$\mathrm{E}\, y = \sum_{k \geqslant 1} k\, e^{-\mu} \frac{\mu^k}{k!} = \mu \sum_{k \geqslant 1} e^{-\mu} \frac{\mu^{k-1}}{(k-1)!} = \mu, \qquad \mathrm{E}\, y(y-1) = \mu^2 \sum_{k \geqslant 2} e^{-\mu} \frac{\mu^{k-2}}{(k-2)!} = \mu^2,$$

  and $\operatorname{var} y = \mathrm{E}\, y(y-1) + \mathrm{E}\, y - (\mathrm{E}\, y)^2 = \mu$.
- (c) $-(y-\mu)^2/(2\sigma^2) = (y\mu - \mu^2/2)/\sigma^2 - y^2/(2\sigma^2)$, which gives the stated form. By Theorem 11.13, $\mathrm{E}\, y = \mu = b'(\theta)$ and $\operatorname{var} y = \sigma^2 = \phi b''(\theta)$. □

Definition 12.36. A **generalized linear model** (GLM) with canonical link consists of a known $X \in \mathbb{R}^{n \times p}$ with rows $x_1', \ldots, x_n'$, known numbers $o_1, \ldots, o_n$, the **offsets** (often 0), an unknown $\beta \in \mathbb{R}^p$, and independent responses $y_1, \ldots, y_n$; y_i has a distribution of Proposition 12.35 with natural parameter

$$\theta_i = \eta_i = o_i + x_i'\beta,$$

the **linear predictor**. All y_i come from the same family, with the same dispersion ϕ, and y_i has cumulant function $b_i(\theta) = m_i \log(1 + e^\theta)$ (binomial with known m_i), e^θ (Poisson) or $\theta^2/2$ (normal). The mean is $\mu_i = \mathrm{E}\, y_i = b_i'(\theta_i)$, and the inverse g_i of the increasing function b_i' ($m_i \operatorname{expit}$, exp or the identity), which satisfies $g_i(\mu_i) = \eta_i$, is the **canonical link**.

- **Normal linear model**: $o_i = 0$, identity link, $\mu_i = x_i'\beta$, $\phi = \sigma^2$ (Definition 12.25).
- **Logistic regression**: $y_i \sim \mathrm{Bin}(m_i, \pi_i)$ with $\operatorname{logit} \pi_i = x_i'\beta$ ($o_i = 0$); for $m_i = 1$ the responses are binary. Here $\pi_i = \operatorname{expit}(x_i'\beta)$, $\mu_i = m_i\pi_i$, and the link is $g_i(\mu) = \operatorname{logit}(\mu/m_i)$. The **odds** are $\pi_i/(1-\pi_i) = e^{x_i'\beta}$: increasing x_{ij} by 1, the other entries of x_i fixed, multiplies the odds by e^{β_j}, the **odds ratio**.
- **Poisson regression with exposure**: $y_i \sim \mathrm{Poisson}(\mu_i)$ with $\mu_i = t_i e^{x_i'\beta}$, that is, $\log \mu_i = \log t_i + x_i'\beta$, for known **exposures** $t_i > 0$ (offsets $o_i = \log t_i$), such as policy-years in insurance. The expected count per unit of exposure is $\mu_i/t_i = e^{x_i'\beta}$; increasing x_{ij} by 1 multiplies it by e^{β_j}, the **rate ratio**.

Theorem 12.37. Let $b_1, \dots, b_n : \mathbb{R} \to \mathbb{R}$ be C^2 with $b_i''(t) > 0$ for all $t \in \mathbb{R}$, let $y \in \mathbb{R}^n$, and let

$$\ell_s(\theta) = \sum_{i=1}^n \Big[y_i\theta_i - b_i(\theta_i) \Big] \quad (\theta \in \mathbb{R}^n), \qquad \mu(\theta) = \begin{pmatrix} b_1'(\theta_1) \\ \vdots \\ b_n'(\theta_n) \end{pmatrix}, \qquad W(\theta) = \operatorname{diag}\Big(b_1''(\theta_1), \dots, b_n''(\theta_n) \Big)$$

(a) ℓ_s is C^2 on $\mathbb{R}^n$, and $\nabla \ell_s(\theta) = y - \mu(\theta)$.

(b) $H\ell_s(\theta) = -W(\theta)$, which is negative definite. Hence ℓ_s is strictly concave on $\mathbb{R}^n$, and a stationary point of ℓ_s, if there is one, is its unique global maximizer.

Proof. (a) Only the ith term of ℓ_s involves θ_i, so $\partial \ell_s/\partial\theta_i = y_i - b_i'(\theta_i)$, which is continuous. By Remark 9.1(F1), ℓ_s is differentiable and $\nabla \ell_s(\theta) = y - \mu(\theta)$.

(b) $\partial^2 \ell_s/\partial\theta_j\partial\theta_i$ is $-b_i''(\theta_i)$ for $j = i$ and 0 for $j \neq i$; these functions are continuous, so ℓ_s is C^2 and $H\ell_s(\theta) = -W(\theta)$. For $v \neq 0$, $v'W(\theta)v = \sum_i b_i''(\theta_i)v_i^2 > 0$, so $-W(\theta)$ is negative definite. Apply Theorem 9.19(c) with $U = \mathbb{R}^n$. □

Proposition 12.38 (Affine change of variables)**.** Let f be C^2 on an open set $U \subseteq \mathbb{R}^m$, let $A \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}^m$, and let $g(x) = f(Ax + c)$ on $V = \{x \in \mathbb{R}^n : Ax + c \in U\}$.

(a) V is open, g is C^2 on V, and

$$\nabla g(x) = A' \nabla f(Ax + c), \qquad Hg(x) = A'\, Hf(Ax + c)\, A \qquad (x \in V)$$

(b) If U is convex, so is V. If $Hf(y)$ is negative definite for every $y \in U$ and $\operatorname{rank} A = n$, then g is strictly concave on V.

Proof. (a)

- V is open. Let $x \in V$ and $y = Ax + c$. There is $\delta > 0$ such that $z \in U$ whenever $\|z - y\| < \delta$. If $\|h\| < \delta/(\|A\|_F + 1)$, then $\|A(x+h) + c - y\| = \|Ah\| \leqslant \|A\|_F\|h\| < \delta$ (Lemma 9.4), so $x + h \in V$.
- Gradient. With $y = Ax + c$, $\mathrm{d}y = A\,\mathrm{d}x$ (Theorem 9.2). By the chain rule (Remark 9.1(F2)), $\mathrm{d}g = \nabla f(y)'\mathrm{d}y = \nabla f(y)'A\,\mathrm{d}x$, so $\nabla g(x) = A'\nabla f(Ax + c)$ (Remark 9.1(F1)).
- Hessian. The map $x \mapsto \nabla f(Ax + c)$ has the differential $Hf(y)\,\mathrm{d}y = Hf(y)A\,\mathrm{d}x$ (chain rule, with $D(\nabla f) = Hf$ by Remark 9.1(F3)), so $\mathrm{d}\nabla g = A'Hf(y)A\,\mathrm{d}x$ by (F2). The entries of $A'Hf(Ax+c)A$ are sums of constants times second partial derivatives of f evaluated at $Ax+c$; they are continuous (Remark 9.1), so g is C^2 and $Hg(x) = A'Hf(Ax + c)A$ (F1, F3).

(b) If $x, z \in V$ and $0 \leqslant t \leqslant 1$, then $A\big((1-t)x+tz\big)+c = (1-t)(Ax+c)+t(Az+c) \in U$, so V is convex. By Theorem 8.16(b) applied to $-Hf(y)$, $A'(-Hf(y))A$ is positive definite; apply Theorem 9.19(c) on V. □

Theorem 12.39. In a GLM with canonical link (Definition 12.36) let $\mu = (\mu_1, \dots, \mu_n)'$ and $W = \operatorname{diag}(b_1''(\theta_1), \dots, b_n''(\theta_n))$, both evaluated at β. The log-likelihood, as a function of β with ϕ fixed, is

$$\ell(\beta) = \frac{1}{\phi}\sum_{i=1}^{n}\Big(y_i\theta_i - b_i(\theta_i)\Big) + \sum_{i=1}^{n} c_i(y_i, \phi), \qquad \theta_i = o_i + x_i'\beta,$$

with c_i the function c of Proposition 12.35 for y_i.

(a) $\nabla\ell(\beta) = X'(y-\mu)/\phi$, the **score**, and $H\ell(\beta) = -X'WX/\phi$ (the Hessian, Definition 9.15).

(b) ℓ is concave, and $\hat\beta$ is an MLE if and only if it solves the **likelihood equations**

$$X'(y-\hat\mu) = 0, \qquad \hat\mu_i = b_i'(o_i + x_i'\hat\beta),$$

which do not involve ϕ. If $\operatorname{rank} X = p$, there is at most one MLE.

(c) Let $\operatorname{rank} X = p$. Newton's method (Definition 9.24) replaces ℓ near the current β by its second-order Taylor polynomial and moves to the maximizer of that polynomial, which is

$$\begin{aligned}\beta^{\text{new}} &= \beta - \Big(H\ell(\beta)\Big)^{-1}\nabla\ell(\beta) = \beta + (X'WX)^{-1}X'(y-\mu)\\ &= (X'WX)^{-1}X'Wz, \qquad z = X\beta + W^{-1}(y-\mu)\end{aligned}$$

This is the weighted least-squares estimate (Proposition 12.23) of the **working response** z with weight matrix W; repeating the step is **iteratively reweighted least squares** (IRLS).

Proof. • By independence, the probability of the observed y is the product of those of the y_i (Remark 11.2(P1), applied to the indicator functions of the observed values); in the normal case the density of y is the product of those of the y_i (P4). Its logarithm is ℓ.

- (a) Every b_i is C^2 with $b_i'' > 0$ (Proposition 12.35). With ℓ_s as in Theorem 12.37, $\ell(\beta) = \phi^{-1}\ell_s(o + X\beta) + \sum_i c_i(y_i, \phi)$ with $o = (o_1, \dots, o_n)'$. By Theorem 12.37(a),(b), $\nabla\ell_s(\theta) = y - \mu$ and $H\ell_s(\theta) = -W$, and Proposition 12.38(a) with $A = X$ and $c = o$ gives $\nabla\ell(\beta) = X'(y-\mu)/\phi$ and $H\ell(\beta) = -X'WX/\phi$.
- (b) $W^{1/2} = \operatorname{diag}\big(b_i''(\theta_i)^{1/2}\big)$ is invertible and $X'WX = (W^{1/2}X)'(W^{1/2}X)$, which is positive semidefinite, and positive definite if $\operatorname{rank} X = p$, as then $\operatorname{rank}(W^{1/2}X) = p$ (Theorems 8.16(a) and 4.15(d)). So $H\ell(\beta)$ is negative semidefinite for every β, and negative definite if $\operatorname{rank} X = p$. By Theorem 9.19(c) on $U = \mathbb{R}^p$, ℓ is concave, every stationary point is a global maximizer, and there is at most one maximizer if $\operatorname{rank} X = p$. Conversely, every maximizer is a stationary point (Theorem 9.16(a)); and by (a), $\nabla\ell(\hat\beta) = 0$ reads $X'(y-\hat\mu) = 0$.
- (c) By (b), $X'WX$ is positive definite. The Taylor polynomial $q(b) = \ell(\beta) + \nabla\ell(\beta)'(b-\beta) + \frac{1}{2}(b-\beta)'H\ell(\beta)(b-\beta)$ has gradient $\nabla\ell(\beta) + H\ell(\beta)(b-\beta)$ (Theorem 9.2) and the negative definite Hessian $H\ell(\beta)$. By Theorem 9.19(b) applied to $-q$, the only maximizer of q is its stationary point $\beta - (H\ell(\beta))^{-1}\nabla\ell(\beta) = \beta + (X'WX)^{-1}X'(y-\mu)$, in which ϕ cancels. As $\beta = (X'WX)^{-1}X'WX\beta$, this is $(X'WX)^{-1}X'W\big(X\beta + W^{-1}(y-\mu)\big) = (X'WX)^{-1}X'Wz$, the estimate of Proposition 12.23 for the response z and the positive definite W. □

The theorem does not assert that an MLE exists. For instance, in a logistic regression with an intercept and binary responses that are all 0, the likelihood equation $\mathbf{1}_n'(y-\hat\mu) = 0$ would require $\sum_i \operatorname{expit}(x_i'\hat\beta) = 0$, which is impossible. Nor does the theorem assert that the IRLS iterates converge.

Logistic and Poisson regression

Proposition 12.40. Let $X = [\mathbf{1}_n\ d]$, where $d \in \mathbb{R}^n$ is the indicator of group 1 (entry 1 for the $n_1 \geqslant 1$ observations of group 1, entry 0 for the $n_0 \geqslant 1$ observations of group 0), so that $x_i'\beta = \beta_0 + \beta_1 d_i$.

(a) In logistic regression with binary responses, let s_l be the number of responses 1 in group l, with $0 < s_l < n_l$ $(l = 0, 1)$. The MLE is $\hat\beta_0 = \operatorname{logit}(s_0/n_0)$ and $\hat\beta_1 = \operatorname{logit}(s_1/n_1) - \operatorname{logit}(s_0/n_0)$; so

$$e^{\hat\beta_1} = \frac{s_1/(n_1 - s_1)}{s_0/(n_0 - s_0)},$$

the sample odds ratio. At $\hat\beta$, the $(2,2)$ entry of $(X'WX)^{-1}$ is $\frac{1}{s_0} + \frac{1}{n_0 - s_0} + \frac{1}{s_1} + \frac{1}{n_1 - s_1}$.

(b) In Poisson regression with exposures t_i, let Y_l be the total count and T_l the total exposure of group l, with $Y_l > 0$. The MLE is $\hat\beta_0 = \log(Y_0/T_0)$ and $\hat\beta_1 = \log\big((Y_1/T_1)/(Y_0/T_0)\big)$, the logarithm of the ratio of the observed rates. At $\hat\beta$, the $(2,2)$ entry of $(X'WX)^{-1}$ is $\frac{1}{Y_0} + \frac{1}{Y_1}$.

If some s_l is 0 or n_l in (a), or some $Y_l = 0$ in (b), there is no MLE.

Proof. • $\operatorname{rank} X = 2$, as both groups are nonempty, so d is not a multiple of $\mathbf{1}_n$. The likelihood equations $\mathbf{1}_n'(y - \hat\mu) = 0$ and $d'(y - \hat\mu) = 0$ (Theorem 12.39(b)) say that $\sum_i (y_i - \hat\mu_i) = 0$ over group 1 and, subtracting, over group 0.

- (a) In group l every $\hat\mu_i$ equals $\hat\pi_l = \operatorname{expit}(\hat\beta_0 + l\hat\beta_1)$, so the equations read $s_l = n_l\hat\pi_l$, that is, $\hat\beta_0 + l\hat\beta_1 = \operatorname{logit}(s_l/n_l)$ $(l = 0, 1)$. These values solve the likelihood equations, so they give an MLE, and the only one (Theorem 12.39(b)); and $e^{\hat\beta_1} = e^{\operatorname{logit}(s_1/n_1)}/e^{\operatorname{logit}(s_0/n_0)}$ is the stated ratio. If s_l is 0 or n_l, the equation $n_l \operatorname{expit}(\beta_0 + l\beta_1) = s_l$ has no solution, as $0 < \operatorname{expit} < 1$, so no MLE exists.
- The weights are $\hat\pi_l(1 - \hat\pi_l)$ in group l. With $w_l = n_l\hat\pi_l(1 - \hat\pi_l) = s_l(n_l - s_l)/n_l$,

$$X'WX = \begin{pmatrix} w_0 + w_1 & w_1 \\ w_1 & w_1 \end{pmatrix}, \qquad \big[(X'WX)^{-1}\big]_{22} = \frac{w_0 + w_1}{w_0 w_1} = \frac{1}{w_0} + \frac{1}{w_1}$$

(Proposition 2.13), and $n_l/(s_l(n_l - s_l)) = 1/s_l + 1/(n_l - s_l)$.

- (b) In group l, $\hat\mu_i = t_i e^{\hat\beta_0 + l\hat\beta_1}$, so the equations read $Y_l = T_l e^{\hat\beta_0 + l\hat\beta_1}$; they are solved by the stated values, which therefore give the only MLE. If $Y_l = 0$, the equation $T_l e^{\beta_0 + l\beta_1} = 0$ has no solution. The weights are the $\hat\mu_i$, with sum Y_l over group l, so $X'WX = \left(\begin{smallmatrix} Y_0 + Y_1 & Y_1 \\ Y_1 & Y_1 \end{smallmatrix}\right)$, and the $(2,2)$ entry of its inverse is $(Y_0 + Y_1)/(Y_0 Y_1)$. □

Definition 12.41. In a binomial or Poisson GLM ($\phi = 1$, Definition 12.36), let $\ell_i(\mu)$ be the logarithm of the probability of the observed value of y_i when y_i has mean μ ($0 < \mu < m_i$, respectively $\mu > 0$), and let $\ell_i^* = \sup_\mu \ell_i(\mu)$, the **least upper bound** of the values $\ell_i(\mu)$; it is the value of the **saturated model**. The **deviance** of the fitted means $\hat\mu_i = b_i'(\hat\theta_i)$ of a GLM (which lie in these ranges) is

$$D = 2\sum_{i=1}^{n} \big(\ell_i^* - \ell_i(\hat\mu_i)\big) \geqslant 0$$

If $\hat\beta$ and $\hat\beta_R$ maximize ℓ without and with a constraint on β, with deviances D and D_R, the **likelihood-ratio statistic** for the constraint is $2\big(\ell(\hat\beta) - \ell(\hat\beta_R)\big) = D_R - D$, as the ℓ_i^* cancel.

Proposition 12.42. In logistic regression with $y_i \sim \operatorname{Bin}(m_i, \pi_i)$, with the convention $0 \log 0 = 0$,

$$D = 2\sum_{i=1}^{n} \left(y_i \log \frac{y_i}{\hat\mu_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - \hat\mu_i} \right)$$

In the normal linear model with known σ^2, the same construction gives $2\sum_i \big(\ell_i^* - \ell_i(\hat y_i)\big) = \mathrm{SSE}/\sigma^2$.

Proof. • Write $k = y_i$ and $m = m_i$. By Proposition 12.35(a) with $\pi = \mu/m$, $\ell_i(\mu) = \log\binom{m}{k} + k\log\frac{\mu}{m} + (m-k)\log\frac{m-\mu}{m}$.

- If $0 < k < m$, then for $0 < \mu < m$, by Lemma 12.27,
$$\ell_i(\mu) - \ell_i(k) = k\log\frac{\mu}{k} + (m-k)\log\frac{m-\mu}{m-k} \leqslant k\Big(\frac{\mu}{k} - 1\Big) + (m-k)\Big(\frac{m-\mu}{m-k} - 1\Big) = 0$$
So $\ell_i^* = \ell_i(k)$, and $\ell_i^* - \ell_i(\hat\mu_i) = k\log\frac{k}{\hat\mu_i} + (m-k)\log\frac{m-k}{m-\hat\mu_i}$, the ith summand of the formula.
- If $k = 0$, then $\ell_i(\mu) = m\log(1-\mu/m) < 0$ increases to 0 as $\mu \to 0$, so $\ell_i^* = 0$ and $\ell_i^* - \ell_i(\hat\mu_i) = m\log\frac{m}{m-\hat\mu_i}$, again the ith summand. The case $k = m$ is the same with μ and $m-\mu$ exchanged.
- Normal: $\ell_i(\mu) = -\frac12\log(2\pi\sigma^2) - (y_i-\mu)^2/(2\sigma^2)$ has $\ell_i^* = \ell_i(y_i)$, so $2\sum_i(\ell_i^* - \ell_i(\hat y_i)) = \sum_i(y_i - \hat y_i)^2/\sigma^2$. □

Remark 12.43 (Imported large-sample facts)**.** The following facts are used without proof. Consider a GLM with canonical link and $\operatorname{rank} X = p$, let $\hat W$ be the matrix W of Theorem 12.39 evaluated at the MLE $\hat\beta$, and let "approximately" refer to large n, under regularity conditions on X (for instance, bounded rows x_i and $X'WX/n$ converging to a positive definite matrix).

(L1) With probability close to 1 the MLE exists, and $\hat\beta$ is approximately $N_p\big(\beta, \phi(X'\hat WX)^{-1}\big)$. Accordingly, the standard error of $c'\hat\beta$ is taken to be
$$\operatorname{SE}(c'\hat\beta) = \sqrt{\phi\, c'(X'\hat WX)^{-1}c}$$

(L2) (**Wald**) If $R \in \mathbb{R}^{q\times p}$ has rank q and $H_0\colon R\beta = r$ holds, then
$$Q = \frac{1}{\phi}(R\hat\beta - r)'\Big(R(X'\hat WX)^{-1}R'\Big)^{-1}(R\hat\beta - r)$$
is approximately χ^2_q; it is the statistic of Corollary 11.24(b) for the approximate normal law of $R\hat\beta$ in (L1). For $q = 1$ and $R = c'$, the Wald statistic $z_W = (c'\hat\beta - r)/\operatorname{SE}(c'\hat\beta)$ is approximately $N(0,1)$, and $c'\hat\beta \pm z_{1-\alpha/2}\operatorname{SE}(c'\hat\beta)$ are the endpoints of an approximate confidence interval of level $1-\alpha$ for $c'\beta$.

(L3) (**Likelihood ratio**) Under the same H_0, $2\big(\ell(\hat\beta) - \ell(\hat\beta_R)\big)$ is approximately χ^2_q, where $\hat\beta_R$ maximizes ℓ subject to $R\beta = r$; for $\phi = 1$ this is $D_R - D$.

(L4) (**Goodness of fit**) In a binomial or Poisson GLM whose m_i, respectively μ_i, are all large, the deviance D is approximately χ^2_{n-p} if the model holds.

Example 12.44. Four batches of 8 insects are exposed to an insecticide at coded doses $x = (-1,0,0,1)'$, and $y = (2,3,5,6)'$ of them die. Fit the logistic regression $y_i \sim \operatorname{Bin}(8,\pi_i)$, $\operatorname{logit}\pi_i = \beta_0 + \beta_1 x_i$: carry out one Newton step from $\beta = 0$, find the MLE, and test $H_0\colon \beta_1 = 0$ at approximate level α.

Solution. Here $X = [\mathbf{1}_4\ x]$, $\operatorname{rank} X = 2$, $\mu_i = 8\pi_i$ and $W = \operatorname{diag}\big(8\pi_i(1-\pi_i)\big)$ (Proposition 12.35(a)).

- At $\beta = 0$, $\pi_i = \frac12$, $\mu = 4\mathbf{1}_4$, $W = 2I_4$, $X'WX = \operatorname{diag}(8,4)$ and $X'(y-\mu) = X'(-2,-1,1,2)' = (0,4)'$; Theorem 12.39(c) gives $\beta^{(1)} = (0,1)'$.
- The likelihood equations (Theorem 12.39(b)) are $\sum_i\mu_i = \sum_i y_i = 16$ and $\sum_i x_i\mu_i = \sum_i x_i y_i = 4$. For $\beta = (0,b)'$, $\operatorname{expit}(-b) = 1 - \operatorname{expit}(b)$ gives $\pi = (1-\operatorname{expit}(b), \frac12, \frac12, \operatorname{expit}(b))'$; the first equation holds, and the second reads $8(2\operatorname{expit}(b) - 1) = 4$, so $b = \operatorname{logit}\frac34 = \log 3$. So $\hat\beta = (0, \log 3)'$ is the MLE (Theorem 12.39(b), $\operatorname{rank} X = 2$), with $\hat\pi = (\frac14, \frac12, \frac12, \frac34)'$: each unit of dose multiplies the odds of death by 3.
- Under H_0, $\sum_i(y_i - 8\pi) = 0$ gives $\tilde\pi = \frac12$. In Proposition 12.42, batches 2, 3 contribute the same terms to D and D_R, and batches 1, 4 contribute 0 to D; so $D_R - D = 2\cdot 2\big(2\log\frac24 + 6\log\frac64\big) = 4(6\log 3 - 8\log 2) = 8\log\frac{27}{16}$, and the likelihood-ratio test (L3) rejects H_0 if $8\log\frac{27}{16} > \chi^2_{1,1-\alpha}$.

Exercises

Exercise 12.1. Five observations of a variable x and a response y are

x_i	1	2	3	4	5
y_i	1	9	12	15	13

Fit the simple regression $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

(a) Compute $X'X$, $X'y$, $(X'X)^{-1}$ and $\hat{\beta}$.

(b) Compute $\hat{y}$ and e, and verify that $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.

(c) Compute SSE, SST and SSR, verify SST = SSR + SSE, and compute R^2.

Solution. (a) $X = [\mathbf{1}_5\ x]$, $\sum_i x_i = 15$, $\sum_i x_i^2 = 55$, $\sum_i y_i = 50$ and $\sum_i x_i y_i = 1+18+36+60+65 = 180$. With $\bar{x} = 3$ and $S_{xx} = 55 - 45 = 10$, Proposition 12.5(b) gives the entries $\frac{1}{5} + \frac{9}{10} = \frac{11}{10}$, $-\frac{3}{10}$ and $\frac{1}{10}$:

$$X'X = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 50 \\ 180 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{10}\begin{pmatrix} 11 & -3 \\ -3 & 1 \end{pmatrix},$$

and $\hat{\beta} = \frac{1}{10}(550 - 540,\ -150 + 180)' = (1, 3)'$. Check with Corollary 12.7: $\bar{y} = 10$ and $S_{xy} = 180 - 150 = 30$, so $\hat{\beta}_1 = 3$ and $\hat{\beta}_0 = 10 - 9 = 1$.

(b) $\hat{y}_i = 1 + 3x_i$: $\hat{y} = (4, 7, 10, 13, 16)'$ and $e = y - \hat{y} = (-3, 2, 2, 2, -3)'$. Then $\sum_i e_i = 0$ and $\sum_i x_i e_i = -3 + 4 + 6 + 8 - 15 = 0$ (Proposition 12.4).

(c) SSE $= 9 + 4 + 4 + 4 + 9 = 30$, SST $= \sum_i (y_i - 10)^2 = 81 + 1 + 4 + 25 + 9 = 120$ and SSR $= \sum_i (\hat{y}_i - 10)^2 = 36 + 9 + 0 + 9 + 36 = 90$, and $90 + 30 = 120$ (Theorem 12.13), and $R^2 = \frac{90}{120} = \frac{3}{4}$.

Exercise 12.2. Let $X_{\cdot 1} = (2, 0, 2, 0)'$ and $X_{\cdot 2} = (3, 3, 1, 1)'$ be the last two columns of the design matrix X of Example 12.6, and add their sum as a fourth column: $\tilde{X} = [\mathbf{1}_4\ X_{\cdot 1}\ X_{\cdot 2}\ X_{\cdot 1} + X_{\cdot 2}] \in \mathbb{R}^{4\times 4}$, and let $y = (9, 3, 5, 3)'$.

(a) Find $\operatorname{rank} \tilde{X}$ and a basis of $\mathcal{N}(\tilde{X})$.

(b) Solve the normal equations $\tilde{X}'\tilde{X}\beta = \tilde{X}'y$ by row reduction.

(c) Show that all least-squares estimates give the fitted values $\hat{y} = (8, 4, 6, 2)'$ of Example 12.6, and find SSE.

Solution. (a) The columns are $\mathbf{1}_4$, $(2, 0, 2, 0)'$, $(3, 3, 1, 1)'$ and $(5, 3, 3, 1)'$. The fourth is the sum of the second and the third, so $\tilde{X}(0, 1, 1, -1)' = 0$; the first three are the columns of X, which are independent (Example 12.6). So $\operatorname{rank} \tilde{X} = 3$, $\dim \mathcal{N}(\tilde{X}) = 4 - 3 = 1$ (Theorem 4.12), and $(0, 1, 1, -1)'$ is a basis of $\mathcal{N}(\tilde{X})$.

(b) The entries of $\tilde{X}'\tilde{X}$ and $\tilde{X}'y$ are inner products of columns (Theorems 1.4(d) and 1.9):

$$\left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 4 & 8 & 8 & 16 & 28 \\ 8 & 8 & 20 & 28 & 44 \\ 12 & 16 & 28 & 44 & 72 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 4 & 4 & 8 & 12 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 4 & 4 & 4 \end{array}\right) \xrightarrow[R_1 \leftarrow \frac{1}{4}R_1,\ R_2 \leftarrow \frac{1}{4}R_2,\ R_3 \leftarrow \frac{1}{4}R_3]{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The last matrix is an echelon form with pivots in columns $1,2,3$ and none in the last column, so the system is consistent and $\beta_3 = t$ is free (Theorem 2.22). Back substitution: $\beta_2 = 1-t$, $\beta_1 = 2-t$ and $\beta_0 = 5-\beta_1-2\beta_2-3\beta_3 = 5-(2-t)-2(1-t)-3t = 1$. So the solutions are

$$\beta = \begin{pmatrix} 1 \\ 2-t \\ 1-t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} \qquad (t \in \mathbb{R}),$$

a particular solution plus $\mathcal{N}(\tilde{X}'\tilde{X}) = \mathcal{N}(\tilde{X})$, as Theorem 12.1(c) says.

(c) These solutions are the least-squares estimates (Theorem 12.3(a)), and each gives $\tilde{X}\beta = X(1,2,1)' + t\tilde{X}(0,-1,-1,1)' = (8,4,6,2)'$. So $\hat{y}$ and $e = (1,-1,-1,1)'$ are those of Example 12.6, and SSE $= 4$.

Exercise 12.3. Let $X = [\mathbf{1}_5 \;\; X_0]$, where the columns of X_0 are $x_1 = (1,2,3,4,5)'$ and $x_2 = (3,0,2,4,1)'$, and let $y = (9,4,4,14,9)'$.

(a) Compute $\bar{x}$, $S_c = X_0'C_5X_0$ and $(X'X)^{-1}$ (Proposition 12.5).

(b) Compute the leverages h_{ii}, and check $\sum_i h_{ii} = 3$ and $h_{ii} \geqslant \frac{1}{5}$.

(c) Compute $\hat{\beta}$, e, $s^2 = \text{SSE}/(n-p)$, and the **standardized residuals** $e_i/(s\sqrt{1-h_{ii}})$.

Solution. (a) $\bar{x} = (3,2)'$, and the columns of C_5X_0 are $a = (-2,-1,0,1,2)'$ and $b = (1,-2,0,2,-1)'$, with $a'a = b'b = 10$ and $a'b = -2+2+0+2-2 = 0$. So $S_c = (C_5X_0)'(C_5X_0) = 10I_2$ (Proposition 1.13), and Proposition 12.5(b), with $\frac{1}{5} + \bar{x}'S_c^{-1}\bar{x} = \frac{1}{5} + \frac{13}{10} = \frac{3}{2}$, gives

$$(X'X)^{-1} = \frac{1}{10}\begin{pmatrix} 15 & -3 & -2 \\ -3 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

(b) $h_{ii} = \text{row}_i(X)(X'X)^{-1}\,\text{row}_i(X)'$ (Theorem 1.4(d)). For a row $(1,u,v)$,

$$(1,u,v)(X'X)^{-1}(1,u,v)' = \tfrac{1}{10}\Big(15 - 6u - 4v + u^2 + v^2\Big) = \tfrac{1}{10}\Big((u-3)^2 + (v-2)^2 + 2\Big),$$

and $(u-3, v-2)$ is row i of C_5X_0, that is, (a_i, b_i). As $a_i^2 + b_i^2 = 5$ for $i \neq 3$ and 0 for $i = 3$, $h_{ii} = \frac{7}{10}$ for $i \neq 3$ and $h_{33} = \frac{1}{5}$. Then $\sum_i h_{ii} = 4 \cdot \frac{7}{10} + \frac{1}{5} = 3 = p$, and every $h_{ii} \geqslant \frac{1}{5}$, as Theorem 12.2(f) requires.

(c) $X'y = (40,\ 130,\ 100)'$, so $\hat{\beta} = \frac{1}{10}(600 - 390 - 200,\ -120 + 130,\ -80 + 100)' = (1,1,2)'$. Then $\hat{y} = (8,3,8,13,8)'$, $e = (1,1,-4,1,1)'$, SSE $= 20$ and $s^2 = 20/2 = 10$. The standardized residuals are $1/\sqrt{10 \cdot \frac{3}{10}} = \frac{1}{\sqrt{3}}$ for $i \neq 3$ and $-4/\sqrt{10 \cdot \frac{4}{5}} = -\sqrt{2}$ for $i = 3$: the observation of smallest leverage has the largest residual.

Exercise 12.4. Six plants of three varieties are grown under two treatments, one plant for each combination; y_{ij} is the yield of variety i under treatment j, and $y = (y_{11}, y_{12}, y_{21}, y_{22}, y_{31}, y_{32})'$. The **two-way additive model** $y_{ij} = \mu + \alpha_i + \gamma_j + \varepsilon_{ij}$, with $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2)'$, has the design matrix

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Assume the linear model.

(a) Show that $z_1 = (-1,1,1,1,0,0)'$ and $z_2 = (-1,0,0,0,1,1)'$ lie in $\mathcal{N}(X)$, that columns 2 to 5 of X are linearly independent, and deduce that $\operatorname{rank} X = 4$ and that z_1, z_2 form a basis of $\mathcal{N}(X)$.

(b) Which of $\alpha_1 - \alpha_2$, $\gamma_1 - \gamma_2$, $\mu + \alpha_3 + \gamma_1$, α_1, μ are estimable?

(c) Show that the BLUE of $\alpha_1 - \alpha_2$ is $\frac{1}{2}(y_{11} + y_{12}) - \frac{1}{2}(y_{21} + y_{22})$, with variance σ^2, and that the BLUE of $\gamma_1 - \gamma_2$ is $\frac{1}{3}(y_{11} + y_{21} + y_{31}) - \frac{1}{3}(y_{12} + y_{22} + y_{32})$, with variance $\frac{2}{3}\sigma^2$. (If $X'a = c$ and $a \in \mathcal{C}(X)$, then $P_X a = a$.)

(d) Evaluate both BLUEs for $y = (8,6,7,3,9,5)'$.

Solution. (a) Let $a_1, \ldots, a_6$ be the columns of X. Each plant has one variety and one treatment, so $a_2 + a_3 + a_4 = a_1 = a_5 + a_6$, that is, $Xz_1 = Xz_2 = 0$ (Theorem 1.4(b)). If $t_2a_2 + t_3a_3 + t_4a_4 + t_5a_5 = 0$, rows 2, 4, 6 give $t_2 = t_3 = t_4 = 0$, and row 1 then gives $t_5 = 0$. As $a_1 = a_2 + a_3 + a_4$ and $a_6 = a_1 - a_5$, the columns $a_2, \ldots, a_5$ span $\mathcal{C}(X)$ and form a basis of it: $\operatorname{rank} X = 4$. So $\dim \mathcal{N}(X) = 2$ (Theorem 4.12), and the independent vectors z_1, z_2 (entries 2 and 5 of $t_1z_1 + t_2z_2$ are t_1 and t_2) form a basis of $\mathcal{N}(X)$ (Theorem 4.5(d)).

(b) By Theorem 12.20(iv), $c'\beta$ is estimable if and only if $c'z_1 = -c_1 + c_2 + c_3 + c_4 = 0$ and $c'z_2 = -c_1 + c_5 + c_6 = 0$. So $\alpha_1 - \alpha_2$, $\gamma_1 - \gamma_2$ and $\mu + \alpha_3 + \gamma_1$ ($-1 + 1 = 0$ twice) are estimable; α_1 ($c'z_1 = 1$) and μ ($c'z_1 = -1$) are not.

(c)

- $a = \frac{1}{2}(1,1,-1,-1,0,0)' = \frac{1}{2}(a_2 - a_3) \in \mathcal{C}(X)$, and $X'a$ has entries 0 (for μ), 1, -1, 0 (for the α's) and $\frac{1}{2}(1-1) = 0$ twice (for the γ's): $X'a = c$ for $c'\beta = \alpha_1 - \alpha_2$.
- $P_X = P_{\mathcal{C}(X)}$ (Theorem 12.8(a)), so $P_X a = a$ (Theorem 6.19(b)). By Proposition 12.21 with $b = a$, the BLUE is $c'\hat{\beta} = (P_X a)'y = a'y$, with variance $\sigma^2 a'a = \sigma^2 \cdot 4 \cdot \frac{1}{4} = \sigma^2$.
- Likewise $a = \frac{1}{3}(1,-1,1,-1,1,-1)' = \frac{1}{3}(a_5 - a_6)$ has $X'a = (0,0,0,0,1,-1)'$; the BLUE of $\gamma_1 - \gamma_2$ is $a'y$, with variance $\sigma^2 \cdot 6 \cdot \frac{1}{9} = \frac{2}{3}\sigma^2$.

(d) $\frac{1}{2}(8+6) - \frac{1}{2}(7+3) = 7 - 5 = 2$, and $\frac{1}{3}(8+7+9) - \frac{1}{3}(6+3+5) = 8 - \frac{14}{3} = \frac{10}{3}$.

Exercise 12.5. Let $y_i = \beta x_i + \varepsilon_i$ ($p = 1$, $X = x$) with all $x_i > 0$ and uncorrelated errors with $\operatorname{var} \varepsilon_i = \sigma^2 x_i$, so that $V = \operatorname{diag}(x_1, \ldots, x_n)$ in Theorem 12.24.

(a) Show that $\tilde{\beta} = \sum_i y_i / \sum_i x_i$, $\operatorname{var} \tilde{\beta} = \sigma^2 / \sum_i x_i$, and $\operatorname{var} \hat{\beta} = \sigma^2 \sum_i x_i^3 / (\sum_i x_i^2)^2$ for the least-squares $\hat{\beta} = \sum_i x_i y_i / \sum_i x_i^2$.

(b) For $x = (1,2,3,4)'$ and $y = (3,7,12,8)'$, compute $\tilde{\beta}$, $\hat{\beta}$, $\operatorname{var} \tilde{\beta}$ and $\operatorname{var} \hat{\beta}$.

(c) Compute the error sum of squares SSE^* of the transformed model of the proof of Theorem 12.24, $s^2 = \mathrm{SSE}^*/(n-1)$, and the standard error $s/\sqrt{\sum_i x_i}$ of $\tilde{\beta}$.

(d) Assuming $\varepsilon \sim N_n(0, \sigma^2 V)$, give a confidence interval of level $1 - \alpha$ for β.

Solution. (a) $V^{-1} = \operatorname{diag}(1/x_i)$, so $x'V^{-1}x = \sum_i x_i$ and $x'V^{-1}y = \sum_i y_i$; Theorem 12.24 gives $\tilde{\beta}$ and $\operatorname{var} \tilde{\beta} = \sigma^2 (x'V^{-1}x)^{-1}$, and (c) there gives $\operatorname{var} \hat{\beta} = \sigma^2 (x'x)^{-1} x'Vx (x'x)^{-1} = \sigma^2 \sum_i x_i^3 / (\sum_i x_i^2)^2$.

(b) $\tilde{\beta} = \frac{30}{10} = 3$ and $\hat{\beta} = \frac{3+14+36+32}{30} = \frac{17}{6}$. With $\sum_i x_i = 10$, $\sum_i x_i^2 = 30$ and $\sum_i x_i^3 = 100$, $\operatorname{var} \tilde{\beta} = \frac{\sigma^2}{10} < \frac{\sigma^2}{9} = \frac{100\sigma^2}{900} = \operatorname{var} \hat{\beta}$.

(c) $T = V^{-1/2} = \operatorname{diag}(x_i^{-1/2})$, so $y_i^* = y_i/\sqrt{x_i}$ and $x_i^* = \sqrt{x_i}$; the transformed model is a linear model with least-squares estimate $\tilde{\beta}$, and

$$\mathrm{SSE}^* = \sum_i \frac{(y_i - 3x_i)^2}{x_i} = \frac{0}{1} + \frac{1}{2} + \frac{9}{3} + \frac{16}{4} = \frac{15}{2},$$

so $s^2 = \frac{5}{2}$ and, as $\operatorname{var} \tilde{\beta} = \sigma^2 / \sum_i x_i$, the standard error is $\sqrt{\frac{5}{2} \cdot \frac{1}{10}} = \frac{1}{2}$.

(d) $T\varepsilon \sim N_n(0, \sigma^2 TVT) = N_n(0, \sigma^2 I_n)$ (Theorem 11.14(a)), so the transformed model is a normal linear model with $n - p = 3$, and Theorem 12.29(b) gives $\left[3 - \frac{1}{2}t_{3,1-\alpha/2},\ 3 + \frac{1}{2}t_{3,1-\alpha/2}\right]$.

Exercise 12.6. For the data of Exercise 12.1 assume the normal linear model, and put $t^* = t_{3,1-\alpha/2}$.

(a) Compute s^2, $\mathrm{SE}(\hat\beta_0)$, $\mathrm{SE}(\hat\beta_1)$ and the statistics T_0 for $\beta_0 = 0$ and for $\beta_1 = 0$.

(b) Give the confidence interval of level $1 - \alpha$ for β_1.

(c) Compute the overall F statistic and check that it equals the square of T_0 for β_1.

(d) At $x_0 = 3$ and at $x_0 = 6$, give the confidence intervals of level $1 - \alpha$ for $\beta_0 + x_0\beta_1$ and the prediction intervals of level $1 - \alpha$ for a new observation.

Solution. By Exercise 12.1, $\hat\beta = (1, 3)'$, $\mathrm{SSE} = 30$, $\mathrm{SSR} = 90$ and $(X'X)^{-1} = \frac{1}{10}\left(\begin{smallmatrix} 11 & -3 \\ -3 & 1 \end{smallmatrix}\right)$; $n - p = 3$.

(a) $s^2 = 10$, $\mathrm{SE}(\hat\beta_0) = \sqrt{10 \cdot \frac{11}{10}} = \sqrt{11}$ and $\mathrm{SE}(\hat\beta_1) = \sqrt{10 \cdot \frac{1}{10}} = 1$; the statistics are $1/\sqrt{11}$ and 3.

(b) $[3 - t^*,\ 3 + t^*]$ (Theorem 12.29(b)); $H_0\colon \beta_1 = 0$ is rejected exactly when $t^* < 3$.

(c) $F = (\mathrm{SSR}/1)/(\mathrm{SSE}/3) = 90/10 = 9 = 3^2$ (Theorem 12.31(c),(d)).

(d) For $x_0 = (1, 3)'$: $x_0'\hat\beta = 10$ and $x_0'(X'X)^{-1}x_0 = \frac{1}{10}(11 - 18 + 9) = \frac{1}{5}$, so the intervals are $10 \pm t^*\sqrt{10 \cdot \frac{1}{5}} = 10 \pm t^*\sqrt{2}$ and $10 \pm t^*\sqrt{10 \cdot \frac{6}{5}} = 10 \pm 2\sqrt{3}\,t^*$. For $x_0 = (1, 6)'$: $x_0'\hat\beta = 19$ and $x_0'(X'X)^{-1}x_0 = \frac{1}{10}(11 - 36 + 36) = \frac{11}{10}$, so the intervals are $19 \pm t^*\sqrt{11}$ and $19 \pm t^*\sqrt{21}$ (Theorem 12.29(b),(c)): both widen away from $\bar x = 3$.

Exercise 12.7. For the four units of Example 12.6, consider $H_0\colon \beta_1 = \beta_2$, that is, $R\beta = 0$ with $R = (0, 1, -1)$.

(a) Use Theorem 12.30 to compute V, $\hat\beta_R$ and $\mathrm{SSE}_R - \mathrm{SSE}$.

(b) Check (a) by fitting y on $\mathbf{1}_4$ and $X_{\cdot 1} + X_{\cdot 2}$ (Corollary 12.7).

(c) Assuming normal errors, compute the statistic F of Theorem 12.31 and check $F = T_0^2$.

Solution. (a) With $(X'X)^{-1}$ and $\hat\beta = (1, 2, 1)'$ of Example 12.6, $(X'X)^{-1}R' = \frac{1}{4}(1, 1, -1)'$, $V = R(X'X)^{-1}R' = \frac{1}{4}(1 + 1) = \frac{1}{2}$ and $R\hat\beta = 1$. So

$$\hat\beta_R = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \frac{1}{4}\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \cdot 2 \cdot 1 = \begin{pmatrix} 1/2 \\ 3/2 \\ 3/2 \end{pmatrix}, \qquad \mathrm{SSE}_R - \mathrm{SSE} = 1 \cdot 2 \cdot 1 = 2,$$

and $\mathrm{SSE}_R = 4 + 2 = 6$.

(b) Under $\beta_1 = \beta_2 = b$, $X\beta = \beta_0\mathbf{1}_4 + bu$ with $u = X_{\cdot 1} + X_{\cdot 2} = (5, 3, 3, 1)'$. Here $\bar u = 3$, $S_{uu} = 4 + 0 + 0 + 4 = 8$ and $S_{uy} = 2 \cdot 4 + 0 + 0 + (-2)(-2) = 12$, so $b = \frac{3}{2}$ and $\beta_0 = 5 - \frac{9}{2} = \frac{1}{2}$, as in (a). The fitted values are $(8, 5, 5, 2)'$, the residuals $(1, -2, 0, 1)'$, and $\mathrm{SSE}_R = 6$.

(c) $n - p = 1$, so $F = (2/1)/(4/1) = \frac{1}{2}$. With $c = (0, 1, -1)'$, $\mathrm{SE}(c'\hat\beta) = s\sqrt{c'(X'X)^{-1}c} = 2\sqrt{1/2} = \sqrt{2}$ and $T_0 = 1/\sqrt{2}$, so $T_0^2 = \frac{1}{2} = F$ (Theorem 12.31(d)).

Exercise 12.8. (a) A regression with intercept and $k = 4$ further columns on $n = 105$ observations has overall $F = 20$. Find R^2.

(b) A regression with intercept and 6 further columns on $n = 40$ observations has $R^2 = \frac{1}{2}$; without three of these columns, $R^2 = \frac{2}{5}$. Compute the statistic F for the hypothesis that the three coefficients are 0, and state when the test of level α rejects it.

(c) A simple regression on $n = 18$ observations has SSE $= 160$ and $R^2 = \frac{9}{25}$. Find SST, SSR, the overall F statistic and $|T_0|$ for $\beta_1 = 0$.

Solution. (a) By Theorem 12.31(c), $F(1-R^2)/(n-k-1) = R^2/k$, so $kF(1-R^2) = (n-k-1)R^2$ and $R^2 = kF/(kF+n-k-1) = 80/180 = \frac{4}{9}$.

(b) $p = 7$, $n - p = 33$, $q = 3$; by Theorem 12.31(c), $F = \frac{(1/2-2/5)/3}{(1-1/2)/33} = \frac{1/30}{1/66} = \frac{11}{5}$. The test rejects if and only if $\frac{11}{5} > F_{3,33,1-\alpha}$ (Theorem 12.31(b)).

(c) SST $=$ SSE$/(1-R^2) = 160 \cdot \frac{25}{16} = 250$ and SSR $=$ SST $-$ SSE $= 90$ (Theorem 12.13); $F = 90/(160/16) = 9$, and $|T_0| = \sqrt{9} = 3$ (Theorem 12.31(d)).

Exercise 12.9. In a trial, 24 of 36 treated patients and 12 of 36 untreated patients recover. Fit the logistic regression $\operatorname{logit}\pi_i = \beta_0 + \beta_1 d_i$ with binary responses ($y_i = 1$ for recovery), where $d_i = 1$ for treated patients.

(a) Find the MLE and the odds ratio $e^{\hat\beta_1}$, and interpret it.

(b) Find $\mathrm{SE}(\hat\beta_1)$, the Wald statistic for $H_0\colon \beta_1 = 0$, and the approximate confidence interval of level $1-\alpha$ for e^{β_1}.

(c) Compute the likelihood-ratio statistic for H_0.

Solution. (a) By Proposition 12.40(a) with $s_0 = 12$, $n_0 = 36$, $s_1 = 24$, $n_1 = 36$: $\hat\beta_0 = \operatorname{logit}\frac{1}{3} = \log\frac{1}{2} = -\log 2$ and $\hat\beta_1 = \operatorname{logit}\frac{2}{3} - \operatorname{logit}\frac{1}{3} = \log 2 + \log 2 = \log 4$. The estimated odds of recovery are $\frac{24}{12} = 2$ with treatment and $\frac{12}{24} = \frac{1}{2}$ without, and their ratio is $e^{\hat\beta_1} = 4$.

(b) By Proposition 12.40(a) and Remark 12.43(L1), $\mathrm{SE}(\hat\beta_1)^2 = \frac{1}{24}+\frac{1}{12}+\frac{1}{12}+\frac{1}{24} = \frac{1}{4}$, so $\mathrm{SE}(\hat\beta_1) = \frac{1}{2}$, and $z_W = 2\log 4 = 4\log 2$ (L2). With $z = z_{1-\alpha/2}$, the interval for β_1 is $\log 4 \pm \frac{z}{2}$, and for the odds ratio $[4e^{-z/2},\ 4e^{z/2}]$.

(c) For binary responses $\ell = \sum_i \big(y_i \log\pi_i + (1-y_i)\log(1-\pi_i)\big)$. At the MLE, $\ell(\hat\beta) = 24\log\frac{2}{3} + 12\log\frac{1}{3} + 12\log\frac{1}{3} + 24\log\frac{2}{3} = 48\log 2 - 72\log 3$. Under H_0 the design is $\mathbf{1}_n$, and its likelihood equation $\sum_i (y_i - \pi) = 0$ gives $\tilde\pi = \frac{36}{72} = \frac{1}{2}$ (Theorem 12.39(b)), so $\ell(\hat\beta_R) = -72\log 2$. The likelihood-ratio statistic is $2(120\log 2 - 72\log 3) = 48\log\frac{32}{27}$ (L3).

Exercise 12.10. A motor portfolio has two rating classes. Policy i is observed for t_i years (its exposure) and has y_i claims; class A has total exposure 1000 policy-years and 24 claims, class B 500 policy-years and 48 claims. Assume the Poisson regression $\log\mu_i = \log t_i + \beta_0 + \beta_1 d_i$, with $d_i = 1$ for class B.

(a) Show that, apart from a term free of β, the log-likelihood depends on the data only through the class totals; find the MLE and the rate ratio of class B to class A.

(b) Find $\mathrm{SE}(\hat\beta_1)$, the Wald statistic for $H_0\colon \beta_1 = 0$, and the approximate confidence interval of level $1-\alpha$ for the rate ratio e^{β_1}.

(c) Compute the likelihood-ratio statistic for H_0.

Solution. (a) Let Y_l and T_l be the total count and exposure of class l ($l = 0$ for A, $l = 1$ for B). By Proposition 12.35(b),

$$\ell(\beta) = \sum_i \Big(y_i(\log t_i + \beta_0 + \beta_1 d_i) - t_i e^{\beta_0+\beta_1 d_i} - \log y_i!\Big) = \sum_{l=0}^{1}\Big(Y_l(\beta_0 + l\beta_1) - T_l e^{\beta_0 + l\beta_1}\Big) + \sum_i (y_i \log t_i - \log y_i!)$$

By Proposition 12.40(b), $\hat\beta_0 = \log\frac{24}{1000} = \log\frac{3}{125}$ and $\hat\beta_1 = \log\Big(\frac{48}{500}\Big/\frac{24}{1000}\Big) = \log 4$: class B has 4 times the claim frequency of class A ($\frac{12}{125}$ against $\frac{3}{125}$ claims per policy-year).

(b) $\mathrm{SE}(\hat\beta_1) = \sqrt{\frac{1}{24}+\frac{1}{48}} = \frac{1}{4}$ (Proposition 12.40(b), Remark 12.43(L1)), and $z_W = 4\log 4 = 8\log 2$ (L2). With $z = z_{1-\alpha/2}$, the interval for the rate ratio is $[4e^{-z/4},\ 4e^{z/4}]$.

(c) At the MLE, $T_l e^{\hat\beta_0+l\hat\beta_1} = Y_l$, so the sum in (a) is $\sum_l (Y_l\log(Y_l/T_l) - Y_l)$. Under H_0 the likelihood equation $\sum_i (y_i - t_i e^{\beta_0}) = 0$ gives the common rate $e^{\tilde\beta_0} = \frac{72}{1500} = \frac{6}{125}$, and the sum is $72\log\frac{6}{125} - 72$. The likelihood-ratio statistic is

$$2\Big(24\log\frac{3/125}{6/125} + 48\log\frac{12/125}{6/125}\Big) = 2(-24\log 2 + 48\log 2) = 48\log 2 \quad \text{(L3)}$$

Problems

Problem 12.1. Let $X \in \mathbb{R}^{n\times(k+1)}$ have full column rank, with hat matrix H and residual maker M.

(a) In the model $y = X\beta + \varepsilon$, show that $e = M\varepsilon$ and $\mathrm{SSE} = \varepsilon' M\varepsilon$.

(b) Let $\mathbf{1}_n \in \mathcal{C}(X)$. Show that $H\mathbf{1}_n = \mathbf{1}_n$, $H\bar J_n = \bar J_n H = \bar J_n$ and $MC_n = C_n M = M$.

(c) Under (b), show that $H - \bar J_n$ is symmetric and idempotent of rank k, that $(H-\bar J_n)M = 0$, and that $\mathrm{SSR} = y'(H-\bar J_n)y$.

Solution. (a) $e = My$ (Theorem 12.3(c)) and $MX = 0$ (Theorem 12.2(c)) give $e = MX\beta + M\varepsilon = M\varepsilon$, and $M = M'M$ (Theorem 12.2(b)) gives $\mathrm{SSE} = e'e = \varepsilon' M'M\varepsilon = \varepsilon' M\varepsilon$.

(b) $\mathbf{1}_n = Xc$ for some c, so $H\mathbf{1}_n = HXc = Xc = \mathbf{1}_n$ (Theorem 12.2(c)). Hence $H\bar J_n = \frac{1}{n}H\mathbf{1}\mathbf{1}' = \bar J_n$, and $\bar J_n H = (H\bar J_n)' = \bar J_n$, as H and $\bar J_n$ are symmetric. Then $M\bar J_n = \bar J_n - H\bar J_n = 0$, so $MC_n = M(I_n - \bar J_n) = M$, and $C_nM = (MC_n)' = M$.

(c) Let $U = \mathrm{span}\{\mathbf{1}_n\}$ and $W = \mathcal{C}(X)$; $U \subseteq W$, as $\mathbf{1}_n \in \mathcal{C}(X)$. Then $P_U = \bar J_n$ (Corollary 6.21(a)) and $P_W = H$ (Theorem 12.2(a)). By Theorem 6.25(b), $H - \bar J_n$ is the projection matrix onto $W \cap U^\perp$, so it is symmetric and idempotent of rank $\dim W - \dim U = (k+1) - 1 = k$ (Theorem 6.19(b)). Also $M = I_n - P_W = P_{W^\perp}$ (Theorem 6.19(b)), and $W \cap U^\perp \perp W^\perp$, as $W \cap U^\perp \subseteq W$; so $(H-\bar J_n)M = 0$ (Theorem 6.24). Finally $\hat y - \bar y\mathbf{1} = Hy - \bar J_n y = (H-\bar J_n)y$ (Example 6.22), so $\mathrm{SSR} = \|(H-\bar J_n)y\|^2 = y'(H-\bar J_n)y$ (Theorem 6.19(b)).

Problem 12.2. Let $X \in \mathbb{R}^{n\times p}$ with $\mathbf{1}_n \in \mathcal{C}(X)$, let $x \in \mathbb{R}^n$, let $X_+ = [X\ x]$ be X with the column x added, and let $y \in \mathbb{R}^n$ with $\mathrm{SST} > 0$. Let SSE and R^2 belong to the least-squares fit with X, and SSE_+ and R^2_+ to the fit with X_+. Show that $\mathrm{SSE}_+ \leqslant \mathrm{SSE}$ and $R^2_+ \geqslant R^2$: adding a column cannot decrease R^2.

Solution.

- Every column of X is a column of X_+, so $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$, and both contain $\mathbf{1}_n$.
- The fitted values $\hat y_+$ of X_+ form the vector of $\mathcal{C}(X_+)$ closest to y (Theorems 12.3(b) and 6.15), and the fitted values $\hat y$ of X lie in $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$. Hence $\mathrm{SSE}_+ = \|y - \hat y_+\|^2 \leqslant \|y - \hat y\|^2 = \mathrm{SSE}$.
- SST does not depend on the design, so by Theorem 12.13, $R^2_+ = 1 - \mathrm{SSE}_+/\mathrm{SST} \geqslant 1 - \mathrm{SSE}/\mathrm{SST} = R^2$.

Problem 12.3. Let $X = [\mathbf{1}_n\ X_0] \in \mathbb{R}^{n\times(k+1)}$ have full column rank ($k \geqslant 1$), let x_i' be row i of X_0, let $\bar{x}$ and S be the mean vector and the sample covariance matrix of X_0 (Definition 8.32), and let h_{ii} be the leverages of X.

(a) Show that S is positive definite and that $h_{ii} = \frac{1}{n} + \frac{d_S(x_i,\bar{x})^2}{n-1}$ ($1 \leqslant i \leqslant n$), with d_S the Mahalanobis distance (Definition 8.35).

(b) Deduce that $d_S(x_i,\bar{x})^2 \leqslant (n-1)^2/n$ and that $\sum_i d_S(x_i,\bar{x})^2 = (n-1)k$.

(c) Check (a) for Example 12.6 and for Exercise 12.3.

Solution. (a) $n \geqslant k+1 \geqslant 2$ (Theorem 4.13(a)). Let $S_c = X_0'C_nX_0 = (n-1)S$. By Proposition 12.5, S_c is invertible, hence so is S; S is positive semidefinite (Proposition 8.33(b)), hence positive definite (Corollary 8.15(c)). Row i of X is $(1, x_i')$, and $h_{ii} = (1,x_i')(X'X)^{-1}(1,x_i')'$ (Theorem 1.4(d)). By Proposition 12.5(b), the block product formula, and $x_i'S_c^{-1}\bar{x} = \bar{x}'S_c^{-1}x_i$,

$$h_{ii} = \frac{1}{n} + \bar{x}'S_c^{-1}\bar{x} - 2x_i'S_c^{-1}\bar{x} + x_i'S_c^{-1}x_i = \frac{1}{n} + (x_i-\bar{x})'S_c^{-1}(x_i-\bar{x}) = \frac{1}{n} + \frac{d_S(x_i,\bar{x})^2}{n-1},$$

since $S_c^{-1} = \frac{1}{n-1}S^{-1}$.

(b) $h_{ii} \leqslant 1$ (Theorem 12.2(f)) gives $d_S(x_i,\bar{x})^2 \leqslant (n-1)(1-\frac{1}{n}) = (n-1)^2/n$. Adding (a) over i and using $\sum_i h_{ii} = k+1$ (Theorem 12.2(f)) gives $k+1 = 1 + \frac{1}{n-1}\sum_i d_S(x_i,\bar{x})^2$.

(c) In Example 12.6, $S_c = 4I_2$, $S^{-1} = \frac{3}{4}I_2$, and the rows of C_4X_0 are $(\pm1,\pm1)$; so $d_S(x_i,\bar{x})^2 = \frac{3}{2}$ and $h_{ii} = \frac{1}{4}+\frac{1}{2} = \frac{3}{4}$. In Exercise 12.3, $S_c = 10I_2$, $S^{-1} = \frac{2}{5}I_2$, and the rows (a_i,b_i) of C_5X_0 have $a_i^2+b_i^2 = 5$ ($i \neq 3$) or 0 ($i=3$); so $h_{ii} = \frac{1}{5}+\frac{2}{4} = \frac{7}{10}$ and $h_{33} = \frac{1}{5}$.

Problem 12.4. Let $X \in \mathbb{R}^{n\times p}$ have any rank, $y \in \mathbb{R}^n$, and $\beta^+ = X^+y$ (Theorem 10.15).

(a) Show that β is a least-squares estimate if and only if $X\beta = XX^+y$, and that β^+ is one.

(b) Show that every least-squares estimate β satisfies $\|\beta\|^2 = \|\beta^+\|^2 + \|\beta-\beta^+\|^2$. Deduce that β^+ is the only least-squares estimate of minimum length, and the only one in $\mathcal{C}(X')$.

(c) Find β^+ for Example 12.11.

Solution. (a) XX^+y is the orthogonal projection of y onto $\mathcal{C}(X)$ (Theorem 10.15(d)), and by Theorem 12.3(a),(b) the least-squares estimates are the β with $X\beta$ equal to this projection. $X\beta^+ = XX^+y$, so β^+ is one.

(b) By (a), $X(\beta-\beta^+) = 0$, so $\beta-\beta^+ \in \mathcal{N}(X) = \mathcal{C}(X')^\perp$ (Corollary 6.6). Also $\beta^+ = X^+XX^+y = X^+X\beta^+$ (Theorem 10.15(a)) is the projection of β^+ onto $\mathcal{C}(X')$ (Theorem 10.15(d)), so $\beta^+ \in \mathcal{C}(X')$. Pythagoras (Proposition 6.2) gives the identity, so $\|\beta\| > \|\beta^+\|$ unless $\beta = \beta^+$. If $\beta \in \mathcal{C}(X')$, then $\beta-\beta^+ \in \mathcal{C}(X') \cap \mathcal{C}(X')^\perp = \{0\}$ (Theorem 6.5(b)).

(c) The least-squares estimates are $(-t,\ 7+t,\ 4+t,\ 1+t)'$, and $\mathcal{C}(X') = \mathcal{N}(X)^\perp$ with $\mathcal{N}(X) = \text{span}\{(-1,1,1,1)'\}$ (Corollary 6.6, Theorem 12.10(b)). By (b), β^+ is the estimate orthogonal to $(-1,1,1,1)'$: $t+(7+t)+(4+t)+(1+t) = 0$, $t=-3$, and $\beta^+ = (3,4,1,-2)'$, of squared length 30.

Problem 12.5. In the linear model let $X = [\mathbf{1}_n\ X_0]$ have rank $p = k+1$ ($k \geqslant 1$), $n > p$, and write $\beta = (\beta_0, \beta_*')'$ with $\beta_* = (\beta_1,\ldots,\beta_k)'$. Show that

$$\text{E SSR} = k\sigma^2 + \|C_nX_0\beta_*\|^2,$$

and that $\text{E}(\text{SSR}/k) \geqslant \sigma^2 = \text{E}\,s^2$, with equality if and only if $\beta_* = 0$. (Only second moments are used.)

Solution.

- By Problem 12.1(c), SSR $= y'Py$ with $P = H - \bar{J}_n$ symmetric and idempotent of rank k.
- By Example 11.8 with $\mu = X\beta$, E SSR $= k\sigma^2 + \beta' X'PX\beta = k\sigma^2 + \|PX\beta\|^2$, as $P = P'P$.
- $HX = X$ (Theorem 12.2(c)), so $(H - \bar{J}_n)X\beta = C_nX\beta = C_n(\beta_0\mathbf{1}_n + X_0\beta_*) = C_nX_0\beta_*$, as $C_n\mathbf{1}_n = 0$ (Proposition 1.13).
- Hence $\mathrm{E}(\mathrm{SSR}/k) = \sigma^2 + \|C_nX_0\beta_*\|^2/k \geqslant \sigma^2 = \mathrm{E}\, s^2$ (Theorem 12.16(c)). Equality holds if and only if $C_nX_0\beta_* = X_0\beta_* - (\bar{x}'\beta_*)\mathbf{1}_n = X(-\bar{x}'\beta_*,\ \beta_*')'$ vanishes, where $\bar{x} = \frac{1}{n}X_0'\mathbf{1}_n$; as $\operatorname{rank} X = p$, this means $\beta_* = 0$ (Theorem 4.13(b)).

So the numerator of the overall F statistic estimates σ^2 plus a nonnegative term that vanishes exactly under the hypothesis $\beta_* = 0$.

Problem 12.6. (a) In the one-way layout of Theorem 12.10, use the design $X_1 = [d_1\ \cdots\ d_g]$, with $\beta = (\mu_1, \ldots, \mu_g)'$, the group means, and assume the normal linear model with $n > g$. Show that $H_0\colon \mu_1 = \cdots = \mu_g$ is $R\beta = 0$ for the $(g-1) \times g$ matrix R with rows $(e_l - e_{l+1})'$, of rank $g-1$, and that

$$F = \frac{\sum_l n_l(\bar{y}_l - \bar{y})^2/(g-1)}{\sum_l \sum_{i\in l}(y_i - \bar{y}_l)^2/(n-g)},$$

where $\bar{y}_l$ is the mean of group l, n_l its size, and $i \in l$ runs over its observations.

(b) For Example 12.11, compute F from this formula and from $R\hat{\beta}$, and state when the test of level α rejects H_0.

Solution. (a)

- $R\beta = 0$ says $\mu_l = \mu_{l+1}$ for $l = 1, \ldots, g-1$, that is, H_0. The first $g-1$ columns of R form an upper triangular matrix with diagonal entries 1, which is invertible (Theorem 5.15(a), Theorem 5.19(a)); so these $g-1$ columns are linearly independent, and $\operatorname{rank} R = g-1$, as R has $g-1$ rows (Theorem 4.13(a)).
- By Theorem 12.10(a), X_1 has full column rank g, $X_1'X_1 = \operatorname{diag}(n_1, \ldots, n_g)$ and $X_1'y$ holds the group totals; so $\hat{\beta} = (\bar{y}_1, \ldots, \bar{y}_g)'$, each fitted value is the mean of its group, and SSE $= \sum_l \sum_{i\in l}(y_i - \bar{y}_l)^2$.
- The β with $R\beta = 0$ are $\beta = \mu\mathbf{1}_g$, with $X_1\beta = \mu\mathbf{1}_n$. By Theorem 12.30(a), $\mathrm{SSE}_R = \min_\mu \|y - \mu\mathbf{1}_n\|^2 =$ SST (Example 6.22). As $\mathbf{1}_n \in \mathcal{C}(X_1)$, Theorem 12.13 gives $\mathrm{SSE}_R - \mathrm{SSE} = \mathrm{SSR} = \sum_i(\hat{y}_i - \bar{y})^2 = \sum_l n_l(\bar{y}_l - \bar{y})^2$. Theorem 12.31(a) with $q = g-1$ and $p = g$ gives the formula.

(b) By Example 12.11, $\sum_l n_l(\bar{y}_l - \bar{y})^2 = 30$ and SSE $= 10$, so $F = \frac{30}{2}\big/\frac{10}{3} = \frac{9}{2}$. From $R\hat{\beta}$: the group means are $7, 4, 1$, so $R\hat{\beta} = (3, 3)'$,

$$R(X_1'X_1)^{-1}R' = \begin{pmatrix} \frac{1}{3} + \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} + 1 \end{pmatrix}, \qquad \Big(R(X_1'X_1)^{-1}R'\Big)^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{6} \end{pmatrix}$$

(determinant $\frac{5}{4} - \frac{1}{4} = 1$), and $(3,3)\Big(R(X_1'X_1)^{-1}R'\Big)^{-1}(3,3)' = 9\Big(\frac{3}{2} + \frac{1}{2} + \frac{1}{2} + \frac{5}{6}\Big) = 30$, the same $\mathrm{SSE}_R - \mathrm{SSE}$. The test rejects H_0 if and only if $\frac{9}{2} > F_{2,3,1-\alpha}$ (Theorem 12.31(b)).

Problem 12.7. Regard the normal linear model as a GLM ($b_i(\theta) = \theta^2/2$, $o_i = 0$, $\phi = \sigma^2$), with $\operatorname{rank} X = p$. Show that $W = I_n$, that the likelihood equations are the normal equations, and that one IRLS step from any β gives the least-squares estimate $\hat{\beta}$.

Solution. $b_i'' = 1$, so $W = I_n$, and $\mu = X\beta$, as $b_i'(\theta) = \theta$. The likelihood equations $X'(y - X\hat{\beta}) = 0$ (Theorem 12.39(b)) are the normal equations $X'X\hat{\beta} = X'y$. The working response is $z = X\beta + (y - X\beta) = y$, so the IRLS step (Theorem 12.39(c)) gives $(X'X)^{-1}X'y = \hat{\beta}$, whatever the starting value β.

Problem 12.8. (a) For an observed value $k \in \{0, 1, 2, \dots\}$ let $\ell(\mu) = -\mu + k \log \mu - \log k!$ ($\mu > 0$), the logarithm of $\mathrm{P}(y = k)$ for $y \sim \mathrm{Poisson}(\mu)$. Show that $\sup_\mu \ell(\mu) = -k + k \log k - \log k!$ (with $0 \log 0 = 0$), and deduce that the deviance of a Poisson GLM is

$$D = 2 \sum_{i=1}^{n} \Big(y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \Big)$$

(b) Show that if $\mathbf{1}_n \in \mathcal{C}(X)$, then the MLE satisfies $\sum_i (y_i - \hat{\mu}_i) = 0$, so that $D = 2 \sum_i y_i \log(y_i / \hat{\mu}_i)$.

Solution. (a) If $k \geqslant 1$, Lemma 12.27 gives, for $\mu > 0$, $\ell(\mu) - \ell(k) = k \log \frac{\mu}{k} - \mu + k \leqslant k\Big(\frac{\mu}{k} - 1\Big) - \mu + k = 0$, so the supremum is $\ell(k) = -k + k \log k - \log k!$. If $k = 0$, $\ell(\mu) = -\mu < 0$ increases to 0 as $\mu \to 0$, so the supremum is 0, the value of the formula. Hence, by Definition 12.41, $\ell_i^* - \ell_i(\hat{\mu}_i) = -y_i + y_i \log y_i + \hat{\mu}_i - y_i \log \hat{\mu}_i = y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i)$, and D is twice the sum of these terms.

(b) If $\mathbf{1}_n = Xa$, the likelihood equations (Theorem 12.39(b)) give $\sum_i (y_i - \hat{\mu}_i) = a' X' (y - \hat{\mu}) = 0$.